The Institute of Chartered Accountants of Bangladesh (ICAB)

Strategic Business Management & Leadership

Volume II



Workbook
For CA Advanced Level Exams

CA BANGLADESH

THE INSTITUTE OF CHARTERED ACCOUNTANTS OF BANGLADESH

www.icab.org.bd

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The Institute of Chartered Accountants of Bangladesh

First edition 2009

Second edition 2017

Third edition 2024

Fourth edition 2026

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Questions within the Workbook should be treated as preparation questions, providing you with a firm foundation before you attempt the exam-standard questions.

**** Italic denotes changes***

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Introduction

Learning outcomes

- Explain and analyse an organisation's current position and performance using both financial and non-financial data, presented in different formats, applying appropriate statistical and data analysis tools
- Explain and evaluate the causes and effects of different types of data distributions and data trends using appropriate statistical and data analysis tools, including the implications for business risk; determining and explaining sensitivity in a range of scenarios
- Structure, assimilate and evaluate historic and estimated data in appropriate ways, using appropriate statistical and data analysis tools, to support business decisions
- Analyse and evaluate the key types of business risks using relevant quantitative and qualitative data and assess their implications within a given scenario, for business strategy and corporate reporting disclosures
- Advise on the risks involved in business and organisational plans and show how these risks can be managed by assurance procedures and other forms of risk mitigation, including managing the strategic, operating and financial risks arising from climate change
- Explain the responsibility of those charged with governance for managing risk and assess the role of assurance in risk mitigation
- Assess the impact of risk on a variety of stakeholders
- Explain and assess the various steps involved in constructing a business risk management plan, by establishing context, identifying risks and the assessment and quantification of risk
- Evaluate and explain the limitations of business risk management
- Assess and explain enterprise risk management, evaluating its framework and its benefits; and analyse, structure and assimilate data provided to evaluate business risks under a range of complex scenarios using appropriate statistical and data analysis tools, recognising various types of data bias in a variety of scenarios
- Analyse, structure and assimilate historic and forecast data provided to evaluate performance, position and risk using relevant statistical tools and spreadsheets, recognising the sensitivity of forecasts to underlying assumptions and changes in estimates
- Assimilate, structure and analyse transactions and other granular data provided, using spreadsheets.

Knowledge brought forward

Business Strategy & Technology covered the risk assessment and management process and we revise the main stages briefly in sections 5 and 6.

Syllabus links

A key point about the enterprise risk management process discussed here is its strong links to the strategy-setting process that we have already touched on in earlier chapters.

Examination context

If you have to analyse business decisions or situations, the assessment may well include identification and evaluation of risks. You may need to make recommendations about risk management, either dealing with specific risks or recommending an overall framework that is appropriate for the business.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Business risks** Risk is one of the penalties entities pay for being in business. Every organisation – be it a multinational company or a small sole trader – faces risk every day. In a world where risk and its effects on corporate objectives are viewed with increasing concern, entities need to have an effective risk management programme in place to alleviate some of the more serious effects. Boards also need to manage positive or upside risk in order to maximise shareholder wealth.	**Approach** This chapter starts with a review of the main risks organisations face. Pay close attention to the case studies in the chapter as they show how risk management has been applied (or not applied) in practice. **Stop and think** How difficult is it for directors to assess their risk appetite?	In the exam you could be asked to discuss the risks arising from major strategic decisions. You may need to discuss the threats to reputation arising from adverse events in the organisation, poor attitudes to ethics or poor corporate social responsibility.	**IQ1: Nature and extent of risks** This question is designed to encourage idea generation and to get you thinking about business risk in the real world. You will find many of the exam questions easier if you develop your business acumen through wider business reading.

CHAPTER 7

Topic	Practical significance	Study approach	Exam approach	Interactive questions
2	**Enterprise risk management** Enterprise risk management provides a framework for organisations to deal with risk. The framework is designed to identify potential risk events and to manage these risks in accordance with the organisation's risk appetite	**Approach** Section 2 provides a template that you can use to discuss the effectiveness of an organisation's overall risk management framework. Also note the point highlighted in COSO's ERM framework that it is important for organisations to consider risk when developing strategies and setting their objectives, as well as considering how risks could affect their ability to implement a strategy successfully. **Stop and think** Who determines the risk appetite of an organisation?	In the exam you could be asked to recommend ways in which specific risks can be managed. The exam will not require a detailed analysis of the COSO ERM Framework. Instead you may be asked to identify strengths and weaknesses of the framework an organisation has in place to manage risks. You can refer to the COSO framework to help you generate ideas.	
3	**Risk management responsibilities** Consideration of risk issues should be an integral part of board agendas. The board should ensure that effective control systems are in place through its various committees and processes.	**Approach** In this section you should focus particularly on the responsibilities of senior managers and risk specialists, who have most influence on how effectively an organisation manages its risks. Ensure you understand the key role of the board in setting the tone	In the exam you could be asked to discuss the responsibilities of directors and staff for managing risks.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
		and culture for appropriate for risk management. **Stop and think** How can risk management and control systems be embedded within business processes?		
4	**Stakeholders and risk** An organisation's attitude to risk will be influenced by stakeholder priorities and it is therefore important for an entity to assess how much leverage stakeholders have over it.	**Approach** Make sure you understand the needs and concerns of key stakeholder groups when it comes to risk. **Stop and think** Should shareholder concerns take priority over other stakeholders when managing risk?	Stakeholder concerns and priorities will differ from scenario to scenario. You should be comfortable identifying stakeholder needs so that you can assess if risk is being managed appropriately and recommend improvements where there are any weaknesses.	
5	**Risk assessment** Identifying risks that an organization faces should be a continuous process and organisations should have formal methods to collect information on risk and response. Statistical methods such as the standard deviation may also be used to evaluate particular risks.	**Approach** This section explores the different stages of the risk analysis and management process. **Stop and think** On what basis should a business decide which risks to avoid and which risks to reduce?	It is unnecessary to rote learn and repeat the risk management process in the exam. Instead, you require a general awareness of the activities an organisation must undertake to identify and manage risks so that you can recommend suitable improvements to the process. You may need to interpret statistical information and do	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
			calculations in spreadsheets.	
6	**Risk response** The ways in which organisations may respond to similar risks can differ significantly. Risk responses vary depending on factors such as the potential impact of risk on the organisation and on management's attitude towards risk.	**Approach** This section looks at the different ways organisations can respond to risks. Review the risk response framework and practise applying it to an entity by attempting IQ2. **Stop and think** Consider an organisation you are familiar with. Which risks have they chosen to accept rather than to manage? Why is this the case?	It is important in the exam that you are able to identify, prioritise and suggest responses to a range of risks. Your recommendations should be practical and tailored to the circumstances in the exam scenario.	**IQ2: VSYS** This comprehensive question asks you to assess risks associated with a decision to outsource, recommend how to control the risk and outline the assurance work to be carried out on suppliers. This is a typical SBM&L question drawing on several areas of the syllabus and requiring application of knowledge to a scenario. **IQ3: Budget airline** This short scenario-based question focuses on suitable risk strategies for an airline. Your answer should be practical drawing on your commercial acumen. To help you, download an airline's annual report to review their risks and associated risk management strategies. **IQ4: LP** This is another practical question asking you to

Topic	Practical significance	Study approach	Exam approach	Interactive questions
				explain and manage the risks present in the scenario; the second part of the question introduces some corporate reporting.
7	**ESG and climate risk management** In the last two decades, risks relating to ESG factors have become more prominent, and organisations need to ensure that these are considered in the risk management process.	**Approach** Read through this section to build up an awareness of the types of risk associated with ESG factors and climate change. Review the summary of guidance to enterprise risk management for ESG. **Stop and think** Why have risks relating to ESG become more critical over the past two decades?	Exam questions may require you to identify and evaluate risks, and these could include risks relating to ESG or climate change. Questions may also ask you to discuss the company's approach to managing ESG risks.	

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Business risks

Section overview

- Risk, and internal management's attitude towards it, has a considerable bearing on the way in which different organisations conduct their business – that is, their business strategy.
- The risk of an organisation, whether genuine or perceived, has a direct effect on a firm's cost of capital, the rates of interest it pays on its loans, and therefore the types of projects it can pursue.

1.1 Risk and uncertainty

Risk and uncertainty must always be taken into account in strategic planning. Many areas of risk and uncertainty are exogenous – that is, outside the control of the organisation.

1.1.1 Risk

Risk is sometimes used to describe situations where outcomes are not known, but their probabilities can be estimated. (This is the underlying principle behind insurance.)

1.1.2 Uncertainty

Uncertainty is present when the outcome cannot be predicted or assigned probabilities. For example, many insurance companies exclude 'war damage, riots and civil commotion' from their insurance cover.

1.2 Risk and business

Risk is bound up with doing business.

It may not be possible to eliminate risks without undermining the whole basis on which the business operates, or without incurring excessive costs and insurance premiums. Therefore, in many situations, there is likely to be a level of residual risk that is simply not worth eliminating.

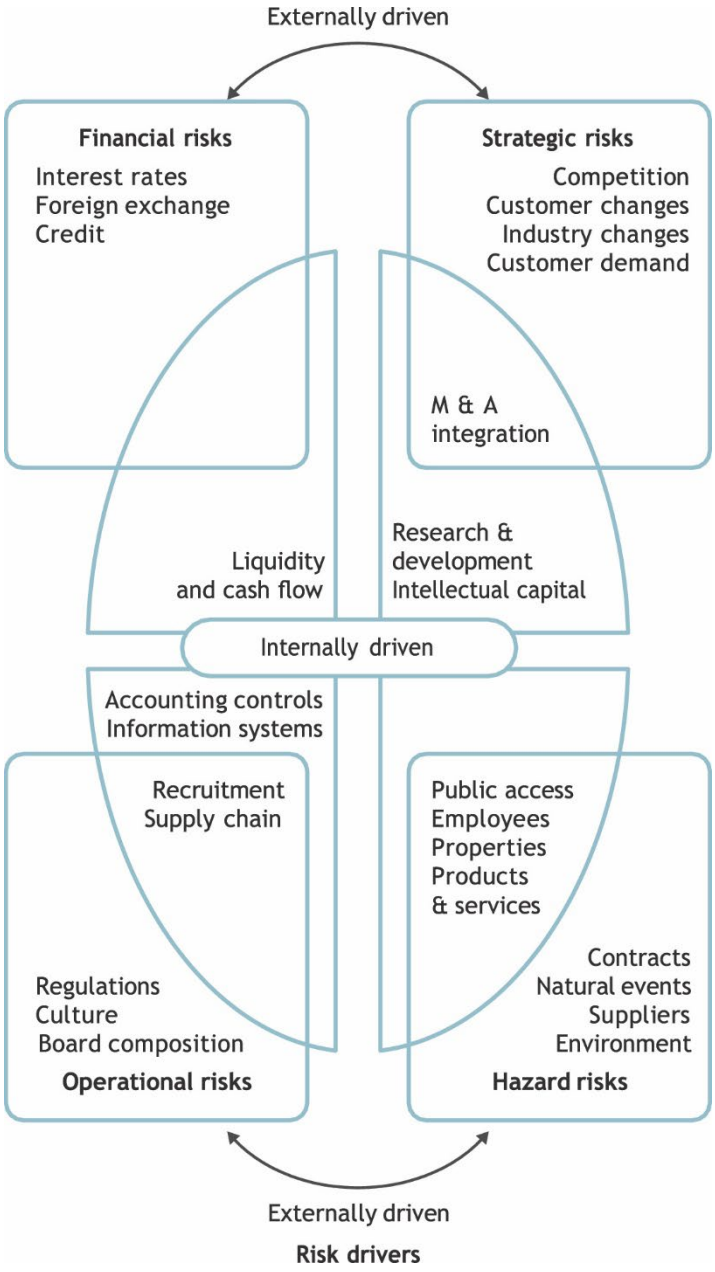
There are some benefits to be derived from the management of risk, possibly at the expense of profits, such as:

- predictability of cash flows
- limitation of the impact of potentially bankrupting events
- increased confidence of shareholders and other investors

However, boards should not just focus on managing negative risks; they should also seek to limit uncertainty and to manage speculative risks and opportunities in order to maximise positive outcomes and hence shareholder value.

In its *Risk Management Standard*, the Institute of Risk Management linked key value drivers for a business with major risk categories.

Figure 7.1: The links between value drivers and risk categories



(Source: Institute of Risk Management – *A Risk Management Standard*)

1.3 Risk and managers

It is worth noting that shareholders and managers have different approaches to risk.

- Shareholders can spread risk over a number of investments.
- Managers' careers tend to be bound up with the success or failure of one particular company, so managers are therefore likely to be more risk-averse than shareholders might be.

1.4 Risk appetite

Since risk management is bound up with strategy, how organisations deal with risk will not only be determined by events and the information available about events, but also by **management perceptions or appetite** to take risk. These factors will also influence risk **culture**, and the values and practices that influence how an organisation deals with risk in its day-to-day operations.

1.4.1 Personal views

Surveys suggest that managers acknowledge the **emotional satisfaction** from successful risk taking, although this is unlikely to be the most important influence on appetite.

1.4.2 Response to shareholder demand

Shareholders demand a level of return that is consistent with taking a certain level of risk. Managers will respond to these expectations by viewing risk taking as a key part of decision-making.

1.4.3 Organisational influences

Organisational influences may be important, and these are not necessarily just a response to shareholder concerns. Organisational attitudes may be influenced by **significant losses** in the past, **changes in regulation and best practice**, or even **changing views** of the benefits that risk management can bring.

1.4.4 National influences

There is some evidence that national culture influences attitudes towards risk and uncertainty. Surveys suggest that attitudes to risk vary nationally according to how much people are shielded from the consequences of adverse events.

1.5 Risk appetite and attitudes



Definitions

Risk appetite: is the nature and strengths of risk that an organisation is prepared to bear.

Risk attitude: is the directors' views on the level of risk that they consider desirable.

Risk capacity: describes the nature and strengths of risk that an organisation is able to bear.

Different businesses will have different attitudes towards taking risk. The risk appetite of directors was discussed in Corporate governance.

Risk-averse businesses are **not** businesses that are seeking to avoid risks. They are businesses that are seeking to obtain sufficient returns for the risks they take. Some risks may be an unavoidable consequence of operating in a particular business sector. However, there will be upper limits to the risks they are prepared to take, whatever the level of returns they can earn.

Risk-seeking businesses are likely to focus on maximising returns and may not be worried about the level of risks that have to be taken.

Most risks must be managed to some extent, and some should be eliminated as being outside the business. Risk management is an integral part of **strategy**, and involves analysing what the **key value drivers are** in the organisation's activities, and the risks tied up with those value drivers.

For example, a business in a high-tech industry, such as computing, which evolves rapidly, has to accept high risks in its research and development activities, so perhaps should not add to that risk by speculating on interest and exchange rates within its treasury activities.

Another issue is that organisations that seek to **avoid risks** (for example, public sector companies and charities) do not need the elaborate and costly control systems that a risk-seeking company (such as those that trade in derivatives, volatile share funds or venture capital companies) may have.

1.6 Conformance and performance

The International Federation of Accountants (IFAC) has highlighted two aspects of risk management which can be seen as linking in with risk aversion and risk seeking.

(a) **Conformance** focuses on controlling pure (only downside) strategic risks. It highlights compliance with laws and regulations, best practice governance codes, fiduciary responsibilities, accountability and the provision of assurance to stakeholders in general. It also includes ensuring the effectiveness of the risk analysis, management and reporting processes, and that the organisation is working effectively and efficiently to achieve its goals.

(b) **Performance** focuses on taking advantage of opportunities to increase overall returns within a business. It includes policies and procedures that focus on alignment of opportunities and risks, strategy, value creation and resource utilisation, and guides an organisation's decision-making.

IFAC guidance states that risk management should seek to **reconcile performance and conformance**
– the two enhance each other. Case studies and surveys commissioned by the IFAC have shown that many people believe that organisations focus too much on compliance, and not enough on strategy and building a business.



Interactive question 1: Nature and extent of risks

In the context of a major confectionery and non-alcoholic beverage company, identify the nature and potential extent of **six** risks that the company might face. (These risks should be specific to the industry in question.)

See **Answer** at the end of this chapter.

1.7 Sources of risk

1.7.1 Sources of risk and uncertainty

Risk	Comment
Physical	Earthquakes, fire, flooding, equipment breakdown. In the long term, climatic changes: global warming, drought (relevant to water firms).
Economic	Assumptions about the economic environment may be incorrect. Not even the government forecasts are always correct.
Business	Lowering of entry barriers (eg, new technology); changes in customer/supplier industries, leading to changed relative power; new competitors and factors internal to the firm (eg, culture); management misunderstanding of core competences; volatile cash flows; uncertain returns; changed investor perceptions, increasing the required rate of return.
Product life cycle	Different risks exist at different stages of the life cycle.

Risk	Comment
Political	Nationalisation, sanctions, civil war, political instability – all these can have an impact on the business.
Financial	Can be affected by changes in interest rates, economic climate, gearing, bad debt risk, liquidity, insolvency.
Reputation	Loss of reputation caused by the adverse consequences of another risk. The loss of reputation will be usually perceived by external stakeholders, and may have serious consequences, depending on the strength of the organisation's relationship with them.

1.7.2 Strategic risks



Definition

Strategic risk: Potential volatility of profits caused by the nature and type of the business's activities.

The most significant risks are focused on the **strategy** the organisation adopts, including concentration of resources, mergers and acquisitions and exit strategies. The market segments that the business chooses will be a significant influence. These will have major impacts on **costs, prices, products and sales** and also the **sources of finance** used. Risks are likely to be greatest for those in start-up businesses or cyclical industries. However, one of the most notable victims of the financial crisis in 2007–8, Lehman Brothers, was not immune to business risks, even after 158 years of operating.

Organisations also need to guard against the risks that **business processes and operations** are **not aligned** to **strategic goals**, or are disrupted by events that are not generated by business activities.

Strategic risks can usefully be divided into:

- threats to profits, the magnitude of which depends on the decisions the organisation makes about the products and services it supplies, and how it supplies them
- threats to profits that are not influenced by the products or services the organisation supplies

Risks to products and services include long-term **product** obsolescence. **Changes in technology** also have long-term impacts if they change the production process, or the way that products and services are distributed to customers (eg, e-commerce versus physical shops). Long-term **macroeconomic changes**, for example a worsening of a country's exchange rate, are also a threat.

Non-product threats include risks arising from the long-term **sources of finance** chosen and risks from a collapse in trade because of an **adverse event**, an accident or natural disaster.

1.7.3 Operational risks



Definition

Operational risk: The risk of loss through a failure of business and internal control processes.

Operational risks include:

- losses from internal control systems or audit inadequacies
- non-compliance with regulations or internal procedures

- information technology failures
- human error
- loss of key-person risk
- fraud
- business interruptions

The main difference between strategic and operational risks is that strategic risks relate to the organisation's **longer-term** place in, and relations with, the **outside environment**.

Operational risks are what could go wrong on a **day to day basis**, and are not generally very relevant to the key strategic decisions that affect a business, although some can have a major impact on the business's future.

(a) For example, the **legal risk of breaching laws in day to day activities** (for example, an organisation's drivers exceeding the speed limit) would be classed as an **operational risk**. However, the legal risk of stricter health and safety legislation forcing an organisation to make changes to its production processes would be classed as a **strategic risk**, as it is a long-term risk impacting seriously on the way the business produces its goods.

(b) The same is true of information technology risks. The risk of a **system failure, resulting in a loss of a day's data**, would clearly be an **operational risk**. However, the risk from using **obsolete technology** would be a **strategic risk**, as it would affect the organisation's ability to compete with its rivals.



Professional skills focus: Assimilating and using information

In order to assess the scale of risk you should use a range of data types and sources to inform your analysis. If an exam scenario presents financial or numerical data it is advisable to use this to provide quantification of the risk, as well as using qualitative analysis to review the impact on the business.

1.8 Business risk and financial risk

1.8.1 Business risk

Business risk, as the name suggests, is the risk associated with the day to day operations of a particular company. It relates to the variability of operating cash flows, the company's exposure to markets, competitors, exchange rates and so on. It is part of the company's overall systematic (or undiversifiable) risk.

1.8.2 Financial risk

Financial risk can be seen from different points of view:

(a) **The company as a whole**. If a company borrows excessively, it may have insufficient funds to meet interest and capital repayments, which may eventually force it into liquidation.

(b) **Lenders**. If a company to whom money has been lent goes into liquidation, lenders may not be paid in full. Companies considered to be a risky investment will be charged higher rates of interest to compensate lenders for the possibility of default.

(c) **Ordinary shareholders**. This group is at the bottom of the list for payment in the event of a company winding up. The lower the profits, and the higher the level of gearing, the greater the risk that is faced by ordinary shareholders.

1.8.3 Relationship between business and financial risk

Business risk is borne by both the firm's equity holders and providers of debt, as it is the risk associated with investing in the firm in whatever capacity. The only way that either party can get rid of the business risk is to withdraw its investment in the firm.

Financial risk, on the other hand, is borne entirely by equity holders, payment to debt holders (ie, interest) taking precedence over dividends to shareholders. The more debt there is in the firm's capital structure, the greater the financial risk to equity holders.

1.9 Compliance risk

Compliance risk is the risk arising from an organisation's failure to comply with laws, regulations, codes of conduct, or standards of practice. Such risks have both an operational and a strategic aspect.

- Operationally, compliance risks are risks that an organisation (and its employees) will fail to comply with established procedures that are designed to ensure compliance with a law or regulation. There may also be a risk that established procedures for achieving compliance are ineffective and do not achieve their intended purpose, so that non-compliance occurs.
- Strategically, compliance risks arise from the possibility that laws or regulations will be changed, so that new compliance procedures must be devised and implemented. In some industries that are heavily regulated, there is also a risk that a regulator will decide that the organisation is failing to comply with regulations.

If an organisation fails to comply with laws and regulations, it could face direct legal consequences (for example, fines or penalties), but compliance risks could also have a wider impact:

- Legal impact: Legal or regulatory action brought against a company, or its management, could result in fines or other financial penalties, or imprisonment.
- Financial impact: The negative impact of regulatory or legal action could affect a company's profits, share price, potential future earnings, and consequently investor confidence in it.
- Reputational impact: The negative publicity arising from non-compliance may damage an organisation's reputation or brand, and could lead to a loss of customer trust in it.
- Business impact: failure to comply with regulatory requirements could lead to an organisation's factories being shut down, or embargoes being placed on its products, thereby significantly disrupting its ability to operate (thereby reinforcing the negative financial impact).

1.9.1 The General Data Protection Regulations (GDPR)

There is growing concern among customers that any data they provide to organisations could be misused, or that their privacy will suffer. This presents a challenge for organisations in determining how best to address their customers' concerns. Organisations also have to comply with the provisions of the General Data Protection Regulations (GDPR), which came into force in May 2018.

As we noted in Corporate governance, GDPR imposes strict requirements over the collection, storage and use of personal data.

GDPR

Companies covered by the GDPR are accountable for their handling of people's personal information. This includes having data protection policies, data protection impact assessments and having relevant documents on how data is processed. Under GDPR, data breaches must be reported to a country's data protection regulator where it could have a detrimental impact. This can include financial loss, confidentiality breaches and damage to reputation.

GDPR is discussed in more detail in Information strategy.

Context example: Facebook and Cambridge Analytica

In early 2018 it was widely reported that Cambridge Analytica, a UK based data analytics and political consulting firm, had improperly gained access to the data of millions of Facebook users.

BBC (2018) reported that the number of people affected by the scandal could be as high as 87 million. The allegations against Cambridge Analytica centred around the alleged use of data from millions of Facebook users to build a system that could target them with personalised political messages, during the UK's Brexit referendum campaign and the 2016 US presidential election, won by Donald Trump.

The way Cambridge Analytica collected the information, without authorisation, violated Facebook's data security policies, and was cited as one of the company's biggest ever data breaches. Facebook users were unaware that companies were selling their data.

Amid claims that Facebook had been aware of the data breach for a number of years, Facebook's CEO Mark Zuckerberg explained that "clearly we should have done more, and we will going forward" (BBC, 2018). He pledged to change Facebook's policy to prevent similar breaches, and published a personal letter in various newspapers apologising for the breach. It is believed the data breach occurred when Facebook users installed an app which captured their data, which was subsequently sold onto Cambridge Analytica.

In the weeks following news of the scandal Facebook's share price fell, and Mark Zuckerberg appeared before the influential US House of Commerce Committee to answer questions relating to the allegations.

In July 2018, the Information Commissioner's Office (which enforces GDPR in the UK) announced that it intended to fine Facebook £500,000 over the data scandal, saying that the company had contravened the law by failing to safeguard people's information.

Source:

BBC (2018). Facebook scandal 'hit 87 million users'. [Online]. Available from: www.bbc.co.uk [Accessed 30 April 2018]

Greenfield, P. (26 March 2018). The Cambridge Analytica files: the story so far. The Guardian. [Online]. Available from: www.theguardian.com [Accessed 30 April 2018]

1.10 Continuous vs event risk

Continuous risk, as the name suggests, is risk that companies face all the time, simply by virtue of being in business. Multinationals, for example, face the continuous risk of foreign currencies moving in the wrong direction and the political risks of operating in different countries. These risks must be continuously monitored as part of the company's general risk management policy.

Event risk is the risk of suffering excessive financial losses due to severe and sudden shocks arising from, for example, human error, natural disasters and stock market crashes. Event risks are difficult to predict but once these events have happened, there will be inevitable consequences, such as liquidity problems. Event risk is often characterised by contagion – that is, one event can precipitate other events whose effects spread across markets and end up affecting everyone. Companies can prepare for event risk by carrying out regular stress testing, which involves generating credible worst case scenarios that show how particular events could affect all relevant markets. It is essential for companies to have crisis management processes in place, covering such crucial areas as communication and leadership.

Context example: The global credit crunch

A credit crunch is a crisis caused by banks being too nervous to lend money to customers or to each other. When they do lend, they will charge higher rates of interest to cover their risk.

By August 2007, credit turmoil had hit financial markets around the world. In September 2007 in the UK, Northern Rock applied to the Bank of England for emergency funding after struggling to raise cash. This led to Northern Rock savers rushing to empty their accounts as shares in the bank plummeted. In February 2008, the UK Chancellor of the Exchequer at the time, Alistair Darling, announced that Northern Rock was to be nationalised.

Years of lax lending on the part of the financial institutions inflated a huge debt bubble as people borrowed cheap money and ploughed it into property. In the US, billions of dollars of 'Ninja' mortgages (no income, no job or assets) were sold to people with weak credit ratings (sub-prime borrowers). The idea was that if these sub-prime borrowers had trouble with repayments, rising house prices would allow them to remortgage their property. In June 2004, following an interest rate low of 1%, rates in the US started to climb and house prices fell in response. Borrowers began to default on mortgage payments and the seeds of a global financial crisis were sown.

The global crisis stemmed from the way in which **debt was sold onto investors**. The US banking sector packaged sub-prime home loans into mortgage-backed securities known as **collateralised debt obligations** (CDOs). These were sold onto hedge funds and investment banks that saw them as a good way of generating high returns. However, when borrowers started to default on their loans, the value of these investments plummeted, leading to huge losses by banks on a global scale.

In the UK, many banks had invested large sums of money in sub-prime backed investments and had to write off billions of pounds in losses. On 22 April 2008, the day after the Bank of England unveiled a £50 billion bailout scheme to aid banks and ease the mortgage market, Royal Bank of Scotland (RBS) admitted that loan losses had hit £1.25 billion in just 6 weeks. In August 2008, RBS reported a pre-tax loss of £691 million (after writing down £5.9 billion on investments hit by the credit crunch) – one of the biggest losses in UK corporate history. At the beginning of 2009, RBS announced that it expected to suffer a loss of up to £28 billion as a result of the credit crunch. On 3 March 2008, it was reported that HSBC was writing off sub-prime loans at the rate of $51 million per day.

In September 2011, the Independent Commission on Banking in the UK issued a report recommending that UK banks' domestic retail operations (operations concerned with customer deposits, business lending and the transmission of money) should be ring-fenced from their wholesale and investment operations. Retail banking activities should be carried out by separate subsidiaries within banking groups, with the ring-fenced part of the bank having its own board and being legally and operationally separate from the parent bank. Ring-fenced banks should have a capital cushion of up to 20%.

Banks were given until 2019 to implement these requirements fully. The time period was set to coincide with the international capital requirements changes being introduced by the Basel regulators.

Under the provisions of the Financial Services (Banking Reform) Act 2013, the UK Treasury has the power to ring-fence the retail operations of large banks from their investment divisions, and to ensure that depositors recover their money before unsecured creditors if a bank becomes insolvent.

1.11 Managing risk in business strategy and financial strategy

Traditionally, management teams tend to be risk averse – that is, they prefer less risk and are prepared to take steps to reduce any potential risks arising from either being in business in general or from specific projects that the company undertakes. The objective of risk management is ultimately to have procedures in place that will reduce these risks to a level that is acceptable to the company and its shareholders. However, setting up and maintaining these procedures takes time, money and human resources, all of which are limited within any organisation.

Risk-averse managers may be willing to accept exposure to greater risks, but only if the expected returns are higher and sufficient to justify the additional risk.

1.12 Corporate reporting consequences

The risks businesses face and the judgements made about those risks have a number of specific consequences for financial reporting.

(a) Under IAS 10, *Events after the Reporting Period*, a business's view of risks will help determine which events are disclosed. Thus, for example, if there are significant exchange rate movements that could result in a risk of material foreign exchange losses, these movements would need to be disclosed. IAS 10 also requires management to make an explicit assessment of the entity's ability to continue as a **going concern** by considering a number of financial, operating and other indicators. Indicative of inability to continue as a going concern would be major restructuring of debt, adverse key financial ratios, substantial sale of non-current assets not intended to be replaced, loss of key staff or major markets. Going concern is considered in Section 2.6 of this chapter in the context of the Risk Guidance.

(b) Part of risk assessment is an analysis of the external business environment, including economic, technological and legal aspects. Adverse changes that are identified may not only require risk management action to be taken; they may also provide evidence of loss of value of assets that needs to be accounted for under the provisions of IAS 36, *Impairment of Assets*.

(c) Risk assessment is significant in a number of ways when applying the requirements of IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*. IAS 37 requires that a transfer of resources embodying economic benefits needs to be assessed as probable, if a provision is to be made. In addition, if the amount of the transfer may be affected by future events, then these should influence how much provision is made, if they are reasonably expected to occur. Allowance is to be made for uncertainty when the amount of a provision is calculated, so that if there are a large number of possible outcomes, the provision is estimated based on its **expected value**, by weighting outcomes with associated probabilities. If risk assessments are revised subsequently, then the amount of the provision may need to be revised as well.

Worked example: Provision

GreenCo prepares its financial statements in accordance with International Financial Reporting Standards. On 25 June 20X0, GreenCo made a public announcement of a decision to reduce the level of emissions of harmful chemicals from its factories. The average useful lives of the factories on 30 June 20X0 (the accounting reference date) was 20 years. The depreciation of the factories is computed on a straight-line basis and charged to cost of sales. The directors formulated the proposals for emission reduction following agreement in principle earlier in the year.

The directors prepared detailed estimates of the costs of their proposals and these showed that the following expenditure would be required.

- £30 million on 30 June 20X1
- £30 million on 30 June 20X2
- £40 million on 30 June 20X3

All estimates were for the actual anticipated cash payments. No contracts were entered into until after 1 July 20X0. The estimate proved accurate as far as the expenditure due on 30 June 20X1 was concerned. When the directors decided to proceed with this project, they used discounted cash flow techniques to appraise the proposed investment. The annual discount rate they used was 8%. The entity has a reputation of fulfilling its financial commitments after it has publicly announced them.
GreenCo included a provision for the expected costs of its proposal in its financial statements for the year ended 30 June 20X0.

Requirements

1 Summarise the criteria that need to be satisfied before a provision is recognised.

2 Explain the decision of the directors of GreenCo to recognise the provision in the statement of financial position at 30 June 20X0.

3 Compute the appropriate provision in the statements of financial position in respect of the proposed expenditure at 30 June 20X0 **and** 30 June 20X1.

4 Compute the two components of the charge to profit or loss in respect of the proposal for the year ended 30 June 20X1. You should explain how each component arises and identify where in the statements of profit or loss and other comprehensive income each component is reported.

Solution

1 **The criteria that need to be satisfied before a provision is recognised**

IAS 37 states that a provision should not be recognised unless:

- an entity has a present obligation to transfer economic benefits as a result of a past transaction or event;
- it is probable that a transfer of economic benefits will be required to settle the obligation; and
- a reliable estimate can be made of the amount of the obligation.

An obligation can be legal or constructive. An entity has a constructive obligation if:

- it has indicated to other parties that it will accept certain responsibilities (by an established pattern of past practice or published policies); and
- as a result, it has created a valid expectation on the part of those other parties that it will discharge those responsibilities.

2 Two of the three conditions in IAS 37 are very clearly met. GreenCo will **incur expenditure** (transfer of economic benefits is virtually certain) and the directors have prepared **detailed estimates** of the amount.

Although GreenCo is not legally obliged to carry out the project, it appears that it has a **constructive obligation** to do so. IAS 37 states that an entity has a constructive obligation if both of the following apply.

(1) It has **indicated to other parties** that it will accept certain responsibilities (by an **established pattern** of past practice or published policies).

(2) As a result, it has created a **valid expectation** on the part of those other parties that it will discharge those responsibilities.

GreenCo has a reputation of fulfilling its financial commitments once they have been publicly announced. Therefore the obligating event is the announcement of the proposal on 25 June 20X0, the obligation exists at 30 June 20X0 (the year-end) and GreenCo is **required to recognise a provision**.

3 Provision at 30 June 20X0:

		£'000
Expenditure on:		
30 June 20X1	30,000 × 0.926	27,780
30 June 20X2	30,000 × 0.857	25,710
30 June 20X3	40,000 × 0.794	31,760
		85,250

Provision at 30 June 20X1:

		£'000
Expenditure on:		
30 June 20X2	30,000 × 0.926	27,780
30 June 20X3	40,000 × 0.857	34,280
		62,060

4

(1) **Provision at 30 June 20X0:**

Depreciation (85,250,000 × 20)	£4,262,500

This is reported in cost of sales.

The provision of £85,250,000 also represents an **asset** as it gives rise to future economic benefits (it enhances the performance of the factories). This is **capitalised and depreciated over 20 years** (the average useful life of the factories).

(2) Unwinding of the discount (see working)	£ 6,810,000

This is reported as a **finance cost**.

WORKING

	£'000
Provision at 1 July 20X0	85,250
Expenditure on 30 June 20X1	(30,000)
Unwinding of discount (balancing figure)	6,810
Provision at 30 June 20X1	62,060

Alternative calculation

Expenditure on:	**£'000**
30 June 20X1 (30,000 – 27,780)	2,220
30 June 20X2 (27,780 – 25,710)	2,070
30 June 20X3 (34,280 – 31,760)	2,520
	6,810

1.12.1 Principles for risk reporting in the annual report

In 2011 the ICAEW Financial Reporting Faculty produced a report entitled *Reporting Business Risks: Meeting Expectations* which sets out seven principles for better risk reporting. The

principles aim for more useful information about risk, assisting investors and other users of corporate reports to form their own judgements. The seven principles are as follows:

(a) Tell users what they need to know, so that they can make their own assessment.

(b) Focus on quantitative rather than qualitative (descriptive) information.

(c) Information on risk should also be integrated with descriptions of business models, forward- looking disclosures, discussion of past performance, and financial reporting.

(d) Many risks stay the same, while others are highly variable and need more frequent updates. The internet, rather than the annual report, might be the best place for such information.

(e) Lists of principal risks should be short, so that they can be fully understood.

(f) Highlight current concerns. It is likely to be of interest to users to know what risks are currently most discussed within a firm.

(g) Review risk experience. Companies should review their experience of risk. What went wrong in the reporting period? What lessons have been learnt? How does experience match up with the risks that were previously reported?



Context example: Infosys – disclosure of key risks

In its 2016–17 annual report, Infosys, an Indian multinational corporation that provides consulting, information technology and outsourcing services, describes its risk management framework.

> "Our industry and company are in significant transformation, and this has naturally resulted in heightening of risks related to strategic choices, strategy execution along with traditional operational and compliance related risks."

The following risk categories are identified:

Strategy

Risks inherent to the industry and competitiveness are analysed and mitigated through strategic choices of target markets, market offerings, business models and the talent base. Potential risks to the long-term scalability and sustainability of the organisation are also analysed and mitigated – for example, societal risks relating to the impact of strategy on the environment, local communities, and conservation of essential resources.

Operational

Risks arising out of internal and external factors affecting policies, procedures, people and systems in support functions which impact service delivery, compromise core values or are not in accordance with generally accepted business practices – for example, risks of business activity disruptions due to natural calamities, terrorist attacks or war or regional conflicts, or disruptions in telecommunications, system failures, virus attacks or breaches of cyber security.

Legal and compliance

Risks arising out of threats posed to financial, organisational, or reputational standing resulting from violations or non-conformance with laws, regulations, codes of conduct or organisational prescribed practices or contractual compliances – for example, risks of potential litigations, breach of contractual agreement, noncompliance with regulations, potential risk arising out of major regulatory/geo- political changes, potential risk arising out of strategic or business or operational decisions.

Later in this chapter, we shall examine the ways in which Infosys managed these risks.

1.12.2 Risk assurance

Changes in the external environment (eg, new regulations; development of new technologies) mean that organisations could face a changing – and increasingly complex – risk environment.

In turn, one of the challenges this presents for boards is making sure they identify the key risks facing their business, and have those risks under control. This could be particularly important in the context of cyber risks, hacking and data breaches where, in line with GDPR requirements, businesses need to be able to demonstrate they have appropriate safeguards in place if a data breach should occur.

An important implication of this is whether a board has adequately assessed the risks its company faces. Potentially, an additional 'risk' could be that a board's assumptions about the risk a company faces do not accurately reflect the risk realities.

In this respect, independent assessment – either by specialist internal auditors, or external advisors – could be very valuable: providing comfort that the board has a complete view of the risks relevant to its business. Having this complete view of the risks helps to ensure that the board is then in a good position to develop appropriate risk mitigation strategies.

Sustainability and climate change

During 2023, IFRS Sustainability Disclosure Standards were issued which require the disclosure of sustainability and climate change-related risks and opportunities relevant to investors. These disclosures are to be presented within the same annual report that contains the audited financial statements and other information required to fulfil legal and regulatory compliance.

Although sustainability- and climate change-related risks and opportunities are expected to follow the methodology adopted by the IFRS Sustainability Disclosure Standards, there will inevitably be crossover between this information and the existing risks disclosures discussed above.

You can learn more about these standards in the section on ESG and climate risk anagement.

2 Enterprise risk management



Section overview

- Enterprise risk management provides a coherent framework for organisations to deal with risk, based on such components as internal environment, strategy- and objective-setting and event identification.
- The framework is designed to identify potential events that may affect the entity and manage risks to be within its risk appetite.

2.1 Nature of enterprise risk management



Definition

Enterprise risk management (ERM): 'The culture, capabilities, and practices that organisations integrate with strategy-setting and apply when they carry out that strategy, with the purpose of managing risk in creating, preserving and realising value.'

(COSO, (2017) Executive Summary: Enterprise Risk Management – Integrating with Strategy and Performance.)

The Committee of Sponsoring Organisations of the Treadway Commission (COSO) is one of the global leaders in developing guidance on risk management and internal controls, designed to improve organisational performance and governance.

In 2004 COSO published its *Enterprise Risk Management – Integrated Framework*, and the underling logic of that framework was that value is maximised when management sets strategy and objectives to strike an optimal balance between growth and return goals and related risks, and efficiently and effectively deploys resources in pursuit of the entity's objectives.

In 2017, COSO published a revised guidance document *Enterprise Risk Management – Integrating with Strategy and Performance* which provides greater insight into the links between strategy, risk and performance. In particular, the revised guidance highlights the need for organisations to consider risk in the strategy-setting process, as well as considering the ways risks could affect an organisation's ability to deliver a strategy successfully.

Traditionally, organisations have viewed enterprise risk management as a way of identifying, assessing and managing risks and threats to their existing strategies. This approach focuses primarily on managing downside risk (threats to executing a strategy successfully). However, COSO highlights that enterprise risk management (ERM) potentially has greater value to organisations through the way it can enhance strategy selection.

Choosing a strategy requires structured decision-making that analyses the potential risks involved in that strategy. As such, ERM can play a very important role in helping organisations deliver value to their stakeholders – by ensuring that management evaluates alternative strategies properly before choosing one. For example, does management understand the risks and opportunities presented by different strategic options? Does the organisation have the resources and capabilities to be able to implement a strategy successfully?

As COSO's definition suggests, one of the key purposes of ERM is to help entities to actively manage risk to acceptable levels in order to achieve objectives and to deliver value.

Every decision an organisation takes either increases, preserves or erodes value. Equally, though, risk is integral to the pursuit of value, and value is a function of risk and return. Consequently, companies shouldn't try to eliminate risk completely (because doing so will prevent them from seizing opportunities which enhance value). Instead they need to manage risk exposure across all of their operations so that they undertake the 'right degree' of the 'right kinds' of risk to pursue their strategy goals effectively.

2.2 COSO's ERM Framework

COSO's ERM Framework is a set of 20 principles, which represent the things an organisation should do as part of its ERM practices. These principles are organised into five groups ('components'):

- governance and culture
- strategy and objective-setting
- performance
- review and revision
- information, communication and reporting

2.2.1 The Components and Principles of COSO's ERM Framework Governance and culture

Governance and culture together form the basis for all the other components of ERM. Governance sets the organisation's tone, including reinforcing the importance of ERM and establishing oversight responsibility for it.

Culture encompasses an organisation's ethical values and desired behaviours, as well as an organisation's understanding of risk. An organisation's culture is typically reflected in its decision- making.

(a) **Exercises board risk oversight** – The board of directors provides oversight of the strategy, for example, reviewing management's proposed strategies and risk appetite to ensure they align with the organisation's mission and core values.

(b) **Establishes operating structures** – The organisation establishes appropriate operating structures in the pursuit of its strategy and business objectives.

(c) **Defines desired culture** – The organisation defines the desired behaviours that characterise its desired culture.

(d) **Demonstrates commitment to core values** – Where the organisation demonstrates a commitment to its core values then this will generally pervade throughout all decision making, improving ethical values and risk management.

(e) **Attracts, develops and retains capable individuals** – The organisation is committed to building human capital, in alignment with its strategy and business objectives.

Strategy and objective-setting

ERM is integrated into an organisation's strategic plans through the process of setting its strategy and business objectives. By gaining an understanding of business context, an organisation can gain insight into internal and external factors (eg, strengths, weaknesses, opportunities, threats) and their impact on risk. An organisation sets its risk appetite in conjunction with setting its strategy. Business objectives allow strategy to be put into practice, and to shape the organisation's day-to-day operations and priorities.

(a) **Analyses business context** – The organisation considers the potential effects of the internal and external environment on its risk profile.

(b) **Defines risk appetite** – The organisation defines risk appetite in the context of creating, preserving and realising value.

(c) **Evaluates alternative strategies** – The organisation evaluates alternative strategies and their potential impact on its risk profile. (The suitability of potential strategies could be influenced by the organisation's risk appetite.)

(d) **Formulates business objectives** – The organisation considers risk while establishing the business objectives, at various levels, that align and support strategy.

Performance

An organisation needs to identify and assess the risks that may affect its ability to achieve its strategy and business objectives. Risks should be prioritised according to their severity, and considering the organisation's risk appetite. The organisation then needs to select appropriate risk responses. In this way, the organisation develops a portfolio view of the amount of risk it has assumed in the pursuit of its strategy and business objectives. The results of this process are reported to key risk stakeholders.

(a) **Identifies risk** – The organisation identifies risk that affects the performance of strategy and business objectives.

(b) **Assesses severity of risk** – An assessment of the severity of each risk will help define a suitable response to each risk.

(c) **Prioritises risks** – The organisation prioritises risks as a basis for selecting responses to risks.

(d) **Implements risk responses** – The organisation identifies and selects appropriate risk responses.

(e) **Develops portfolio view** – The organisation develops and maintains a comprehensive view of all risks and how they could collectively impact across the business.

Review and revision

An organisation needs to review its performance in order to consider how well the ERM components are functioning, over time and in the light of substantial changes (for example, changes in the business environment). The organisation also needs to consider what revisions

may be needed to its strategy and business objectives as a result of the changes, for example investing in new business areas, or scaling back its operations in other areas.

(a) **Assesses substantial change** – The organisation identifies and assesses changes that may substantially affect its strategy and business objectives.

(b) **Reviews risk and performance** – The organisation understands the relationship between risk and future performance so it can take preventative steps to avoid specific scenarios.

(c) **Pursues improvement in enterprise risk management** – An objective of continuous improvement to enterprise risk management improves the organisation's agility to respond to risk events.

Information, communication and reporting

ERM requires a continual process of obtaining and sharing necessary information – from internal and external sources – which flows up, down and across the organisation.

(a) **Leverages information and technology** – The organisation leverages its information systems and technology to capture, process and manage data and information to support ERM.

(b) **Communicates risk information** – The organisation uses communication channels to support ERM by sharing information.

(c) **Reports on risk, culture and performance** – The organisation reports on risk, culture and performance at multiple levels and across the entity.

2.3 Benefits of ERM

COSO's ERM Framework document (2017) highlights the following benefits that organisations can gain from integrating enterprise risk management across their entities. These benefits highlight the fact that risk should not be viewed solely as a constraint or challenge to setting and carrying out a strategy. Instead, the way an organisation responds to challenges could be a source of strategic opportunities and ways for it to differentiate itself from its competitors and achieve a competitive advantage over them.

Increase the range of opportunities. By considering all reasonable possibilities – both positive and negative aspects of risk – management can identify new opportunities, as well as challenges associated with current opportunities.

For example, a food company identified that its primary consumers were becoming increasingly health conscious and changing their diet. This change indicated a potential decline in future demand for the company's current products. In response, management identified ways to develop new products and make existing ones healthier, which allowed the company to maintain revenue from existing customers (preserving value) as well as creating additional revenue by appealing to a wider customer base (creating value).

Identify and manage risk entity-wide. Every organisation faces a number of risks, that can affect different parts of the organisation. Sometimes, a risk can emanate from one part of an organisation, but have an effect on another part. However, by looking across the organisation as a whole, management can identify and manage these risks to help sustain and improve performance.

Increase positive outcomes and advantage while reducing negative surprises. ERM helps an organisation to improve its ability to identify risks and establish appropriate responses, thereby reducing negative surprises (and their related costs or losses) and allowing the organisation to benefit from advantageous developments.

Having discussions about alternatives and possibilities doesn't mean that an organisation won't face challenges and threats, but it should mean the organisation is less surprised by them.

Reduce performance variability. The challenge for some companies has less to do with surprises and more to do with variability in performance. Performing significantly ahead of

schedule or beyond expectations, can cause problems just as performing below expectations does.

For example, a public transportation system can aim for better 'on-time' performance, but if a train departs 10 minutes early this could cause greater problems for passengers than if it departs 10 minutes late (because it could mean passengers miss the train). To manage such variability, transit schedulers build natural pauses into the timetable.

In the same way, ERM allows organisations to anticipate risks of over- or under-performance and take action to minimise disruption.

Improve resource deployment. Having information about risks (and which risks are most important) allows an organisation to assess resource needs and enhance resource allocation.

For example, a gas distribution company recognised that its ageing infrastructure increases the risk of a gas leak occurring. By monitoring trends in gas leak-related data, the company was able to assess the areas of greatest risk across its distribution network. Management subsequently developed a plan to replace worn-out infrastructure, and repair those sections that had useful life remaining. This approach allowed the company to maintain the integrity of the infrastructure while spreading the demand for resources over a longer period of time.

Having a greater focus on resources should mean an organisation uses its resources – money, people, time etc – more efficiently, and in turn this should improve its ability to generate value for its stakeholders.

Enhance resource resilience. An organisation's medium- and long-term viability depends on its ability to anticipate and respond to change, not only to survive but also to evolve and thrive (as the food company did in response to concerns about consumers' changing diets). This ability to respond effectively to change becomes increasingly important as the pace of change accelerates and business complexity increases.

2.4 Risk architecture

In a slightly different framework to that of ERM, the IFAC argued that developing a risk architecture is not just a response to risk but also marks an organisational shift, changing the way the organisation:

- organises itself
- assigns accountability
- builds risk management as a core competence
- implements continuous, real-time risk management

Best practice, the IFAC argued, is to develop a highly integrated approach to risk management, using a common language, shared tools and techniques and periodic assessments of the risk profile for the entire organisation. Integration is particularly important when most units have **many risks in common**, and when there is **significant interdependency** between units. It is vital when managers are trying to achieve a **shared corporate vision**.

The risk architecture developed by the IFAC has eight components:

- acceptance of a risk management framework
- commitment from executives
- establishment of a risk response strategy
- assignment of responsibility for risk management process
- resourcing
- communication and training
- reinforcing risk cultures through human resources mechanisms
- monitoring of the risk management process

The IFAC identified four components of risk management:

- **Structure** – to facilitate the identification and communication of risk
- **Resources** – sufficient to support implementation
- **Culture** – reinforcing decision-making processes
- **Tools and techniques** – developed to enable organisation-wide management of risk

2.5 The Risk Guidance

The UK Turnbull report aimed to provide guidance on risk management and control systems to supplement the outlines set out in the UK Corporate Governance Code. Turnbull emphasised the importance of the **evolution** of a system of internal control to take account of new and emerging risks, control failures, market expectations or changes in the company's circumstances or business objectives. The Turnbull guidance is now contained in the 2014 FRC publication *Guidance on Risk Management, Internal Control, and Related Financial and Business Reporting*.

Internal Control, and Related Financial and Business Reporting (the Risk Guidance). The guidance is primarily directed at companies applying the UK Corporate Governance Code.

2.5.1 Objectives of the Risk Guidance

The objectives of the Risk Guidance are to encourage companies to adopt a **risk-based approach to establishing a system of internal control**, ie, to manage and control risk appropriately rather than eliminate it. This should enable companies to achieve their business objectives while being responsive to the risks they face.

The Risk Guidance emphasises the importance of an embedded and ongoing process of identifying and responding to risks. Thus a company must:

- establish **business objectives**
- identify the **key risks** associated with these
- agree the **controls** to address the risks
- set up a **system to implement the decision**, including regular feedback

The guidance aims to reflect sound business practice, as well as to help companies comply with the internal control requirements of the Code.

2.5.2 Risk management responsibilities of directors, management and employees

The responsibilities of directors, management and employees to implement the Risk Guidance are as follows.

Directors

- Ensure the design and implementation of appropriate risk management and internal control systems.
- Determine the nature and extent of the principal risks faced and the organisation's risk appetite.
- Ensure that appropriate culture and reward systems have been embedded throughout the organisation.
- Agree how the principal risks should be managed or mitigated.
- Monitor and review the risk management and internal control systems, and the management's process of monitoring and reviewing.
- Ensure sound internal and external information and communication processes are in place and communicate with external stakeholders on risk management and internal control.

The Risk Guidance sees a close relationship between the setting of a company's risk appetite, its risk management and internal control, and how it approaches issues of going concern and longer term sustainability. Therefore, as part of the responsibilities outlined above, the board

has responsibility for determining the organisation's going concern XE "Going concern" status and making related disclosures in the financial statements.

Management

- Implement board policies on risk and control.
- Provide the board with timely information.
- Establish clear internal responsibilities and accountabilities at all levels of the organisation.

Employees

- Acquire the necessary knowledge, skills and authority to establish, operate and monitor the system of internal controls.

2.5.3 Going concern: viability statement in UK

The 2018 UK Corporate Governance Code contains a provision requiring a 'viability statement':

> "Taking account of the company's current position and principal risks, the board should explain in the annual report how it has assessed the prospects of the company, over what period it has done so and why it considers that period to be appropriate. The board should state whether it has a reasonable expectation that the company will be able to continue in operation and meet its liabilities as they fall due over the period of their assessment, drawing attention to any qualifications or assumptions as necessary." [Provision 31]

The guidance on the viability statement suggests that the period selected often appears to be based on a company's medium-term business plan. However, the nature of the business, its stage of development, the industry it operates in should all be taken into account when determining an appropriate period, as well as the company's investment and planning periods.

The length of this period will be for the directors to decide. However, the guidance also notes that it may be useful for companies to discuss with investors their information needs to help inform the period selected.

More generally, however, viability statements serve to reiterate the point that the long-term success of a company is dependent on the sustainability of its business model and its management of risk. As the guidance on the viability statement points out: 'Decisions made by the board will have a direct impact on the viability of the company, over differing time periods.'

2.6 Risk resourcing

Whatever the division of responsibilities for risk management, the organisation needs to think carefully about how risk management is resourced; sufficient resources will be required to implement **and** monitor risk management (including the resources required to obtain the necessary information). Consideration will be given not only to the **expenditure** required, but also to the human resources in terms of **skills and experience**.

3 Risk management responsibilities



Section overview

- This section discusses the underlying features of risk and control systems.
- Consideration of risk issues should be an integral part of board agendas.
- The board's risk committee and the risk management function are also key players in managing risk.

- There are various methods that can be used to promote awareness of risk and control issues within a company.
- It is important that risk management and control systems are embedded within business processes.

3.1 Board responsibilities

Ownership of the risk management and internal control system is a vital part of the chief executive's overall responsibility for the company. The chief executive must consider, in particular, the **risk and control environment**.

As well as explicit responsibilities, the board's role in '**setting the tone'** and demonstrating clearly that the directors respect the need for effective control systems is a very important part of risk management. This includes respecting the need for separation of duties between managers carrying out executive duties, and non-executive directors and staff responsible for monitoring them.

3.1.1 Review of board's role

In 2010, the Financial Reporting Council undertook a review of how boards were approaching their responsibilities.

The main points arising from the consultation included the following:

- Boards should aim for **better risk taking**, but this does not necessarily mean less risk taking, as risk taking is essential to entrepreneurship.
- **Different board committees** are appropriate for different industries. A risk committee is appropriate for companies in the financial sector. Separate committees are commonly used by companies in the pharmaceutical and extractive industries, which are exposed to significant safety, environmental or regulatory risks. Examples in these industries include compliance committees and corporate responsibility committees.
- Responsibility for monitoring internal controls and risk management could be delegated to board committees, but the whole board should retain strategic responsibility for risk decision-taking. Boards need to understand how risk exposure might change as a result of changes in strategy and the operating environment.
- Boards need to focus on individual risks capable of undermining the strategy or long-term viability of the company or damaging its reputation. Reputation risk requires greater attention, partly because failures can be publicised widely and quickly in the global information environment. Boards need to have robust crisis management plans.
- Boards should not just focus on net or residual risk, but also need to understand exposure to the combination of risks faced, before risk management policies are implemented.
- It could be difficult to decide how much information about risks boards need and, in particular, when a particular risk should be brought to the board's attention.
- Organisations need transparency and clear lines of reporting and accountability.
- Investors are increasingly seeking more meaningful reporting on risk; for example, an integrated discussion of business model, strategy, key risks and mitigation. Investors also want to know how companies' exposure to risk is changing.

3.2 Risk committee

Boards need to consider whether there should be a separate board committee, with responsibility for monitoring and supervising risk identification and management. If the board doesn't have a separate committee, under the UK Corporate Governance Code, the audit committee will be responsible for risk management.

Consideration of risk certainly falls within the remit of the audit committee. However, there are a number of arguments in favour of having a separate risk committee.

(a) A risk management committee can be **staffed by executive directors**, whereas an audit committee under corporate governance best practice should be staffed by non-executive directors. However, if there are doubts about the competence and good faith of executive management, it will be more appropriate for the risk committee to be staffed by non-executive directors.

(b) As a key role of the audit committee will be to liaise with the external auditors, much of their time could be **focused on financial risks.**

(c) A risk committee can take the lead in **driving changes in practice**, whereas an audit committee will have a largely monitoring role, checking that a satisfactory risk management policy exists.

Having a separate risk committee can aid the board in its responsibility for ensuring that adequate risk management systems are in place. The application of risk management policies will then be the responsibility of operational managers, and perhaps specialist risk management personnel.

3.2.1 Risk committees in the financial sector in UK

The UK Walker Report (published in 2009) recommended that FTSE100 bank or life insurance companies should establish a risk committee. Reasons for this recommendation included the need to avoid overburdening the audit committee, to draw a distinction between the largely backward- looking focus of the audit committee and the forward-looking focus of determining risk appetite; and from this, monitoring appropriate limits on exposures and concentrations. The committee should have a majority of non-executive directors.

Walker recommended that the committee should concentrate on the fundamental prudential risks for the institution: leverage, liquidity risk, interest rate and currency risk, credit/counterparty risks and other market risks. It should advise the board on current risk exposures and future risk strategy, and the establishment of a supportive risk culture.

The committee should regularly review and approve the measures and methodology used to assess risk. A variety of measures should be used. The risk committee should also advise the remuneration committee on risk weightings to be applied to performance objectives incorporated within the incentive structure for executive directors.

3.3 Risk management personnel

3.3.1 Risk manager

The risk manager will need technical skills in **credit, market and operational risk**. Leadership and persuasive skills are likely to be necessary to overcome resistance from those who believe that risk management is an attempt to stifle initiative. The risk manager is typically responsible for:

(a) **Providing the overall leadership, vision and direction** for enterprise risk management (ERM).

(b) **Establishing an integrated risk management framework** for all aspects of risk across the organisation, integrating ERM with other business planning and management activities, and framing authority and accountability for ERM in business units.

(c) **Promoting an ERM competence** throughout the entity, including facilitating the development of technical ERM expertise, helping managers align risk responses with the entity's risk tolerances and developing appropriate controls.

(d) **Developing risk management policies**, including the quantification of management's risk appetite through specific risk limits, defining roles and responsibilities, ensuring

compliance with codes, regulations and statutes, and participating in setting goals for implementation.

(e) **Establishing a common risk management language** that includes common measures around likelihood and impact, and common risk categories. Developing the analytical systems and data management capabilities to support the risk management programme.

(f) **Implementing a set of risk indicators and reports** including losses and incidents, key risk exposures and early warning indicators. Facilitating managers' development of reporting protocols, including quantitative and qualitative thresholds, and monitoring the reporting process.

(g) **Dealing with insurance companies**: An important task because of increased premium costs, restrictions in the cover available (will the risks be excluded from cover?) and the need for negotiations with insurance companies if claims arise. If insurers require it, demonstrating that the organisation is taking steps to actively manage its risks. Arranging financing schemes such as self-insurance or captive insurance.

(h) **Allocating economic capital to business activities** based on risk, and optimising the company's risk portfolio through business activities and risk transfer strategies.

(i) **Reporting to the chief executive on progress** and recommending action as needed. Communicating the company's risk profile to key stakeholders such as the board of directors, regulators, stock analysts, rating agencies and business partners.

3.3.2 Risk management function

Larger companies may have a bigger risk management function whose responsibilities are wider than a single risk manager. The Institute of Risk Management's *Risk Management Standard* lists the main responsibilities of the risk management function:

- setting policy and strategy for risk management
- primary champion of risk management at a strategic and operational level
- building a risk-aware culture within the organisation, including appropriate education
- establishing internal risk policy and structures for business units
- designing and reviewing processes for risk management
- coordinating the various functional activities which advise on risk management issues within an organisation
- developing risk response processes, including contingency and business continuity programmes
- preparing reports on risks for the board and stakeholders

3.3.3 Assurance procedures

The assurance work carried out by internal audit will play a significant part in the organisation's risk management processes, internal audit being required to assess and advise on how risks are countered. Internal auditors will be concerned to see that managers have made adequate responses to risks, have designed robust risk management processes and that these mitigate the risks. This approach can be refined to focus on particular areas, for example start-ups and other future- orientated activities, where core controls have not developed and which thus carry higher risks.

The starting point for a risk audit is to **identify business objectives** and the **risks** that may prevent the organisation from achieving those objectives. Internal audit's work will be influenced by the organisation's **appetite** for bearing risks. Internal audit will assess:

- the **adequacy of the risk management and response processes** for identifying, assessing, managing and reporting on risk
- the risk management and control **culture**

- the **appropriateness of internal controls** in operation to **limit risks**
- the **operation and effectiveness** of the **risk assessment and management processes**, including the internal controls, with a focus on processes aimed at risks that are classified as key risks
- the **reliability of reporting on risks and controls**

The areas that auditors will concentrate on will depend on the **scope** and **priority** of the assignment and the **risks identified**. Where the risk management framework is insufficient, auditors will have to rely on their own **risk assessment** and will focus on **recommending an appropriate framework**.

Where a framework for risk management and control is embedded in operations, auditors will aim to use **management's assessment of risks** and concentrate on **auditing the risk management processes**.

A key part of internal audit's role in control systems is to provide feedback that influences the **design and operation** of **internal control systems**. Internal audit recommendations need to be seen in the context of the organisation's **strategic objectives** and **risk appetite**.

3.4 Procedures for embedding risk awareness

Employees cannot be expected to avoid risks if they are not aware that they exist in the first place. Embedding a risk management frame of mind into an organisation's culture requires top-down communications on what the risk philosophy is and what is expected of the organisation's employees.



Context example: Internal communications programme

Here is an example of an internal communications programme slightly adapted from an example in the original (2004) COSO framework.

Internal communications programme

- Management discusses risks and associated risk responses in regular briefings with employees.
- Management regularly communicates entity-wide risks in employee communications such as newsletters, or via the intranet.
- ERM policies, standards and procedures are made readily available to employees, along with clear statements requiring compliance.
- Management requires employees to consult with others across the organisation as appropriate when new events are identified.
- Induction sessions for new employees include information and literature on the company's risk management philosophy and ERM programme.
- Existing employees are required to take workshops and/or refresher courses on the organisation's ERM initiatives.
- The risk management philosophy is reinforced in regular and ongoing internal communication programmes and through specific communication programmes to reinforce tenets of the company's culture.

3.4.1 Human resource procedures

There are also organisational measures which can be taken to spread ownership of risk management.

(a) Management need to make sure that staff receive clear and consistent messages that **managing risk is part of all employees' daily responsibilities**, and is very important to an organisation's success and survival.

(b) Personnel should understand the need to **resist pressure from superiors** to participate in improper activities, and channels outside normal reporting lines should be available to permit reporting such circumstances.

(c) Managers should **provide appropriate incentives**. This may entail setting performance targets and tying results to performance pay.

3.4.2 Training

Aside from practical matters like showing employees which buttons to press or how to find out the information they need, training should include **explanations** of why things should be done in the way that the trainer recommends. If employees are asked to carry out a new type of check but are not told why, there is every chance that they won't bother to do it, because they don't understand its relevance. Instead, it will just seem to mean more work for them and to slow up the process for everyone.

3.4.3 Risk policy statement

Organisations ought to have a statement of risk policy and strategy that is distributed to all managers and staff.

3.5 Control systems

The 2014 FRC guidance on risk management emphasises the importance of control systems in effectively managing risks. Internal control consists of the policies, processes, tasks, behaviours and other aspects of a company that when taken together:

(a) **Facilitate its effective and efficient operation** by enabling it to respond appropriately to significant business, operational, financial, compliance and other risks to achieving the company's objectives. This includes the **safeguarding of assets** from inappropriate use or from loss and fraud and ensuring that **liabilities are identified and managed**.

(b) **Help ensure the quality of internal and external reporting**. This requires the maintenance of proper records and processes that generate a flow of timely, relevant and reliable information from within and outside the organisation.

(c) **Help ensure compliance with applicable laws and regulations**, and also with internal policies with respect to the conduct of business.

Internal control systems should:

- be embedded in the operations of the company and form part of its culture
- be capable of responding quickly to evolving risks within the business
- include procedures for reporting immediately to management, significant control failings and weaknesses, together with control action being taken

The system should include control activities, information and communication processes and methods for monitoring the continued effectiveness of the system of internal control. A sound system of internal control reduces, but does not eliminate, the possibilities of losses arising from poorly judged decisions, human error, deliberate circumvention of controls, management override of controls and unforeseeable circumstances.



Context example: Three lines of defence model

The three lines of defence model is one approach to safeguarding the internal control framework.

First line of defence

This describes the **controls** an organisation has in place to deal with its day to day business. Controls are designed into systems and processes and compliance with process should ensure an adequate control environment.

Second line of defence

This describes the **committees and functions** that are in place to provide an oversight of the effective operation of the internal control framework. The effectiveness of the second line is determined by the oversight committee structure, their terms of reference, the competence of the members and the quality of the management information and reports that are considered by these oversight committees.

This second line is reinforced by the advisory and monitoring functions of risk management and compliance. Risk management defines the financial and operational risk assessment processes for the business; maintains the risk registers; and undertakes regular reviews of these risks. Compliance advises on all areas of regulatory principles, rules and guidance.

These functions report on their work undertaken and significant findings to the appropriate executive risk oversight committees in the second line.

Third line of defence

This describes the **independent assurance** provided by the **board audit committee**, a committee of non-executive directors chaired by the senior independent director, and the **internal audit function** that reports to that committee.

Internal audit undertakes a programme of risk-based audits covering all aspects of both the first and second lines of defence. Internal audit may well take some assurance from the work of the second line functions and reduce or tailor its checking of the first line.

4 Stakeholders and risk



Section overview

Organisations' attitudes to risks will be influenced by the priorities of their stakeholders and how much influence stakeholders have. Stakeholders that have significant influence may try to prevent an organisation bearing certain risks.

We discussed generally in Strategic analysis how stakeholders can influence objectives and strategies. This section focuses on the impacts that stakeholders can have on the strategies that businesses develop for managing their risks.

Businesses have to be aware of stakeholder responses to risk. They may take actions, or events could occur, that may generate a response from stakeholders. This response could have an adverse effect on the business.

To assess the importance of stakeholder responses to risk, the organisation needs to determine how much leverage its stakeholders have over it.

4.1 Shareholders

Shareholders can affect the market price of shares by selling them. They also have the power to remove management. It would appear that the key issue for management to determine is whether shareholders:

- **Prefer a steady income from dividends**, in which case they will be alert to threats to the profits that generate the dividend income, such as investment in projects that are unlikely to yield profits in the short term.
- **Are more concerned with long-term capital gains**, in which case they may be less concerned about a short period of poor performance, and more worried about threats to long-term survival that could diminish or wipe out their investment.

However, the position is complicated by the different risk tolerances of shareholders themselves. Some shareholders will, for the chances of a higher level of income, be prepared to bear greater risks that their investments will not achieve that level of income. Therefore, some argue that because the shares of listed companies can be freely bought and sold on stock exchanges, if a company's risk profile changes, its existing shareholders will sell their shares, but the shares will be bought by new investors who prefer the company's new risk profile. The theory runs that it should not matter to the company who its investors are.

However, this makes the assumption that the investments of all shareholders are actively managed and that shareholders seek to reduce their own risks by diversification. These are not necessarily true in practice. It is also unlikely that the directors will be indifferent to who the company's shareholders are.

Shareholders' risk tolerance may depend on their views of the organisation's risk management systems, how effective they are and how effective they should be. Shareholder sensitivity to this will increase the pressures on management to ensure that a risk culture is **embedded** within the organisation.

4.2 Debt providers and creditors

Debt providers are most concerned about threats to the amount the organisation owes. They can take various actions, with potentially serious consequences such as denial of credit, higher interest charges and, ultimately, putting the company into liquidation.

When an organisation is seeking credit or loan finance, it will obviously consider what action creditors will take if it does default. However, it also needs to consider the ways in which debt finance providers can limit the risks of default by, for example, requiring companies to meet certain financial criteria, provide security in the form of assets that can't be sold without the creditors' agreement, or personal guarantees from directors.

These mechanisms may have a significant impact on the development of an organisation's risk strategy. There may be a conflict between strategies that are suitable from the viewpoint of the business's long-term strategic objectives, but are unacceptable to existing providers of finance because of threats to cash flows, or are not feasible because finance suppliers will not make finance available for them, or will do so only on terms that are unduly restrictive.

4.3 Employees

Employees will be concerned about threats to their job prospects (money, promotion, benefits and satisfaction) and, ultimately, threats to the jobs themselves. If the business fails, the impact on employees will be great. However, if the business performs poorly, the impact on employees may not be so great if their jobs are not threatened. Employees will also be concerned about threats to their personal wellbeing, particularly health and safety issues.

The variety of actions employees can take include pursuit of their own goals rather than shareholder interests, industrial action, refusal to relocate or resignation.

Risks of adverse reactions from employees will have to be managed in a variety of ways:

(a) Legislation requires that some risks, principally threats to the person, should be avoided.

(b) Businesses can limit employee discontent by good pay, conditions etc.

(c) Businesses can take out insurance against key employees leaving. However, they may decide to accept that some employees will be unhappy but believe the company will not suffer a significant loss if they leave.

4.4 Customers and suppliers

Suppliers can provide (possibly unwillingly) short-term finance. As well as being concerned with the possibility of not being paid, suppliers will be concerned about the **risk of making unprofitable sales**. Customers will be concerned with **threats to their getting the goods or services** that they have been promised, or not getting the value from the goods or services that they expect.

Suppliers can respond to risk of non-payment by refusing credit. Customers can shop elsewhere. The impact of customer-supplier attitudes will also depend on how much the organisation wants to **build long-term relationships with them**. A desire to build relationships implies involvement of the staff that are responsible for building those relationships in the risk management process. It may also imply a greater degree of disclosure about risks that may arise to these long-term partners in order to maintain the relationship of trust.

4.5 The wider community

Governments, regulatory and other bodies will be particularly concerned with risks where the organisation does not act as a **good corporate citizen**, implementing, for example, poor employment or environmental policies. A number of the resulting actions that might be taken could have serious consequences. Government can impose tax increases or regulation or take legal action. Pressure group tactics can include publicity, direct action, political sabotage or campaigning.

Although the consequences can be serious, the risks that the wider community are concerned about are rather less easy to predict than for other stakeholders, being governed by varying political pressures. This emphasises the need for careful monitoring of changing attitudes and likely external responses to the organisation's actions as part of the risk management process.

Professional skills focus: Concluding, recommending and communicating

When formulating recommendations to deal with risk it is important to consider the reaction of different stakeholder groups. Organisations do not operate in isolation so your recommendations should be pragmatic taking into account the influence of powerful stakeholder groups on any strategies that an organisation implements.

CHAPTER 7

5 Risk assessment

Section overview

- This section covers a commonly used framework for assessing and managing risk.
- The initial stage will include establishing both the internal and external contexts, the risk criteria and the structure for risk analysis.
- Identifying the risks that the organisation faces should be a continuous process. As the business environment changes, so do the risks faced by organisations operating in that environment.

- The fundamental difficulty in assessing risk is determining how often this particular risk may occur. Information may not be available on all past events.
- The other main element of risk assessment is assessing the impact on the organisation if a risk materialises.
- For physical assets, quantification is often fairly straightforward. However, exposure of financial and intangible assets is more difficult to put a monetary figure on.
- Grouping of risks according to their likelihood and potential impact supports the establishment of priorities for risk mitigation.
- The individual risks that have been identified in different parts of the business must be consolidated (ie, aggregated) to establish the risk at the corporate level.

5.1 The risk management process

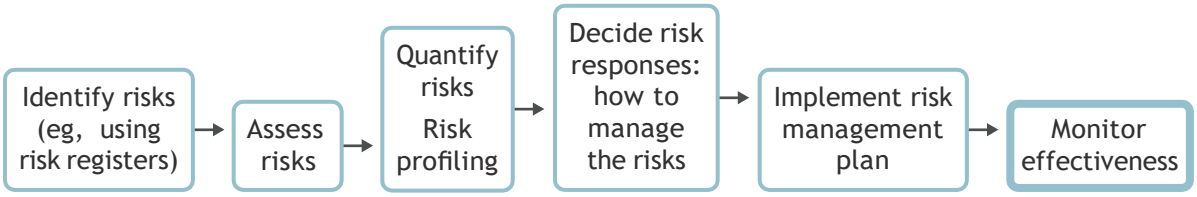
A commonly used framework for assessing and managing risk involves the following processes.

- establishing the context
- risk identification
- risk assessment
- risk quantification
- risk profiling
- risk consolidation
- risk responses

In real life, none of these processes are easy – what is considered to be a risk by some might not be seen as such by others, which can lead to disputes over which risks should be given priority.
Quantification of an item that is essentially subjective is never going to be straightforward – in itself, putting a value on different risks is a subjective process.

Having devised responses to risk (making decisions about how risks should be managed), the risk management plan should be implemented. Once implemented, there should be regular monitoring of the risk management system to check that it is working effectively, as planned.

Figure 7.2: The risk management process



5.2 Risk register

Organisations should have formal methods of collecting information on risk and response. A risk register **lists and prioritises the main risks** an organisation faces, and is used as the basis for decision-making on how to deal with risks. It acts as a basic component of the risk and control assessment process in order to record identified risks. The existence of a risk register illustrates that risks have been identified and to what activities they relate.

The risk register will typically include the following:

- description of the risk
- when the risk might occur
- the impact, assuming that the risk does occur
- an assessment of the risk's likelihood or probability of occurrence
- a priority rating or score, obtained from the impact and probability assessment
- the management's strategy as to how the risk will be addressed
- the containment strategy, defining what exactly will happen if the risk occurs

In compiling the risk register, the firm must describe the risk in a clear manner so that everyone in the firm who needs to be aware of the risk understands it, not just the person who is most directly involved in its management.

A risk register would take an individual event and state what the **loss of value** might be, should the risk occur, and then will apply an impact assessment to it.



Context example: Risk register

Consider, for example, a case of potential fraud where the risk impact might be assessed at say, £5 million with a 15% chance of its occurrence in any one year. Multiplying these together (£5m × 15%) would give a severity of £0.75 million in any one year. Of course, this would not mean that the company would expect a loss of £5 million or even £0.75 million. The actual cost would be based on the distribution of losses. So the loss could be at least £0.25 million with a 75% likelihood, or up to £2 million with a 25% likelihood.

Any assessments are based on actual knowledge from real loss events that have occurred, together with additional assessment made by management and risk specialists regarding the quality of the control environment, along with control and risk self-assessment key performance indicators.

The severity could be calculated by assessors who might be asked to choose from a number in the range of 1 to 10 for each of the risk issues that they are addressing. These expectation (or likelihood) levels would then be multiplied to come up with a severity rating. Hence, **expectation** multiplied by **impact** would deliver the severity figure.

The risk manager, or the person administering the risk register and the assessment, would be the person who would then take the different expectations and convert them into something more meaningful in terms of real values, rather than levels of severity. An expectation of 6 could equate to, say, 30% and an impact of 5 could equate to £10 million. This, then, would give the severity of £3 million ie, 30% of £10 million.

5.3 Establishing the context

Before any risk can actually be identified, the context within which that risk will be assessed must be established. For example, management may only be interested in identifying financial risks, which means that those responsible for gathering information will concentrate on that area of risk only.

5.3.1 Establishing the internal context

Risk is essentially the chance that an event will occur which will prevent the company from meeting its objectives. Therefore, in order to understand the risks, you must first identify the objectives. By doing so you will ensure that decisions taken to reduce risk still support the overall goals of the organisation, thus encouraging long-term and strategic thinking.

Context example: Owner-managed restaurant

A small restaurant employs a full-time chef, an apprentice chef and two waiters. Business is slow and cash flow is becoming a problem, so the owner decides that he will therefore have to let the apprentice chef and one of the waiters go.

Four months later, the restaurant wins a five-year contract to supply lunches to a local firm and business in general picks up, with the opening of a new office block in close proximity to the restaurant's premises. The restaurant owner was in the process of preparing for the new contract, having tendered for it several months before paying off the apprentice chef and the waiter, and was also aware of the imminent opening of the office block.

The owner then had problems of a different kind. The chef could not cope with both the contracted lunches and the general improvement in business, nor could the remaining waiter deal with a full restaurant on his own. The apprentice chef had already found alternative employment, as had the previous waiter. The owner was forced to take on a less experienced apprentice, who then had to be trained by the already overworked chef and another waiter, who was unfamiliar with the general operations of the restaurant.

In this context, the restaurant owner would have been better off trying to find an alternative means of dealing with what he must have known to be a temporary cash flow problem, rather than getting rid of staff in whose training time and money had already been invested.

When trying to establish the internal context, business owners should also consider such issues as:

- internal culture: are staff likely to be resistant to change?
- existing business capabilities, such as people, equipment and processes

5.3.2 Establishing the external context

The external context is the overall environment in which the business operates, including an understanding of the perceptions that clients or customers have of the business. This could take the form of a SWOT analysis. It should also cover such issues as external regulations that the business must comply with.

5.3.3 Establishing the risk management context

In order to correctly identify risks associated with a project, you must first define the project's limits, objectives and scope.

Context example: Accountancy body

Due to expansion of staff and necessary documentation, the student support and education departments of a professional accountancy body are moving into larger office premises. Before the move takes place, the head of administration, who is taking charge of the move, undertakes a risk assessment of the relocation.

The risk management context includes:

- the main objective: to move staff, furniture, equipment and documentation to the new premises with minimum disruption to student services and production of examination papers
- ensure the security and confidentiality of the examination papers held at the current premises which will be transferred in locked safes to the new location
- an overall timeframe (including planning and liaison with telephone companies) of three months
- a budget of £25,000 for the use of external relocation support

This risk management context provides sufficient information with which the head of administration can assess the risks associated with the relocation, particularly those that impact on the primary objective of minimising disruption to student services and the production of examination papers.

5.3.4 Developing risk criteria

This step allows the business to identify unacceptable levels of risk or, looking at it another way, to define acceptable levels of risk for a specific project. These risk levels can be more closely defined as the process progresses.

In the case study above, for example, it would be completely unacceptable for the confidentiality and security of the examination papers to be compromised, so this documentation must therefore be kept in locked safes and transported using professional safe movers.

5.3.5 Defining the structure for risk analysis

The final stage in the establishment of context is to define the structure for risk analysis. This involves isolating the risk categories that need to be managed, which can then be assessed individually. This will allow for greater depth and accuracy when identifying important risks.

5.4 Risk identification

No one can manage a risk without first being aware that it exists. Some knowledge of perils, what items they can affect and how, is helpful to improve awareness of whether **familiar risks** (potential sources and causes of loss) are present, and the extent to which they could harm a particular person or organisation. The risk manager should also keep an eye open for **unfamiliar risks** which may be present.

Actively identifying the risks before they crystallise makes it easier to think of methods that can be used to manage them.

Risk identification is a **continuous process**, so that new risks and changes affecting existing risks may be identified quickly and dealt with appropriately, before they can cause unacceptable losses.

5.4.1 Risk conditions

Means of identifying conditions leading to risks (potential sources of loss) include:

(a) **Physical inspection**, which will show up risks such as poor housekeeping (for example, rubbish left on floors, for people to slip on and to sustain fires)

(b) **Enquiries**, from which the frequency and extent of product quality controls and checks on new employees' references, for example, can be ascertained

(c) **Checking** a copy of every letter and memo issued in the organisation for early indications of major changes and new projects

(d) **Brainstorming** with representatives of different departments

(e) **Checklists** ensuring risk areas are not missed

(f) **Benchmarking** against other sections within the organisation or external experiences

5.4.2 Event identification

A key aspect of risk identification is identifying events that could affect the implementation of strategy or achievement of objectives.

Events analysis includes identification of:

(a) **External events** such as economic changes, political developments or technological advances

(b) **Internal events** such as equipment problems, human error or difficulties with products

(c) **Leading event indicators**. By monitoring data correlated to events, organisations identify the existence of conditions that could give rise to an event; for example, customers who have balances outstanding beyond a certain length of time being very likely to default on those balances.

(d) **Trends and root causes**. Once these have been identified, management may find that assessment and treatment of causes is a more effective solution than acting on individual events once they occur.

(e) **Escalation triggers**, certain events happening or levels being reached that require immediate action

(f) **Event interdependencies**, identifying how one event can trigger another and how events can occur concurrently. For example, a decision to defer investment in an improved distribution system might mean that downtime increases, and operating costs go up.

Once events have been identified, they can be classified horizontally across the whole organisation and vertically within operating units. By doing this, management can gain a better understanding of the interrelationships between events, gaining enhanced information as a basis for risk assessment.

5.4.3 Key risk indicators

COSO provides guidance on key risk indicators (KRIs), metrics that some organisations use to provide an early signal of increasing risk exposure. At their simplest, they can be **key ratios** that management uses as indications of evolving problems requiring actions. Sometimes they may be more elaborate, involving the aggregation of several risk indicators.

KRIs are derived from specific events or root causes that can prevent performance goals from being achieved. Examples include the introduction of a new product by a competitor, a strike at a supplier's plant, proposed changes in the regulatory environment and input-price changes. Importantly though, KRIs are measures designed to **identify changes to existing risks**. Therefore it is important to distinguish risk indicators from performance measures, which are typically retrospective in nature.



Context example: Key risk indicator

A government agency wants to retain competent staff. To support this aim, it has a target to maintain staff turnover rates at less than 5% per year.

A key risk indicator for the agency would be a percentage of staff eligible to retire within five years. Anything higher than 5% indicates a risk to the target.

A key performance indicator is the actual turnover, because this is based on historical performance, rather than providing an early signal of potential future risks.

The guidance points out that effective KRIs should be developed by risk managers and business unit managers working together. They should be developed in concert with strategic plans. Determining the frequency of reporting of KRIs will be important – operational management may need to see them in real time; senior management on a less frequent, aggregated basis.

The guidance highlights the following elements of well-designed KRIs:

- based on established practices or benchmarks
- developed consistently across the organisation
- provide an unambiguous and intuitive view of the highlighted risk
- allow for measurable comparisons across time and business units
- provide opportunities to assess the performance of risk owners on a timely basis
- consume resources effectively

5.5 Risk assessment

It is not always simple to forecast the financial effect of a possible disaster, as it is not until **after** a loss that extra expenses, inconveniences and loss of time can be recognised. Even then, it can be difficult to identify all of them.

Organisations will probably keep more detailed records of their activities and the unit costs involved, but it is unlikely that any organisation can predict the full cost of every loss that might befall it with certainty.

5.6 Risk quantification

Risks that require more analysis can be quantified, where possible results or losses and probabilities are calculated and distributions or confidence limits added on. From this exercise is derived the following key data to which the organisation could be exposed by a particular risk:

- average or expected result or loss
- frequency of losses
- chances of losses
- largest predictable loss

The risk manager must also be able to estimate the effects of each possible cause of loss, as some of the effects that they need to consider may not be insured against.

The likely frequency of losses from any particular cause can be predicted with some degree of confidence, from studying available records. This confidence margin can be improved by including the likely effects of changed circumstances in the calculation, once they are identified and quantified. Risk managers must therefore be aware of the possibility of the increase of an existing risk, or the introduction of a new risk, affecting the probability and/or possible frequency of losses from another cause.

Often, quantification of losses will not involve statistical techniques, but a simple single estimate of what would be lost if adverse events or circumstances occur. For example, if an accountancy firm had a client that generated a fixed fee each year, the loss would be their contribution (fees lost less labour and other variable costs saved).

Ultimately, the risk manager will need to know the frequency and magnitude of losses that could place the organisation in serious difficulty.

Professional skills focus: Structuring problems and solutions

Businesses are exposed to a wide range of risks and constantly have to make decisions about which risks to prioritise. In the exam you may be tested on your ability to identify risks within a scenario and to prioritise key issues in order to make the best use of an organisation's resources.

5.6.1 Exposure of physical assets

Exposures with physical assets may include:

- **total value of the assets**, for example, the value of items stolen from a safe
- **costs of repair** if, for example, an accident occurs
- **change of value of an asset**, for example, property depreciating in value because of a new airport development nearby
- **decrease in revenues**, for example, loss of rent through a rental property being unlettable for a period

- **costs of unused capacity**, costs incurred by spare capacity that is taken as a precaution but does not end up being used

5.6.2 Exposure of financial assets

While the risk of trading shares and most forms of debt might be that their values fall to zero, this is not necessarily true of futures (where losses could be unlimited) and options (whose losses are limited to the option premium). In addition, anyone who is exposed to loss as a **result of price rises** is, in theory, exposed to the risk of **infinite loss**, since prices could rise indefinitely.

5.6.3 Exposure of human assets

The most severe risk to employees is the risk of death or serious injury. The loss to the employee's family, for which the organisation may be liable, could be the **future value** of their **expected income stream**, mitigated by any benefits available but enhanced by other losses that arise as a result of death; for example, loss of any available tax allowance. Alternatively, it could be measured by the expenditure required to fulfil the **needs** of the deceased's dependent family. For less serious injuries, the costs of medical care may be the relevant figure.

Certain individuals may make a significant contribution to the office because of their knowledge, skills or business contacts. One measure of this loss will be the present value of the individual's contribution (attributable earnings less remuneration). Indirect costs may include the effect on other staff of the loss of the key person (decreased productivity or indeed the costs of their own departure).

If a director, partner or senior employee dies or departs, there may be costs of having to cope with the disruption, perhaps even including the costs of dissolution, if local law requires termination of a partnership on the departure of a single partner.

5.7 Risk profiling

This stage involves using the results of a risk assessment to group risks into risk families. One way of doing this is a likelihood/consequences matrix.

Note: The detailed entries in the matrix are illustrative only.

Figure 7.3: Likelihood/consequences matrix

Likelihood	Consequences: Low	Consequences: High
Low	Loss of suppliers	Loss of senior or specialist staff Loss of sales to competitor Loss of sales due to macroeconomic factors
High	Loss of lower level staff	Loss of key customers Failure of computer systems

This profile can then be used to set priorities for risk mitigation.

5.8 Risk consolidation

Risk that has been analysed or quantified at the division or subsidiary level needs to be aggregated at the corporate level and grouped into categories. This aggregation will be required as part of the overall review of risk that the board needs to undertake. The process of risk categorisation also enables the risks categorised together to be managed by the use of common control systems.

5.9 Probability

Probability is a measure of **likelihood** and can be stated as a percentage, a ratio, or more usually as a number from 0 to 1. It is a measure of the likelihood of an event happening in the long run, or over a large number of times.

$$\text{Probability of achieving the desired results} = \frac{\text{Number of ways of achieving desired result}}{\text{Total number of possible outcomes}}$$

5.9.1 Independent mutually exclusive events



Definitions

Mutually exclusive: Outcomes where the occurrence of one of the outcomes excludes the possibility of another (eg if a score of six is achieved when a dice is rolled once, it is not possible to score a four (or any other value) from the same roll.

Independent events: Two events are independent if the probability of one event occurring is not affected by the second event occurring (eg if a dice is rolled twice, the probability of scoring a six the second time is independent of the number scored on the first roll)

The probability of either one or another mutually exclusive events occurring can be calculated as the sum of the probabilities of the two events. For example, the probability of rolling a dice once and scoring either a five or a six can be determined by adding together the probability of rolling five and the probability of rolling six:

1/6 + 1/6 = 2/6

5.9.2 Independent not mutually exclusive events

The **general rule of addition** for two events, A and B, which are not mutually exclusive, is as follows: Probability of (A or B) = P (AUB) = P(A) + P(B) – P(A and B)

For example, in a standard pack of 52 playing cards, what's the probability of getting an ace or spade?

Ace = 4/52

Spade = 13/52

Ace of spades = 1/52

Therefore, the probability of selecting an ace or a spade is:

4/52 + 13/52 – 1/52 = 16/52 = 4/13

5.9.3 Dependent events

Dependent events are where the probability of one event is affected by the other, for example, the probability that sales exceed budget on a particular day may depend on the weather. Conditional probabilities are calculated as follows, where A and B are two dependent events:

$$P(A \text{ and } B) = P(A) \times P(B|A)$$

Where P(B|A) means the probability of B occurring given that A has occurred.

Worked example: Independent non mutually exclusive events

P(A) = the probability that it is sunny on any particular day = 0.6 P(B) = the probability that daily sales of ice creams exceed £500
P(B|A)= the probability that sales of ice creams exceed £500 given that it is sunny = 0.8

Requirement

What is the probability that is it both sunny and sales of ice creams exceed £500?

Solution

The probability that it is both sunny and sales exceed £500 is calculated as:

P (A and B) = P(A) x P(B|A) = 0.6 x 0.8 = 0.48

The formula above can be reorganised as follows:

$$P(B|A) = \frac{P\ (A \text{ and } B)}{P\ (A)}$$

Worked example: Dependent events

During the last 100 days, it is not sunny on 40 days. Sales of ice cream exceeded £500 on 55 days. 50 of these were sunny days and 5 were not sunny.

Requirement

What is the probability that sales exceed £500 when it was not sunny?

Solution

The probability that sales exceeded 500 when it was not sunny can be calculated as follows:
P(A) = the probability that it is not sunny = 0.4

P(B) = the probability that sales of ice cream exceed £500

P(B|A) = the probability that sales of ice cream exceed £500 given that it is not sunny

P(A and B) = the probability that it was not sunny and sales of ice cream exceeded £500 = 0.05 (five days out of 100)

$$P(B|A) = \frac{P\ (A \text{ and } B)}{P\ (A)} = \frac{0.05}{0.4} = 0.125$$

Interactive question 2: Dependent events

The internal auditors of Hacket Ltd reviewed the quality of production output for the month of November 20X1. They found a number of faults in products and particularly in the output produced by one machine operator, Alan.

The internal auditor discovered that overall there were faults in 3% of all products made by Hacket in November 20X1. However, of the faults discovered, 12% were due to Alan's mistakes. Alan had processed 4% of all products in November 2021.

Requirement

Management wants to consider disciplining Alan, but it first wishes to know what percentage of the items produced by Alan contained faults

See **Answer** at the end of this chapter.

5.9.4 Expected values

An expected value is a weighted average value that you might expect to get if you take some action where the possible outcomes are variable, but you can estimate the probability of each outcome occurring. Imagine if you are planning to sell an item on eBay. You are told that there is a 20% chance that you will not sell the item, a 30% chance that you will achieve a price of £30 and a 50% chance that you will sell the item for £50. Your expected selling price is £34, being (20% × 0) + (30% × 30) + (50% × 50).

Although the outcome of a decision may not be certain, there is some likelihood that probabilities could be assigned to the various possible outcomes from an analysis of previous experience. You may also be given a probability distribution



Definition

Probability distribution: A probability distribution is a statistical function that describes the possible values and associated probabilities that a variable may take.



Worked example: Expected value

Ziggy Ltd is about to launch a new confectionery product, Stardust. Market research has indicated that annual demand for Stardust is uncertain. The following spreadsheet, showing the probability distribution, has been provided by the market research team. This shows five possible levels of contribution from selling Stardust, along with their probabilities:

	A	B	C
1	Outcome	Probability	Contribution £
2	Very bad	0.1	(1,000)
3	Bad	0.2	1,000
4	Neutral	0.4	5,000
5	Good	0.25	12,000
6	Very good	0.05	15,000

Requirement

Calculate the expected value of contribution

Solution

D7 =SUM(D2:D6)

	A	B	C	D
1	Outcome	**Probability**	**Contribution £**	**Contribution × Probability**
2	Very bad	0.1	(1,000)	(100) [1]
3	Bad	0.2	1,000	200
4	Neutral	0.4	5,000	2,000
5	Good	0.25	12,000	3,000
6	Very good	0.05	15,000	750
7	Expected value			**5,850**

[1] =B2*C2

CHAPTER 7

The concepts of probability and expected value are vital in **business decision-making**. The expected values for single events can offer a helpful guide for management decisions.

- A project with a positive EV should be accepted.
- A project with a negative EV should be rejected.
- When choosing between options the alternative which has the **highest EV of profit** (or the **lowest EV of cost**) should be selected.

Where probabilities are assigned to different outcomes we can evaluate the worth of a decision as the **expected value**, or weighted average, of these outcomes. The principle is that when there are a number of alternative decisions, each with a range of possible outcomes, the optimum decision will be the one which gives the highest expected value.

Expected values can be built into decision trees to aid decision making. The amount of expected profit is likely to be conditional on the result of various decisions. We will look at this in more detail below. First, let us briefly consider some limitations of using expected values as a basis for decisions.

Limitations of expected values

Evaluating decisions by using expected values has a number of limitations.

(a) The **probabilities** used when calculating expected values are likely to be estimates. They may therefore be **unreliable** or **inaccurate**.

(b) Expected values are **long-term averages** and may not be suitable for use in situations involving one-off decisions. They may therefore be useful as a **guide** to decision making.

(c) Expected values do not consider the **attitudes to risk** of the people involved in the decision- making process. They do not, therefore, take into account all of the factors involved in the decision.

(d) The **time value of money may not be taken into account**: £100 now is worth more than £100 in 10 years' time.

Probabilities and expected values can be represented diagrammatically using **decision trees** to aid decision making.

5.9.5 Decision trees

Decision trees are diagrams which illustrate the choices and possible outcomes of a decision.

Decision trees can incorporate both the probabilities and values of expected outcomes. It is important to note that **in the SBM&L exam you will *not* be required to draw a decision tree**.

Decision trees provide a clear and logical approach to problem solving by:

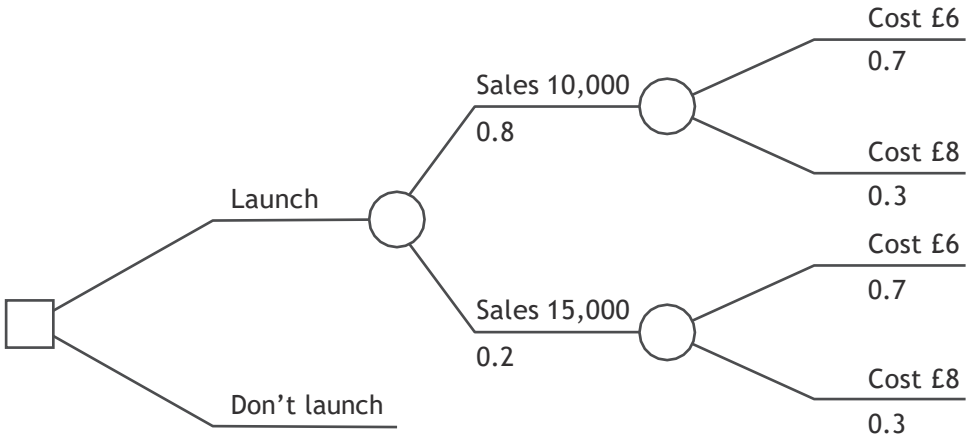
- showing all possible **choices** as **branches** on the tree (squares)
- showing all possible **outcomes** as **subsidiary branches** on the tree (circles)

Decision trees begin with a decision point, usually represented by a **square**, and the various choices branch off from this, flowing from left to right.

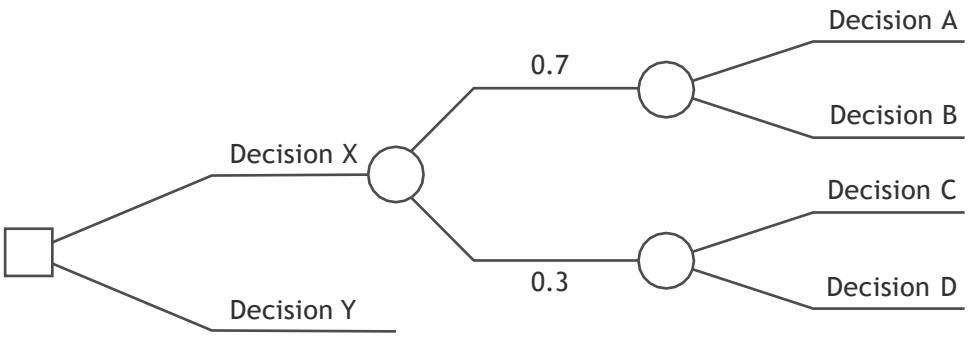
If the outcome for any of those choices is certain, then the branch of the decision tree for that alternative is complete.

If the outcome of a particular choice is uncertain, then the various possible outcomes must be shown. This is done by inserting an **outcome point** on the branch. This is symbolised by a **circle**. Each possible outcome will then branch out from that outcome point. These are known as **subsidiary branches**.

The **probability** of each outcome occurring should be written on the branch of the tree that represents that outcome.



Sometimes a **decision taken now** will lead to **other decisions** being taken in the future. When this situation arises, the decision tree can be drawn as a **two-stage tree**, as shown below.



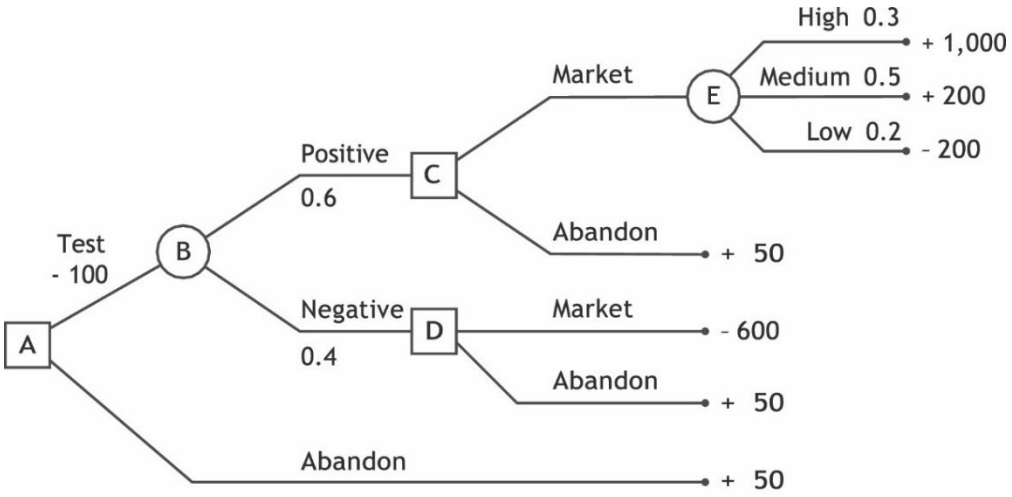
Evaluating decisions with a decision tree

The EV of each decision option can be evaluated, using the decision tree to help with keeping the logic properly organised. Once the basic decision tree (ie, choices, outcomes and probabilities) has been drawn from left to right chronologically, the basic rules for attaching values to the decision process are as follows.

(a) We start on the right-hand side of the tree and work back towards the left hand side and the current decision under consideration. This is sometimes known as the 'rollback' technique or 'rollback analysis'.

(b) Working from right to left, we calculate the EV of revenue, cost, contribution or profit at each outcome point on the tree.

Consider the decision tree below which has been prepared for a new product that has been developed. The decision is whether the new product should be test marketed or abandoned. The outcomes are high, medium or low demand and are dependent on whether the result of the test marketing is positive or negative.



The right-hand-most outcome point is point E, and the EV is as follows.

D5 =SUM(D2:D4)

	A	B	C	D
1	Outcome	**Profit £'000**	**Probability**	**Profit × Probability £'000**
2	High	1,000	0.3	300 [1]
3	Medium	200	0.5	100
4	Low	(200)	0.2	(40)
5	Expected value			360

[1] =B2*C2

This is the EV of the decision to market the product if the test shows a positive response. It may help you to note the EV on the decision tree itself, at the appropriate outcome point (point E).

(a) **At decision point C**, the **choice** is as follows.

(1) Market, EV = +360 (the EV at point E)

(2) Abandon, value = +50

The choice would be to market the product, and so the EV at decision point C is +360.

(b) **At decision point D**, the **choice** is as follows.

(1) Market, value = – 600

(2) Abandon, value = +50

The choice would be to abandon, and so the EV at decision point D is +50.

The second stage decisions have therefore been made. If the original decision is to test market, the company will market the product if the test shows a positive customer response, and will abandon the product if the test results are negative.

The evaluation of the decision tree is completed as follows.

(c) **Calculate the EV at outcome point B**.

0.6 × 360 (EV at C)

+ 0.4 × 50 (EV at D)

= 216 +20 = 236

(d) **Compare the options at point A**, which are as follows.

(1) Test: EV - EV at B minus test marketing cost = 236 – 100 – 136

(2) Abandon: Value = 50

The choice would be to test market the product, because it has a **higher EV of profit**.
Evaluating decisions by using a decision tree has a number of limitations:

- The time value of money may not be taken into account.
- Decision trees are not suitable for use in complex situations (eg, probabilities could be continuous in the form of a normal distribution, rather than only two possible outcomes).
- The outcome with the highest EV may have the greatest risks attached to it. Managers may be reluctant to take risks which may lead to losses.
- The probabilities associated with different branches of the tree are likely to be estimates, and possibly unreliable or inaccurate.

Professional skills focus: Applying judgement

It is important to remember that calculations, such as probabilities and expected values, have limitations so can only guide management decisions rather than providing a definitive solution. Apply your judgement to calculations when making recommendations.

5.10 Statistics

Definition

Descriptive statistics: Descriptive statistics describes the properties of sample and population data

Definition

Inferential statistics: Inferential statistics uses properties from descriptive statistics to test hypotheses and draw conclusions

Where outcomes associated with a risk can be predicted reliably, and the probabilities of those outcomes can be estimated, statistical techniques can be used to analyse the risks. The statistical methods covered here are:

- mean (or expected value)
- standard deviation
- co-efficient of variation

While the material covered here is provided in the context of measurement of risks, it should be noted that the knowledge of statistics is relevant throughout this exam, for example it could be used in questions involving analysis of the performance of a business.

5.10.1 Mean

The mean (or average) of a set of data is calculated by taking the sum of all the values and dividing by the number of values in the distribution.

Context example: Mean

The daily takings of a shop last week were:

	A	B
1		Takings (£)
2	Monday	5,000
3	Tuesday	5,500
4	Wednesday	7,250
5	Thursday	9,300
6	Friday	12,000
7	Saturday	15,325
8	Sunday	12,700
9	Total weekly sales	**67,075** [1]
10	Average (mean) daily sales	**9,582** [2]

[1] =SUM(B1:B8)

[2] =AVERAGE(B1:B8)

5.10.2 Expected values

Expected values was covered above. It should be noted that the expected value is a type of mean, as it represents the mean outcome that is expected if a decision is repeated many times.

5.10.3 The mean or expected value does not indicate the level of risk associated with an event. There could be two risks that both have the same mean or expected value, but very different risk profiles.

Risk means the variability of outcomes. If the potential outcomes of an event are all close to the expected value, there is less risk than if the potential outcomes are very different. Consider two potential investments, both with the same expected value:

	A	B	C	D
1	State of market	**Probability**	**Project 1**	**Project 2**
2	Poor	0.3	(20,000)	10,000
3	Medium	0.5	40,000	29,000
4	High	0.2	55,000	37,500
5	Expected value		25,000 [1]	25,000

[1] =(C2*B2)+(C3*B2)+(C4*B4). This can then be copied into cell D5.

Although both projects have the same expected cash flows of £25,000, Project 1 is more risky, as there is a wider variability of potential returns than for Project 2. The potential outcomes are more widely dispersed.

One method commonly used to measure dispersion is the standard deviation.

Definition

Standard deviation: A measure of the amount of variation or dispersion in a data set

Context example: Standard deviation

The shop takings for last week have been recorded in the spreadsheet below.

The mean and standard deviation of takings has been calculated using the formula in the spreadsheet:

	A	B
1		Takings (£)
2	Monday	5,000
3	Tuesday	5,500
4	Wednesday	7,250
5	Thursday	9,300
6	Friday	12,000
7	Saturday	15,325
8	Sunday	12,700
9	Mean	9,582 [1]
10	Standard deviation	3,910 [2]

[1] =AVERAGE(B2:B8)

[2] =STDEV(B2:B8)

5.10.4 Standard deviation in probability distributions

The formula for the standard deviations of a probability distribution is:

$$\text{Standard deviation} = \sqrt{\sum(X - \overline{X})^2 \times Pr(X)}$$

Where: $\overline{X}$ is the expected value and Pr(X) is the probability of the distibution taking th e value X

Context example: Example standard deviation of probability distribution

Ziggy Ltd is about to launch a new confectionery product, Stardust. Market research has indicated that annual demand for Stardust is uncertain. The following probability distribution has been provided by the market research team, showing five possible levels of contribution from selling stardust, along with their probabilities:

	A	B	C
1	Outcome	**Probability**	**Contribution £**
2	Very bad	0.1	(1,000)
3	Bad	0.2	1,000
4	Neutral	0.4	5,000
5	Good	0.25	12,000
6	Very good	0.05	15,000
7	Expected value		5,850 [1]

[1] =(C2*B2)+(C3*B3)+(C4*B4)+(C5*B5)+(C6*B6)

The calculation of the variance and standard deviation is as follows:

	A	B	C	D	E	F
1	Outcome	**Probability**	**Contribution**	$(x-\bar{x})$	$(x-\bar{x})^2$	$(x-\bar{x})^2 Pr(x)$
2	Very bad	0.1	(1,000)	(6,850) [1]	46,922,500 [2]	4,692,250 [3]
3	Bad	0.2	1,000	(4,850)	23,522,500	4,704,500
4	Neutral	0.4	5,000	(850)	722,500	289,000
5	Good	0.25	12,000	6,510	37,822,500	9,455,625
6	Very good	0.05	15,000	9,150	83,722,500	4,186,126
7		**Expected value**	5,850		Variance	23,327,500 [4]
8					Standard deviation	4,830 [5]

[1] =C2-C$7

[2] =D2^2

[3] =E2*B2

[4] =SUM(F2:F6)

[5] =F7^0.5

The standard deviation is therefore $\sqrt{23,327,500}$ = £4,830

5.10.5 Interpreting the standard deviation

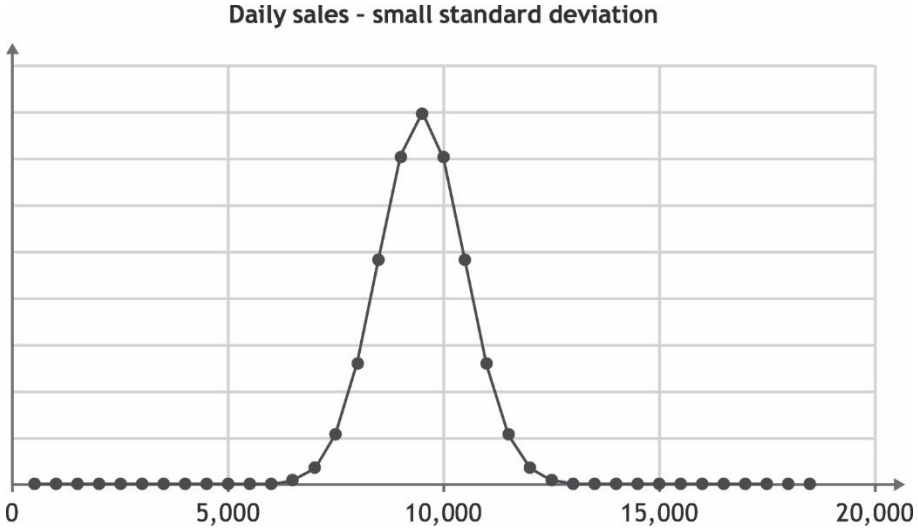
The bigger the standard deviation, the more widely dispersed the possible outcomes of an event are. A bigger standard deviation means a higher risk.

In the example (standard deviation) above, the standard deviation of shop takings was £3,620. This recognises that although the expected takings from the shop are £9,582 per day, the actual takings each day will differ from this. Sometimes they will be more than the expected value, sometimes less. On average, the difference between the actual daily takings and the expected value will be £3,620.

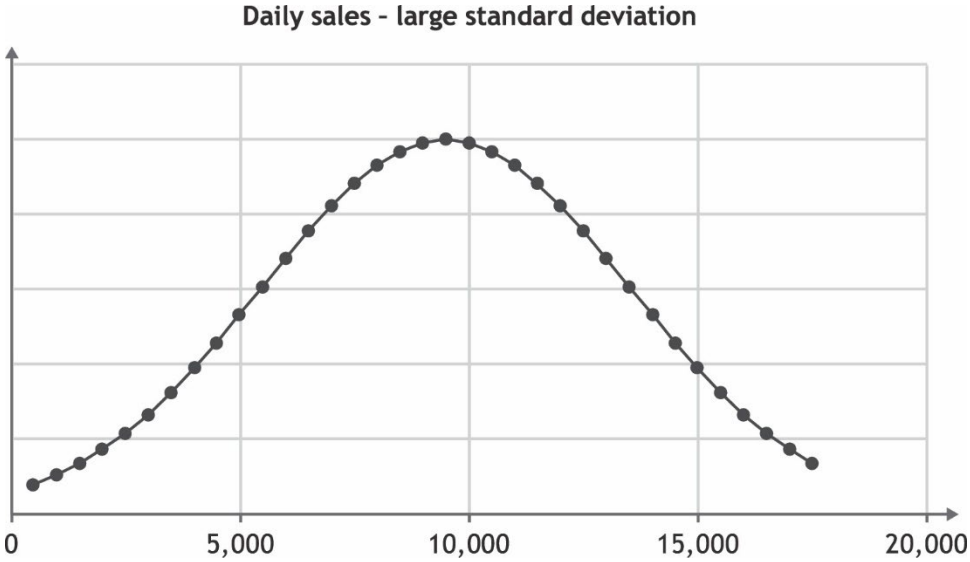
If distributions have a smaller standard deviation, it means that the actual outcomes will generally be closer to the expected outcome, while a larger standard deviation means the outcomes will be further away from the expected value.

The meaning of standard deviation can be illustrated using the following diagrams. The diagrams are based on the example of the daily sales for the shop above. The charts show the probability distribution of daily sales, where the mean is £9,500 per day.

In the first chart, the standard deviation is a relatively low £1,000. On most days, sales are pretty close to the mean of £9,500 and are not widely dispersed.



The second chart shows that when the standard deviation is larger, daily sales are much more widely dispersed, in other words they vary more widely from day to day.



The relevance of standard deviation to risk is that a higher standard deviation means a greater level of risk.

5.10.6 Use of standard deviation in risk management

Standard deviations can be used when evaluating the risks associated with particular strategies or decisions if the information is available. Strategies may be rejected if the standard deviation is considered too high, given the risk appetite of the organisation.

In the investment industry, standard deviations are calculated for investments, for example as a measure of the volatility of share prices on the stock markets. Investment decisions are taken based on the expected return and standard deviation of the investment returns. If the standard deviation is high, a higher expected return will be required by investors.

In the exam, you would not be required to calculate the standard deviation. However, you may be given a standard deviation and asked to interpret it.

5.10.7 Co-efficient of variation

It is difficult to interpret the standard deviation without considering the size relative to the size of the data. The standard deviation may be larger, simply because the values of the data in a distribution are higher. A more meaningful measure is the coefficient of variation, which relates the standard deviation to the expected value of a distribution.

Definition

Co-efficient of variation: The standard deviation of a distribution divided by the mean or expected value.

Context example: Co-efficient of variation

A company is considering launching either one or both of two potential products. The following probability distribution has been provided by the market research team, showing five possible levels of contribution from selling each product, along with their probabilities:

	A	B	C	D
1	Outcome	**Probability**	**Product 1 – contribution**	**Product 2 – contribution**
2	Very bad	0.1	(1,000)	16,000
3	Bad	0.2	1,000	18,000
4	Neutral	0.4	5,000	22,000
5	Good	0.25	12,000	29,000
6	Very good	0.05	15,000	32,000
7	Expected contribution		5,850 [1]	22,850 [2]
8	Standard deviation		4,830	4,830
9	Co-efficient of variation		82.6% [3]	21.1% [4]

1 =(B2*C2)+(B3*C3)+(B4*C4)+(B5*C5)+(B6*C6)

2 =(B2*D2)+(B3*D3)+(B4*D4)+(B5*D5)+(B6*D6)

3 =C8/C7

4 =D8/D7

Note: The calculation of the standard deviation has not been shown here.

Both products have the same standard deviation, which may suggest that they bear the same level of risk. However, the differences in contribution for product 1 are relatively much larger. Contribution at 'very good' is 3 times larger than contribution at 'neutral' for product 1, while for product 2, it is only 45% greater.

The co-efficient of variation of product 2 is much lower than product 1. Although the standard deviations of the two products are equal, the level of contribution of product 2 is much higher than that of product 1. The variations as a percentage of the level of contributions (represented by the expected value) are much lower for product 2 than product 1. This explains the coefficient of variation.

For the co-efficient of variation (CoV) it is worth noting that CoV cannot be used where the data can be negative (eg not for NPV but it is valid for PV of inflows). It also cannot be used for

interval scale (eg, questionnaire scores). Also zero must be meaningful (so a Fahrenheit scale is inappropriate).

5.11 Probability distributions

Probability distributions were defined above as a statistical function that describes the possible values and associated probabilities that a variable may take. There are two forms of probability distribution:

- Discrete distributions, where the possible values that a variable can take are discrete (ie specific, exact values) and the probability of each value occurring can be stated.
- Continuous distributions, where the set of outcomes are values within a continuous range (eg, the age of a population would be defined by different bands, such as 20-30 years, 30-40 years etc).

The discussion of probabilities in sub sections 1 and 2 above related to discrete probabilities. This section focusses on a continuous probability distribution, the normal distribution.

5.11.1 Normal distribution

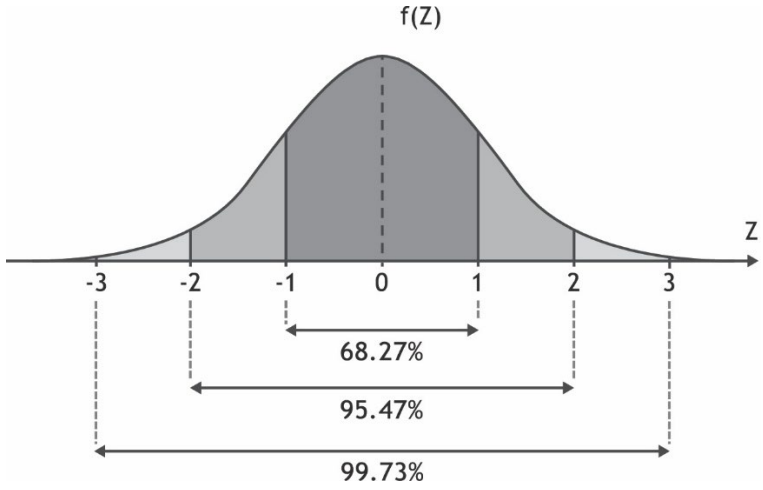
Many large probability distributions in the real world approximate the normal distribution. The normal distribution is a bell-shaped curve, and the probabilities of variables taking particular values is represented as the area under the curve between the two values.

The diagram below shows the normal distribution:

μ is the mean of the distribution

σ represents a standard deviation

The area under the curve shows the probabilities of being within certain ranges of the mean, where distance from the mean is measured in standard deviations.



Properties of the normal distribution:

The normal distribution has the following consistent properties:

- The mean of the distribution = the median = the mode
- The distribution is symmetrical – the probability of identifying a value as equal to or below the mean is 50% and the probability of it being equal to or above the mean is also 50%.
- The probability of being within particular ranges of the mean depends on the standard deviation:

(1) The probability of a value being between the mean, and one standard deviation above the mean is 34.1%

(2) 68.2% of values lie between one standard deviation below and one standard deviation above the mean

(3) 95.4 % of values lie between two standard deviations below and two standard deviations above the mean

(4) 99.7% of values lie between three standard deviations below and three standard deviations above the mean.

Some other useful values are:

- 90% of values lie within 1.645 standard deviations above and 1.645 standard deviations below the mean.
- 95% of values lie with 1.96 standard deviations above and 1.96 standard deviations below the mean.
- 99% of values lie within 2.58 standard deviations above and 2.58 standard deviations below the mean.

5.11.2 Z-score

A Z-score shows how many standard deviations above the mean a particular point is. This is useful when performing calculations with the normal distribution. Having calculated a Z-score for a particular point, it is possible to look up the value in normal distribution tables to ascertain the probability of a variable taking a value between the mean and that point.

The Z score for a particular value, X is calculated as:

$$\frac{X-\mu}{\sigma}$$

Where μ is the mean of the distribution and σ is the standard deviation.



Worked example: Z-score

The average number of units produced during a particular day is normally distributed, with a mean of 1,000 and a standard deviation of 50 units.

Requirement

The production manager wants to know the probability that output will exceed 1,100 units.

Solution

The Z-score is calculated as (1,100 − 1,000)/50 = 2.

This means that 1,100 units is 2 standard deviations above the mean.

From the normal distribution tables above, the probability of taking a value higher than 2 standard deviations above the mean is 2.26%. Therefore, the probability that output will exceed 1,100 units is 2.2%. Note: This can be seen from the diagram above as follows: 50% of values lie below the mean. 95.47% of values lie within 2 standard deviations of the mean, therefore 47.74% (95.47%) lie in the range between the mean and 2 standard deviations above the mean. In total therefore 97.74% of values lie in the range between 0 and 2 standard deviations above the mean. The probability of being above 2 standard deviations above the mean is therefore 1 − 97.74% = 2.26%

Interactive question 3: Z scores 1

The mean number of units produced by a machine is 1,000 per day, with a standard deviation of 25 units.

Requirement

The production manager wishes to know what is the probability of producing between 950 and 1,000 units per day.

See **Answer** at the end of this chapter.

Interactive question 4: Z scores 2

The number of units produced by a machine is 1,000 per day, with a standard deviation of 25 units.

Requirement

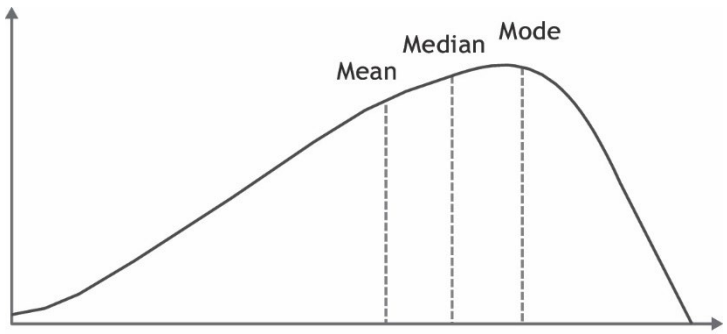
The production manager wishes to know what is the probability of producing between 975 and 1,025 units per day.

See **Answer** at the end of this chapter.

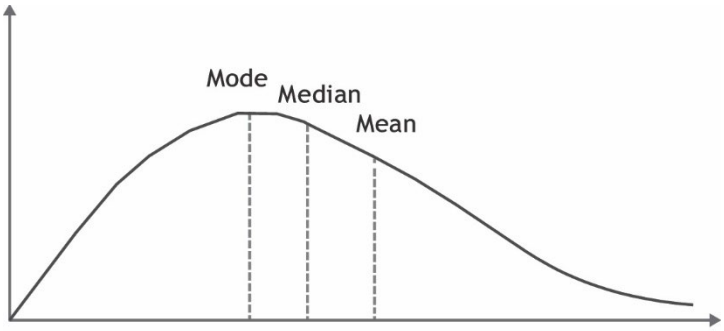
5.11.3 Skewness

- A left-skewed (negatively skewed) distribution has the majority of values concentrated on the right-hand side of the distribution. There are fewer values on the left-hand side of the distribution but these are more spread out, so the curve has a long left-hand tail but appears to lean slightly to the right. The mode typically occurs at the highest point in the distribution, and typically the median is to the left of the mode (so it has a lower value than the mode) and the mean is to the left of the median (so it has a lower value than both the mode and the median).
- A right-skewed (positively skewed) distribution has the majority of values concentrated on the left- hand side of the distribution. There are fewer values on the right-hand side of the distribution but these are more spread out, so the curve has a long right-hand tail but appears to lean slightly to the left. Again the mode typically occurs at the highest point in the distribution, and typically the median is to the right of the mode (so it has a higher value than the mode) and the mean is to the right of the median (so it has a higher value than both the mode and the median).
- The normal distribution is not skewed, and the mean = the median = the mode at the highest point of the distribution.

Skewness can be illustrated by the following diagrams:



Left skewed



5.11.4 Sampling and confidence intervals

Definition

Population: the entire set of data from which a sample is selected for analysis (eg sales to all customers in the last year.

Definition

Sampling: analysing a sample of data from a population, and based on this, making inferences about the population.

Developments in information technology and the growth of big data over the past 20 years have made it more feasible to analyse whole populations of data.

It is often unfeasible or even impossible to find out information about every item in a population. In order to perform statistical analysis therefore, **samples** are taken and **inferences** are made about the population based on analysis of the sample. (Eg in order to find out the mean salary of ICAB members, a sample of ICAB members could be taken. The mean salary of the members of the sample would be used as an estimate for the mean salary of all ICAB members.)

When making inferences, it is probable that the **statistics obtained from the sample will not be exactly the same as the population**, so it has to be recognised that **they are an estimate**. In order to make estimates more reliable:

- the sample should be selected randomly, to avoid introducing bias (see Chapter Data analysis)
- larger samples are likely to be more representative of the population than small samples.

5.11.5 Sampling distributions

One sample of ICAB members is likely to have a different average salary to another sample. If a third sample was taken, this might have a third average salary, and so on. The means of the samples follow a probability distribution, known ats a sampling distribution.

Definition

Sampling distribution : A sampling distribution is a probability distribution of a statistic taken from all possible samples of a given size from a population (eg the sampling distribution of the mean is a probability distribution that shows the distribution of sample means of a given size).

The central limit theorem states that if all samples of a given size, n, are taken from a population, as n gets larger, the sampling distribution of the sample means approaches a normal distribution, with the mean of the distribution being equal to the population mean, and a standard deviation (referred to as a standard error) equal to the population standard deviation divided by the square root of the sample size.

Worked example: Sampling distribution

In Beeland, the mean height of males is 180 cm with a standard deviation of 15 cm.

Requirement

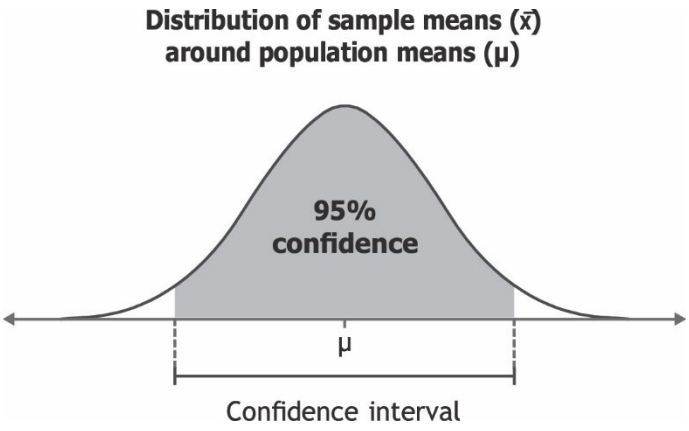
If all possible samples of men, with a sample size of 40 were taken, what would be the mean and standard error of the sampling distribution of the sample mean?

Solution

The sample means would be normally distributed, with a mean of 180cm (population mean) and a standard error of 15/(√40) = 2.37.

5.11.6 Confidence intervals

When we use a sample to estimate the population mean, it is not likely that our sample mean will be exactly the same as the population mean. If we know the population mean and standard error of sample means, we can define a range within which a certain proportion of sample means would lie. This is a confidence interval. When we define a confidence interval, we define the probability – eg, a 95% confidence interval would contain 95% of sample means.



Confidence intervals can be calculated by using the properties of the normal distribution. If we know the population mean, and standard error of the sampling distribution of the mean, then we can calculate a range of values within which a certain portion of sample mean lies. In the normal distribution, 95% of values lie within 1.96 standard deviations of the mean. Therefore our confidence interval at 95% is:

Population mean plus or minus 1.96 standard errors

For a 90% confidence interval, the interval is population mean plus or minus 1.645 standard errors

Worked example: Confidence intervals

In Beeland, the mean height of males is 180 cm with a standard deviation of 15 cm. If all possible samples of men, with a sample size of 40 were taken, the sample means would be

normally distributed, with a mean of 180cm (population mean) and a standard error of 15/(√40) = 2.37.

Requirement

What would the confidence interval of sample means be at 95%?

Solution

A 95% confidence interval for sample means of size 40 is: 180 ± 1.96 × 2.37 = 180 ±4.65

= 175.35 to 184.65 cm

This means that 95% of samples of size 40, taken from the male population of Beeland would have a mean between 175.35 and 184.65 cm.

The same value can be achieved using the spreadsheet function CONFIDENCE

B5 =CONFIDENCE(B1,B2,B3)

	A	B
1	Alpha (100% -95%)	0.05
2	Standard deviation	15
3	Sample size	40
4	Mean	180
5	Confidence value	**4.65**

The confidence interval is therefore 180 + 4.65 = 175.35 to 184.65

In practice, when we take a sample from a population, we can use this to estimate the population mean and the population standard deviation. We can also define confidence intervals within which the true population mean lies.

Estimate of population mean = sample mean

Estimate of population standard deviation = standard deviation of sample. Estimate of standard error of the sampling distribution =

$$\frac{\text{sample standard deviation}}{\sqrt{n}}$$

where n is the sample size.



Interactive question 5: Sampling and confidence intervals

A statistician is trying to estimate the mean height of males in Beeland. He has taken a sample of size 40, and calculated a mean of 178 cm for this sample with a standard deviation of 28.46. He has estimated the standard error of the sampling distribution of sample means to be 4.5 cm (28.46/(√40)).

Requirement

What is the 95% confidence interval for the population mean? Explain the meaning of it.

See **Answer** at the end of this chapter.

5.12 Hypothesis testing

Hypothesis testing means using statistics to see if a predetermined idea (hypothesis) is correct.

A common type of hypothesis test relates to the mean of a population, where the analyst has a pre- determined idea (a hypothesis) about the value of a mean, and wants to test whether the hypothesis is correct. The following steps are taken:

(a) The hypothesis is formulated (eg the population mean is X). An alternative hypothesis is also formulated, being the conclusion that will be reached if the hypothesis is disproved (the mean is not X).

(b) The level of significance is chosen. This relates to the probability that will be accepted that the null hypothesis is rejected when it is correct. For a 5% significance level for example, there is a 5% chance that the hypothesis will be rejected when it is correct.

(c) A confidence interval is calculated for the population mean, based on the hypothesis mean, standard error and the significance level. If the significance level is 5% for example, 95% confidence intervals are used.

(d) If the value of the sample mean lies within the confidence intervals, then the null hypothesis is accepted. If the value lies outside the confidence interval, then the null hypothesis is rejected and the alternative hypothesis accepted.



Interactive question 6: Hypothesis testing

Alpha produces steel ball bearings of various sizes. One of the company's products is a ball bearing with a diameter of 5cm. Due to variations in operating conditions, the diameter of the bearings changes slightly from batch to batch, but customers will tolerate a small variation in size.

The production director has tested a sample of 40 batches to see if the average size really is 5cm. He has defined the null and alternative hypotheses for the test:

- null hypothesis: the average size of the ball bearings is 5cm
- alternative hypothesis: the average size of ball bearings is not 5cm.
- level of significance chosen: 90%

The sample gave a mean of 5.2cm. Based on the sample, the production director has estimated that the standard deviation of sample means is normally distributed with a standard error of 0.1cm.

Normal distribution tables show that 90% of values within a normal distribution lie within 1.65 standard deviations of the mean.

Requirements

6.1 Calculate the confidence intervals for the production director's test at the 90% confidence level.

6.2 State whether the null hypothesis would be accepted or rejected at the 90% level given the sample mean achieved.

6.3 What is the conclusion of the production director's test?

See **Answer** at the end of this chapter.

5.13 Regression and correlation

Exploratory data analysis aims to identify relationships between items in the data. The basic techniques for exploratory data analysis are regression analysis and correlation.

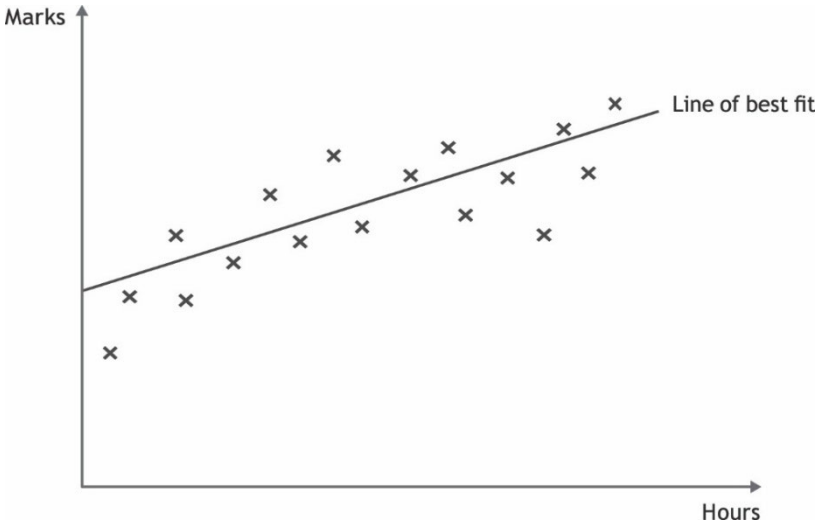
Definitions

Regression analysis: Regression analysis aims to specify the relationship between two or more variables. One of these variables is the dependent variable, whose value depends on the independent variable(s).

Correlation: Correlation is a measure of the extent to which changes in the dependent variable are explained by changes in the independent variable.

5.13.1 Regression and correlation

An accountancy training company believes that there is a relationship between the number of hours of studying, and the marks that students achieve in their exam. It has asked a sample of students to record the number of hours that they spend studying for their exam, and has then plotted these against the marks that they achieve in the exam on a simple graph or line chart:



Based on the analysis, a "line of best fit" has been drawn in by eye on the chart to show the trend in how the variables relate to each other. The line of best fit can be converted into a mathematical equation to show the relationship between the dependent variable (marks in exams) and the independent variable (hours of studying).

The line of best fit has an equation of the form $y = ax + b$, where:

- y is the predicted value of the dependent variable, given the value of the independent variable, x;
- a shows how much the dependent variable changes for a unit change in the independent variable (how may additional marks would a student obtain for one extra hours studying);
- b shows what the value would be if the value of the independent variable is zero (how many marks would a student who has done zero hours of studying achieve in the exam).

Tutorial Note

You would not be expected to perform regression analysis in the SBM exam, but you need to beaware of what regression means and how it is used.

Interactive question 7: Greendot

Greendot is considering acquiring one of two companies, Lila and Plume. Both companies have two divisions, each selling similar products in different markets. Greendot is looking to invest in a growth company with steady cashflows. You have been provided with monthly revenue data for each company in a pre-populated spreadsheet.

	A	B	C	D	E	F	G
1		**Lila Revenues £'000**			**Plume Revenues £'000**		
2	Month	Division A	Division B	**Total**	Division A	Division B	**Total**
3	1	10	40	**50**	10	40	**50**
4	2	12	45	**57**	12	45	**57**
5	3	14	50	**64**	14	50	**64**
6	4	16	55	**71**	16	55	**71**
7	5	18	60	**78**	18	60	**78**
8	6	20	65	**85**	20	65	**85**
9	7	22	70	**92**	22	70	**92**
10	8	24	75	**99**	24	75	**99**
11	9	26	80	**106**	26	80	**106**
12	10	28	85	**113**	28	85	**113**
13	11	30	90	**120**	30	90	**120**
14	12	32	95	**127**	32	86	**118**
15	13	34	100	**134**	34	82	**116**
16	14	36	105	**141**	37	78	**115**
17	15	38	110	**148**	40	74	**114**
18	16	40	115	**155**	43	70	**113**
19	17	42	120	**162**	46	66	**112**
20	18	44	125	**169**	49	62	**111**
21	19	46	130	**176**	52	58	**110**
22	20	48	135	**183**	55	54	**109**
23	21	50	140	**190**	58	50	**108**

Requirement

Using spreadsheet functionality calculate the correlation and standard deviation for the data provided in the pre-populated spreadsheet. Comment on which company best fits with Greendot's acquisition goals.

See **Answer** at the end of this chapter.

5.13.2 Correlation coefficient

The correlation co-efficient is a statistical measure which indicates how well regression analysis explains the data. It answers the question: what portion of the change in the dependent variable can be explained by changes in the independent variable? If all the points in the data

lie exactly on the line of best fit, this indicates that 100% of the change in the dependent variable are due to changes in the independent variable.

The correlation coefficient can take a value between -1 and +1. If the value is minus, the dependent variable and independent variable are inversely related, which means that an increase in the dependent variable leads to a decrease in the independent variable. At a value of -1 or +1, all changes in the dependent variable are explained by the independent variable. As the value moves toward zero, the relationship becomes weaker, with the changes in the dependent variable explaining a lower portion of the changes in the independent variable.

Interactive question 8: Regression and correlation

8.1 A shop owner has analysed the relationship between daily sales of soft drinks and the outside temperature. They have calculated that daily sales have the following equation:

Y = 500 + 10 X, where Y is daily sales from drinks in £, and X is the temperature in degrees Celsius at midday. The correlation coefficient is 0.7.

Requirements

What is the dependent variable, and what is the independent variable in the analysis?

8.2 If the temperature at midday is 20 degrees Celsius, what is the predicted sales of drinks for that day?

8.3 What does a correlation coefficient of 0.7 mean?

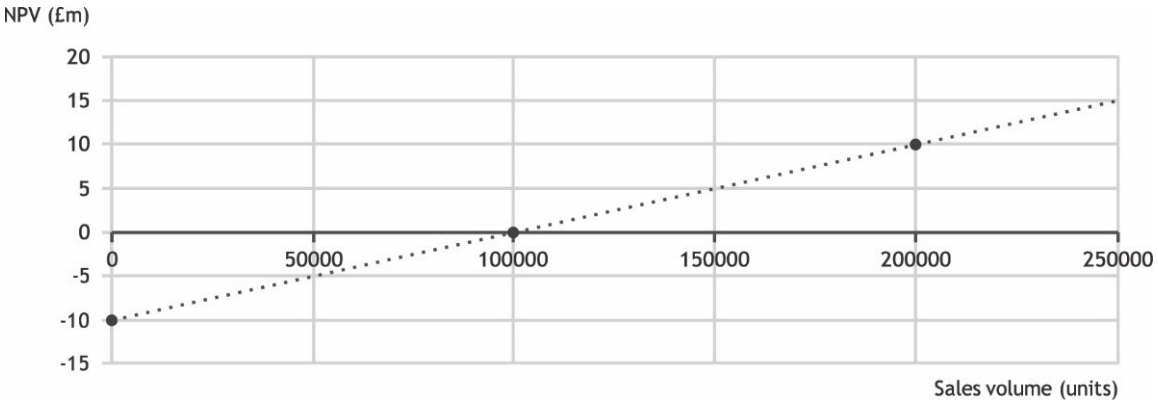
See **Answer** at the end of this chapter.

5.13.3 Investment appraisal and regression analysis

Regression analysis can be useful in investment appraisal to identify a set of factors that have a strong link to the returns from a project.

The regression equation helps to build an understanding of the sensitivity of a project's NPV to changes in these factors.

For example, the impact of sales volume on NPV could be modelled.



Here this equation is approximately y = –10 + 0.0001x which means that starting from –10 (£m), NPV increases by approximately 0.0001 (£m) for each forecast unit of sales. This could also be used, as part of sensitivity analysis, to compare the impact of changes in one variable (such as the sales volume) to the predicted NPV of a project.

A regression equation is unlikely to fully explain the relationship between y and x. The failure of the regression equation to fully explain the relationship between the dependent (y) and independent (x) variable results in an error variable, so the regression equation becomes y = a + bx + error. The larger the error term is, the less certain is the regression line.

5.13.4 Multiple regression analysis

Multiple regression analysis involves exploring whether identifying more than one independent variable reduces the error term and identifies the most important independent variables that reduce the error term as low as possible. This aims to provide a stronger regression line that quantifies the key independent variables that are associated with changes in the value of the dependent variable.

Multiple regression analysis will result in a more complex formula. For example, if two key independent variables (x1 and x2) had been identified then the formula would be y = a + bx1 + cx2+ error.

From the previous example, if the two independent variables affecting sales had been identified as time (X1) and economic growth (X2) then the regression equation could be y = 10 + 0.9X1 + 0.05X2.

The increasing availability of Big Data also makes it possible to identify the relationship between non-financial factors (eg, rainfall, social media followers, website hits) and the revenue from a project or even the overall NPV.

Multiple regression analysis can also be used to help make predictions of the value of a company which can be useful in evaluating a takeover. For example, some of the variables that could affect share prices include company profits, forecast revenue growth, the gearing ratio, and the number of competitors in the industry.

Advantages of linear regression

- Linear regression models are simple to use and easy to explain to non-financial managers.
- Linear models can be used to predict the impact of expanding variables beyond current estimates (such as identifying the impact of sales volumes or material costs being higher than predicted).

Limitations of linear regression

- There will not always be a linear relationship between variables and outcomes.
- Basic linear regression models can only consider the impact of one variable at a time. More complex multi-linear models are required to consider additional variables at the same time.
- Linear models may identify spurious relationships between variables and outcomes as they do not consider the difference between correlation and causation.
- Results will be less meaningful if the data collected is inaccurate or if the error term is large.

5.13.5 Correlation vs causation

A cause and effect relationship (also known as a causal relationship) exists between two variables when a change in one **causes** the change in the other.

For example, if staff are paid hourly, then as hours worked increase, wage costs will increase. The increase in hours worked has **caused** the increase in wage costs. There is a positive correlation between the number of hours worked and wage costs that, in this case, can be described as a cause- and-effect relationship.

However, **correlation does not necessarily mean that a cause-and-effect relationship exists**. There may be a reason for the correlation that is not **causal**.

For example, when the sales of sun cream increase, the sales of ice cream also increase. The increase in sun cream sales is not causing the increase in ice cream sales. There is a third variable, namely the weather, influencing both types of sales. This variable is known as a confounding variable.

Also, correlation may occur by pure chance; this is more likely to happen with a small set of data.

This highlights the need to exercise professional scepticism when analysing data and drawing conclusions.

Questions to consider might include:

- Does the relationship seem plausible?
- Could the relationship be because of chance or could a third variable be involved?

5.14 Diversification

Risks can be reduced by diversification, which involves taking on additional risks, where the outcomes respond in a different way to external factors. During the lockdown resulting from the coronavirus pandemic, companies that provided home delivery of foods saw a large increase in demand, while companies with activities related to hospitality saw a huge decrease in their income, so the impact of the virus on these two businesses was opposite.

The sum of two distributions may have a lower standard deviation than the standard deviations of the individual distributions if they react differently to external factors.



Context example: Real life example

An investment company is considering investing in two new businesses, a luxury hotel and a discount food retailer. The annual cash flows of the businesses depend on the state of the economy, which can be poor (with a probability of 25%), good (with a probability of 50%) or very good (with a probability of 25%). The cash flows of the two businesses, and the total cash flows of the two businesses are forecast to be as follows, along with calculations of the expected cash flows and standard deviations:

State of economy	Prob	Annual cash flows - hotel £	Annual cash flows - food retailer £	Annual cash flows - combined
Poor	0.25	400,000	1,300,000	1,700,000
Good	0.50	1,000,000	1,000,000	2,000,000
Very Good	0.25	1,300,000	800,000	2,100,000
Expected value		925,000	1,025,000	1,950,000
Standard deviation		375,000	205,649	170,783

Commentary

The table above demonstrates the following important points about combining two distributions:

(a) The expected value of the combined distribution is the sum of the expected values of the two individual distributions.

(b) The standard deviation of the combined distribution is less than the standard deviations of both the individual distributions.

While the first point will always be the case, it is not always the case that combining two distributions will lead to a lower standard deviation than each of the individual distributions. The reason it has occurred here is that the two distributions are **negatively correlated**: as the

economy improves, people have more money to spend on luxury hotels, so the cash flows of the hotels increases, but demand for discount food retailers falls as people can afford to shop at more luxurious supermarkets.

Diversification works by taking on a combination of risks which are negatively correlated.

5.14.1 Covariance and correlation

The correlation co-efficient shows how strong the relationship is between two variables. The co- efficient can take any value between 1 and −1. A value of one means perfect correlation, ie all the changes in one variable are explained by changes in the other variable. A value of −1 also means perfect correlation, except that as one variable increases, the other variable falls. As the coefficient moves towards zero, the relationship between the two variables is weaker, and a co-efficient of zero suggests that changes in one variable have no impact on the other variable.

The covariance is a measure of how two distributions vary with each other. Where the covariance is positive, the two distributions vary positively (ie, as the value of one distribution increases, the value of the other increases). Where the covariance is negative, the opposite is the case. The covariance is determined by the correlation between the two distributions.

In the diversification example above, the covariance of the cash flows from the two businesses would be negative, because as cash flows earned by the luxury hotel rise, cash flows earned by the discount retailer fall. Where the covariance is negative, the standard deviation of the combined distributions is lower than the standard deviations of the two individual distributions. Diversification is most effective at reducing risk where the covariances between the risks is negative.

You would not be required to calculate the covariance or the correlation coefficient in an exam, but need to be aware of their meaning, and how they impact on diversification.

6 Risk response



Section overview

- Although organisations operating in the same industry may face similar risks, the ways in which they respond to these risks can differ significantly.
- Risk responses depend on such factors as the potential impact of the risk on the organisation and management's attitude towards risk.
- The four main responses to risk are: transfer, avoid, reduce and accept.
- Implementation of the risk management process should be treated as a separate project with clear objectives and success criteria.
- Just because risk management procedures are in place does not mean that companies are immune from the effects of risk. Such procedures may reduce the impacts of risk but will not eliminate them completely.
- Risks must be continually monitored to determine any change in profile that may lead to procedures that control those risks being changed. This stage is effectively an audit of the overall risk management process, where expected and actual results are compared; and recommendations made for remedial actions.

Once risks have been identified, assessed and quantified, decisions must be taken as to how to respond to these risks. Methods of dealing with risk include avoidance, reduction, acceptance (retention)and transfer.

Risk response can be linked into the likelihood/consequences matrix and also the organisation's appetite for risk taking. This is known as the TARA framework: **T**ransfer, **A**void, **R**educe, **A**ccept.

Risk responses

		Consequences	
		Low	**High**
Likelihood	**Low**	**Accept or absorb** Risks are not significant. Keep under review, but costs of dealing with risks unlikely to be worth the benefits.	**Transfer** Insure risk or implement contingency plans. Reduction of severity of risk will minimise insurance premiums.
	High	**Reduce or manage** Take some action, eg, self-insurance to deal with frequency of losses.	**Avoid or control** Take immediate action to reduce severity and frequency of losses, eg, insurance, charging higher prices to customers or ultimately abandoning activities.

6.1 Transfer of risk

Risks can be transferred – to other internal departments or externally to suppliers, customers or insurers. Risk transfer can even be to the state.

Decisions to transfer risk should not be made without careful checking to ensure that as many influencing factors as possible have been included in the assessment. A decision not to rectify the design of a product, because rectification could be as expensive as paying any claims from disgruntled customers, is, in fact, a decision to transfer the risk to the customers without their knowledge: it may not take into account the possibility of courts awarding exemplary damages to someone injured by the product, to discourage people from taking similar decisions in the future.

Internal risk transfer can also cause problems if it is away from departments with more 'clout' (for example, sales) and towards departments, such as finance, that may be presumed to downplay risks excessively.

6.1.1 Legal and other restrictions on transferring risks

The first restriction is that a supplier or customer may **refuse** to enter a contract unless the organisation agrees to take a particular risk. This depends on the trading relationship between the firms concerned, and not a little on economics: how many suppliers could supply the item or service in question, for example, and how great is the need for the item?

6.1.2 Risk sharing

It is rare to be able to transfer all the risk to a third party, and in many cases risks can be partly held and partly transferred to someone else. An example is an insurance policy, where the

insurer pays any losses incurred by the policy holder above a certain amount.

Risk sharing arrangements can be very significant in business strategy. For example, a **joint venture** arrangement allows the venture partners to share the risk associated with the venture.

6.2 Avoidance of risk

Organisations will often consider whether risk can be avoided and, if so, whether avoidance is desirable – that is, will the possible savings from losses avoided be greater than the advantages that can be gained by not taking any measures and running the risk?

An extreme form of avoiding business risk is terminating operations altogether – for example, operations in politically volatile countries where the risks of loss (including loss of life) are considered to be too great; or the costs of security, too high.



Context example: Toyota

We mentioned in Strategic marketing and brand management the problems Toyota has faced in recent years regarding concerns about the safety of its cars. Toyota responded to these concerns by employing a risk avoidance strategy. Sales of a number of models were suspended in the US. Although Toyota's actions aimed to resolve the risks to health and safety, it may have been less effective in mitigating the risks to its reputation. Commentators highlighted an initial reluctance to admit the problem, along with poor communication of what it intended to do to regain control of the situation. The impact threatened car sales and share price, with investors reluctant to hold Toyota shares because of the level of uncertainties involved.

6.3 Reduction of risk

Often, risks can be avoided in part, or reduced, but not avoided altogether. This is true of many business risks, where the risks of launching a new product can be reduced by market research, advertising and so on.

Other risk reduction measures include contingency planning, loss control, internal control and, in the case of some financial risks, hedging. Financial risk management is covered in Financial risk management.

6.3.1 Contingency planning

Contingency planning involves identifying the **post-loss needs** of the business, **drawing up plans** in advance and **reviewing them regularly** to take account of changes in the business. The process has three basic constituents.

Information	How, for example, are the sprinklers turned off once the fire is extinguished? All the information that will need to be available during and after the event should be gathered in advance.
Responsibilities	The plan should lay down what is to be done by whom.
Practice	Unless the plan has been tested, there is no guarantee that it will work. A full- scale test may not always be possible; simulations, however, should be as realistic as possible and should be taken seriously by all involved.

6.3.2 Loss control

Control of losses also requires careful advance planning. There are two main aspects to good loss control: the physical and the psychological.

(a) There are many physical devices that can be installed to minimise losses when harmful events actually occur. Sprinklers, fire extinguishers, escape stairways, burglar alarms and machine guards are obvious examples. It is not enough, however, to install such devices. They will need to be inspected and maintained regularly.

(b) The key psychological factors are awareness and commitment. Every person in the business should be made aware that losses are possible and that they can be controlled.

6.3.3 Internal controls

Operational risks, financial reporting risks and compliance risks are reduced through internal control systems and internal controls. The purpose of controls is to:

- reduce the risks of failure to achieve organisational goals or targets (preventive controls)
- identify variations between actual performance and target (detective controls)
- take corrective action when actual performance varies adversely from target (corrective controls)

The work of internal auditors, and the review of internal control as part of an external audit, are examples of assurance procedures relating to risk and risk management.

6.3.4 Assurance procedures

A firm of accountants may be engaged to give advice on the adequacy of controls within a business plan, or to carry out a review of the adequacy and effectiveness of existing controls.

An assurance engagement to assess the effectiveness of controls or a control system involves:

- establishing the purpose and nature of the control system and of the controls within the system
- assessing the effectiveness of the design of the controls, and whether the controls as designed are sufficient for achieving the stated objectives of the control system
- assessing whether the controls, if suitably designed, are implemented effectively, or whether there appear to be failures in the operation of the controls

By assessing the effectiveness of the design and application of controls, assurance engagements should help to manage risks by reducing the likelihood or the impact of adverse risk events.

6.4 Accepting risks

Risk acceptance or retention is where the organisation bears the risk itself and, if an unfavourable outcome occurs, it will suffer the full loss. Risk retention is inevitable to some extent. However good the organisation's risk identification and assessment processes are, there will always be some unexpected risk. Other reasons for risk retention are that the risk is considered to be insignificant, or the cost of avoiding the risk is considered to be too great compared with the potential loss that could be incurred.

The decision of whether to retain or transfer risks depends firstly on whether there is anyone to transfer a risk to. The answer is more likely to be 'no' for an individual than for an organisation, because:

- individuals have more small risks than do organisations, and the administrative costs of transferring and carrying them can make the exercise impractical for the insurer
- the individual has smaller resources to find a carrier

In the last resort, organisations usually have customers to pass their risks or losses onto, up to a point, while individuals do not.

Professional skills focus: Applying judgement

One of the skills tested in the ACA exams is your ability to apply judgement and to assess the interaction of information from different sources. An organisations' response to risk depends upon the effect of the risk on the business as well as management's attitude towards risk so read the scenario carefully to determine the most appropriate strategy in the specific circumstances.

Interactive question 9: VSYS

VSYS Inc manufactures a range of computer products from its single factory located in a medium- sized town in central US. About 20% of the working population are employed at VSYS, and the company has a reputation for being a good employer with specific focus on maintaining and enhancing benefits for its employees.

Although the company is profitable, the recent management accounts show falling margins with the possibility of a loss being made next year – the first in the 25-year history of the company. The main reasons for the falling profits have been identified as increasing competition from manufacturers in the Far East, and ongoing quality control issues with several key manufacturers. A recent feasibility study shows that moving production to a Far Eastern country would enable VSYS to take advantage of lower labour costs and proximity to suppliers of high-quality components. The administration and marketing functions would remain at their current location.

Movement of production systems to the Far East is seen as a particular problem for VSYS. Specific areas of concern include:

- obtaining and maintaining supplies from new suppliers
- setting up production lines with new workforce and new machinery
- maintaining sufficient inventory of materials to meet demand when the delivery times are uncertain
- implementing any necessary revisions to the management accounting systems

However, the board is confident that the move will be successful and looks forward to a positive response from workers and shareholders.

Requirement

Assess the risks associated with the decision to outsource to the Far East, briefly recommend ways in which these risks can be controlled and briefly describe the assurance work that VSYS should carry out on potential suppliers.

See **Answer** at the end of this chapter.

6.5 Risk pooling and diversification

Risk pooling and diversification involves using portfolio theory to manage risks. You may remember that portfolio theory is an important part of an organisation's financial strategy, but its principles can be applied to non-financial risks as well.

Risk pooling or diversification involves creating a portfolio of different risks based on a number of events, some of which may turn out well while others will turn out badly, the average outcome of which will be neutral. What an organisation has to do is to avoid having all its risks positively correlated, meaning that everything will either turn out extremely well or extremely badly.

One means of diversification may be geographical – spreading risk across countries at different stages of the trade cycle.

In addition, although diversification may sound good in theory, the company may have insufficient expertise in the product or geographical markets into which it diversifies, leaving it vulnerable to competition from other companies that focus on a specific market or product type.

Interactive question 10: Budget airline

A budget airline that offers low-cost short-haul flights is considering the provision of flights to a country with a volatile political environment, where public spending on such facilities as airports is often withdrawn without warning. There is also a history of foreign planes being grounded for no apparent reason and being forbidden to leave the country for several days. Market research has shown that despite these problems, there is considerable demand for low-cost flights to this country.

Requirement

Identify any potential risk strategies that could be adopted by the airline's management.

See **Answer** at the end of this chapter.

6.6 Implementation of risk management plans

Implementation of the risk management process helps to clarify what ongoing actions should be taken beyond the planning stage to ensure that the system is implemented in a manner that is beneficial to the management team. The implementation process helps to ensure that you get the best risk protection for the amount invested in the risk and other management processes.

The implementation process focuses on the process itself rather than on the risks of a particular project. This step should be ongoing, focusing on the performance of the risk management process and the way in which it is integrated with other processes relevant to the project in question.

Organisations need to treat the implementation stage as a separate project with clear objectives and success criteria, clear planning, proper resourcing and effective monitoring and control.

6.7 Monitoring of risk management plans

All risk management plans must be monitored to ensure that they are achieving the desired results and that changes to the project's risk profile are reflected.

To assess the effectiveness of risk management plans, standards and benchmarks must be established against which results should be measured. Standards can come from such sources as industry regulations or the industry leader.

Once the standards and benchmarks have been established, the risk management plan can be measured continually against them over time to allow actual performance to be compared with expected performance, which in turn can be used to adjust below-standard results.

In order to determine when adjustments to performance should take place, a business needs to establish a threshold (or 'trigger point') which represents a sufficient change in risk exposure to warrant another risk analysis being carried out. Risk management is a continuous process and those responsible for handling risk should be prepared to treat it as such.

As with any process, evaluation of risk management plans is essential to ensure they are performing to expectations. Managers and stakeholders in the risk management process should consider such areas as:

- How successful was the plan and were the benefits and costs at the predicted level?
- In the light of the above, are any changes needed to improve the plan?

- Would the plan have benefited from the availability of additional information?

Risk monitoring is similar to an audit of the risk management process. Various tests will be carried out to determine whether individual controls are working properly and recommendations made in the light of results. However, unlike auditing, risk management monitoring does not take place only on an annual basis. Risk monitoring is a continuous process.



Context example: Infosys – risk management

We have already discussed the key risks faced by Infosys according to its 2016–17 annual report. The report went into detail about its risk management activities.

During the year, risk management practices were primarily focused on the effectiveness of strategic programs in improving competitive position, new initiatives, preparedness to address incidents that could cause disruption to physical and technological infrastructure, strengthening internal controls to detect fraudulent activity, leadership development, leadership succession planning, and monitoring the impact of changes in the regulatory environment.

The company carried out the following risk management activities:

- Regularly assessed progress on the execution of strategic programs, specifically progress on the growth of new software enabled services, impact of automation, performance of subsidiary businesses, leadership succession planning and operating cost optimisation.
- Regularly assessed the business environment including client concentration, client technology spend, growth of top clients and revenue bookings from large outsourcing engagements.
- Reviewed key operational risks and actions based on the internal risk register, external assessments, internal audit findings and incidents. Reviewed operational risk areas including client service delivery, information security (cyber-attacks and threat intelligence), women's safety, physical security, succession planning, capital expenditure on infrastructure and business continuity management.
- Monitored key developments in the regulatory environment, especially of the UK and USA, relating to immigration laws, minimum wages and impact to businesses of our clients.
- Monitored the availability of natural resources, such as water and power, and its impact on operations.

Source: https://www.infosys.com/investors/reports-filings/annual-report/annual/Documents/AR-2017/financials/pdf/Infosys_AR17_Risk_Management_Report.pdf

6.8 Board monitoring of control systems

As well as monitoring specific plans, the board needs to review the effectiveness of systems taken as a whole. Boards should regularly receive and review reports and information on internal control, concentrating on:

- what the risks are and strategies for identifying, evaluating and managing them
- the effectiveness of the management and internal control systems in the management of risk, in particular how risks are monitored and how any weaknesses have been dealt with
- whether actions are being taken to reduce the risks found
- whether the results indicate that internal control should be monitored more extensively

In addition, when directors are considering annually the disclosures they are required to make about internal controls they should conduct an annual review of internal control. This should be wider ranging than the regular review; in particular it should cover:

- the changes since the last assessment in risks faced, and the company's ability to respond to changes in its business environment
- the scope and quality of management's monitoring of risk and internal control, and of the work of internal audit, or consideration of the need for an internal audit function if the company does not have one
- the extent and frequency of reports to the board
- significant controls, failings and weaknesses which have, or might have, material impacts on the accounts
- the effectiveness of the public reporting processes

6.9 Reporting on risk management

There are minimum expected guidelines for disclosure on risk management and corporate governance. Publicly traded companies should report on the risks they face, and outline these risks in more detail.

- The governing body of the company (generally the board of directors) should acknowledge responsibility for internal control systems.
- An ongoing system should be in place for identifying, evaluating and managing significant risks.
- An annual process should be in place for reviewing the effectiveness of the internal control systems.
- There should be a process to deal with the internal control aspects of any significant problems disclosed in the annual report and accounts.

6.10 Limitations of risk management plans

As with all business processes, regardless of their quality, risk management plans have their limitations. They are only as good as the information that is used to construct them. If risks are not assessed properly, then a great deal of time and resources could be wasted in dealing with the risk of losses that, in fact, are highly unlikely to occur – time and resources that could have been more gainfully employed elsewhere. Some organisations overestimate what risk management processes should be able to achieve, to the extent that work is suspended until the risk management process is considered to be complete.

At the other extreme, there are organisations who believe themselves to be immune from losses resulting from business risk, simply because they have risk management processes in place.

Complacency is one of the worst enemies of successful businesses. As mentioned above, risk management processes must be continually monitored for any weaknesses – just like the business environment itself, they are not static instruments.

Such is the focus on risk and its consequences in today's business that there is a danger of management spending so long thinking about the negative aspects of projects that they forget about the positive aspects.



Interactive question 11: LP

LP manufactures and supplies a wide range of different clothing to retail customers from 150 stores located in three different countries in the Eurozone.

In order to increase sales, a new internet site is being developed which will sell LP's entire range of clothes using 3D revolving dummies to display the clothes on screen. The site will use some new compression software to download the large media files to purchasers' PCs so that the clothes can be viewed. This move is partly in response to environmental scanning which indicated that a new competitor, PVO, will be opening an unknown number of stores in the next six months.

As a cost-cutting move, the directors are considering delaying LP's new range of clothes by one year. Sales are currently in excess of expectations and the directors are unwilling to move away from potentially profitable lines.

A retail customer of LP's has recently brought legal proceedings against LP for loss of business through one of the chemicals used to waterproof some garments releasing toxic fumes after prolonged exposure to sunlight. The case is due to come to court in two weeks' time but LP's lawyers think that it could be a very lengthy case and believe that LP will eventually lose it. LP's board has made a number of estimates. The directors believe that the best outcome for LP will be damages of £300,000 payable in one year's time. The worst possible outcome would be for the case to continue for 3 years, in which case the estimate of damages and costs is £2,500,000, payable in 3 years' time. A further estimate, between these two extremes, is that damages of £900,000 will be payable in 2 years' time. Management's estimates of probabilities are best outcome 30%, worst case outcome 10% and middle ground outcome 60%. No provision or any disclosure has been made for this court case in the draft financial statements that are due to be finalised over the next few weeks. Prior to the legal claim being made, LP had already stopped using the chemical in its manufacturing process.

Requirements

11.1 Explain the business risks facing LP and briefly describe how these risks can be managed.

11.2 Explain how the possible losses arising from the legal claim should be dealt with in LP's financial statements.

See **Answer** at the end of this chapter.

7 ESG and climate risk management



Section overview

Over the past two decades, risks associated with ESG issues and climate change have become much more significant, both in terms of their likelihood and in terms of their impact.

There is no universally accepted definition of what is meant by ESG risks – narrow definitions may focus on the environment or social issues, while broader definitions focus on a range of issues.

Guidance has been provided to apply the COSO enterprise risk management model to ESG risks.

Climate change has brought additional risks to organisations, both at a strategic level and at an operational level.

7.1 Nature of ESG risks

Since 2006, the World Economic Forum has carried out its annual Global Risks Perception survey, in which it asks leaders what they perceived as being the most significant risks over the next ten years. The following quotation from the 2021 report shows how significant risks related to the environment and climate change are.

"Among the highest likelihood risks of the next ten years are extreme weather, climate action failure and human-led environmental damage… Among the highest impact risks of the next decade, infectious diseases are in the top spot, followed by climate action failure and other environmental risks;"

(World Economic Forum, *The Global Risks Report 2021 16th edition*. Available from: https://www3.weforum.org/docs/WEF_The_Global_Risks_Report_2021.pdf (Accessed 14 July 2022)

There is no universally accepted definition of ESG risks but, broadly speaking, risks relating to environment, poor social responsibility and poor governance can be categorised as ESG risks.

Recent high profile ESG events include the following:

- **Environmental**: Events caused by climate change such as extreme weather causing disruption to supply chains.
- **Social**: Reputational harm caused by events, such as OXFAM's alleged cover up of sexual harassment by employees in Haiti in 2018.
- **Governance**: Corporate failures resulting from fraud or false accounting at the highest level, such as the failure of Carillion which was due to poor management in taking on loss making contracts and failing to perform adequate risk assessments.

A much broader approach to defining ESG related risks was developed by MCSI (*quoted in COSO, WBCSD Enterprise Risk Management, Applying enterprise risk management to environmental, social and governance-related risks):*

Pillars	Themes	ESG Key issues
Environment	Climate change	Carbon emissions
		Product carbon footprint Financing environmental impact
		Climate change vulnerability
	Natural resources	Water stress
		Biodiversity and land use
		Raw material sourcing
	Pollution and waste	Toxic emissions and waste
		Packaging materiality and waste
		Electronic waste
	Environmental opportunities	Opportunities in clean tech
		Opportunities in clean building
		Opportunities in renewable energy
Social	Human capital	Labour management

Pillars	Themes	ESG Key issues
		Health and safety
		Human capital development
		Supply chain labour standards
	Product liability	Product safety and quality
		Chemical safety
		Financial product safety
		Privacy and data security
		Responsible investment
		Health and demographic risk
	Shareholder opposition	Controversial sourcing
	Social opportunities	Access to communications
		Access to finance
		Access to health care
		Opportunities in nutrition and health
Governance	Corporate governance	Board
		Pay
		Ownership
		Accounting
	Corporate behaviour	Business ethics
		Anti-competitive practices
		Tax transparency
		Corruption and instability
		Financial system instability

7.2 Enterprise Risk Management for ESG risks

COSO and the World Business Council for Sustainable development (WBCSD) have provided guidance on how to apply the COSO enterprise risk management model (see Section 2 above) to ESG risks. The guidance is based around the five components of the COSO Framework:

1. **Governance and culture for ESG related risks:** The governance of an organisation will determine how ESG risks are managed. The board and executive management should be aware of ESG related risks and should foster a culture of collaboration between those responsible for risk management and those responsible for ESG issues.
2. **Strategy and objective setting for ESG related risks:** There must be a strong understanding of the impacts the organisation has on nature and society, as well as the dependencies the business has on nature and society. Organisations should examine the value creation process to understand these dependencies in the short-, medium- and long-term, and consider them when setting strategic objectives.

3. **Performance for ESG-related risks: This involves three steps**:
    - **Identifying risks** – various tools can be used to identify ESG related risks, and there is an emphasis on collaboration between the risk managers and sustainability practitioners, to ensure that ESG related risks are taken into account during the risk management process.
    - **Assessing and prioritising risks** – the severity of risks must be assessed in a manner that managers understand. It is important that emerging or longer-term risks are not ignored.
    - **Implementing risk responses** – this will determine in the longer term how effectively the organisation can preserve or create value over the longer term.
4. **Review and revision for ESG-related risks**: A process must be in place for monitoring the effectiveness of the ERM activities, and if necessary, taking action to modify them. Metrics must be used to alert management of the need for changes in the identification, assessment and responses to risks.
5. **Information, communication and reporting:** Reporting both internally and externally to support risk-informed decision making.

7.3 Risks related to climate change

According to the United Nations, "Climate change is the defining issue of our time and we are at a defining moment".

Climate change refers to long-term shifts in temperatures and weather patterns. Some of these shifts occur naturally, such as due to variations in the solar cycle but, since the 1800s, human activities have been the main driver of climate change.

Human activities contribute to climate change through the creation of greenhouse gases, principally carbon dioxide, methane, nitrous oxide and ozone. These are produced by activities such as the burning of fossil fuels (eg, oil and coal), deforestation and the use of landfill sites for waste disposal. Greenhouse gasses accumulate in the earth's atmosphere and trap the heat from the sun. Climate change has many adverse effects, including severe weather conditions and melting polar ice caps leading to rising sea levels.

The Paris Agreement of 2015 is a legally binding treaty in which 193 parties (192 countries plus the EU) have committed to reduce their carbon emissions. The aim of the agreement is to limit increases in global temperatures by the end of this century to 2.0 degrees Celsius above pre-industrial levels. Scientists believe that this can be achieved but will require a huge reduction in carbon emissions and the use of technology to remove carbon from the atmosphere.

In the UK, the government has committed to a legally binding target of net zero emissions by 2050. Climate change can potentially bring many risks to organisations.
The Task Force on Climate-related Financial Disclosures (TCFD) identifies **several classes of risks and opportunities** relating to climate change:

Risks	Opportunities
Acute: extreme weather events **Chronic**: changing weather patterns and rising mean temperature and sea levels	
Policy and legal: these relate to changes in regulations and exposure to litigation	**Resource efficiency**: cost savings due to more efficient use of resources and recycling

Technology: relates to risk of existing products and services being replaced with lower emissions options, and unsuccessful investment in new technologies	**Energy sources**: use of new sources of energy and use of government incentives
Market: changing costs of raw materials and changing customer behaviour	**Products and services**: development of new, low emission goods and services
Reputation: risk of loss of reputation among customers and other stakeholders (eg, investors)	**Markets**: access to new markets

7.3.1 Strategic and operational risks related to climate change

Strategic risks are risks that will impact an organisation over the longer term. In terms of climate change, these risks identified by the TCFD are strategic in nature. Examples of strategic risks related to climate change are:

- Being required to withdraw goods and services that contribute to greenhouse gasses from the market – for example, car manufacturers are moving into the production of electric cars and will stop producing diesel- and petrol-powered cars in the next two decades.
- Changing laws relating to greenhouse gas emissions – such as requirements to reduce carbon emissions. These may bring additional compliance costs.
- Reputational risk for organisations that are deemed to cause excessive amounts of greenhouse gasses
- Changing weather patterns may make it unfeasible to continue production, particularly in agriculture, if there is an increase in droughts or floods.

Operational risks are those risks that impact operations. Examples of operational risks related to climate change are:

- Disruption to operations caused by extreme weather conditions such as floods, droughts or wildfires. These disruptions may affect the organisation directly or may impact on parties further up the value chain – for example, if a supplier of a vital component has to close down for a period due to an extreme weather.
- Shortage of water. Water is used in many industries and water shortages may lead to the rationing or lack of availability of water.
- Fines and penalties for failure of systems that monitor an organisation's greenhouse gasses, leading to exceeding legal limits.

Summary

Tick off

Risk management tools and plans have become increasingly important over the last few years. More and more organisations are recognising the benefits of having formal plans for dealing with the risk. ☐

As with all business processes, risk management plans require detailed groundwork to ensure they fulfil the objectives. A commonly used framework is: ☐

- Establishing the context
- Risk identification
- Risk assessment
- Risk profiling
- Risk quantification
- Risk responses

Implementation of plans

Each part of the framework must be dealt in with detail to allow management to fully understand what they want the risk management plan to do. ☐

Management should also be aware of the limitations of risk management, including guarding against complacency at the one extreme and trying to build a perfect system at the other. ☐

Consider the risks relating to climate change and ESG factors and ensure that these are considered in the risk management process. ☐

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, complete the following knowledge diagnostic and check you are able to confirm you possess the following essential learning from this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
1.	What are the main sources of risk and uncertainty for businesses? (Topic 1)
2.	What principles does the COSO ERM framework set out? (Topic 2)
3.	What are the responsibilities of the Board with regard to risk management? (Topic 3)
4.	What are the main concerns of stakeholder groups regarding risk and uncertainty? (Topic 4)
5	What do the standard deviation and coefficient of variation tell us about a particular probability distribution? (Topic 5)
6.	How can risks be prioritised and managed? (Topic 6)
7.	What are the main types of ESG risk? (Topic 7)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are helpful to further topic understanding and guide skills application before you proceed.

Question	Learning benefit from attempting this question
2 HOOD	This question is a good example of the need to apply your knowledge to a scenario. It is a full risk question examining HOOD's risks, the effect of those risks on the organisation and the mitigation strategies that the company could employ to minimise the risk. Use your technical knowledge and commercial awareness to provide a practical response.
4 Bush Council	Risk management is also important in the not-for-profit sector which is covered in this question. Be sure to understand risk management in profit seeking and not-for-profit entities for the SBM&L exam. Here you are asked to examine the opinions and risk attitudes of two board members. You are also required to recommend suitable assurance procedures for internal audit. Remember that assurance represents 10% of the syllabus and can be the source of some straightforward marks in the exam.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your ueries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Technical reference

1 IAS 10, *Events After the Reporting Period*

- Outlines the requirements as to when events after the end of the reporting period should be adjusted in the financial statements for that period. Adjusting events are those which provide evidence of conditions existing at the end of the reporting period, whereas non-adjusting events are those which indicate conditions arising after the reporting period.

2 IAS 36, *Impairment of Assets*

- Seeks to ensure that an entity's assets are not carried at more than their recoverable amount (being the higher of fair value less costs of disposal, and value in use). An annual impairment test is required for goodwill and certain intangible assets, but for the majority of assets an impairment test is only required where there is an indication of impairment of an asset.

3 IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*

- Outlines the accounting for provisions (liabilities of uncertain timing or amount), as well as contingent assets (possible assets) and contingent liabilities (possible obligations and present obligations which are not probable or are not reliably measurable). Provisions are measured at the best estimate of the expenditure required to settle the present obligation, and reflect the present value of expenditures required to settle the obligation where the time value of money is material.

Self-test questions

Answer the following questions.

1 ANG

ANG is a road haulage contractor. The company specialises in collection and delivery of large or heavy items, such as railway locomotives and sections of bridges, from the manufacturer to the customer. The company owns 49 road vehicles of different sizes to enable transportation of the different goods.

ANG's risk management policy is based on taking out insurance. As well as the standard employer and third-party liability classes of insurance, ANG insures against damage to road infrastructure such as bridges and tunnels from its own vehicles, or as a result of goods being carried becoming unstable and falling off ANG's lorries.

ANG's terms and conditions of carriage note that radioactive goods will not be transported under any circumstances. Explosives are carried, but only where the owner accepts liability on their own insurance.

Contingency planning is limited; the board of ANG believes that if any risks do occur, then ANG has sufficient vehicles to continue operations.

The board of ANG is also considering a new venture for the same-day delivery of goods where the distance to travel is more than its existing fleet of road vehicles could travel in one day. This venture involves the purchase of surplus Hercules transport planes from the army. The board has recently decided to make the purchase of the planes because they are being offered at a substantial discount. Marketing activities will commence next month.

Requirements

1.1 Explain the elements of a risk management framework in an organisation.

1.2 Explain the risk management strategies available to an organisation.

1.3 Evaluate the risk management strategy of ANG, explaining any amendments that you think are necessary.

2 HOOD

HOOD sells a wide range of coats, anoraks, waterproof trousers and similar outdoor clothing from its 56 stores located in one country. The company is profitable, although the gross profit in some stores has declined recently for no apparent reason.

Each store uses EPOS to maintain control of inventory and provides the facility to use EFTPOS for payments. However, about 55% of all transactions are still made in cash. Details of sales made and inventories below reorder levels are transferred to head office on a daily basis where management reports are also prepared.

Inventory is ordered centrally from head office, details of requirements being obtained from the daily management information provided by each store. Orders are sent to suppliers in the post, with stock arriving at each store approximately 10 days after the reorder level is reached.

Requirements

2.1 Identify the different risks facing HOOD, placing the risks into suitable categories.

2.2 Discuss the potential effect of each risk on the organisation, describing how the impact of that risk may be minimised.

3 The LinesRUs Company

The LinesRUs Company is responsible for maintaining the railway infrastructure for the rail network in a large European country. Main areas of responsibility for the company include:

- ensuring that the railway tracks are safe
- ensuring its signalling equipment is installed correctly and works properly
- maintenance of overhead power lines for electric trains

Income is fixed each year dependent on the number of train services being operated and is paid via a central rail authority. The company is granted a sole franchise each year to provide services on the rail network.

Work is scheduled in accordance with the amount of income, and to provide LinesRUs with an acceptable operating profit. Any additional work, over and above standard maintenance (eg, due to foreseen factors such as bridges being damaged by road vehicles and unforeseen factors such as car drivers falling asleep and driving their cars onto railway tracks), is negotiated separately and additional income obtained to repair the infrastructure in these situations.

A lot of maintenance work is relatively simple (eg, tightening nuts and bolts holding railway tracks together) but is extremely important, as an error may result in a train leaving the rails and crashing. The board of LinesRUs is aware of many of these risks and attempts to include them in a risk management policy.

However, recently a train was derailed, causing the death of 27 passengers. Initial investigations show that faulty maintenance was the cause of the derailment. One of the unforeseen consequences of the crash has been a fall in the numbers of people using trains, with a subsequent fall in income for train operators. LinesRUs are being sued by the train operators for loss of income, and the national press is suggesting LinesRUs must be incompetent and are calling for a re-evaluation of the method of providing maintenance on the rail network.

One of the risks that LinesRUs faces is the risk of injuries to staff during working hours. Currently the average number of injuries experienced per year is 300. An injury is only recorded if a member of staff requires medical treatment. The average cost of an injury in terms of lost days and compensation to the injured workers is £25,000. The company is considering investing in one of two safety programmes that would reduce the expected number of expected injuries. Statistics compiled by rail maintenance companies in other countries show the number of expected injuries as a result of implementing the programmes are as follows:

	Programme 1	Programme 2
Expected injuries	250	225
Standard deviation	30	80
Additional costs	£1 million	£1.5 million

Requirements

3.1 Advise the directors of LinesRUs of the main stages of a structured risk analysis approach that will be appropriate to the company's needs.

3.2 Using the 'Transfer-Avoid-Reduce-Accept' framework, construct four possible strategies for managing the risk that rail crashes could occur. Your answer should describe each strategy and explain how each might be applied to this case.

3.3 Discuss the relative advantages of the two safety programmes for staff.

4 Bush Council

Bush Council is the local government authority responsible for the running of public services in a district of approximately 300 square miles and with a population of over 400,000. The Bush district comprises a mixture of towns, villages and rural areas.

The council employs approximately 16,000 staff in a wide variety of occupations. The council is responsible for the maintenance of the entire public infrastructure in its area of responsibility, including the roads and sewerage systems. The council also manages education and care for vulnerable residents. The council has a divisional structure, with each division taking responsibility for specific matters such as education and roads throughout the Bush district.

Injury statistics

Employment law requires that every employer, including Bush Council, must maintain a register of all workplace injuries sustained by employees. There is no precise definition of a reportable injury, but council guidelines indicate that anything that requires a dressing, such as a bandage or sticking plaster, must be reported as minor injuries. Injuries are classified as 'serious' if they require the victim to be absent from work for more than three days and 'severe' if they require admission to hospital or involve a fatality.

The latest injury statistics show that there were 175 injuries during the year ended 31 December 20X0, of which 25 were serious injuries and 3 were severe. The council's director of operations is satisfied with these figures because the number of injuries is no worse than in previous years. He holds the view that such figures are to be expected, given the diverse range of jobs, many of which are risky, throughout the council. The chief executive of the council does not share these views: he thinks that the council should try to prevent all injuries by eliminating accidents in the workplace. The chief executive is also concerned about the accuracy of the injury statistics and wants the council's internal audit department to ascertain whether the figures are reliable.

Requirements

4.1 Discuss the director of operations' view that it is impossible to prevent all workplace injuries and discuss the chief executive's view that it is unacceptable for Bush Council to tolerate any injuries.

4.2 Recommend assurance procedures that internal audit could use to verify the figures for the number of injuries in Bush Council's workplaces.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

Note: This is not an exhaustive list – you may have thought of different examples that are equally relevant.

Six risks:

- **Seasonality of business** – Most purchases are likely to be associated with seasons. Easter and Christmas are major seasons for the confectionery business, but there may be dips at other times of the year.
- **Impulse buying** – A large proportion of confectionery purchases are made on impulse. If economic changes reduced the amount of impulse buying – due to consumers having less money to spend – this could have a major effect on profits.
- **Supply of raw materials** – Sugar is a major raw material used in the manufacture of confectionery and non-alcoholic beverages. There may be a risk of relying too heavily on one major source of supply of sugar and other raw materials necessary for the continued production of products.
- **Competition** – The confectionery and non-alcoholic beverages markets are highly competitive. If there is a particular product that contributes a large proportion of sales revenue, there is a considerable risk that a rival company will bring out a similar product and take some of the market share. The extent of this risk will depend very much on the power of the brand.
- **Role of food in public health** – With lots of publicity about levels of obesity, children's eating habits, heart disease and diabetes, there is a significant threat to the confectionery and fizzy drinks markets. There is potential for governments to restrict advertising of certain products and to impose additional taxes on confectionery and fizzy drinks, which could make marketing more difficult. This could have a significant downward effect on sales and profits. Consumer tastes may change for health-related reasons. If the company is unable to respond, this will also result in declining sales or margins.
- **Product recalls and incorrect labelling of merchandise** – The confectionery industry is particularly susceptible to the risk of product recalls and incorrect labelling. The necessary publicity given to the potential consequences of nut allergies, for example, has led to much stricter regulation of labelling information. There have been instances of products being recalled due to failure to include warnings of nut content on labels. It is not just the product recall itself that is expensive – the potentially damaging effect on the company's reputation could have an even greater impact. Although product recalls are infrequent, their considerable impact is such that very tight internal controls are necessary to prevent their occurrence.

Answer to Interactive question 2

Probability that product is faulty, given that it was produced by Alan

Let P(A) = probability that product was faulty

Let P(B) = probability that product was processed by Alan

P(A and B) = 3% x 12% = 0.36%

Probability that product is faulty, given that it was produced by Alan

$$= P(A|B) = \frac{P(A \text{ and } B)}{P(A)} = \frac{0.36\%}{4\%} = 9\%$$

Therefore 9% of the products produced by Alan are faulty.

Answer to Interactive question 3

Z score is (950 − 1,000)/25 = −2.

950 units is therefore 2 standard deviations below the mean of 1,000 units. We are therefore looking at the probability of being in the range between the mean and 2 standard deviations below the mean.

Using the diagram above, we can see that there is a 95.47% chance of a value occurring between 2 standard deviations above and 2 standard deviations below the mean. There is therefore a 47.7% probability of being between the mean and 2 standard deviations below it (95.47% ÷ 2).

There is therefore a 47.7% chance that the machine will produce between 950 and 1,000 units per day.

Answer to Interactive question 4

In this case, the mean is in the middle of the range and we are looking for the probability of being between 1 standard deviation below and 1 standard deviation above the mean. Referring to the normal distribution diagram above, we can see that the probability of being in the range from 1 standard deviation below to 1 standard deviation above the mean is 68.2%.

Note: This has a much higher probability than being between the mean and two standard deviations from the mean.

There is therefore a 68.2% chance that production will be between 975 and 1025 units per day.

Answer to Interactive question 5

The 95% confidence intervals for the for the population mean are:

Confidence intervals = sample mean ± 1.96 × standard error

= 178 ± 1.96 × 4.5 = 169.2 cm to 186.8 cm

The meaning is that there is a 95% chance that the actual population mean lies within the interval 169.2 to 186.8.

Note - when using excel for this type of problem, enter the standard deviation of the sample, not the standard error of the distribution:

B5 =CONFIDENCE(B1,B2,B3)

	A	B
1	Alpha (100% -95%)	0.05
2	Standard deviation	28.46
3	Sample size	40
4	Mean	178
5	Confidence value	**8.82**

The confidence interval is therefore 178± 8.82 = 169.18 to 186.82

Answer to Interactive question 6

6.1 The 90% confidence interval is the expected mean ± 1.65 standard deviations = 5 ±(1.65×0.1) = 4.835 to 5.165

6.2 The sample mean is 5.2cm, which lies outside the confidence interval. The null hypothesis is therefore rejected at the 90% level.

6.3 The conclusion is that the mean of the ball bearings is not 5cm. The sample had a mean of 5.2cm which is outside of the confidence interval. It is very unlikely that a sample would have a mean of 5.2cm if it came from a population with a mean of 5.0cm.

Answer to Interactive question 7

Analysis of the pre-populated spreadsheet shows that for Lila company, both division's monthly revenues increase in parallel. There is therefore near perfect correlation shown by the correlation of 1(Cell B25) between the revenues of the two divisions. Lila fulfils Greendot's acquisition criteria in that both divisions are showing growth. Although the standard deviation of 43.43 shows variability in monthly revenues, it is due to growth and is therefore a largely predictable variability.

Plume company shows increasing revenue for Division A across the 21 months of data provided. Division B increases to month 13 and then falls to month 21. The result is therefore a low correlation of 0.12 (Cell F25) between divisional monthly revenues. The total revenue of the company has a much lower SD (Cell G24, SD = 21.76) than the sum of the SDs of the two divisions (Division A SD = 14.71, Division B SD = 14.35) because risk is offset between the two divisions in months 12 to 21 where revenues move in opposite directions. Despite lower variability in cashflows, as shown by the lower standard deviation for Plume (21.76 compared to 43.43 for Lila), revenues are falling in Division B so Plume does not fulfil Greendot's acquisition criteria.

	A	B	C	D	E	F	G
1		**Lila Revenues £'000**			**Plume Revenues £'000**		
2	Month	Division A	Division B	**Total**	Division A	Division B	**Total**
3	1	10	40	**50**	10	40	**50**
4	2	12	45	**57**	12	45	**57**
5	3	14	50	**64**	14	50	**64**
6	4	16	55	**71**	16	55	**71**
7	5	18	60	**78**	18	60	**78**
8	6	20	65	**85**	20	65	**85**
9	7	22	70	**92**	22	70	**92**
10	8	24	75	**99**	24	75	**99**
11	9	26	80	**106**	26	80	**106**
12	10	28	85	**113**	28	85	**113**
13	11	30	90	**120**	30	90	**120**
14	12	32	95	**127**	32	86	**118**
15	13	34	100	**134**	34	82	**116**

	A	B	C	D	E	F	G
16	14	36	105	**141**	37	78	**115**
17	15	38	110	**148**	40	74	**114**
18	16	40	115	**155**	43	70	**113**
19	17	42	120	**162**	46	66	**112**
20	18	44	125	**169**	49	62	**111**
21	19	46	130	**176**	52	58	**110**
22	20	48	135	**183**	55	54	**109**
23	21	50	140	**190**	58	50	**108**
24	Standard deviation	12.41 **1**	31.02	43.43	14.71	14.35	21.76
25	Correlatio n Lila	1 **2**			Correlatio n Plume	0.12 **3**	

1 =STDDEV(B3:B23) The formula can be copied in to Cells C24:G24

2 =CORREL(B3:B23,C3:C23)

3 =CORREL(E3:E23,F3:F23)

Answer to Interactive question 8

8.1 The dependent variable is daily sales of drinks. The independent variable is the temperature at midday. (Hint: sales depend on the temperature.)

8.2 If the temperature at midday is 20 degrees Celsius, then predicted sales of drinks are 500 + (10× 20) = £700.

8.3 A correlation coefficient of 0.7 shows that:

The value is positive, so the relationship between the dependent and independent variable is also positive (sales of drinks go up when the temperature goes up).

The coefficient is closer to 1 than 0, so quite a high portion of daily changes in sales of drinks is due to changes in the external temperature. The regression analysis explains the data comparatively well.

Answer to Interactive question 9

Risks

Possible non-compliance with laws – US

The possible reduction in the workforce in the US will mean that many employees will be entitled to some redundancy pay. There is the possibility of breach of employment law in the processing and payment of final salaries. Internal audit will need to review any redundancy calculations on a test basis to ensure that employment law has been complied with.

Possible non-compliance with laws – Far East

Establishing a factory in the Far East will mean that VSYS will have to comply with the laws and regulations of a foreign country. The directors must ensure that the law in the country is understood, possibly by hiring local solicitors.

Overall risks from new systems

Setting up a new factory in the Far East will also mean establishing new management and financial accounting systems in that country. Risks inherent in establishing those systems may

be minimised by exporting the systems currently being used in the US. However, it is unlikely that no modifications will be necessary, as new systems will be necessary to meet the specific situation in the new location.
New systems always provide a risk of failure or incorrect reporting, due to lack of adequate testing or implementation problems. Internal audit will need to review the systems in detail to try to minimise the errors that occur.

Communication risks – Far East to US

Establishing a new production location will mean that regular management and other reports will be sent between two geographically diverse locations. This new communication system will run risks such as communications being lost or intercepted en route. The board will need to ensure that appropriate encryption systems are introduced across the communication system to minimise these risks.

Board control

Geographical distance from the US to the Far East may limit the board's ability to maintain appropriate control of the new production location. The risk is that the new factory may manufacture the computer components correctly, but fail to meet its own constraints regarding mix of components produced or timescales for production. To maintain adequate control, a director may have to be appointed to be in residence at the new factory to ensure both locations are attempting to meet the objectives of VSYS.

ESG issues

The issue of redundancy has been discussed above, but there is a wider issue that the transfer of manufacturing away from where VSYS is based will have significant social implications. It is not clear what proportion of the workforce is employed in manufacturing, but as 20% of the town's working population is employed at VSYS, many jobs may be at risk, placing a strain on the rest of the town and its local economy.

The company appears to have taken its governance responsibilities seriously by considering its longer-term viability but the adverse reputation from this strategic change may still have negative consequences.

It is also unclear how the Far East manufacturing arrangement would work in practice, but it seems likely that there will be greater amounts of travel involved for management, supervision and the supply chain getting finished components back to the US market, all of which has an adverse carbon impact. The sourcing of raw materials for components that go into the company's products may also present sustainability challenges, especially if the country where the new factory is located does not comply with US regulations.

Assurance work

- Review financial statements of supplier for evidence of financial health.
- Discuss with supplier's directors their future plans and forecasts, to see if there is a possibility of future financial problems or whether the supplier is at risk of overtrading.
- Review ethical code, if supplier has one, and ascertain how the supplier ensures adherence to this code.
- Review evidence of supplier's employment terms and, if feasible, inspect suppliers' factories for evidence of working conditions.
- Search for evidence in media or on the internet of whether the supplier has been accused of poor employment practices or unethical behaviour.
- Review terms of contract and assess strength of guarantees and effectiveness of remedies if supplier fails to perform in accordance with contract.
- Ensure that contract includes sufficient protection for VSYS's intellectual property.

- Obtain evidence of supplier's capacity and assess whether supplier may have trouble meeting orders, particularly during periods of peak demand.
- Examine samples of production to see if supplier reaches required quality standards.

Answer to Interactive question 10

Risks:

- **Risk avoidance**
  - The airline could abandon plans to offer services to this country.
- **Risk reduction**
  - Invite the host government to be a partner in the venture.
  - Seek written assurances from the host government that the airport to be used will be fully maintained to international safety standards and that planes will not be prevented from taking off without good – and communicated – reasons.
  - Have contingency plans in place to adequately deal with any operational problems while in the country – for example, establish contacts with a local coach company to ensure transport to hotels if passengers have to spend another night in the country before the plane can take off; negotiate deals with local hotels to provide any necessary accommodation.
  - If the new service is likely to boost the country's economy due to, for example, an increase in visitors, it may be worth trying to lobby the country's Government for a change in policy.
- **Risk transfer**
  - Have adequate insurance in place to cover the cost of any planes being grounded.
  - Consider setting up an alliance with an airline located in the country in question, with the service being offered in that airline's name but with a codeshare arrangement.
- **Risk retention**
  - Accept the possibility that airports may not be properly maintained or planes may be grounded, and any accompanying costs.

Answer to Interactive question 11

11.1 Part (a)

Business risks

These are risks that LP's **performance could be better or worse than expected**.

The new business venture to sell clothes on the internet using 3D models to display the clothes.

There is the risk that demand will be far short of that anticipated or that costs of developing the internet site will significantly exceed budget.

LP should have assessed the 3D project for feasibility. Budgets should have been established and actual expenditure regularly compared with budgets. If actual expenditure is unavoidably and significantly in excess of budget, the board should consider whether the project should continue. Thorough testing procedures should have been built into the plan, and these should ensure that the website is capable of coping with anticipated demand. Once the website is operational, LP should monitor the level of sales generated by obtaining customer feedback through the site, and comparing sales generated with the costs of keeping the website updated.

Product obsolescence

The decision to lengthen the time of sale for each product may appear to decrease development costs. However, the board of LP must also take into account demand for the goods. The fashion industry tends to issue new clothes and designs every few months and, certainly in temperate climates, fashions will change according to the season. There is a risk that not amending the style of products sold will reduce sales far in excess of the reduction in expenditure. The overall going concern of the company may also be adversely affected if customers perceive the clothes to be 'out of date' and change to other suppliers.

LP should monitor the performance of products in detail, and look for evidence of falling sales and other signs that its products are viewed as old-fashioned, for example adverse customer or press comment. The board should also consider whether work on developing new products should continue to some extent, so that new lines can be launched quickly if demand falls.

New competition

The new company PVO appears to be aggressively attacking LP's marketplace position. While the overall effect of the new competitor is difficult to determine, having a new range of clothes available is likely to attract customers with little, if any, brand loyalty to LP.

LP should make sure that competitor activity is carefully monitored and responses are made to known or predicted competitor activity, for example, an advertising campaign to counter new products being launched by the competitor. LP's board should also review very regularly the performance of products which are most vulnerable to competitor activity and decide whether to invest more in these or concentrate on other, less vulnerable products.

Environmental legislation

Although LP is preoccupied with the legal case brought by their retail customer, there may also be exposure to litigation from the regulatory authority if it is found to have breached any environmental legislation regarding the use of toxic chemicals in its production processes. Such litigation could lead to fines and/or penalties that could be costly to the business.

If it has not already done so, the management of LP should conduct a regulatory audit to confirm that the company is compliant with any environmental legislation and if it discovers any risk of non-compliance, it should take remedial action as a priority and consider documenting this in case it is approached by the regulatory authority.

11.2 Part (b)

Provision

According to IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*, a provision shall be recognised when:

- An entity has a present obligation as a result of a past event.
- It is probable that an outflow of resources embodying economic benefits will be required to settle the obligation.
- A reliable estimate can be made of the amount of the obligation.

If these conditions are met, then a provision must be recognised.

The assessment of a provision for a legal claim is always a difficult area as it will be based on the evidence available but it could also be argued that any provision or disclosure could be prejudicial to the court case itself.

In this case, it would appear that the lawyers and management are fairly certain that damages and costs will be payable. The problem is the amount of any provision to be made. As there is a timescale involved here, the first stage will be to calculate the present value of each of the outcomes. Management have also assigned probabilities to each of the three possible

outcomes, so a further decision must be made as to whether to calculate an expected value or take the value of the most likely outcome. IAS 37 states that where a single obligation is being measured, the individual most likely outcome may be the best estimate of the liability, although in some circumstances, the range of outcomes may mean that a higher figure is required.

Outcome		Discount factor @ 10%	Present value	Probability	Expected value
			£'000		£'000
Best	300	1/1.10	273	30%	82
Most likely	900	$1/1.10^2$	744	60%	446
Worst	2,500	$1/1.10^3$	1,878	10%	188
					716

IAS 37 requires the estimated value of the provision to be the amount that the entity would rationally pay to settle the obligation. Arguably, in this case that could either be £744,000 as the most likely outcome or £716,000 as the expected value. As the directors are likely to want as low a provision as possible, they are likely to choose the expected value.

Answers to Self-test questions

1 ANG

1.1 Risk management structure

The organisation needs a structure **to facilitate and communicate information about risks**. A system such as an intranet or groupware product would be suitable, as it connects all the individuals in an organisation, allowing access to shared databases where information about risks can be stored.

Resources

Sufficient resources are required to support **effective risk management**. This means that the board of an organisation must allocate **an appropriate budget for risk management**, and then the budget should be spent on appropriate areas. The appointment of a risk management officer will help to ensure that budgeted amounts are spent appropriately.

Risk culture

The **culture** of the organisation should be developed as far as possible to ensure employees are **aware of risk and to act to avoid risks where possible**. Having a risk avoidance culture will help to ensure that management decisions taken focus on, and avoid, important risks.

Tools and techniques

Appropriate tools and techniques should be available in the organisation to enable **the efficient and consistent management of risks across an organisation**. Tools and techniques available may include obtaining **appropriate insurance** against risks and having **a clear risk management policy in place**.

1.2 Avoidance

In this situation, the organisation attempts to determine whether the possible **losses avoided from not undertaking a risky activity are greater than the advantages that can be gained from carrying out the activity**. If the losses avoided appear to outweigh the benefits of carrying out the activity, then the activity may not take place. In an extreme situation, entire sections of the business may be closed down if the risk or loss is considered to be too great.

Reduction

Risks are avoided in part but not reduced to zero. For example, the risk of launching a new product can be reduced by obtaining market research on possible demand for the product prior to manufacture and launch.

Risk reduction will also involve **contingency planning** to ensure that **if a risk does crystallise, the damage from that risk is minimised**. For example, most companies will have a contingency plan against their computer systems failing. Files will be backed up regularly, and alternative processing locations will be available if one centre becomes unavailable, eg, due to fire or flood.

Acceptance

Risk retention is where the **organisation bears the risk itself**. This means that if the unfavourable outcome occurs, then the organisation will suffer the full loss of that event.

Risk retention normally occurs in two situations. First, where some **risk occurs** which the organisation's **risk management policy did not detect**. Second, where risk was **classified as insignificant** or **the cost of eliminating the risk** was deemed to be **too great compared to the likelihood of that risk occurring**.

Risk retention may also involve **self-insurance**. This means that funds are placed into some fund against risks actually occurring.

Transfer

The last risk management strategy is to **transfer the risk to a third party**. The most commonly used risk transfer policy is to take out **insurance** against a risk occurring. However, **risks may also be transferred to other third parties, often without the knowledge of that party**. For example, there may be a minimal risk of errors occurring in some software. The cost of carrying out additional testing may be more than any compensation that may be payable if the error occurs and the customer makes a successful complaint. In this situation, risk has been transferred to the customer without the customer's knowledge.

1.3 **Transfer**

The **overall risk management strategy** of ANG appears to be one of **risk transfer**. This is the policy adopted by most businesses and is wholly **appropriate**, given that the **likelihood of many risks occurring is low; but if they do occur, then significant expenditure would be involved**. For example, if a load did fall off one of ANG's lorries, then the damage caused could be considerable, not only to the load itself, but also to other vehicles, people and even the roads being used. **ANG would not be able to operate legally without this insurance**, and so it is essential to obtain it.

Insurance

Whether ANG needs to insure against **damage to roads, bridges** and so on is unclear. The Government of the country is normally responsible for maintaining the transport infrastructure. ANG could probably withdraw this insurance and effectively **transfer the risk to the Government**. Some cost savings would accrue from this move.

Self-insurance

It appears that ANG effectively **self-insures against loss of vehicles** in respect of being able to provide a replacement vehicle at short notice. This may be acceptable in the case of individual losses. However, it may be **inappropriate** in situations where, for example, **a significant number of ANG's vehicles are destroyed** in a fire or flood. Where haulage contracts are signed for time-critical delivery of goods, then some **reciprocal agreement with another haulage company may be appropriate**.

Avoidance

The decision by ANG to **avoid risk** completely in **the transfer of hazardous materials** seems sensible. There has been some bad publicity about the transfer of radioactive goods by road, and the potential for claims, particularly if an accident occurred in an area of high population density, could be excessive and the damage to ANG's reputation would be considerable. In the case of explosives, ANG would need to **ensure** that the contract for carriage clearly stated that the **owner of the goods was responsible for insurance**. ANG may also want to obtain a copy of the insurance contract to confirm this.

Risk acceptance

There appears to **be some risk in purchasing the transport planes** prior to any market appraisal of the new venture. Normally the **risk** of a new venture would be **reduced by carrying out market research prior to significant expenditure being incurred**. The board would usually be advised to check whether there was a demand for this service prior to expenditure being committed.

Published accounts, internal accounts and budgets are a useful source of information. The more carefully costs are allocated to each department, the easier it should be to calculate additional costs that would be incurred in a given emergency, which budgets will be affected and how quickly necessary funds can be made available.

Detailed plans of site and buildings will show potential bottlenecks in fire escape routes, obstructions which might create difficulties for fire engines; and problems of access, which could occur for both the site and its neighbours from fire, explosion or escaping gas.

Physical inspection of buildings and machines (and perhaps of personnel, in the medical sense, and of their working practices) should take place on a regular basis.

2 HOOD

2.1 Risk can be defined as the possibility that events or results will turn out differently from what is expected.

The risks facing the HOOD Company are outlined below.

Operational risks

These are risks relating to the business's day to day operations.

Accounting irregularities

The unexplained fall in gross profit in some stores may be indicative of **fraud** or **other accounting irregularities.** Low gross profit in itself may be caused by **incorrect inventory values** or loss of **sale income**. Incorrect stock levels, in turn, can be caused by **incorrect inventory counting** or **theft of inventory** by employees. Similarly, loss of sales income could result from **accounting errors** or employees **fraudulently removing cash** from the business rather than recording it as a sale.

Systems

Technical risks relate to the **technology** being **used by the company** to run its business.

Backup

Transferring data to head office at the end of each day will be inadequate for backup purposes. Failure of computer systems during the day will still result in loss of that day's transaction data.

Delays in inventory ordering

Although stock information is collected using the EPOS system, reordering of inventory takes a significant amount of time. Transferring data to head office for central purchasing may result in some discounts on purchase. However, the average 10 days before inventory is received at the store could result in the company running out of inventory.

Non-business risks

These are risks that arise for reasons beyond the normal operations of the company or the business environment within which it operates.

Event

HOOD may be vulnerable to losses in a warehouse fire.

Business Risks

External risks relate to the business; they are essentially uncontrollable by the company.

Macroeconomic risk

The company is dependent on one market sector and vulnerable to competition in that sector.

Product demand

The most important social change is probably a change in fashion. HOOD has not changed its product designs for four years, indicating some lack of investment in this area. Given that

fashions tend to change more frequently than every four years, HOOD may experience falling sales as customers seek new designs for their outdoor clothing. HOOD may also be vulnerable to seasonal variations in demand.

Corporate reputation

Risks in this category relate to the overall perception of HOOD in the marketplace as a supplier of (hopefully) good-quality clothing. However, this reputation could be damaged by problems with the manufacturing process and a consequent high level of returns.

Profiling

By identifying and profiling the effects of the risks, HOOD can assess what the consequences might be, and hence what steps (if any) are desirable to mitigate or avoid the consequences.

Sustainability of products

As there is a variety of source materials used to manufacture HOOD's products, the company may be exposed to reputational damage if products are found to have come from unsustainable sources. For example:

- *Natural products such as cotton can be farmed using significant amounts of water and requiring the use of dangerous chemicals.*
- *Synthetic products may be sourced from unsustainable sources such as vinyl which have an adverse carbon impact.*
- *Even sustainable sources may come from locations which require significant logistical support (eg air travel) to get them to market.*

2.2 The potential effects of the risks on HOOD and methods of overcoming those risks are explained below.

Operational risks

Accounting irregularities

The potential effect on HOOD is **loss of income**, either from inventory not being available for sale or cash not being recorded. The overall amount is unlikely to be significant, as employees would be concerned about being caught stealing.

The risk can be minimised by introducing additional controls, including the necessity of producing a **receipt for each sale** and the **agreement of cash received** to the **till roll** by the shop manager. Loss of inventory may be identified by more frequent inventory checks in the stores or closed-circuit television.

Systems

Backup

The potential effect on HOOD is relatively minor; **details of one shop's sales** could be lost for part of one day. However, the cash from sales would still be available, limiting the actual loss.

Additional procedures could be implemented to **back up transactions** as **they occur**, using online links to head office. The **relative cost of providing these links**, compared to the likelihood of error occurring, will help HOOD decide whether to implement this solution.

Delays in inventory ordering

The potential effect on HOOD is **immediate loss of sales**, as customers cannot purchase the garments that they require. In the longer term, if stock-outs become more frequent, customers may not visit the store because they believe goods will not be available.

The risk can be minimised by letting the stores order goods directly from the manufacturer, using an extension of the EPOS system. Costs incurred relate to the provision of internet

access for the shops and a possible increase in cost of goods supplied. However, this may be acceptable compared to overall loss of reputation.

Non-business risks

Event

The main effects of a warehouse fire will be a **loss of inventory** and the incurring of costs to replace it. There will also be a **loss of sales** as the inventory is not there to fulfil customer demand, and perhaps also a loss of subsequent sales as customers continue to shop elsewhere.

Potential losses of sales could be avoided by holding contingency inventory elsewhere, and losses from the fire could be reduced by **insurance**.

External risks

Macroeconomic risk

The potential effect on HOOD largely depends on HOOD's **ability to provide an appropriate selection of clothes**. It is unlikely that demand for coats etc will fall to zero, so some sales will be expected. However, an increase in competition may result in **falling sales** and, without some diversification, this will automatically affect the overall sales of HOOD.

HOOD can minimise the risk by **diversifying into other areas**. Given that the company sells outdoor clothes, commencing sales of other outdoor goods, such as camping equipment, may be one way of diversifying risk. It can also look to reduce **operational gearing**, fixed cost as a proportion of turnover.

Product demand

Again, the **risk of loss of demand and business to competitors** may undermine HOOD's ability to continue in business.

This risk can be minimised by having a **broad strategy** to **maintain** and **develop** the **brand** of HOOD. Not updating the product range would appear to be a mistake, as the brand may be devalued if products do not satisfy the changing tastes of customers.

The board must therefore allocate appropriate investment funds to updating the products and introduce new products to maintain the company's image.

Corporate reputation

As well as **immediate losses of contribution from products** that have been returned, HOOD faces the consequence of loss of future sales from customers who believe its products no longer offer quality. Other clothing retailers have found this to be very serious; a **reputation for quality**, once lost, undoubtedly **cannot easily be regained**.

The potential effect of a drop in overall corporate reputation will be falling sales for HOOD, resulting eventually in a **going concern problem**.

HOOD can guard against this loss of reputation by **enhanced quality control procedures**, and introducing processes such as **total quality management**.

3 The LinesRUs Company

3.1 Risk identification

Risks cannot be managed without first realising that they exist. Managers need to maintain a **list of known or familiar risks** and the extent to which they can harm the organisation or people within it. Managers also need to be aware that unfamiliar risks may exist and maintain vigilance in case these risks occur. **Risk identification** is an **ongoing process** so that new risks and changes affecting existing risks may be identified quickly and dealt with appropriately, before they result in unacceptable losses.

LinesRUs appears to have **identified some risks** in its risk management policy. However, other risks do occur and managers within LinesRUs must be able to identify and respond to those risks quickly.

Risk assessment

It may be difficult to forecast the financial effects of a risk until after a disaster has occurred. Areas such as **extra expenses, inconvenience and loss of time** can then be recognised, even if they were not thought of in initial risk analysis. In a severe situation, damage to the company's reputation could result in LinesRUs becoming bankrupt.

In this situation, there has been a loss of confidence in the company, the extent of which may not have been foreseen. This has resulted in **additional expense in terms of lost passengers** – legal advice will be needed to determine whether LinesRUs is liable and whether the company's insurance meets this liability. It is also uncertain what the **additional time and cost of repairing the track** will be and whether LinesRUs can claim additional income for this work.

Sources of information to ensure that the risk can be minimised may include obtaining regular reports from train operators on the state of the rail infrastructure, and **monitoring news feeds** such as Reuters for early indication of potential disasters. LinesRUs should **file appropriate reports** of physical inspection of track as evidence of maintenance work carried out.

Risk profiling

This stage involves using the results of risk assessment in order to group risks into families. A consequence matrix is one method of doing this.

		Consequences	
		Low	**High**
Likelihood	**Low**	Loss of lower-level staff	Loss of senior staff
	High	Loss of suppliers	Major rail disaster affecting reputation of company Loss of computer data on maintenance work Loss of franchise

The analysis will be incomplete for LinesRUs because not all risks can be identified.

Risk quantification

Risks that require more detailed analysis can be quantified and, where possible, results and probabilities calculated. The result of calculations will show average or expected result or loss, frequency of losses, chances of losses and largest predictable loss to which LinesRUs could be exposed by a particular risk.

Unfortunately, **many of the risks facing LinesRUs** are **significant**. So while quantification can be enhanced by past events such as drivers falling asleep, they appear to be one-off situations, meaning that the actual event may not occur again. However, the adverse effects of the risk in terms of costs necessary to repair the rail infrastructure will be helpful, enabling LinesRUs to ensure that appropriate insurance is available – effectively guarding against loss by transferring the risk.

Risk consolidation

Risks analysed at the divisional or subsidiary level need to be **aggregated at the corporate level**. This aggregation will be required as part of the overall review of risk that the board needs to undertake. **Systems** should **identify changes in risks as soon as they occur**, enabling management to monitor risks regularly and undertake annual reviews of the way that organisation deals with risk.

There is no information on the **organisational structure** of LinesRUs. Given the risky nature of the company's business, LinesRUs is likely to be an independent legal entity to ensure that no other companies are adversely affected should LinesRUs go out of business.

3.2 Risk responses

Transfer

The risk is transferred to a third party. As noted above, this may not be possible if insurers are **not willing to accept the risk**. Alternative methods of risk transfer may have to be considered, including asking the State for some form of insurance.

Avoidance

LinesRUs may consider whether the risk can be **avoided**. However, given that maintenance work must continue and that errors are always possible, then the risk may **crystallise**. Avoidance is not possible.

The only method of avoidance would appear to be **termination of operations**. This again may not be appropriate, given that this would close LinesRUs's business.

Reduction

The risk can be **reduced by taking appropriate measures**. In the case of LinesRUs, these will include **regular training** for maintenance staff. Management should use other methods such as newsletters to **raise awareness** of the importance of work being carried out and the potential consequences of error. There should be **maintenance and enforcement of appropriate disciplinary procedures** where breaches of work practices have been identified.

LinesRUs may also consider loss control options. These may include hiring of lawyers to defend LinesRUs and release of publicity material on the work of LinesRUs, showing the extent of maintenance work normally carried out. LinesRUs may also consider loss control options. These may include hiring of lawyers to defend LinesRUs and release of publicity material on the work of LinesRUs, showing the extent of maintenance work normally carried out.

Acceptance

This is where the organisation retains the risk and, if an unfavourable outcome occurs, it will suffer the full loss. In the case of the rail crash, LinesRUs may have to **retain the risk** if **suitable insurance cannot be found**. Given the uncertainties regarding the costs resulting from the unfavourable outcome, insurers may be unwilling to insure for this type of event.

3.3 Both programmes would reduce the expected number of injuries. From a financial perspective the expected savings would be as follows:

	Programme 1	Programme 2
Expected reduction in injuries	50	75
Cost per injury	25,000	25,000
Expected injury costs saved	1,250,000	1,875,000
Cost of programme	£1 million	£1.5 million
Financial benefits of programme	250,000	375,000

Both programmes are worth investing in from a financial perspective, as the expected cost savings from the lower number of injuries is more than the cost of the programme. In terms of which programme is better, the financial benefits of programme 2 are higher than programme 1, so at first sight, it seems that programme 2 would be the better of the two. It should be noted that programme 2 is more risky than programme 1 however, as shown by the standard deviation, which is 80 for programme 2, compared to only 30 for programme 1. This means that under programme 2, the actual number of injuries in a particular year will vary from the

expected number to a much greater extent than they would for programme 1. The management will therefore need to decide if they wish to accept this higher volatility in exchange for enjoying a lower expected number of accidents.

4 Bush Council

4.1 Points in favour of Director of Operations' view

Human error

Even if Bush has strong risk management systems in place, they may still be **undermined by human error**. An isolated lapse in concentration could result in an accident.

Credible policies

In order to minimise or eliminate risks, more onerous health and safety procedures may be introduced, including investigation of the factors that have led to injuries. However, staff **may not take these procedures seriously** if they feel they are impractical. Staff failing to operate onerous procedures properly may result in greater risk than staff operating less strict procedures effectively.

Points against Director of Operations' view

Complacency

The director's view appears to be **complacent**. The current injury statistics seem to be high. There is scope for reducing injuries towards zero, even if Bush can never prevent all injuries.

Reduction measures

Practical measures can be taken to reduce injuries. **Health and safety training** can be improved. Bush can introduce **requirements for staff** performing certain tasks, for example lifting heavy objects.

Negligence claims

The director's toleration of an 'acceptable' level of injuries may leave the council **vulnerable to legal claims**. Staff who have been injured could use the director's statements as evidence of a negligent attitude by senior management towards employee safety.

Points in favour of zero tolerance

Consequences of breaches

A strong argument in favour of zero tolerance is the **consequences of accidents**, possibly serious injury or death. Although a lapse may only have resulted in a minor injury on one occasion, the same lapse another time could have much more severe consequences.

Duty of council

However health and safety law is drafted, the council has a **clear moral duty** to ensure its employees' safety.

Safety culture

Aiming towards eliminating injuries can **help promote a strong culture of safety**. If staff understand that there is no such thing as an acceptable level of injuries, they are unlikely to become complacent and will take steps to reduce the level of accidents further.

Points against zero tolerance

Employee involvement in hazardous activities

The extent of the council's responsibilities makes it inevitable that some staff will become **involved in hazardous activities**. This will mean that there will always be a risk of injuries occurring, even if it can be reduced to very small levels.

Costs

Some risk prevention procedures, for example, requiring staff to wear cumbersome clothing, may be **impractical**. The costs and time taken to investigate minor problems may be excessive.

4.2 Assurance work

- Review HR records of the number and type of accidents and injuries in the workplace.
- Review accident and injury records from a sample of locations and establish whether the categorisation of reportable and serious injuries is being applied correctly.
- Enquire whether the council has received any health and safety visits. Review documentation from any of these for evidence of serious accidents and injuries.
- Talk to employees to identify any accidents not recorded in accident book.

Chapter 8
Data analysis

Introduction

Learning outcomes

Examination context and syllabus links

Chapter study guidance

Learning topics

1 Data and analysis

2 Strategic, financial and operational data

3 Strategic data analysis

4 Financial data analysis

5 Operational data analysis

6 Sustainability reports

7 Big data and data analytics

8 Obtaining more information

9 Data analysis in the Strategic Business Management exam

Summary

Further question practice

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Explain, demonstrate and evaluate how data from multiple sources can be selected, captured and analysed to provide management information, recognising the causes and effects of different types of data bias, data omissions, data limitations and data trends and applying an appropriate degree of professional scepticism
- Explain and analyse an organisation's current position and performance using both financial and non-financial data, presented in different formats, applying appropriate statistical and data analysis tools
- Explain and demonstrate how management information can be used to select from proposed strategies, taking account of limitations of data, including data bias
- Undertake appropriate quantitative and qualitative data analysis, statistical analysis, business analysis and financial statement analysis
- Assess and explain enterprise risk management, evaluating its framework and its benefits; and analyse, structure and assimilate data provided to evaluate business risks under a range of complex scenarios using appropriate statistical and data analysis tools, recognising various types of data bias in a variety of scenarios
- Explain financial and operational data and other management information, drawing inferences relating to its completeness, accuracy and credibility, as a basis for a meaningful analysis of the position, future prospects and risks for a business
- Demonstrate how suitable financial, strategic and operational analysis techniques can be used to analyse financial and operational data and to evaluate business position, prospects and risks, including the analysis and benefits of 'Big Data', artificial intelligence and machine learning
- Communicate an explanation (stating any reservations regarding transparency and objectivity of data and information) of the position, prospects and risks of a business, based on analysis of financial and operational data and information, including data analytics and assess the extent to which limited assurance and reasonable assurance engagements can identify and mitigate information risks in this context
- Apply professional scepticism to data sources and data capture in interpreting quantitative and qualitative information
- Analyse, structure and assimilate historic and forecast data provided to evaluate performance, position and risk using relevant statistical tools, data analysis and spreadsheets, recognising the sensitivity of forecasts to underlying assumptions and changes in estimates
- Assimilate, structure and analyse transactions and other granular data provided, using spreadsheets
- Evaluate the ethical implications of an organisation's selection, capture, analysis and use of data

Examination context and syllabus links

This chapter reviews data analysis skills that you have covered already in Business Strategy & Technology at Professional Level. Data analysis is even more important at the Advanced Level, because of the requirement to analyse more complex scenarios. In the exam, you may need to analyse financial and/or non-financial data, as well as financial statements.

It is important to note these different sources of data or financial information:

A company's **financial statements** are a vital source of information about the company's performance and so, in your exam, you should be prepared to analyse and interpret published financial statements. You should also be prepared to consider how well this information from a company's financial statements correlates with any other information you have been given about the company (including, possibly, management's own statements about the company in their 'Management Commentary' in an Annual Report). Equally, a company's integrated report could provide information about its strategy and future prospects – but again it is important to consider how well this forward- looking information fits with the company's current position.

However, as well as interpreting published information about a business, you might also need to **analyse internal data** or **management information** in your exam. Remember, this data could be non- financial as well as financial. For example, if a company is using a balanced scorecard approach to performance measurement, it might be necessary to analyse its performance in relation to customer service, process efficiency, and learning and development, as well as its financial performance.

The reference to the balanced scorecard (which we discussed in more detail in Strategic performance management) highlights that you may need to undertake data analysis in the context of **performance measurement**. For example, you might need to analyse performance information in order to assess how well an entity is performing, for example, against its objectives, or against its key performance indicators (KPIs). In turn, performance measurement might involve **benchmarking performance** between different divisions or different companies, or it might involve analysing whether managers have achieved their performance targets and therefore qualify for a bonus.

Alternatively, you may find a situation where the management of an entity think that the entity is performing well, but analysis of the management accounting information does not support this. For example, the KPIs which management are monitoring may not cover the entity's **key processes**, and performance in those areas may be poor (while performance in the areas actually being measured has remained quite good).

However, the scenarios where data analysis could take place will not be limited to performance measurement only, and you should be prepared to face a wide range of scenarios in your exam. For example, other possible scenarios might include:

(a) **Valuation of a business**, for example, either by a company considering acquiring the business, or by a group considering the disposal of one of its companies. We will look at valuation methods in detail in Business and securities valuation of this Workbook.

(b) **Evaluation of capital projects**, for example, using net present value (NPV) calculations

(c) **Evaluation of proposals** to outsource a business division or a process

(d) **Evaluating performance of divisions or strategic business units**, for example, using return on capital employed (ROCE) or economic value added techniques

(e) **Cost analysis**, for example, the mix between fixed and variable costs (operational gearing), or issues around overhead apportionment across different products, and the link between costs and prices. Alternatively, cost analysis could be used in relation to strategic choices. For example, if an organisation wants to pursue a cost leadership strategy, how well do its costs – in relation to competitors' costs – fit with that strategy?

Please note, however, that this is only a small selection of the possible types of scenario you **might** face in your exam. The types of scenario where data analysis could take place will not be limited to these, though. Remember that, in Strategic choice, we discussed how potential strategic choices should be evaluated in relation to the suitability, acceptability and feasibility for an organisation.

Financial analysis, and an evaluation of an organisation's financial resources and financial constraints, is likely to be a key part of evaluating the feasibility of a proposed strategy.

Also, note that data analysis could be required more generally in the context of strategic management accounting issues: for example, in order to identify an entity's market share, or to calculate the rate of market growth.

Feedback from past exams shows that the lack of meaningful analysis remains a common weakness in candidates' scripts at Advanced Level.

A key point to remember is that the data analysis you perform should be linked to the wider strategy, position or issue in the scenario. Numbers will form part of the analysis, but the numbers should act as a 'peg' on which to hang discussion. You must demonstrate an understanding of the story behind the numbers. Explanation and evaluation of any numerical analysis is therefore an important element of data analysis.

Note also that learning outcome 8(a) identifies that you may need to undertake qualitative analysis as well as quantitative analysis. Assessing qualitative factors may be crucial in understanding the numbers in a scenario. For example, an organisation's revenue may be falling due to declining customer satisfaction with its products. In this context, note again the potential significance of 'big data' (for example, customer opinions from social media) which could be an important source of qualitative data for an organisation.

In Strategic choice we have already discussed some of the ways the insights gained from big data and data analytics can be a source of competitive advantage for organisations: through customer insight, product innovation and operational improvements. We will look at the characteristics of big data in more detail in this chapter, as well as reiterating the importance of 'data led' decision-making for organisations, rather than having to make decisions based on intuition or 'gut feeling'.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Data and analysis** In any organisation, managers need information for control and decision-making purposes. However, in order to get that information, the underlying data must be analysed in order to identify key patterns and trends. Analysing data is therefore vital for understanding how well an entity is	**Approach** This section starts by looking at the data analysis skills you could be expected to demonstrate in the exam. Note in particular the need to apply the skill of professional scepticism and question the quality and reliability of any data you are given. **Stop and think** What impact can	In the examination, you could be asked to assimilate data from internal and external sources; consider the implications of the data; consider the reliability of data and potential for data bias; and recommend how additional data may be obtained to enhance your analysis.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	performing.	data bias have on decision making?		
2	**Strategic, financial and operational data** Data or information in reports may be strategic, financial or operational in nature. An entity's financial statements offer a vital source of information about its performance. Internal information (such as management accounts) is equally important; particularly if it includes a balance of non-financial performance indicators as well as financial ones.	**Approach** This section highlights the different types of data and reports (strategic, financial and operational) you could be expected to analyse. Read through the approach to analysing data in reports to ensure that you consider all relevant issues in the exam. **Stop and think** What are the key differences between strategic and operational reports?	In the exam, you may be required to undertake appropriate quantitative and qualitative data analysis, business analysis and financial statement analysis. You could also be asked to analyse financial and operational data and other management information as a basis for evaluating a business's current position and future prospects	
3	**Strategic data analysis** Strategic decision making is concerned with the formulation and review of strategies for achieving the objectives of the organisation. Analysis of strategic reports allows for the identification of issues relating to environmental change, the state of the industry and the level of competitiveness.	**Approach** This section suggests an approach to analysing strategic data. The strategic models are revision from previous studies but are now being used as a framework to analyse strategic data. Allow yourself time to work through all the examples and Interactive questions. **Stop and think** Which model is most appropriate to	One of the challenges in the exam may be to identify the strategic, long-term implications of information in a report or statement. Strategic models are not awarded marks in isolation but could be used to help you structure your answer and to analyse data.	**IQ1: Treadway** Using the scenario, you are asked to identify the strategic implications of the information. Consider the long-term strategy and objectives of the company. You also need to use your judgement and professional scepticism to identify any information that would help to provide a fuller picture.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
		assess the level of competition and profitability in an industry?		
4	**Financial data analysis** When assessing company performance, analysis of financial statements and other such financial data is always a good starting point. However, it is important to be aware of the limitations of financial analysis which needs to be balanced by non-financial analysis.	**Approach** Ensure you are happy with financial analysis and that you know how to perform relevant ratios and calculations. This may mean revisiting previous studies. It is also important to be able to critique the data and demonstrate your understanding of the limits of financial analysis. **Stop and think** Why is profit not always the best measure of business success?	Exam questions will test your ability to identify strategic issues highlighted by financial data. You will be required to conduct financial analysis and comment on the implications of your analysis for business decisions.	
5	**Operational data analysis** Operational data analysis involves reviewing internally produced reports, many of which are non-financial in nature.	**Approach** When reading this section, make sure you understand how to use operational data to ocus on three key issues for an entity; efficiency, effectiveness and cost control. **Stop and think** What are the conflicts between being efficient and effective?	In the exam, operational data analysis can be used to highlight problems with the efficiency and effectiveness of the business. In turn you can use this analysis to identify areas of potential strategy improvement.	**IQ2: The Eatwell Restaurant** In this question you are required to compare performance with other restaurants by conducting analysis on the financial and operational data provided. You are also asked to identify other information you would need to enhance your analysis.
6	**Sustainability reports** Sustainability reports are now	**Approach** Read though the section and note the additional	Exam questions may include sustainability	**IQ3: BPC** In this question you are given a sustainability report

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	becoming increasingly common due to both mandatory reporting requirements and voluntary disclosures. You must be familiar with the types of metrics used in such reports and be able to analyse them.	challenges of analysing information in sustainability reports.	information or sustainability reports and ask you to interpret them or discuss the information provided.	prepared for a fictional pharmaceuticals company called BPC. You are asked to evaluate the performance of BPC and discuss the report, including identifying any additional information you would like.
7	**Big data and data analytics** The volume and detail of data available to organisations is increasing ever more rapidly, and the concept of 'big data' reflects this, in conjunction with AI and machine learning. The ability to analyse increasingly large data sets, and to uncover previously hidden trends could enable organisations to make decisions which are better informed, and could also be a source of competitive advantage (for example, through having a superior understanding of customer needs and behaviours).	**Approach** Understand the value and opportunities that Big Data can bring but make sure you also appreciate the challenges which it could present for organisations. **Stop and think** Why is it important to apply professional scepticism to data?	Big data and data analytics are becoming increasingly important for businesses which is likely to be reflected in your SBM&L exam. You could be asked to communicate an explanation of a business's current position and future prospects based on the analysis of financial and operational data and other information available, including 'big data', artificial intelligence and machine learning. You should also be able to question the reliability of data analysis being aware of how wrong conclusions can be reached.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
8	**Obtaining more information** Data analysis may be weakened as a result of incomplete or unreliable source information. The accountant should always consider whether the quality of information, and therefore analysis, can be improved.	The issues of additional information requirements and gaining assurance over information are addressed in this section. The syllabus specification grid suggests that assurance could be worth up to 10% the marks in the SBM&L exam. Although we cover assurance in other chapters in relation to specific issues, this chapter covers the core material about assurance engagements, so make sure you read this section carefully.	In the exam you may be required to assess the completeness, accuracy and credibility of the data and management information available. You may also be asked to communicate any reservations about the transparency and objectivity of data and information provided. Regarding assurance, it is certain to appear in at least one of the two exam scenarios. Whilst practising assurance style questions is essential, remember that SBM&L is an open-book exam so be sure you have access to different assurance questions in the exam to provide you with inspiration.	**IQ4: Company-wide cost control** This interactive question is good practice of assurance issues. It asks you to consider using either an assurance report giving a conclusion or an 'agreed upon procedures' assignment.
9	**Data analysis in the Strategic Business Management exam** A lack of meaningful analysis is one of the most common weaknesses in candidates' answers. It is important to evaluate data in order to draw meaningful	**Approach** This is an important section, so work through it carefully. You should be familiar with the 'What – How – Why – When – So what' approach from your previous studies, but evidence from past exams suggests that	Read carefully through the key weaknesses in answers so you know what to avoid. To improve you answers, revisit the recommended approach in order to get a balance of numerical and descriptive analysis.	**IQ5: WG plc** The requirement simply asks to you analyse the main issues facing WG. This task involves analysis of strategic and operational information presented in the scenario, in order to reach meaningful conclusions about

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	conclusions and recommendations that may assist in strategy development. Data in exam exhibits may be presented in different formats including spreadsheets, data tables, graphs and charts.	students don't always apply the model effectively. As a result, there is a lack of meaningful analysis. Answering the Interactive questions and Self-test questions for this chapter is particularly important. **Stop and think** What is the best way of presenting your numerical analysis?		the key problems the company is facing.

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Data and analysis

Section overview

This section considers the problems with using information in reports or statements, because of the characteristics or quality of the data in them. Professional accountants have to use their common sense and judgement when they analyse data. They are often required to draw conclusions or make recommendations on the basis of information in business reports and financial statements. The analysis of such data is normally both quantitative and qualitative. It is important that accountants should be aware of the limitations of any data they are using when they make such conclusions or recommendations.

The word 'data' has several meanings. It is commonly associated with input to a computer, or 'raw data' which is processed to obtain meaningful information. For the purpose of this chapter, a useful definition of data is: "Facts from which other information may be inferred".

Professional accountants are often presented with reports and statements, from which they are expected to identify issues and draw conclusions. In other words, they have to analyse the data and consider its implications.

Reports and statements vary in nature. They may be **internally produced** business reports or financial reports, as well as published financial statements:

(a) Internally produced business reports may cover internal operational issues, such as:
   - performance reports; or
   - external issues relating to markets and competition, or the business environment generally.

Internal reports may be produced by any department section or unit of the business and may contain historical data or forward-looking forecasts and plans.

(b) Internally produced financial reports may be management accounting reports, particularly performance reports on costs and profitability. The nature of management accounting reports varies with the type of business and the type of cost and management accounting system in use.

Data may also be provided in **externally produced reports**, such as reports by consultancy firms, government departments and international bodies.

1.1 Requirements for data analysis

The Strategic Business Management & Leadership examination expects you to be able to do the following:

(a) Study data on any business-related or finance-related topic, from internal or external sources.

(b) Consider the implications of what the data appears to show, make inferences and draw tentative conclusions.

(c) Consider the reliability of the data, and therefore your confidence in the conclusions you have drawn.

(d) Where appropriate, recommend how additional data may be obtained to test the validity of your conclusions: where information may not be conclusive, the challenge is to identify what more you need to know and how to set about finding it.

This chapter does not set out to provide more technical content and knowledge. Instead, it suggests methods of approaching data analysis and using the technical knowledge that you already have.

The skills required for data analysis may be summarised as follows:

(a) choosing analytical tools that are appropriate in the context of the question, eg, financial ratios, KPIs, breakeven calculations

(b) carrying out the relevant calculations

(c) interpreting the resulting information to demonstrate an understanding of the story behind the numbers and communicating that analysis succinctly

(d) analysing the wider consequences and implications of the numerical data; for example, a fall in production costs, perhaps harming product quality

(e) exercising judgement to draw conclusions and/or produce sensible recommendations, taking potential data bias into consideration

(f) looking beyond the information provided at what additional information may be useful to generate a better analysis/understanding and at any reservations regarding the data/techniques/assumptions applied

(g) linking different pieces of data to explain trends or outcomes

(h) highlighting weaknesses or omissions in the data provided

(i) discussing cause and effect relationships – eg, identifying underlying causes of changes in the data



Professional skills focus: Assimilating and using information

You need to be able to interpret information in various formats to draw conclusions and develop solutions. To demonstrate this skill effectively you need to have a good approach to data analysis, combining your technical knowledge with a review of the scenario, whilst managing the time constraints of the exam.

1.2 Characteristics of data

A useful starting point is an appreciation of the characteristics or qualities of data. Information should be reliable; but data often **lacks reliability**, for any of the following reasons.

(a) **Incomplete**. Data is often incomplete, in the sense that it does not tell the user everything that he or she needs to know. Incomplete information is a source of evidence, but not enough for the evidence to be conclusive. The user should want to learn more before reaching a conclusion.

Incompleteness of data can be a particular problem with external reports, whose purpose may be only indirectly related to the interests and concerns of the report user.

(b) **Lacks neutrality**. Information may lack neutrality. A report may contain opinions and recommendations that reflect the opinions and bias of the report writer. Professional scepticism may need to be applied in interpreting such data or placing reliance on it.

(c) **Inaccurate**. The data in a report or statement may be inaccurate, or the user of the report or statement may suspect that it is inaccurate. Alternatively, data may be insufficiently accurate for the requirements of the user. Without confidence in the accuracy of data, the user cannot make reliable conclusions.

(d) **Unclear**. Information may lack clarity, especially when it comes from an external source. Lack of clarity may be due to:

(1) **poor expression of ideas** in an external report by the report author, or lack of clarity about the assumptions on which information in the report is based

(2) **deliberate lack of transparency** by the information provider. An example of this might be press releases by a competitor organisation, whose statements about a particular item of news may be deliberately obscure without being untruthful

(e) **Historical**. Historical data may be used to make forecasts or conclusions about the future. However, any historical-based prediction is inevitably based on the assumption that what has happened in the past is a valid guide to what will happen in the future. This may not be the case.

(f) **Not up to date**. When events in the business environment are changing rapidly, information may get out of date very quickly. There is a risk that any data in a report or statement is no longer accurate because it is no longer up to date.

(g) **Not verifiable**. Some data or information may not be verifiable. Management may want corroboration of a fact or allegation, but there may not be an alternative source for checking its accuracy. This is often the case in employment disputes at work: two individuals may contest claims made by the other, and there may be no way of checking whose allegations are correct.

(h) **Source**. Information may come from a source that is not entirely reliable. This may be a particular problem with secondary data from external sources.

Accountants must use the data that is available to them, even though it is not 100% reliable. They may have to qualify their opinions or judgements according to their view about how much reliance they can place on it. A major problem is often incompleteness.

Data may **lack relevance** as well as reliability.

(a) There may be a risk of drawing unjustified conclusions from available data, and interpreting data in ways that the facts do not properly justify. The data user may imagine that there is evidence to justify a conclusion, when the evidence from the data is not at all conclusive.

(b) With financial data, there may be a risk of using financial statements prepared under the accruals concept to make conclusions when cash flows and incremental costs should be used.

An accountant may want to use data to make comparisons, such as comparing the performance of different companies or different segments of a business. Unfortunately, data may not be properly comparable. For example, comparing sets of data about the performance of two rival companies may not be entirely reliable because the available data for the two companies:

- has been collected in different ways;
- is based on different assumptions; or
- is presented differently, under different headings.

Note: These issues around the characteristics of data also highlight the importance of **professional scepticism** when analysing data. Professional scepticism is one of the professional skills you will be expected to demonstrate in your SBM&L exam, and some of the ways you could demonstrate it are: recognising bias, or varying quality in data and evidence; identifying gaps in evidence; or identifying inconsistences and contradictory information.

1.3 Making judgements about the quality of data

If the information is not entirely reliable, you may have to make a judgement on the degree to which it may be trusted. It is not sufficient to decide that information is either 'reliable' or 'unreliable': there are **degrees of reliability** between the two ends of the scale.

Having made a judgement about the reliability of data, the accountant must:

- reach a conclusion, but possibly with some reservations; or
- consider what additional information can be obtained before making a firm conclusion.

Professional skills focus: Applying judgement

One of the skills tested in the ACA exams is your ability to apply judgement by identifying omissions, inconsistencies or bias in data. You should therefore apply professional scepticism to data sources and data capture when interpreting quantitative and qualitative information. When analysing data in the exam be prepared to make judgements about the quality or source of the data and factor this in to your conclusions.

Context example: Like for like retail sales

In its report *Audit Insights: Retail* (2013) the ICAEW Audit and Assurance Faculty notes that, although like for like sales figures are the most prominent KPI for the retail sector, they are not comparable on a consistent basis.

Instead, retailers use their judgement to calculate movements in like for like sales by identifying and removing any distorting elements from the calculation. There is no standard basis for calculating like for like sales, nor is there any standard agreement as to what factors should be classified as 'distortions' and therefore excluded from like for like sales.

Figures may not be calculated in the same way between different retailers. Perhaps even more importantly, the basis of the calculation may change from year to year within the same business.

As the report points out, "Without an understanding of the adjustments and judgements in each case, the public may place greater significance on comparisons between like for like sales than is warranted".

Factors that may lead to stores being excluded from like for like sales include:

- stores undergoing a refurbishment or a refit
- stores undergoing a resize (eg, where adjustment to floor space is greater than 5% of the original floor space)
- stores due to be closed in the near future
- stores opened during the period under review
- stores suffering from a major disruption to trading (eg, flood; fire; roadworks in the immediate vicinity leading to a significant decline in footfall; redevelopment; or opening of a direct competitor's store nearby)
- impact of one-off events (eg, impact of the Olympic Games on sales in London in 2012)

The judgement involved in determining which stores classify as 'like for like', and the lack of standardisation, means the performance measure is less valuable to the public than it might otherwise be.

The ICAEW report suggests that the fundamental judgements required to develop an approach to like for like sales in each period depend on a fully developed understanding of sale activity across the business. Unfortunately, however, one reason for the variations in like for like calculations may simply be the range of data available to different retailers.

Finally, the report suggests that the usefulness of like for like sales as a performance measure is further reduced because the core relationship between like for like sales and profitability has changed, due to increasingly widespread deep discounting. As a result, sales numbers can be pushed upwards at the expense of profitability. But retailers rarely link data on profitability (such as movements in profit margins) to like for like sales.

In summary, the report suggests that, 'Greater transparency around the method used to calculate like for like sales, for example disclosure of calculation methods alongside the results, would benefit retailers, investors and customers alike'. The key question, however, is whether retailers could collectively develop, and implement, an agreed calculation standard.

Implications for assurance

In the scenarios described by the ICAEW report, the like for like sales figures appear not to be an audited section of the retailers' annual reports. Nonetheless, trends in sales highlighted by the like for like figures could be useful for readers of the accounts in understanding the performance of the retailers.

In this respect, given that the like for like sales figures are a prominent KPI, it could be beneficial for them to be subject to an external assurance report which, for example, could evaluate the methodology used to calculate the figures, ensure that the figures have been presented in accordance with the methodology, and ensure that the methodology remains consistent from one period to the next within the same entity.

Note: These considerations for an assurance engagement are the same as the ones we highlighted in Strategic performance management in relation to assurance over KPIs.

This last point about consistent methodology is crucial for ensuring the comparability of the figures from one period to the next. If the basis for calculating the KPIs changes over time, then any apparent trends in the figures could be a reflection of the changes in the methodology, rather than trends in underlying performance.

1.4 Other aspects of data analysis

It has already been suggested that an important aspect of data analysis is an understanding of the limitations of the data or information available for analysis. Other issues may also be relevant for consideration.

(a) Materiality and priorities

When analysing data, it may be necessary to keep a sense of perspective. Some issues may be relatively insignificant and so immaterial for the purpose of analysis.

When a report or statement raises several different issues, you may need to identify which is the most important, and prioritise them.

(b) Uncertainty

When available data consists of numerical estimates, you should question the accuracy and reliability of the estimates. It may be possible to undertake some sensitivity analysis, such as considering a 'worst possible' scenario or a 'best possible' scenario, and commenting on these, as well as the 'most likely' scenario (based on the estimates provided).

1.5 Data and assurance

In section 1.3, we noted that accountants may need to make judgements about the quality or reliability of data. In essence, data only becomes valuable to an entity when it is converted into actionable information – which can be used to drive revenue goals, increase cost efficiencies, or inform business decisions more generally. However, before entities convert data into information they need to be confident in the data and the processes which produce it.

Assurance reporting can be used to help provide this confidence. An independent professional accountant, with relevance experience, applying the highest standards to examine data, processes or information, and expressing an assurance conclusion on them, can provide a strong signal of reliability.

We look at assurance in more detail in section 7of this chapter, but the following extract from the ICAEW *Assurance Sourcebook – A Guide to Assurance Services* provides a useful overview summary:

'Owners, management, investors, government, regulators and other stakeholders need to rely on the successful conduct of business activities, sound internal processes and the production

of credible information. These operational and reporting processes enable users to make decisions and develop policies. Confidence diminishes when there are uncertainties around the integrity of information or of underlying operational processes.'

There could be a range of potential subject matters over which different stakeholders might require assurance, but they can be summarised into three broad categories:

- Data – extracted or calculated volumes, values or other items
- Processes and controls – a series of organised activities designed to meet defined objectives
- Reporting – all, or part, of a written report which may contain a combination of data, design of processes and narrative, including any assertions the reporting organisation has made

2 Strategic, financial and operational data



Section overview

Data or information in reports may be strategic, financial or operational in character. This section provides a brief overview of the types of report you may be expected to analyse or comment on.

You may be required to analyse information in a report or statement, and state opinions or reach judgements on the basis of what the report or statement contains. A suitable approach depends on the nature of the report or statement. These may conveniently be classified into three types:

(a) strategic level reports

(b) financial reports

(c) operational reports

These classifications are useful for considering an approach to analysis, but it is possible to have strategic level financial reports and financial reports on operational aspects of a business.

2.1 Strategic level reports

Strategic level reports contain information that is used to make strategic judgements or to guide strategic thinking by senior management. Some strategic reports are produced internally, but many are reports produced externally, which may have some relevance for an organisation's strategic thinking and choice of strategies.

Examples of strategic reports are:

(a) a report produced by an independent research organisation or consultancy firm on the state of a particular industry and the future challenges facing that industry

(b) a government report on the national economy and on monetary and fiscal policies for management of the economy

(c) a report from an international organisation such as the World Trade Organization on the current state of international trade

(d) a general report on the business and financial markets infrastructure in a particular country or region

(e) a market research report from an independent research agency into conditions in a particular product-market area

These are examples, but there are other types of report that may contain information of strategic relevance and interest. What they have in common is that the information within the report can be used to formulate strategic thinking within an organisation.

2.2 Financial reports

Financial reports can have either strategic or operational value.

(a) Published reports and accounts of rival quoted companies can provide useful insights into competitor organisations – their strategies, revenues, profitability. Comparisons of a business with a rival company may suggest areas of difference where one company or the other appears to have a competitive advantage or enjoys superior performance.

(b) A company should not have to rely on its own published financial statements for relevant information. More detailed financial reports should be produced internally for management. These may be strategic reports, linked to aspects of strategy but financial in nature. Most organisations have regular reports relating to annual budgets and monthly or quarterly budgetary control reports. Other financial reports may be operational in nature, such as reports on product costs and profitability, segmental profitability reports or customer profitability analysis.

2.3 Operational reports

Accountants may produce or be required to analyse a range of operational reports. Accountants may produce non-financial reports as well as financial reports, or reports that mix financial and non- financial data. In your examination, you may be required to analyse the information in an internally produced performance report, which may have implications for the efficiency or effectiveness of operations. Reports could include spreadsheet data, data tables, graphs and charts.

2.4 An approach to analysing data in reports

A basic approach to analysis should be the same for each type of report:

(a) assess the information in the report or statement
(b) identify the issues that it raises
(c) consider the implications
(d) consider the reliability of the data and whether it is sufficient to make conclusions with confidence
(e) consider the additional information that should be obtained before reaching a conclusion or making a recommendation
(f) when reaching conclusions or making recommendations, qualify your views by recognising the limitations of the data on which your views are based

The approach to analysing data should differ to some extent according to the nature of the report or statement: strategic, financial or operational. This is because the issues that it raises may be strategic, financial or operational in nature.

3 Strategic data analysis



Section overview

This section suggests an approach to analysing strategic data. The challenge in an exam scenario may be to identify the strategic implications of information in a report or statement, and this section highlights how you could use strategic models as a way of identifying key issues.

Strategic decision-making is concerned with the formulation and review of strategies for achieving the objectives of the organisation. In a rational approach to planning, strategies are developed after analysing the external business environment and the internal resources and competences of the organisation.

If you are required to analyse a strategic level report, you may be expected to identify any of the issues relating to environmental change, the state of the industry, competitiveness, resources and competences.

3.1 Changes in the business environment: PESTEL analysis

A report may contain strategic information about changes in the wider business environment. In such a context, PESTEL analysis can provide a useful framework when reading a report and considering its strategic implications. Changes in the environment may prompt a change in strategic thinking.

(a) **Political change**. Political change may have implications for business and business strategy:

  (1) A change of government could have implications for the Government's fiscal policy and spending plans. This may be significant for companies that earn substantial revenues from government contracts. In the UK, government policy towards privatisation of public services has implications for private sector service providers.

  (2) Political change in another country may affect strategic plans for investing in the country. Liberalisation of politics may attract more foreign investment: greater autocracy in government or the threat of nationalisation of foreign businesses is likely to deter foreign investment.

  (3) International political cooperation may have implications for business: examples may be changes in international attitudes to free trade and free trade agreements, and international efforts to restrict opportunities for tax avoidance by international companies.

(b) **Economic change**. Changes in economic conditions have obvious implications for business strategy. As one example, the slow economic recovery after the global financial crisis in 2008 may have persuaded some companies to defer new investment decisions, whereas others may have had difficulty in obtaining new finance to invest.

(c) **Social change**. Social change may have implications for business strategy. For example, as a result of an ageing population, companies may be affected by changes in demand for certain types of product such as healthcare, and there may be implications for retirement age in the working population. Very high and sustained levels of youth unemployment could have implications for social unrest and security.

(d) **Technological change**. Some technological changes, with associated social change, continue to have a major impact on businesses, especially those producing technology-related products such as smartphones, tablets, broadband services and software products. With some technological changes, an important strategic concern is the speed of adoption by consumers, and so the need of companies to respond very quickly to changes as they happen.

(e) **Environmental change**. Environmental change affects most companies in some ways and to some extent, and many large companies have environmental policies aimed at reducing levels of pollution and waste. Changes in government attitudes to environmental protection, such as changes in carbon pollution regulation, 'green' energy sources, the use of fracking to extract shale gas and the use of genetically modified food products, can have long-term strategic implications for many companies in many different industries.

(f) **Legal change**. Legal and regulatory change can also have important strategic implications. In the UK, for example, it seems probable that the insurance industry has been affected by the development of a 'compensation culture' and increased legal actions for compensation by consumers.

The examples given here are indicative of issues that may be raised in a strategic level report. When change is identified, organisations should consider the implications of the change for strategy. When the report relates to the wider business environment, PESTEL is a useful checklist of areas to consider. It is a framework for analysis, however, and does not provide automatic answers.



Worked example: Energy costs

An accountant working for a leading German car manufacturer has read a report from a consultancy group about the future of energy and energy sources. The report includes analysis of two developments:

- Reports that the UK authorities are willing to permit energy exploration companies to tap shale reserves. This is seen as a key driver of falling energy costs in the UK over the next few years.
- A policy of the German Government to phase out nuclear energy by 2030 and replace it with renewable energy sources.

Requirement

What are the strategic implications of this information?

Solution

Some points you could suggest are:

Exploiting shale reserves in the UK will reduce energy costs in the UK, and the UK could presumably become an exporter of energy in the future.

The replacement of nuclear energy with 'green' energy might lead to an increase in energy costs in Germany.

Manufacturing companies such as car producers are likely to have high energy costs as a proportion of their total production costs. These changes could make German car producers less competitive than UK car producers, but there is insufficient information to assess the scale of the threat.

However, it may force German producers to consider the location of their production facilities and relocate out of Germany, to a lower-cost country.

3.2 Industry analysis

A strategic report may focus on conditions in a particular industry, rather than changes in the broader business environment. A report may indicate threats to the industry as a whole, or opportunities for the industry to develop and expand as a whole.

You may be asked to analyse a report on decline in a particular industry, where decline may be due to:

- limited natural resources on which the industry depends, and the challenge of securing supplies of essential resources
- overconsumption of resources and threats to sustainability of the industry
- decline in total market demand for the product
- the emergence of new products or services that threaten to make an entire industry obsolete: here the strategic challenge may be to redefine the industry in which a company operates

Interactive question 1: Treadway

Treadway Stores is a UK-based international supermarket group. The following information has been extracted from an internal report to management on the position of the group within its industry.

Until last year, the company invested heavily in new store space in all regions. Opening new stores had the advantage of helping the group to increase annual turnover. However, this was at the expense of smaller operating margins. More recently, new investment has slowed down: consumer spending in UK supermarkets is expected to continue to grow over the next few years, but only at the rate of general price inflation.

Growth rates in Asia and the rest of Europe are better. In recent years, every £1 spent by the group in Asia has produced annual sales of £0.85, and every £1 spent in Europe outside the UK has produced additional revenue of £0.40.

There is strong evidence that consumers have become much more price conscious. Revenues of discount stores have increased sharply, and Treadway customers are now buying more low-price or discounted products. Treadway has responded to growing competition in the industry by hiring more staff to improve customer service, cut prices and change the products that it sells. Last year, profit margins fell to their lowest level in over 20 years.

The UK operation continues to invest in smaller local convenience stores, where average prices are 5% higher than in larger supermarkets, and the group is expanding its online shopping services.
Although online sales in food and drinks products have been good, the group is unable to compete successfully with Amazon in online sales of non-food products.

With the reduction in capital investment on new superstores, the group is now trying to make its existing stores more effective. A result of the decision to reduce capex is expected to be a significant increase in free cash flow.

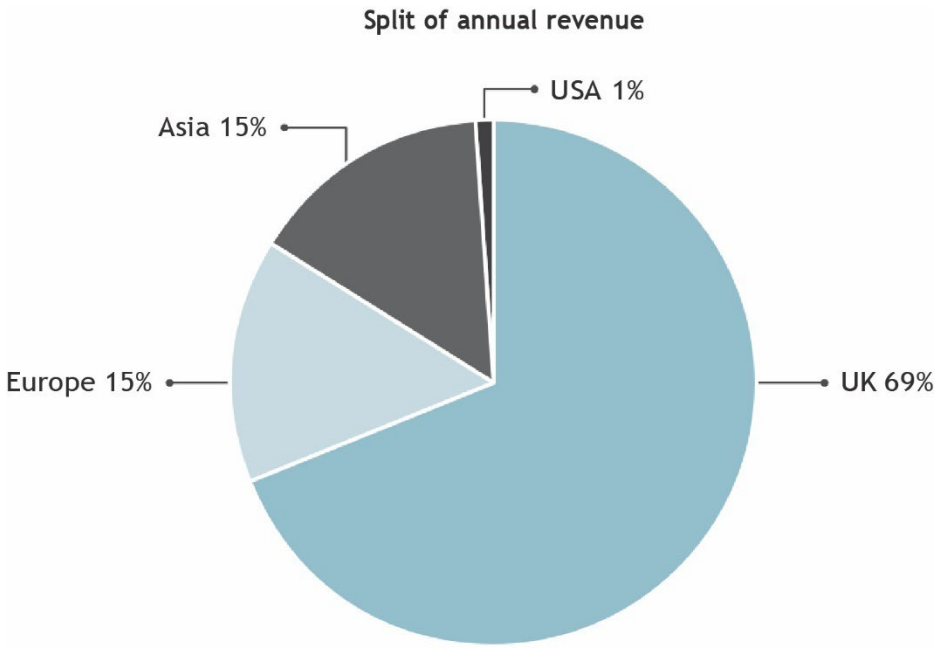
Treadway also intends to improve return on capital employed (ROCE). Both ROCE and the company's P/E ratio are below those of major global competitors such as Walmart.

Requirements

1.1 What do you consider to be the strategic implications of this information?

1.2 What additional information do you think the group should try to obtain in order to carry out a review of current strategy?

Exhibit: Annual revenue split



See **Answer** at the end of this chapter.

3.3 Competition analysis

Reports on competition within a particular industry or market may focus on profitability, revenues and market share. You may be required to comment on aspects of competition and profitability, and the implications for the future of the business.

Several models could provide a useful checklist of issues that may be relevant for analysis and comment.

(a) **Porter's five forces model** identifies five factors within an industry or market that affect the strength of competition within the market and so the potential for profitability. Profitability will be limited in markets where the following forces are strong.

  (1) **Competition between firms** that are already in the market: both price and non-price competition affects profit margins.

  (2) **Low barriers to entry**. When new entrants are able to enter a market easily, it will be impossible for existing firms to sustain high levels of profitability for long. High profits will attract new entrants, and the added competition will reduce prices and profit margins.

  (3) **Supplier strength**. When a market has a small number of dominant suppliers, or a single monopoly supplier, costs of supply are likely to be high and opportunities for supply flexibility are low.

  (4) **Buyer strength**. When a market is dominated by a small number of buyers, the buyers are able to put pressure on companies in the industry to sell to them at low prices. An example is the ability of large supermarket companies to demand low prices from their suppliers and squeeze profit margins in industries such as food manufacturing and farming.

(5) **Availability of substitute products**. Profit margins in a market may be low when consumers have a choice of available alternatives, and can switch between the different products or services. For example, prices of tickets to live entertainment events may be restricted by the alternative that consumers have to watch the event on subscription television.

Note: Although Porter's original model included these five forces, it has subsequently been amended to include a sixth force: 'complementors'.

(b) A strategic report may discuss a company's product portfolio.

The Boston Consulting Group (BCG) matrix provides a method of analysing the product portfolio of a company into cash cows, stars, question marks and dogs. A company needs a sufficient number of cash cows to generate profits and funding for new investments. It also needs 'stars' that currently require substantial investment, but are expected to be successful and the cash cows of tomorrow. For 'question marks', the strategic problem may be to decide whether to stop investing in the product and use resources elsewhere

(c) A strategic report may discuss the position of an individual product in terms of sales growth, profitability and investment requirements. **Product life cycle analysis** can be used to assess the position within its life cycle where a product has reached:

(1) Early introduction and early growth phases, where revenues are small, investment requirements are high, the product is making a loss and cash flows are negative

(2) A rapid growth phase where new investment is still required, but revenues grow strongly and the product eventually becomes profitable and generates positive cash flows

(3) A maturity phase, where the product becomes a cash cow

(4) A decline phase, where sales revenues fall, and a decision for the company is whether to withdraw from the product market, or whether to remain in the market, but operating at lower levels of output

The checklist of issues in the above list provides a basis for analysing strategic reports on competition within an industry and product-market area. As stated earlier, checklists do not provide answers, but a framework for analysis.

3.4 Risk analysis

Strategic data analysis typically involves analysis of strategic risk. Risk may be described as the risk to an organisation from developments in the broad business environment, the industry in which the organisation operates or the competitive environment, such as the risk from strategic initiatives by major competitors.

Where a report indicates the existence of a risk, you may be required to consider the:

- scale or significance of the risk;
- need for urgent action to address the risk; or
- potential implications of doing nothing about it.



Worked example: Ferry routes

In July 20X0, Ferry purchased exclusive rights to operate a car and passenger ferry route until December 20X9. This offers an alternative to driving an additional 150 km via the nearest bridge crossing. There have been several ambitious plans to build another crossing but they have failed through lack of public support and government funds.

Ferry refurbished two 20-year-old roll on, roll off ('Ro-Ro') boats to service the route. The boats do not yet meet the emission standards of Environmental Protection Regulations which come

into force in two years' time, in 20X6. Each boat makes three return crossings every day of the year, subject to weather conditions, and has the capacity to carry approximately 250 passengers and 40 vehicles. The service operates 360 days each year.

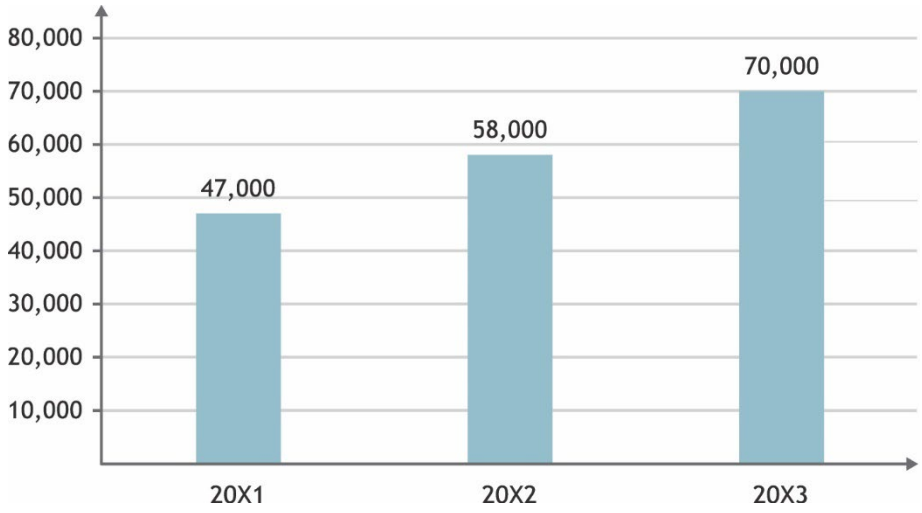
Hot and cold refreshments and travel booking facilities are offered on the one-hour crossing. These services are provided by independent businesses on a franchise basis.

Ferry currently receives a subsidy from the local transport authority as an incentive to increase market awareness of the ferry service and its efficient and timely operation. The subsidy increases as the number of vehicles carried increases and is based on quarterly returns submitted to the authority.

Ferry employs 20 full-time crew members who are trained in daily operations and customer service, as well as passenger safety in the event of personal accident, collision or breakdown.

The management of Ferry is planning to apply for a recognised Safety Management Certificate (SMC) in 20X5. This will require a ship audit, including the review of safety documents and evidence that activities are performed in accordance with documented procedures. An SMC valid for five years will be issued if no major nonconformities have been found.

Annual passenger numbers



Requirement

Identify and explain the business risks facing Ferry which should be assessed.

Solution

The following table summarises business risks that may be identified in the data:

Risk	Comments
Political and regulatory	Risk that the Government may decide to build a new bridge, in spite of previous failures Risk that local transport authority may remove subsidy
Environmental	Environmental regulations come into force in two years' time: risk that the boats will not meet minimum regulatory standards

Legal	The business is protected by exclusive rights that run out in about six years' time. Risks that rights will not be extended beyond December 20X9
Industry analysis	Is there sufficient demand for the service? Risk of insufficient demand Evidence: Lack of public support for bridge Annual capacity = 2 boats × 6 crossings per day × 40 vehicles × 360 days = 172,800 car journeys. Actual sales demand in 20X3 = 70,000 = 40.5% of capacity. Is this sufficient to sustain a profitable business, even allowing for future growth (20.7% in 20X3) Risks of high cost of health and safety. Risk of accidents and injury. Risk of failure to obtain Safety Management Certificate
Inadequacy of data	More information is needed about revenues, costs and demand forecasts in order to assess the business risk more confidently

3.5 Internal analysis: strengths and weaknesses

Although strategic data analysis may focus mainly on external analysis of the environment, industry and competition, internal analysis of resource strengths and weaknesses may also be relevant.

Data may indicate strengths or weaknesses in the resources and competences of an organisation, which can give them a competitive advantage over rival organisations, or expose them to serious competitive disadvantage.

Strengths and weaknesses could exist in any key resource, such as:

- intellectual property
- management
- location or distribution channels
- employee skills
- business experience

In some industries, **intellectual property** is a major strategic asset. Patents help to protect sales and profits against inroads by competitors, provided that the patent can be enforced effectively or until an improved technology is developed by a competitor – and patented.

When companies depend on the **knowledge and talents of individual employees**, they are exposed to the risk of defection by employees to rival organisations. During the early 2000s, for example, it was fairly common for 'star traders' in the financial markets to defect from one bank to another, to obtain higher remuneration, often taking a whole team with them.

Depending on the nature of a strategic report, you may find it useful to think about the implications of the information in the report using **SWOT analysis** – identifying internal strengths and weaknesses and external threats and opportunities that could have strategic implications for the organisation.

4 Financial data analysis

Section overview

Financial analysis of information in reports or statements covers issues such as profitability, revenue and costs; investment; cash flow; and funding and capital structure. You should be familiar with financial analysis from your previous studies, but you should expect an examination question to test your ability to identify potential strategic issues which are highlighted by financial data, as well as recognising any weaknesses in the available data itself.

At this stage of your studies, you should already be familiar with the basic tools of financial analysis, including key ratio analysis. If you are asked to comment on the implications of information in a financial statement or report, you will be expected to identify which ratios may be relevant and interpret the significance of any ratio that you measure. Even so, these would be basic tasks at this level of your studies. You could be expected to analyse financial data about any of the following:

(a) the financial markets. You may be asked to comment on data about conditions in the financial markets, such as interest rates or exchange rates, and implications of changes in market conditions for the organisation

(b) revenue, profitability and costs – and pricing

(c) cash flow or liquidity

(d) capital structure

If you are given financial data for analysis, you should consider the adequacy or limitations of the information and be aware of what the information does not tell you. What is missing could be more important than what the report or statement contains.

(a) Data about profitability may present product profitability, when you should be more concerned with customer profitability, distribution channel profitability or market segment profitability.

(b) Data about profitability may be provided, when you should be more concerned about cash flow and funding.

(c) Cost and management accounting information may be presented in a traditional format, such as an absorption costing or marginal costing statement, when you may consider that another approach to presenting information is needed – for example, an activity-based costing statement, or information about particular aspects of cost that traditional statements do not analyse, such as quality costs.

The challenge with analysing financial information may be not so much to demonstrate your knowledge of financial analysis as to demonstrate your understanding of the limits of financial analysis when insufficient or inappropriate data is available.

Worked example: Wizard Ltd

Wizard Ltd is a specialist component manufacturer for the aerospace industry employing 54 people. It has two main customers located in North America. The relative success of Wizard over the last few years has attracted interest from a number of potential industry buyers. One of Wizard's main customers, Draco plc, is now considering making a bid for the entire share capital of Wizard, effectively bringing Wizard's services in-house. Draco is concerned that the specialist products that Wizard supplies it with allow it to charge, in the words of the Draco purchasing manager, 'outrageous prices'. The financial adviser to Draco has obtained the following information relating to Wizard.

Extracts from the financial statements of Wizard for 20X5

	$'000
Revenue	14,730
Cost of sales	8,388
Other costs	5,202
Profit before tax	1,140
Profit after tax	798
Dividend paid	390
Non-current assets	5,364
Inventories	1,392
Receivables	876
Cash	192
Payables	1,464
Equity share capital	600
Retained earnings	5,760

Information obtained from the Aeronautical Trade Association

Average P/E ratio (for quoted companies)	9.0
Average annual growth in reported post-tax profits (20X4–20X5)	3.0%
Average pre-tax profit margin	5.1%
Average pre-tax ROCE	13%
Average receivables days	78
Average payables days	34
Average revenue per employee	$154,200

The finance director of Draco has provided the following summary of Draco's recent performance:

	20X5 $m	20X4 $m	20X3 $m	20X2 $m
Revenue	58.75	55.60	50.30	50.50
Pre-tax profit	4.40	7.15	7.75	10.05
Dividend paid	0.40	2.50	2.50	2.50

Requirement

Analyse the financial position and performance of Wizard as at the end of 20X5.

Solution

Measure	Industry	Wizard	Workings
Gross ROCE		99.7%	(14,730 – 8,388)/(600 + 5,760)
Pre-tax ROCE	13%	17.9%	1,140/(600 + 5,760)

Gross profit rate		43.1%	(14,730 – 8,388)/14,730
Pre-tax profit rate	5.1%	7.7%	1,140/14,730
Non-current assets turnover		2.75	14,740/5,364
Receivables days	78	22	(876/14,730) × 365
Payable days	34	64	(1,464/8,388) × 365
Inventory days		61	(1,392/8,388) × 365
Revenue per employee $	154,200	272,778	(14,730/54) × 1000
Pre-tax profit per employee $	7,864	21,111	(1,140/54) × 1000
Dividend cover		2.05	798/390
Current ratio		1.68	(1,392 + 876 + 192)/1,464
Quick ratio		0.73	(876 + 192)/1,464

Analysis

Pre-tax ROCE and pre-tax profit rate – These are 37% and 51% higher than industry average, which supports the view that Wizard is able to charge high prices. This would appear to be a result of the specialism of the services that Wizard provides. Additionally, there may be strict cost control within Wizard, further allowing it to generate higher margins. Should Wizard be acquired by Draco, then the products will be available at 'cost', thereby saving Draco money, while allowing it to potentially benefit from the premium prices it can charge to Wizard's other main customer.

Receivables days – At 22, these are exceptionally low compared to the industry average. This is probably due to the fact that Wizard only has two main customers, making it possible to form close working relationships. Given the specialism that Wizard provides, it is likely that its customers do not want to sour this relationship by delaying payment. There is no reason to believe that this will change if Draco acquires the company.

Payable days – At 64, this is almost twice the industry average and reflects either a strict cash management policy within Wizard, or potentially a cash flow problem. Given the high profitability within Wizard, and its healthy balance sheet, it would appear that Wizard has squeezed its suppliers quite hard. Once acquired by Draco, this strategy may need to change to bring it in line with company policy.

Inventory days – At 61, this indicates the time that inventory is held by Wizard. This demonstrates that the production process within Wizard is about two months and may be a reflection of the complexity of the manufacturing process that it undertakes. It may be a result of the safety checks, which are a key feature of supply in the aerospace industry, and the time taken to do this may contribute to the 61-day figure.

Revenue and profit per employee – These are, respectively, 76% and 168% higher than industry average, which is a further reflection of the profitability and revenue generation abilities of Wizard. This is further evidence of its ability to charge high prices and possibly control costs. Interestingly, we are told nothing about the salaries within Wizard and it may be that as a smaller firm, their salaries may be different to those within Draco. Should Wizard's salaries be higher than Draco's, this could lead to demands for higher wages among Draco's workforce. In terms of costs, it may well be that once Wizard is acquired, the greater purchasing power which a larger company would have may lead to further economies of scale and even cheaper supplies.

Dividend cover – As Wizard is not a listed company, its dividend ratio is not strictly comparable with Draco's. What is relevant is that it evidences Wizard's ability to pay dividends and hence

generate cash. This is potentially good news for Draco as it has recently cut its own dividends, which will have disappointed shareholders.

Liquidity ratios – Without industry statistics, these are in themselves fairly meaningless. However, the current ratio is greater than one, indicating good liquidity, and the quick ratio is close to one. Allied with its low receivables and high payables days, this indicates that Wizard does not appear to have cash flow problems.

Overall – It would appear from the above analysis that Wizard is a profitable company that does not appear to have any liquidity or working capital concerns.

4.1 Users of financial analysis

Different stakeholders have different expectations of a company, and therefore will require different information about its performance:

(a) **Shareholders** – Shareholders will be interested in the quality of their investment and the returns they can expect through dividends and capital growth. Ratios which could be particularly important to them are: earnings per share, price/earnings (P/E) and dividend yield.

(b) **Bankers and debt holders** – Bankers and debt holders will be primarily interested in the risk attached to their investment in a company. Ratios which therefore could be important to them are: capital gearing, and interest cover.

(c) **Suppliers and employees** – Suppliers and employees both depend on the company continuing in business so that it continues to be a customer or an employer in the future. These stakeholders will be interested in a company's ability to meet its short-term liabilities. Ratios such as the current ratio and acid test ratio could be important in this respect.

(d) **Management** – They may need to consider all the ratios noted above, because they are important to (and are therefore likely to be monitored by) other stakeholders. Additionally, the company's management need to consider the company's performance in relation to profitability, cost control and working capital management. Ratios such as gross and net profit margins, inventory turnover, and receivable and payable days could therefore be important performance measures.

Managers may also want to measure the performance of different divisions within an organisation. ROCE and residual income (RI) are two methods of measuring divisional performance. However, there are potential problems with using ROCE and RI for evaluating and controlling business divisions:

(a) They are based on annual profit figures, and so disregard the future earnings of the division. Using BCG terminology from Strategic analysis, a cash cow might present a high ROCE; and a star, a low one, which would be a misleading guide to their true financial value, if assessed as the NPV of future earnings.

(b) To boost ROCE or RI, assets with low book values will be used in preference to new assets. This could lead to short-termism, with divisions preferring not to invest in new assets, even though such an investment would be better for their longer-term future performance.

However, it is important that managers in an organisation do not focus solely on financial performance measures. As we have noted in Strategic performance management, it is desirable to have performance metrics which look at key aspects of non-financial performance as well as financial performance. In the same way, it is advisable to use performance metrics which act as leading indicators, rather than focusing solely on lagging indicators.

4.2 Approach to analysing financial data

If you are given financial data for analysis, you should expect to carry out some numerical analysis. You will have to decide yourself how to do the analysis.

(a) If you are given data for more than one year, you should measure changes over time. If you are given financial data about a competitor, you should try to make a comparative analysis.

(b) There may be value in carrying out cost-volume-profit analysis (breakeven analysis) on data that you are given, but you will need to state your assumptions about fixed and variable costs.

(c) If you are given information about historical performance and targets, you should try to carry out numerical analysis of the extent to which the organisation is on track for meeting its targets.

Show all your numerical workings and state clearly the assumptions you have made.



Professional skills focus: Structuring problems and solutions

In order to draw conclusions about the position and performance of an organisation you will need to analyse financial and non-financial data presented in scenario exhibits. Ensure that you take care to present any calculations professionally, stating your assumptions and cross referencing your numerical analysis in the body of your discussion.

5 Operational data analysis



Section overview

Analysis of operational data is likely to use internally produced management reports, which are produced primarily for control purposes rather than planning purposes. Much operational data is non-financial rather than financial.

Although operational data may be financial, non-financial or a combination of both, much of it is likely to be non-financial.

Data about operations should normally come from internal sources within the organisation.

When analysing operational performance, it may be useful to have a checklist of areas of performance. Any problems or issues arising out of an operational report are likely to raise questions about one or more of the following areas.

5.1 Efficiency and effectiveness

Efficiency is concerned with getting the maximum output from a given quantity of resources or achieving a given quantity of output with the minimum of resources. Efficiency is also known as productivity, and typical productivity measures are:

- output per worker/hour
- output per machine per hour
- sales per square metre of floor space
- average time to produce a unit or complete a task
- average number of tasks completed per day
- quantity of materials per unit of output

- waste per unit of output
- production cycle time

Effectiveness is concerned with achieving objectives with the resources that are used. Measures of effectiveness depend on what the organisation is trying to achieve, and they compare planned targets with actual achievements. Important aspects of effectiveness may be:

- **Quality**: Product quality, including product design and performance reliability, may be a key factor in providing customer satisfaction.
- **Delivery**: Effectiveness in delivery of a product or service may relate to factors such as speed of delivery and reliability of delivery.

Resources may be used efficiently but ineffectively. Similarly, resources may be used effectively, but in an inefficient way.

5.2 Balanced scorecard

A balanced scorecard approach to operational performance analysis links performance targets and performance measures through all levels of an organisation, from operational level to strategic level. The four perspectives of performance in the balanced scorecard (as discussed in Strategic performance management) are:

- **Financial perspective** – achieving financial objectives in both the short and long term
- **Customer satisfaction perspective**, and aspects of performance that have the biggest effect on providing customer satisfaction
- **Internal business perspective**, and critical aspects of the internal operations of an organisation
- **Innovation and learning perspective** – issues such as product innovation and innovation in service delivery, and the acquisition of learning and knowledge by employees

Note, however, that none of the perspectives of Kaplan and Norton's balanced scorecard link directly to aspects of social responsibility or sustainability, which are becoming increasingly important elements of an organisation's overall performance. In this respect, in an article for ICAEW's Finance and Management Faculty ('The new thinking on key performance indicators', May 2006), David Parmenter suggested that in order to achieve a properly balanced view of performance, the number of perspectives of the balanced score card should be increased to six: financial; customer; internal process; **employee satisfaction**; learning and growth; and **environment and community**.

5.3 Cost

Cost is a financial aspect of performance. It is difficult to assess operational performance without also considering the cost incurred by a business.

Kaplan and Norton (who devised the balanced scorecard) recommend that activity-based costing should be used to produce cost measures for important internal business processes. These costs, in conjunction with measurements about speed/time and quality, should be monitored over time, and **benchmarked** with a view to continuous improvement or process re-engineering.

Benchmarking would allow managers to see not only how an organisation's costs vary over time (historical benchmarking), but also how costs vary in different parts of an organisation, or how an organisation's costs compare to competitors' costs. Monitoring costs and process efficiency against competitors could be particularly important for an organisation pursuing a low-cost (or cost leadership) strategy.

Information about costs could also play an important part in any decisions about whether to outsource certain functions or processes, or whether to retain them in-house.

Interactive question 2: The Eatwell Restaurant

The owners of The Eatwell Restaurant have diversified business interests and operate in a wide range of commercial areas. Since buying the restaurant in 20X0, they have carefully recorded the data below.

	20X1	20X2	20X3	20X4
Total meals served	3,750	5,100	6,200	6,700
Regular customers attending weekly	5	11	15	26
Number of items on offer per day	4	4	7	9
Reported cases of food poisoning	4	5	7	7
Special theme evenings introduced	0	3	9	13
Annual operating hours with no customers	380	307	187	126
Proposals submitted to cater for special events	10	17	29	38
Contracts won to cater for special events	2	5	15	25
Complimentary letters from satisfied customers	0	4	3	6
Average number of customers at peak times	18	23	37	39
Average service delay at peak times (mins)	32	47	15	35
Maximum seating capacity	25	25	40	40
Weekly opening hours	36	36	40	36
Written complaints received	8	12	14	14
Idle time	570	540	465	187
New meals introduced during the year	16	8	27	11

Financial data

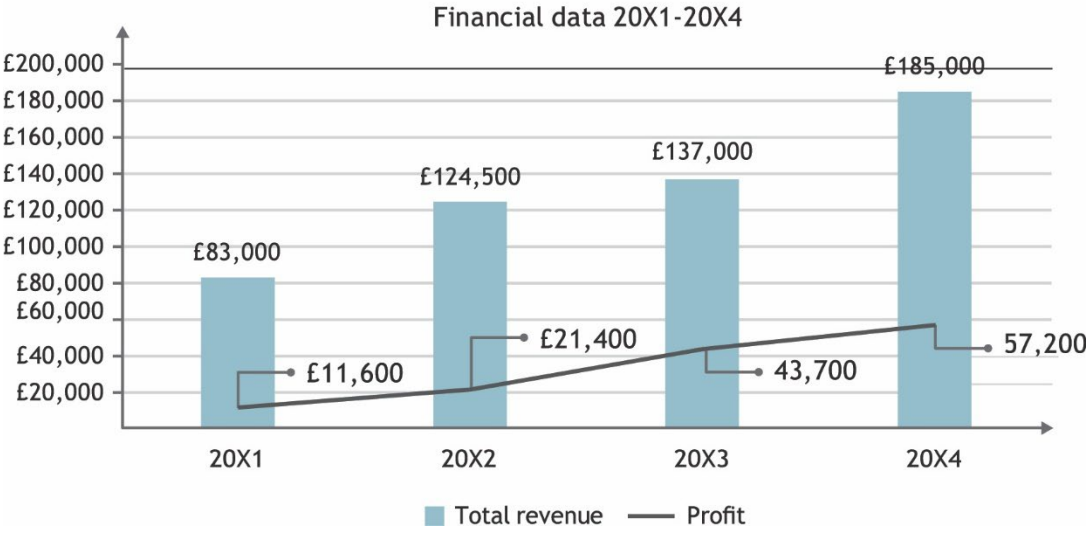
	£	£	£	£
Average customer spend on wine	3	4	4	7
Revenue from special events	2,000	13,000	25,000	55,000
Value of food wasted in preparation	1,700	1,900	3,600	1,450
Total revenue of all restaurants in locality	895,000	1,234,000	980,000	1,056,000

Requirements

2.1 Assess the overall performance of the business and submit your comments to the owners. They wish to compare the performance of the restaurant with their other business interests and require your comments to be grouped into the key areas of performance, such as: competitive performance, financial performance, quality of service, flexibility, and resource utilisation.

2.2 Identify any additional information that you would consider to be of assistance in assessing the performance of The Eatwell Restaurant in comparison with another restaurant. Give reasons for your selection and explain how they would relate to the key performance area categories used in (a).

Exhibit: Revenue and profit 20X1-20X4



See **Answer** at the end of this chapter.

6 Sustainability reports

Section overview

As sustainability reporting becomes more common, accountants may be asked to analyse them. Some additional challenges are presented, such as the fact that the reports may contain much qualitative information. This section discusses these issues and how to deal with them.

6.1 Approach to analysis

Alongside financial information, many organisations are now providing information about their wider impact on ESG matters. As discussed in Strategic performance management, there are now increasing mandatory disclosure requirements for organisations. There are also a number of voluntary reporting frameworks for reporting on sustainability, such as the GRI standards, referred to in Finance awareness.

Accountants will be expected to analyse the information provided in the same way as we analyse financial information. The approach suggested to analysing reports in section 2 applies equally to sustainability reports.

6.2 Users and their needs

The focus of analysis of sustainability reports will depend on who the analysis is for:

- Investors are interested in matters that may impact on the value of their investment over the short, medium and longer term. In particular, risks and opportunities related to climate change, the viability of the existing business model, and the reputation of the industry.
- Society as whole may be more interested in the organisation's impact on sustainability, and the extent to which it is making a positive or negative impact on society – for example, the contribution the business is making towards achieving the UN sustainable development goals (referred to in Strategic performance management).

When analysing sustainability reports, do take into account who the analysis is for.

6.3 Additional considerations

Professional scepticism should also be exercised when analysing sustainability reports. Some organisations may be providing sustainability reports purely as a marketing tool, hoping that simply producing a report will show that they are concerned about sustainability, even though their underlying activities may suggest otherwise. If the report adopts a more honest tone – recognising the challenges that the organisation faces, without giving the impression that management have all the answers – it will be more credible.

Sustainability reports may include forward-looking information, including targets that the organisation hopes to achieve (eg reductions in carbon emissions). These forward-looking targets are more credible if supported by historic data showing achievements already made (eg, how much have carbon emissions fallen in the recent past?).

As with financial information, information in sustainability reports is much more useful if it can be compared to other similar organisations or benchmarks. The use of accepted standards such as GRI standards can make information more comparable.

Is the information in the report consistent with the financial information? If the company mentions the importance of a particular activity in the sustainability report, which is not mentioned at all in the financial report, this can cast doubt on the claims in the sustainability report (eg the business claims that it expects to meet future clean-up costs, but there is no provision in the financial statements).

Best practice sustainability reporting

In terms of evaluating sustainability reports, it is also useful to consider what a good report should include, to help you assess how well any given report demonstrates those characteristics. A good sustainability report should include:

- *An overview of an organisation's ESG strategy, demonstrating how that fits with the organisation's overall purpose and strategy.*
- *A descriptions of the organisation's ESG priorities, goals and metrics.*
- *An evaluation of the organisation's progress towards those goals.*

This information could either be disclosed in a standalone sustainability report or integrated into an organisation's annual report and investor presentations.



Interactive question 3: Better Pharmaceuticals Company (BPC)

You are reviewing the following sustainability report on behalf of an investor.

Better Pharmaceuticals Company (BPC) – Sustainability Report – 20X2

Society – we provide effective treatments for patients at reasonable prices. Our treatments are developed using our expert research and development team, supported by artificial intelligence to ensure we can find new treatments that will enable people to recover from, or live more comfortably with chronic ailments.

Environment – we are aiming to reduce the harmful impact that we have on the environment by 35% over the next ten years. We are also aiming to reduce our greenhouse gas emissions to zero within 40 years.

Economy – we aim to provide a good return to our shareholders, while providing long-term careers to the people who work for us and provide high-quality products to patients.

We have worked with our stakeholders to identify the topics that are most important to them and, based on these, we have we have identified 8 key areas of focus. We measure our performance based on these 8 key areas of focus. Our performance for the year ended 31 December 20X2 was as follows:

Society	
Area of focus	**20X2 performance**
Treating patients	1.2 million patients treated with BPC products
Developing new effective medicines	9 new products passed clinical trials and were launched commercially during the year
Diversity of board	33% of the board are female
Employee engagement	75/100 employee engagement score

Environment	
Greenhouse gas emissions	15% decrease in greenhouse gas emissions since 20W5
Renewable energy use	56% of our energy is from renewable resources

Economy	
Long-term investment	R&D spend accounts for 18% of sales
Shareholders return	Average total shareholder value added of 56% since 20W7

Requirement

(1) In so far as the information allows, evaluate the impact that BPC has had during the year ended 31 December 20X2

(2) State what other information would help you to make a better assessment of BPC's impact

(3) Critically evaluate the report and how useful it is for showing the impact of BPC

See **Answer** at the end of this chapter.

6.3.1 Challenges associated with ESG information

The sub-section above focuses on potential issues around analysing sustainability and ESG reports. However, related to this, it is also important to consider the challenges associated with ESG information. In particular, it tends to be less robust than financial information, and often hasn't been through the same rigorous reporting processes. Although the development of standards such as ISSB's climate-related disclosure standards (IFRS S1 and S2) helps to improve consistency and comparability, there could still be a greater degree of uncertainty around measurement processes than is the case with financial information.

Equally, for accountants involved in collecting data for ESG reporting, one of the challenges they face will be identifying who within the organisation can provide the data required. The processes for collecting ESG information will often not be mapped out in the same way that they are for financial information, but it will be important to record the data sources to ensure continuity (eg, if the manager who provided a particular piece of information one year leaves the company before the following year's report is due).

The International Federation of Accountants (IFAC) recognises that accountancy professionals play a key role in providing high-quality, sustainability-related reporting and assurance services. A report written by IFAC 'Equipping Professional Accountants for Sustainability' (2024) outlines how professional accountants need to be systems-thinkers, recognising the connectivity of

finance and sustainability. IFAC stresses that sustainability is not a topic to be considered in isolation and professional accountants, at all stages in their development, should consider the impact of sustainability on business models, value chains and strategies, as well as on governance, risks and opportunities.

7 Big data and data analytics



Section overview

- The amount of data available to organisations is increasing ever more rapidly, and big data – with its characteristics of volume, velocity and variety – encapsulates both the opportunities and challenges which new sources of data present to organisations.
- The ability to analyse these large, and unstructured, data sets and to uncover previously hidden patterns of information could be an important element of an organisation's competitive advantage.

Whereas intuition historically played a large part in business decisions (in the absence of reliable and timely data) organisations now expect business decisions to be based on robust data analytics and data-driven insight, supported by intuition and experience.

One of the key challenges facing today's business managers is how they can use data and information about current performance to leverage additional value throughout an organisation's value chain (for example, through identifying greater operational efficiencies, or enhancing customer relationships) or to identify new sources of competitive advantage.



Context example: Efficiency

In an article for McKinsey, Mayhew *et al* (2016) discuss a consumer goods company that sought to increase margins on one of its well-known breakfast brands. The company deconstructed the entire manufacturing process into sequential elements, and then – using advanced analytics – scrutinised each element to see where it could unlock value. In this case, the solution was found in the oven: adjusting the baking temperature by a tiny fraction not only made the product taste better, but also reduced the cost of production.

Mayhew, H *et al* (2016) *Making data analytics work for you – instead of the other way around*http://www.mckinsey.com/business-functions/digital-mckinsey/our-insights/making-data- analytics-work-for-you-instead-of-the-other-way-around

7.1 Data analytics

Although much of the focus in this section of the chapter is on the increasing amounts of data available to organisations – so called 'big data'– it is important to stress that **data analytics does not relate only to 'big data'**.

Data analytics is the process of collecting and examining data in order to extract meaningful business insights, which can be used to inform decision-making and improve performance. As such, data analytics can be applied to internal data as well as external data. For example, data analytics techniques could be applied to sales or purchases data to discover patterns or trends in them. However, sales or purchases data – as part of a company's traditional accounting records – could not really be described as 'big data'.

The words of Bernard Marr, one of the leading writers on big data, are instructive here:

> "…[O]n one hand Big Data is changing the world because we now have so much more data and new data formats. But on the other nothing much has changed because we are still seeking to use data and information to inform corporate decision-making. The only real difference is that we now have new data formats that we can use and new technology to actually analyse that data and leverage it." (Bernard Marr (2015), Big Data: using smart big data, analytics and metrics to make better decisions and improve performance; Wiley; Chichester.)

As such, the process of analysing data, reporting results, and use those results to make decisions remains crucial for business managers, whether they are using internal data, or external, 'big data' data sets.

Having said that, the primary focus of section 6 of this chapter is on big data, and the potential benefits and insights it can provide to organisations.

7.2 The internet of things

Organisations today potentially have access to more data than ever before – about their customers and suppliers and about their operations. The growth of the internet, multimedia, wireless networks, smartphones, social media, sensors and other digital technology have all helped to fuel a data revolution – increasing the number of data sources available, their volume and their complexity. The rise of the 'Internet of things' – as discussed in Strategic analysis – has been particularly significant in this respect.

Definition

Internet of things: A situation in which everyday objects have network connectivity, allowing them to send and receive data over the internet.

In the 'Internet of things', sensors embedded in physical objects such as mobile phones, motor vehicles, smart energy meters, RFID tags, tracking devices and traffic flow monitors all create and communicate data which is shared across wired and wireless networks that function in a similar way to the internet. The timing and location of cash withdrawals from ATM machines could also be a potential source of data, as could footage from data from webcams or surveillance cameras.

Consumers using smartphones, laptops and tablets to browse the internet, to search for items, to make purchases and to share information with other users (for example through social media) all create trails of data. All of these are potentially sources of big data.

7.3 Big data and analytics

We have highlighted the key characteristics of 'big data' in Strategic choice earlier in this text (**volume, velocity, variety**, and **veracity**).

Crucially, however, the potential value of big data for organisations lies in its ability to help them make better decisions, and thereby improve performance.

In this respect, 'big data analytics' is likely to be crucial to making use of the potential value of big data.

Big data analytics refers to the process of collecting, organising and analysing large sets of data ('big data') to discover patterns and other useful information which an organisation can use in its future business decisions.

Big data analytics should help an organisation to reveal insights in data which had previously been too difficult or costly to analyse – due to the volume and variability of the data involved. These insights can be historical, real-time or predictive.

Being able to extract insights from the data available is crucial for organisations to benefit from the availability of big data – for example, to help them understand the complexity of the environment in which they are operating, and to respond swiftly to the opportunities and threats presented by it; or to develop new insights and understanding into what customers need or want.

7.4 Making use of big data

Historically, only the largest corporations have had sufficient resources to be able to process big data. Now, however, it is becoming possible for all organisations to access and process the volumes of big data potentially available to them, due to cost-effective approaches such as cloud-based architectures and open source software.

McKinsey's report *Big Data: the next frontier for innovation, competition and productivity* suggests that "Big Data has now reached every sector in the global economy. Like other essential factors of production such as [physical] assets and human capital, much of modern economic activity simply couldn't take place without it".

This suggests that the ability to capture and analyse big data, and the information gained by doing so, have become important strategic resources for organisations. Making effective use of big data could confer competitive advantage for an organisation. Alternatively, in time, competitors who fail to develop their capabilities to use big data and information as strategic resources could be left behind by those who do.

While these might initially seem to be quite bold claims, big data can certainly create value for organisations through its ability to drive innovation and by helping organisations gain greater and faster insights into their customers.

Similarly, analysing data from as many sources as possible when making decisions can also increase the amount of useful information available to managers when they are making those decisions.

However, the distinction between simply having 'data' and having 'useful information' is important here. Simply having more data available to them does not, in itself, benefit organisations or provide them with any competitive advantage. Instead, organisations only benefit if that data is converted into valuable information and managers then use that information to make effective decisions – which, for example, improve operational efficiency and customer experience, or which enable new business models to be created.



Context example: Big data and the logistics industry

Logistics providers, such as DHL, manage a massive flow of goods around the globe, and at the same time create vast data sets; for example, from recording the origin and destination, size and weight of millions of shipments every day, and then from tracking their location across global delivery networks.

Big data can provide a number of potential benefits to logistics providers, and this case example highlights three of these.

Last-mile optimisation

A major constraint on the levels of operational efficiency in a distribution network is the so-called 'last mile' – the final stage of a supply chain in which goods are handed over to the recipient. This final stage can often be one of the most expensive in the supply chain (for

example, if a delivery driver becomes stuck in a traffic jam, or if a recipient is not present when a courier attempts to deliver their package, and so the delivery has to be rearranged).

However, big data analysis can help to increase last-mile efficiency, through route optimisation aimed at saving time in the delivery process. Rapid processing of real-time information assists route optimisation in a number of ways.

When the delivery vehicle is loaded and unloaded, sensors detect the destinations of the packages and the optimal delivery sequence is then calculated for the driver.

Once the delivery vehicle has begun its journey, telematics databases are used to automatically change delivery routes according to current traffic conditions. For example, if heavy traffic is causing delays on the route originally scheduled, a revised route is generated for the driver to avoid those delays.

In addition, routing intelligence makes use of availability and location information posted by recipients in order to avoid unsuccessful delivery attempts.

The 'big data' aspects of this scenario are that the driver's control systems make use of a number of different optimisation procedures, fed by correlated streams of real-time events, to dynamically re- route vehicles while they are on the road. As a result, each driver receives instant driving direction updates from their onboard navigation system, guiding them to the next best point of delivery.

Capacity planning

Effective capacity planning can be an important competitive advantage for logistics providers. Excess capacities reduce profitability, while capacity shortages affect service quality and potentially jeopardise customer satisfaction (eg, by increasing delivery times).

Resource planning is therefore an important consideration for logistics planners at both strategic and operational levels. At a strategic level, the topology (links, nodes etc) and the capacity of the distribution network need to be adapted according to anticipated future demand. The results from capacity planning often lead to capital expenditure, such as investments in warehouses, distribution centres and vehicles. Therefore, more precise capacity demand forecasting can increase efficiency and lower the risks attached to investing in storage and fleet capacity (either by overinvesting or underinvesting).

Big data can support such capacity planning by analysing data about the historical capacity and utilisation rates of transit points and transportation routes. In addition, it can take account of seasonal factors and emerging freight flow trends (which are affected by external economic information, such as industry or regional growth forecasts).

As well as identifying potential under capacity, planners can expose any potential overcapacity, and identifying this should provide a trigger to try to increase sales volumes – for example, by using dynamic pricing mechanisms.

Risk evaluation and resilience planning

Business to business logistics providers know their customers' supply chains in great detail, and customers rely on the providers' predictive risk assessment.

In order to be able to identify potential threats to the supply chain, logistics providers need to maintain a model which not only describes all the elements of the supply chain topology but also monitors the forces which could affect the performance of that supply chain – for example, local developments in politics, economy, or events in the natural environment.

Data on these environmental factors can come from a number of sources (eg, weather forecasts, new sites, social media and blogs) but must then be aggregated and analysed. Much of this data stream is unstructured and continuously updating, so the power of big data analytics in dealing with velocity and variability could be invaluable in detecting supply chain risks. For example, if there is a tornado warning in the region where a transhipment point is

located, the logistics company will need to alert its customer to this risk, as well as suggesting suitable counter-measures to mitigate against potential disruption (for example, either replanning the transport route, or else increasing substitute supplies from another area).

Based on a research paper produced by DHL: *Big Data in Logistics*, www.dhl.com

As well as the specific example of risk management highlighted in the DHL case study, better quality data should also improve managers' ability to monitor and control business activities in an entity more generally. For example, it should lead to more accurate reporting, particularly when supported by dashboards that enable continuous monitoring and real-time insight into operations.

7.5 The value of big data

The case study above (about big data in the logistics industry) has illustrated some of the ways big data can help an organisation, but more generally, DHL's report *Big Data in Logistics* suggested that one of the main ways in which big data can create value for organisations is through improving operational efficiency. Data can be used to make better decisions, to optimise resource consumption and to improve process quality and performance. In this respect, big data provides similar benefits to automated data processing, although big data can increase the level of transparency in the data.

McKinsey's *Big Data* report also highlights its ability to create transparency as one of five broad ways in which big data can create value for organisations:

Creating transparency – Making data more easily accessible to relevant stakeholders, in a timely manner, can create value in its own right – for example, by revealing insights from data which had previously been too costly or complex to process. This transparency could relate to data within an organisation as well as external data – through better integration and analysis of data produced by different parts of an organisation. For example, within a manufacturing company, integrating data from research and development, engineering and manufacturing units to enable concurrent engineering could significantly reduce time to market as well as improving quality.

However, in many cases the increased transparency resulting from big data is likely to relate to external data. For example, analysing shoppers' transactions, alongside social and geographical data, can reveal peer influence among customers – ie, the extent to which shoppers' choices are shaped by their friends and neighbours as well as by the marketing efforts of the company itself.



Context example: T-Mobile and customer retention

Customer churn (the loss of customers over a period of time) is a problem for all telecommunications providers. To try to reduce churn, providers typically analyse the usage patterns of individual subscribers and their own service quality. Providers also offer specific rewards to try to ensure the loyalty of some customers – based on parameters such as customer spending, usage and length of subscription.

Traditionally, the focus of these retention efforts has been based on the value of individual customers to the provider. While they have yielded some increases in customer loyalty, customer churn remains a major concern to the providers.

In order to predict customer behaviour more effectively, T-Mobile US has started including the social relationship **between** subscribers in its churn management model. In particular, the company has tried to identify so-called 'tribe leaders' – people who have a strong influence within larger, connected groups. T-Mobile's analysis has identified that if a 'tribe leader' switches to a competitor's service, it is likely that a number of their friends and family members will also switch.

This insight has prompted T-Mobile to change the way it calculates customer value. Customer value now reflects not only a customer's lifetime subscription spending on their mobile phone services but also the size of their social network or 'tribe'. Moreover, this means that T-Mobile can focus its retention activities on retaining the 'tribe leaders' rather than on trying to retain individual subscribers.

Historically, the data analysis T-Mobile used for its customer retention activities came primarily from its own billing systems. Creating this new perspective on its customers has required the company to ingest approximately one petabyte of raw data – including information from web clickstreams and social networks – into the information systems it uses to monitor customer churn. However, in the first quarter in which T-Mobile used its new churn management model, churn rates were reduced by 50% compared to the equivalent quarter the previous year.

Based on a case study included in DHL's research paper: *Big Data in Logistics*, www.dhl.com

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Another important context in which transparency can be valuable for an organisation is in relation to **fraud**. For example, having real-time information available from a variety of sources could help an organisation expose fraud and irregular business practices among customers, employees, suppliers or other partners more quickly than it would otherwise have been able to do.

(We noted in Strategic choice that fraud detection is one of the potential applications of machine learning in financial services. More generally, AI and machine learning can help organisations analyse and use data more efficiently than ever before –as we will discuss later in this chapter.)

Performance improvement – The increasing amount of transactional data they store in digital form provides organisations with an increasing amount of accurate and detailed performance data – in real or almost real time. By analysing variability in performance – and the causes of that variability – organisations then manage performance to higher levels. For example, identifying what customers are saying in social media about an organisation's products or its customer service could help the organisation identify how well it is meeting customers' needs. Customers' conversations could help the organisation identify potential changes which are needed to its products, or the way they are delivered, in order to meet customers' needs more effectively – and thereby to increase sales.

In addition to increasing revenues, cutting costs is also likely to be a key component of increasing profits. Having a complete picture of operational activities – and a detailed picture of costs – can help an organisation identify patterns that indicate wasteful or inefficient processes. Similarly, understanding the dynamics of the supply chain in more detail could help an organisation optimise its costs, capacity and inventory levels, as well as the service it provides customers (for example, through improved product availability).

Market segmentation and customisation – The volume and variety within big data enables organisations to create highly specific segments within its markets and to tailor its products and services precisely to meet those needs.

The idea of market segmentation (which we discussed in Strategic marketing and brand management) is already a key concept within strategic marketing. However, big data could facilitate the real-time micro-segmentation of customers for targeted promotions and advertising – for example, by sending tailored (and possibly even personalised) recommendations to customers' mobile devices while they are in the right area to take advantage of the offers.

The ability to perform precise customer segmentation and targeting could be used to help organisations improve customer loyalty and retention, as well as in attracting new customers.

UK supermarkets were at the forefront of the big data revolution, and their store card databases have provided them with data which they can mine to gain insight into customers and their purchases.

McKinsey's report highlights that big data could also be valuable in segmenting public sector markets. Traditionally, public sector markets have not segmented citizens (service users) in the same way that private sector companies have segmented customers and potential customers. However, big data could enable public sector organisations also to tailor products and services more effectively.

Decision making – The sophisticated analytics tools which are used to uncover previously hidden patterns and trends in data could also be used to improve decision-making. For example, trends identified by a retailer in in-store and online sales – in real time – could be used to manage inventories and pricing. In some cases, decisions will be made by managers in store (based on analytics from the data sets) but in other cases the decisions themselves could even become automated. So for example, a retailer could use algorithms to optimise decisions about inventory levels and pricing in response to current and predicted sales data (ie, 'dynamicie,'dynamic pricing,' which we mentioned in Strategic marketing and brand management). (This again links in to the ideas of predictive analytics, and machine learning.)

Having a more detailed understanding of customer behaviour can also be useful in relation to strategic decisions. For example, understanding the customer base of a target business could be an important element in an investment decision: customer loyalty rates could give a valuable insight into the value a business can generate from its customers (and therefore how much a potential acquirer should be prepared to pay for it).

New products and services – Entities can use data about social trends and consumer behaviours to create new products and services to meet customers' needs, or to enhance existing products and services so that they meet customers' needs more exactly. For example, the emergence of real-time location data, from traffic light sensors and satellite navigation systems, could enable insurance companies to refine the pricing of their insurance policies according to where, and how, people drive their cars.

More generally, big data could also provide new business opportunities in their own right. For example, Facebook's advertising business incorporates analysis of a user's actions as well as their friends' actions. Equally, Amazon could be seen as an example of a company which has built its business – and serves its customers – using data and analytics; for example, through the way it makes recommendations for customers linked to the purchases made by other customers with similar interests.

7.6 Data and customers

So far in this section we have looked at the way organisations can use big data to improve their understanding of what their customers want. However, Web 2.0 technologies (which we discussed in Strategic marketing and brand management) mean that data is increasingly available to customers as well as organisations.

For example, online customer reviews are now commonplace, and smartphone applications ('apps') now enable customers to evaluate and compare product prices in real time. This increased

availability of data creates a new market transparency which can give customers a greater insight into what they are buying, and who they are buying from.

In this respect, the data helps customers to base purchasing decisions not only on price, but also on a company's social reputation – for example, in terms of customers' feedback in relation to the quality of service they have received.

7.7 Data and insight

As the fourth 'V' characteristic of big data – veracity – highlights, being able to trust data is crucial in order for that data to be valuable for an organisation.

Similarly, in order to benefit from data, businesses need to move from simply 'having it' to getting insight from it, and doing something differently as a result.

The 'Data analytics' section on PwC's website suggests seven key questions which an organisation should be asking of its data, in order to help the organisation and release insight from its data.

Question	Comments
What value exists in your data?	Management need to consider what data is currently available, how good the quality of that data is, and what they can (or can't) do with it.
Can you trust your data?	It is essential to be able to trust the data, and the systems that collect and hold it. Accountants can play an important role in providing assurance in this area: • reviewing data sources and data management practices to give management confidence in the data • reviewing business systems to ensure the right controls and monitoring practice are in place • re-performing business processes to check the accuracy of the information being reported
What happened and why?	Business information needs to be presented through user-friendly interfaces (eg, dashboards) to show management what is happening in their business. Using analytics, data sets can be combined to reveal trends, patterns, triggers and causal relationships to help management understand 'why' the organisation is performing as it is.
What might happen next?	Using data patterns from the past, combined with industry and entity knowledge, can help management accurately anticipate the future (and plan more effectively as a result). Organisations can also use real-time (or close to real-time) predictive analytics solutions, which can be automated and embedded into business processes (for example, making additional product recommendations for a customer based on current purchases). Analytics could also be very useful for identifying and predicting specific hidden activities, such as fraud or market manipulation.
What is the right action for your business?	Management can use descriptive and predictive insights available to help them identify the appropriate courses of action, subject to their resource constraints and business priorities.
Is insight being delivered to the right people at the right time?	In order for an organisation to maximise the value it obtains from its data, it needs to ensure the right data and insights are being delivered to the right people at the right time. An important consideration here will be tailoring output to meet the different needs of different audiences – for example, summary dashboards for senior management, drill-down solutions for operational managers, or key customer insights delivered to the screens of call centre staff.

Question	Comments
How do you embed data analytics into your organisation?	There are a number of factors which could affect an organisation's ability to use data analytics effectively; including: Culture – the organisation will need to embrace fact-based decision making (rather than decisions based on intuition); based on data which provides insight into what, why and what next. People and skills – will the organisation need to recruit new staff; what training will existing staff need (either to analyse data or to interpret the results of data analytics)? Technology strategy – can the organisation's information systems analyse the data effectively? Where is the organisation storing its data – in house, or via a third-party data hosting service (eg, cloud storage)?

https://www.pwc.co.uk/data-analytics/data-building-blocks.html" https://www.pwc.co.uk/data-analytics/data-building-blocks.html [Accessed: 10 August 2016]

However, when thinking about the ways they can use data and analytics, it is also important for organisations to consider whether they have sufficient, appropriately skilled staff to perform analytics.

7.8 Big data and automation

In its publication *Big data and analytics – what's new?*, the ICAEW Information Technology Faculty highlights another way in which big data can be used in organisations: to automate non-routine decision-making.

Driverless cars illustrate the way data from a wide variety of sensors, mapping applications and satellites can be used to enable technology to make decisions (and thereby to navigate a car). Very few accidents have so far been reported, suggesting that the technology is very accurate in its decision-making, and not prone to fatigue or carelessness (as 'human' drivers might be).

The potential implications of this in an organisational context are that computers will be able to take on more tasks which had previously been difficult to automate. For example, healthcare companies are beginning to exploit machine learning techniques to automate medical diagnosis. Computers can hold far more information than humans, and can quickly and accurately work through the possible scenarios based on the symptoms present to identify the most likely causes for them.

However, as ICAEW's publication identifies, there are still clear risks relating to the use of automation. If something goes wrong, how will a business know? How will it correct potential errors? And, perhaps more fundamentally, there still remains questions around the extent to which computers are better decision-makers than humans, compared to scenarios when human knowledge remains vital.

7.9 Artificial intelligence (AI) and machine learning

One of the recurring themes in any discussion about data, is that the volume of data potentially available to businesses is greater than ever before, and is continuing to increase. However, this data only really becomes valuable to businesses through the insights it can give them. Although the volume of data may be increasing, the volume of insights a business can extract from it will be limited, if a business only has its existing staff available to analyse the data.

In this context, AI and machine learning (which we discussed in Strategic choice) are increasingly mentioned alongside big data, as ways in which technology can help organisations gain insight from data.

AI is a broad concept where machines behave and think more like humans. Machine learning is an application of AI that uses statistics and historical data to identify patterns and automatically improve the efficiency of performing a given set of tasks over time.

A machine's ability to identify patterns in data, and to generate models and recommendations based on them means that machine learning can be used to help make predictions and prescriptions. (Predictive analytics anticipates what will happen; prescriptive analytics provides recommendations, for example, in the way internet retailers recommend future purchases for customers based on their past purchases.) The following examples illustrate some of the ways AI and machine learning could help improve an organisation's performance:

Predictive maintenance – machine learning can use data from sensors throughout an operating system to identify performance issues in the system. Having identified any potential problems, management can take action to reduce downtime and operating costs while improving yield. For example, monitoring engine vibration and 'internet of things' sensor data alongside maintenance history machine learning can identify anomalies in the performance of an engine, and – in turn – can identify if any maintenance may be required. Predictive maintenance like this can help to improve performance and extend the life of assets.

Real-time forecasts – One commonly used application of machine learning is in the real-time optimisation of delivery routes for logistics companies (as mentioned in the DHL case example earlier in this section). This can help not only to reduce delivery times, but also improve fuel **efficiency**.

Product recommendations – Combining customer demographic and past transaction data with social media monitoring can help generate individualised product recommendations (as used by Amazon and Netflix) which help to increase the conversion rate between the number of 'hits' on a website and the number of sales generated.

Personalised marketing – For example, data about individual drivers' driving patterns and distances driven can be used by insurance companies to customise premiums. (More generally, across the retail sector, using customer data to personalise promotions can help retailers increase sales.)

Supply chain – Forecasting inventory requirements based on causal drivers of demand, rather than relying on prior outcomes or historic trends, can improve forecasting accuracy. This could help companies reduce their inventory costs, by matching inventory to product demand more accurately. If all the parties in a supply chain hold the optimum level of inventory, this should lead to a reduction in 'out of stock' or 'over-stocked' situations.

Anomaly detection – As well as contributing to predictive and prescriptive analytics, machine learning can also be valuable in analysing current (real time) data sets to highlight unexpected patterns and outliers (anomalies) in them. The earlier example about predictive maintenance is, in effect, an application of this – because the performance of units in need of repair and maintenance will be anomalous to the expected performance. However, identifying anomalies in data sets could also be particularly useful in helping organisations to identify fraudulent transactions (which do not fit with the regular pattern of transactions) or to identify potential malicious attacks on a website (evidenced by a sudden and unexpected increase in traffic on the website).

7.10 Potential limitations of big data

Some critics have argued that big data is simply a buzzword, a vague term which has turned into an obsession in large organisations and the media. These critics argue that very few instances exist where analysing vast amounts of data has resulted in significant new discoveries or performance improvements for an organisation.

Correlation not causation – The primary focus within big data is on finding correlations between data sets, rather than focusing on the cause of any trends and patterns. It can often

be easier to identify correlations between different variables than to determine what – if anything – is causing that correlation. Correlation does not necessarily imply causality.

Similarly, if an organisation does not understand the factors which give rise to a correlation, it will equally not know what factors may cause the correlation to break down.



Context example: Google Flu Trends

Google Flu Trends was presented as a means of tracking and predicting the spread of influenza across the US.

The program used algorithms which identified correlations between the symptoms people searched for online and flu symptoms.

However, after providing a swift and accurate account of flu outbreaks for several winters, in the 2012–13 season Flu Trends overstated the spread of flu-like illnesses across the US by almost a factor of two.

The cause of this problem was that ultimately Google did not know what linked the search terms with the spread of flu, and Google's algorithms weren't designed to identify what caused what. They were simply finding statistical patterns in the data; and as such they focused on correlation rather than causation.

One explanation of the Flu Trends failure in 2012–13 is that there were a number of news stories in December about the dangers of flu, and these provoked internet searches by people who were healthy.

Sample population – While the data sets available through big data are often very large, they are still not necessarily representative of the entire data population as a whole. For example, if an organisation uses 'tweets' from the social networking site Twitter to provide insight into public opinion on a certain issue, there is no guarantee the 'tweets' will accurately represent the view of society as a whole. (For example, according to the Pew Research Internet Project, in 2013, US-based Twitter users were disproportionately young, urban or suburban, and black.)

Data vs relevant information – More generally, in their review article *Two dogmas of Big Data*, Deloitte note that there is a misconception that "more bytes yields more benefits". In other words, management decisions should be based on relevant information, not raw data. Therefore, by itself, increasing the volume of data available to an organisation does not necessarily provide managers with better information for decision making.

Deloitte's article also suggests that, given the time and expense involved in gathering and using big data, entities need to consider whether "big data yields commensurately big value". As the article points out, the paramount issue when gathering data is not volume, variety or velocity per se, but "gathering the right data that carries the most useful information for the problem at hand".

Data silos – Furthermore, an organisation's ability to maximise the value it obtains from big data could be restricted by 'data silos' within the organisation (although this issue represents a problem with the ways organisations share data and information between departments rather than a specific limitation of the value of big data). For example, insurance companies are aware that big data could have a significant effect on their industry – through helping to combat fraud or through enabling them to understand customers better and to price premiums more accurately. Equally, however, insurers are aware that data silos persist within their organisations which reduce the value they can extract from the data, with communication channels between the risk department and the sales and marketing departments, in particular, often being inadequate.

Data cleansing – The ICAEW Information Technology Faculty report *Big data and analytics – what's new?* identifies that when analysing any data there is a risk that the data may be

inaccurate, inconsistent, or out of date. However, these problems could be magnified in relation to big data as the 'new' sources of data (such as social media) could be unreliable or become outdated very quickly.

David Hand, quoted in the February 2016 edition of *economia* cautions that: "Blind analysis of large data sets is likely to lead to… some gross mistakes. Awareness of the issues of data quality is becoming critically important in the big data world… Accountants have tools for exploring and checking data quality and these will be particularly important."

Where data is being relied upon to make important decisions about specific individuals (eg, customers) or organisational resources, it remains very important to ensure that the quality of the underlying data is fit for purpose. High quality data is achieved by removing inaccurate records before analysis is carried out, a process called **data cleansing**.

On the other hand, in situations where data is being used to identify more general trends, the sheer volume of data means that the granular quality of individual pieces of data is less important. Analysis of the data will still correctly identify general trends, even if some of the individual data items are unreliable.

7.11 Ethics and governance

7.11.1 Potential ethical issues

Although big data can help entities gather more information about their customers and understand customer behaviour more precisely, gathering this data could also raise significant ethical and privacy issues. In particular, to what extent should information about individuals remain private (or confidential) rather than being shared across analytical systems?

Organisations' demand for big data has led to data itself becoming a business – with entities such as data brokers collecting massive amounts of data about individuals, often without their knowledge or consent, and being shared in ways they don't want or expect. Critics argue that in order for big data to work in ethical terms, the individuals whose data is being collected need to have a transparent view of how their data is being used or sold.

On the one hand, if organisations have access to personal data – such as health records and financial records – this could help them to pinpoint the best medical treatment for a patient or the most appropriate financial products for a customer. On the other hand, however, these categories of personal data are those which consumers regard as being the most sensitive. In this respect, big data raises questions around how organisations and individuals will manage the trade-offs between privacy and utility of data.

If companies are using big data properly, it will be vital for them to consider data protection and privacy issues. On the one hand, they must ensure they comply with any legislation about these areas (for example, in relation to the EU's General Data Protection Regulation (GDPR). But even if they comply with prevailing laws, the large-scale collection and exploitation of data could still arouse public debate, which could subsequently damage corporate reputation and brand value.

The issue of **data security** is also closely linked to issues of privacy. What steps – and technologies – are organisations taking to prevent breaches of data security which could expose either personal consumer information or confidential corporate information? (The example of Facebook and Cambridge Analytica which we mentioned in Business risk management highlights the implications of some of these issues.)

The significance of data security as a potential issue could also be increased if organisations want to use third-party cloud service providers to store their data. Although using an external service provider could be an effective way of increasing the volume of data an organisation can store, the organisation needs to be sure that the provider can protect their data. (We will look at this issue of cloud computing and data security in more detail in the next chapter.)

7.11.2 Potential governance issues

The use of big data also requires organisations to maintain strong governance on data quality. For example, the validity of any analysis using the data is likely to be compromised unless there are effective cleansing procedures to remove incomplete, obsolete or duplicated data records.

Similarly, it is very important for organisations to ensure that the overall data quality from different data sources is high because the volume, variety and velocity characteristics of big data all combine to make it difficult to implement efficient procedures for validating data or adjusting data errors.

7.12 Reliability of data analysis

It is clear that there is increasing use of data analytics to inform business decisions. Users of the information produced by data analytics should take steps to ensure the analysis is reliable, and apply some professional scepticism. Scepticism does not mean that the users assume that the data or its conclusions must be wrong; rather it means being aware that data analysis is not always accurate for several reasons:

- There may be bias inherent in the data that is analysed. This may be intentional or unintentional.
- The data may have been intentionally manipulated during the analysis process.
- The data may have been analysed accurately, but the presentation of the data, or the conclusions drawn from it may have been designed to mislead the users.

7.12.1 Unrepresentative samples

Data analysis often involves sampling. This involves taking a sample of data and based on analysis of the sample, making inferences about the population. This includes hypothesis testing, whereby a hypothesis (eg 95% of our customers are happy) is tested by taking a sample. If the results of the sample are not significantly different from the hypothesis, then the hypothesis is accepted.

If sample data is used to make inferences about the population then it is important that the sample is representative of the population, and free from bias. If not, wrong conclusions may be reached.

The first problem may be that the sample used is too small. The larger the sample is, the more likely it is to be representative of the population. However, increasing sample sizes can increase the costs of collecting data significantly, so a balance needs to be struck between the size of the sample and the costs of data collection. Users of data should question the size of the samples used.

Another issue relating to samples is bias. This is where certain classes of data in the population are less likely to occur in the sample than expected.

7.12.2 Data bias

When analysing data for decision making it is important to be aware of data bias.



Definition

Data bias: Data is biased when it is not representative of the population that is being analysed. Bias can be inherent in the data collected or introduced by those analysing the data.

With the proliferation of data available, managers are becoming increasingly reliant on data to support strategic decisions. However, such decisions are not guaranteed to be accurate simply

because they are based on data. Depending on how the data is collected, who collects the data and what data is available for collection certain parameters may be over or underestimated, resulting in misinterpretation and erroneous decisions.

When assessing and interpreting data is it is important to be aware of the main types of bias that could impact the final decision taken.



Definition

Selection bias: Selection bias refers to situations where the method used to select a sample results in data with certain characteristics being less likely, or having no chance of being selected. As a result, the sample would not be representative of the population.

Selection bias often occurs during election campaigns, when pollsters ask a sample of voters about their voting intentions. Some voters prefer not to disclose their voting intentions, or cannot be selected because they do not have internet access (if the poll is being conducted online). This means that such voters are not included in the samples, and wrong conclusions are reached about how people will vote.

Self-selection bias occurs when individuals select themselves. For example, a customer satisfaction survey is most likely to be answered by individuals who are either extremely happy or extremely unhappy with the service received. The majority of customers will not reply so the results are incomplete.

Survivorship bias is a type of selection bias, whereby only items that survived some previous event are included in the sample. An accounting firm might decide to do a survey to find out how good its programme for trainee accountants is, by surveying a sample of trainees who have worked for the firm for one year. Such a survey would exclude trainees who left the firm before the end of the first year who were presumably not very happy with the programme.

Observer bias relates to interpretation and occurs when observing and recording results. The researcher allows their assumptions (which may be unconscious) to influence their observations.

Omitted variable bias occurs when a variable is excluded from the data model and therefore the cause of a change in one variable is incorrectly attributed to another variable in the model. For example, analysis of sales volumes incorrectly attributes a sudden increase to a new advertising campaign when it is in fact caused by an increase in the price of a competitor product.

Cognitive bias relates to human perception and describes the way individuals interpret and understand information differently based on their own background, experiences and beliefs.

Confirmation bias: A type of cognitive bias that occurs when the person performing the analysis has already reached their conclusion, and is using data to support that conclusion. They may intentionally ignore data that they know will not support their conclusion.

If market research is being performed to assess the popularity of a new product, that the company has spent a lot of time and money developing, there may be pressure on the market research department to conclude that the product is *not* likely to fail. This is likely to lead the marketing team to ignore any data suggesting that the product is not popular.



Context example: Confirmation bias and Brexit

According to The British Psychological Society 'Confirmation bias may explain why the Brexit result was such a shock to many'. All interested parties on both sides of the political debate, from politicians, to campaign leaders and voters 'would have been prone to the human tendency to seek out information - from newspapers and TV analysis, but also friends,

colleagues and social media, that agreed with their pre-existing beliefs and reject information that didn't.' Whether a 'Leaver' or a 'Remainer', individuals on either side of the debate could find a plethora of strong arguments to confirm their own beliefs and perceptions. The example demonstrates that the existence of confirmation bias can make it difficult to see alternative points of view, affecting the quality and objectivity of any decisions taken.

Source:

The British Psychological Society. (n.d) Making better decisions: How understanding our psychology can stop us falling into the bias trap. [Online] Available from: https://www.bps.org.uk/sites/bps.org.uk/files/Policy/Policy%20-%20Files/Changing%20Behaviour%20-%20Making%20Better%20Decisions.pdf [Accessed 8 June 2021]



Professional skills focus: Applying judgement

Before drawing conclusions from any information, ensure that you have considered possible bias within the data. In particular, take into account the source of the data and whether it is complete and relevant to the decision being made. Consider whether further information is needed before a sound conclusion can be drawn.

7.13 Reaching the wrong conclusions

7.13.1 Spurious correlation

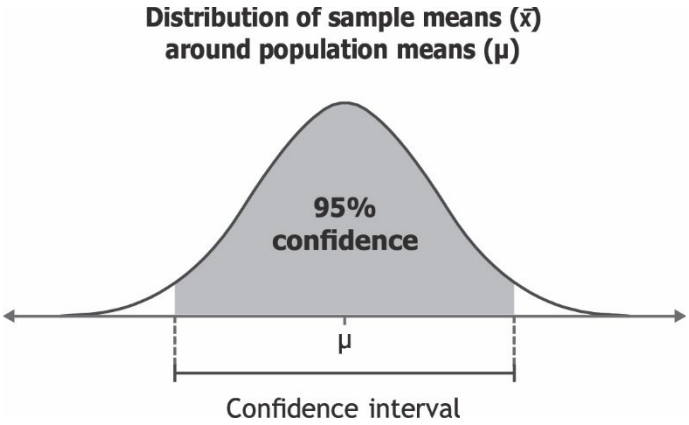
Even if the underlying data is sound, analysis may still lead to questionable conclusions. This can occur intentionally or unintentionally. One common problem is spurious correlation, where two variables show strong correlation, but this does not mean that one causes the other. An example of this is the increase in the number of cases of diabetes and the increase in sales of smartphones over the last twenty years. These two show a strong correlation, but this does not prove that smart phones cause diabetes.

7.13.2 Type I and Type II errors

Hypothesis testing is based on formulating a hypothesis, and then taking a sample of data to find out if that hypothesis is correct. The hypothesis typically relates to the mean of a population, μ.

To test the hypothesis, a sample is taken and the sample mean is calculated. Hypothesis testing recognises that even if the hypothesis is true, not every sample mean will be equal to the mean of the population. In fact, if we were to take every sample of a given size from the population, and note down the sample means, then sample means would have their own distribution. It is often assumed that the sample means themselves are normally distributed with a mean that is equal to the mean of the population, and a standard deviation (or standard error) that depends on the standard deviation of the population and the sample size.

The hypothesis would only be rejected if the sample shows a result that is statistically significantly different from that expected by the hypothesis. Statistical significance uses confidence intervals. You would not need to calculate confidence intervals, but it is useful to have an awareness of them. We may set a confidence interval of 95%. This means that we identify the upper and lower limits within which 95% of samples means would lie. If our sample mean lies outside this range, we reject the hypothesis. If it lies within the range, we accept it.



Hypothesis testing can reach the wrong conclusions. A type I error occurs where the hypothesis is correct, but because the sample mean is outside of the confidence interval, the hypothesis is rejected. A type II error occurs when the hypothesis is incorrect, but the hypothesis is accepted because the sample mean lies within the 95% confidence interval.

In medical testing, the terms "false positive" and "false negative" are often used:

- A false positive is where a test incorrectly indicates the presence of a disease or illness when the illness is not present. This is a type I error.
- A false negative is where a test incorrectly fails to indicate the presence of a disease of illness when it actually does exist. This is a type II error.

7.14 Trend analysis

Trend analysis aims to uncover patterns in a company's data in order to predict future trends and support strategic decision making. Trend analysis involves collecting information from multiple time periods and plotting the information on a graph to assess whether the data points to an upward or downward trend. Examples of trend analysis include:

- Reviewing monthly revenue and cost information from the income statement to identify inconsistencies or fraud, eg a spike in revenue in one period should be investigated to understand the cause.
- Forecasting future revenues and costs, by extrapolating the trend line, for the purpose of budgeting.
- Analysing and extrapolating historical data to support future strategic decisions such as investment in a new product or expansion into a new market.

Trend analysis and prediction of future trend data can be performed using spreadsheet functionality. Note that TREND functionality only looks for linear relationships in data.

Worked example: Trend analysis

As part of the monthly re-forecasting process, you have been asked to forecast the next 5 months of revenue for a flooring range sold by carpet and flooring retailer Home Comfort. You have been given the data below, relating to actual sales of Zermatt, a laminate flooring product available in a variety of colours, for Months 1–9.

	A	B
1	Month	Zermatt revenue (£'000)
2	1	20
3	2	32
4	3	45

5	4	46
6	5	51
7	6	53
8	7	65
9	8	85
10	9	75

Requirement

Using spreadsheet functionality, calculate forecast sales for Months 10–15.

Solution

To calculate forecast sales, you can use the spreadsheet function TREND. The formula used in cell E2 is =TREND(B2:B10, A2:A10, D2:D7)

	A	B	C	D	E
1	Month	Product A revenue (£'000)		Month	Zermatt revenue (£'000)
2	1	20		10	88 **1**
3	2	32		11	95
4	3	45		12	102
5	4	46		13	109
6	5	51		14	116
7	6	53		15	123
8	7	65			
9	8	85			
10	9	75			

1 TREND FUNCTION is used to calculate Months 10-15

Tutorial Note

In the Trend Analysis example above, the data shows a time series trend, with one variable beingmonths. The spreadsheet function TREND can examine the relationship between any two variables where there is a meaningful causal link. For example, TREND could be used to analyse the relationship between air temperature and precipitation levels. It is expected that as averagetemperatures at the Earth's surface rise, more evaporation occurs resulting in increased precipitation.

7.15 Data visualisation

Data visualisation is the use of charts and diagrams to present information. The advantage of these is that high level information can often be more quickly understood if presented in the form of charts and diagrams than just providing numerical data.

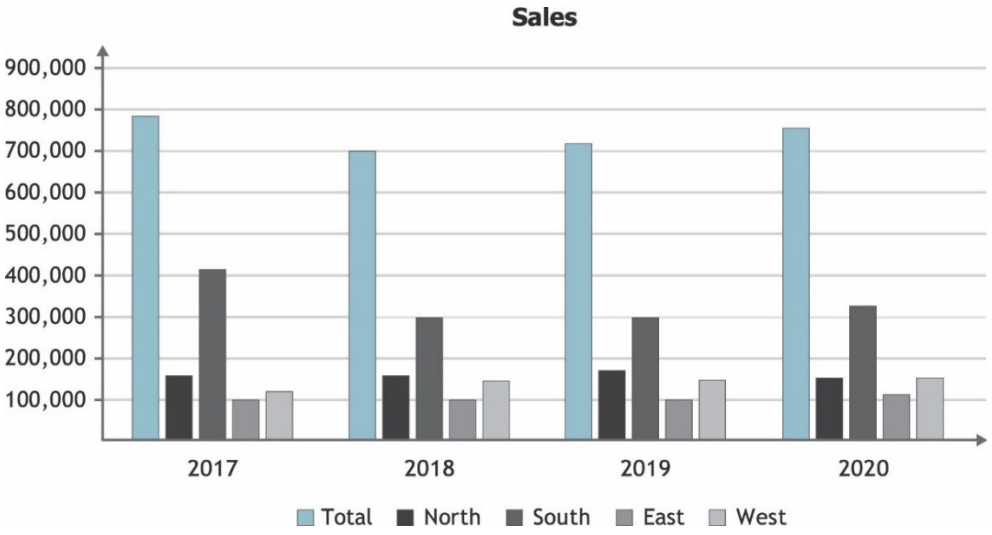
Many such visualisations are provided on "dashboards" which provide users with up to date summaries of key data. In performance management, performance dashboards are often used by senior management to monitor the performance of the organisation using key performance indicators that reflect the critical success factors of the business.

You may be provided with visualisations in an exam question and asked to interpret or discuss them. Some of the more common visualisations are shown below.

7.15.1 Bar charts

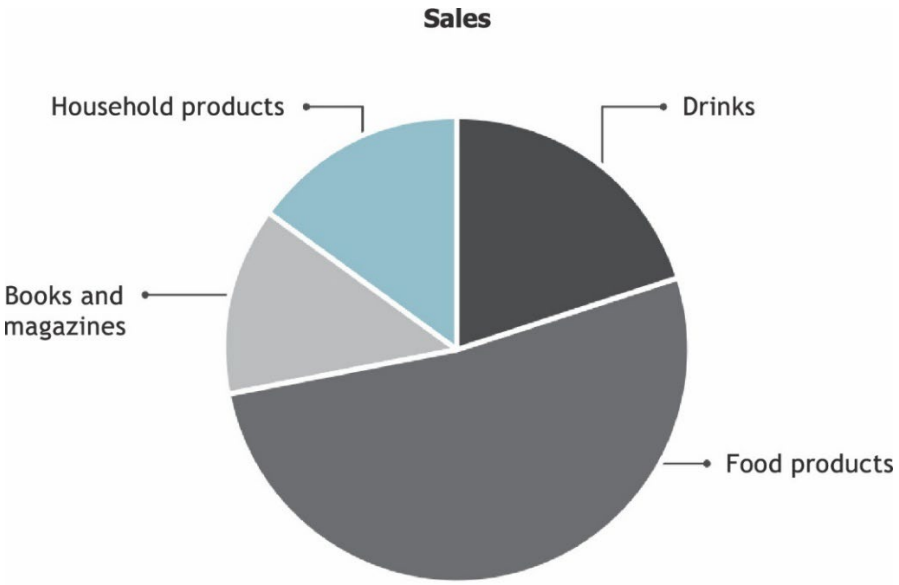
Bar charts present information where the length of the bars represents the value of the data. Bar charts are useful for presenting discrete data where comparisons are made between different data sets - for example, sales in different periods. Clustered bar charts show the components that make up the total as well as the total itself. They are useful provided that there are not too many sub- components.

If there are too many, the charts begin to become too cluttered and may confuse users:



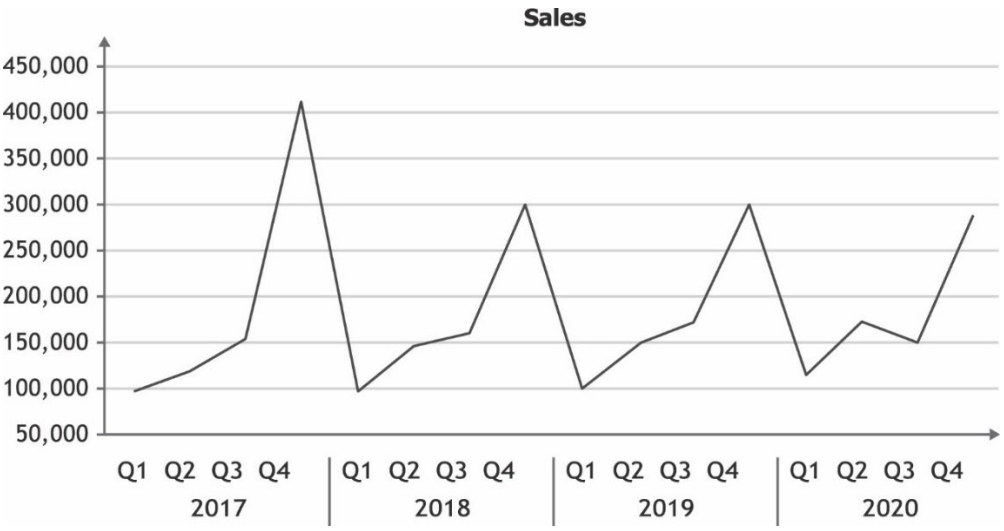
7.15.2 Pie charts

Pie charts are a useful way of showing the components that make up a total.



7.15.3 Line charts

Line charts are useful for showing trends in a data series (e.g. sales over time).

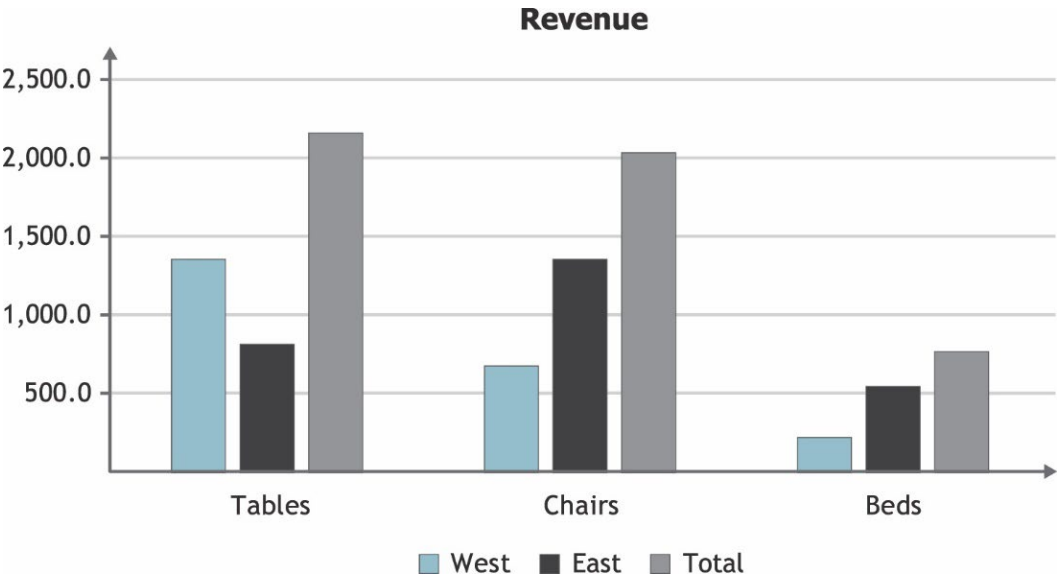


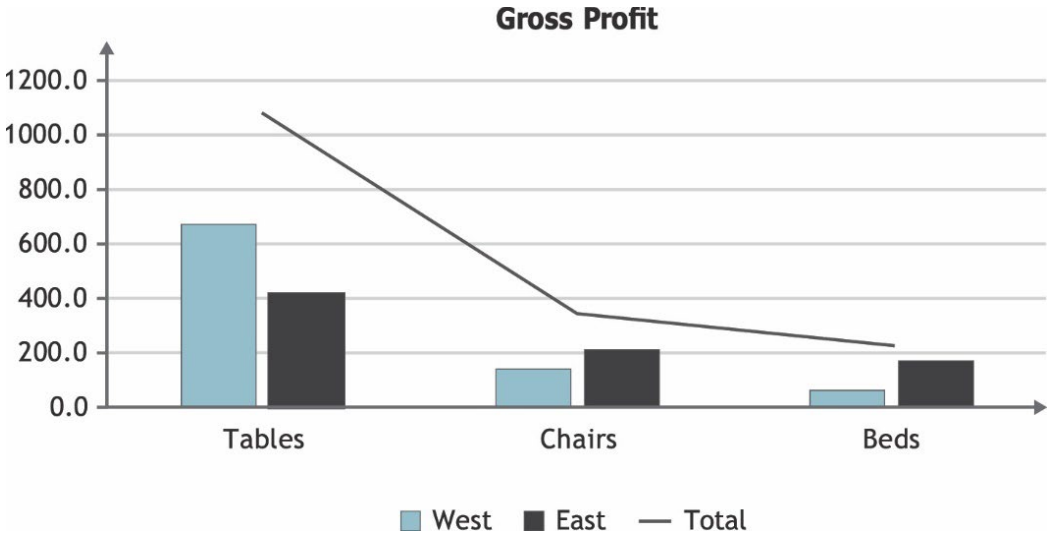


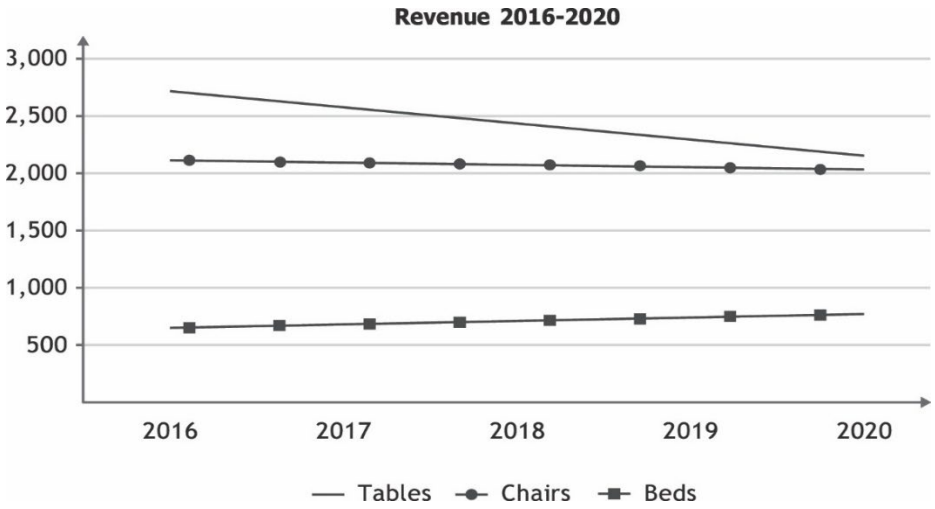
Context example: Visualisations example

The Big Furniture Group (BFG) sells three main product groups; tables, chairs and beds. The company operates through two divisions, West and East, each of which have a similar number of stores.

The following information has been provided by the sales manager:







Based on the above, the following can performance review has been written:

East has the highest total revenues of £2,700 compared to £2,200 for West. However, West's gross profit is higher, at £840 compared to £780 for East. West has achieved an overall gross profit margin of 38.2% compared to 28.9% for West.

The gross margins are not consistently higher for West. East is achieving a higher margin for tables and beds. The reason for the higher overall margin achieved by West relates to the mix of sales. More than half of West's revenues (63.6%) are from tables, and these have a much higher profit margin than the other two products. Only 29.6% of East's revenue comes from sales of tables. It appears that West has been selling tables for a lower price than East, given the lower margins achieved, but this has been a smart move as the 3.6% drop in gross margin has led to almost double the sales achieved by East for tables.

East has achieved 133% more revenue from the sale of chairs than West. However, the difference in gross margin for chairs is very large - 25% for West compared to 14.3% for East. This suggests that very large discounts were given for chairs by East, which would explain the higher revenues. It is questionable whether this was a good decision. Although East did achieve revenue from chairs that was £800 higher than West, it only achieved £50 more gross profit, which suggests that excessive discounting was needed to make the additional sales.

Also, gross margins on chairs are much lower than gross margins on tables, so East has focussed its sales effort on the wrong product.

Where East does appear to have done better than West is in the sale of beds, where it achieved higher revenues and higher gross margins. However, since the sale of beds accounted for only 18.5% of East's revenues, this has not compensated for the lower performance on the sale of tables.

Looking at the longer term trends for total company revenues, in 2016 sales of tables were around 2,600, sales of chairs 2,100 and sales of beds around 600. Since then there has been a slight but steady decline in sales of tables. Sales of chairs remains fairly constant, and sales of beds have been increasing slightly. It would be worth identifying the reason for the decline in sales from tables, as these have the highest margin, so if the decline can be averted, that would help keep profit margins from declining.

In summary, West has performed better than East in terms of achieving higher gross profits, due to focussing on the sale of tables, which have a higher margin than chairs. East should be encouraged to focus more on the sale of tables in future and not offer such large discounts for chairs.

WORKINGS

It is difficult to read precise amounts from the graphs, so the following figures are approximate:

	Tables	Chairs	Beds	Total
West revenue	1,400	600	200	2,200
East revenue	800	1,400	500	2,700
West gross profit	650	150	40	840
East gross profit	400	200	180	780
West gross margin	46.4%	25%	20%	38.2%
East gross margin	50%	14.3%	36%	28.9%

8 Obtaining more information



Section overview

In the examination, you may be required to comment on the limitations of the data or information which you are provided with, and draw "inferences relating to its completeness, accuracy and credibility". If you feel that the data available is unreliable, you should also be prepared to suggest how it could be improved, or what other sources of data might be available.

You may be required to analyse a statement or report and, on the basis of the information available, provide an explanation of the position, prospects and risk of a business. Having made your analysis or given your explanation, you should go on to consider the risk that your explanation may be incorrect because of limitations in the data or information available.

You would need to explain what these limitations are.

Data available for analysis may be unreliable, possibly because it is incomplete or because it comes from an unreliable source. In this situation, the accountant should consider whether the quality of the information can be improved. The learning outcomes for this subject call for an ability to "assess the extent to which the limited assurance and reasonable assurance engagements can identify and mitigate information risks in this context".

In other words:

- What additional information might you be able to obtain?
- Where would the information be obtained?
- How reliable would it be? What would be the limitations of any additional information you can obtain? Would your additional information be able to provide reasonable assurance, or only limited assurance?

8.1 Limitations of the available data

Even though the exam question is unlikely to ask you specifically to comment on limitations in the data provided, you should be prepared to demonstrate that you are aware of any weaknesses in your analysis due to unreliable/incomplete information.

You should also be prepared to indicate what information you would like to obtain, but make sure that your suggestions are realistic.

(a) It is inappropriate to suggest the need for information that could not be obtained for practical reasons or which would be too expensive to obtain and not worth the cost.

(b) Any data obtainable on the industry and competitors will help to provide a benchmark for the performance of the business. However, the amount of information about competitors may be limited and you might need to indicate the sources of any such additional data.

Example

The financial information in a case study or scenario is likely to be in the form of a summary. You might recommend that:

- more detailed information would be useful, such as a breakdown of revenues or profits by product, country or business unit
- where you have been provided with historical information for analysis purposes at five-yearly intervals, data for the years in between would help assess the trend more accurately
- where average figures have been given, information about variations around the average might be useful, to indicate variability and risk

Professional skills focus: Concluding, recommending and communicating

The ability to make recommendations based on valid evidence is one of the professional skills tested in ACA exams. When formulating recommendations, it is important to state the limitations of the data you have been provided with, where relevant, even if the exam requirement does not specifically require this. You may also wish to suggest additional information sources to help strengthen your recommendations.

Analysing data

Data may be presented in a pre-populated spreadsheet, requiring further analysis. Spreadsheet functionality can be used to interrogate such data and draw conclusions. For example, SUMIFS adds the values in a range of cells that meet multiple criteria and is used in

the illustration below (XY data) to identify sales revenues per country per month. Guidance on how to use SUMIFS can be found in Spreadsheet Formulae for Strategic Business Management.

Worked example: XY data

XY is a small company that makes and sells two specialist components: the A and the B. Sales areas are in Spain, Germany and Italy but production occurs in the UK. Two years ago a sales management system was introduced in an attempt to improve management information.

Sales are made in Euros but are recorded in the sales management system in £. The sales manager for each area receives an annual bonus in September (after the European summer holidays have finished) based on these £ revenues.

XY's production team estimate monthly production requirements by reviewing last year's sales data. They keep zero inventory. Recently, the production team has been criticised because components are out of stock when the sales team try to ship them to customers. The production team complain that this is the fault of the sales team because the sales information is not accurate.

Data available in the sales system is shown below:

	A	B	C	D
1	Month	Sales area	Product	Sales revenue £'000
2	Jun	Spain	B	54
3	Aug	Germany	B	72
4	Jun	Germany	B	12
5	Jun	Italy	B	23
6	Aug	Italy	B	78
7	Jun	Italy	A	32
8	July	Germany	A	3
9	Jun	Spain	B	31
10	Aug	Germany	A	105
11	July	Italy	A	6
12	Aug	Spain	A	44
13	Aug	Italy	A	79
14	Jun	Germany	A	31
15	Jun	Italy	**B**	23
16	Aug	Spain	B	104
17	July	Spain	A	7
18	Aug	Germany	B	13

Requirement

Using SUMIFS functionality, analyse the data provided and assess the problems with the current approach to recording and using sales data.

Identify additional information that would be useful in assessing the sales data.

Solution

An analysis of the data has been carried out using the SUMIFS functionality. To ensure that formulae have captured all the relevant data, it is good spreadsheet practice to reconcile spreadsheet numbers. For example total sales revenue in D19 should reconcile to Cell K8. The formula in Cell H2 looks at the sales revenue (column D) for June (column A), for Spain (column B) and for Product A (column C).

=SUMIFS($D$2:$D$18,$A$2:$A$18,"Jun",$B$2:$B$18,"Spain",$C$2:$C$18,"A")

	A	B	C	D	E	F	G	H	I	J	K
1	Month	Country	Product	Sales revenue £'00 0		**Country**	**Product**	**June**	**July**	**Aug**	**Total**
2	Jun	Spain	B	54		**Spain**	A	0	7	44	51
3	Aug	Germany	B	72			B	85	0	104	189
4	Jun	Germany	B	12		**Italy**	A	32	6	79	117
5	Jun	Italy	B	23			B	46	0	78	124
6	Aug	Italy	B	78		**Germany**	A	31	3	105	139
7	Jun	Italy	A	32			B	12	0	85	97
8	July	Germany	A	3		**Total**		206	16	495	**717**
9	Jun	Spain	B	31							
10	Aug	Germany	A	105							
11	July	Italy	A	6							
12	Aug	Spain	A	44							
13	Aug	Italy	A	79							
14	Jun	Germany	A	31							
15	Jun	Italy	B	23							
16	Aug	Spain	B	104							
17	July	Spain	A	7							
18	Aug	Germany	B	13							
19	**TOT AL**			**717**							

Problems with XY's sales data

(1) Incomplete data

Some products appear to have no sales at all in July, suggesting that the sales data recorded by the sales managers could be incomplete. For example, in Spain there are no sales revenues for product B in July despite there being £85,000 and £104,000 in June and August respectively. The lack of sales may be attributable to the annual European holidays but could also be caused by delays in sales managers recording the sales in the system.

(2) Overstated data

The sales figures in August are significantly higher than in the months of June and July. Whilst sales teams may have made more sales in August because the summer holidays are over, it could also suggest that as many sales as possible are being pushed through in August in order to secure a higher bonus.

(3) Production problems

The production team are relying on data that is potentially inaccurate. For example, if sales occur in July but are not recorded on the system due to sales manager holidays, components will be out of stock when customers require them. Inaccurate and incomplete data could result in the loss of customers in the long term.

(4) Lack of integrated objectives

The system has been set up to meet the needs of the sales team but does not address the requirements of the production team who are struggling to forecast production requirements. The current system produces inaccurate management information affecting decision making and may also result in dissatisfied customers if stock is not available when required.

Additional information

- Annual sales figures for the last three years will help XY to understand overall sales trends to assess whether the sample data provided is in line with previous years. For example, are sales always lower in July than in August?
- Exchange rate information used to record the sales revenue would allow XY to assess whether all the sales managers are recording their sales at the same Euro/£ exchange rate. This could influence the bonus received.
- Clarity should be sought over the different objectives and responsibilities in relation to the system. For example, are sales managers measured on the accuracy and timeliness of the information they input to the system? Is the production team penalised for failing to meet customer demand?

Tutorial Note

Be prepared to question the data provided and apply professional scepticism and judgement. After using spreadsheet functionality to analyse the data, assess whether it appears complete and accurate. For example, is it realistic that some products have no sales in July even though they have large sales in June and August? Consider why sales suddenly peak in August. Could the databe manipulated in any way? The scenario hints at problems with the sales data so use these to guide your answer.

8.2 Assurance engagements



Definition

Assurance engagement: An assurance engagement is one in which a practitioner aims to obtain sufficient appropriate evidence to express a conclusion designed to enhance the degree of confidence of the intended users other than the responsible party about the outcome of the evaluation or measurement of an underlying subject matter against criteria.

The most common type of assurance engagement is the external audit. However, there are a range of other assurance engagements an accountant can undertake (as we have seen in earlier chapters, for example in the context of assurance over third party operations, or due diligence (Strategic choice)). The basic principles and procedures for the performance of these assurance engagements are provided by International Standard on Assurance Engagements (ISAE) 3000 (Revised), *Assurance Engagements Other than Audits or Reviews of Historical Financial Information*, issued by the International Auditing and Assurance Standards Board (IAASB). You should already be familiar with this standard from the Audit and Assurance paper at the Professional Level; however, we will include a brief reminder of its key points here.

ISAE 3000 (Revised) distinguishes between two types of assurance engagement:

- **reasonable assurance engagements**, which result in a **positive expression of opinion** and where the level of assurance given is deemed to be high (eg,'the management has operated an effective system of internal controls'); and
- **Limited assurance engagements**, which result in **negative assurance** and where the level of assurance given is deemed to be moderate (eg,'nothing has come to our attention that indicates significant deficiencies in internal control').

Assurance engagements performed by professional accountants are normally intended to **enhance the credibility of information** about a subject matter by evaluating whether the subject matter conforms in all material respects with suitable criteria, thereby improving the likelihood that the information will meet the needs of an intended user. In this regard, the level of assurance provided by the professional accountant's conclusion conveys the degree of confidence that the intended user may place on the credibility of the subject matter.

There is a broad range of assurance engagements, which may include any of the following areas:

- engagements to report on a wide range of subject matters covering financial and non-financial information
- engagements intended to provide high or moderate levels of assurance
- **attestation engagements** (where the underlying subject matter has not been measured or evaluated by the practitioner, but the practitioner concludes whether or not the subject matter information is free from material misstatement) and **direct engagements** (where the underlying subject matter has been measured and evaluated by the practitioner, and the practitioner then presents conclusions on the reported outcome in the assurance report)
- engagements to report internally and externally
- engagements in the private and public sector

Specific examples of assurance assignments include:

- assurance attaching to special purpose financial statements
- adequacy of internal controls
- reliability and adequacy of IT systems
- environmental and social matters
- risk assessment
- regulatory compliance
- verification of contractual compliance

8.2.1 Preconditions for an assurance engagement

Before accepting an assurance engagement, a practitioner needs to establish that the preconditions for an assurance engagement are present.

This means that the roles and responsibilities of the parties are suitable in the circumstances, and the engagement has the following characteristics:

- The underlying subject matter is appropriate.
- The criteria to be applied in preparing the subject matter information are suitable and will be available to the intended users.
- The practitioner will have access to the evidence needed to support their conclusion.
- The practitioner's conclusion is contained in a written report.
- There is a rational purpose for the engagement.

In addition, the practitioner should only accept an assurance engagement where there is no reason to believe that relevant ethical requirements – including independence – will not be satisfied, and provided that the persons who will perform the engagement have the appropriate competences and capabilities to do so.

8.2.2 Elements of an assurance engagement

An assurance engagement will normally exhibit the following elements:

- A **three party relationship** involving:
  - a professional accountant (the auditor or 'practitioner')
  - a responsible party (the client company)
  - an intended user (eg, investors, regulators)
- **Subject matter** (ie, the information or issue to be attested)
- **Suitable criteria** (ie, standards or benchmarks to evaluate the subject matter)
- An **engagement process** (the terms of the engagement and process)
- A **conclusion** (ie, a written assurance report)

Planning

Planning an assurance engagement will normally include considering the following:

- terms of the engagement
- characteristics of the subject matter and the identified criteria
- the engagement process and sources of evidence
- understanding the entity, the environment and the risks
- identifying intended users and their needs
- personnel requirements to complete the assignment

The practitioner should also:

- obtain an understanding of the subject matter
- assess the suitability of the criteria to evaluate or measure the subject matter
- consider materiality and engagement risk

Obtaining evidence

The practitioner should obtain sufficient appropriate evidence on which to base the conclusion. This may include:

- obtaining representations from responsible parties
- considering the effect of subsequent events

Conclusions

The professional accountant should express a conclusion that provides a level of assurance as to whether the subject matter conforms, in all material respects, with the identified suitable criteria.

The ISAE **does not require a standardised format for reporting**. However, it states that the assurance report will normally include the following elements:

- a title that indicates the report is an independent assurance report
- an addressee
- an identification of the subject matter (eg, period covered, qualitative vs quantitative, objective vs subjective)
- an identification of the criteria (assertions, measurement methods, interpretations, regulations)
- inherent limitations
- specific users and intended purposes
- responsible parties and responsibilities
- statement that the work was performed in accordance with ISAEs
- summary of work performed
- conclusion
- report date
- name and location of practitioner giving the report

The professional accountant's conclusion provides a level of assurance about the subject matter, either reasonable or limited. **Absolute assurance is generally not attainable** as a result of such factors as:

- the use of selective testing
- the inherent limitations of control systems
- the fact that much of the evidence available to the professional accountant is persuasive, rather than conclusive
- the use of judgement in gathering evidence and drawing conclusions, based on that evidence
- in some cases, the characteristics of the subject matter

Therefore, professional accountants ordinarily undertake engagements to provide one of only two distinct levels: a reasonable or limited assurance. These engagements are affected by various elements; for example, the degree of precision associated with the subject matter, the nature, timing and extent of procedures, and the sufficiency and appropriateness of the evidence available to support a conclusion.

8.2.3 Consulting engagements

Assurance engagements are associated with engagements that are initiated and paid for by one party, for the purpose of providing an independent opinion to someone else.

In the context of data analysis, obtaining better-quality information could be a **consulting engagement** rather than an assurance engagement. The key issue remains: What additional information can be obtained, how reliable will it be, and how would the additional information enable me to reconsider or adjust my conclusions?

8.2.4 Illustration: Analysis of a financial forecast

An accountant may be asked to comment on the implications of a financial forecast that has been prepared by the operations management of a client company.

The initial judgement of the accountant may be to explain the financial or strategic implications of the forecast, but question the reliability of the forecast. It may be possible to carry out a consultancy engagement to assess the reliability of the forecast.

The aim of the engagement would be to obtain sufficient appropriate evidence as to whether:

- management's best-estimate assumptions on which the prospective financial information is based are **not unreasonable**; and
- the prospective financial information is **properly prepared** on the basis of these assumptions.

The accountant would need to consider:

- the likelihood of material misstatement
- the competence of management regarding the preparation of financial forecasts (based, perhaps, on previous experience)
- the extent to which the forecast is affected by management's judgement
- the adequacy and reliability of the underlying data

The accountant should have **sufficient knowledge** of the business to be able to evaluate the significant assumptions that management have made.

Information in a financial forecast is subjective information. It is impossible for an accountant to give the same level of assurance regarding forecasts as for historical financial information.

Limited assurance can be given in the form of a negative opinion – there is nothing to suggest that the forecast is inappropriate.

Where an assurance engagement is for the benefit of management of the client company, a consultancy engagement might be more appropriate. In this type of engagement, the

accountancy firm assesses the degree of reliability in the forecast and perhaps provides an alternative forecast based on different assumptions.

With any forecast, however, it is important to understand the assumptions and recognise that these may well prove incorrect.

8.2.5 Assurance over prospective financial information

Financial forecasts – whether they are produced for internal management purposes or for external use – are the most common type of prospective financial information.

ISAE 3400, *The Examination of Prospective Financial Information* deals with the issues an auditor should consider when accepting an assurance engagement of this sort, and the examination procedures they should carry out.

We discussed ISAE 3400 in more detail in Strategic implementation earlier in this Workbook, in the context of assurance over business plans.

8.3 Assurance and non-financial information

Although the focus of corporate reporting has traditionally been on financial reporting, companies now disclose a much broader range of information; information which goes far beyond traditional financial reporting. For example, companies now regularly report non-financial information including:

(a) the strategic report or **management commentary** (see Strategic analysis of this Workbook), and other statements which are contained in their annual report: corporate governance statements; information on risk management policies; internal controls or wider operating data

(b) corporate responsibility reporting on environmental, social and economic performance ('triple bottom line')

(c) reporting on matters of public interest: for example, carbon emissions, or quality of service provision. For example, reporting requirements imposed by regulators in the UK require water companies to disclose detailed operating data relating to water quality, leakages and customer service.

Moreover, although this non-financial information does not form part of a company's audited financial statements, stakeholders want to know that the information is **credible** and **reliable**. Therefore, there is also demand for external assurance over this non-financial information.

Providing this assurance over non-financial information will also help prevent an **expectation gap**. Such a gap could arise if stakeholders perceive that the auditors' work, and ultimately their opinion, extends beyond the information in the accounts to non-financial information contained in other elements of the annual report. Such information could include disclosures about oil and gas reserves, research and development (R&D) pipelines (particularly for pharmaceutical companies) and audience size (for entertainment and media companies). This information is not audited but, nevertheless, it relates to key performance areas of the companies.

Management information

This demand for a broader range of non-financial information also means that companies may need to review or revise their management information systems. If, historically, these systems have been designed to provide financial information, they may not capture the additional non-financial information which companies now need to monitor and report.

Therefore, information systems may be another area in which external companies can provide assurance. In this case, the assurance sought could be that the technology systems, and the processes they support, are functioning as intended.

We will look at information and information systems in more detail in the next chapter of this Workbook. However, in the context of data analysis, it remains important to consider whether the information systems that are generating the data being analysed appear robust and reliable.

Worked example: Television networks

One of the major US television networks has seen external reports that indicate the following changes in TV viewing habits of US consumers:

- The Nielsen ratings for TV viewing show a large year on year fall in television ratings for programmes of all the major US networks.
- An independent consulting group has reported a survey finding that live and same-day viewing by people aged between 18 and 49 has fallen by 10% in one year, for programmes broadcast by the major TV networks in the US.
- The TV companies believe that TV viewing is at an all-time high. The decline in live watching is offset by strong growth in viewing through digital recorders, on-demand videos and online streaming to computers and mobile devices.

Some major advertisers are now arguing that falling numbers of live viewers means that rates for advertising should be reduced. The TV companies are inclined to argue that changing viewing habits, from live viewing to time-shifting, means more viewers, and advertising rates should therefore increase.

Requirement

What additional information may help to resolve the differences in opinion between advertisers and TV networks, and where might this information be obtained?

Solution

It would be useful to know:

- whether viewers who watch programmes at a later time, rather than live or same-day, are in the habit of fast-forwarding through advertising breaks, and whether this affects viewing numbers for advertisements (but not sponsorship of TV programmes)
- whether advertisers have effective alternatives to TV as a medium for advertising

Information on these issues could be obtained by means of independent research, but the quality of the research and the information it produces could be challenged and disputed. In this example, additional information may therefore add little to the argument.

Context example: Assurance at Channel 4

The remit of Channel 4 television in the UK, as laid out in legislation by Parliament, requires it to be innovative, experimental, distinctive and diverse.

Channel 4 is unique in the way it is set up. It operates as a publicly owned, commercially self-sufficient, not for profit entity. This operating model differentiates Channel 4 from the publicly funded BBC and from commercially driven competitors, whose primary concern is with delivering shareholder value.

By contrast, Channel 4's sole concern is with the delivery of its remit, and its commercial and financial strategy is designed to support that remit.

Under the Digital Economy Act, Channel 4 is required to publish a Statement of Media Content Policy (SMCP), outlining how it has delivered against its remit in the preceding year, and how it plans to do so in the year ahead.

The SMCP is included in Channel 4's Annual Report, and includes information and key measures relating to the corporation's spending on productions, and details about the programmes produced, and how they fulfil Channel 4's brief for diversity and innovation. Channel 4 has developed a methodology to cover specific definitions, how data for measures are selected, and how measures are calculated.

Although the SMCP does not form part of Channel 4's audited financial statements, the information in it is independently assured by Deloitte, under a limited assurance engagement (as defined in ISAE 3000 (Revised)).

In their Assurance report, Deloitte highlight that their primary procedures in order to gain assurance over the report consisted of:

- interviewing managers at Channel 4's head office, including those with operational responsibility for preparing the assured disclosures
- evaluating the processes and controls for managing, measuring, collating and reporting the assured disclosures, including the application of the methodology within Channel 4's internal guidelines to the underlying assumptions; and
- testing the compilation of a representative sample of Channel 4's SMCP data, selected on the basis of their inherent risk and materiality to Channel 4.

One of the features of Channel 4's SMCP report is that it covers a number of different disclosures, clearly identified in a methodology document. Having such a document is important, because it ensures that Channel 4 reports its performance across a range of key measures, consistently applied from year to year, rather than selectively only including those measures in which it has performed well. This highlights one of the key criteria for assurance engagements: **neutrality**. The criteria selected to measure performance should be free from bias.

Another important criterion for assurance engagements is **reliability** – selecting measures which allow consistent evaluation of information. For example, if similar entities use different criteria to assess the same aspect of performance, it will make it very difficult to make any meaningful (or reliable) comparison of performance between the two entities. This issue reiterates the points made in the case example earlier in this chapter about 'like for like' sales. If different stores calculate 'like for like' sales on different bases, and if the calculations vary from year to year, it will be very difficult to benchmark performance on a reliable basis.

8.4 Agreed-upon procedures

Agreed-upon procedures (AUP) assignments are dealt with by International Standard on Related Services (ISRS) 4400, *Engagements to Perform Agreed-Upon Procedures Regarding Financial Information*, also issued by the IAASB.

(Note, however, that ISRSs have not been adopted in the UK.)

In an engagement to perform AUP, an auditor is engaged to carry out those specific procedures of an audit nature to which the auditor and the entity and any appropriate third parties have agreed, and to report on the **factual findings** of those procedures. The report also describes the purpose of the engagement and describes the procedures undertaken in sufficient detail to communicate the nature and extent of the work performed.

Crucially, though – in contrast to engagements in which an auditor provides an opinion or a conclusion – no assurance is expressed in an AUP engagement, and the recipients of the report **must form their own conclusions** from the report by the auditor. The report is restricted to those parties that have agreed to the procedures to be performed, since others, unaware of the reasons for the procedures, may misinterpret the results.

The value of AUP comes from the auditor objectively carrying out procedures and tests with relevant expertise, thus saving the engaging party from having to carry out the procedures and tests themselves. AUP are most effective where the engaging party is knowledgeable enough to identify the key matters to focus on, to discuss and agree the procedures to be performed, and to interpret the findings of the AUP in their own decision-making.

Interactive question 4: Company-wide cost control

You are a trainee Chartered Accountant at Miles, Smith and Bowring LLP (MSB), a firm of accountants and business advisers.

Along with your manager, you have recently attended a meeting at Hopman Heavies plc (HH), a company operating in the mechanical engineering sector, which manufactures heavy lifting equipment such as cranes, bulldozers and hydraulic elevation platforms. HH is not audited by MSB.

During the meeting, the CEO of HH explained that he was concerned about cost control within the company:

"Costs have been allowed to increase without adequate controls and this must change. For example, a contract with a new supplier of hydraulic systems, Zydrau plc, is being considered, and the method of determining the contract price needs to be reviewed closely, as the hydraulic systems are a key component item for all the types of equipment we manufacture."

The CEO continued by providing some more details about the contract with Zydrau plc:

"There has been preliminary agreement by HH on a three-year contract with Zydrau. Given the significance of hydraulics as a component cost for HH, we are anxious not to pay more than is necessary to suppliers. Unfortunately, however, the nature of each piece of hydraulics equipment supplied varies, and so agreeing a fixed price is not possible.

"A provisional contract has been drawn up using a pricing formula of: direct cost incurred on the contract plus 8%. Direct cost is defined in the provisional contract as the incremental costs to Zydrau caused by supplying equipment to HH."

The CEO explained that the contract would require independent scrutiny of Zydrau's management accounts in order to substantiate the amount of direct costs applicable to the contract with HH. He has requested MSB to carry out this engagement, but admitted he is not sure of the most appropriate basis for the engagement. He said the interim Finance Director at HH had suggested that the engagement could be carried out as either an assurance assignment or on an 'agreed-upon procedures' basis. However, the CEO admitted he was not sure what this meant.

Zydrau has agreed, in principle, to either of these types of arrangement.

Requirement

Advise the CEO, with reasons, whether: (1) an assurance assignment with a report giving a conclusion; or (2) an 'agreed-upon procedures' assignment would be more appropriate.

See **Answer** at the end of this chapter.

8.5 Assurance in the Strategic Business Management exam

We have referred to assurance engagements on several occasions throughout this Workbook, and it is worthwhile considering how you might approach a requirement about an assurance engagement in your exam.

The following are some general points which you may find useful to consider in relation to a question about assurance engagements:

(a) What is the purpose of the assurance engagement?

(b) What is the nature of the subject matter for which assurance is required?

(c) Who is providing the information on which assurance is required, and who is the information user?

(d) What are the benchmarks or standards against which the information will be judged? (Or in the case of financial forecasts, what are the assumptions made in the forecast and what is the reliability of any basic information used as a basis for the forecast?)

(e) How would the assurance work be carried out? Where/how would the accountant find evidence?

(f) Can the accountant provide reasonable assurance, or just limited assurance? (Or if the accountant can provide no assurance at all, should they withdraw from the engagement, perhaps?)

9 Data analysis in the Strategic Business Management exam



Section overview

This chapter concludes with a suggested approach to the analysis of the data you may undertake in the case study scenario in your exam. The 'What-How-Why-When-So what?' method described should already be familiar to you from your Business Strategy and Technology studies, but it is illustrated in the suggested solution to the interactive question which follows.

9.1 Key weaknesses in answers

One of the common weaknesses in candidates' answers in the Advanced Level exams is a lack of **meaningful analysis.**

The following list highlights weaknesses commonly identified in exam answers and suggests ways to address these:

(a) Restating facts or numbers without applying them to the context of the question

A common failing is to explain **what** has happened rather than **why**, eg, stating that sales have grown by 15% in the period but not indicating why they have done so.

Solution: Including the word 'because' in your answer changes the 'what' into a 'why'. In most cases a good answer passes the 'because' test, eg,"Market share has fallen from 35% to 29% **because** of the entry of a new lower-cost competitor".

The 'because' should be related to specific information in the scenario, demonstrating that you have understood the relationship between the financial/quantitative information and the business issues.

(b) Failing to use the additional information from the scenario in answers, resulting in generic answers that could apply to any company, rather than the one in the scenario

Solution: Make specific points, focusing on the particular organisation and relating to the circumstances in the scenario.

Eg,"Falling R&D expenditure may be a problem for XYZ Ltd because it has built market share on the basis of its innovative products."

(c) Interpreting figures/results in isolation

Solution: Link the figures/analysis, eg, if market share has increased but gross margins have decreased, the company may have made a decision to reduce the selling price as part of a market penetration strategy.

(d) Focusing on a narrow range of measures

Financial measures alone will not provide the full picture and are often the result of other factors, not the cause.

Solution: Your answer should, where possible, address a variety of performance indicators. Use the balanced scorecard headings to help you consider a wider range of measures. Remember that these measures will often help you understand what is causing the strategy to succeed or fail.

(e) Failing to use numerical analysis to support the rest of your answer

The data analysis element may be one of the first requirements. Conclusions that you draw from this will help in answering later parts of the question.

Solution: Consider where else in your answer the analysis may be relevant/how you can 'make the numbers talk'.

Eg, if the data analysis shows that the business is currently loss making and that sales and profitability are forecast to decline further, then the business cannot afford to do nothing. Any new strategy that is expected to address this decline or increase returns should be acceptable to the shareholders.

(f) Failing to explain trends in the data by identifying cause and effect relationships

Solution: Examine the information from different perspectives. This may, for example, include analysing information into ratios or percentages based on the data provided.

Eg, where sales revenues are growing by (say) 10% per year, but the number of branches/outlets is growing by 15% per year, then calculating the sales per branch/outlet will show that sales per branch is, on average, falling. Thus, growth in overall sales revenue may be due to investment in more outlets, rather than generic growth in sales per branch because of improved efficiency or stronger market conditions. Alternatively, sales revenue growth might be analysed in relation to volume growth, changes in selling prices and changes in sales mix. Analysis of this data may reveal the relative causes (quantitatively) of sales revenue growth from each of these underlying factors.

(g) Failing to achieve a reasonable balance between numerical and descriptive analysis

Some weak answers are almost entirely descriptive. Other weak answers include enough calculations, but their descriptive analysis is little more than stating which numbers have gone up and which have gone down.

Solution: Both numerical and descriptive analysis are important and need due emphasis. Two possible approaches are:

(1) Set out a comprehensive numerical analysis at the beginning of the answer or in an appendix with workings (eg, in a table). Then produce the descriptive interpretation of these numbers.

(2) Mix the numerical analysis with the descriptive analysis by producing calculations as each issue arises.

In general, approach (a) tends to produce better answers with a more systematic evaluation of the issues. The numerical analysis tends to be more comprehensive and better thought out, with clearer workings. However, if you use approach (a) make sure that you are careful with your time allocation. If you spend too much time on calculations, you may not have enough time to produce sufficient descriptive analysis.

(h) Failing to understand how IFRS reported profit may fail to reflect the underlying performance of the business

Solution: Question whether changes in profit, as measured by IFRS, reflect changes in underlying performance. Consider, for instance, that good strategic decisions may take some time to be reflected in reported profit, which may even fall in the short term.

9.2 Recommended approach

In the light of these weaknesses, the following is a suggestion for the approach to adopt when tackling data analysis:

Step 1: Review scenario and requirements

Step 2: Decide what analysis is appropriate

Step 3: Produce the necessary calculations

Step 4: Interpret your analysis

Step 5: State the additional information required

The steps that cause most problems are Steps 3 and 4.

9.3 WHAT-HOW-WHY-WHEN-SO WHAT analysis

It is difficult to produce a universal approach, but one tool which includes both numerical and descriptive elements is: **WHAT-HOW-WHY-WHEN-SO WHAT analysis**.

WHAT	Look at WHAT has happened overall (eg, revenue has increased by 21%).
HOW	'HOW' seeks to identify the reasons why the 'WHAT' element has occurred (eg, sales prices have risen by 10% and monthly sales volumes have risen by 10% following the price change).
WHY	Look for the underlying **causes** of the HOW element, which may be part of the data provided in the question (eg, there have been significant product improvements introduced during the year, with additional features compared to competitors. This has meant that a higher price can be charged but has also resulted in an increase in demand, despite this increased price). 'WHY' is ultimately more important than 'WHAT' in terms of analysis. Explaining what has happened is a necessary starting point, but you then need to use your analytical skills to identify reasons and provide explanations for what has happened. Look for the links between cause and effect. Also, do not simply consider facts in isolation. Think about whether two or more different issues may be attributable to a single cause, or whether a number of different factors are contributing to a single issue/outcome.
WHEN	If you're assessing the impact of changes in strategy over time or in making comparisons, it is important to know WHEN changes occurred (eg, if the price and volume changes above occurred halfway through the year, then the increase in sales revenues for the current year may be limited to, say, 10.5%, but the changes will be more significant in next year's figures).

SO WHAT	The above steps analyse and interpret the nature of the data provided and attempt to identify and explain the underlying causes of any changes in the data. The next stage is to ask the question, '**So, what** are the **consequences** of our analysis for deciding on the future business strategy?' (eg, what are the consequences for profit of the 10% increases in sales price and sales volume after considering the variable cost increases arising from the product improvements and volume increases? How have competitors responded with price changes and improvements in their own products, which may make the consequences next year different from those which occurred this year?).

Also remember, more generally, that when analysing data, it is important to apply **professional scepticism and judgement**. For example, are there any gaps in the data you have been provided? Are there any inconsistencies between that data and other sources? How does the data you have been given interact with data and information from other sources?

9.4 Issues with accounts

When carrying out data analysis, you will need to use what you've learnt specifically about analysing financial statements, in particular:

- distortions and creative accounting policies, such as income smoothing and understated provisions
- the factors determining important figures in the accounts; in particular, operating profit

Adjustments may be needed to the figures reported in the financial accounts before data analysis can be carried out. These may include remeasurement to market value and recognising assets or liabilities that are not included in the accounts. If you are analysing the income statement, you may need to strip out non-operating or non-recurring items from results to be able to make a fairer comparison over time.



Interactive question 5: WG plc

Introduction

WG plc was formed four years ago, following the merger of two large pharmaceutical companies. Prior to the merger, the two companies had been competitors: they believed that by combining forces, the shareholders of each company would benefit from increased profits arising from the rationalisation of manufacturing facilities, distribution networks, and concentration of resources towards more focused research and development (R&D).

With operating outlets in Europe, Asia, the US and Africa, WG plc regards itself as a global company. It employs approximately 50,000 people worldwide and has developed a varied portfolio of products. Its profits before tax last year increased by 20% and represented approximately 35% of revenue. The company declared that its earnings and dividends per share in the same period increased by 15% over the previous financial year.

All manufacturers of pharmaceutical products claim that their pricing policies need to be set at a level to achieve high profitability in order to attract funds from investors. They argue that this is necessary to meet their high R&D commitments. In recent years, WG plc and other pharmaceutical manufacturers have encountered public and governmental challenges to their high levels of profitability.

WG plc encounters strong competition from other world-class pharmaceutical manufacturers but these are few in number. High R&D costs present a major obstacle to potential competitors tempted to enter the industry.

Mission and objectives

The directors of WG plc have defined their overall corporate mission as being to "combat disease by developing innovative medicines and services and providing them to healthcare organisations for the treatment of patients worldwide".

The directors have confirmed their main objective is to sustain profitability while achieving the company's overall mission. They have also explained that WG plc aims to work towards eliminating those diseases for which the company is engaged in providing treatments. Achievement of the profitability objective is continually threatened by patents coming to the end of their lives. Patents give the sole right to make, use and sell a new product for a limited period.

Product development

A large proportion of the company's turnover in recent years has been derived from one particular drug. The patent for this drug expires next year and it is expected that its sales at that time will represent no more than 10% of total revenue. Four years ago, the sales of this drug produced almost half the company's entire revenue.

A new product, Coffstop, has now completed its rigorous development phases and is being marketed to pharmaceutical stores throughout the world by WG plc. It is in competition with a similar drug, Peffstill, produced and marketed by a direct competitor of WG plc. Medical research and opinion has concluded that Coffstop is generally more effective than Peffstill in treating the condition for which they are intended. Both drugs are available over the counter from pharmacies. The directors of WG plc are optimistic that Coffstop will become very popular because of its improved effectiveness over other market products.

The retail market price of Coffstop is £1.50 per bottle, compared with £10 per bottle of Peffstill. However, the recommended dosage of Coffstop is six times more than that for Peffstill. The bought-in costs per bottle to the retail pharmacist are £0.50 and £7.40 for Coffstop and Peffstill respectively.
Initial indications to the management of WG plc are that retail pharmacists tend to prefer to stock Peffstill on the basis that it achieves 2.6 times the level of gross contribution per bottle compared with Coffstop.

It is estimated that the cost to the retailer of holding Coffstop is £0.40 per bottle; and Peffstill, £0.80. The availability of shelf space is a limiting factor for most retailers. The shelf area occupied by each bottle of Coffstop is 18 square centimetres; and 60 square centimetres for each bottle of Peffstill.
Early indications show that the average weekly sales volume for retail outlets stocking both products is 120 bottles of Coffstop and 20 bottles of Peffstill.

Market development

WG plc has experienced slow growth in its mature markets of Western Europe, North America and Japan. These markets contribute 80% of overall revenue but their governments have reduced expenditure on pharmaceutical products in recent years. The company has encountered a rapid sales increase in its expanding markets of Eastern Europe, South America, the Asia Pacific region, India, Africa and the Middle East. The directors of the company hold the view that increasing population growth in these markets is likely to provide substantial opportunities for the company over the next two decades.

R&D

Almost 15% of WG plc's revenue last year was spent on R&D. WG plc has the largest R&D organisation of all pharmaceutical companies worldwide.

Much research is sponsored by national governments and world health organisations. A major piece of research which has recently been undertaken relates to new treatments for malaria, as the disease is now demonstrating some resistance to existing treatments. WG plc has established a 'donation programme' for the new drug in virulent areas for the disease. This means that the company is donating batches of the drug to the health organisations in these areas. The cost of this programme is offset by the sales of the new drug in other areas of the world by making it available to people proposing to travel to the regions where malaria is widespread.

Requirement

On the basis of the information in this report, analyse the main issues facing WG.

See **Answer** at the end of this chapter.

Summary

	Tick off
Financial statements provide a vital source of information about a company's performance, but data analysis might also involve internal data or management information, or data from externally produced reports. The data to be analysed could be non-financial as well as financial.	☐
When analysing data, it is important to consider how reliable that data is, and therefore how much reliance can be placed on it.	☐
The data or information in reports can be classified into three broad types: strategic, financial and operational.	☐
Strategic data analysis could be particularly important in the context of strategic planning: for example highlighting changes in the business environment; the state of the industry; competitor analysis; analysis of resources and competences; and benchmarking.	☐
Financial analysis could relate to profitability, cash flow, funding and capital structure, or investment appraisal. In addition to performing relevant calculations, it is very important to discuss the strategic or operational significance of your findings from your calculations.	☐
The 'internet of things' and big data mean that the volume and variety of data available to organisations is greater than ever before. Being able to analyse this data to uncover previously hidden patterns of information and insight could be a source of competitive advantage for organisations.	☐
The 'What - How - Why - When - So what' method is suggested as an approach for analysing data in the context of the exam case study scenarios.	☐
Although some operational data will be financial (eg, activity costs) much operational data is likely to be non-financial (eg, process efficiency, quality management). Operational data is very important from a control perspective.	☐
Companies are now disclosing an increasing amount of non-financial information which does not form part of their audited financial statements. Accordingly, there is an increase in demand for external assurance over this information, to increase stakeholders' confidence about the credibility and reliability of the information.	☐
Analyse the metrics used in sustainability reports.	☐

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, complete the following knowledge diagnostic and check you are able to confirm you possess the following essential learning from this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
1.	What are the characteristics of good data and information? (Topic 1)
2.	What is a basic approach for analysing data in reports? (Topic 2)
3.	Who are the main users of financial analysis and what are their different needs? (Topic 4)
4.	What is the difference between efficiency and effectiveness and how can they both be calculated and measured? (Topic 5)
5.	What value does Big Data bring to an organisation? What are the problems associated with Big Data? (Topic 6)
6.	What are the factors that could lead to the results of data analysis being unreliable? (Topic 6)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
2 BTH plc	This question predominantly focuses on financial analysis and therefore requires the calculation of ratios. The chapter content does not list all the ratios you could use so take time to revisit your previous studies and recap on profitability, liquidity, gearing and investor ratios. SBM&L is an open book exam so do not feel you need to commit all the ratios to memory since you can take such information into the exam with you. Remember that this exam is not testing your ability to perform calculations. Instead you need to analyse your numbers and discuss the implications of what they are showing.
4 Malvo Ltd	Data could be formatted in charts or graphs or appear as a data dashboard, as per this question. You need to be comfortable assimilating and interpreting data irrespective of how it is presented. In this question you are asked to use the data to assess the effectiveness of a new IT implementation. Marks will not be awarded for simply conducting data analysis. You must make sense of what the information reveals and draw conclusions in order to answer the question set.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your ueries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Self-test questions

Answer the following questions.

1 Cumulus Limited

Cumulus Limited has obtained finance to convert a former industrial site in the North of England into an outsourced data centre. The company has received a report from consultants containing the following information.

Another company that opened a larger outsourced data centre four years ago in North Wales has so far leased about one-eighth of its capacity.

Private equity firms are investing in outsourced data centres amid increasing interest in cloud computing. Most outsourced data centres (providing over 80% of total capacity) are located in London, where connectivity is fastest.

The perceived benefit of locating a centre in a region where average temperatures are cooler is that colder temperatures reduce the high costs of keeping IT equipment cool. Lower energy costs, lower costs of land and lower salaries are all reasons why Cumulus expects to offer its services at a lower price than London rivals.

The business strategy of Cumulus is to sell IT capacity wholesale to a small number of large users. A recent trend in the market is growth in demand for co-location facilities, where larger numbers of users are willing to rent smaller quantities of storage space alongside other users.

Technological developments continue to reduce the physical size of storage capacity.

At the moment, Cumulus has no customers signed up, and does not expect to win any until its centre is open and functional.

Requirement

Analyse the implications of this data for the senior management of Cumulus.

2 BTH plc

BTH plc has the following results for the last two years of trading.

BTH plc – INCOME STATEMENT FOR THE YEAR ENDED 31 DECEMBER

	20X7	20X8
	£'000	£'000
Revenue	14,400	17,000
Less cost of sales	11,800	12,600
Gross profit	2,600	4,400
Other expenses	1,000	2,000
Finance costs	200	–
Profit before tax	1,400	2,400
Income tax expense	420	720
Profit for the year	980	1,680

BTH plc – STATEMENT OF FINANCIAL POSITION AS AT 31 DECEMBER

		20X7		20X8
	£'000	£'000	£'000	£'000
Non-current assets		2,500		4,000
Current assets				
Inventories	1,300		2,000	
Trade receivables	2,000		1,600	
Cash and cash equivalents	2,400		820	
		5,700		4,420
Total assets		8,200		8,420
Equity and liabilities				
Share capital (2.4 million ordinary shares of £1 each)		2,400		2,400
Revaluation surplus		500		500
Retained earnings		1,200		2,100
		4,100		5,000
Non-current liabilities				
Long-term borrowings		2,600		–
Current liabilities				
Trade payables	1,080		2,700	
Current tax payable	420	1,500	720	3,420
		8,200		8,420

Notes

1 Finance costs represent interest payable on debt.

2 Tax is charged at 30% of profit.

3 Long-term borrowings represent 10% loan stock.

4 Dividends amounting to £520k were paid in 20X7, and £780k was paid in 20X8.

5 The average rate of inflation during the year was 3%.

Requirement

Assess BTH plc's financial performance in 20X8, calculating any ratios that you consider to be useful to help your assessment.

3 Flyway Airline

Flyway Airline (Flyway) is the national airline of Ostland. It was originally owned by the Government but was listed on the local stock exchange after it was sold to private investors more than 20 years ago. The company's objective is to be the best premium global airline.

Flyway provides long- and short-haul services all over the world and is based at its hub at Ostcity airport. Flyway has been hit by a worldwide reduction in air travel due to poor economic conditions. The most recent financial results show a loss and this has caused the board to reconsider its position and take action to address the changed environment.

Flyway has cut its dividend in order to conserve cash and it is trying to rebuild profitability by reducing costs by 14%. The airline is capital intensive, as it maintains a large fleet of modern aircraft. Two major costs for the airline are staff and fuel. In trying to renegotiate working conditions and pay, the management have angered the unionised workforce. There has already been some strike action organised by the unions representing the aircraft crew and

ground staff and more is threatened. The staff are upset about changes to pension provisions which will require them to make larger contributions and also a reduction in the number of crew on each aircraft. They believe this will require them to work harder and so they want a compensating pay rise, but nothing has been offered.

Additionally, the board has been considering taking advantage of new technology in aircraft engines by making a large investment (£450 million) in new low-noise, fuel-efficient aircraft in an effort to reduce the environmental complaints surrounding air travel and also to cut costs.

The management accountant has provided the board with some data on Flyway and two of its main competitors (see figures below). Sudland Air is a government-owned and -run airline in the neighbouring country of Sudland. It has a similar mix of business to Flyway and targets a similar market. Eazee Air is currently one of the most successful of the new privately-owned airlines that have gained significant market share over the last 20 years by offering a cheap but basic short-haul service to customers in and around Ostland. Eazee Air subcontracts many of its activities in order to remain flexible.

Data provided by the board (based on figures for the most recent calendar year):

	A	B	C	D	E
1			Flyway	Sudland Air	Eazee Air
2	Passengers ('000)		23,649	38,272	35,624
3	Passenger kilometres (millions)		79,618	82,554	40,973
4	Revenue	£m	5,430	7,350	2,170
5	Costs				
6	Fuel	£m	1,480	1,823	535
7	Staff	£m	1,560	2,998	238
8	Staff numbers		32,501	56,065	5,372
9	Operating profit	£m	630	54	127
10	Number of aircraft		182	361	143
11	Average aircraft size (seats)		195	163	125
12	Seat kilometres (millions)		100,654	105,974	46,934

Note: A seat kilometre is generated for every one kilometre flown by an **available** seat on the company's aircraft.

In preparation for the next board meeting, the CEO has asked you to calculate some suitable performance measures and explain the results.

The CEO also believes that Flyway could be making more use of Big data. He has recently returned from a conference about 'Big data in the Airline industry' where one of the speakers talked about the benefits of big data in relation to four key areas:

- identifying trends in passenger demand and using this to set prices
- understanding and influencing the customer's selection process (in particular, reducing the number of potential customers who look at the details of a flight, or start booking a flight, online, but do not go on to complete the transaction)
- boosting revenue from in-flight sales by optimising the on-board store for individual flights
- understanding customer sentiment and improving customer satisfaction.

Flyway currently offers a standard selection of in-flights products on all of its flights. The CEO has asked you to explain how using big data could help Flyway improve its performance.

Requirements

3.1 Using the data provided, analyse the performance of the three airlines using appropriate performance indicators, and comment on your results.

3.2 Explain how big data could be used to improve Flyway's performance, in relation to the four key areas identified at the conference.

4 Malvo Ltd

Malvo Ltd ('Malvo') is an online retailer which sells a range of electronics products to customers. Malvo doesn't manufacture any products itself but has established good relationships with leading manufacturers. Malvo's management team have identified that fulfilling customer orders quickly and accurately is crucial to customer satisfaction and retaining customers. They have also identified the importance of effective inventory management. Last year, Malvo installed a new computer program which captures data about customer orders and inventory levels. The program produces detailed data which can be analysed in a variety of different ways to identify trends or to produce management information. The dashboard below shows one configuration of the data. However, the dashboard also enables users to track the real-time status of the current day's open ('live') orders – for example, whether they have shipped, or whether they are delayed. Prior to the new computer program being installed, Malvo's management team had to rely on weekly summaries to monitor inventories and sales.

Data dashboard – customer orders and inventory

Volumes – current day

Customer orders open	2,113
Items to ship	2,792
Overdue shipments (to customers)	197
Shipments due from suppliers	126
Late shipments (from suppliers)	11

Top 3 stock items

Product	Stock Code	In stock	Customer orders
40" television	154011	86	42
Radio	190127	79	23
Fridge	212016	68	61

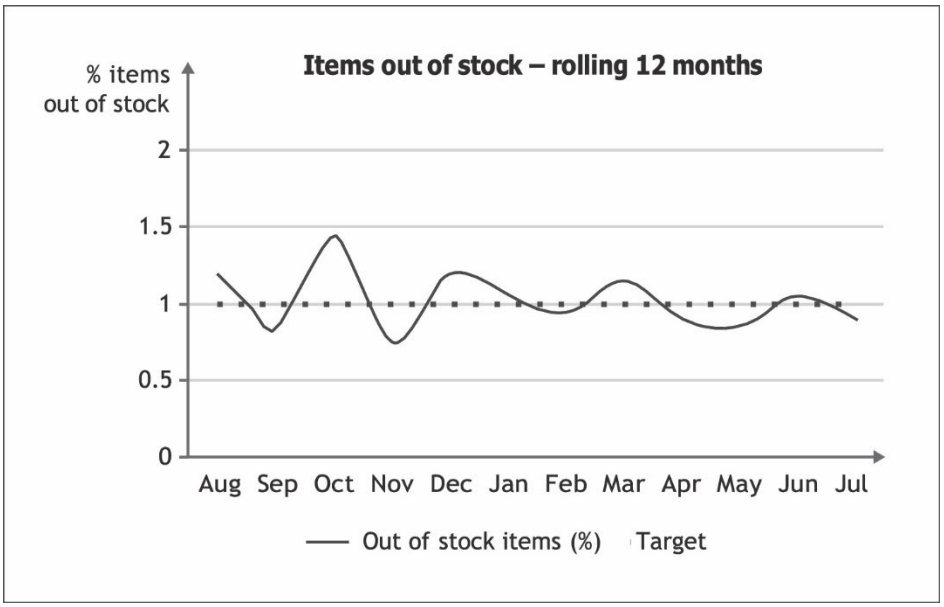
KPIs

	Current month	Last month
Perfect order rate (%)	94.2	94.5
Back order rate (%)	8.5	8.1

Items out of stock - see graph

Top 3 orders

Product	Stock Code	Customer orders	In stock
Electric fans	413072	76	54
Fridge	212016	61	68
Tablet computer	681308	52	39



Back orders are customer orders that cannot be fulfilled when presented, but which the customer is prepared to wait for. The back order rate (%) is the proportion of total customer orders which are back orders.

Perfect order rate: the perfect order rate is the % proportion of total orders which are fulfilled on time, and in full (ie, if multiple products are ordered, then all of them are available and despatched together).

Items out of stock reflects the number (%) of items that are out of stock at the time a customer places an order for that item.

Requirement

Using the dashboard data, analyse how the new computer program helps Malvo to manage its inventory more effectively. Identify further information that could usefully be extracted from the data captured.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 **Note:** The suggested solutions to this question are illustrative and based on the author's assessment. Your views may be different, but they should address the key issues.

Here, the condition of the UK industry is almost certainly the key issue because this is currently the source of most of the group's revenue.

Strategic implications

The most significant information in the report is probably that the UK business of the group, which accounts for over two-thirds of total revenue, is under pressure.

Customers are more price conscious and discount stores seem to be successful.

Treadway has cut prices and hired more staff, but sales are flat in real terms and profit margins are falling.

The positive aspects of UK business appear to be the performance of smaller convenience stores and online food shopping, but there is no information on revenues or profit margins for these aspects of operations.

The management of Treadway needs to consider whether the supermarket industry in the UK may have reached, or be nearing, capacity.

If it is, the challenge is to maintain profitability in the UK business until alternative strategies for growth can be developed and implemented.

1.2 **Other information to consider**

Other aspects of the information that should be considered are:

- The potential for growth in either Asia or the rest of Europe. Asia may seem to offer higher growth prospects, but there is insufficient information about conditions or prospects in either of these two regions to make a firm judgement.
- The US accounts for just 1% of group sales globally and is making losses. It would seem appropriate to consider disinvestment and pulling out of the US market. Presumably this would result in write-off costs, but we do not know what these might be, or whether a buyer could be found for the US business.
- By cutting back investments in superstores, the group expects to increase free cash flows. What should be done with the money? One option would be to increase annual dividend payouts, which may boost the share price and so the group's P/E ratio.
- Lower investment in new stores may improve annual operating margins, by reducing operating costs. Depending on what the group does with its spare cash, there may be prospects for improvements in return on capital.

These judgements have been based on incomplete information and assumptions about future sales growth in the UK that may be too pessimistic.

Sales growth may recover and consumer preferences for discount products may be relatively short term in nature.

However, the risk of a long-term slowdown in the supermarket industry in the UK suggests that the group should urgently review its strategy and consider alternative ways of achieving long-term growth. More information is needed about:

- market conditions and growth prospects in the rest of Europe and Asia
- the profitability of online sales and convenience stores in the UK, and prospects for growth
- prospects for the UK supermarkets industry. The forecast of no real growth may be correct, and the switch by consumers to lower-price or discount products may well have occurred. If this is the case, other supermarket groups in the UK should be suffering in the same way as Treadway. Competitor analysis, and comparisons of performance with other supermarket groups, could provide useful information in support of the assumptions in the report.

Answer to Interactive question 2

2.1 Competitive performance

	20X1	20X2	20X3	20X4
Market share	(83/895) 9%	(124.5/1,234) 10%	(137/980) 14%	(185/1,056) 18%

The restaurant is therefore taking an increasing proportion of the area's restaurant business, doubling its market share over the four-year period.

The number of proposals submitted to cater for special events has increased dramatically, from 10 proposals submitted in 20X1 to 38 submitted in 20X4, while the **percentage of contracts won as a percentage of proposals submitted** has shown remarkable **growth**.

	20X1	20X2	20X3	20X4
Contracts won as % of proposals submitted	20%	29%	52%	66%

The restaurant appears to be **increasingly effective in winning business** in this developing area.

Financial performance

	20X1	20X2	20X3	20X4	20X1–20X4
Change in revenue		+50%	+10%	+35%	+123%
Change in profit		+84%	+104%	+31%	+393%
Profit margin	14%	17%	32%	31%	

The analysis above shows **continuous growth in revenue** and an even **stronger growth in profitability**. The **increase in profit margins** may be a result of **improved resource utilisation**, with fixed costs as a percentage of revenue falling.

It is clear that **20X2** was a **successful** year compared with 20X1, and that **20X3** results were **even better**. While there was a significant increase in revenue in **20X4**, the increase in profitability was less than in previous years and the **profit margin fell** (admittedly only by 1%). This could indicate the need for **tighter cost control**.

Quality of service

Just under 7% ((5 × 52)/3,750) of meals served in 20X1 were to regular customers, compared with over 20% ((26 × 52)/6,700) in 20X4. The business, therefore, has a **growing number of regular customers** who can be assumed to be happy with the price, level of service, quality of food or, indeed, the total package offered by the restaurant.

The data about **complimentary letters, written complaints** and **cases of food poisoning** does **not paint a clear picture** about quality of service as no definitively clear trends are evident, even when the number of meals served is taken into account.

	20X1	20X2	20X3	20X4
Meals served per complimentary letter	3,750	1,275	2,067	1,117
Meals served per written complaint	469	425	443	479
Meals served per reported case of food poisoning	938	1,020	886	957

Without a yardstick such as rates achieved by competitors, it is therefore **difficult to draw firm conclusions** on the quality of service provided by the restaurant, especially as the number of customers almost doubled over the period. More accurate information could possibly be gathered from a large-scale customer satisfaction survey.

Flexibility

One measure of a business's flexibility is **how well it copes with varying levels of demand**. The restaurant's average service delay at peak times shows no clear trend but has fluctuated widely from 47 minutes in 20X2 to less than a third of that in 20X3. When these figures are analysed in conjunction with the average number of customers at peak times, however, it is clear that performance was particularly poor in 20X2 (with a low level of customers but the longest delay), while performance in 20X3 was better. Overall, however, it is clear that there are **problems in flexing resources to meet demand at peak times**.

The number of **items on offer each day**, the **new meals introduced** during the year, the **special theme evenings introduced** and the **weekly opening hours** also indicate improving levels of flexibility, reflecting the **increasing choice available to customers**. The number of items on offer has more than doubled over the four-year period, from 4 to 9, the number of new meals introduced has varied between 8 and 27, the number of special theme evenings has increased from 0 to 13, and opening hours increased in 20X3.

Resource utilisation

This is usually measured in terms of **productivity** (output relative to some form of input). Given the information available and assuming the restaurant is open 52 weeks a year, one measure of productivity is **total meals served/opening hours**. This ratio has steadily increased from 2 in 20X1 to 3.6 in 20X4.

Levels of non-productive time (measured by idle time rates and proportion of operating hours with no customers) **declined**.

	20X1	20X2	20X3	20X4
Idle hours	570	540	465	187
Opening hours (weekly × 52)	1,872	1,872	2,080	1,872
Idle time %	30%	29%	22%	10%
Operating hours with no customers as % of opening hours	20%	16%	9%	7%

In conjunction with the increase in the number of meals served (the year on year increases being 36%, 22%, and 8%), these measures would tend to indicate overall improvements in resource utilisation.

The **increase in capacity** by 60% in 20X3 allowed more customers to be seated during peak times (although we do not know if this was due to increasing floor space or to seating more customers in the same space), but it was **not matched by similar increases in overall activity level**, and did in fact correspond with a **drop in the number of meals served per seat**.

	20X1	20X2	20X3	20X4
Meals served per seat	150	204	155	168

Weekly **opening hours** were **increased** in 20X3 but, as the figures above demonstrate, there was **no corresponding increase in meals served per seat**.

Innovation

The business appears to have been **particularly successful in this area**, with attempts at innovative ways of satisfying customer needs, including the introduction of **special theme evenings**, **increased items on offer** and the successful development of **catering for special events**.

A number of **new meals** have also been **introduced**, although the degree of experimentation has varied considerably from year to year.

2.2 Additional information for assessing performance Competitiveness

(1) Any similar data from one or more restaurants in the locality would enable the business to determine how well it was performing in relation to competitors.

(2) It would also be useful to have data about total meals served in all restaurants in the same price band in the locality in order to assess market share in terms of volume.

(3) More general information about national trends in eating out and restaurant prices, and market research (particularly customer surveys) on similar restaurants would provide a broader context to the performance assessment.

(4) Details of the cost of catering for special events would allow the profitability (or otherwise) of this area of business to be determined.

Financial performance

(1) Cost data on labour, food and overheads, which is missing at the moment, would enable a more in-depth profitability analysis.

(2) Details of assets would enable the calculation of ROCE.

Quality of service

(1) Especially useful would be any customer feedback received or systematically collected by the restaurant (in addition to the complaints and compliments already detailed).

(2) Any reviews of the restaurant that might have appeared in guides, newspapers and so on would provide an expert's analysis of the service offered.

(3) Data on intangible factors, such as the courtesy of staff and ambience of the restaurant, would enable a fuller assessment of the quality of service.

Flexibility

(1) Details of the ease with which the restaurant deals with requests for non-menu items (such as those connected with special dietary needs) would give additional information with which to assess this area.

(2) It would be useful to know whether any staff training to promote multi-skilling (which should improve the business's ability to cope with fluctuations in demand) has ever taken, or could take, place.

Resource utilisation

(1) A number of useful measures could be calculated if information about staffing levels was provided (eg, meals served per hour per member of the waiting staff or revenue per member of staff).

(2) If information about floor area was also provided, measures such as revenue per square metre could be calculated.

(3) It would be useful to know how seat numbers were increased; for example, increasing the restaurant seating space available, or adding more seats in the same space.

Answer to Interactive question 3

3.1 Performance

BPC has made a contribution to society in terms of treating patients and developing new drugs which will allow patients to be cured. It is not clear from the report how true it is that the company sells the drugs for "reasonable prices" as no information is given about the prices charged or prices for similar products in the market. Only 33% of the board are female, which suggests that the company has not performed particularly well in terms of gender equality. An ideal would be 50%. Employee engagement of 75% sounds like a good score so employees do seem to enjoy working for BPC.

In terms of environment, the reduction in greenhouse gasses of 15% since 20W5 represents a simple average fall of 2.1% per year. This does not seem particularly good, given the context of the Paris Agreement to reduce climate change. The aim to reduce greenhouse gasses to zero in 40 year also appears slow. Climate change is a major global concern, and therefore BPC has a moral obligation to do their part in helping to reduce it. Failing to do more could harm BPC's reputation, leading to a boycott of its products. The company must do better in this regard. In terms of using renewable energy, the company is using over 50%, which is good.

From an economic perspective, the company is doing reasonably well – a return of 56% to shareholders over five years is a good steady return. A high portion of revenues is being invested back into R&D which does suggest that the company is thinking of the longer term, so the business does appear to be sustainable from an economic perspective.

Overall, therefore, based on the limited information provided, the society and economy aspects of sustainability are doing well, but the company must do more in terms of environmental sustainability.

3.2 Additional information

In order to make a more valid assessment of BPC's performance, comparatives for the previous year would be useful for the number of patients treated in order to see if the number of patients treated is growing. It would also be useful to have some information about the prices charged for the medicines to see if the claim of "reasonable prices" is borne out.

Some indicator of the size of the company would be useful – for example, the number of employees or total revenue – as this would allow the calculation of ratios such as patients per employee, or revenue per patient. This could then be compared with information from competitors to get a better idea of whether the society is good.

Many of the scores would benefit from having an industry comparator – an industry average, or a similar sized company to compare performance to. This would be useful for evaluating the diversity of the board, employee engagement, renewable energy use, long term investment and shareholder return.

For greenhouse gas emissions, the absolute emissions in metric tonnes of CO_2 equivalent would be useful and this could be used to calculate emissions per £ of revenue or per employee, which could then be compared to others in the industry to make a judgement about whether the absolute level of emissions is high.

3.3 Evaluation of the report

The report highlights a small number of key areas, which is a strength. Highlighting key areas enables users to focus on the important areas, without getting lost in the detail. The report also gives the high-level performance indicators, with a link to the company's web site for more detail, which may enable users to obtain more detailed information.

The narrative before the performance indicators helps users to understand what the company is aiming to achieve. However, the narrative does contain many statements such as

"reasonable prices" and "aiming to reduce the harmful impact that we have on the environment." Without further clarification about exactly what these statements mean, the impression is given that the report is trying to provide an overly good view of the performance.

The report does cover the three categories of sustainability, society, environment and economy, and so does give a broad picture of the overall impact of the company.

Answer to Interactive question 4

Zydrau contract – assurance or agreed-upon procedures

Agreed-upon procedures

In an agreed-upon procedures (AUP) engagement, MSB will provide a report of factual findings from the procedures and tests performed, which need to be agreed with both HH and Zydrau. The procedures and tests required should be sufficiently detailed so as to be clear and unambiguous, and discussed and agreed in advance with both HH and Zydrau staff, so that the factual findings are useful and appropriate to the cost-plus contract.

When performing an AUP engagement on historical financial information MSB, as practitioners, is required, as a minimum, to comply with ISRS 4400, Engagements to Perform Agreed-upon Procedures Regarding Financial Information.

Our report for an AUP engagement will not express a conclusion and, therefore, it is not an assurance engagement. It will not provide recommendations based on the findings.

We would request that HH and Zydrau review the procedures and findings in our report and use the information to draw their own conclusions.

The value of an AUP engagement comes from MSB, as practitioners, objectively carrying out procedures and tests with relevant expertise thus avoiding the need for HH to carry out the procedures and tests themselves and therefore protecting confidentiality for Zydrau.

AUP engagements are most effective in situations, such as this, where there is a clear matter to focus on in the form of the costs plus pricing contract.

The benefit to HH of having an AUP engagement is therefore that it provides evidence for HH that Zydrau is complying with the terms of the contract in identifying, measuring and allocating only costs appropriate to the pricing contract and consistent with that contract. This prevents excessive, and otherwise unobservable, costs being added by Zydrau thereby increasing the price of goods to HH in a manner that is inconsistent with the contract.

Zydrau may be more likely to allow MSB to carry out this task as a professional accountant than perhaps it would with HH, due to the commercial sensitivity of other information that may be obtained in the process. In this context, ISRS 4400 requires compliance with the applicable requirements of the Code of Ethics for Professional Accountants.

Assurance assignment

A key feature that distinguishes an assurance assignment from an AUP assignment is the requirement to express a conclusion.

Discussion with both parties when scoping an assurance engagement will be an important part of the process. This will help MSB to obtain a sound understanding of the objectives of the engagement, the nature of the subject matter (cost structure and costing processes of Zydrau) and the requirements of both HH and Zydrau.

This understanding will help to plan procedures to gather sufficient and appropriate evidence to support a conclusion that is useful in the light of the needs of the parties involved.

It should be possible to obtain the assurance required, provided that the scope of the assurance engagement is in line with the relevant standards; eg, the method of measuring or

evaluating the subject matter is appropriate, criteria are suitable, and sufficient and appropriate evidence exists.

However, from HH's perspective, an assurance engagement may not represent the optimal approach from a cost-benefit perspective, as the amount of evidence to support a conclusion may be much greater than for an AUP assignment.

Answer to Interactive question 5

Suggested approach to a solution

An approach to developing a solution is suggested here. You may prefer to take a different approach, but you should be able to consider all the relevant information provided, identify the most important issues and draw conclusions.

Stage 1: Make notes of the relevant facts

Making notes is a method of reviewing the facts. In the table below, the notes are organised according to the checklists suggested in this chapter. In this example, however, issues in the broader business environment do not seem to be a significant issue

External environment Political Economic Social Technological Environmental Legal	 - - - - - Increase in legal actions against drug companies
Industry environment	High R&D costs Patents to protect intellectual property: importance of patent protection for the business of major drugs companies Rigorous development and testing for new products Government and public challenges to high levels of profit National governments and health organisations sponsor most research: donation programmes
Competition	Strong competition between firms, but few firms High barriers to entry No substitute products for patented drugs Peffstill a serious competitor for Coffstop
Internal strengths and weaknesses	Main revenue-earning drug: patent ends next year. Loss of up to 10% of revenue? No other major product, apparently. Why is R&D not more effective in producing innovative products? More information needed about patents and product portfolio More spending on R&D than competitors Difficulty in finding retail outlets for Coffstop
Financial	Fall in revenue growth to 5% annually from 15%. Are there any forecasts for future sales? Profit before tax 35% of sales, profit after tax 15%. So tax 20% of sales? This is surprising. Is the data reliable? Fall in revenue but 15% increase in earnings and dividends? How

	was this achieved? R&D costs = 15% of revenue last year Detailed figures for profitability of Peffstill and Coffstop for retailers: calculations of profitability required
Operational	Data on performance restricted to financial data, see above

Deciding the method of analysis

The notes should give you some ideas about the approach to take to the analysis.

Here, we suggest that the main issue is the need for WG to continue to develop successful new products which can be patented. We know that a 'best seller' will go out of patent next year, so that other companies will be able to make and sell the product, probably more cheaply. Issues such as the performance of R&D and competition will be included in the analysis.

A further issue to consider is the financial performance of the company and prospects for the future.

Another issue, although perhaps one of lesser importance, may be the risk of action by national governments to reduce the large profits of drugs companies.

Calculations

Detailed calculations can be produced for the relative profitability of Coffstop and Peffstill. The focus here should be on profitability for retailers, and identifying the key measure of performance. It is not profitability per unit sold. It is the gross profit per week per square centimetre of shelf space.

	Coffstop	**Peffstill**	
	£	**£**	
Recommended retail price	1.50	10.00	
Bought-in cost	0.50	7.40	
Gross contribution per bottle	1.00	2.60	= the 1: 2.6 ratio in the data
Holding cost per bottle	0.40	0.80	
Net contribution per bottle	0.60	1.80	
Sales per week (bottles)	120	20	
Total net contribution per week	£72	£36	
Shelf space per bottle (square centimetres)	18	60	
Contribution per square centimetre of shelf space	£4.00	£0.60	

Interpretation

The 'WHAT-HOW-WHY-WHEN-SO WHAT' approach to analysis can be used.

WHAT?	(1) Concern about future profitability: need for successful new products (2) Possible doubts about future growth in revenue, earnings and dividends (3) Possible risk of government action to reduce profits
HOW?	Patent of successful product running out Problem with selling Coffstop to retailers Fall in rate of revenue growth, but stronger earnings and dividend growth last year

	Government and public concern about high profit margins: governments are sponsors of R&D
WHY?	Analyse problems of product portfolio and failure of R&D to develop major new drug, in spite of high spending on R&D. Is this a key strategic problem for the company? Failure of company to explain higher profitability of Coffstop to retailers. Need to correct this failing, but why did it happen? In spite of high profit/sales ratio, it is not clear how earnings could increase by 15% when the rate of sale growth slowed to 5%. Is it possible that the company cut R&D spending? Or possibly gains from favourable currency movements? Some discussion of the apparently high rate of tax also appropriate. Governments may be concerned about high profitability of drugs companies. Legislation against patent unlikely. Governments may cut sponsorship of R&D or may ask for better terms in donation programmes. But at the moment, this is not the main problem for the company. Need to keep the matter under review.
WHEN?	Action to promote Coffstop is urgent, especially if it may become a major product for the company. An assessment of future financial prospects and the effectiveness of R&D are also urgently required.
SO WHAT?	The company needs a continual cycle of innovation, product development and successful patenting. Without this, its future financial stability and survival could come into question.

Limitations in the data

More information would be useful for analysis:

- A more detailed analysis of revenues and profitability in recent years, including an explanation of tax charges
- More information about the company's current product portfolio, sales and profitability of each product (historical and projected) and remaining patent lives
- A report from the head of R&D about development projects in hand and an analysis of historical performance – new products developed and product histories. (It is important to establish whether the R&D department is as effective as it should be.)

Conclusion

Your views on this example may differ, and you may prefer to approach an answer in a different way. However, this suggested solution tries to demonstrate the advantages of a structured approach to data analysis.

Answers to Self-test questions

1 Cumulus Limited

The major question is whether the current strategy of Cumulus will succeed, and whether the company will survive. If not, it should look for ways to either change its strategy or cancel the project entirely.

The company is targeting customers who will rent large quantities of space in its centre. The big growth in demand could be for co-location services in cloud computing.

The company hopes to attract customers by offering lower prices. London-based service providers can offer faster connectivity. This may be important for users with regular links to institutions in London's financial centre.

Reductions in the physical size of storage may raise questions about whether a large centre will ever be filled to capacity.

The slow uptake in demand for the services of the company suggests that sales revenue growth for Cumulus may be slow and insufficient for breakeven for quite a long time.

2 BTH plc

Ratios

Profitability ratios	**20X7**	**20X8**
ROCE		
$\frac{1,600}{6,700}$ $\frac{2,400}{5,000}$	23.9%	48%
Gross profit margin		
$\frac{2,600}{14,400}$ $\frac{4,400}{5,000}$	18.1%	25.9%
Operating profit margin		
$\frac{1,600}{14,400}$ $\frac{2,400}{17,000}$	11.1%	14.1%
Asset turnover		
$\frac{14,400}{6,700}$ $\frac{17,000}{5,000}$	2.1x	3.4x
Liquidity ratios	**20X7**	**20X8**
Trade receivables		
$\frac{2,000}{14,400} \times 365$ $\frac{1,600}{17,000} \times 365$	51 days	34 days

Liquidity ratios	**20X7**	**20X8**
Inventories		
$\frac{1,300}{11,800} \times 365$ $\quad \frac{2,000}{12,600} \times 365$	40 days	58 days
Trade payables		
$\frac{1,080}{11,800} \times 365$ $\quad \frac{2,700}{12,600} \times 365$	33 days	78 days
Cash operating cycle	58 days	14 days

Financial gearing	**20X7**	**20X8**
$\frac{2,600}{6,700}$ $\quad \frac{0}{5,000}$	38.8%	-

Analysis

Revenue has increased by 18.1% (£17m vs £14.4m). This is substantial when compared to inflation at 3%, suggesting growth in volume.

Non-current assets have also increased significantly (from £2.5m to £4.0m) suggesting BTH plc has either expanded or invested new equipment (which has supported the revenue increase). The revenue increase has also occurred alongside an increase in gross profit margin (from 18.1% to 25.9%), which again suggests the growth is due to investment rather than a strategy of lowering prices to increase sales volume.

Operating profit margin has increased less significantly than gross profit margin though, as 'other expenses' have doubled between 20X7 and 20X8. Despite the increase in revenue, this seems a significant increase. There may have been 'one-off' cost increases associated with the business expansion, but it will be important for BTH plc to maintain its control over operating costs in the future to help increase margins.

In addition to the revenue growth, BTH plc has redeemed its loan stock (£2.6 million) during the year. As the loan stock was classified in non-current liabilities at 31 December 20X7, this seems unexpected. However, as a result, financial gearing has decreased from 38.8% to nil.

Repaying the loan stock during the year could potentially have put pressure on liquidity, but BTH plc still has a positive cash balance at 31 December 20X8.

This appears to have been helped by the company's working capital management policies though, which have seen receivable days reduced by a third (from 51 to 34 days) while payable days have more than doubled (from 33 to 78 days). It is possible that the increase in trade payable days could reflect revised payment terms and BTH's increasing bargaining power over suppliers as it has increased in size. However, if the increase simply reflects slow payment on BTH's behalf, this could be damaging to the company in the longer term, because it could reduce the number of suppliers prepared to supply to it, and could damage the company's credit rating.

The increase in inventory days could be a cause for concern if it continues, because it reduces the efficiency of BTH's working capital management. However, if BTH has expanded significantly during the year, this could lead to some distortion in the working capital ratios

(because the sales and cost of sales figures for the year 20X7–20X8 won't accurately reflect the level of these figures going forward).

Overall, however, BTH appears to be performing well with increased revenues and profits, and it appears to be generating significant amounts of cash (eg, to redeem the loan stock, to increase non- current assets, while increasing its dividend payment (by 50%, from £520k to £780k) and still retaining a positive cash balance).

It will be important that the business doesn't focus on short-term profits and cash flow at the expense of a longer term, sustainable growth strategy. However, there is insufficient information available to assess whether or not it is doing that.

3 Flyway Airline

3.1 The following performance indicators could be used to analyse the three airlines:

	A	B	C	D
13		Flyway	Sudland Air	Eazee Air
14	Operating profit margin	630/5,430	54/7,350	127/2,170
15		11.6%	0.7%	5.9%
16	Operating profit margin RANK	1 [1]	3	2
17	Capacity utilisation (load factor)	79,619/100,654	82,554/105,974	40,973/46,934
18		79.1%	77.9%	87.3%
19	Load factor RANK	2 [2]	3	1
20	Revenue/staff member (£'000)	5,430 m / 32,501	7,350 m / 56,065	2,170 m / 5,372
21		167	131	404
22	Revenue/staff member RANK	2 [3]	3	1
23	Fuel cost/seat kilometre (£)	1,480 m / 100,654 m	1,823 m / 105,974 m	535 m / 46,934 m
24		0.015	0.017	0.011
25	Fuel cost/km RANK	2 [4]	3	1

[1] RANK(B15,$B$15:$D$15,0). The formula can be copied in C16:D16.

[2] RANK(B18,$B$18:$D$18,0). The formula can be copied in C19:D19.

[3] RANK(B21,$B$21:$D$21,0). The formula can be copied in C22:D22.

[4] RANK(B24,$B$24:$D$24,1). The formula can be copied in C25:D25.

Operating margin – Flyway has the highest operating margin of the three airlines (11.6%), which suggests it is being run efficiently overall. We might expect Flyway to achieve a relatively high margin because it appears to be pursuing a **differentiation strategy**. However, Sudland Air, which appears to be pursuing a similar strategy, generates an operating profit margin of less than 1%, despite being a larger company than Flyway.

Capacity utilisation (load factor) – By showing, on average, how full each airline's aircraft are, this indicator shows how well the airlines are using their asset base (ie, their aircraft). Flyway and Sudland's performance is similar in this respect, but Eazee's is significantly better. This is likely to be because Eazee (a low-cost airline) is pursuing a **cost leadership** strategy. Flyway might consider reducing its prices to try to improve capacity utilisation, but it needs to do so in the context of its overall strategy. If it reduces prices too much, it may end up compromising the quality and service it offers to passengers, but these elements are crucial to its strategy as a differentiator.

Revenue per staff member – This is an important measure in the context of the recent disputes over working conditions and pay. Flyway's staff appear to be performing better than Sudland's which, in turn, might strengthen their claims for a pay rise.

The comparison between Flyway's and Eazee Air's performance for this measure may be less meaningful. Eazee **outsources** many of its activities, meaning its staff numbers will be significantly lower than Flyway, which carries out the corresponding activities **in-house**.

Fuel costs – The board's interest in new fuel-efficient aircraft indicates that reducing fuel costs is an important concern for Flyway. Again, Eazee ranks first in terms of controlling its fuel costs with Flyway and Sudland ranking second and third respectively. This might be because it has more **fuel-efficient planes**, which would support the board's argument for Flyway investing in new aircraft. Alternatively, however, Eazee may have negotiated more favourable fuel contracts with its suppliers, or may be using lower grade fuel.

Tutorial Note

It is important to use fuel cost per **seat** kilometre as the performance indicator here rather than fuel cost per **passenger** kilometre, because we are looking to monitor the fuel efficiency of the aircraft, rather than the airline's ability to fill its aircraft with passengers.

3.2 Flyway's business strategy, as a premium airline, coupled with the difficult trading conditions, mean that it is increasingly important for Flyway to provide its customers with the best services and experiences possible, so that they choose to fly with it in preference to another carrier. Increasing the data it holds about customers, and potential customers should help Flyway's management make better decisions, and thereby should help the company achieve this.

Detecting key trends – Analysing conversations on social media could help Flyway identify potential trends in customer demand; for example, if there are major events taking place in a particular destination, or if certain resorts are increasing (or decreasing) in popularity as holiday destinations. Being able to forecast demand more accurately – and almost in 'real time'

- could help Flyway boost revenue, through applying **dynamic pricing**. For example, Flyway could keep prices high on flights which are going to be popular, but could reduce prices on flights which look like they are going to have a lower capacity utilisation in order to try to boost demand for those flights.

Customer selection process – One of the key issues here is for airlines to understand the reason why potential customers have not completed their transaction – for example, this could be due to price, seat availability, difficulties in the booking process itself. In this respect, if Flyway was able to capture data about the stage in the booking process which causes potential customers to abandon their booking, it could then look at ways to tackle the problem

- for example, if there was a confusing user interface on Flyway's website this could be amended to make booking easier. Equally, applying the 'velocity' aspect of Big data, if a customer abandons a transaction, and Flyway already has their contact details, the customer could immediately be sent an incentive to try to encourage them to complete the purchase.

In-flight sales – Flyway currently offers a standard selection of in-flight products across all its flights. However, the amount of products it sells (and therefore the amount of in-flight revenue

it generates) is likely to depend on how well the products offered meet customer needs. By analysing the purchasing patterns of different passengers (or different types of passengers) on different routes, Flyway could customise the range of products it offers on different flights, to focus on the products which are most likely to appeal to the passengers on a particular flight.

Customer satisfaction – In the same way that conversations on social media could help to identify trends in demand, they could also help to indicate how satisfied customers are (or aren't) with their flights and the service they receive from Flyway. However, although knowing whether customers are satisfied or not is useful, perhaps the greater value will come from identifying any factors which are reducing satisfaction levels (eg, comfort of the planes, quality of customer service received) so that Flyway can then address the causes of any problems and take steps to improve its performance in those areas.

4 Malvo Ltd

Tutorial Note

The solution below is only a suggested solution showing some of the points you could have made. You may have thought of others – and (in an exam situation) you would receive credit forvalid points you made, even if they are not included in the suggested solution.

Quality of information

In general terms, the dashboard data has two key benefits over the information previously available to Malvo's management:

- It is available in real time, so managers can identify issues as they arise, rather than having to wait until the end of the week for the weekly summary to be available.
- The granularity of the data means that managers can identify, and respond to, specific issues rather than general trends.

Overdue items

Being able to review data in real time could be particularly useful in relation to monitoring overdue items, and for identifying potential problems with them.

Although having visibility over the number of overdue items is important, potentially the useful information for managers could be the reasons for any delays – for example, whether there are problems in the warehouse which are delaying shipment, or whether there have been problems in getting products from suppliers. If managers can identify the reasons for items becoming overdue, this should improve their ability to take the actions necessary to keep shipping operations running as smoothly as possible.

In this respect, the ability to track the real-time status of all orders could be particularly useful. If Malvo filtered the orders to isolate all the delayed orders, it could then assess any common factors – for example, if there is a problem with a particular product, or with products from a particular supplier.

Late shipments

The dashboard identifies the number of late shipments, but it could be useful for the managers to drill down further – for example, to see which suppliers are affected, and what may have caused the problems (for example, bad weather delaying transport). If shipments from a specific supplier are repeatedly late, and therefore Malvo cannot rely on that supplier, it could be necessary to look for an alternative supplier.

Back orders

The percentage of items which are back ordered is an important measure of the effectiveness of the company's inventory management, and will also influence customer satisfaction levels.

Although it is almost inevitable that there will be some products which customers have to wait for, if customers regularly have to wait – or have to wait for extended periods of time – it is likely that they will look to purchase from one of Malvo's competitors instead. In this respect, as well as monitoring the back order rate it could also be useful to track the number of days which customers have to wait for back orders.

Perfect order rate

The 'perfect order' rate will also influence customer satisfaction levels, so Malvo should aim to keep this figure as high as possible. The current month's figure (94.2%) means that about 1 in 20 orders (5%) are not completed perfectly (ie, on time, and in full).

Once again, though, while having visibility over the figure is useful, potentially the more valuable insights to managers would come from identifying any factors which cause orders to be imperfect – for example, if there are particular products which are repeatedly out of stock.

Out of stock items

The significance of 'out of stock' items is that they potentially result in lost sales. If a customer tries to purchase an item, but that item is out of stock, the customer might look to purchase it from one of Malvo's competitors instead. (Malvo's target is to keep out of stock items below 1%, but there was a noticeable peak in October at around 1.4%, suggesting Malvo may have suffered supply chain problems then.)

This highlights a more general aim for Malvo's managers here, to use management information (and analytics) to try to balance supply and demand as effectively as possible; because this should help to minimise the risk of stock-outs, and also avoid having excess stocks which it is difficult to sell.

Having detailed information about stock levels and order levels for individual items could be very useful here in tracking imbalances between supply and demand – although having this information only becomes useful if it leads to actions to remove any imbalances.

Top 3 stock items

Although Malvo has a high level of fridges in stock, this appears consistent with the level of orders. However, the amount of televisions and radios in stock appears high for the level of orders.

This might suggest that Malvo needs to reduce the price of these items or introduce some kind of special offer (to increase demand), or that it should reduce the amount of these items it buys in future. However, there may be specific circumstances which have led to the current imbalance (for example, if Malvo has recently made bulk orders of the items to get a trade discount from their supplier).

Top 3 orders

Fridges (stock code 212016) are the second most popular product by customer orders, as well as being the product which Malvo has the third highest stock of. As noted above, this suggests that Malvo has correctly anticipated the level of demand for fridges, and planned its stock levels accordingly*.

However, there appears to be a danger of stock-outs for fans and tablet computers because the quantity of orders is greater than the amount in stock. An important consideration here is whether there have been problems in Malvo's supply chain which have meant products have been delayed, and Malvo has now less stock than it planned to have; or whether the imbalance is due to unexpectedly high demand. For example, demand for fans might have been increased by a period of hot weather.

*: An alternative explanation could be that, in response to high levels of stock, Malvo has reduced the price of its fridges to boost demand.

Chapter 9
Information strategy

Introduction

Learning outcomes

Knowledge brought forward

Examination context and syllabus links

Chapter study guidance

Learning topics

1 Information technology and strategy

2 Information for strategic planning and control

3 Management information systems (MISs)

4 The value of information

5 Evaluating management information and performance data

6 Cybersecurity

7 Using information to develop competitive advantage

8 Using digital skills to develop competitive advantage

Summary

Further question practice

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Outline proposals and advise on outline requirements for information technology applications to support business strategy, for example in the context of e-commerce, e-business, virtual arrangements, artificial intelligence, digital assets and cloud computing, including assurance issues in relation to data security
- Use management accounting information (for example, costs, prices, budgets, transfer prices) and management accounting tools (for example, break-even, variances, limiting factors, expected values, ABC, balanced scorecard) to evaluate short- and long-term aspects of strategy
- Explain and appraise how management information systems can provide relevant quantitative and qualitative data to analyse markets, industry and performance, including the capture and analysis of big data
- Demonstrate and explain methods for determining the value of information in the context of developing an information strategy
- Assess financial and operational data, and information from management information systems, drawing inferences relating to its completeness, accuracy and credibility, and provide an evaluation of assurance procedures in evaluating information risks, including those relating to cybersecurity
- Demonstrate and explain how businesses capture, analyse and utilise information to develop competitive advantage
- Evaluate the impact of cloud computing and the borderless business on the provision of strategic management information, including the use of cryptocurrencies
- Explain and appraise corporate strategies for ensuring security of data and prevention of attacks against data in the context of cybersecurity
- Explain the impact of regulations relating to data security (eg, GDPR) including the issues incurred in complying with such regulation

Knowledge brought forward

The Business Strategy & Technology syllabus at Professional Level looked at the way management information and information systems support an organisation's overall business strategy and competitive advantage. It also highlighted that organisations need information for a range of purposes, including planning, control, performance measurement and decision making.

In the chapter Strategic performance management of this Workbook, we highlighted the importance of having relevant and reliable performance information in order to manage performance. In this chapter, we will now consider the information systems which could provide that information, and how the resulting information can be used by staff and managers in organisations.

Examination context and syllabus links

It is unlikely that a question will focus solely on information systems in their own right, although information systems and e-business may play a key role in the successful implementation of a strategy.

However, while information **systems** themselves are important to an organisation, the **information** which they provide is perhaps even more important. Managers need information for decision making and control. For example, in order to measure performance against targets or key performance indicators (KPIs) (see the chapter Strategic performance management) managers need to have appropriate information about the areas of performance under review.

In this context, you might be expected to comment on the information available for decision making or control in a scenario – is there enough information available? Is the information at the correct level (eg, strategic, tactical or operational)? What other information is needed? Where might the business obtain relevant information? Is the information system adequate to fulfil the functions required of it by the business? How can the organisation use information to generate competitive advantage?

An underlying consideration for information strategy is that an organisation's information systems should provide the appropriate type and amount of information which management need to select, implement and control its chosen business strategy. This also means that the information strategy needs to be aligned to the business strategy, in terms of the type of information available. (For example, if an organisation is pursuing a differentiation strategy based on the high quality of its product then information about aspects of product quality will be very important to it.)

In addition, the level of detail, the form of the information and its timing should be appropriate to the role of the person(s) who receives it.

Make sure you do not overlook the importance of IT/IS to strategy. Very often an organisation's ability to deliver a strategy may depend on having sufficient IT capabilities to do so. Equally, an organisation's IT capabilities may be instrumental in shaping its strategy. For example, does it have sufficient IT infrastructure and capabilities to support a digital strategy (as discussed in the chapter Strategic choice)?

Remember, in the chapter Strategic analysis we noted how an organisation's resources and capabilities contribute to its competitive advantage. In this context, it is worth noting that the way an organisation manages and uses information could, in itself, become a source of competitive advantage – for example, if the organisation is able to respond to market trends or opportunities more quickly than its rivals on the basis of the information it gathered about those opportunities. (The way an organisation makes use of big data, the internet of things, and machine learning could all contribute to its capability in this respect as well.) Equally, gathering data and information about customers and customer requirements could also be useful in increasing the value an organisation's customers place on its products and services, as we noted in the context of customer relationship management (CRM) (covered in the chapter Strategic marketing and brand management of this Workbook).

Conversely, problems relating to an organisation's IT systems could have a significant impact on its performance, and potentially also on its reputation. This could be particularly true in relation to cybersecurity. We have looked at cybersecurity in the context of the supply chain in the chapter Strategic implementation, but we will look at cybersecurity, and the ways organisations can ensure the security of their data, again more generally in this chapter.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Information technology and strategy** Strategic information is used to plan the objectives of an organisation and to assess whether they are being met. It is therefore important that an organisation has the necessary information systems in place to support its information requirements.	**Approach** This section looks at IS and IT from a strategic perspective; for example, how could they shape corporate strategy or deliver competitive advantage. It is very important to understand how the emergence of digital technologies (digital transformation) - such as the growth of cloud computing- is re-shaping business models. **Stop and think** What impact has the Covid-19 pandemic had on business reliance on cloud computing?	Remember that SBM&L is a business management exam, not primarily an IT exam. You do not need to know the details of a range of different information systems; but rather you need to appreciate how systems and information could be used to measure and manage performance in an entity and how they could support business strategy. You may therefore be required to advise management about the use of information technology applications (including artificial intelligence, digital assets and cloud computing) to support an organisation's business strategy.	**IQ1: Antiques dealer** This question requires application of the value chain to an antiques retailer to assess the strengths and weaknesses of the current business model. You are then asked to evaluate the impact on the value chain of introducing e-commerce. The question demonstrates the breadth of knowledge that could be tested in one question.
2	**Information for strategic planning and control** Managers need information to make effective decisions and to control and coordinate the activities of anorganisation.	**Approach** This section looks at the importance of information for planning and control. Much of the section (costs, prices, variances etc) should be revision, in which case you can skim through it relatively	In the exam you should use calculations, such as those seen in the interactive questions, to support your discussion. Questions may expect you to use a variety of management	**IQ2: Absorption and marginal costing** Use this question to re-cap how to calculate profit using absorption and marginal costing. This topic is revision from the Management Information (MI)

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	Much of the data required will be internally generated financial information, for example costs, prices and variances. However externally generated information, such as customer demand levels, can influence management decisions, for example decisions concerning pricing and inventory requirements.	quickly. **Stop and think** Why are budgets useful for an organisation? What are the potential problems associated with budgeting?	accounting tools to evaluate aspects of an organisation's strategy or its performance.	paper. **IQ3: Activity based costing** This question is also revision from the MI paper. Determine the most appropriate cost drivers and calculate total product cost for four different products. **IQ4: Breakeven analysis** Breakeven analysis is a useful calculation for the exam since it allows you to quickly establish whether a company should pursue a new product/service strategy. Practise the technique using this question. **IQ5: Multi-product breakeven analysis** Work through this question making sure that you know how to calculate the breakeven point and breakeven revenue for multiple products.
3	**Management information systems (MISs)** As discussed in previous sections, information is critical for an	**Approach** This section examines the different types of management information systems an entity	The content of this section is unlikely to be tested in isolation; you may need to form a judgement on the quality of data and	**IQ6: Health Club** This question encourages you to consider the main purpose of a new MIS and the benefits it could

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	entity's decision making and control and can even be a source of competitive advantage. To capture information at the strategic, tactical and operational levels of the organisation various Management Information Systems are required.	could use. However, remember your focus should be on the information which the systems can provide for management, rather than on the technical details of the systems themselves. **Stop and think** Consider a company with which you are familiar. What do you think their CSFs are? What are their information requirements as a result of these CSFs?	information produced by an organisation's management information systems, as well as the possible risks associated with using it.	bring to the company.
4	**The value of information** Information is recognised as a valuable resource in a company's quest to achieve competitive advantage. However, good information is not free so organisations must understand the trade-off between the cost of information and the value it delivers.	**Approach** This is a short and relatively self-contained section. You can read through it relatively quickly, make sure you understand the value and the cost of information. **Stop and think** How might good information result in competitive advantage?	In the exam you may need to assess the value which an organisation could gain from having better information and information systems, it is important to adopt a practical and commercial approach. Remember that information comes at a cost to the business so this should be factored into any discussion about a company's information needs.	
5	**Evaluating management information and performance data**	**Approach** The SBM&L syllabus refers	An exam question may require you to assess the quality of an entity's	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	The objective of management accounting and management accounting systems is to provide information for managers to use for planning, control and performance measurement. However, information is only useful to managers if it adds to their understanding of a situation.	specifically to issues surrounding the completeness, accuracy and credibility of data and information, so this topic is important. Assurance over information and information systems is another important area, and one which is becoming increasingly important in the context of cybersecurity (discussed in a later section). **Stop and think** What are the impacts on a business of using poor quality information to make decisions?	information or to suggest improvements or alternative data sources. You may also be asked to outline assurance procedures around the reliability and adequacy of an entity's IT systems.	
6	**Cybersecurity** A key challenge of new technologies is the increased cybersecurity risk they pose. As businesses continue to adopt digital technologies, the protection of systems, networks and data in cyberspace is becoming increasingly important. The importance of data security is increased further	**Approach** We have already discussed cybersecurity in the context of supply chains in the chapter Strategic Implementation, but in this chapter we look in more detail at the measures organisations can take to protect themselves against cyber-risks. When thinking about cybersecurity, a useful approach	As cybersecurity issues become more prevalent in the real-world this is likely to be reflected in the SBM&L exam. You may be required to assess the threats to an organisation's information system (including cyberthreats) and the organisation's subsequent response to them. You may also be asked to explain the impact of	**IQ7: Cybersecurity** In this question you are required to explain the issues that the board should consider in relation to cybersecurity and to make recommendations regarding any actions it should take. Your response should be practical and applicable to the business in the scenario.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	by legislation such as the General Data Protection Regulation (GDPR), relating to the use and processing of personal data.	might be to consider three broad aspects: What are the causes of cyber-risks? What are the consequences of cyberattack? What can organisations do to protect against the risks (and how effectively are they doing this)? By keeping up to date with business news you will see real examples of cybersecurity breaches and responses. **Stop and think** There have been several well-publicised cybersecurity breaches. What caused them? What actions did the companies take?	regulations relating to data security, and the implications for businesses in complying with them.	
7	**Using information to develop competitive advantage** The information an entity holds (particularly about customer requirements and its supply chain) could play an important role in helping that entity develop competitive advantage over its rivals.	**Approach** This is another important section because of the way information links to other elements of the syllabus we have already covered; for example, CRM, supply chain management, and big data. **Stop and think** To improve strategic decision making,	In the exam you may be required to advise both profit seeking and not-for- profit entities how to strengthen their position relative to competitors. You may therefore be asked to explain how capturing, analysing and using information can help an organisation to develop competitive	**IQ8: Data and knowledge** This is a scenario-based question dealing with a range of topics. You are required to explain the advantages of integrating CSFs into performance management systems using KPIs, advise on the benefits of a knowledge management

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	Information technology has had a strategic impact on organisations' business strategies - in particular through the development of e-business and e-commerce.	organisations should increase the amount and quality of data available to them. However, what problems could this also cause for an entity?	advantage.	strategy and explain how data mining can be used. Your responses must be applied to the scenario rather than simply listing technical knowledge points.

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Information technology and strategy



Section overview

- Strategic information is used to plan the objectives of an organisation, and to assess whether those objectives are subsequently met. Therefore it is important that an organisation has an information systems strategy so that it can meet its information requirements.
- Organisations often have to consider three different strategies in relation to information: information systems (IS) strategy, information technology (IT) strategy and information management (IM) strategy.
- Not only have developments in IT facilitated the introduction of a range of management information systems, but they have also had a significant impact on corporate strategy itself (for example, by transforming the value chain).

1.1 Information and strategy

Strategic information is used to plan the objectives of the organisation, and to assess whether the objectives are being met in practice. Therefore it is important that organisations have an information systems strategy so that it can meet its information requirements.

As we have mentioned in the chapter Strategic performance management, organisations have to consider three different aspects of their information strategies: information systems (IS) strategy, information technology (IT) strategy and information management (IM) strategy.

We can summarise these three strategies in very simple terms by saying that IS strategy defines **what** is to be achieved; IT strategy determines **how** hardware, software and telecommunications can achieve it; and the IM strategy describes **who** controls and uses the technology provided.

However, the relationship between IS, IT and IM is important. In order for a company's IS strategy to be successful, the company will need sufficient IT resources (including hardware, software, network resources and suitably skilled staff) in order to implement and support the strategy.

A company does not necessarily have to own resources in-house, though; an alternative would be to outsource IT services and departments to a specialist IT company. However, before making such a decision, an organisation needs to consider how critical its IT services are to its overall operations. If IT services are critical to the organisation's operations, and to its competitive advantage, the organisation should try to keep its IT services in-house rather than outsourcing them.

1.2 IS, IT and strategy

An organisation's information systems may not only **support** business strategy; they may also help

determine corporate/business strategy. In particular:

(a) IS/IT/IM may provide a possible source of **competitive advantage**. This could involve new technology not yet available to others or simply using existing technology in a different way.

(b) Information systems may help in formulating business strategy by **providing information** from internal and external sources.

(c) Developments in IT may provide **new channels** for distributing and collecting information and/or for conducting transactions. The most fundamental illustration of this has been the way the internet has opened up opportunities for e-business and e-commerce.

When considering IS/IT in a strategic context it could be useful to consider how they contribute to an organisation's current strategic position. One way of doing this could be through a SWOT analysis.

For example:

Strength: A business might have a sophisticated customer database which allows it to send out targeted marketing messages to customers. A retailer might have an inventory management system which automatically reorders stock lines according to sales being recorded in the shop tills, thereby allowing it to minimise the levels of inventory it needs to hold.

Weakness: A business's ordering system (either online or manual) is unreliable, and so customers cannot be confident that the goods they order will be delivered correctly or within an acceptable timescale.

Opportunity: If a business does not currently have a website which allows customers to purchase items online, the opportunity to develop such a website could provide a significant boost to its sales.

Threat: Conversely, if a competitor has recently upgraded their IT systems with the result that they can now offer a greater range of services to their customers and provide them with improved levels of service, this could pose a threat to an organisation because customers may switch to using the competitor instead.



Context example: British Airways

Although the main focus of this chapter is on the way that IT and information could enhance competitive advantage, it is also important to remember that, in today's businesses, IT systems are crucial to maintaining everyday operations. If the systems fail, then businesses may not be able to operate at all.

This was demonstrated in May 2017 when British Airways (BA) had to cancel all flights from London Heathrow and Gatwick on the Saturday of a busy, bank holiday weekend, due to a major IT systems failure. In total, about 800 flights – to worldwide locations – were cancelled, stranding approximately 75,000 passengers. The problems are expected to cost the airline about £58million.

The root cause of the problem was a power supply issue at a data centre near Heathrow, but as BA's Chief Executive admitted, the crash affected 'all of our check-in and operational systems'. In addition, it appears that BA's back-up and disaster recovery systems also failed to step in during the incident.

Commentators noted that the problem highlighted just how interconnected the airline's systems are, and as a result, how susceptible airlines' operations can be to any one element failing (for example, airlines rely on IT systems for operating flights, ticketing, boarding, and for their websites and smartphone apps).

Unions claimed that the problem was a result of staff cuts and BA's decision to outsource its IT requirements to India in an attempt to save money. However, BA's chief executive denied those claims. He also said there was no evidence that the system failure was the result of a cyberattack, and he emphasised that passenger data had not been compromised.

BA promised to compensate all customers affected by what it called the 'catastrophic' failure.
Sources:

Batchelor, T. (2017) 'BA flights grounded: Apologetic CEO Alex Cruz denies catastrophic computer failure was caused by job cuts', *The Independent*, 29 May 2017, www.independent.co.uk

Cox, J. (2017) 'British Airways' IT meltdown over May bank holiday cost the airline around £58m', *The Independent*, 28 July 2017, www.independent.co.uk

1.2.1 Developing an IT strategy

When formulating an overall IT strategy, the following aspects should be taken into consideration:

(a) What are the key business areas which could benefit most from an investment in information technology, what form should the investment take, and how could such strategically important units be encouraged to use such technology effectively?

(b) How much would the system cost in terms of software; hardware; management commitment and time; education and training; conversion; documentation; operational manning; and maintenance? The importance of lifetime application costs must be stressed – the costs and benefits after implementation may be more significant than the more obvious initial costs of installing an IT function.

(c) What criteria for performance should be set for IT systems? Two areas can be considered: the technical standard the information system achieves and the degree to which it meets the perceived and often changing needs of the user.

(d) What are the implications for the existing workforce – has it the requisite skills to use the new systems, can it be trained to use the systems, and will there be any redundancies?

1.2.2 IT and its effect on management information

The use of IT has permitted the design of a range of information systems. Executive information systems (EISs), management information systems (MISs), decision support systems (DSSs), knowledge work systems and office automation systems can be used to improve the quality of management information.

IT has also had an effect on **production processes**. For example, computer integrated manufacturing changed the methods and cost profiles of many manufacturing processes. The techniques used to **measure and record costs** have also adapted to the use of IT.

1.3 How IT is changing corporate strategy

It should be obvious that IS and IT should **support** corporate strategy, but there are also a number of ways IS/IT can **influence** corporate strategy.

In 1985, the *Harvard Business Review* published an article by Michael Porter and Victor Millar aimed at general managers facing the changes resulting from the rapid and extensive development of IT. Although it was written a number of years ago, the article is still relevant to the **strategic employment of IS** and the use of IT. It dealt with three main interlinked topics:

- the ways in which IT had become **strategically significant**
- how the **nature of competition** had changed
- **how to compete** in the new, IT-influenced environment

1.3.1 The strategic significance of IT

IT transforms the **value chain**.

Porter and Millar's article remarks that each of the value chain activities has both **physical** and **informational** aspects. While historically technical advances concentrated on physical aspects, more recently improvements in information management and data management have become increasingly significant (as for example the references to big data, internet of things, AI and machine learning in the chapters Strategic analysis and Strategic choice of this text identify).

Simple improvements are made by faster and more accurate processing of existing forms of data, and more dramatic ones by creating new flows of previously unavailable information. This has a particular effect on the linkages between the various activities and extends the company's **competitive scope**, which is the range of activities it can efficiently undertake.

Slack *et al* in their text, *Operations Management*, also highlight that e-business more generally has had an impact in many areas of operations management:

- **Design** – customer feedback on requirements, testing and information exchange on new products/services, data mining to understand consumer behaviour better
- **Purchasing** – ordering (through electronic data interchange (EDI)), funds transfer, supplier selection and supplier portals
- **Production** – production planning and control, scheduling, inventory management, quality monitoring and control, enterprise resource planning (ERP)
- **Marketing/sales and customer servicing** – new sales and distribution channels, online help desks, reduced cycle time, third-party logistics, scheduling systems

However, as well as thinking specifically how IT can affect the value chain, you should be prepared to consider how IT and e-business have affected business more generally.

For example:

The use of computer-aided design can lead to the faster production of new products and designs. Organisations could either use this speed as a basis for making designs cheaper (cost leadership) or, for example, in the clothing and fashion industry, as a means of getting the latest fashions to market more quickly than their rivals (differentiation).

Websites and email have changed the nature of communication between organisations and customers.

The internet has also changed the nature of the supply chain and channel structure – for example, by allowing customers to book flights and hotel rooms for their holidays directly from the airline company and the hotel online, rather than having to use travel agents; by allowing customers to buy a vast range of products through a single retail platform (eg, Amazon) rather than having to buy different products from different retailers according to the type of product.



Interactive question 1: Antiques dealer

GLS is a long-established retailer which specialises in the sale of antiques. GLS is owned by a married couple who both work in the business. They have no employees. Their premises consist of a large modern shop and there is an apartment above this in which the owners live. Over the last five years the local area has become very fashionable and the shop is now surrounded by smart restaurants, cafes and upmarket fashion outlets. This area has also become a very popular place to live which has meant that property values have increased substantially. The owners believe that if they disposed of their premises they would make a substantial capital gain. The owners have noticed that the fixed costs of their property, including insurance, local tax, security and maintenance, have risen very sharply during the last five years.

Since establishing the business 30 years ago the owners have developed their expertise. They now have a national reputation in the antiques trade and many repeat customers. They traded profitably each year from the start of the business until two years ago, but in the last year have made an operating loss for the first time. The owners are often consulted by other antiques traders and collectors by letter and telephone and they have developed a considerable income stream by charging for their advice. However, they have found that their business location is becoming increasingly problematic. Although the popularity of their area of town has increased and led to many more people living and visiting the area, unfortunately for the

owners most of these people are not interested in antiques. They are young people who like the area but do not have the disposable income to spend on antiques.

A further problem is that the shop is not situated in a large city and it is very inconvenient for many antiques traders and collectors to visit. The owners believe the location has recently restricted the success of their business. The owners know that a very popular development in the antiques trade has been the establishment of 'Antiques Fairs' where antiques are bought and sold. Some of these have established international reputations and have many thousands of visitors. However, because of GLS's location and the need to keep their shop open, the owners do not attend these. The owners recently set up a website which has basic information about their business on it, such as their address, telephone number and the opening times of their shop. The website has received a large number of hits but it does not seem to have increased sales.

Requirements

1.1 Analyse the strengths and weaknesses of GLS's current business using the value chain model.

Note: You are not required to draw a value chain diagram in any part of your answer to this question.

1.2 Evaluate how the introduction of e-commerce could affect GLS's value chain.

See **Answer** at the end of this chapter.

1.3.2 IT and competitive strategy

IT enhances competitive advantage in two principal ways:

- by **reducing costs**
- by making it easier to **differentiate products**

One example of IT-driven cost reduction is the way service industries (eg, shops, airlines) have introduced self-service tills or check-in facilities in place of 'staffed' facilities.

However, although IT can be used to reduce costs, it is perhaps debatable whether this generates a long-term competitive advantage. For example, businesses increasingly use virtual conferencing as a means of cutting costs and imposing operational efficiency due to the ease with which data can be shared. However, if all the firms in an industry start using virtual conferencing, will this actually generate any competitive advantage for any individual firms in that industry? Similarly, with the example of self-service check-in facilities, most, if not all, airlines now offer this facility and so it is no longer a source of competitive advantage.

Differentiation. One way an organisation might seek to differentiate itself from its competitors is by meeting customers' needs and requirements more closely than their competitors. The greeting card company Moonpig has adopted such an approach by allowing customers to design their own cards online.

IT could also enhance competitive advantage by forming the basis of completely **new businesses**. It makes new businesses technically feasible, it creates derived demand for new products and it creates new businesses inside old ones. The impact of Apple's iPod illustrates all three of these effects. The device itself is based on the MP3 file format, a large iPod ecosystem of accessories has been created, and the product represented a departure from Apple's previous hardware and software strategies.

We looked at the impact digital technologies can have on strategies and business models in the chapter Strategic choice, but will revisit some of the key issues around digital transformation here.

1.4 Digital transformation

The potential impact of IT on businesses and industries has become more topical in recent years, in relation to the concepts of digital disruption and digital transformation, which we introduced in chapter Strategic choice earlier.

Digital disruption relates to the impact that new digital technologies (such as broadband and wireless communications, social media, cloud computing) have on 'business as usual', and **digital transformation** refers to the way those impacts can be proactively managed.

In general terms, business transformation is about developing new business models and could include, amongst other things, new products or services, new production processes, new delivery channels, approaching new markets, working collaboratively with suppliers.

By contrast, the primary focus of digital transformation is understanding the existing information and data flows in a business, and then reshaping the business processes and systems to maximise the use of digital technologies.

1.4.1 Technological factors enabling change

The ICAEW publication *Digital transformation – the next steps: A business guide to digital change management* highlights that digital transformation has only been possible due to the emergence of a number of underlying, enabling technologies. These include:

- ever-increasing processing power
- ever-increasing data storage
- improved operating systems and networking infrastructure
- the internet as a platform
- the increasing reach and bandwidth of broadband
- the emergence of open source software
- miniaturisation and the rise of mobile

Coupled with the falling costs of technology, these developments have helped to generate an environment in which it is increasingly easy for entrepreneurs to explore new applications of technology and to challenge existing business applications and business models.

1.4.2 Benefits of digital transformation

A successful digital transformation can help a business not only to improve its existing business model (eg, through process improvements, reducing costs and achieving closer relationships with customers) but also to create new opportunities (new markets, new revenue streams, new supplier relationships).

Amazon and eBay are two high profile examples of how a traditional business model can be transformed by a new company using disruptive technologies. Both companies have harnessed technological innovation – faster processing, better networking, increased bandwidth, cheaper technology and diverse devices – to enhance their customer offering, improve their back office systems (eg, CRM, logistics, accounting) in order to expand into market leading companies.

However, a word of caution is required. Digital transformation will not mean that all technology start- ups will be successful. Ultimately – as with any markets or business models – many will still fail, while others will manage to keep going but doing so will be a struggle.

Importantly though, digital transformation is not solely about cost reduction. The greater areas of benefit arise from the way 'digital' can add value – through increased customer engagement and retention, personalisation of products and services, integration of systems and collaboration with suppliers to speed up fulfilment, and improved quality control processes.

Context example: Uber

The ICAEW publication *Digital transformation – the next steps* includes a number of the ways digitalisation has led to new business models, across a range of industries. One of the examples featured is that of the taxi service, Uber.

Uber's growth and success provides an example of how traditional business models (ie, taxi services) can be challenged by innovative new approaches.

Uber began in 2009 as a luxury car service in San Francisco, but by the end of 2018 it operated in more than 700 cities worldwide, and reportedly had more than 90 million users a month. At the time of its stock market listing in May 2019, it was valued at $82.4 billion, making it one of the largest initial public offerings (IPOs) of all time.

In order to compete successfully against existing taxi services, Uber analysed the customer experience and used digital technologies to overcome the negative aspects of hiring a taxi.

Problems in the existing business model:

- Getting a taxi is not always an easy experience, as it involves trying to find and hail a taxi, or having to wait after phoning to book one.
- Once in the taxi, the customer experience can be poor – disagreements over the route, whether the meter was started, agreeing a price, getting a receipt etc.

Uber's solution:

- Uber changed the private transportation market by creating a digitally enabled personal transport service.
- Marketing through local networks and social media.
- Use of mobile technologies and Google to improve customer convenience and to enable a better customer experience.

Digital transformation:

- The Uber smartphone app is integrated with Google Maps so that customers can identify how close they are to the nearest cars, set a meeting point on the screen, and hail a car.
- Driver's information (including customer satisfaction ratings) is provided in real time.
- Uber drivers call or text customers to confirm they are on the way to collect them. The app charges the customer's bank card on arrival at the destination, so cash, change, tips or receipts are not needed.

Source: ICAEW (2016) *Digital transformation – the next steps: A business guide to digital change management*

However, despite the potential benefits of digital transformation, it is important to remember that not all companies get it right, and some fail to recognise the need for digital transformation at all.
Remember the example we mentioned, in the chapter Strategic choice, of Blockbuster which failed to recognise the importance of digital transformation, and subsequently failed as a company.

Given the increasing importance of digital technologies, an important consideration for organisations will be to evaluate their current position, in order to determine the priority areas for them to change.

Some of the key areas to consider are:

Knowledge management	How well is knowledge managed in the business? Is there an intranet? Can information (eg, information about business performance) be shared easily?
Efficiency and productivity	Does the business use ERP or manufacturing resource planning systems? Does the business use cloud technology to be more efficient, flexible and mobile?
Payment and finance	Can the business accept telephone, mobile and online payments? Does the business use online banking? Do managers have access to accounts and finance at anytime, anywhere? Is the business' e-commerce system integrated with back office functions (eg, finance; inventory; delivery)?
Customers and marketing	Does the company's website engage customers and add value? Does the website work on tablets and mobile devices? Does the business deliver products/services online? Does the business have a search engine optimised website? Do you regularly evaluate competitors' products, websites and social media? Do you have a CRM system? Do you regularly send out e-communications to customers? Does the business offer personalised products or services? How responsive do customers think the business is? Does the business use social media? Does it have a good online interaction with its customers? Does the business check all online channels for customer feedback and any issues which could affect brand reputation? Does the business use information gathered from the website and social media to identify potential new ways of working or potential new products/services?
IT systems	How does the business manage its computer network? Has it implemented cloud technologies for its infrastructure? Does the business use technology to collaborate with other businesses (eg, sharing information with business partners)?
Governance	Does the business have an IT policy (covering use of hardware and software, data protection, and internet usage)? Does the business have a disaster recovery plan? (Is the plan regularly updated?) Are the business' systems secure (ie, protected from viruses/malware; hackers)? Does the business have measures in place to enable staff to use their own devices securely when accessing organisational data?

Summarised from: ICAEW (2016) *Digital transformation – the next steps: A business guide to digital change management*

1.5 IT networks

'Information technology' includes any devices which collect, manipulate, store or distribute information.

Networking IT

In an IT context, networking means linking two or more devices together in some way so that data can be shared among them.

Networks allow decentralised computers and devices to communicate with each other. This enables databases and applications software to be shared, and provides flexibility (for example, by allowing users to access information from different places).

For example, home workers can connect up to an organisation's systems using **virtual private network** links which treat the home workers as if they were on site.

However, there are also constraints and risks around IT networks. The costs of installing wired networks can be very high, while there may be security issues with wireless networks.

The increased importance of hardware, software and networks in supporting companies' strategies also highlights the importance of having proper controls over them – for example, to safeguard the integrity of corporate networks and databases.

1.5.1 Technology and data

When considering the IT architecture in a company, it is important to distinguish between the following components:

Technology platform – the internet, intranets, extranets and other computer systems and software which provide a platform which supports the strategic use of IT for e-business and e-commerce

Data resources – databases which store data and information for business processes and decision support

When considering whether a company's IT applications can support its business strategy, it is important to consider not only the technology platform but also its data resources.

Nonetheless, developments in IT have had a significant impact on the operations, costs, work environments and competitive positions of many companies.

Enterprise software could be particularly useful to organisations in a business management context in relation to **enterprise resource planning**, **supply chain management** and **CRM. Business intelligence** software (eg, data mining, data analytics) could also be particularly useful for providing management with information.

Evaluating enterprise software

The increased importance of IT and software to business operations means that selecting the right systems and software is becoming increasingly important for organisations.

Aspects which an organisation should consider when selecting IT systems/software include:

- **Reliability** – assurance about the integrity and consistency of the application and all its transactions
- **Interoperability** – the system's ability to interface and share data with other systems (including external systems)
- **Leveragability** – the ability to access stored data and other system resources at all times and from everywhere within the enterprise
- **Scalability** – the ability to continue to provide the required quality of service as the load or usage increases

- **Security** – the ability to allow certain users access to application functions and data while denying access to other users
- **Maintainability and manageability** – the ability to correct flaws in the system and to ensure the continued health of the system without adversely affecting other components of the system
- **Portability** – the ability of the software to run on a variety of hardware and operating systems

1.5.2 Digital assets

We discussed digital assets in the chapter Strategic analysis. In the chapter Strategic choice, we looked at an organisation's digital strategy which involves them using digital technology and digital assets to challenge existing ways of doing things, and we noted that an organisation's digital assets can often be strategically important to it.

In this respect, the data organisations hold can often constitute an important digital asset in its own right – for example, data about customers, or internet of things (IoT) data.

However, the fact that companies are collecting and analysing increasing amounts of data also reinforces two key points:

- **Data quality** – As the volume and variety of data increases, companies need processes in place to ensure they have good quality data. To be valuable to an organisation, data needs to be accurate, consistent and complete.
- **Data security** – Organisations need an effective strategy to keep data secure, particularly that data which is perceived to be most important to the company. This point is central to the issue of cybersecurity (which we look at in Section 6).

In an article, *Protecting your critical digital assets*, the management consultancy firm McKinsey identifies a five step approach which it recommends to clients for protecting their critical digital assets:

(a) Identify and map digital assets, including data, systems and applications across the business value chain

(b) Assess the risks for each asset, and its importance to the business (in order to identify the business' key assets – its 'crown jewel' assets)

(c) Identify potential attackers, the availability of assets to users, and current controls and security measures protecting the systems through which access can be gained to the assets

(d) Locate where security is weakest around crown jewel assets and identify controls that should be in place to protect them

(e) Create, and implement, a set of initiatives to address the high-priority risks and control gaps. (This list of initiatives should be regularly refreshed to reflect new data, systems, applications and risks.)

1.6 Cloud computing

The ICAEW IT Faculty paper *Security and Assurance in the Cloud* (published in 2013) highlights that industry globalisation, coupled with the recent economic climate and the resulting fiscal pressures this places on businesses have resulted in increased demands for the availability, scalability and efficiency of business IT solutions.

One of the most significant consequences of this has been the growth of cloud computing.



Definition

Cloud computing: the delivery of on-demand computing resources – everything from applications to data centres – over the internet on a pay-for-use basis.

(IBM)

Although cloud services are often provided by an external provider and used by multiple customers on the same infrastructure, this is not always the case. Different types of cloud can be identified:

Private cloud – cloud services are supplied on an infrastructure that belongs to only one customer. The service can be managed by the customer themselves, or by the supplier.

Public cloud – the cloud service is owned and provided by a supplier who serves multiple organisations through the same infrastructure system.

Community cloud – a cloud shared by a particular community of organisations with common interests or data protection concerns.

Hybrid cloud – a cloud that is a combination of two or more distinct cloud infrastructures (private, community or public).

Cloud computing technologies have changed the ways in which organisations store and manage their data. An increasing amount of organisational data is now held in servers operated by cloud- based service providers. In effect, cloud computing is the access to business services through the internet.

There could take the form of:

- **Software as a service** – eg, anti-virus software. The software is hosted in the cloud, but appears on users' devices with full functionality. This type of cloud service is the most common, and is often aimed directly at the end user (eg, Hotmail; Gmail).
- **Platform as a service** – eg, Windows Azure. For a fee, the cloud provider offers virtual space in which customers can host and develop their own applications.
- **Infrastructure as a service** – eg, data storage and back-up. Cloud storage can handle all kinds of structured and unstructured data.

1.6.1 Benefits and risks of cloud computing

Cloud computing ('public cloud') can provide an organisation with a number of benefits:

- **Cost effectiveness** – using cloud computing services may be more cost effective than operating in-house technology. Moreover, cloud computing could remove the capital costs required for buying hardware; replacing them with revenue costs for the cloud-based services that an organisation uses.
- **Flexibility** – cloud computing offers companies the capacity to scale up their capacity as required, and so can be ideal for businesses with growing – or fluctuating – service demands. Operational agility is one of the main drivers for cloud adoption.

  In a similar vein, establishing a cloud-based approach to data storage and management can be done faster than establishing the technology in-house.
- **Accessibility** – storing organisational data in the cloud means that it is accessible anywhere around the world where there is internet connectivity.

  Cloud-based services can also be very useful for **increased collaboration** between teams working in different locations. Similarly, this can also improve document control. Instead of having to send files as email attachments to be worked on by colleagues (with the associated risk of conflicting versions, formats etc.) the cloud enables all files to be stored centrally, so that everyone can see the same version of a file.
- **Availability** – cloud computing is available both to large organisations and much smaller entities alike. This could be particularly important for small companies looking to establish

themselves in a market – because the cloud gives them access to cutting-edge technology, without the need for significant capital investment.

- **Automatic software updates** – cloud service providers roll out regular software updates – including security updates – so organisations don't have to worry about updating systems software or systems themselves.

As the ICAEW IT Faculty paper notes, 'Many sources claim that Cloud computing can help organisations lower the total cost of ownership, provide higher return on investment and increase efficiency with features such as dynamic provisioning and utility-like pay-as-you-go services.'

As the services are typically provided using shared computer resources, this helps to achieve economies of scale and lower management costs. The on-demand nature of the service can also lower cost-of-entry barriers to new organisations, because it removes the need for large capital budgets to fund and operate IT data centres, equipment and staff.

In effect, using a public cloud enables an organisation to outsource its IT resources, and so it can be seen as an extension of the business process outsourcing activities we discussed in the chapter Strategic implementation of this Workbook.

However, as with outsourcing in general, an organisation needs to consider the potential risks associated with Cloud computing as well as its benefits. In particular, the increased risks associated with entrusting company information assets to the cloud must be clearly understood and managed by appropriate stakeholders. Perhaps the most obvious concern with using a public cloud is whether the service provider's infrastructure is secure so that an organisation's data and applications are protected while they are stored on the cloud.

If an organisation stores its data on the cloud, then it has to give up control of its data to an external party (ie, the cloud-based service provider). As such, there is a risk that data held by the service provider may be stolen, lost or corrupted. Although this risk might most obviously relate to hacking, there is also a danger that the service provider's own staff might interfere with data stored on its servers.

Although we have mentioned that 'Accessibility' is one of the potential benefits of cloud computing, it is nonetheless important to recognise that there will inevitably be times when the cloud-based services will suffer outages or other technical issues. Even the best cloud service providers can suffer technical issues – or there could be wider connectivity issues preventing an organisation from connecting to the cloud – so it is important to consider the potential impact that any such disruption could have for an organisation, if it adopts a cloud-based model.

Similarly, although we have cited cost effectiveness as a potential benefit of the cloud, it is also important to remember that cloud providers charge for the amount of resource used, so an organisation could be locked into increasing costs as it grows (compared to the diminishing marginal cost of an in-house investment).

The ICAEW paper *Security and Assurance in the Cloud* provides a useful summary of the key concerns an organisation should consider before transferring information to a public cloud:

Concern	Comment
Data privacy, risk and security	An organisation should consider carefully what information it is prepared to place on a cloud. If the information is critical to the business, or an organisation can't be reassured that information will be adequately protected, then it should probably not be moved to the cloud. Is there a practical option to encrypt sensitive information before it is

Concern	Comment
	uploaded it to an organisation's cloud provider? (Again, though, if the information is sensitive and needs to be encrypted, then perhaps the organisation should consider whether that information should be placed in the cloud in the first place.)
Access and identity	An organisation should ask potential cloud service providers how access to data that they will be hosting in the cloud will be controlled. As well as identifying how unauthorised access to its data will be prevented, an organisation also needs to identify how other customers, and its own staff, will be granted access to the systems which are hosting its information. Data held in the cloud is typically in a shared (multi-tenancy) environment alongside data from other clients, so there must be effective access separation and confidentiality protection.
Resilience and business continuity	Organisations should maintain backups and have contingency plans in place in case of business interruption or disaster. Before the move to the cloud, they need to ensure their service provider also addresses the need for resilience. Does the service provider have sufficient capacity and a planning process to deal with any outages they may experience? How would the organisation's own business be affected by these? An alternative consideration here is whether the cloud could provide an organisation with a better approach to its own resilience planning, through cost-effective off-site storage of its own data.
Regulatory compliance	Are there any regulatory or legal restrictions which could affect a move to the cloud? For example, are there any restrictions which require you to process information in a specific jurisdiction or country? Another potential issue which could be affected by risk and regulatory constraints is choosing which deployment model is appropriate for an organisation (public, private, community or hybrid).
Assurance	Depending on the nature of an organisation's business, and the range of stakeholders who might have an interest in a move to the cloud, an organisation should consider seeking assurance from potential service providers around key aspects of the service they provide (for example, their plans for data security, service outage and recovery) – before entering into a contract with them. As with any strategic procurement decision, an organisation should not simply engage the first Cloud Service Provider (CSP) it finds, but instead it should work through a comparative evaluation of the providers on the market, focusing on the aspects of service which are key to its needs and identifying the CSP which best fulfils those needs.

1.6.2 Assurance over cloud services

As the fifth concern listed in the table highlights, before becoming a customer of a CSP, an organisation may wish to gain further assurance about how the CSP runs their business –for example, how the CSP manages its own risk and governance and how they select and screen their own staff.

Similarly, aspects of physical security, IT service management, incident management, and business continuity and resilience are all areas which an organisation should consider obtaining assurance over before engaging with a CSP.

It is possible that the CSP may have already engaged an accountant to provide a report on their controls in place at their organisation which can then be made available to all potential customers. However, if such a report is not available, a potential customer should consider asking an accountant of their own to conduct an assurance engagement on the control environment and the controls in place at the CSP.

As we noted in chapter Strategic implementation in relation to outsourcing, ISAE 3402, *Assurance Reports on Controls at a Service Organisation* provides guidance for assurance engagements carried out on service organisations, where the service organisation is responsible for the design of suitable controls.

ISAE 3402 states that the objectives of the service auditor are: to obtain reasonable assurance about whether, in all material respects, based on suitable criteria:

(a) The service organisation's description of its system fairly presents the system as designed and implemented throughout the specified period or as at a specified date
(b) The controls related to the control objectives stated in the service organisation's description of its system were suitably designed throughout the specified period
(c) Where included in the scope of the engagement, the controls operated effectively to provide reasonable assurance that the control objectives stated in the service organisation's description of its system were achieved throughout the period

Security

In its report 'Cloud adoption – Understanding the risk of cloud services' ICAEW's IT faculty provides the following advice in relation to security in the cloud:

It is impossible to guarantee that data can be totally secure. Vulnerabilities are found in software all the time, and any platform that is publicly accessible carries more inherent risk than one that is not. From a security perspective, the following questions should be asked of the service provider to ensure they follow the best practice principles.

- Is the platform regularly given to third-party 'penetration testers' for potential vulnerabilities, who vigorously test the platform to determine whether an attacker could gain unauthorised access?
- What is the process in the event of a security breach being determined?
- Is data held on the platform stored in an encrypted format?
- Is payment data held on a platform which complies with the payment card industry's data security standard (PCI DSS)?
- Are there recognised, standard working processes and procedures in place and adopted (for example, ISO accreditations held)?
- Is the platform protected against 'denial of service' attacks, where attackers could prevent access to the service indefinitely by flooding the platform with erroneous traffic or requests for information?

The level of security deemed necessary will depend on the value of the data in the users' eyes, but all users of a service should feel satisfied that efforts are being made to secure the platform against unauthorised third parties that may seek to disrupt access to the service, or obtain a copy of the data for malicious intent.

Note: As a general point, ensuring the security of a platform is particularly important in the context of the General Data Protection Regulation (GDPR) – discussed later – which requires organisations to have appropriate security measures in place to protect any personal data they hold.

1.6.3 Cloud computing, software and big data

The ICAEW IT Faculty (2014) publication *Big data and analytics – what's new?* Highlights that the enormous growth in computer power and storage in recent years has been the core

enabler of big data –by making it possible to capture and process entire data sets, regardless of their size and complexity.

The publication also notes that the cloud computing model further supports the widespread use of big data. By using a cloud, a business doesn't need to buy all the computing resources it might use, but instead simply uses them when needed. Therefore the cloud model potentially provides a business with access to vast computing resources – as would be needed to collect and process big data – on an efficient and flexible basis.

Software advances have also complemented the development in processing and storage capability. For example, new types of software support large and unstructured data sets better than traditional database management systems. Software such as Apache Hadoop supports the management of very large data sets by splitting the processing between many computers.

Similarly, capabilities in handling unstructured data, such as video and text, have also improved.

Importantly, as the document *Big data and analytics – what's new?* points out, these increases in computing power and capabilities make it economically viable for businesses to collect and process data from many sources:

- The internet provides a variety of clickstream data, such as searches, site visited, and goods viewed, as well as actual transaction
- Social media has created new types of data, including status updates, comments and likes, photos and videos
- Mobile technology is providing more opportunity to create social media and internet data, and generates new data about the location of individuals
- Open data – relating primarily to public sector data, such as geo-spatial data or transport data
- The 'internet of things' – resulting from computer chips and sensors being embedded in physical assets (such as machines or domestic appliances)

As we have already discussed in the chapter Data analysis, businesses can then use the resulting data to enhance their understanding of their operations and their customers – for example, by using more detailed data about customers to understand their preferences, and then exploiting the real-time nature of big data to improve services, for example by personalising the offers made to customers.

However, as we noted in the chapter Data analysis also, it is important to remember that data analytics and 'big data' are not the same thing, and data analytics does not relate solely to 'big data'.

Although increases in technology make it possible for business to collect and examine unstructured, external data (ie, big data), organisations can still use analytics techniques to identify patterns and trends in internal data.

1.6.4 Cloud computing vs owned technology

In the preceding sections, we have looked at the potential benefits, and risks, associated with cloud computing. The balance between these benefits and risks becomes critical for senior management within organisations when they are deciding whether their organisation should pursue a cloud-based approach to data management, or whether to continue to manage data in-house using owned hardware and software.

The answer to such a question could depend on a number of factors. Organisations with IT staff that possess the levels of expertise required to manage IT systems may prefer to retain data storage and data management in-house. Similarly, complex data compliance requirements, or a risk averse attitude among senior management about allowing external parties to control organisational data, will make in-house data management more likely.

Conversely, for organisations, particularly small and medium-sized entities that need a multi-national or global presence but lack the necessary IT expertise and resources to manage their data and infrastructure in-house, a cloud-based approach could be an appropriate choice.

A related question for organisations to consider is: what data will they store in the cloud? In particular, will personal data be part of the data accessed or stored in a cloud computing service? Personal data is subject to data protection laws (such as the EU's General Data Protection Regulation (GDPR)), and an organisation must still comply with those laws even if it uses external providers.

Typically, in law, the cloud service provider is viewed as a data processor, acting on the instructions of the data controller (the organisation using the cloud platform). As such, the data controller is required to ensure that its agreement with the cloud provider meets data protection standards, and that it minimises the risks of data misuse or loss.

1.7 The impact of cloud computing and borderless businesses on strategic management information

In chapter Strategic choice, when we discussed borderless business, we identified that one of the key factors underpinning internationalisation has been the flow of information, supported by developments in communications.

However, developments in technology – particularly in relation to cloud computing – are equally important in providing information to help manage borderless businesses. On the one hand, this can relate to the timeliness and quality of information which is available for management. On the other hand, the data which can be transmitted through the cloud can also drive innovation and improve the performance of business operations.

1.7.1 Performance information

By definition, a borderless business is one which has operations in multiple countries. However, in order for senior management to know how the business as a whole is performing, they will need information about all of the operations – in different countries, and in different industries or market segments.

Historically, local operations are likely to have produced their own management information packs and then emailed them to head office at the end of each period.

Cloud-based data processing products enable managers to access real-time (or near-time) financial information for all the operations, providing the business with much timelier financial management information than was previously available.



Context example: Specsavers

The following case study is summarised from an article in ICAEW's Finance & Management publication (October 2015). Although it relates to a multi-site business, rather than a multinational one, it still illustrates the way cloud computing can help to improve management information for planning and control.

For any growing business, the ability to manage data in multiple silos is challenging. The problems of collecting the data, reconciling it, and having confidence that the consolidated information represents reality, can slow down decision making and increase risk.

Yet this is the inevitable consequence of multi-location business, for example, a chain of retailers. That was the problem faced in 2014 by Specsavers, which managed budgeting and forecasting for its 2,000 locally-run retail businesses using Excel spreadsheets. The group planning and reporting manager at Specsavers described the situation as 'a world of Excel silos… a version control nightmare among numerous stakeholders and contributors'. He admitted that 'inaccuracies increasingly threatened to throw business goals off course'.

The solution was to use a single cloud-based application. The cloud-based software has the look and feel of a spreadsheet, but it creates a standardised version of budgets, forecasts and reports that is distributed to all users. Anyone with the correct privileges, who logs in, can analyse business performance at that moment, across all the locations.

The financial benefit is measured through better, quicker, more accurate planning… Another process benefit is that many Specsavers senior executives now do their own financial reporting without waiting for consolidated results from 2,000 spreadsheets. As the group planning and reporting managers put it, 'The company is now all on the same financial page.'

Source: Phillips, T. (2015) Head in the Clouds, *Finance & Management*, October 2015

1.7.2 Availability and accessibility of information

As we noted earlier in our discussion of cloud computing earlier in this section, storing data in the cloud means it is accessible from anywhere in the world (where there is internet connectivity), and at any time of day.

This could be particularly important for senior managers in borderless businesses, who will travel around the world, and consequently may need to work from anywhere, and access information from different places. Cloud-based applications will enable information from all parts of the business to be instantly available to employees and managers who need it – regardless of their location or time zone.

Equally importantly, borderless firms will need to coordinate functions and activities in different locations (for example, coordinating the flow of components or finished goods across the supply chain). Here again, a cloud-based system will enable users in different locations to access the information they need. Moreover, this information could also be shared with suppliers and distributors to enable a business to integrate supply chains distributed across different countries.

More generally, cloud computing improves cross-border data flows, and organisations can benefit from this by streamlining their operations, and improving their products and services.



Context example: Rio Tinto

Rio Tinto is a leading mining and metals company with operations in over 40 countries, including a strong presence in Australia and North America.

To make its mining operations more efficient, Rio Tinto created its 'Mine of the Future' programme to identify the size, location and quality of ore it is mining, by aggregating the data it collects in real time. The company collects this data from the trucks and from the drills it uses in its mines all around the world. In addition it collects information from process surveillance cameras, control systems and maintenance system logs.

This information is then processed at its Processing Excellence Centre (PEC) in Brisbane, Australia, with 20 different analytical systems. The 'Mine of the Future' programme has generated millions of dollars in savings across Rio Tinto's global operations by rooting out inefficiencies.

Rio Tinto is able to use this efficiency to be more environmentally friendly, and to promote safer mining in each of its locations. More efficient mining, in turn, also leads to cost benefits.

Adapted from: Castro, D. & McQuinn A. (2015) Cross-Border Data Flows Enable Growth in All Industries, *Information Technology & Innovation Foundation*, February 2015

1.7.3 Scalability

Another important characteristic of the 'cloud' is that it is effectively infinite in size, and so firms can increase their data storage capacity as required.

The dynamics of global competition mean that firms have to meet heightened customer expectations in terms of quality, price and value. As a result, firms are typically holding more data – about their customers, and about detailed sales trends – than ever before. In this respect, the ability to store large data sets – especially, unstructured 'big data' sets, could be very useful in providing borderless businesses with insights into their customers in different countries and different markets.

Similarly, being able to store large data sets could also be important in enabling them to monitor the detailed performance of their suppliers



Context example: ING

Banks use data to improve decision making in a number of ways, including improving the customer experience, developing new products, and informing marketing decisions.

The Dutch bank ING serves over 48 million individual and institutional clients in more than 40 countries, and has a workforce of over 75,000 employees.

ING was an early adopter of big data analytics to analyse its financial data, website interactions, customer service calls, user feedback, and other data to segment its consumers and use predictive marketing.

ING collects detailed information about how customers – across different countries – interact with the bank in person, online and over social media. It then uses that information to inform its marketing strategy, brand development, customer relations, and even the products it offers. In turn, this allows ING to deliver tailored services to suit an individual consumer's needs, such as personalised bank alerts, more robust cybersecurity, better fraud detection, lower interest rates for borrowers (due to reduced credit card fraud) and top-quality customer service.

To support this analysis, ING has rebuilt its cloud computing infrastructure to help accommodate the volume of data it collects.

Adapted from: Castro, D. & McQuinn A. (2015) Cross-Border Data Flows Enable Growth in All Industries, *Information Technology & Innovation Foundation*, February 2015

1.7.4 Cryptocurrencies

As we noted in the chapter Strategic analysis, one of the potential benefits of cryptocurrencies is that they can be used by people anywhere in the world to transact with each other, in almost real-time, thereby removing the time and cost of making cross-border payments.

However, as also noted in the chapter Strategic analysis, if an organisation starts using cryptocurrencies then it will also need to consider how they should be accounted for, because there are currently no specific accounting requirements for them. Nonetheless, it will be essential that the organisation records its cryptocurrency transactions in its management information, so it will need to decide the accounting treatment it is going to apply, and then apply it consistently.

One further issue to consider is the potential volatility in the value of the cryptocurrencies themselves, so the use of cryptocurrency could increase an organisation's balance sheet exposure, depending on the process of converting between cryptocurrency and fiat currency (£, $, € etc). Again though, this volatility could increase the importance of disclosing information about an organisation's cryptocurrencies transactions and balances – particularly if the organisation makes a significant number of cryptocurrency transactions.

1.8 The internet, intranets and extranets

The internet has been the most influential technology in the last few decades: e-commerce could simply not exist without the internet.

Intranets use the same technologies as the internet, but access to them is restricted to computer networks within an organisation. The purpose of intranets is to allow the secure sharing of information to any part of an organisation, its system and its staff.

Extranets link organisations together through secure business networks using internet technology.

Extranets are particularly useful for various aspects of **supply chain management**; for example, details of orders placed with suppliers, orders received from customers, payments to suppliers and payments received from customers can all be transmitted through extranets. Similarly, banks can also be incorporated in these networks.

Using networks in this way is often called **electronic data interchange** (EDI).

1.8.1 E-business and e-commerce

E-business is the use of internet-based technology to either support existing business processes or to create entirely new business opportunities.

Where these internet-based operations or processes are connected to a buying or selling activity, they become **e-commerce**.

E-commerce strategy

Most experts agree that a successful strategy for e-commerce cannot simply be bolted on to existing processes, systems, delivery routes and business models. Instead, management groups have, in effect, to start again by asking themselves fundamental questions.

- What do customers want to buy from us?
- What business should we be in?
- What kind of partners might we need?
- What categories of customer do we want to attract and retain?

In turn, organisations can visualise the necessary changes at **three interconnected levels**:

Level 1 – the simple **introduction of new technology** to connect electronically with employees, customers and suppliers (eg, through an intranet, extranet or website)

Level 2 – **reorganisation** of the workforce, processes, systems and strategy in order to make best use of the new technology

Level 3 – **repositioning** of the organisation to fit it into the emerging e-economy So far, very few companies have gone beyond Levels 1 and 2. Instead, pure internet businesses such as Amazon have emerged from these new rules. Unburdened by physical assets, their competitive advantage lies in **knowledge management** and **customer relationships**.

1.8.2 M-commerce

In addition to e-business, we should recognise the increasing importance of accessing computermediated networks from mobile devices while on the move (eg, mobile phones, personal digital assistants and tablet computers).

Mobile commerce (M-commerce), or M-business, refers to any commercial transactions conducted on a mobile device such as a smartphone or tablet. There are three main areas of m-commerce:

- *Browser/web based – websites where you can buy, order or subscribe to products or services, for example online stores (Tesco, Marks and Spencer) and marketplaces (Amazon and Ebay).*

- *Apps – these allow users to interact in a range of ways from marketplaces (Vinted) and shopping apps (ASOS) to banking apps (Barclays), payment apps (PayPal) and customer loyalty apps (Starbucks). Apps allow users to store personal data such as card details to make transactions quick and easy.*
- *In-person – payment methods that rely on mobile phones to process the payment, for example mobile wallets (Google and Apple Pay) and QR code payments.*

For many businesses, such as digital banks, M-commerce is their only channel to interact with customers.



Context example: Digital banks

Digital banks, such as Starling and Monzo, are app-based banks that have no physical premises. Sometimes known as disruptor banks, for their ability to challenge the long-held status quo of the banking industry, digital banks have lower overheads and pass this onto customers in the form of lower fees and higher interest rates. Due to a combination of speed, convenience and ease of use, mobile banking apps have become common place and customers are increasingly being attracted to digital banks. For example in 2022 over 85% of adults in the UK used online banking compared to just 52% in 2012 and this trend, alongside growth in digital banks, is set to continue.

Source: Statista (n.d), Digital banks in the UK - statistics & facts. Available from: https://www.statista.com/topics/5996/disruptor-banks-uk/ [Accessed on 24 July 2024]

1.8.3 Virtual arrangements

Network technologies are used by organisations to integrate workers across sites and working at home. Developments in broadband, particularly improved capacity and better data security, have improved the ability to communicate across sites and from **home**. Broadband telecommunications systems allow 'remote' computer users to communicate with each other, and to send and receive information.

A virtual organisation can be seen as an extension of the idea of network organisations, although truly virtual organisations do not have any physical presence at all.

There is some disagreement among academics as to a precise definition of the virtual organisation, but a consensus exists with regard to **geographical dispersion** and the centrality of **IT** to the production process.

Virtual organisations use networks to link people, assets and ideas, enabling a virtual organisation to ally with other organisations to create and distribute products and services without being limited by traditional organisation boundaries or physical locations. In a virtual network, one organisation can use the capabilities of another without being physically tied to that organisation.

The virtual organisation model is useful when a company finds it is cheaper to acquire products, services or capabilities from an external vendor, or when it needs to move quickly to exploit new market opportunities but lacks the time and/or resources to respond to the opportunities on its own.

An organisation is not a virtual organisation merely because it uses IT extensively and has multiple locations. Nevertheless, the ability to share information between members of a virtual organisation is likely to be critical to its operation.



Context example: Amazon

Amazon.com could be viewed as a virtual organisation.

Customers come to the Amazon website via internet service providers, often from links on other (affiliate) websites. Although Amazon processes the customers' orders, it does not hold much inventory itself. If a customer orders a book through Amazon it is likely the book will be

dispatched from the publisher's warehouse, and the delivery will be handled by a logistics or mail company. Nonetheless, the customer feels they are dealing with one organisation (Amazon) and not many different companies.

But for this relationship to work, Amazon needs information from its partners – for example, it needs to know inventory availability and an estimate of delivery times so that it can provide this information for the customer when they make their order. Equally, Amazon needs to be confident that its partners will deliver the service they have agreed to provide (for example, if a partner says inventory will be available in 48 hours, then it needs to be available in 48 hours).

Note: In the chapter Strategic choice, we looked at platform-based business models in the context of digital strategies. In the sense that platform-based business models work by bringing producers and consumers together, rather than by an organisation making and selling a product or service itself, we can suggest that these might also be seen as examples of virtual organisations.

1.9 E-business strategies

We can identify three broad types of e-business strategy which a company can employ to help it gain a competitive advantage:

Cost and efficiency improvements: Focus on improving efficiency and lowering costs by using the internet and other digital technologies as a fast, low-cost way to communicate and interact with customers, suppliers and business partners (eg, use of email to communicate with customers; and EDI to communicate with suppliers)

Performance improvement in business effectiveness: Make major improvements in business effectiveness – for example, the use of intranets can substantially improve information sharing, collaboration and knowledge management within a business or with its trading partners

Product and service transformation: Developing new internet-based products and services, or supporting entry into new markets (including e-commerce which enables access to a global marketplace)

1.10 IT and competitive advantage

Throughout this section, we have alluded to the potential which IT has as a source of competitive advantage for companies. However, it is important to note that companies only achieve competitive advantage by doing something different from their competitors, and they only achieve sustainable competitive advantage by doing something which their competitors cannot replicate over time.

For example, the first airline company to introduce self-service check-in kiosks gained a competitive advantage (or source of differentiation) by doing so. However, all the major airline companies now have self-service check-in kiosks so they are no longer a source of competitive advantage. By contrast, not having them would place a company at a competitive disadvantage.

Equally, when the first logistics and shipping company allowed customers to track their packages via the web, this innovation was seen as a source of competitive advantage. Now, however, we expect to be able to track orders with all logistics and shipping companies.

Both the examples above illustrate the importance of IT to modern business strategy. However, they also illustrate that the majority of innovations which confer competitive advantage on the companies which adopt them first are subsequently shared and become routine.

In this respect, we can divide IT investments into two broad categories: strategic IT and utility IT.

Strategic IT represents spending on new capabilities which directly supports new business strategies and innovations; all the remainder of IT spending is utility spending.

Customised applications and application platforms

In the same way that we can distinguish between strategic IT and utility IT, we also need to distinguish between customised applications and generic software.

Although packaged software is vital to many businesses, gaining any competitive advantage or differentiation from a generic package is very difficult (because competitors can also use the package). Therefore, strategic IT investments are most often based on custom applications.

Crucially, a company needs its application platform to support current technologies. Therefore, a company's choice of application platform (such as the Microsoft application platform) can be a key part of creating competitive advantage.

1.10.1 Artificial intelligence

We have already discussed (in the chapters Strategic choice and Data analysis) the potential for organisations to use artificial intelligence (AI) and machine learning to improve their performance.



Context example: H&M

The 'fast fashion' industry is highly competitive, and in response to falling sales the multi-national retailer H&M is using insights from AI and big data to try to improve performance.

To be successful, fast-fashion retailers need to predict what the market wants to avoid a bad product cycle. Because the price points are already very low, it is tough for fast-fashion retailers to recover from bad purchase decisions and to move unwanted inventory.

The insights provided by data can help to build a more flexible and faster supply chain, facilitate trend detection, manage inventory and set prices.

Inventory for individual stores – Traditionally, all H&M stores would carry very similar merchandise (whether located in Germany, the UK, the US or Sweden). However, this meant H&M regularly need to cut prices to clear unsold inventory from its stores around the world (more than 4,000 of them). In an effort to stock individual stores with merchandise local customers want, H&M is using big data and AI to analyse receipts, returns and loyalty card data to tailor the merchandise for each store.

Automated warehouses – In order to get customised inventory to each store, H&M invested in automated warehouses that will ultimately result in next-day delivery for 90% of its European market. The company is also rolling out RFID technology to its stores to improve efficiencies in its supply chain.

Customer experience – H&M already offers personalised recommendations for online shoppers, but it is bringing that capability to its 'bricks-and-mortar' stores. When they are in a store, customers can explore suggestions for merchandise selected for them by algorithms.

H&M is also working on a better integration between the online and offline shopping experience; for example, enabling customers to find out if an item they discovered online is available at a physical store nearby.

Customised design – As a result of a partnership with Google and H&M's digital fashion house, Ivyrevel, the company has introduced Coded Couture. This is an Android application that monitors customers' activity and lifestyle and then custom designs and produces a dress – the 'Data Dress'. Depending on how an individual spends their day, the app translates how active they are, where they eat, and the typical weather in the area to make decisions about colour, materials and added details for a bespoke look, promising to 'create one-of-a-kind designs' based on how individuals live their lives.

Based on: Marr B. (2018) *How Fashion Retailer H&M is Betting on Artificial Intelligence and Big Data to Regain Profitability* [Online] Available from: https://www.forbes.com/sites/bernardmarr/2018/08/10/how-fashion-retailer-hm-is-betting-on- artificial-intelligence-and-big-data-to-regain-profitability/#6e93e6565b00 [Accessed 14 August 2018]

However, while AI and machine learning can be very powerful, there are still potential risks and issues around using them which organisations need to be aware of:

Data accuracy – Organisations need to ensure that the AI systems they use produce correct, precise and reliable results. In many cases, machine learning involves 'supervised learning' in which a machine 'learns' a process or task from labelled training data. However, in order for the machine to learn accurately, the training data needs to be free from bias and systematic error – for example, arising from an unfair sampling of a population.

Interpretability – As AI algorithms become more complex, it becomes increasingly difficult for humans to make sense of how they work. In some cases, AI applications are a 'black box' where not even engineers can decipher why a machine made a certain decision. However, this could be a cause of concern for businesses and the public. The use of 'black box' algorithms makes it difficult not only to identify when things go wrong, but also to determine who is responsible in the event that they do (for example, if a 'driverless' car is involved in an accident).

Accountability – Accountability is a central element of corporate governance, and implies that there should be a line of responsibility for business actions. However, as the point about 'interpretability' identifies, AI systems introduce an additional level of complexity into any discussions about accountability. Who is responsible for the decision-making process of a machine? This question is complicated further because AI is largely outsourced by companies, rather than being developed in house. So, if an algorithm goes wrong, who is accountable: the designer, or the company which is using it? And if it is the company, then who within the company is accountable? These questions are a source of considerable debate, and no clear answers have yet emerged.

However, as the Institute of Business Ethics notes in its briefing paper *Business Ethics and Artificial Intelligence*: 'Some business leaders have indicated that responsible organisations need to redefine how they interact with technology to be able to be seen as trustworthy in the age of artificial intelligence. People tend to trust those… institutions that operate with openness and take account of the public interest. Working with regulators and policy makers, businesses have the opportunity to make a significant contribution to agree on a framework of ethics and norms in which AI can thrive and innovate safely,'

2 Information for strategic planning and control



Section overview

- Managers need information (including management accounting information) for three main reasons:
  - To make effective **decisions** (for example, how much of a product to make, what price to charge for it, and what distribution channels to use)
  - To **control** the activities of an organisation. There are four steps to this control process: establish measurable goals, measure actual performance, compare actual performance against the goals, and then take corrective action if necessary
  - **Coordinating** the activities of different departments and divisions (for example, by coordinating the flow of materials, semi-finished goods, and finished goods throughout the supply chain)

2.1 Management information and strategy

Managers need information for three main reasons:

- To make effective **decisions**
- To **control** the activities of the organisation
- to **coordinate** the activities of the organisation

2.1.1 Information and decisions

Decision making is a key element of management. For example, a marketing manager must decide what price to charge for a product, what distribution channels to use and how to promote the product. Equally, a production manager must decide how much of a product to make, while a purchasing manager must decide how much inventory to hold and whom to buy inputs from.

At a more strategic level, senior managers must decide how to allocate scarce financial resources among competing projects, how the organisation should be structured, or what business-level strategy an organisation should be pursuing.

In order to make effective decisions, managers need information from both inside and outside the organisation. For example, when deciding how to price a product, marketing managers need information about the way consumer demand will vary in relation to different prices, the cost of producing the product and the organisation's overall competitive strategy (since its pricing strategy will need to be consistent with this overall strategy).

2.1.2 Information and control

The management control process can be summarised in four key steps:

- Establish measurable standards of performance or goals
- Measure actual performance
- Compare actual performance against established goals
- Evaluate the results and take corrective action where necessary

As we have identified in the chapter Strategic performance management, and as section 2.2 below reminds us, management accounting information plays a key role in this respect – by enabling managers to measure performance and to control the business, by comparing actual performance against goals and targets.

2.1.3 Information and coordination

The third key element of management is coordinating the activities of individuals, departments or divisions in order to achieve organisational goals.

For example, in the chapter Strategic implementation we looked at supply chain management, and one area where coordination is particularly important is in relation to managing global supply chains. Organisations are using increasingly sophisticated IT systems to coordinate the flow of materials, work in progress and finished products throughout the world. (Such control and coordination is particularly important for organisations which employ a 'just in time' production system, where there is little or no 'buffer stock' to draw on in the event of any delay or disruption in the supply chain.)

2.2 Management accounting information

Management accounting information is used by managers for a variety of purposes:

(a) **To measure performance**. Management accounting information can be used to analyse the performance of the business as a whole, and of the individual divisions, departments or products within the business. Performance reports provide **feedback**, most frequently in the form of comparison between actual performance and budget.

(b) **To control the business**. Performance reports are a crucial element in controlling a business. In order to be able to control their business, managers need to know the following:

(1) What they want the business to achieve (targets or standards; **budgets**)

(2) What the business is actually achieving (**actual performance**)

By comparing the actual achievements with targeted performance, and identifying **variances**, management can decide whether corrective action is needed, and then take the necessary action when required.

Much control information is of an accounting nature because costs, revenues, profits and asset values are major factors in how well or how badly a business performs.

(c) **To plan for the future**. Managers have to plan, and they need information to do this. Much of the information they use is management accounting information.

(d) **To make decisions**. As we have seen, managers are faced with several types of decision:

(1) **Strategic decisions** (which relate to the longer-term objectives of a business) require information which tends to relate to the organisation as a whole, is in summary form and is derived from both internal and external sources.

(2) **Tactical and operational decisions** (which relate to the short or medium term and to a department, product or division rather than the organisation as a whole) require information which is more detailed and more restricted in its sources.

In the remainder of this section, we will look briefly at some of the management accounting information and management accounting tools which managers can use to evaluate aspects of business strategies.



Professional skills focus: Assimilating and using information

One of the challenges of the SBM&L exam is to draw conclusions from data, facts, calculations and your own analysis. Scenario exhibits may include different types of data which you should use to perform further analysis. This will enable you to provide the business in the scenario with clear and practicable advice.

2.3 Costs

One of the most crucial elements of an organisation's performance management and control is to ensure that the value created by its activities is greater than the cost of carrying out those activities. (This is the point highlighted by the 'margin' element in Porter's value chain model.)

2.3.1 Costs and strategy

A key element of performance measurement and control for a company will be ensuring that its costs remain under control. However, cost information is also important from a strategic perspective.

For example:

A company's choice of generic strategy interacts with cost and value. A company which is pursuing a **cost leadership** strategy needs to maintain a strict control of costs (and, wherever possible, also needs to benchmark its costs and the efficiency of its processes against its competitors to ensure its costs remain lower than theirs).

However, a **differentiation** strategy will also have cost implications – for example, associated with product quality and customer service or after-sales service.

Moreover, the structure of costs and value creation is likely to change over time, as illustrated, for example, in the **product life cycle**. As sales and production rise in the growth stage of the life cycle, unit costs could be expected to fall due to **economies of scale**. In the mature stage, prices become increasingly sensitive as firms compete with one another to try to increase their share of the market. Therefore cost reductions may be required to help firms sustain their profits.

Costs and decision making

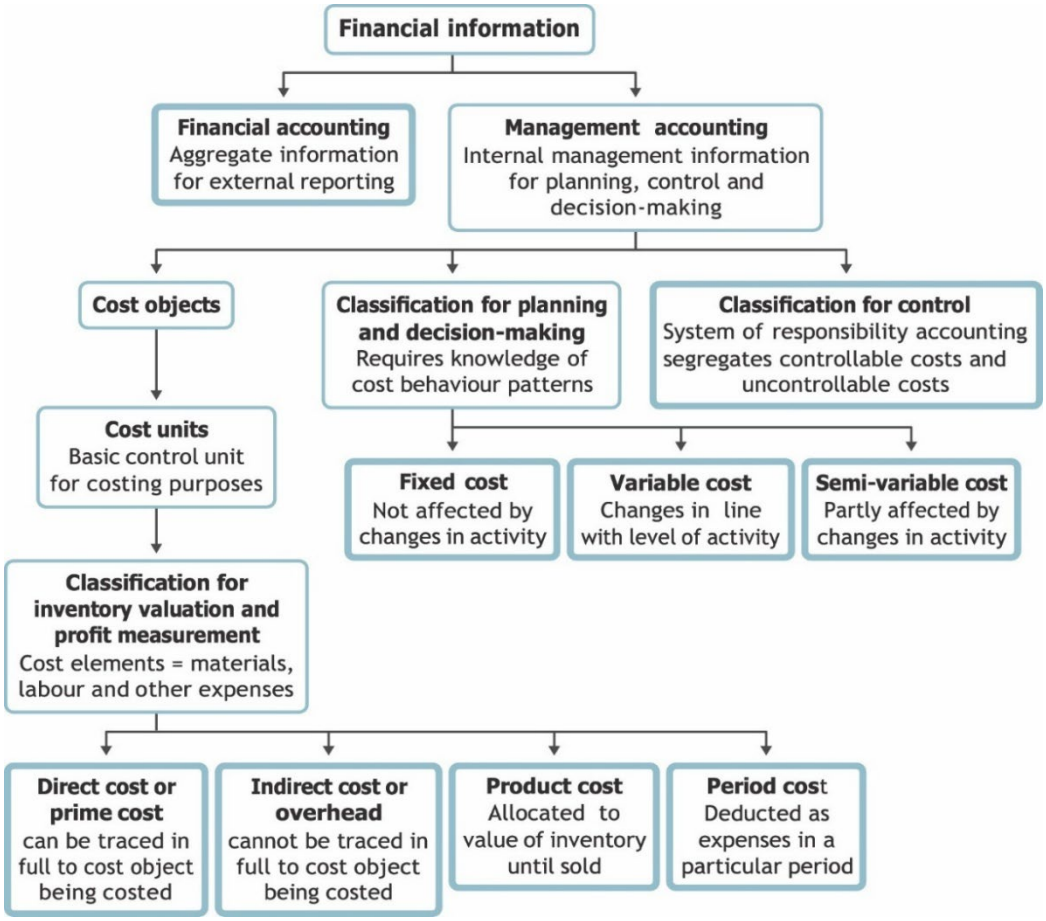
Managers also need to know the cost of producing different products and services because they cannot make informed decisions about pricing or about whether or not to continue producing them without having accurate cost information.

Sub-optimal decision making – Importantly, if managers do not have accurate and reliable cost information, this is likely to lead to sub-optimal decision making. For example, if the costs attributed to producing a product are understated, then a company may continue producing that product when it would actually be better advised to stop producing it.

Short- vs long-term trade-off – Equally, managers may find themselves under pressure to reduce costs. However, it is important that decisions taken to reduce costs in the short term don't hinder an organisation's ability to achieve its strategic goals. For example, if marketing expenditure is reduced to cut costs in the short term, this could be detrimental to an organisation's strategic goal to increase market share.

2.3.2 Cost classification

You should already be familiar with cost classification from your earlier studies, but here is a summary brought forward from the Management Information paper.



Make sure you are familiar with all these classifications.

2.3.3 Costing systems

One of the purposes of the following costing systems is to calculate the cost of a unit of output which can ultimately be used to set the selling price.

Absorption costing

With absorption costing, a unit of output is valued at **full cost** – that is, the prime cost plus an absorbed share of production overhead costs.

Marginal costing

Marginal costing is an alternative to absorption costing where only variable costs are included in the valuation of units. All fixed costs are treated as period costs and written off against sales revenue in the period in which they are incurred.

The difference in unit valuations using the two methods lies in the treatment of the fixed costs. The absorption cost of sales will include some fixed costs from a previous period (included in opening inventory) while all fixed costs are written off as expenses in the year of incurrence with marginal costing.



Interactive question 2: Absorption and marginal costing

Hamilton Ltd manufactures and sells a single product, the Feronda, which has a selling price of £150 per unit and variable costs of £70 per unit. During the months of July and August, the following details are available.

	July	August
Fixed production costs	£110,000	£110,000
Production	2,000 units	2,500 units
Sales	1,500 units	3,000 units

Hamilton Ltd normally expects to produce 2,200 units and fixed production costs were budgeted at £110,000 per month, which are absorbed on a per unit basis. There were no opening inventories in July.

Requirements

2.1 Determine the profit for each of July and August using:

(a) Absorption costing

(b) Marginal costing

2.2 Demonstrate why the profits under each method are different.

See **Answer** at the end of this chapter.

2.3.4 Activity-based costing (ABC)

Activity-based costing (ABC) is an alternative approach to absorption costing. Cost drivers – that is, those activities that cause costs in the first place – are identified and overheads are assigned to products or services based on the number of the cost drivers generated by each. More than one cost might have the same cost driver, so costs associated with the same driver are gathered into cost pools and then allocated using the appropriate driver. The product costs resulting from ABC should be more accurate than those under absorption costing, as overheads are allocated on a more objective basis. Try the question below to make sure you remember the principles and procedures of ABC.

Interactive question 3: ABC

Tammy plc currently makes and sells four products, cost and output details of which are below.

Product	Alpha	Bravo	Echo	Oscar
Output (units)	500	300	400	200
Cost per unit (£)				
Material	60	42	80	100
Labour	32	20	35	50
Activities				
Number of set ups	20	15	30	35
Number of times materials handled	3	4	2	6
Number of orders	11	12	16	25
Number of spare parts required	15	20	10	15

Total overhead costs

	£
Set-up costs	25,000
Material handling	69,000
Ordering costs	32,000
Engineering costs	45,000
	171,000

Requirements

3.1 Using the information given, what is the most suitable cost driver for each overhead cost?

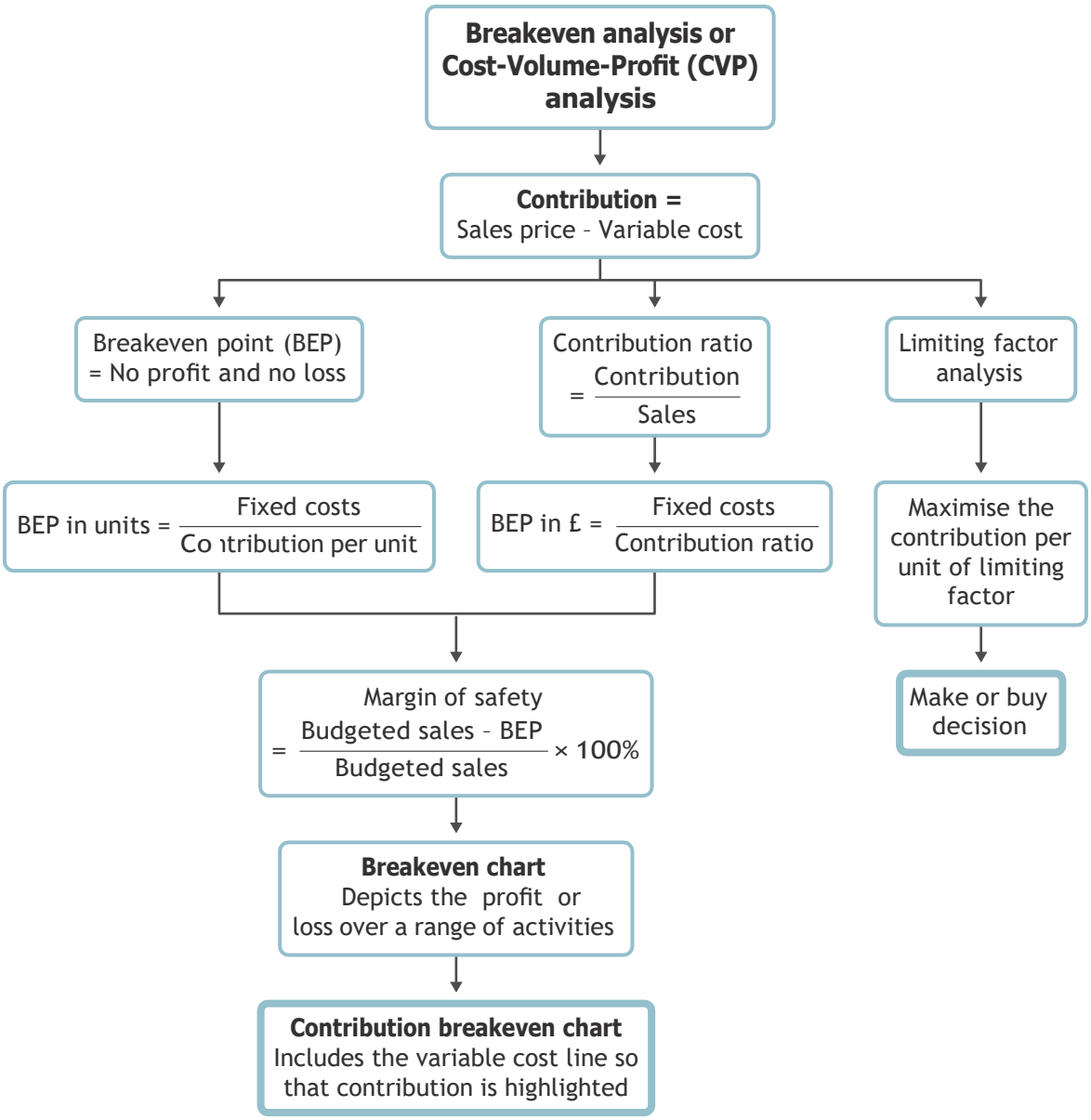
3.2 Calculate the total product cost for each of the four products, showing all workings.

See **Answer** at the end of this chapter.

2.4 Breakeven analysis

Breakeven analysis is the study of the interrelationships between costs, volume and profit at various levels of activity. The study of breakeven analysis requires understanding, application and interpretation of various formulae which are summarised below.

You will notice from the chart that breakeven analysis is also used for decision making purposes where there are limiting factors, such as make or buy decisions. Remember that the most important figure for decision making is **contribution** – if a product is making a contribution towards fixed costs then production should continue, as overall profit will be reduced (or loss increased) if this contribution is lost.



Interactive question 4: Breakeven analysis

Bonzo plc manufactures and sells roller skates. Information regarding the current sales volume, price and costs as well as anticipated changes resulting from a planned marketing campaign can be seen in the spreadsheet extract below:

	A	B	C
1		Current	After campaign
2	Sales price per pair	55	65
3	Variable cost per pair	25	25
4	Fixed costs:		
5	Marketing	75,000	115,000
6	Salaries	200,000	200,000

7	Other	150,000	150,000
8	Annual sales volume (pairs)	20,000	20,000

Requirements

4.1 Using the information provided in the spreadsheet extract, calculate how many pairs of roller skates have to be sold in one year for Bonzo plc to break even, before the marketing campaign?

4.2 Using the information provided in the spreadsheet extract, calculate the current margin of safety in terms of number of pairs of skates sold?

4.3 Following the marketing campaign, if Bonzo plc wishes to make a profit of £25,000, how much sales revenue must be generated from the sale of roller skates? Use the relevant information from the spreadsheet extract.

See **Answer** at the end of this chapter.

2.5 Multi-product breakeven analysis

The key issue with multi-product breakeven problems is determining the mix in which the products are sold and reducing it to the lowest common denominator. For example, if 500 units of Product X and 250 units of Product Y are sold, then the mix in which the two products are sold is 2:1. This mix is used to define a standard batch which can be used to determine a breakeven point in terms of number of batches. Once this has been determined, the number of batches can then be converted into the number of units of each product that must be sold at the breakeven point.

Worked example: Multi-product breakeven analysis

Toodiloo Ltd manufactures and sells two types of microwave oven: the Silver Star oven and the Gold Senator combination oven and grill. The following information is available for the two models.

	Silver Star	Gold Senator
Sales volume in units	5,000	3,000
Per unit:	£	£
Selling price	65.00	90.00
Variable costs	45.00	60.00

Direct fixed costs are £40,000 for the Silver Star and £30,000 for the Gold Senator. General fixed costs, which can only be avoided if neither model is sold, are £63,000.

Requirement

Calculate how many units of each model would have to be sold for Toodiloo Ltd to cover all its fixed costs.

Solution

As with all breakeven questions, the first thing to do is to establish the contribution per unit:

	Silver Star	Gold Senator
	£	£
Selling price	65.00	90.00
Variable costs	45.00	60.00
Contribution per unit	20.00	30.00

The key to multi-product breakeven analysis is to look at the product mix. In this case we are told that Toodiloo sells 5,000 units of the Silver Star and 3,000 units of the Gold Senator, giving a product mix ratio of 5:3. A standard batch can therefore be assumed to comprise five Silver Star and three Gold Senator, which will give a total contribution towards total fixed costs of £190 (5 × £20.00 + 3 × £30.00).

Using the total contribution per standard batch, we can calculate the number of batches that have to be sold in order to break even:

$$\text{Breakeven point in batches} = \frac{\text{Total fixed costs}}{\text{Contribution per batch}}$$
$$= \frac{133{,}00}{190}$$
$$= 700 \text{ batches}$$

How many units of each model will have to be sold in order to break even? Remember that in every batch, five units of the Silver Star are sold and three units of the Gold Senator. Therefore, in order to break even, Toodiloo will have to sell:

Silver Star: 5 × 700 = 3,500 units Gold Senator: 3 × 700 = 2,100 units

Check:

	Silver £	Gold £	Total £
Contribution per unit	20.00	30.00	
Total contribution	70,000	63,000	133,000
Fixed costs:			
Direct	40,000	30,000	70,000
General			63,000
Total profit/(loss)			NIL

Note: You will have probably noticed that this breakeven number of batches and units will only work if the product mix remains at 5:3. As soon as the product mix changes, you will have to calculate a new breakeven point. As you might expect, an increase in the proportion of sales of products with a higher contribution will normally reduce the breakeven point, while an increase in the proportion of sales of products with a lower contribution will normally increase the breakeven point.

Interactive question 5: Multi-product breakeven analysis

Netcord Ltd produces and sells three different types of tennis racquet: the McEnrova, the Grafassi and the Federdal. The sales and costs forecast for the next period are as follows.

	McEnrova	Grafassi	Federdal	Total
Sales units	8,000	6,000	4,000	
	£	£	£	£
Sales revenue	320,000	360,000	400,000	
Variable costs	176,000	228,000	280,000	
Contribution	144,000	132,000	120,000	396,000
Total fixed costs				306,900
Net profit				89,100

Total fixed costs can only be avoided if all models are eliminated.

Requirement

How many units of each model must be sold in order for Netcord Ltd to break even and what is the breakeven sales revenue?

See **Answer** at the end of this chapter.

2.6 Price

Price is a key element of the marketing mix, and you should have covered pricing and pricing issues in the Business Strategy and Technology syllabus at Professional level. However, when evaluating pricing strategies, it is vital to consider how 'price' information fits with the other elements of the marketing mix.

Price directly affects how well an organisation performs competitively, and is also closely linked to the perceptions of value for money held by the customers.

If the price is too high, the exchange may not be perceived as worthwhile, and customers may not buy the product.

If the price is too low, then one of two things could happen:

(a) The product sells well; however, the revenue per item is lower than it could be if the price were higher, so less revenue is earned which in turn means less profit than may be possible with a higher price.

(b) The consumer perceives the price to be too low and interprets this as meaning quality has been compromised. The customer does not buy the product at all.

Getting this balance right (and not setting a price either too high or too low) can be difficult and the organisation will have to take many factors into account when setting the price for its products and services.

The following factors should be considered when evaluating a firm's pricing strategy:

(a) What are the **pricing objectives** (for example, budget pricing and premium pricing)?

(b) What are the firm's **target markets**, and what will they be prepared to pay for the firm's products?

(c) How will demand for the firm's products or services vary with price (ie, what is their price elasticity)?

(d) What is the relationship between demand, cost and profit (eg, how does the breakeven point vary for a range of different prices)?

(e) How do the firm's prices compare to **competitors' prices**?

(f) What is the basis for pricing?

- **Cost-based pricing** (marginal cost-plus, full cost-plus pricing, mark-up pricing, target-return pricing)
- **Demand-based pricing** (set prices high when demand is high, and low when demand is low)
- **Competition prices** (set prices in relation to competitors' pricing, depending on the organisation's strategy)
- **Marketing-orientated pricing** (set prices to reflect the marketing strategy: ie, target market profile, brand positioning and sales targets)

(g) What is the pricing strategy?

- For example:
- **Differential pricing** (charging different prices to different customers for the same quality and quantity of product)
- **New product pricing** (price skimming, penetration pricing)
- **Psychological pricing** (eg, everyday low prices; bundle pricing, where two or more complementary products are sold together for a single price which is lower than the aggregate price if both were purchased separately; prestige pricing, where prices are set artificially high to give a product a 'quality' image)
- **Promotional pricing** (eg, selling products below the usual mark up for a short-term period)

Prices have to be revised in response to market conditions and trends (eg, recession, new entrants). In the short term, firms can also use **dynamic pricing** to better reflect the current levels of demand and supply.

However, when making short-term revisions to price, it is important that an organisation also considers any potential longer-term implications for the brand, or the relationship between demand, costs and profits.



Context example: McDonald's in China

In February 2009, McDonald's lowered the price of four of its meals in China by up to one-third. The price of the Filet-o-Fish, double cheeseburger, chicken fillet sandwich and pork burger meals was reduced to 16.50 yuan (or $2.40).

McDonald's said that the price reduction follows the Chinese Government's direction to stimulate domestic demand and help build a stronger economy, in the light of concerns about the impact which the global economic slowdown was having on China.

McDonald's felt that it could do its part by stimulating domestic demand in the restaurant sector. However, McDonald's was not the only food retailer to have lowered prices in China. Others, like KFC, had also started promotions as consumers in China began to worry about slowing growth and rising unemployment.

Only a year earlier, the prices of staple items like pork, rice and cooking oil were soaring, lifting inflation and threatening to overheat the Chinese economy which had recorded double-digit growth for a number of years. Once growth started to slow, the pressure shifted to retailers to entice shoppers with special deals.
However, China remains an attractive location because of the size of the market and because its growth rates are still ahead of most of the rest of the world. Therefore, McDonald's price cut

was also influenced by its plans to expand in the country, which was one of its fastest-growing markets.
McDonald's had 1,050 outlets in China at the start of 2009, but by 2014 this figure had increased to over 2,000.
In this respect, McDonald's had been a beneficiary of the global economic downturn, as shoppers around the world switched to lower-priced goods.

2.7 Budgets

Budgets should provide an organisation with short-term targets within the framework of longer-term strategic plans. Budgets represent the short-term targets which need to be achieved in order to fulfil strategic objectives.

Budgets also provide a mechanism for **controlling performance** as they provide a yardstick against which to assess performance. This means finding out why actual performance did not go according to plan, and then seeking ways to improve performance for the future.

Budgets enable managers to manage by exception; that is, focus on areas where things are not going to plan (ie, the exceptions). This is done by comparing the actual performance to the budgets to identify the **variances**.

However, the reason a budget is not achieved may sometimes be because the budget itself was unrealistic. If this is the case, the budget may need to be revised. Only realistic budgets can form a credible basis for control.

2.8 Variance analysis

When actual performance is compared to standards and budgeted amounts, there will inevitably be **variances**. They may be favourable or adverse depending on whether they result in an increase to, or a decrease from, the budgeted profit figure.

2.8.1 Sales variances

Sales volume variance is the difference between the original and flexed budget profit figures. This is an important variance because losing sales generally means losing profit as well. If it has the effect of making profit lower than budgeted it is **adverse** and if it makes profit higher than budgeted it is **favourable**.

Sales price variance is the difference between actual sales revenue and actual volume at the standard sales price. Higher sales prices (if all else remains constant) mean an increase in profit.

2.8.2 Materials variances

Total direct materials variance is the difference between the actual and direct materials cost and the direct materials cost according to the flexed budget. If the actual material cost is higher than budget, it has an adverse effect on profit.

Direct materials usage variance is the difference between actual usage and budgeted usage for the actual volume of output, multiplied by the standard materials cost. If actual usage is higher than budgeted usage, there will be an adverse effect on profit.

Direct materials price variance is the difference between actual materials cost and the actual usage multiplied by the standard materials cost. Again, if actual costs are higher than those budgeted, there will be an adverse effect on profit.

2.8.3 Labour variances

Total direct labour variance is the difference between the actual direct labour cost and the direct labour cost according to the flexed budget. If more is spent on labour than was budgeted, there will be an adverse effect on profit.

Direct labour efficiency variance is the difference between the actual labour time and budgeted time, for the actual volume of output, multiplied by the standard labour rate. It looks at the actual versus the budgeted number of hours used to produce the output. If actual time is greater than budgeted time, the effect on the profit will be adverse. The faster people work, the more profit can be made.

Direct labour rate variance is the difference between the actual labour cost and the actual labour time multiplied by the standard labour rate. This means it compares the actual cost of the hours worked against the anticipated cost based on a standard hour. Where actual costs exceed the standard, profit will be adversely affected.

2.8.4 Fixed overhead variances

Fixed overhead spending variance is the difference between the actual and budgeted spending on fixed overheads. Higher than budgeted overheads lead to less profit and have an adverse effect.



Worked example: Flexible budgets and budgetary control

Penny manufactures a single product, the Darcy. Budgeted results and actual results for May are as follows.

	B	C	D	E	F
		Budget	Actual	Variance	Adv/Fav
1	Production and sales of the Darcy (units)	7,500	8,200		
2		£	£	£	
3	Sales revenue	75,000	81,000	6,000	Fav [1]
4	Direct materials	22,500	23,500	1,000	Adv [2]
5	Direct labour	15,000	15,500	500	Adv
6	Production overhead	22,500	22,800	300	Adv
7	Administratio n overhead	10,000	11,000	1,000	Adv
8		70,000	72,800	2,800	Adv
9	Profit	5,000	8,200	3,200	Fav

[1] =IF(D3>C3,'Fav','Adv'). The formula can be copied into cells analysing income eg, F9.

[2] =IF(D4<C4,'Fav','Adv'). The formula can be copied into cells analysing costs eg, F5:F8.

Note: (Fav) denotes a favourable variance and (Adv) an unfavourable or adverse variance.

In this example, the variances are meaningless for the purposes of control. All costs were higher than budgeted but the volume of output was also higher; it is to be expected that actual variable costs would be greater than those included in the fixed budget. However, it is not possible to tell how much of the increase is due to **poor cost control** and how much is due to the **increase in activity**.

Similarly, it is not possible to tell how much of the increase in sales revenue is due to the increase in activity. Some of the difference may be due to a difference between budgeted and actual selling price but we are unable to tell from the analysis above.

For control purposes we need to know the answers to questions such as the following:

- Were actual costs higher than they should have been to produce and sell 8,200 Darcys?
- Was actual revenue satisfactory from the sale of 8,200 Darcys?

Instead of comparing actual results with a fixed budget which is based on a different level of activity to that actually achieved, the correct approach to budgetary control is to compare actual results with a budget which has been **flexed** to the actual activity level achieved.

Suppose that we have the following estimates of the behaviour of Penny's costs:

(1) Direct materials and direct labour are variable costs.
(2) Production overhead is a semi-variable cost, the budgeted cost for an activity level of 10,000 units being £25,000.
(3) Administration overhead is a fixed cost.
(4) Selling prices are constant at all levels of sales.

Solution

The **budgetary control analysis** should therefore be as follows.

	B	C	D	E	F	G
		Fixed Budget	Flexible budget	Actual	Variance	Adv/Fav
1	Production and sales of the Darcy (units)	7,500	8,200	8,200		
2		£		£	£	
3	Sales revenue	75,000	82,000 [1]	81,000	1,000	Adv [2]
4	Direct materials	22,500	24,600 [3]	23,500	1,100	Fav [4]
5	Direct labour	15,000	16,400 [5]	15,500	900	Fav
6	Production overhead	22,50023	23,200 [6]	22,800	400	Fav
7	Administrat ion overhead	10,000	10,000 [7]	11,000	1,000	Adv
8		70,000	74,200	72,800	1,400	Fav
9	Profit	5,000	7,800	8,200	400	Fav

[1] See Working (1)
[2] =IF(E3>D3,'Fav','Adv'). The formula can be copied into cells analysing income eg, G9.
[3] See Working (2)
[4] =IF(E4<D4,'Fav','Adv'). The formula can be copied into cells analysing costs eg, G5:G8.
[5] See Working (3)
[6] See Working (4)
[7] See Working (5)

WORKINGS

(1) Selling price per unit = £75,000/7,500 = £10 per unit Flexible budget sales revenue = £10 × 8,200 = £82,000
(2) Direct materials cost per unit = £22,500/7,500 = £3 Budget cost allowance = £3 × 8,200 = £24,600
(3) Direct labour cost per unit = £15,000/7,500 = £2 Budget cost allowance = £2 × 8,200 = £16,400

(4) Variable production overhead cost per unit = £(25,000 – 22,500)/(10,000 – 7,500)
= £2,500/2,500 = £1 per unit
∴ Fixed production overhead cost = £22,500 – (7,500 × £1) = £15,000
∴ Budget cost allowance = £15,000 + (8,200 × £1) = £23,200

(5) Administration overhead is a fixed cost and hence budget cost allowance = £10,000

Comment

(1) In selling 8,200 units, the expected profit should have been not the fixed budget profit of £5,000, but the flexible budget profit of £7,800. Instead, actual profit was £8,200 ie, £400 more than we should have expected.

One of the reasons for this improvement is that, given output and sales of 8,200 units, the cost of resources (material, labour etc) was £1,400 lower than expected.

These total cost variances can be analysed to reveal how much of the variance is due to lower resource prices and how much is due to efficient resource usage.

(2) However, the sales revenue was £1,000 less than expected because the price charged was lower than budgeted.

We know this because flexing the budget has eliminated the effect of changes in the volume sold, which is the only other factor that can affect sales revenue. You have probably already realised that this variance of £1,000 (A) is a **selling price variance**.

The lower selling price could have been caused by the increase in the volume sold (to sell the additional 700 units the selling price had to fall below £10 per unit). We do not know if this is the case but without flexing the budget we could not know that a different selling price to that budgeted had been charged. Our initial analysis above had appeared to indicate that sales revenue was ahead of budget.

The difference of £400 between the flexible budget profit of £7,800 at a production level of 8,200 units and the actual profit of £8,200 is due to the net effect of cost savings of £1,400 and lower than expected sales revenue (by £1,000).

The difference between the original budgeted profit of £5,000 and the actual profit of £8,200 is the total of the following:

(1) The savings in resource costs/lower than expected sales revenue (a net total of £400 as indicated by the difference between the flexible budget and the actual results).

(2) The effect of producing and selling 8,200 units instead of 7,500 units (a gain of £2,800 as indicated by the difference between the fixed budget and the flexible budget). This is the **sales volume contribution variance**.

A **full variance analysis statement** would be as follows:

	£	£
Fixed budget profit		5,000
Variances		
Sales volume	2,800 (F)	
Selling price	1,000 (A)	
Direct materials cost	1,100 (F)	
Direct labour cost	900 (F)	
Production overhead cost	400 (F)	
Administration overhead cost	1,000 (A)	
		3,200 (F)
Actual profit		8,200

If management believes that any of the variances are large enough to justify it, they will investigate the reasons for their occurrence to see whether any corrective action is necessary.

2.8.5 Reasons for variances

Variances may occur for a number of reasons. One possible reason is that the budget itself was not realistic. Unless they are achievable, budgets are not a useful method of control. However, there are many other reasons why variances may arise, as shown by the table below.

Variance	Possible reason for variance
Sales volume	Poor performance by sales staff Deterioration in market conditions between the time the budget was set and the actual event Lack of goods or services to sell as a result of a production problem
Sales price	Poor performance by sales staff Deterioration in market conditions between the time the budget was set and the actual event
Direct materials usage	Poor performance by production department staff, leading to high rates of scrap Substandard materials, leading to high rates of scrap Faulty machinery, causing high rates of scrap
Direct materials price	Poor performance by the buying department staff Using higher-quality material than was planned Change in market conditions between the time the budget was set and the actual event
Direct labour efficiency	Poor supervision A worker with a low skill grade taking longer to do the work than was envisaged for the correct skill grade Low-grade materials, leading to high levels of scrap and wasted labour time Problems with a customer for whom a service is being rendered Problems with machinery, leading to labour time wasted Dislocation of materials supply, leading to workers being unable to proceed with production
Direct labour rate	Poor performance by the human resources department Using a higher grade of worker than was planned Change in labour market conditions between the time of setting the budget and the actual event
Fixed overhead spending	Poor supervision of overheads General increase in costs of overheads not taken into account in the budget



Professional skills focus: Applying judgement

Variances may occur for a number of performance reasons but may also arise as a result of incomplete or inaccurate data. In order to make sense of variances you will need to draw inferences from further qualitative and quantitative analysis of the scenario information as well as taking into consideration any errors or omissions in the data used to prepare the calculations.

2.9 Limiting factors

Every organisation operates under resource **constraints**.

Usually, an organisation's output is restricted by the level of demand rather than the organisation's ability to produce.

However, sometimes there is a limit to the amount which can be produced due to a limiting factor within the organisation.

Examples of limiting factors are:

- a shortage of production capacity
- a limited number of key personnel, such as salespeople with technical knowledge
- a restricted distribution network
- limited shelf space or display space (for a retailer)
- too few managers with knowledge about finance or overseas markets
- inadequate research design resources to develop new products or services
- a poor system of strategic intelligence
- lack of money
- a lack of adequately trained staff

Once the limiting factor has been identified, the planners should:

- in the short term, make best use of the resources available
- try to reduce the limitation in the long term

The most profitable combination of products will occur where the **contribution per unit of the scarce factor** is maximised.

When evaluating strategic plans, managers need to assess what the limiting factors (if any) are, and then assess whether the company is producing the combination of products which allows it to maximise its profits.

For example, if labour is scarce then the company's priority should be on producing the products which generate the highest profit per unit of labour.



Worked example: Scarce resources

A business makes three different products (A, B and C), as follows:

	A	B	C	D
1	Product	A	B	C
2	Selling price per unit (£)	25	20	23
3	Variable cost per unit (£)	10	8	12
4	Weekly demand (units)	25	20	30
5	Machine time per unit (hours)	4	3	4

Fixed costs are not affected by the choice of product because all three products use the same machine. Machine time is limited to 148 hours a week.

Requirement

Which combination of products should be manufactured if the business is to produce the highest profit?

Solution

	A	B	C	D
1	Product	A	B	C
2	Selling price per unit (£)	25	20	23
3	Variable cost per unit (£)	(10)	(8)	(12)
4	Contribution per unit (£)	**15**	**12**	**11**
5	Machine time per unit (hours)	4	3	4
6	Contribution per machine hour	£3.75	£4.00	£2.75
7	Order of priority	2 [1]	1	3
8	Production schedule:	HOURS		
9	20 units Product B	60		
10	22 units Product A	88		
11		148 [2]		

[1] =RANK(B6,$B$6:$D$6,0). The formula can be copied in C7:D7.

[2] =SUM(B9:B10)

Therefore produce:

This leaves unsatisfied the market demand for a further three units of product A and 30 units of product C.

Limiting factors could also be important in determining the feasibility of an organisation's strategy. If an organisation has a limited amount of labour available, the total amount of sales it can generate will be restricted by this labour. If the organisation is looking to grow, but isn't also looking to increase its staff levels, its strategy will not be feasible (unless it is able to prevent staff levels being a limiting factor in some other way – for example, by automating some processes which are currently carried out manually).

Limiting factors and make or buy decisions

The issue of limiting factors could also have implications on a decision about whether to make or buy. If a factor is scarce and is preventing growth, then buying additional units of that resource might be justified, even if a traditional make or buy decision would not otherwise justify it.

Limiting factors and capacity

The reference to limiting factors and capacity also highlights that organisations may need to adjust their capacity by some means. For example, if labour (or staff hours available) is the limiting factor, an organisation should consider how it can increase its labour capacity.

Overtime – The quickest and most convenient way of increasing capacity is to increase the number of hours worked by the existing staff, by offering them overtime payments to work additional hours.

However, this method is only useful if the timing of the extra capacity matches that of the demand. For example, there will be no benefit from asking retail staff to work longer hours in the evenings if the excess demand is occurring during normal daytime working hours.

Conversely, at a micro level an organisation might be able to solve capacity issues by building **flexibility** into **job design and job roles** so that staff could be transferred from less busy areas into the busiest areas for short periods of time. For example, because they found that peak

times for registering new customers coincided with the least busy times in the kitchen and restaurant areas, the hotel chain Novotel trained some of its kitchen staff to also escort customers from the reception area up to their rooms.

Adjusting the size of the workforce – If capacity is largely governed by the size of the workforce, then one way to increase capacity is to take on extra staff in periods of high demand. These might often be temporary or part-time staff.

A variation on this approach could be to use **subcontractors**. **Demand management** Instead of looking to resolve capacity issues through supply side solutions, it might also be possible to address them through demand management activities.

Such an approach is popular among travel operators who offer cheaper holidays and flights at less busy times in order to try to stimulate off-peak demand and curtail peak demand.

2.10 Expected values

An **expected value** (or **EV**) is a weighted average value based on probabilities. The EV for a single event can offer a helpful guide for management decisions.

Although the outcome of a decision may not be certain, there is some likelihood that probabilities could be assigned to the various possible outcomes from an analysis of previous experience.

If the probability of an outcome of an event is p, then the expected number of times that this outcome will occur in n events (the EV) is equal to n × p.

The concepts of probability and EV are vital in **business decision making**. The EVs for single events can offer a helpful guide for management decisions:

- A project with a positive EV should be accepted.
- A project with a negative EV should be rejected.
- When choosing between options, the alternative which has the **highest EV of profit** (or the **lowest EV of cost**) should be selected.

Where probabilities are assigned to different outcomes we can evaluate the worth of a decision as the **EV**, or weighted average, of these outcomes. The principle is that when there are a number of alternative decisions, each with a range of possible outcomes, the optimum decision will be the one which gives the highest EV.

EVs can be built into **decision trees** in order to aid decision making. The amount of expected profit is likely to be conditional on the result of various decisions.

However, remember the limitations of using EVs as a basis for decisions.

2.10.1 Limitations of EVs

Evaluating decisions by using EVs has a number of limitations.

(a) The **probabilities** used when calculating EVs are likely to be estimates. They may therefore be **unreliable or inaccurate.**

(b) EVs are **long-term averages** and may not be suitable for use in situations involving one-off decisions. They may therefore be useful as a **guide** to decision making.

(c) EVs do not consider the **attitudes to risk** of the people involved in the decision-making process. They do not, therefore, take into account all the factors involved in the decision.

(d) The time value of money may not be taken into account: £100 now is worth more than £100 in 10 years' time.

2.11 Transfer pricing

You should have looked at issues around transfer pricing in Business Strategy and Technology, but it is important to remember that transfer pricing can have important strategic implications.

Sales and prices

Transfer prices can determine the overall price and sales of a product.

Suppose Division A supplies an intermediary product to Division B for £12,000. Division B does additional work to the product (at a cost of £5,000) and the product's market price is set at a 15% mark-up on cost.

If Division A transfers the intermediary product at cost, the final price will be £19,550, ie, (£12,000 £5,000) × 1.15.

However, if Division A sets a transfer price of £15,000 for the intermediary product, the final price would be £23,000, ie, (15,000 + 5,000) × 1.15.

The difference in price is likely to have a significant effect on the volume of products sold, and therefore company profits.

Tax liabilities

If a multinational company has divisions in countries with different tax rates, the level at which transfer prices are set will influence the overall level of tax the company has to pay.

Dysfunctional decision making

Transfer pricing could lead to dysfunctional decision making.

In the illustration above, if Division A believes it can achieve a higher price by selling on the open market (rather than selling to Division B) it should take the open market price. However, this might mean that Division B does not have a product to sell if it cannot obtain the component elsewhere.

Equally, however, if Division B can obtain supplies of an equivalent product more cheaply from an external supplier than from Division A it should buy the product from the external supplier. However, this might leave Division A with unsold stock.

Transfer pricing and control

As a means of control, the main concern is reconciling the need for transfer prices to be set at an **appropriate level to assess performance** with the need for **goal congruence** within the organisation. Ideally, pricing levels should be set at a level to avoid the complications and loss of resources of departments dealing unnecessarily with external sources rather than with internal 'suppliers'.

Transfer pricing may also be a significant issue if one division of a company makes an **investment** that benefits all the other divisions.

A **transactions cost approach**, taking into account costs associated with setting and administering the transfer price and also time commitments and obligations, is an appropriate way to determine which transactions should take place within an organisation and which transactions should occur in the outside world.

Transfer pricing and multinational companies

More generally, transfer pricing could be an important issue for multinational companies, and we will look at international transfer pricing in more detail in the chapter 'International financial management' of this Workbook.

2.12 Balanced scorecard

We have already discussed the balanced scorecard in the chapter Performance management of this Workbook.

The **scorecard** was devised as a way of integrating the traditional **financial indicators** with **non-financial measures**, such as operational performance quality, customer satisfaction and staff

potential. It is balanced in the sense that managers are required to think in terms of all four perspectives in order to prevent improvements being made in one area at the expense of another.

The aspects of the balanced scorecard can be as effective as **financial measures (as indicators of long-term profitability)**, control mechanisms, business trends or benchmarks against other organisations. They can act as **targets** for employees, and will be more effective if linked to the organisation's reward schemes. The range of perspectives they provide can be a **better link** with strategy than a few financial measures.

Perspective	Addresses	Examples
Customer	Measures relating to what actually matters to customers (time, quality, performance of product)	Customer complaints On-time deliveries
Internal business	Measures relating to the business processes that have the greatest impact on customer satisfaction (quality, employee skills)	Average set-up time Quality control rejects
Innovation and learning	Measures to assess the organisation's capacity to maintain its competitive position through the acquisition of new skills/development of new products	Labour turnover rate % of revenue generated by new products
Financial	Measures that consider the organisation from the shareholders' viewpoint	Return on capital employed Earnings per share

2.13 Information on sustainability

Increasingly, organisations need to manage and report on their sustainability performance. Information on sustainability will be required at the strategic level so the board and stakeholders are aware of the organisation's overall performance and can report these to stakeholders. Metrics used for ESG reporting were discussed in the chapters Strategic performance management and Data analysis. Information is also needed to assist management in managing the risks related to sustainability and ESG issues. Risk management, including risks related to ESG, was discussed in chapter Business risk management.

At the tactical and operational levels of the organisation, it will be necessary to set targets and measure performance, to ensure that the strategic goals are achieved. Examples of tactical and operational measures include greenhouse gas emissions at local level, headcount information, and surveys on staff satisfaction and information about breaches of ethical codes.

A big problem with sustainability information is that many organisations' information systems do not capture relevant data

Context example: Accenture survey

According to Accenture: 'In our ESG Measurement Study, only 26% of surveyed finance leaders could agree that they had clear, reliable data to underpin their ESG KPIs. But there seems to be a clear reason for this: most companies have not yet extended their measurement capabilities and infrastructure to ESG data. In less than one third of cases (31%), ESG reporting is fully embedded in core operational and management systems, with just under 70% still claiming that manual or semi- automated processes were used.'

Source: Accenture (2022) *Measuring sustainability. Creating value, Time to rethink performance and redefine success* [Online] available from https://www.accenture.com/_acnmedia/PDF-179/Accenture-Sustainability-Measurement-Report.pdf#zoom=40 [Accessed 18 July 2022]

The use of manual or semi-automated processes to process ESG-related data causes a risk of inaccurate processing, which can lead to errors in the reported metrics. At an operational and tactical level, this may mean that incorrect decisions are made, or control processes are not activated when they should be. At the reporting level, this can lead to inaccurate reporting to stakeholders.

An additional challenge related to ESG data is scarcity of staff with the knowledge and skills to collate and analyse the data on ESG issues.

3 Management information systems (MISs)



Section overview

- There are many different types of MIS available according to the level of information required. These range from operational systems (such as transaction-processing systems, which process large volumes of data from routine transactions) to strategic-level systems (such as EISs, whose focus is on identifying high-level issues and long-term trends).
- If the information that senior managers obtain from EISs enables them to respond to opportunities and threats more quickly than their competitors, this could be a source of competitive advantage for an organisation.
- An organisation's critical success factors should help determine its information requirements, because an organisation needs to know how well it is performing in relation to its key operational goals.

In the chapter 'Strategic performance management' of this Workbook, we noted the idea of a hierarchy of decision making within organisations, meaning that information is required for planning and control at strategic, tactical (management) and operational levels.

The existence of this hierarchy means that different types of MIS are required to provide the different types of information required at the different levels – although the higher level (strategic and tactical) applications, to some extent, still make use of data which has been produced by operational systems.

In the chapter Data analysis, and in section 1 of this chapter, we have also highlighted the potential importance of big data as a source of new insights and performance information. One of the important characteristics of big data analytics is that managers may no longer be confined to analysing individual data sets, but instead data from a number of different sources can all be linked.

Therefore, although we are looking at specific systems in this section, it is also important to remember that in practice, data could come from a variety of different sources; for example, external data showing customers' reactions to a new product could be very useful when updating an organisation's forecasts or budgets. Equally, however, as we noted in the chapter Data analysis and section 1.6.3 of this chapter, data analytics techniques are not only used in relation to big data.

Organisations can still use data analytics to identify trends and gain insight from internal data (such as accounting information).

3.1 Information systems

In chapter Strategic performance management, we also mentioned some of the types of information systems which firms can use to provide them with information about their performance and their processes:

- Executive Information Systems (EISs)
- Management Information Systems (MISs)
- Decision Support Systems (DSSs)

Different types of system are typically used to provide different levels of information (strategic, tactical or operational) for different user groups in an organisation. However, as Laudon and Laudon note (in their text *Management Information Systems*), although different systems serve different management groups in a business, all of them 'provide business intelligence that helps managers… make more informed decisions'.

3.1.1 Operational level



Definition

Transaction processing systems: A transaction processing system (TPS) performs and records the daily, routine transactions necessary to conduct business – for example, sales order entry and hotel reservations.

TPSs provide operational level data. Operational managers need systems which keep track of the everyday activities and transactions in an organisation, such as sales, receipts, payroll, and the flow of materials in and out of inventory. The principal purpose of TPSs is to provide answers to routine questions (for example, how many units of a product are in stock?) and to track the flow of transactions through an organisation (for example, what has happened to a supplier's payment?).

However, TPSs are often vital for the successful running of a business. For example, how would airline companies operate without their computerised reservation systems, and how would supermarkets operate without their computerised EPoS tills?

3.1.2 Tactical level

Management information systems (MISs) provide middle managers with reports on an organisation's current and historical performance. MISs summarise data from TPSs and enable it to be presented in reports which can be used to monitor and control the business. Typically, MISs provide answers to routine questions which have been specified in advance and which have a predefined procedure for answering them. For example, MIS reports could be used to compare monthly sales figures for different products to planned targets. MISs have the following:

- support **structured** decisions at operational and management control levels
- designed to report on **existing** operations
- have little analytical capability
- relatively **inflexible**
- have an **internal** focus

By contrast to MISs, **decision support systems (DSSs)** can be used for less routine decision making. They focus on problems which are unique or rapidly change, and which may not have a predefined procedure for finding their solution. For example, DSSs might be used to assess the impact on production schedules if sales were double for a month.

Although DSSs use internal information from TPSs and MISs, they often also incorporate information from external sources, such as competitors' product prices.

DSSs have more analytical power than other systems, enabling them to analyse and condense large volumes of data into a form that helps managers make decisions. The objective is to allow the manager to consider a number of **alternatives** and evaluate them under a variety of potential conditions.

3.1.3 Strategic level

Senior managers need information systems which address strategic issues and long-term trends, both within an organisation and in the external environment. They are concerned with questions such as where does our organisation fit in the industry? What new products should we be offering? What new acquisitions would help protect us from cyclical business swings?

EISs XE 'Executive information system (EIS)' or executive support systems (ESSs) help senior managers make these decisions. ESSs incorporate **external data** (about industry trends, forecasts, or external events such as tax changes or the emergence of new competitors) as well as summarised **internal information** from MISs and DSSs.

An EIS is likely to have the following **features**:

- flexibility
- quick response time
- sophisticated data analysis and modelling tools

The outputs from ESSs are often presented as graphs or charts, and are increasingly presented in the form of **digital dashboards**.

Interactive question 6: Health club

KLL is a large health and fitness complex located in a capital city. Started seven years ago, the business has been profitable. The introduction of a much wider range of activities over the past few years has led to increased complexity of administration and difficulty in interpreting the rapidly growing basic data generated daily. This data remains largely unstructured and this in turn leads to uncertainty in decision making.

The present MIS is able to produce monthly reports on the performance overall but can only break down the key indicators of revenue and gross profit into six broad categories: water sports, sports hall activities, fitness training, beauty treatments, squash courts and outdoor sports. Thus there is no detail on specific activities such as table tennis, sauna room, badminton and soccer.

The Managing Director and the board cannot distinguish the profitable activities from the unprofitable ones. The Managing Director tells the board, 'We must have a management information system that can cope with our complex business; there are so many variables it is becoming impossible to make decisions with confidence. Sometimes we have detail we cannot interpret and sometimes we simply do not have enough good information.' The Finance Director points out that 'the staff are doing their best but they have limited technical knowledge and the software support company is often slow to help'.

The board have recognised that it is important to build an MIS to serve the company well into the future, and are currently reviewing various proposals for new systems.

Requirement

Prepare a memorandum to the board explaining the main purposes of a new MIS and the benefits the company could expect such a system to bring.

See **Answer** at the end of this chapter.

3.2 Enterprise applications

One of the key issues facing an organisation and its managers is the question of how to manage all the information in the different systems within the organisation. In particular, if the organisation has a number of different operating systems, how can these systems share information and how can work or activities be coordinated?

One solution is to implement enterprise applications – systems which span different functional areas and focus on executing business processes across the organisation.

There are four main enterprise applications:

- enterprise resource planning systems
- supply chain management systems
- CRM systems
- knowledge management systems

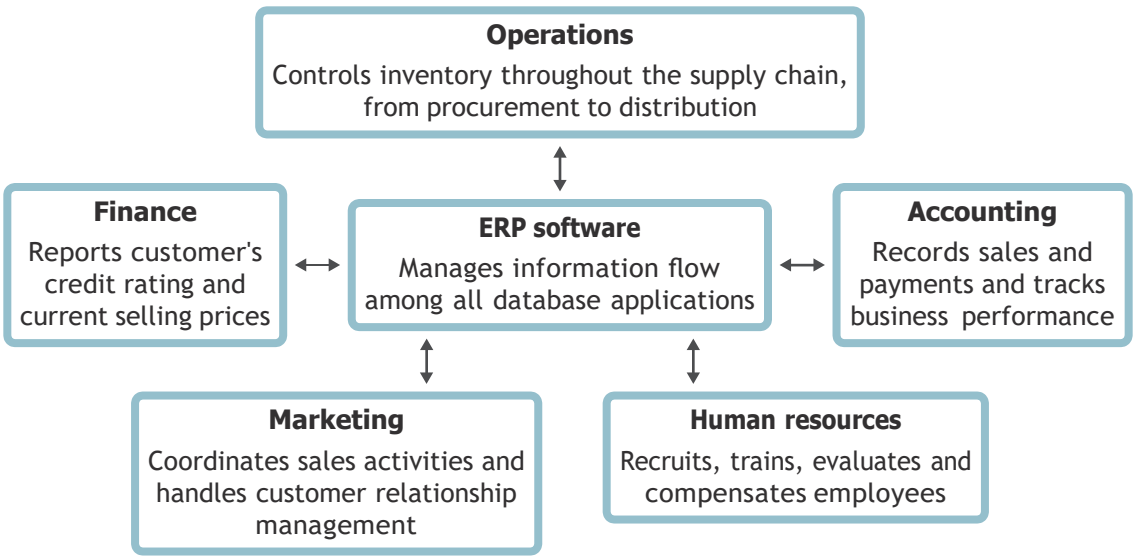
The presence of these integrated systems also serves as a reminder that management accounting information does not exist in isolation, but is part of the wider information system in an organisation.

3.2.1 Enterprise resource planning systems

Enterprise resource planning systems (ERPSs) are software systems designed to support and automate the business processes of medium-sized and large enterprises. ERPSs are accounting- orientated information systems which aid in identifying and planning the enterprise-wide resources needed to schedule, make, account for and deliver customer orders. They facilitate the flow of information between all business functions within an organisation, and they manage connections to outside stakeholders (such as suppliers).

ERPSs handle many aspects of operations, including **manufacturing, distribution, inventory, invoicing** and **accounting**. They also cover support functions such as **human resource management** and **marketing**. **Supply chain management** software can provide links with **suppliers** and CRM can help build relationships with **customers**.

Figure 9.1: Integration of information systems in an enterprise resource planning system (ERPS)



ERPSs thus operate **over the whole organisation** and **across functions**. All departments that are involved in operations or production are **integrated** into one system. In this way, adopting ERPSs makes firms more agile in the way they use information, meaning they can process that information better and integrate it into business procedures and decision making more effectively.

Some ERPS software is custom-built, and ERPS software is now often written for organisations in particular industries. ERPSs can be configured for organisations' needs and software adapted for circumstances. The data is made available in data warehouses, which can be used to produce customised reports containing data that is consistent across applications. Data warehouses can also **support performance measures**, such as the balanced scorecard, and strategic planning.

ERPSs should result in **lower costs** (for example, through workforce analytics and workforce redeployment) and lower **investment required** in assets. ERPSs should increase **flexibility** and **efficiency of production**, for example by coordinating procurement and logistics functions. They should also increase **customer to cash processes**, and thereby improve control of cash flow.

Their disadvantages include cost, implementation time and lack of scope for adaptation to the demands of specific businesses. Also, a **problem** with one function can affect all the other functions. ERPSs linked in with supply chains can similarly be vulnerable to problems with any links in the chain, and switching costs may be high. The blurring of boundaries can also cause accountability problems.



Context example: Enterprise Systems

(a) ERPSs

Case study

Laudon and Laudon (in Management Information Systems) offer the following illustration to show how organisations can benefit from enterprise systems.

Imagine a company has 10different major product lines, each produced in separate factories and each with separate, and incompatible, sets of systems controlling production, warehousing and distribution.

As a result, it will be difficult for managers to understand what is happening in the business as a whole, and it is likely that their decision making could be based on manual hard-copy reports, many of which will be out of date.

At the time they place an order, sales personnel might not know whether the items being ordered are in stock, and manufacturing staff will not easily be able to use sales data to plan for new production.

The company could benefit from an ERPS which collects data from the different product lines and factories, as well as from a number of key business processes – not just in manufacturing and production (including inventory management) but also in sales and marketing, finance and accounting, and human resources.

The benefit of such an integrated system is that when new information is entered by one process, such information is immediately made available to other business processes.

For example, imagine if the company makes automobile components. If a sales representative places an order for tyre rims on behalf of a customer, the system verifies the customer's credit limit, schedules shipment of the parts to the customer, and reserves the necessary items from inventory. If inventory stock is not sufficient to fulfil the order, the system schedules the manufacture of more rims, and orders any material or components needed from suppliers. Sales and production forecasts are immediately updated to reflect

the customer order. General ledger and cash levels are automatically updated with the revenue and cost information from the order.

Users across the company could log into the system and find out the status of the order at any time. Additionally, management could obtain information at any point in time about how the business was operating. The system could also generate company-wide data for management analyses for product cost and profitability.

(b) Supply chain management

Supply chain management systems can have similar benefits for organisations.

Laudon and Laudon highlight the example of the high-end bicycle manufacturer Cannondale.

Cannondale has supply and distribution chains which span the globe, and Cannondale offers its customers the option of 'made to order' models of bicycle. As a result, Cannondale has to manage over 200,000 individual parts, some of which have to be ordered from speciality vendors.

Managing the availability of parts in a constantly changing product line affected by volatile customer demand requires a great deal of manufacturing flexibility.

However, historically, Cannondale's material requirements planning system did not give it the flexibility it needed. The company's legacy material requirements planning system for planning production and controlling inventory could only produce reports on a weekly basis, meaning that the company was forced to substitute parts to meet demand, and sometimes lost sales as a result of these substitutions.

To overcome this issue, Cannondale purchased a new supply chain software system which collates data from operations at multiple sites and provides detailed, accurate, up to date information via an easy to use interface.

In addition, supply chain participants from different locations are able to model manufacturing and inventory data in 'what if' scenarios to see the impact of changes in demand across the entire supply chain.

The improved information from the new system enabled Cannondale to respond to customer orders much more rapidly with lower levels of inventory and without the need to hold extra 'safety stock' just in case it was needed.

Moreover, the lead times for producing bicycles have been reduced, and Cannondale's dates for promising deliveries have become more reliable and accurate.

3.3 Financial statement information

So far in this chapter, we have concentrated on the hierarchy of data and information in the context of MISs – for example, in the way that MISs summarise data from TPSs so that it can be presented in reports which can be used to monitor and control an organisation.

However, the data from TPSs not only feeds into management information and management accounts; it also forms the basis for the organisation's financial statements. For example, the sales figures from a retailer's shop tills will ultimately feed into the revenue figures in its financial statements.

Therefore, while an organisation needs to be able to capture and summarise transactions data accurately in order that managers can make informed decisions and control the business effectively, it equally needs accurate data to populate its financial statements.

This not only highlights the inherent links between management accounting information and financial accounting information; it also reinforces the importance of the internal controls in organisations over their TPSs to ensure that the data collected in them is accurate and reliable.

3.4 Sources of information

On several occasions in this chapter we have noted that different levels of decision may require data or information from either internal or external sources.

If an organisation is not able to capture reliable and accurate raw data then, even if it has sophisticated information systems, its managers will not have the quality of information they need to make informed decisions.

3.4.1 Internal information

Capturing data and information from **inside** the organisation involves designing a system for collecting or measuring data and information which sets out procedures for:

- what data and information is collected
- by whom
- how frequently
- by what methods
- how data and information is processed, filed and communicated

The accounting ledgers provide an excellent source of information regarding what has happened in the past. This information may be used as a basis for predicting future events.

TPSs and enterprise systems can also be sources of internal information, such as EPoS tills in retail stores.

Equally, sales teams deal with customers and so are in a good position to obtain information about customers and competitors.

Many companies also conduct market research. Although this generally deals with specific issues, it can indicate general environmental concerns (eg, consumers' worries).



Context example: EPoS Systems

Virtually all major retail stores now use electronic point of sales (EPoS) systems, which give them a fast and convenient way of transacting sales, while also recording vital business information.

At the most basic level, EPoS systems total up a customer's bill, calculate any change due and issue receipts in the same way that cash tills have historically done.

However, EPoS systems can keep track of inventory levels and can also record customer information. This ability to manage inventory and promote customer relationship management (CRM) helps EPoS systems improve a retailer's performance.

For example, by keeping track of the products sold, an EPoS system can assist inventory management by ensuring that the retailer has adequate supplies of a product to meet demand, and reorders top performing products as necessary. However, the system could also highlight which product lines are not selling very well, such that the retailer may question whether it wants to continue selling them, or whether it discounts their price to try to encourage demand.

Equally, if management wants to change the price of an item or run a special offer on it, this can be done very easily with an EPoS system. Importantly, from a performance measurement perspective, the system can also record data on how price changes have affected sales.

The data from EPoS systems can be used for marketing purposes, particularly when used in conjunction with store loyalty cards (such as Tesco's Clubcard). The systems can record trends and patterns in individual customers' behaviour, and in doing so they can provide valuable data for personalised marketing campaigns.

3.4.2 External information

Sources of information from outside sources include the following:

(a) **Media**. Newspapers, periodicals and television offer environmental information.

(b) Sometimes more detailed country information is needed than that supplied by the press. **Export consultants** might specialise in dealing with particular countries, and so can be a valuable source of information. The **Economist Intelligence Unit** offers reports into particular countries.

(c) Academic or **trade journals** might give information about a wide variety of relevant issues to a particular industry.

(d) **Trade associations** also offer industry information.

(e) Consultancy firms or analysts (eg Mintel) can also provide industry reports, or reports about different market sectors.

(f) The Government can be a source of statistical data relating to employment and unemployment levels, wage rates, overseas trade (including the balance of trade) and so forth, which is often summarised in newspapers. Official statistical sources also include government censuses, and demographic and expenditure surveys.

(g) Sources of technological environmental information can include the national patents office (because patents for new products are registered with the patent office).

(h) Stockbrokers produce investment reports for their clients which involve analysis into particular industries.

(i) The **internet** (for example, 'current awareness services' where subscribers can register particular key words related to their industry with media vendors and then receive automatic emails of articles and announcements that include those key words as tags). The websites of rival firms may also give an insight into their mission, objectives, strategy and financial performance.

(j) **Big data.** The growth of **social media** and the rise of the 'internet of things' have dramatically increased the amount of 'big data' which is available to organisations – as we discussed in the chapter Data analysis.

(k) **Annual reports** of competitors, suppliers or firms in a potential target market can also provide useful information.

3.5 Strategic management accounting and external information

We discussed strategic management accounting briefly in the chapter Strategic performance management of this Workbook.

Two of the key features of strategic management accounting are the importance it places on **external information** and on **non-financial information** in addition to internally generated financial information.

This external orientation highlights the extent to which customers, competitors and the external environment, as well as the actions of the organisation itself, affect an organisation's performance. So, for example, whereas a traditional management accountant would report on an organisation's own revenues, the strategic management would report on **market share** or trends in market size and growth.

(a) **Competitive advantage is relative**. Understanding competitors is therefore of prime importance.

For example, knowledge of competitors' costs as well as a firm's own costs could help inform strategic choices; a firm would be unwise to pursue a cost leadership strategy without first analysing its costs in relation to the cost structures of other firms in the industry.

This also highlights the importance of **benchmarking** – comparing performance to competitors.

(b) **Customers** determine if a firm has competitive advantage.

Some **examples** of strategic management accounting issues are provided below.

Item	Comment
Competitors' costs	What are they? How do they compare with ours? Can we beat them? Are competitors vulnerable because of their cost structure?
Financial effect of competitor response	Have sales fallen?
Product profitability	A firm should want to know not just what profits or losses are being made by each of its products, but also why one product should be making good profits while another equally good product might be making a loss.
Customer profitability	Some customers or groups of customers are worth more than others.
Pricing decisions	Accounting information can help to analyse how profits and cash flows will vary according to price and prospective demand.
Value of market share	A firm ought to be aware of what it is worth to increase the market share of one of its products.
Capacity expansion	Should the firm expand its capacity and, if so, by how much? Should the firm diversify into a new area of operations, or a new market?
Brand values	How much is it worth investing in a 'brand' which customers will choose over competitors' brands?
Shareholder wealth	Future profitability determines the value of a business.
Cash flow	A loss-making company can survive if it has adequate cash resources, but a profitable company cannot survive unless it has sufficient liquidity.

3.6 Linking strategic and operational information

One of the key challenges that organisations face is linking their (long-term) strategy to their day to day operations. For example, a strategic plan might set revenue growth targets for an organisation over the next five years, but the operational plan will need to consider what practical steps will be taken to generate these revenue increases, in effect creating a road map that defines the detail of how the overall strategies are going to be put into action. (We looked at the idea of strategy maps as an aid to implementing the balanced scorecard, in the chapter Strategic performance management.)

One of the underlying aims of multi-dimensional performance measurement systems, such as the balanced scorecard, is to help organisations align their operational objectives and initiatives with their overall corporate strategy and goals – to help them achieve those goals.

Importantly, however, an organisation needs to have systems in place to be able to capture the information it needs to assess how well it is performing.

As the context of strategic management accounting highlights, this performance information is likely to include non-financial performance, and could also include external elements (such as

competitor performance and market growth) as well as information about the organisation's own financial performance.

3.7 Critical success factors (CSFs) and information requirements

The use of **critical success factors (CSFs)** can help to determine the information requirements of an organisation.

CSFs are a small number of key **operational goals** vital to the success of an organisation. If these operational goals are achieved, the organisation should be successful. CSFs are **measured** by KPIs.

The CSF approach is sometimes referred to as the **strategic analysis** approach. The philosophy behind this approach is that managers should focus on a small number of objectives, and information systems should be focused on providing information to enable managers to monitor these objectives.

Two separate types of CSF can be identified:

(a) **Monitoring** CSFs are important for **maintaining** business. A **monitoring** CSF is used to keep abreast of existing activities and operations.

(b) **Building** CSFs are important for **expanding** business. A **building** CSF helps to measure the progress of new initiatives and is more likely to be relevant at senior executive level.

One approach to **determining the factors** which are critical to success in performing a function or making a decision is as follows:

- List the organisation's corporate objectives and goals
- Determine which factors are critical for accomplishing the objectives
- determine a small number of **KPIs** for each factor

Note that most KPIs will be quantitative. It is quite possible that CSFs will be quantitative as well.

One of the **objectives** of an organisation might be to maintain a high level of service direct from inventory without holding uneconomical inventory levels. This is first quantified in the form of a **goal**, which might be to ensure that 95% of orders for goods can be satisfied directly from inventory, while minimising total inventory holding costs and inventory levels.

CSFs might then be identified as the following:

- **supplier performance** in terms of quality and lead times
- Reliability of **inventory records**
- **forecasting** of demand variations

Deciding on the **KPIs** for each of these CSFs is not necessarily straightforward. Some measures might use **factual**, objectively verifiable data, while others might make use of **'softer' concepts** such as opinions, perceptions and hunches.

For example, the reliability of inventory records can be measured by means of physical inventory counts, either at discrete intervals or on a rolling basis. Forecasting of demand variations will be much harder to measure.

Where measures use quantitative data, performance can be measured in a number of ways:

- In **physical quantities**, for example units produced or units sold
- In **money terms**, for example profit, revenues, costs or variances
- in **ratios** and **percentages**

Sequence	Explanation	Example (1) UK National Health Service	Example (2) Mobile phone operator
Organisational goal	Overall strategy	Improve healthcare	Increase sales by entering new markets
CSFs	Operational goal: must be achieved for the overall strategy to be on track	Measurable reduction in time between booking an operation and receiving it	Establish network coverage in two countries in a year's time
KPIs	Data sharing performance on CSF	For example % of patients seen after waiting: less than one month less than three months less than six months more than six months	% of country covered and date, reported monthly
Critical information requirements	Information requirements to generate KPI	Booking and operations data to enable accurate KPI to be compiled	Information about masts installed

3.7.1 Data sources for CSFs

In broad terms, we can identify four **general sources** of CSFs (based on Rockart's work in this area in the 1970s and 1980s).

(a) The **industry** that the business is in. For example, in the supermarket industry, having the right product mix available in each store, and having products actually available on the shelves for customers to buy, will be prerequisites for an organisation's success, regardless of the detailed strategy it is pursuing.

(b) The **company** itself and its situation within the industry (eg, market leader or small company; competitive strategy; geographical location).

(c) The **external environment**, for example consumer trends, the economy, and political factors of the country in which the company operates (PEST factors).

(d) Temporary organisational factors, which are **areas of corporate activity** that are **currently unacceptable** and represent a cause of concern, such as high inventory levels. New laws or regulations could also be seen as temporary factors: eg, if a regulator has recently fined a financial services company for mis-selling its products, then a possible CSF for the company would be: to ensure that similar mis-selling does not occur again in the near future.

More specifically, possible internal and external data sources for CSFs include the following:

(a) **The existing system**. The existing system can be used to generate reports showing **failures to meet CSFs**.

(b) **Customer service department**. This department will maintain details of **complaints** received, **refunds** handled, **customer enquiries** etc. These should be reviewed to ensure all failure types have been identified.

(c) **Customers**. A survey of customers, provided that it is properly designed and introduced, would reveal (or confirm) those areas where **satisfaction** is high or low.

(d) **Competitors**. Competitors' operations, pricing structures and publicity should be closely monitored.

(e) **Accounting system**. The **profitability** of various aspects of the operation is probably a key factor in any review of CSFs.

(f) **Consultants**. A specialist consultancy might be able to perform a detailed review of the system in order to identify ways of satisfying CSFs.

3.8 MISs and competitive advantage

In section 2 of this chapter we noted that managers need information to make effective decisions and to control the activities of their organisation.

MISs have a critical role in providing them with this information. As such, the main purpose of MISs is to provide the right information, to the right people, at the right time.

In this respect, we can identify the following key benefits from MISs:

(a) **Implementation of management by objectives**: MISs allow management and staff to view, analyse and interpret useful data to set goals and objectives, and then to assess performance against those objectives.

(b) **Identify strengths and weaknesses**: Performance reports (for example, revenue reports, cost and productivity reports) can help managers identify strengths and weaknesses within an organisation, and consequently also improve its **business processes** and **operations**.

(c) **Generate competitive advantage**: If implemented properly, MISs can provide a wealth of information to allow management to construct effective plans to enable their organisations to outperform their competitors – for example, by enabling an organisation to produce products more quickly or more cheaply than rival firms.

Similarly, using customer data effectively can help an organisation align its products and its business processes to the needs of its customers. The effective management of customer data can also help an organisation with the segmentation and targeting aspects of its strategic marketing (which we discussed in the chapter Strategic marketing and brand management).

(d) **Fast reaction to market changes**: As the environment in which organisations operate becomes increasingly complex and dynamic, the speed with which an organisation can respond to opportunities and threats in that environment could become a source of competitive advantage.

As such, this speed of response is a **dynamic capability** for the organisation (as discussed in chapter Strategic analysis). An organisation's potential to sense opportunities and threats, to make timely and market-orientated decisions in response to the changing environment, and to change its resource base accordingly can be a source of competitive advantage to it.

An organisation's MISs are likely to be crucial in providing an effective platform which enables it to make timely and informed decisions in response to environmental changes.

If there is a sudden change in customer tastes or trends, firms with higher timely decision making capacity can grasp the opportunities this presents more quickly than competitors. As such, the quicker the relevant information becomes available to management, and the faster the decision making process, the more likely the organisation is to grasp the opportunities to gain competitive advantage over its rivals.

In order for managers to be able to respond rapidly and appropriately to change in their external environment, however, they must be capable of collecting internal and external information, identifying key strategic issues and making strategic decisions in a timely manner.

In this respect, by providing access to various external databases, EISs enable senior managers to search and retrieve a large amount of external information about suppliers, customers, competitors, and regulatory bodies and other stakeholders in a timely manner.

EISs can also transform traditional management reporting systems in order to provide senior management with more non-financial performance information in critical areas of their organisations.

As such, the internal and external information which EISs can provide for senior managers can lead to improved productivity, improved decision making in terms of quicker identification of potential problems and opportunities and the more successful introduction of new products.

Some writers have also noted that the use of EISs, along with DSSs, helps strategic decision makers to generate and analyse a greater number of alternatives – thereby increasing the comprehensiveness of the decision making process.

4 The value of information



Section overview

In this section we look at the factors which make information valuable. We also briefly examine the potential trade-off between the cost of obtaining information and the benefit gained from having that information.

4.1 Factors that make information a valuable commodity

Information is now recognised as a valuable resource and a **key tool in the quest for a competitive advantage**.

Easy **access** to information, the **quality** of that information and **speedy methods of exchanging** the information have become essential elements of business success.

Organisations that make **good use of information** in decision making and which use new technologies to access, process and exchange information are likely to be **best placed to survive** in increasingly competitive world markets.

4.2 The value of obtaining information

In spite of its value in a general sense, information which is **obtained but not used** has no actual value to the person who obtains it. It is only the **action taken** as a result of a decision which realises actual value for a company. An item of information that leads to an actual increase in profit of £90 is not worth having if it costs £100 to collect.

Businesses may also try to assess the costs of not having the information and also whether alternative (cheaper, more convenient) sources may be used instead.

4.3 Costs of information

Costs of information include the costs of system development and set-up, day to day running and storage costs.

Effective budgeting may be required to keep costs under control, particularly in the purchase of new equipment. An activity-based approach may be appropriate.

4.4 The value of information

There are a number of theoretical models which can be used to value information. However, most of them highlight the same factors in determining the value of information:

- The extent of uncertainty faced by decision makers
- The benefit of making the optimal decision compared to making a decision which is not optimal in the light of better information
- The cost of making use of the information and incorporating it into decisions

Importantly, information only has a value if alternative courses of action are available, and if these alternative courses of action will result in different results for an organisation. Information is most valuable for an organisation when many alternative courses of action are available but the costs associated with a 'wrong' action are high.

By contrast, if there are no alternative courses of action available or if a 'wrong' decision will not result in any net costs to an organisation, information relating to that decision has no value.

4.5 Cost-benefit analysis

In effect, we can evaluate the value of having better information in the context of a cost-benefit analysis.

Having relevant data and information available should improve the quality of decisions which are made, which in turn should improve an organisation's performance. However, there will also be a cost involved in capturing and analysing the data and information.

The critical question for an organisation is whether the benefits from having the information are greater than costs involved in obtaining that information.

4.5.1 The benefits of a proposed information system

In order to evaluate how the benefits of a proposed information system compare to the costs of the systems, an organisation needs to try to quantify the benefits in some way. Several factors need to be considered when trying to quantify the benefits.

Improved data collection, storage and analysis tools may indicate previously unknown opportunities for sales. Such tools may include software that allows relationships to be discovered between previously unrelated data.

New technology can be used to automate work which was previously manual. This saves staff time and may result in a smaller workforce being required.

Systems such as inventory control can benefit as losses from obsolescence and deterioration are reduced.

Computerised systems that create a more prompt and reliable service will increase customer satisfaction. In some cases it may be that providing decision makers with the most accurate and up to date information possible can have substantial benefits. The main areas of benefit areas follows:

(a) Models can be created to forecast sales trends and the likely effect on costs. Organisations that can make accurate forecasts are in a better position to plan their structure and finances to ensure long-term success.

(b) Organisations facing uncertain times, or those which operate in dynamic, evolving environments, need to make complex decisions (often quickly) to take advantage of opportunities or to avoid threats. Scenario planning models enable a wide range of variables to be changed (such as inflation rates and sales numbers), the overall effect on the business to be identified and a business plan to be constructed.

(c) Modelling can be extended into the market that the organisation operates in. Trends such as sales volumes, prices and demand can be analysed. Relationships between price and sales volume can be identified. These can be used by an organisation when deciding on a pricing strategy. Setting the best price for a product can help drive up sales and profitability.

(d) Organisations will benefit from improved decision making where systems can accurately evaluate a wide range of projects. Investment decisions often involve large capital outlays, and if the system prevents bad decisions it can prevent the organisation wasting large sums of money.

(e) Systems can also prevent an organisation agreeing 'bad' deals. Tenders for suppliers or other long-term contracts can prove costly if the wrong choice is made.

4.6 Strategic implications of information systems

When formulating an overall IT strategy the following aspects should be taken into consideration:

(a) What are the key business areas which could benefit most from an investment in IT, what form should the investment take, and how could strategically important units be encouraged to use such technology effectively?

(b) How much would the system cost in terms of software; hardware; management commitment and time; education and training; conversion; documentation; operational manning; and maintenance?

(c) The importance of lifetime application costs must be stressed – the costs and benefits after implementation may be more significant than the more obvious initial costs of installing an IT function.

(d) What criteria for performance should be set for IT systems? Two areas can be considered: the technical standard the information system achieves and the degree to which it meets the perceived and often changing needs of the user.

(e) What are the implications for the existing workforce – has it the requisite skills; can it be trained to use the systems; will there be any redundancies?

5 Evaluating management information and performance data



Section overview

- Information is only useful to managers or staff if it adds to their understanding of a situation.
- The characteristics of 'good' information can be summarised in the mnemonic 'ACCURATE': accurate, complete, cost-beneficial, user-targeted, relevant, authoritative, timely, and easy to use.

5.1 The objectives of management information

The objective of management accounting and management accounting systems is to provide information for managers to use for planning, control and performance measurement.

In order to evaluate how well the systems are providing this, managers need to assess whether the information available to them gives them what they need to know for planning, control and making decisions.

The management accountant's role is to provide managers with feedback information in the form of periodic reports – suitably analysed and at an appropriate level of detail – to determine whether the business is performing according to plan.

It may be the case that there is too much information available, or the information available is in a format unsuitable for managers to use. For instance, a production manager needs to know about outputs and costs in their department but not primarily about marketing data nor even necessarily summarised data that would go into a board report. Information overload can sometimes be as much of a problem as having too little information.

Accounting information needs to be distilled in a manner that makes it clear and concise and does not overwhelm the user. In this context it is important to highlight that, while management accounting involves the process of transforming data about an organisation's performance into information that managers can use for many reasons, management accounting only produces good information if it is useful and relevant to its users.

5.1.1 Presenting performance information

A number of developments in output reporting from information systems have been driven by the need to provide timely and tailored information, and also to avoid swamping the user with too much information.

Dashboards

Increasingly, companies are looking at ways to reduce the number (and size) of paper reports, and to provide the necessary information to decision makers in an easy to read manner.

One of the ways to do this is by using 'Executive Dashboards' which show current data, pictures, graphs and tables to illustrate how a business is performing and to help managers make better decisions. For example, a coffee and baked goods chain has been looking to expand and is preparing to open a number of new stores. The chain managers use dashboards to see the status of the new stores. The dashboards display geographical areas and the new stores which are being developed. By clicking on an individual store, executives can see details of how the new stores are being constructed and if any are being delayed.

Historically, much of the criticism of information systems and reports has focused on the difficulties users have faced when trying to produce the reports they wanted from the systems available.

Reporting tools tended to be rigid and imposed many requirements about the way reports were produced. However, current reports offer more flexibility, and thereby allow managers to get the reports they actually need or want.

Drill-down reports

Dashboards are often also combined with drill-down reports. Drill-down reports enable users to look at increasingly detailed data about a situation. For example, the sales managers could first look at data for a high level (such as sales for the entire company) and then drill down to a more detailed level (such as sales for individual departments of the company) if he or she is concerned about sales performance. The manager should then also be able to drill down to a very detailed level, possibly to look at sales for an individual sales representative. In this way, the manager can dictate the level of detail and information presented and can avoid being overloaded with too much initial detail.

Exception reports

Another way of managing the amount of information being presented, and thereby **preventing information overload**, is through the use of exception reports. Exception reports are reports that are only triggered when a situation is unusual or requires management action. For example, the parameters could be set so that exception reports are generated for all capital projects which exceed budget by greater than £100,000.

However, the key to using exception reports successfully is setting the parameters carefully. The aim of an exception report is only to highlight the situations which require management action. If the parameters are set too low (for example, all capital projects which exceed budget by over £100) then the manager will end up looking at too many items. Conversely, if the parameters are set too high (for example, capital projects which exceed budget by over £10 million) then situations which should receive management attention will not do so.

Because the aim of exception reports is to highlight situations which require management attention or action, they are best used to monitor aspects of performance which are important to an organisation's success. In this respect, exception reports could be used to report against KPIs, or other aspects of an organisation's performance relating to its CSFs.

Finally, in relation to the outputs of information systems as a whole, **users need to get involved when scoping what they require from their information systems**. If an MIS has immense capacity but does not give users the data they need individually, then the system is making life harder for the user.

5.2 Qualities of good information

The qualities of good information – both financial and operational – are outlined in the following table. You can use the mnemonic 'ACCURATE' to help you remember the qualities of good information. 'ACCURATE' can also be used as a framework when describing how poor information can be improved.

5.2.1 Qualities of good information

Quality	Example
Accurate	Figures should add up, the degree of rounding should be appropriate, there should be no typing errors, items should be allocated to the correct category, and assumptions should be stated for uncertain information.
Complete	Information should include everything that it needs to include, such as external data if relevant, comparative information and qualitative information as well as quantitative. Sometimes managers or strategic planners will need to build on available information to produce a forecast using assumptions or extrapolations.
Cost-beneficial	It should not cost more to obtain the information than the benefit derived from having it. Providers of information should be given efficient means of collecting and analysing it. Presentation should be such that users do not waste time working out what it means.
User-targeted	The needs of the user should be borne in mind; for instance, senior managers need strategic summaries periodically, and operational managers need more detailed performance information.
Relevant	Information that is not needed for a decision should be omitted, no matter how 'interesting' it may be.
Authoritative	The source of the information should be a reliable one (not, for instance, 'Joe Bloggs Predictions Page' on the internet unless Joe Bloggs is known to be a reliable source for that type of information). However, subjective information (eg, expert opinions) may be required in addition to objective facts.

Timely	The information should be available when it is needed. It should also cover relevant time periods, the future as well as the past.
Easy to use	Information should be clearly presented, not excessively long, and sent using the right medium and communication channel (email, telephone, hard-copy report etc).

5.2.2 Improvements to information

As well as being able to identify the qualities of good information, you may need to identify the problems that an organisation has with the information it currently produces, and suggest potential ways that information can be improved.

The table below contains some suggestions as to how poor information can be **improved**.

Feature	Examples of possible improvements
Accurate	Use computerised systems with automatic input checks rather than manual systems. Allow sufficient time for collation and analysis of data if pinpoint accuracy is crucial. Incorporate elements of probability within projections so that the required response to different future scenarios can be assessed.
Complete	Include past data as a reference point for future projections. Include any planned developments, such as new products. Information about future demand would be more useful than information about past demand. Include external data.
Cost-beneficial	Always bear in mind whether the benefit of having the information is greater than the cost of obtaining it.
User-targeted	Information should be summarised and presented together with relevant ratios or percentages. Consider use of graphics or dashboards to summarise data for senior management.
Relevant	The purpose of the report should be defined. Avoid trying to fulfil too many purposes at once. Consider whether several shorter reports would be more effective. Information should include exception reporting, where only those items that are worthy of note – and the control actions taken by more junior managers to deal with them – are reported.
Authoritative	Use reliable sources and experienced personnel. If some figures are derived from other figures the method of derivation should be explained.
Timely	Information collection and analysis by production managers needs to be speeded up considerably, probably by the introduction of better information systems (possibly even systems that can provide real-time information).

Feature	Examples of possible improvements
Easy to use	Graphical presentation, allowing trends to be quickly assimilated and relevant action decided on. Alternative methods of presentation should be considered, such as graphs and charts, to make it easier to review the information at a glance. Numerical information is sometimes best summarised in narrative form or vice versa. A 'house style' for reports should be devised and adhered to by all. This would cover such matters as number of decimal places to use, table headings and labels, and paragraph numbering.



Professional skills focus: Concluding, recommending and communicating

The ability to draw conclusions by distinguishing between the qualities of data provided is one of the professional skills tested in the ICABexams. When formulating recommendations, it is important to recognise the limitations of the data you have been provided with and, where appropriate, you may wish to suggest alternative information sources to help strengthen your recommendations and suggested strategies.

5.3 Completeness, accuracy and credibility of data and information

Another key feature which affects the quality and usefulness of information is the extent to which it accurately reflects real-world objects or events. For example, if an organisation's sales in a period are known to have decreased (such as due to a new competitor launching) but management information shows sales increasing, the management information's accuracy and usefulness for decision making must be called into question.

There are a number of factors which could potentially lead to poor-quality data and information:

Business dynamics change: A company expands into new markets but figures for the new markets are not incorporated into standard reports; a company purchases another company and has to consolidate figures from different software applications.

System design changes: Over time the design of databases may change, such as when new fields are added. This will not prevent transactions being recorded accurately, but can affect management information. For example, it may mean that current information is no longer being compared to historical information on a like for like basis.

Weak control over application changes: Cost (or time) pressure may lead business units to create or modify local applications despite not fully understanding the IT systems or software involved. As a result, these locally modified applications may no longer follow the same standards as applications in the rest of an organisation. This could lead to problems in consolidating or comparing data from the different systems.

Lack of common data standards or metadata: If an organisation doesn't have a standardised way of recording data, inconsistencies could arise if different operators record similar transactions or data differently.

Legacy systems: As companies grow, they start building new systems with new system architectures. However, the way data is structured in these may be different to the way data is held in existing legacy systems. If the 'legacy' systems are not enhanced to bring them in line with the new systems, it will be difficult to manage data between the two systems.

Time decay: Over time, the quality of data will decrease unless that data is updated. For example, customers on a customer database may change address or marital status. If a company is not aware of these changes, then the value of its data for marketing purposes declines.

Data entry issues: In many systems, there will always be a risk of human error, but this can be reduced by well-designed entry forms. For example, the risk of data entry error can be reduced if there are controls that prevent a telephone number entry being posted with insufficient digits.



Context example: Retail IT systems

The 2013 report *Audit Insights: Retail* by ICAEW's Audit and Assurance Faculty highlights that IT systems in retail have historically been developed in-house, resulting in a diverse and often inefficient architecture.

Often, retailers have only invested in IT after their investments in stores had been completed, with the result that few retailers have integrated IT systems and control.

However, the move towards online retail has now provided greater impetus to develop IT systems, and has prompted many retailers to review their overall IT strategies.

The ICAEW report notes that in challenging economic conditions, many retailers may find it difficult to make significant capital investment in IT, despite the risks of making do with less developed systems. Some retailers are now replacing their customer-facing systems to allow for the 'full multichannel experience'. Others are making do with existing systems, delaying the need for costly investment in IT with the associated risks of migrating to a new system. However, the report warns that: 'This is likely to prove a false economy, as retailers which do not adapt and embrace the opportunities of the changing environment may struggle to stay competitive. Furthermore, a lack of access to robust data increases risk across the business.' For example, failure to invest in IT improvement may lead to security and reputation risk if commercially sensitive data is not kept secure and up to date. Equally, the retailers' ability to monitor customers' shopping habits and to understand their needs will be impeded by a lack of data monitoring and analytics capacity.

Crucially, without relevant and reliable data, retailers will not be able to fully understand what gives rise to profits across different channels. A better understanding of the factors which give rise to profits could help retailers make more efficient use of their resources and identify their most profitable products. In turn, this will help improve pricing strategies and target promotional activity to drive profitable growth.

Finally, strong internal data systems will help retailers track and value their inventory more accurately, enabling them to have more detailed control over how much of their working capital is tied up as inventory at any given time.

In this respect, the report notes: 'As customers increasingly look to the internet and other non-traditional sales channels, strong internal data systems will allow different aspects of the business to work together more effectively. This includes inventory management, supplier relationship management, and efficient processing of orders and payments.'

Issues of accountability and quality control might also be relevant when considering the quality of information and data:

Accountability – Are managers held accountable for making sure that procedures (controls) are in place to ensure the completeness and reliability of data, and for making sure that those procedures are followed?

Quality control – Are there any systems tests to check the consistency and accuracy of the outputs from automated systems and databases? Are unexpected results investigated?

A company's internal audit department could play an important role in providing assurance over these areas.

5.3.1 Audit data analytics

In addition to the role a company's internal audit department plays in providing assurance over the quality of its data and information, we should also recognise that data analytics also provides external auditors the opportunity to monitor large, complete data sets, rather than testing samples as they might historically have done. As such, audit data analytics could be used to help the auditor identify trends, correlations and deviations from expected outcomes, and/or potential risks in a client's data or control systems.

One of the other important features of data analytics is the capacity to show results graphically. So, for example, an auditor could use audit data analytics to analyse a client's gross margins and sales, split by product type or region, and the results could be illustrated on a dashboard, with different products being shown in different colours, to highlight unusual results. Moreover, the current year figures could also be compared with prior years to gain a deeper understanding of any outliers, variations or trends, and to help an auditor determine what further audit tests might be needed to gain assurance over the figures.

5.3.2 Business performance management software

Although business performance management should not be primarily about software, organisations need to consider whether their performance management software is suitable for managing performance effectively and efficiently.

Many organisations still rely on office tools (such as Microsoft Excel and PowerPoint) as their main technology to analyse and report performance data. However, particularly for large and complex organisations, spreadsheets may not be appropriate for performance management.

For example, many spreadsheets contain significant **errors**. A lack of version control, and a lack of logging changes over time, lead to errors which could compromise the reliability of data in the spreadsheets and impede management's ability to make decisions based on data from the spreadsheets.

Scalability: Large organisations are likely to find that the amount of data to analyse means that spreadsheets grow into big documents with colour coding, macros, calculations etc. In turn, this causes spreadsheet-based applications to become slow and prone to crashing. Often, there is just too much data and complexity in the spreadsheet.

Equally, spreadsheet-based solutions are often **manually fed and updated**. As well as increasing the risk of human error, this makes them very time consuming, as business analysts have to spend time updating the spreadsheets on a regular basis.

Therefore, organisations should consider whether it may be more appropriate for them to use specialist performance management software rather than relying on standard office tools (such as Excel). For example, performance management software could provide managers with interactive drill-down capabilities to analyse performance data, and it could also provide business intelligence features such as trend analysis, root-cause and impact analysis, and simulation and scenario features.



Context example: The problems with using spreadsheets

Spreadsheets are widely used in businesses around the world but problems can occur, particularly when spreadsheets are used to manipulate large volumes of data. In the middle of the coronavirus pandemic Public Health England admitted that they had missed out 16,000

confirmed cases of coronavirus in their reporting between September 25th and October 2nd 2020. Although the 16,000 confirmed cases were informed of their positive test, the omission had serious consequences since it prevented the NHS Test and Trace system from tracking down close contacts of those who had tested positive. So, what caused the error in the first place? On investigation it became clear that laboratory

data was being transferred to Excel spreadsheets which were reaching their limit in terms of the number of rows that they could process. Whilst current Excel files can handle 1,048,576 rows and 16,384 columns of data, the companies analysing swab tests were using older Excel files that could only handle 65,000 rows of data (approximately 1,400 cases). This error and the knock-on effect on confidence in Public Health England clearly illustrates the limitation of relying on spreadsheets.

Despite its many uses and applications, Excel is not always the right tool for the job and all organisations should ensure their data analysis systems are sufficiently robust to cope with the task at hand.

Source: Clough,P (2020) *Why you should never use Microsoft Excel to count coronavirus cases* [Online] Available from: https://theconversation.com/why-you-should-never-use-microsoft-excel-to- count-coronavirus-cases-147681 [Accessed 5 July 2021]

5.4 Information risks

In 2015, the UK Government published a paper – *Managing Information Risk*– which included a summary of the key areas of information risk an organisation needs to consider. Information risks are risks which affect an organisation's guardianship and management of information. In this respect, it is important to note that information risks are not necessarily the same as IT risks, although managing IT security is likely to be a key component of any strategy to manage information risks.

Risk category	Example of risk
Governance and culture	Lack of comprehensive oversight and control (so anything can go wrong) When something goes wrong, handling it badly and not learning from it (so the problem can happen again) Third parties (eg, suppliers) let you down (eg, because they are not made aware of the standards/timetables required of them) New business processes don't take information risk into account
Information management and information integrity	Critical information is lost (with legal, reputational or financial consequences) Critical information is wrongly destroyed, not kept, or can't be found when needed Lack of basic disciplines in recordkeeping (eg, records are incomplete) Inaccurate information (which causes the wrong decisions to be made or the wrong action to be taken) Electronic information becomes unreadable due to technical obsolescence (with legal, reputational or financial consequences)
The human dimension	Despite having procedures and rules, staff, acting in error, do the wrong thing or make mistakes (meaning things go badly wrong) Despite having procedures and rules, 'insiders', acting deliberately, do the wrong thing (and things go badly wrong)

	External parties get your information illegally (and expose it/act maliciously/defraud you or your customers)
Information availability and use	Inappropriate disclosure of sensitive personal information (causing reputational damage or worse) Failure to disclose critical information when required Failure to utilise the value of the information asset Failure to allow information to get to the right people at the right times (leading to a failure to service customers)

Sources of internal assurance

As well as identifying these risks, the government paper suggests potential sources of internal assurance over them. These include the following:

- identifying a board-level senior information risk owner, supported by a team, to manage the organisation's information and information risks
- identifying key information assets across the organisation (in terms of both information content and information systems)
- producing and regularly updating a risk register for the organisation's information risks with key risks prioritised and action plans in place to address them
- Compliance with legislation and key standards (eg, GDPR); spot checks to ensure data quality is being maintained
- Clear guidance and rules about what information needs to be kept, how long it needs to be kept, and where it can (or cannot) be stored
- Mandatory training in place for asset owners and users of information systems
- Controls in place to restrict access to (or ability to change) key files; audit checks on inappropriate use of key systems, personnel security
- Factoring information management into business and system design processes
- Backups of key information; backup systems held in a secure, separate location
- Strong, regular engagement of the Audit Committee with information risks
- 'Whistleblowing' procedures in place and understood by all staff
- Mapping of key suppliers, their associated information assets linkages, and their risks
- clear standards and contractual obligations for suppliers to meet

5.5 Information systems and assurance

In this chapter we have highlighted a number of ways in which organisations use IS and IT.

Equally, however, we must recognise that the increasing use of computer systems brings certain risks to an organisation, which in turn could have an impact on the organisation's financial statements, or on the decisions made by management. Decision makers need to be confident in the credibility of the information they are using to make decisions.

Two key risks of using computerised systems are:

- The system is put at risk by a virus or some other fault or breakdown which spreads across the system
- the system is invaded by an unauthorised user who could then disrupt the smooth operation of the system, or obtain commercially sensitive information from it

Consequently, it is important for an organisation to ensure that its systems are as reliable as possible, and that they are the best systems at the given cost.

If the organisation has purchased its information systems from an external service provider it might seek these assurances from its service provider. However, the service provider has a vested interest in believing that its system is reliable and the best available, because they are paid to supply it.

Therefore, the organisation might seek an assurance service from its auditors to undertake work to ascertain whether the assertions made by the service provider are correct. In other words, the auditors might be asked to undertake an assurance assignment to report on the reliability and adequacy of an organisation's IT systems.

If a firm of accountants is considering taking on such an assurance engagement, it must ensure that it has staff with sufficient skills and experience to undertake the procedures required. To this end, there must be an IT specialist on the engagement team.

Information subject to assurance

More generally, there is a wide range of information which could be subject to some form of external assurance. The following are examples of areas where an external assurance service might be requested:

(a) **Quantitative information, including non-financial information and performance measures such as KPIs** – the range of information which organisations now disclose (or have to disclose) about themselves has gone far beyond traditional financial reporting. However, if organisations are disclosing this information externally (for example, as part of their Annual Report) it follows that they also need external assurance on the quality of that information. (We discussed assurance in relation to KPIs in chapter Strategic performance management earlier in this Workbook.)

In relation to **business performance measurement** an entity could seek assurance that its performance measurement systems contain relevant and reliable measures for assessing the degree to which its goals and objectives are achieved, or how its performance compares to that of its competitors.

(b) **Environmental information** – For example, if an entity has stated a performance target to reduce greenhouse gas emissions, it will need someone to measure its level of emissions and verify the degree to which they have been reduced. (Again, we discussed assurance over reporting environmental impacts in the chapter Strategic performance management.)

(c) Aspects of IT such as information flows and security over those information flows. In particular, **IS reliability** – assurance that an entity's internal information systems provide accurate and reliable information for operating and financial decisions.

(d) **Electronic commerce** – Assurance that systems and tools used in electronic commerce provide appropriate data integrity, security, privacy and reliability.

(e) Compliance with contractual obligations.

(f) Risk assessment; risk management systems and processes– assurance that an entity's profile of business risks is comprehensive, and an evaluation of whether the entity has appropriate systems in place to effectively manage those risks.

(g) Internal controls and the internal control environment.

(h) Governance, strategy and management processes.

Few organisations can function effectively without IT systems supporting their key business processes, and many cannot function at all if their systems fail. However, organisations' systems may be vulnerable to attack or failure, either as a result of flaws in the design of the systems, failure to apply security patches, or poor security management.

Unauthorised access to an organisation's systems and data could have serious financial or legal implications, as well as potentially damaging the reputation of the company. In this respect, the directors of the organisation might seek an assurance service from their auditors,

or another firm of accountants, that the organisation's technology systems and the processes they support are functioning as intended. For example, are security systems designed to reduce the risk of unauthorised access to systems and data reliable?

Systems audit

An example of an assurance assignment might be a request to report on the adequacy of the internal controls in place. Internal control effectiveness is generally assessed by means of a systems audit.

As part of any external audit, auditors are required to assess the quality and effectiveness of an entity's accounting system, which necessarily includes a consideration of any computer systems in place within the entity which are linked to its accounting system.

However, auditors could also accept an assurance engagement outside of the audit to report specifically on how reliable an entity's information systems are.

Key areas which an assurance engagement is likely to concentrate on are:

(a) Management policy

(1) Does management have a written statement of policy in relation to computer systems and other information systems?

(2) Is that policy compatible with management policy in other areas?

(3) Is that policy sufficient and effective? Is it adhered to?

(4) Is the policy updated when any systems are updated? Does it relate to the current systems?

(b) Segregation of duties

(1) Is there adequate segregation of duties in relation to data input?

(2) Are there adequate systems controls (eg, passwords) to enforce segregation of duties?

(c) Security

(1) Is there a security policy in place covering physical security (locked doors/windows), access security (passwords) and data security (virus shields)?

(2) Is the security policy sufficient and effective? Is it adhered to?

General controls and application controls

When testing the control environment in an entity, an auditor should assess general controls as well as application controls.

The purpose of **general IT controls** is to establish a **framework of overall control** over the computer information system's activities to provide a reasonable level of assurance that the overall objectives of internal controls are achieved.

The areas covered by general IT controls include:

- development of computer applications
- prevention and detection of unauthorised changes to programs
- testing and documentation of program changes
- controls to prevent unauthorised amendments to data files
- controls to ensure continuity of operations (eg, backup and emergency procedures)

The purpose of **application controls** is to establish specific control procedures over accounting applications to provide reasonable assurance that all transactions are authorised and recorded, and are processed completely, accurately and on a timely basis.

Application controls include data capture controls, data validation controls, processing controls, output controls and error controls.

Controls and assurance over e-commerce

Earlier in this Workbook, we noted how e-commerce has provided new growth opportunities for entities. However, it is equally important to note that an entity using e-commerce also needs to have internal controls in place to mitigate against the risks associated with e-commerce.

In particular, these risks include security issues (eg, ensuring customers' transactions are secure) and process alignment (eg, if a website is not automatically integrated with the internal systems of the entity, such as its accounting system and its inventory management system, the entity will need to ensure that it processes transactions completely and accurately).

A key issue with e-commerce is trust. In many cultures, consumers grant their trust to business parties that have a close physical presence. On the internet this physical presence is simply not there. The seller's reputation, the size of the business, and the level of customisation in products and services also engender trust.

Internet merchants need to elicit consumer trust when the level of perceived risk in a transaction is high.

However, research has found that once consumers have built up trust in an internet merchant, such concerns are reduced.

Internet merchants need to address issues such as fear of invasion of privacy and abuse of customer information (about their credit cards, for example) because these issues can stop people even considering the internet as a shopping channel.

The parties involved in e-commerce need to have confidence that any communication sent gets to its target destination unchanged and without being read by anyone else.

WebTrust and SysTrust are examples of assurance services developed in the last few years in relation to e-commerce. The underlying principles of these two services have been combined into one common set of principles known as Trust Services, which allow auditors to evaluate business systems and controls.

Trust Services are based on five principles:

- **Security** – The system is protected against unauthorised access.
- **Availability** – The system is available for operation and use as agreed.
- **Processing integrity** – System processing is complete, accurate, timely and authorised.
- **Online privacy** – Personal information obtained as a result of e-commerce is collected, used, disclosed and retained as agreed.
- **Confidentiality** – Information designated as confidential is protected as agreed.

WebTrust and SysTrust can be used to provide assurance on an organisation's website and on its systems respectively. Such assurance engagements are performed as reasonable assurance engagements in accordance with ISAE 3000 (Revised).

6 Cybersecurity



Section overview

It has always been important for businesses to keep their data and information systems secure. However, as businesses increasingly use digital technologies, cybersecurity – protecting systems, networks and data from cyberthreats – has become a critical issue for them. Cybersecurity failures can cause significant damage, for example through business disruption or reputational damage.

One of the recurring themes throughout this Workbook has been the increasing importance of information technology and digital technology to contemporary organisations; for example, through the way the digital world brings business partners, consumers and suppliers together, and enables organisations to gather and use information (for example, in relation to 'big data').

However, as the ICAEW Information Technology (IT) Faculty report *Audit Insights: Cyber Security* (2018) highlights, businesses need to understand the risks associated with new technologies as well as the opportunities they provide:

[A] key challenge in cybersecurity stems from the continually changing nature of the risk, especially given the fast pace of change in technology. New technologies provide both opportunities and challenges for cybersecurity. Big data, artificial intelligence and all types of automation, for example, create new risks and new targets for attackers. Greater reliance on algorithms and data for all business operations increases the potential impact of breaches and attacks. These technologies also provide greater opportunities to improve cybersecurity through more automated controls, better understanding of anomalies on networks, and better prediction around the behaviour of attackers.

Crucially, as businesses increasingly use digital technology to transform their business operations and the way they engage with customers, so cybersecurity – the protection of systems, networks and data in cyberspace – is also becoming an increasingly important issue for all businesses.

Moreover, as the IT Faculty report suggests, cyber-risks are constantly evolving:

- Threats change as attackers get more sophisticated and find new ways of breaking into systems. New attackers also emerge, as easy-to-use hacking tools reduce the level of skill required to carry out attacks.
- Vulnerabilities change as businesses digitally transform their business models and ways of working, potentially creating new weaknesses in business processes; eg, through mobile ways of working.
- Existing controls and assurance models are superseded by new approaches. For example, moving to cloud-based systems has made traditional assurance models around IT controls more difficult to apply.
- The impact of failures increases, as businesses capture more data and rely more heavily on digitally-based services. A slow or poor response to a major breach is very quickly and publicly shared over social media, increasing at least short-term reputational damage.

6.1 Cybersecurity controls

As the ICAEW IT Faculty report *Audit Insights: Cyber Security* (2013) points out, cybersecurity requires an organisation to have controls in place to address a wider range of risks than the ones traditional 'information security' is designed to safeguard an organisation against.

Traditional approaches to information security have focused on the internal controls needed to maintain the confidentiality, integrity and availability of data. However, controls relating to cybersecurity need to incorporate external factors:

- Potential threats can now come from around the world, and could involve organised criminals, corporate spies and 'hacktivists' as well as disgruntled or careless employees.
- Security weaknesses can be found throughout a supply chain, not just within a single business. Similarly, the disparate nature of data storage across servers, cloud storage and mobile devices provides a range of access points for attackers to exploit.
- The impact of security failures can extend across every aspect of a business, including disruption of operations and customer service, interference with production control systems, damage to brand and reputation, theft of intellectual property or commercially sensitive information and regulatory fines.

Context example: Target Corporation

Target Corporation is the second-largest discount retailer in the United States, with around 1,800 stores there.

A data breach of Target's systems in December 2013 compromised the personal details of up to 110 million customers. One set of data (affecting up to 70 million people) included a mix of names, mailing addresses, phone numbers and email addresses. A second set of data (affecting up to 40 million people) included details of customers' credit and debit card accounts.

Banks that issue credit and debit cards were also concerned that the thieves could use the stolen addresses and phone numbers to contact customers and try to get them to divulge account information. In turn, that could have resulted in fraud losses for the banks.

Target admitted that thieves had broken into its point of sale system, installed malicious software in the system, and stolen credit and debit card data in an attack which lasted for more than two weeks – from 27 November to 15 December 2013.

The timing of the breach (in the run-up to the Christmas holiday season) meant that its impact was particularly significant. Target reported a 46% drop in net profit over that quarter, and incurred $61 million in costs related to the breach (including reimbursements to card networks to cover fraud and the cost of issuing new cards, as well as legal costs associated with the various investigations arising from the security breach).

Perhaps more importantly, the data breach – and the negative publicity resulting from it – reduced customer satisfaction and badly affected Target's efforts to sign up more customers for its in-house credit and debit cards, which were a key part of its strategy to prevent customers from defecting to competitors such as Amazon.

Target's share price also fell 11% after the breach was revealed, and the breach was a key factor in the departure of the CEO and the CIO. The positions of some non-executive directors and members of the audit committee were also threatened, based on their poor level of oversight.

It emerged that attackers accessed Target's data through the compromised systems of a heating, air conditioning and refrigeration supplier, illustrating the risks of integrated supply chains. Another weakness which the attackers exploited was the point-of-sale system, which had been customised to provide enhanced marketing opportunities without fully considering the security risks involved.

In view of the increased importance of cybersecurity, the ICAEW IT Faculty report *Audit Insights: Cyber Security* (2013) raises four 'flags' which it suggests organisations should consider in relation to it:

(a) **Business should consider 'cyber' in all their activities** – Businesses' ability to make use of information about customers, supply chains, competitors and others will increasingly influence business success. But alongside the opportunities provided by digital technology and the internet there are risks around the security of important information and the digital infrastructure. Businesses need to manage these cyber-risks to ensure that they exploit the opportunities in a secure and sustainable way.

As such, ICAEW's report recommends that businesses should consider cybersecurity as a **business risk** (rather than a technical risk) and should consider the implications for cybersecurity of all their strategies and operations. (If we think back to the example of Tesco's business risks we considered in the chapter Strategic analysis of this manual, we can see that Tesco includes 'Data security and privacy' as one of its key business risks.)

Considering cybersecurity as a business risk could be particularly important for businesses hoping to enter a new supply chain, or considering an acquisition. Equally, having a plan in

place to deal with cyberattacks could also provide a competitive advantage against competitors in the market. For example, it is important for organisations in a supply chain to be seen as trusted partners, and having strong security capabilities could be one way in which a supplier develops its reputation with its customers.

(b) **Businesses need to accept that their security will be compromised** – Although businesses still need to apply appropriate preventative controls to protect their information, they need to operate in a way which assumes that some of their information will inevitably be accessed by others.

This means that intelligence and monitoring, detection and response are increasingly important aspects of security.

(1) **Intelligence and monitoring** – Intelligence on potential threats can be gathered from a variety of sources. For example, businesses could monitor social media to identify whether data has been leaked into the public domain; or they could participate in information sharing schemes with trusted partners or the government and its security services to understand attacks experienced by others and to help prepare for similar attacks in future.

(2) **Detection** – Attackers can potentially breach systems for weeks or months before being detected, stealing large amounts of sensitive data in the process. Therefore detecting a breach early is imperative, in order to tackle it and limit the damage caused. Data analytics can be important in this respect – if a business has a clear baseline (which represents normal activity) it will be easier to identify abnormal activity (eg, high levels of data downloading, systems activity at unusual times of day, or access from unexpected places).

(3) **Response** – As well as implementing technical remedies to breaches, a business may also need to manage a variety of stakeholder concerns to limit the overall impact of the breach – for example, communicating with customers who may have been affected by the breach and responding to their concerns; or ensuring compliance with any regulatory requirements; and keeping investors informed.

The growth of social media exposes businesses which do not respond effectively to breaches. News of data breaches or compromises can spread very quickly across platforms such as Twitter. If a business is slow to respond, and only provides limited information to customers, this can amplify the impact of the breach on the business's reputation.

(c) **Businesses should focus on their critical information assets** – It is almost inevitable that businesses will not be able to protect all of their information at all times. So, instead of trying to do this, businesses should prioritise their information assets and focus their security resources on the key assets. In particular, entities should identify where breaches would have a substantial impact on the competitiveness and sustainability of their business, and focus their security in these areas.

(d) **Most businesses don't get the basics right** – ICAEW's report states that up to 80% of security breaches could be prevented by implementing basic good practices in cybersecurity. However, businesses of all sizes, and in all industries, struggle to get the basics right. Complex IT environments, and lack of expertise, can both be reasons for this, but people are the weakest link in implementing effective security, and human failings are increasingly being exploited by attackers to gain access to confidential information.

More generally, the increased risk that cyberthreats pose to businesses also highlights the need for management to have a framework for assessing and treating risks. The International Organisation for Standardisation's ISO 27001 – *Information security management* describes best practice for an information security management system, and identifies the requirements for establishing, implementing, maintaining and continually improving such a system. As such,

complying with ISO 27001 should help an organisation improve its defences against cyberattacks, and, by doing so, increase the confidence that customers and other interested parties have in the organisation's systems.



Context example: CESG – Ten steps to cybersecurity

CESG, the National Technical Authority for Information Assurance in the UK, has found that, in a number of cyberattacks, attackers exploit basic weaknesses in organisations' systems.

Therefore, for many organisations, the first step in implementing an effective risk management strategy should be to remove these basic weaknesses.

CESG has identified 10 key steps which it believes are crucial in helping organisations to protect themselves against the majority of cyberthreats:

(a) **Defining the organisation's risk management regime** – This is central to an organisation's cybersecurity strategy, because it helps to determine the organisation's risk appetite.

(b) **User education and awareness** – Produce user security policies covering acceptable and secure use of the organisation's systems. Establish a staff training programme. Maintain user awareness of the cyber-risks.

(c) **Home and mobile working** – Develop a mobile working policy and train staff to adhere to it. Ensure all devices comply with the minimum requirements of the policy. Protect data in transit and at rest (eg, through storing devices securely).

(d) **Secure configuration** – Apply security patches and ensure that the secure configuration of all ICT systems is maintained.

(e) **Removable media** – Produce a policy to control all access to removable media (eg, USB sticks). Limit media types and use. Scan all media for malware before importing on to the corporate system.

(f) **Managing users' privileges** - Establish account management processes which control user privileges and monitor user activity (eg, monitoring websites users visit).

(g) **Incident management** – Establish an incident response and disaster recovery capability (including the creation of an incident management team). Produce and test incident management plans. Report criminal incidents to law enforcement.

(h) **Monitoring** – Establish a strategy for monitoring all ICT systems and networks, and then continuously monitor them. Analyse logs for any unusual activity that could indicate an attack.

(i) **Malware protection** – Establish anti-malware defences that are applicable and relevant to all business areas. Scan for malware across the organisation.

(j) **Network security** – Protect the organisation's networks against external and internal attack. Manage the network perimeter. Filter out unauthorised and malicious content. Monitor and test security controls.

As with any other risks, cyber-risks need to be managed proactively by the Board. Similarly, part of the corporate governance activity should focus on obtaining assurance that the organisation is pursuing an effective information risk management regime. Is it managing the right information risks and cyber-risks for its business? Are the necessary security controls in place to meet these risks? Are the security controls regularly tested, monitored and reviewed?

In relation to this, CESG suggests that the Cyber Essentials scheme (see next section on cybersecurity and assurance) defines the minimum set of security controls an organisation (large or small) should have in place to mitigate the risks from cyberattacks.

ICAEW's IT faculty has identified its own '10 steps to cybersecurity', which – not surprisingly – reiterate some of the steps identified by CESG:

(a) **Allocate responsibilities** – As with any business activity, it's crucial to identify what must be done, and who will do it. Overall responsibility should rest with a senior manager (for example, the CIO), who has a broad view of all the risks and how to tackle them. Other individuals can then be assigned specific roles – for example, installing security software.

Management should identify the information and technology which are really vital to the business, and therefore where the key risks lie. For example, damage to a business's financial system, or loss of its customer list could lead to the failure of the business.

Identifying the key risk areas, then establishing what security measures already exist, whether they work, and what extra measures may be required will allow an organisation to target its security efforts where they are most needed.

(b) **Protect your computers and your network** – Malicious activity could come from outside a business or from inside. Attacks from outside (eg, from hackers) can be protected by installing a firewall.

The firewall can also be used to manage staff's internet activity, for instance by blocking access to sites where employees might encounter security risks.

(c) **Keep your computers up to date** – Suppliers of software and operating systems (such as Windows) frequently issue software updates ('patches') to fix problems or to improve security.

It's essential to keep all computers and other devices up-to-date with the latest patches. A single, vulnerable computer in a network can put all the others at risk.

(d) **Control employee access to computers and documents** – Although computers should be guarded with a firewall, it is still necessary to protect user accounts and documents with passwords. Each individual user should have a unique user name and password, which can be used to regulate users' access to different parts of the system.

The use of passwords and access rights not only controls staff access to systems and information, it can also provide further security against outside intrusions.

(e) **Protect against viruses** – Ensure that computers are fitted with up-to-date anti-virus software. This software regularly scans a computer in search of malicious software (or 'malware') and deletes any that is found. Regular updates to fend off new threats are key to anti-virus software.

(f) **Extend security beyond the office** – If employees work away from the office – using laptops, phones or tablets – make sure these are also covered by anti-virus software, password protection and (where applicable) a firewall. Ensure that sensitive data is kept in an encrypted are of the computer or device.

Employees should also be reminded of the potential dangers of connecting to unencrypted public wi-fi, because hackers can intercept data.

(g) **Don't forget disks and drives** – Removable disks and drives (such as USB sticks) pose security risks in two ways: they can introduce malware into computers, and they can be mislaid when containing sensitive information.

Ensure that as far as possible, employees only use disks and drives owned by their business are used in their computers, and set up anti-malware software to scan them whenever they are used in the office.

(h) **Plan for the worst** – No system is 100% secure, so it is worth planning how you would react if things go wrong. Do you need external help (eg, a specialist computer company)? Do you need to contact key customers or suppliers to explain there is a problem? Can some functions be continued using other computers while systems are repaired?

Ensure there is a clear plan identifying who is responsible for doing what in an emergency. This may incorporate elements of plans for other disasters, such as a fire on the business' premises.

(i) **Educate your team** – It is vital to tell everyone in the business that security matters, and to ensure that they know how they can help to uphold security (for example, by keeping passwords safe, and changing them regularly).

Training sessions or written policy documents may help remind staff about the importance of security, and the risks they may face. For example, reminding them not to click on web links or email attachments from unfamiliar sources.

Employees also need to watch out for social engineering (or 'phishing') where hackers try to trick them into revealing details that make an organisation's computers vulnerable.

(j) **Keep records** – and test your security. Maintaining security is an ongoing, not a one-off fix, so it's important to keep clear records – for example, keeping a record of the hardware and software an organisation uses will help build up a picture of your business's security status, and spot potential weak points.

Good record keeping will also help you regularly test all your security measures, and ensure that you have functioning, up-to-date software (for example, through keeping a record of software patches applied).

Ultimately, any business is only as secure as its weakest link, and testing will help make sure that no weaknesses get over-looked.

These 'steps' along with CESG's steps could provide a useful framework for appraising an organisation's strategies for ensuring the security of its data and for preventing cyberattacks.

More generally, in the light of the increased challenges which cyber-risks present to organisations, senior management also have to do more to promote an awareness of cybersecurity. ICAEW's 2018 *Audit insights: Cyber Security* recommends that businesses should instigate the following measures:

- **Embed a greater awareness of cyber-risk throughout their strategy and operations**. Ultimately, it is the board's responsibility to ensure that management respond to the risks identified and take appropriate steps to mitigate them.
- **Build a security-focused culture** based on continuous improvement and integrated thinking around cybersecurity, rather than seeing it as a series of one-off tactical activities.
- **Improve cyber-risk reporting**: make cyber-risk reporting more transparent, qualitative and focused on the impact of security investments; for example, threat analysis and the extent to which investments have reduced cyber-risk.
- **Reduce the complexity of the IT environment**: standardising systems where possible and mitigate risks through effective controls between legacy and modern environments. Although using bespoke systems may help meet specific business needs, having a variety of complex and disparate systems makes effective cybersecurity (such as patch management) more difficult. So, from a cybersecurity perspective, businesses should be encouraged to simplify and standardise the IT environment as far as possible. (This could also be an important issue in the context of an acquisition, and integrating legacy systems in the company being acquired.)
- **Manage talent**: develop talent management strategies at all levels of the organisation to cope with skills shortages in relation to managing cybersecurity in order to reduce reliance on contractors and consultants.
- **Apply discipline**: focus on applying strong discipline around security, and consistently complying with good practices.

In addition, management can also help to improve cybersecurity by:

- **Learning from past security breaches.** Following a security breach, senior management should use this as an opportunity to promote the importance of cybersecurity throughout

the organisation, and should look to put in place measures to address the weaknesses that permitted security breaches to occur in the past.

- **Determining the organisation's tolerance to the cyber-risks** is an important step in designing management strategies. Such an exercise may lead to the conclusion that additional funding is required to enhance the cybersecurity features of the organisation's IT/IS infrastructure.
- Ensuring that **non-executive board members play an active role in promoting cybersecurity** during their interactions with the board. This may involve keeping their knowledge about the evolving nature of cyber-risks up to date and challenging the executive directors about the need for following best practice in cybersecurity.



Professional skills focus: Structuring problems and solutions

In the SBM&L exam you will be required to identify a range of creative and pragmatic solutions to business problems identified in the question scenario. Rather than memorising and repeating syllabus knowledge you should use the content in your study materials, along with wider business reading, to develop your commercial acumen thus enabling you to suggest value-adding solutions for the business.



Context example: National Health Service (NHS) in UK

When discussing cyber-risk and cybersecurity, there could be a number of different perspectives to consider:

- Causes: for example, attacks by hackers (using malware, or ransomware); weaknesses in the IT systems (eg, out of date security patches); or human error in opening an attachment which contained malware
- Consequences: for example, potential disruption to operations; loss of data, damage to reputation, and the potential financial consequences of these
- Steps that could have been taken to prevent a cyberattack, or which could be taken to prevent a similar attack recurring in the future

Think about these different elements as you read this case study:

In May 2017, the National Health Service (NHS) in the UK was hit by a large-scale cyberattack which disrupted treatment at a number of hospitals, forcing them to cancel routine appointments and to divert emergency patients to neighbouring hospitals which had not been affected by the attack.

Hospitals reported problems with their email systems, as well as clinical and patient systems, following the ransomware attack which 'locked' computer data and demanded payment (in Bitcoin) for its release.

The attacks did not only target the NHS, but were part of a much larger ransomware campaign, affecting organisations across a range of sectors and in a number of different countries, including Russia, Japan, Germany and Spain (where major companies like Telefonica and Santander were both affected by the attack).

Nonetheless, the attacks raised concerns about cybersecurity across the NHS, as well as ransomware's ability to cause disruption to vital services.

Organisations' abilities to guard against malicious activity rely on having robust security measures in place, as well as educating the workforce.

As one threat prevention expert noted: 'Organisations need to be able to prevent infections [ransomware attacks] taking hold in the first place, by scanning for, blocking, and filtering out

suspicious file content before it reaches their networks. It's also essential that staff are educated about the potential risks of incoming emails from unknown parties, or suspicious-looking emails that appear to come from known contacts.'

Although the NHS wasn't the only organisation affected by the attack, the attack nonetheless raised questions about the resilience of the NHS's IT systems, and highlighted a lack of infrastructure investment, resulting in the fact that the service relies heavily on ageing infrastructure.

Some of the NHS's IT systems can only function using old operating software. For example, some NHS trusts continued to use Windows XP as an operating system, even though Microsoft stopped supporting that system in 2015. The ransomware was able to exploit a flaw in the system's file sharing service, which arose from the lack of updates in Microsoft security patches.

Battling with a lack of funding, some NHS trusts are running unpatched Windows XP machines, leaving them vulnerable to attack.

As one former NHS IT director pointed out, 'Upgrades are costly and there simply isn't enough money in most NHS trusts' budgets [to upgrade their systems].' Another NHS IT director, acknowledged that, although there had been increased support from government to upgrade IT across NHS trusts, investment in infrastructure is still not given enough priority. And many NHS trusts do not have a specific budget for cybersecurity.

Following the attack, opposition politicians were quick to criticise the lack of funding, and the lack of cybersecurity across the health service. The shadow health secretary said 'This incident highlights the risk to data security within the modern health service and reinforced the need for cybersecurity to be at the heart of government planning. The digital revolution has transformed the way we live and work, but we have to be ready for the vulnerabilities it brings, too.'

However, although a lack of funding and a lack of cybersecurity infrastructure have been highlighted as key issues, concerns have also been raised about a lack of awareness of cybersecurity among staff.

Although there is cybersecurity training in place at most NHS trusts, it does not seem to be having much impact yet. In some cases, a lack of properly integrated IT systems means that clinical staff use 'workarounds' to make systems work more easily for them.

The former NHS IT director said, 'Clinical staff… want to do their work as well and as efficiently as possible, and when the systems we use are not allowing them to do so, they find other solutions.' For example, staff keep their computers logged on when they go to the toilet or get a coffee, because if they logged off and came back they would have to start their current session again.

In 2016, the national data guardian, Fiona Caldicott, also said that there were problems with data not always being protected. Her report into data security in the NHS highlighted that examples of poor practice included confidential papers being stored in unlockable cabinets, and the use of unencrypted laptops.

Sources:

Ashford, W. (2017) 'NHS hospitals hit in global ransomware attack', *Computer Weekly*, 12 May 2017, www.computerweekly.com

Evenstad, L. (2017) 'NHS cyberattack shows need for resilient infrastructure and security education',
Computer Weekly, 15 May 2017, www.computerweekly.com

Context example: Reckitt Benckiser

In June 2017, the multinational consumer goods company, Reckitt Benckiser – whose brands include Dettol, Durex, Strepsils and Vanish – realised that its systems had been infected by the 'Petya' malware attack that started in Ukraine (where the company has operations).

Reckitt's chief executive said the virus had 'self-destructed' within one hour of striking, but in that time it had created havoc.

The chief executive also attempted to quantify the impact of the attack. In a market update warning, he noted that second quarter sales would fall by 2% instead of being flat, as had previously been expected.

The attack, which disrupted Reckitt's manufacturing and its ability to issue invoices, and to distribute products in many of the 60 countries in which it operates, was also expected to restrict full year sales growth to 2%, instead of 3% as previously expected.

One analyst estimated that the revised sales forecast equated to a drop of £112 million in annual revenue, and Reckitt admitted: 'The continued production difficulties in some factories mean we expect to lose some further revenue permanently.'

Reckitt's decision to quantify the impact of the attack was very unusual, but the company decided to issue a statement after taking advice from its brokers.

The company's chief executive also pointed out, 'This is new territory for everyone. What we are seeing is that with every cyberattack, the force is getting multiplied.'

Reckitt had safety measures in place, including software patches before the attack, which raised the broader question of how companies can protect themselves from future attacks. Reckitt's IT investment was in line with its peers, and it had prioritised cybersecurity, including creating a senior role to oversee global protection.

One suggestion, made by a cybersecurity partner at PwC, is that multinationals might have to rebuild their networks with boundaries and segregated systems to prevent malware infecting their entire operations.

Although Reckitt Benckiser was the first company to reveal a financial impact of the Petya cyberattack, the malware infected organisations in more than 60 countries, so many other companies were affected, with AP MollerMaersk, WPP, FedEx and its subsidiary, TNT, being among the worst hit.

With increased automation, cyberattacks can have a huge effect on critical infrastructure, leaving companies unable to carry out basic functions, such as accessing emails or documents. Maersk (a leading logistics and shipping company) was unable to take customer orders for two days, and had to re-route ships after IT systems were taken offline.

One analyst estimated that the attack could have cost Maersk about $60 million in lost revenues, which reinforces how sensitive automated systems are to hacker attacks and cybercrime.

Based on: Daneshku, S. (2017) 'Reckitt seeks to quantify havoc of malware attack', *Financial Times*, 6 July 2017

6.2 Cybersecurity and assurance

Although technologies and trends such as cloud computing, the increasing use of mobile devices and social media, and the emergence of big data can have benefits for organisations, they also present significant security challenges. Very few organisations are now immune to cyberthreats, and increasingly potential business partners or suppliers are seeking assurance over an entity's cybersecurity before they are happy to do business with them.

The 2015 *Information Security Breaches Survey*, conducted by PwC on behalf of the UK government, found that 40% of organisations ensure that a provider of services has ISO 27001 certification for services. The survey also found that 13% of organisations require their suppliers to comply with an independent service auditor's report (eg, ISAE 3402) over their security.

Although ISO 27001 certification shows that an organisation has a functioning information security management system, an ISAE 3402report by an independent third party will be a more effective way of checking that a supplier's controls are suitably designed to achieve their objectives, and that they function as they are meant to.

As we noted in section 1.6.2 earlier in this chapter, an ISAE 3402 report will provide an independent opinion on the design, existence and possibly the operating effectiveness of a service provider's controls.

(As we noted in the chapter Strategic implementation, where we discussed ISAE 3402 in more detail, a 'Type 1' report will provide an opinion on the design and existence of the control measures. A 'Type 2' will provide an opinion on these aspects as well as the operating effectiveness of the controls.)

Cyber Essentials

In the UK, a third framework for assessing cybersecurity in an organisation is Cyber Essentials, which the government launched in June 2014, and which forms a pre-requisite for suppliers bidding for many central government contracts.

Cyber Essentials requires an organisation to assess its system security in relation to five key control areas:

- **Service configuration** – ensuring that systems are configured in the most secure way for the needs of the organisation
- **Boundary firewalls and internet gateways** – these are designed to prevent unauthorised access to or from private networks, but good setup of them either in hardware of software form is important for them to be effective
- **Access control** – ensuring only those people who should have access to systems do have access, and at the appropriate level
- **Malware protection** – ensuring that virus and malware protection is installed and is kept up to date
- **Patch management** – ensuring that the latest supported version of applications is used, and all the necessary patches supplied by the vendor have been applied.

The Cyber Essentials assessment can be applied at two levels:

- The **basic Cyber Essentials certification** is awarded on the basis of a verified self-assessment. Organisations assess themselves against the five key control areas via a questionnaire, which is approved by a senior executive such as the CEO. An independent, qualified assessor then verifies the information provided to assess whether an appropriate standard has been achieved for certification to be awarded.

This option offers a basic level of assurance, but can be achieved at low cost.

- **Cyber Essentials PLUS** provides a higher level of assurance. A qualified and independent assessor examines the same five controls, testing that they work in practice by simulating hacking and phishing attacks.

Because of the more resource intensive nature of this process, Cyber Essentials PLUS is much more expensive that Cyber Essentials. However, the level of assurance provided is correspondingly higher.

Interactive question 7: Cybersecurity

Hiway manufactures car infotainment systems, which provide navigation information and information about traffic conditions (for drivers) as well as entertainment (such as music and games) for passengers. It has a contract from two leading car manufacturers to supply the infotainment systems for their vehicles.

A recent report has alleged that serious security flaws in the technology used in Hiway's system make it vulnerable to cyberattack, with the consequence that the electronic functions within the car (eg, such as the speed gauge and parking sensors) or the electronic locks and immobilisers could also be at risk.

Hiway's PR director has strenuously denied the report, and said that none of the company's infotainment systems have been victims of a cyberattack. Nonetheless, Fova plc, one of the manufacturers with whom Hiway has a contract, has called for an urgent meeting to discuss the findings of the report.

The CEO is very concerned about the potential loss of such a significant customer, and has called an emergency board meeting to discuss cybersecurity at Hiway. At the meeting, several of the directors raised concerns that, although Hiway's systems may not have been attacked, the company would struggle to demonstrate to Fova that Hiway has good cybersecurity practices in place.

Requirement

Explain the issues that the board of Hiway should consider in relation to cybersecurity, and make recommendations regarding any actions it should take in this respect.

See **Answer** at the end of this chapter.

6.3 Security and data protection: GDPR

The pressure on cyber security has also been increased, in recent years, by tougher regulations around the protection of personal data, such as the EU General Data Protection Regulation (GDPR).

Note: GDPR is EU legislation so the detailed points of the legislation may not apply to companies in countries outside the EU. Therefore, rather than focusing on detailed points of the legislation, think about the principles the legislation is designed to uphold.

Definition

Personal data: Any data related to a person such as a name, address, date of birth, bank details, medical records, photo, email address, or posts on social networking websites.

The internet has dramatically changed the way people and organisations communicate – but it also means people enter their personal details online – for example, when buying goods, paying bills or subscribing to social media sites.

Equally, the logic of data-driven marketing is that the more data (and information) companies gather about customers, the more targeted and relevant their communications with customers will be, and the better the customer experience will be.

However, it is also important that organisations handle personal data responsibly and securely, and this has been emphasised by data protection legislation, such as GDPR. As we have already mentioned in the chapter Corporate Governance, GDPR requires all organisations to follow strict data protection principles in relation to collecting and using personal data.

GDPR is designed to give individuals more power over their data, and to restrict organisations' ability to collect and use data for monetary gain (eg, by selling individuals' data to third parties).

6.3.1 Principles of GDPR

The GDPR gives customers greater rights about how their data is used, and requires organisations to adhere to the following principles:

- **Consent must be given** – Organisations may not process personal data unless they have been given specific consent to do so by the individual whose data is being processed.
- **The right to be informed** – Individuals must be informed before data is gathered. Customers must consent for their data to be gathered, and consent must be specifically given rather than implied.
- **The right to be forgotten** – If an individual stops being a customer of an organisation, or withdraws their consent, they have the right to have their data deleted.
- **The right to access** – Individuals have the right to request access to their personal data and to how it is being used. The company must provide a copy of the data, free of charge, if requested.
- **The right to have information corrected** – Individuals can have their data updated if it is out of date, or incorrect in any way.
- **The right to object** – Individuals have a right to stop their data being processed and used for direct marketing. There are no exemptions to this rule, and any processing must stop as soon as the request is received.
- **The right to be notified** – If an organisation suffers a data breach which compromises an individual's personal data, the individual should be informed directly and without undue delay; at most **within 72 hours** of the organisation becoming aware of the breach.

When notifying a data subject about a breach, an organisation needs to describe clearly the nature of the breach, and the likely consequences of it. This is to help people whose data is potentially affected by the breach to take steps to protect themselves from the potential effects of the breach.

If a breach occurs, the organisation must also notify the relevant supervisory authority without undue delay (and within 72 hours of becoming aware of it). In the UK, the relevant supervisory authority with respect to GDPR and data protection is the Information Commissioner's Office (ICO).

6.3.2 Business implications of GDPR

Scope – GDPR is very wide ranging. It applies to all businesses and organisations established in the EU. Even if the organisation is not based in the EU, if it offers goods or services to EU citizens, it is subject to GDPR.

Penalties – There are tough financial penalties for non-compliance. Companies and organisations who don't comply with GDPR could face fines of up to 4% of annual global revenue, or 20 million Euros, whichever is greater.



Context example: British Airways

In July 2019, the Information Commissioner's Office (ICO) fined British Airways £183 million – equivalent to 1.5% of the airline's annual turnover – for failing to protect customer data from a hacking attack.

The ICO said that the airline's 'poor security' had allowed the personal details of around 500,000 people to be harvested by criminals in August and September 2018. The data stolen

included log-in, payment card and travel booking details, as well as customers' names and addresses.

Announcing the penalty, the Information Commissioner said, 'People's personal data is just that – personal. When an organisation fails to protect it from loss, damage or theft, it is more than an inconvenience. That's why the law is clear – when you are entrusted with personal data you must look after it. Those that don't will face scrutiny from my office to check they have taken appropriate steps to protect fundamental privacy rights.'

However, in August 2019, BA said that it planned to appeal against the fine, arguing that it had been proved that BA failed to comply with GDPR rules.

Sources: Bentham, M, Record £183m fine for BA over passenger data breach, *Evening Standard*, 8 July 2019, pp 1–2.

Lynch, R. BA owners plans appeal against £183m fine over cyberhacking, *Evening Standard*, 2 August 2019, pg. 40.

Controlling data – An important implication of GDPR is that it forces organisations to get a better understanding of the personal data they hold. Questions an organisation needs to consider are:

- What data does it hold? (Does it need this data, or should it be erased?)
- Where is the data stored?
- Who can access the data?
- Are there any risks to the data?

Data security – One of the key issues in complying with GDPR is that organisations need to develop and implement safeguards to help prevent data breaches, and to contain any breaches which do occur.

Organisations also need to take rapid action to notify individuals affected, and authorities, if a breach does occur. As a result, organisations need to have a **communication plan** in case a data breach occurs.

Companies that process a lot of sensitive personal data have to employ a data protection officer (DPO). This officer has to report to senior members of staff, monitor compliance with GDPR and be a point of contact for employees and customers.

Compliance and accountability – One of the principal aims of GDPR is to make organisations 'accountable' with regards to how data is collected, used and processed.

A particularly important aspect of this accountability is ensuring that an organisation has data protection policies in place which provide an **appropriate level of security** over the personal data it holds.

The 'appropriate' level of security can depend on the specific risks an organisation faces, but some general measures which could help to demonstrate compliance include:

- measures to protect the **confidentiality** of personal data, eg, **encrypting** it
- measures to strengthen the **resilience** of data processing systems, eg, firewalls; anti-virus software
- checking that **consents** obtained meet GDPR requirements, and re-obtaining them where necessary
- **data minimisation** – safeguards to minimise the amount of data collected and the period for which it is stored
- HR-based measures, including **staff training** about data handling
- **internal audits** of data processing activities

- **regularly testing**, assessing and evaluating the effectiveness of technical and organisational measures for ensuring the security of data processing

The ICAEW IT Faculty's report The essential guide to GDPR also highlights that 'accountability' means not only that an organisation complies with the regulation but also that it must show how it complies – for example, by documenting the security measures it has in place. GDPR also encourages a risk-based approach to data protection.
In the UK, the Information Commissioner's Office (ICO) recommends conducting a data protection impact assessment (DPIA), to help document the decision of data collection, usage and processing, and the reasons for them. (A DPIA is a process designed to help organisations systematically analyse, identify and minimise the data protection risks of a project or plan.)

Conducting a DPIA can also help an organisation ensure that it is only using the minimum of personal data necessary for any given purpose. By taking a risk-based approach and documenting how personal data is being used and processed, an organisation will also be able to demonstrate their approach to data protection if the need arises.

This includes explaining when and how the organisation uses encryption. The Data Protection Act 2018 and GDPR do not impose an obligation on organisations to use encryption. However, if encryption is not being used, then an organisation will be expected to demonstrate what other technical and organisational security measures it is using to protect personal data.

6.3.3 GDPR and acquisitions

So far, we have focused on the business implications of data protection within a single company. However, companies should also be aware of the potential data protection implications in an acquisition, or other business transactions.



Context example: Marriott International

In 2016 Marriott International acquired Starwood Hotels and Resorts. In September 2018, Marriott identified that the Starwood business had suffered database hacking over a long period of time, potentially as far back as 2014. As a result, the personal data of more than 300 million guests, worldwide, had been compromised. Although Marriott is based in the US, some of the customers affected are EU nationals, meaning the breach falls under the scope of GDPR.

In July 2019, the UK's ICO announced its intention to fine Marriott £99 million for alleged infringements of GDPR. The ICO commented that its investigation revealed that Marriott had 'failed to undertake sufficient due diligence when it bought Starwood, and should also have done more to secure its systems.'

Source: Ilan, D., Farmer, N. and Wales, T, *UK Regulator intends to fine Marriott £99 million for personal data breach, spotlighting M&A cybersecurity diligence*, Mondaq, 15 July, 2019, [Online], Available from: http://www.mondaq.com/uk/x/825660/Security/UK+Regulator+Intends+To+Fine+Marriott+99+Milli on+For+Personal+Data+Breach+Spotlighting+MA+Cybersecurity+Diligence [Accessed 5 August 2019]

Implications of data protection and GDPR in the context of mergers and acquisitions

As the Marriott example illustrates, purchasers need to review a target business's data protection compliance in their due diligence investigations.

Purchasers also need to consider how they will integrate acquired businesses with their own, and how this integration process could affect the holding and use of personal data.

However, data protection is not only an issue for purchasers. Sellers also need to prepare for a possible sale by reviewing their operational processes to ensure they will not breach any data

protection laws when providing data to a potential purchaser (eg, do they have a right to share any personal data about employees or customers to potential purchases in the course of a deal, or should this data be anonymised?). Equally, sellers need to ensure that access to personal data is only given to those who need it (eg, by using virtual data rooms, with restricted access, as a means of providing data to potential purchasers).

7 Using information to develop competitive advantage



Section overview

- The notions of organisational learning and knowledge management indicate that knowledge should be seen as an important resource for an organisation and, accordingly, knowledge management as an important competence or capability.
- Increasing the amount and quality of data available to an organisation should help support strategic decision making within that organisation.
- In particular, improving the level of data an organisation holds about its customers should help the organisation understand their purchasing behaviour better, such that it can tailor its products and services to their wants (or needs) more closely.

You should already be familiar with the concept of knowledge management from the Business Strategy and Technology syllabus. However, it remains relevant here because knowledge management is becoming increasingly important in helping organisations sustain competitive advantage.

7.1 Knowledge management

Knowledge management refers to the set of business processes developed in an organisation to create, store, transfer and apply knowledge. Knowledge management increases the ability of the organisation to learn from its environment and to incorporate knowledge into its business processes. (Laudon and Laudon, Management Information Systems)

Knowledge management is a relatively new, but increasingly important, concept in business theory. It is connected with the theory of the **learning organisation** and founded on the idea that knowledge is a major source of competitive advantage in business.

Studies have indicated that 20–30% of company resources are wasted because organisations are not aware of what knowledge they already possess. Lew Platt, former Chief Executive of Hewlett-Packard, highlighted this when he said, 'If only HP knew what HP knows, we would be three times as profitable.'

In effect, knowledge management has three phases: capturing knowledge, recording knowledge and disseminating knowledge (across the organisation).

The importance of knowledge

As organisations become more complex, there is more knowledge to manage. Moreover, the importance of capturing and sharing it increases as job mobility increases. If staff leave, there is a danger that knowledge could leave with them if it has not been properly managed within the organisation.

Also, organisations' external environments – technology, competitors, markets – are changing rapidly so organisations need to ensure they have up to date knowledge about these to take account of the opportunities and threats they represent.

Knowledge is thus seen as an important **resource**, and knowledge management may in itself constitute a **competence**; it can certainly **underpin** many competences, and knowledge management should be seen as a strategy to achieve competitive advantage, for example through the sharing of cost reduction ideas across divisions, or through the diffusion of innovation.

Companies are now starting to use web technologies such as blogs and wikis for internal use to foster collaboration and information exchange between individuals and teams. Collaboration tools from commercial software vendors (such as Microsoft SharePoint) can also be used to share information between individuals and teams in an organisation.

In a knowledge management system, an organisation will appoint **knowledge managers** who are responsible for collecting and categorising knowledge and encouraging other people in the organisation to use the available knowledge. The knowledge managers also monitor the use of knowledge in their organisation.

Some companies are now taking the idea of knowledge sharing one stage further and are adopting the practice of **knowledge brokering**. In knowledge brokering, companies look externally to find ways of improving internal business processes. In effect, knowledge brokering resembles benchmarking by allowing companies to find world-class solutions to problems rather than having to invent their own solutions.

For example, a bank faced frequent complaints from customers about the length of the queues in its local branches. The bank staff responsible for reducing queuing times identified three potential sources of brokers: amusement parks, supermarkets and department stores. In each of these environments, it is important to keep queuing times under control. In time, the bank worked with an amusement park and a supermarket to redesign layout of the windows in its branches and change the way it deployed staff between back office and customer-facing windows at busy times.

7.2 Organisational learning

Organisational learning is particularly important in the increasing number of task environments that are both complex and dynamic. It becomes necessary for strategic managers to promote and foster a **culture that values intuition**, **argument from conflicting views, and experimentation**. A willingness to back ideas that are not guaranteed to succeed is another aspect of this culture; there must be freedom to make mistakes.

The aim of **knowledge management** is to exploit existing knowledge and to create new knowledge so that it may be exploited in turn. This is not easy. All organisations possess a great deal of data, but it tends to be unorganised and inaccessible. It is often locked up inside the memories of people who do not realise the value of what they know. This is what Nonaka calls **tacit knowledge**. Even when it is made **explicit** (available to the organisation) by being recorded in some way, it may be difficult and time consuming to get at, as is the case with most paper archives. This is where knowledge management technology can be useful –for example, intranets, databases and other knowledge repositories.

7.2.1 Information, knowledge and competitive advantage

Information and communications technologies have reduced the **cost** of storing and transmitting information, but they have also increased organisations' **capacity** for storing, processing and communicating information.

As access to information becomes easier and less expensive, skills and competences relating to the selection and efficient use of information become increasingly important to organisations. This reinforces the idea that information and information management could become a core competence and a source of competitive advantage for an organisation.

The resource-based approach to strategy – discussed in the chapter Strategic analysis – highlights that a successful organisation acquires and develops resources and competences over time, and exploits them to create competitive advantage.

The ability to capture and harness corporate knowledge has become critical for organisations as they seek to adapt to changes in the business environment, particularly those businesses providing financial and professional services. Therefore, as we have already mentioned, knowledge becomes a strategic asset. And organisation-specific knowledge, which has been built up over time, is a core competence that cannot easily be imitated.

Therefore, knowledge management can help promote competitive advantage through:

- the fast and efficient exchange of information
- effective channelling of the information to:
  - improve processes, productivity and performance
  - identify opportunities to meet customer needs better than competitors
  - promote creativity and innovation

However, the importance of meeting customer needs better than competitors means that organisations need to capture and analyse information about customers and potential customers rather than simply looking at internal processes.



Context example: EuroDisney

The following example illustrates that market understanding is as important as pure market research.

When Disney opened its EuroDisney resort near Paris, the company lost $921 million in its first year. However, the decision to enter the European market had been well supported by the research Disney had undertaken.

Figures showed a growing number of European visitors to US theme parks (suggesting European people like going to theme parks).

The more detailed choice of location had been based on modelling population figures, which showed 17 million people lived within a two-hour drive of the Paris site, and 109 million within a six- hour drive. These potential customer numbers were much greater than for theme parks in the US.

However, while the figures were encouraging, the launch of EuroDisney proved an expensive lesson in market understanding, rather than just market research.

Disney ignored the failure of amusement parks in France; it dismissed anti-Disney demonstrations as insignificant; and it ignored the fact that European holiday patterns are significantly different to those in the US. People in Europe have longer holidays, but spend less on each.

Perhaps crucially, marketing short-sightedness also led Disney to ban alcohol from EuroDisney. Clearly Disney's market research had overlooked the key aspect of French culture that the French like to drink wine at lunchtime. Because Disney prevented them from doing so, customers voted with their feet and stayed away from the park.

In the same way that a company needs to know its customers and serve them well, a company also needs to engage with its suppliers. The more a company engages with its suppliers, the better the suppliers can provide vital inputs.

How a company can really get to know its customers, or its suppliers, is a key challenge facing businesses with millions of offline and online customers.

Our discussions of big data and big data analytics in the previous chapter also highlighted the potential value which organisations can gain from analysing large and complex data sets – in order, for example, to send tailored recommendations to customers' mobile devices, or to predict customer demand for products and thereby to adjust the quantities of inventory ordered and to adjust prices dynamically.

7.3 Knowledge management (KM) systems

Laudon and Laudon highlight that one apt slogan of knowledge management is: 'Effective knowledge management is 80% managerial and organisational, and 20% technology.'

In other words, the culture and patterns of behaviour in an organisation need to support knowledge management. IT cannot support knowledge management by itself. For example, in order for knowledge to be shared between teams, members from different teams have to be prepared to share it.

In terms of actually developing and implementing a KM strategy, there are five main steps to consider:

(a) **Support from senior management**. Senior management support will be needed not only to provide the necessary **resources** and to lead the development of a knowledge-based culture, but also because if senior managers are not seen to be supporting the strategy then other staff will not do so either.

(b) **Installing the IT infrastructure**. IT hardware and software will need to be acquired to ensure that the organisation has the capabilities to **capture, store and communicate knowledge**.

(c) **Developing the databases**. Advanced databases and database management systems may need to be developed (either 'in house' or 'in the cloud'), with the details of their design and structure being tailored to the type of knowledge the organisation is looking to capture.

(d) **Develop a sharing culture**. Knowledge is widely known to represent **power**, and staff are likely to want to hoard the knowledge they have already accumulated rather than to share it. A **culture** of **knowledge sharing** must be developed.

(e) **Capturing and using the knowledge**. Existing knowledge needs to be captured and recorded in the databases. Staff then need to be trained how to use the databases and encouraged to do so.

7.3.1 Potential issues in implementing a KM system

Structure and culture – The current structure and culture of an organisation may not be conducive to sharing knowledge; for example, if there is little communication between departments in an organisation, or if staff are reluctant to share knowledge for fear that it will reduce their power within the organisation. These inherent barriers will have to be overcome in order for the system to be successful.

Technological infrastructure – If an organisation does not have a suitable network which allows information to be stored and accessed, one will have to be installed before knowledge can be shared across the organisation. There may be significant costs associated with installing such a network.

Incompatible systems and sources of information – Problems could arise if some divisions or departments record data or information in systems which are incompatible with those used by other divisions or departments. Such a situation will mean that data or information must be transferred into a new common format before they can be shared, and there is a risk that errors or omissions could result from the resulting conversion process.

Equally, it is possible that some information is not stored in a digital form at all, and so the organisation will have to decide how this material can be indexed and archived such that it can be accessed when needed.

Resistance to change – Staff in different areas of an organisation may already have their own preferred ways of organising data. However, this may not be compatible with the common format in which data is held on the network. Staff may be reluctant to change their current practices, particularly if they are not given adequate training in any new systems or sufficient time to adapt to them.

7.3.2 IT and KM systems

Importantly, although there is an IT element to KM systems and their infrastructure, a KM strategy need not be IT-driven. **IT should support rather than dominate the strategy**. The **cultural aspects** of a KM strategy (particularly encouraging staff to share knowledge) are likely to be just as critical to its success as the IT elements.

Nonetheless, IT elements do play an important role in facilitating knowledge management. In this context, expert systems, databases and data warehouses all help to acquire and store information which can, in turn, be converted into knowledge.

7.3.3 Expert systems

An **expert system** is a computer program that captures **human expertise** in a limited domain of knowledge. Such software uses a knowledge base that consists of facts, concepts and the relationships between them and uses pattern-matching techniques to solve problems. For example, many financial institutions now use expert systems to process straightforward loan applications. The user enters certain key facts into the system, such as the loan applicant's name and most recent addresses, their income and monthly outgoings, and details of other loans. The system will then:

- check the facts against its **database** to see whether the applicant has a good previous credit record
- perform **calculations** to see whether the applicant can afford to repay the loan
- make a **judgement** as to what extent the loan applicant fits the lender's profile of a good risk (based on the lender's previous experience)
- **a decision is then suggested** based on the results of this processing

IT systems can be used to store vast amounts of data in an accessible form. A **data warehouse** receives data from operational systems, such as a sales order processing system, and stores it in its most fundamental form, without any summarisation of transactions. Analytical and query software is provided so that reports can be produced at any level of summarisation and incorporating any comparisons or relationships desired.

The value of a data warehouse is enhanced when **data mining** software is used. True datamining software **discovers previously unknown relationships** and provides insights that cannot be obtained through ordinary summary reports. These hidden patterns and relationships constitute **knowledge**, as described above, and can be used to guide decision making and predict future behaviour.
Datamining is thus a contribution to organisational learning. For example, the relationship between the weather and changes in people's purchasing habits in supermarkets can be viewed as knowledge discovery through data mining.

The concepts of 'analytics' and 'big data analytics' which have been mentioned on a number of occasions throughout this manual, develop these ideas further.

7.3.4 Databases and models

The way in which data is held on a system affects the ease with which that data can be accessed and then analysed. Many modern software packages are built around a database. A database provides a comprehensive set of data for a number of different users.

A database is a collection of data organised to service many applications. The database provides convenient access to data for a wide variety of users and user needs.

A database management system is the software that centralises data and manages access to the database. It is a system which allows numerous applications to extract the data they need without the need for separate files.

Databases can be used in conjunction with a variety of tools and techniques, eg, DSSs, EISs, data warehousing and data mining.

7.3.5 Data warehouses



Definition

Data warehouse: A data warehouse consists of a database containing data from various operational systems and reporting and query tools.

A data warehouse is a large-scale data collection and storage area containing data from various operational systems, plus **reporting** and **query tools** which allow the data to be analysed. The key feature of a data warehouse is that it provides a single point for **storing a coherent set of information** which can then be used across an organisation for **management analysis** and decision making.

Importantly, a data warehouse is **not an operational system**, so the data in it remains static until it is next updated. For example, if a supermarket introduces a customer credit card, the history of customers' transactions on their cards could be stored in a data warehouse so that management could analyse spending patterns.

However, although the reporting and query tools within the warehouse should facilitate management reporting and analysis, data warehouses are primarily used for **storing** data rather than analysing it.

A data warehouse contains data from a range of **internal** (eg, sales order processing system, nominal ledger) and **external sources**. One reason for including individual transaction data in a data warehouse is that, if necessary, the user can drilldown to access transaction-level detail. Increasingly, data is obtained from newer channels such as customer care systems, outside agencies and websites.

Maintenance of a data warehouse is an iterative process that continually refines its content. Data is copied to the data warehouse as often as required – usually daily, weekly or monthly. The process of making any required changes to the format of data and copying it to the warehouse tends to be automated.

The result should be a coherent set of information available for use across the organisation for management analysis and decision making. The reporting and query tools available within the warehouse should facilitate management reporting and analysis.

The reporting and query tools should be flexible enough to allow multidimensional data analysis, also known as **online analytical processing** (OLAP). Each aspect of information (eg, product, region, price, budgeted sales, actual sales, time period) represents a different dimension. OLAP enables data to be viewed from each dimension, allowing each aspect to be viewed by itself and in relation to the other aspects.

Features of data warehouses

A data warehouse is subject-oriented, integrated, time-variant and non-volatile.

(a) **Subject-oriented**

A data warehouse is focused on data groups, not application boundaries. Whereas the operational world is designed around applications and functions such as sales and purchases, a data warehouse world is organised around major **subjects** such as customers, suppliers, products and activity.

(b) **Integrated**

Data within the data warehouse must be consistent in format and codes used – this is referred to as **integrated** in the context of data warehouses.

For example, one operational application feeding the warehouse may represent sex as 'M' and 'F', while another represents sex as '1' and '0'.

While it does not matter how sex is represented in the data warehouse (let us say that 'M' and 'F' is chosen), it **must** arrive in the data warehouse in a **consistent, integrated** state. The data import routine should cleanse any inconsistencies.

(c) **Time-variant**

Data is organised by time and stored in time-slices.

Data warehouse data may cover **a long time horizon**, perhaps from 5 to 10 years. Data warehouse data tends to deal with **trends** rather than single points in time. As a result, each data element in the data warehouse environment must carry with it the time for which it applies.

(d) **Non-volatile**

Data **cannot be changed** within the warehouse. Only load and retrieval operations are made.

Advantages of data warehouses

Advantages of setting up a data warehouse system include the following:

(a) Enhances strategic decision making. The warehouse provides a single source of authoritative data which can be analysed using data mining techniques to support strategic decision making. Basing decisions on data (rather than having to rely on gut instinct) should also enhance the quality of decisions being made.

(b) Data quality and consistency. Having a single source of data available will reduce the risk of inconsistent data being used by different people during the decision-making process. This means managers should be able to have greater confidence in the accuracy of the data on which decisions are being made.

(c) **Saves time**. Data warehousing can enable faster responses to business queries, by storing data in an easily accessible central repository and enabling data to be retrieved from multiple sources. By providing users with data from a number of sources in one place, data warehouses mean that informed decisions can be made more quickly. Users won't need to spend time retrieving data from different sources.

In addition, managers and executives can interrogate the data themselves without relying on the IT department to generate reports for them.

Ultimately, the value of data warehouses is derived from the increases in revenues or reductions in costs which result from decisions made on the basis of the data provided by the warehouse.

Data warehouses have proved successful in some businesses for:

- quantifying the effect of marketing initiatives
- improving knowledge of customers
- identifying and understanding an enterprise's most profitable revenue streams

In this way, data warehouses (and data mining) allow organisations to use the data they hold to help improve their competitiveness.

Limitations of data warehouses

Some organisations find they have invested considerable resources implementing a data warehouse for little return. To benefit from the information a data warehouse can provide, organisations need to be flexible and prepared to act on what they find. If a warehouse system is implemented simply to follow current practice, it will be of little value.

Other limitations exist, particularly if a data warehouse is intended to be used as an operational system rather than as an analytical tool. For example:

- the data held may be outdated
- an efficient regular routine must be established to transfer data into the warehouse
- a warehouse may be implemented and then, as it is not required on a day to day basis, be ignored

Another important consideration is the quality of data in the warehouse. We have noted that one of the advantages of a data warehouse is that it improves decision making, but this will only be the case if the data within the warehouse is accurate. Therefore, it is important that data is 'cleaned' before it is loaded into the warehouse, but a **data cleansing** exercise will take time and require suitably skilled staff.

There is also an issue of **security**. The management aim of making data available widely and in an easily understood form can be at variance with the need to maintain confidentiality of, for example, payroll data.

This conflict can be managed by **encrypting** data at the point of capture and **restricting access** by a system of authorisations entitling different users to different levels of access. For this to work, the data held must be classified according to the degree of protection it requires; users can then be given access limited to a given class or classes of data. Encryption at the point of capture also exerts control over the **unauthorised uploading** of data to the data warehouse.

7.3.6 Data mining

While a data warehouse is effectively a large database which collates information from a wide variety of sources, data mining is concerned with the discovery of meaningful relationships in the underlying data.

Data mining software looks for hidden patterns and relationships in large pools of data. Data mining is primarily concerned with **analysing data**. It uses statistical analysis tools to look for **hidden patterns and relationships** (such as trends and correlations) in large pools of data. The value of data mining lies in its ability to highlight previously unknown relationships.

In this respect, data mining can give organisations a **better insight into customer behaviours**, and can lead to **increased sales through predicting future behaviour**.

When a supermarket customer pays for their shopping using a loyalty card (see example about Tesco's Clubcard in the next section), the supermarket can create a record of the items the customer has bought. The purchasing behaviour of customers can be used to create a profile of what kinds of people the cardholders are.

Data mining techniques could be applied to customers' purchasing information to identify patterns in the items which were purchased together, or what types of item were omitted from

shopping baskets, and how the make-up of customers' baskets varied by different types of customer. The supermarket could then target its promotions to take advantage of these purchasing patterns.

In this way, by identifying patterns and relationships, data mining can **guide decision making**.

True data mining software discovers **previously unknown relationships**. The hidden patterns and relationships the software identifies can be used to guide decision making and to **predict future behaviour**.

In the chapter Strategic marketing and brand management, we considered how obtaining detailed customer information is an important element of **CRM**.

Obtaining detailed customer information allows a firm to identify customer needs and develop improved ways of meeting those needs, as well as targeting marketing campaigns to specific customers, bringing relevant new products or services to their attention.

Customer loyalty/reward cards can provide valuable information about the buying habits and patterns of customers, but more generally the process of gathering and storing data about customers and then analysing patterns is an application of data warehousing and data mining.



Interactive question 8: Data and knowledge

PBB is a UK university. PBB's management board has identified student performance as a critical success factor (CSF). PBB's management board has identified this CSF because it targets an area where it is currently underperforming compared to other UK universities.

PBB is aware of a nearby comparable university, KLN, which had considerable success when several of its departments worked together to improve student performance. KLN has a culture of sharing knowledge and a knowledge management strategy. PBB does not have a culture of knowledge sharing. Within PBB, knowledge is regarded as the personal property of the individual and very few of its staff are prepared to share their knowledge with any of their colleagues. PBB has an abnormally high level of staff turnover compared to other universities, nearly twice as high as that of KLN.

Student performance

Student performance is measured by PBB as the number of students who successfully complete their courses. Those students who do not are described as 'drop-outs'. The number of drop-outs is measured by the drop-out rate. PBB has access to data for all UK universities for student drop-out rates analysed by age and gender.

Drop-out rates vary greatly across PBB's academic departments. In some academic departments the drop-out rate is extremely high; in others it is very low, much better than the national average. Where the drop-out rate is much better than the national average, the departments have operated extensive schemes for student support. For example, students with personal problems can seek help from trained counsellors, students with financial problems have been helped to find part-time work and students with academic problems are given extra individual tuition from the academic staff.

In the departments where the drop-out rates are extremely high, none of these student support schemes is operating. The departments with the extremely high drop-out rates are not aware of how the departments with the very low drop-out rates support their students. The departments with very low drop-out rates are unwilling to share their knowledge about how to reduce the drop-out rates, as they have spent considerable time and effort developing their schemes and regard these as their own property.

PBB has not conducted any systematic analysis into its overall drop-out rate. Within its information systems, PBB has the following information about each of its students:

- Name
- Age
- Gender
- Address
- Educational record prior to joining PBB
- Educational record within PBB
- academic department

PBB is aware that many universities have successfully used data mining to assist them in managing student performance.

Requirements

8.1 Many organisations integrate their CSFs into their performance management systems by converting them to key performance indicators (KPIs).

Briefly explain, using examples, the advantages that PBB would gain by doing the same.

8.2 Advise PBB's management board of three benefits the university and its staff could expect to receive from the implementation of a knowledge management strategy.

8.3 Explain data mining and how the outputs of the analysis could be used by PBB to improve the student drop-out rates.

See **Answer** at the end of this chapter.

7.4 Analytics and business intelligence

In May 2013, ICAEW's Finance and Management Faculty published an article,*10 key business analytics and how they are used to drive value*.

The article highlights that the amount of data available to companies, generated either internally or externally, is greater than at any time in the past. Companies can use analytics to turn this data into insight and thereby improve decision making.

Web analytics – Analytics tools (like Google Analytics) allow companies to track the traffic on their website – for example: what search phrases people used, how they got to the page, what they were looking at and in which sequence, how long they stayed, whether they were converted into customers.

Such information is very useful to enable companies to track their website effectiveness and customer engagement. In turn, if companies can improve the effectiveness of their website and their engagement with their customers, this should help them achieve their strategic goals.

Customer analytics – Customer analytics enable companies to understand which customers are their most loyal, their most profitable or their most expensive to keep. Data rich companies such as telecom companies and retailers are now also looking at understanding customer lifetime value, and even identifying trigger points when customers are likely to cancel their contracts or shop at a rival company.

Predictive business analytics – In the simplest form, predictive analytics allow companies to use their data to forecast and predict future liquidity and cash flows as well as revenue and profit predictions. More sophisticated approaches use the cause and effect logic to understand, for example, the impact of increased customer satisfaction on future loyalty as well as financial performance. In one case, a bank found that even though most of its customers were happy, few actually generated a profit for the bank.

HR analytics – Given that people are important assets for companies, companies need to identify who is performing well, who is performing less well and who needs more support or training.

HR analytics can also be used to identify the best ways to recruit new talent, to monitor development activities, to identify skills gaps in an organisation, for succession planning, and to identify the drivers of staff satisfaction. Google used analytics to identify 10 factors which make a good manager, and it now uses this insight when recruiting and training managers.

Project analytics – Companies use project analytics to track whether projects are running on time and on budget, and also whether they are generating the desired outcomes. In many cases, simple project management software applications can generate most of the analytics which are required, although for more sophisticated project-specific analytics software may be required.

Process analytics – By contrast to project analytics which look at 'one-off' projects, process analytics are concerned with the day to day operations in a company. Process analytics are used to understand which processes are optimised and to identify processes that can be improved or changed to increase efficiency and effectiveness.

Traditional process analytics start with process mapping (often end to end) and an analysis of the effectiveness and efficiency of each of the process steps. Other approaches are TQM, Lean and Six Sigma, which allow companies to measure, analyse and optimise process performance.

Supply chain analytics – Supply chain analytics enable companies to optimise their supply chains; for example, by finding the most efficient delivery routes, the best locations for their warehouses or production plants, and the most intelligent storage approaches.

Supply chain analytics can also be used to analyse data from delivery routes to better understand fuel consumption, and to identify potential risks to the supply chain (for example, from disruption to road or air freight links).

Procurement analytics – The main benefits from procurement analytics will be generating significant cost savings in the purchase of goods and services, and reducing the business risks associated with those purchases. Procurement analytics allow companies to optimise, consolidate or standardize purchasing and contracts, and enable companies to strategically source products and services at the right time, for the right price, and in the right quantities.

Marketing and brand analytics – Applying analytics to marketing and brand-building activities enables companies to understand the effectiveness of their marketing campaigns.

Digitalisation and e-commerce means that companies can track how many customers and how much business each marketing campaign has generated. Analytics allow companies to identify the most effective marketing channels and marketing strategies, for example. Companies can also use analytics to track brand awareness and brand engagement.

Big data analytics – Big data analytics refer to the ability to analyse larger quantities or more unstructured data (ie, data not in a database) such as keywords from conversations people have on Facebook or Twitter, and content they share through media files (photographs and videos).

The aim of big data analytics is to extract insights from unstructured or large volumes of data in order to help organisations make better decisions, as we discussed in the chapter Data analysis.

Context example: Human swarm

Morrisons

The supermarket chain, Morrisons, uses weather forecasts to predict customers' purchasing patterns.

It has analysed five years of sales data and has identified how sales patterns change in line with increases or decreases in temperature. Although people make purchasing decisions as individuals, overall trends in their purchases show that we act as a 'human swarm'.

When temperatures fall during the winter, purchases of 'warming' food such as soup, porridge and ready meals increase. Therefore, when the weather forecast shows a fall in temperature, Morrisons increases the amount of these 'warming' foods it ships from its central distribution centre to its stores.

Similarly, Morrisons has identified that when weather forecasts in summer predict three or more consecutive days of hot, dry weather, demand for barbecue-related foods increases significantly. By reacting to the weather forecast, Morrisons can not only control the quantities of different products it ships from its warehouses to its stores, but it can also ask its suppliers to change the volume of different products it supplies. For example, if hot weather is forecast, Morrisons asks its supplier of minced beef to switch from producing ready meals (such as cottage pie) to producing beef burgers (which will be used for barbecues).

Bravissimo

The lingerie and clothing retailer also uses local weather reports to change its online adverts in different parts of the UK as the weather changes. Bravissimo uses a software solution (called weatherFIT) to enable it to tailor online adverts in real time to respond to local weather data.

In the three months of March to May 2013, Bravissimo reported a 600% growth in revenue from its online adverts compared to the same period the previous year.

Bravissimo's senior marketing manager commented: 'Using [weatherFIT] to fine-tune our … advertising and promotions by taking into account local weather conditions really boosted sales in the crucial run-up to the holiday season'.

Context example: Tesco and customer insights

The ability to collect and analyse data has transformed Tesco from an organisation that **thinks** it knows what customers want to one that has the knowledge and insights into what customers prefer, and how these preferences keep shifting over time. Former Chief Executive Officer (CEO) Sir Terry Leahy stated 'We don't spend a pound or dollar on a store without talking to our customers – they are the best management consultants'.

An essential component of Tesco's performance data is its extensive customer knowledge gathered through its Clubcard loyalty programme, established in 1994 and now with over 14million users.
Although 'Clubcard' was ostensibly introduced as a loyalty programme, its main purpose for Tesco was to provide insights which would help managers improve the way they run the company.

Ultimately, loyalty programmes which are solely used to target customers with discounts and offers are self-defeating, because they reduce the margins which a company earns on its sales. But loyalty programmes can be beneficial to companies when:(a) the potential to generate competitive advantage from the data is recognised, and (b) the capability to mine data, and to make sense of and apply the insights gleaned from that data, is embedded into the organisation as a capability and focus.

It was the decision-support 'potential' of the data that convinced the senior leaders in Tesco to endorse the idea of a loyalty programme.

When Tesco began its Clubcard programme, it outsourced the data analysis to Dunnhumby, an organisation that specialises in data analysis. Tesco realised it didn't have the skills to systematically analyse the mass of data gleaned from its customers, and therefore left it to Dunnhumby to develop the strategy for the data analysis.

However, as it increasingly began to realise that analytics are an important driver of success, Tesco wanted to improve its in-house competences to analyse customer and performance data. Therefore, it created an internal team responsible for analysing data and extracting insights. Tim Mason, Tesco's marketing director and chairman of Tesco.com, explained:

These people are geographers, statisticians who had spent a lot of time applying those skills to understanding how customers would behave. They could crunch through the stuff that came from the Clubcard, see the patterns in it and they could start to help the management of the organisation understand what was going on, but also point towards what should be done about it. They had to find the data, and present it in a way that makes the decisions stark, and clear.

Tesco ensures it maintains the ability to develop common-sense responses. It aims to create processes that enable relevant insights to be used to improve customers' experience.

Presentation of data

Data is presented in different ways in Tesco, such as through insightful performance reports and organisational intelligence applications that provide dashboards and performance reports to the management team. Most reporting is scheduled weekly or monthly. These dashboards and reports allow executives to drill down into the data and perform high-level analysis of their own.

Decisions based on customer insights

Tesco's objective is to never make any changes without first talking to its customers. It also ensures it runs experiments to test ideas before implementing them on a wider scale. The performance data plays a vital role in this process and has enabled Tesco to take new ideas and offers to smaller groups of customers, while using the remaining customers as control groups. This takes a lot of risk out of innovative ideas.

In many ways, the performance and customer data has become a powerful laboratory to test whether new ideas work or not. Tesco's performance information, especially its Clubcard data, is not just about passively observing trends; it is a massive laboratory of customer behaviour. If it launched a new initiative, but it was doing something wrong, it knew about it in days. Equally, if it was doing something right, the new initiative could be implemented nationwide in weeks.

Tesco's marketing director highlighted that, as a company, Tesco has moved from being intuitive to being analytical. 'This is a much more complicated business than it used to be. We don't forget our intuition, but better data led to better thinking, and our data give us the confidence to ask the right questions. You can have all the data you want, but the key is to use them to ask the right questions.'

For example, Tesco is now able to conduct experiments to understand whether new product lines, innovative offers, and price reductions have the desired effects. Using its customer data allows Tesco to track the response immediately, which takes a lot of guesswork out of organisational decision making.

Abridged from a case study in ICAEW Business performance management community: *Driving customer insights from data at Tesco*

However, it is not only large companies, with large IT budgets, like Tesco, that are able to make use of big data. Current technology means that small companies can also benefit from big data analytics.

For example, consider a retailer with two physical shops and a website selling goods. Each shop has a CCTV camera which records footage for crime prevention purposes. If the company buys (some relatively inexpensive) pattern recognition software, then the CCTV footage can be turned into customer analytics. The software can use the data from the camera to provide an analysis of customer footfall, and which areas of the shop people go to, what products they look at, and the conversion rates from looking at a product to buying it.

The company could also use tools such as 'Social Mention' which allows it to track comments made about it on social media, and even provides an analysis of those comments (eg, whether they are positive or negative).

7.4.1 Business intelligence

Business intelligence (BI) refers to technologies, applications and practices for collecting, integrating, analysing and presenting business information.

Analytics relates to the use of (a) data and evidence, (b) statistical, quantitative and qualitative analysis, (c) explanatory and predictive models, and (d) fact-based management to drive decision making.

Together, they include approaches for gathering, storing, analysing and providing access to data that helps users to gain insights and make better fact-based business decisions, to improve performance, to help cut costs or to help identify new business opportunities.

Examples of BI and analytics applications include:

- measuring, tracking and predicting sales and financial performance
- budgeting, financial planning and forecasting
- analysing customer behaviours, buying patterns and sales trends
- tracking the performance of marketing campaigns
- improving delivery and supply chain effectiveness
- CRM
- risk analysis
- strategic value driver analysis

Overall, a company also needs intelligence about its business environment to enable it to anticipate change and design appropriate strategies that will create business value for customers and be profitable in new markets and new industries in the future. Not only does a company have to anticipate the future, but it also needs the capability to react to that future successfully. (This reiterates the point we noted in section 3.8 earlier, that the speed and effectiveness with which an organisation reacts to opportunities and threats could be a dynamic capability for it.)

7.5 Information management in the supply chain

One of the key areas in which information can help generate competitive advantage is the supply chain, and supply chain management, which we discussed in the chapter Strategic implementation of this Workbook.

In relation to supply chain management, Edward Davis and Robert Spekman refer to the notion of the **'extended enterprise'**. This is a concept in which a dominant enterprise 'extends' its boundaries to all or some of its suppliers, developing collaboration within the supply chain. Davis and Spekman argue that organisations can gain competitive advantage through developing collaborative supply chains.

The transition from conventional supplier/buyer relations to an extended enterprise based on collaboration can be seen in the following steps:

Open market negotiations: Price-based discussions and adversarial relationships Cooperation: Fewer suppliers, information sharing, longer-term contracts Coordination: Information linkages, workflow linkages, EDI exchange Collaboration: Supply chain integration, joint planning, technology sharing
In this way, Davis and Spekman argue, firms can achieve sustainable competitive advantage through:

- achieving and maintaining lowest costs
- dealing with changing customer requirements that demand an increasing number of new and innovative products and services

Supply chain management is not only about cost reduction, but also about looking for revenue- enhancing opportunities.

Davis and Spekman's model highlights the importance of information flows within the extended enterprise. The model relies on connectivity between the supply chain partners, with seamless and transparent information flows between them.

7.5.1 Bullwhip effect

We have just suggested that supply chain management and networks should involve companies working together to meet customers' needs more effectively. In theory, collaboration and connectivity should enable this: customers order products, the vendor keeps track of what is being sold and orders enough materials or inputs from supplier to meet customers' demand and replenish inventory levels in line with expected future demand.

However, in practice, the supply chains are not always this coordinated. Suppliers, manufacturers, salespeople and even customers often have an incomplete understanding of what the real demand is. These dynamics create inaccuracy and volatility in production levels, and these (inaccuracies and volatility) increase for operations further upstream in the supply chain.

This increase is known as the bullwhip effect, because the increasingly large disturbances in the chain as they work their way to the end resemble the oscillations in a whip when it is cracked.

Each group in the supply chain (suppliers, manufacturers, sales staff) only has control over part of the chain, but the decisions they take (for example, ordering too much or too little) affect production or inventory levels throughout the whole chain. Furthermore, each group in the chain is influenced by decisions that others are making.

We can illustrate the effect using a very simple model of a supply chain, in which all the producers in the chain work on the principle that they keep one month's inventory in stock at any time. In the model, market (customer) demand has historically been running at 100 units per period (prior to period 1). However, demand starts to fluctuate from period 2. The bullwhip effect can be seen by the fact that the producer and intermediate supplier's monthly production levels fluctuate increasingly more than market demand, in response to changes in market demand.

Period	Customer	Final producer		Intermediate supplier	
	Demand	Production	Inventory	Production	Inventory
1	100	100	100	100	100

2	95	90 (*)	95	80	90
3	95	95	95	100	95
4	100	105	100	115	105
5	100	100	100	95	100

* Production (90) = Closing inventory (95) + Demand (95) – Opening inventory (100)

Although the bullwhip effect in our model is simply caused by movements in and out of stock working their way up the supply chain, the effects can be magnified by problems of communication or coordination in the supply chain. This can be illustrated by the story of a car manufacturer which found itself with surplus inventories of green cars. To help get these sold, the car manufacturer's sales department offered special deals on green cars, so demand for them increased. However, the production departments were unaware of the promotion, and so when they saw the increase in sales they increased the production of green cars.

This simple 'car' example highlights one of the key problems behind the bullwhip effect: each operation in the supply chain only reacts to the orders placed by its **immediate customer**, but they have little overview of what is happening throughout the chain as a whole.

Therefore, in order to improve supply chain performance (and reduce the bullwhip effect) organisations need to improve the coordination of all the activities in the chain and the **knowledge sharing throughout the supply chain**. For example, if retailers make information on sales available to their suppliers, suppliers are more aware of movements in final customer demand and manage production accordingly.

In this context, one way to reduce the bullwhip effect is through better forecasts. However, a more important solution is to make sure that the strategies of all the firms in the supply chain are harmonised, and one way of doing this is through vendor-managed inventory.

7.5.2 Vendor-managed inventory (VMI)

One way of reducing fluctuations in demand and production throughout the supply chain is to allow an upstream supplier to manage the inventories of its downstream customer. This is known as VMI.

Under a VMI model, the (downstream) buyer of a product provides information about customer demand to the (upstream) supplier of that product, and the supplier manages production levels in order to maintain an agreed inventory of the product to meet demand.

VMI encourages a closer relationship and understanding between buyer and supplier, and it helps reduce both the levels of inventory in the supply chain and the risk of stock-out situations. Because the vendor is responsible for supplying the buyer when items are needed, this removes the need for the buyer to hold significant levels of safety stock.

However, a crucial element in VMI working effectively is the use of EDI between the buyer and supplier; for example, so that the supplier knows the quantity of a product the buyer has sold to end- user customers. Effective VMI also uses statistical methodologies and demand-planning tools to help forecast and maintain the correct levels of inventory in the supply chain (taking account of variables such as promotions or seasonality, for example).

Integrated supply chain management packages provide web-enabled visibility to suppliers so that they can review the on-hand inventory at their buyers' warehouses. The packages also allow buyers and suppliers to calculate buffer quantities, and when the buffer level is reached this triggers a new order from the supplier.

A crucial difference between a VMI system and traditional supply chain arrangements is that the vendor receives data from the buyer rather than purchase orders. Instead of the downstream buyer making purchase orders and 'pulling' supply through the system, the upstream supplier now initiates the order (based on the purchase information they receive) and they 'push' supplies through the system.

7.6 Big data and digitalisation in knowledge-based organisations

Our discussions of big data in the chapter Data analysis, coupled with our discussions of knowledge management in this chapter underline how knowledge – and the use of technology to support knowledge creation – are becoming increasingly important components of organisations' potential competitive advantage.

The increasing competitiveness of the global economy means that productivity gains or product improvements made by one company are often rapidly eliminated by their competitors' response.

In this respect, a company's process of **innovation** – and its ability to derive value from the information it has about products and markets – is likely to be critical to maintaining its competitive advantage in the face of a constantly changing market and economic environment. (This also echoes the idea of dynamic capabilities we discussed in the chapter Strategic analysis.)

Equally, in the 'information society' which exists today, information is one of the most valuable assets which organisations have, and information management and knowledge management are also likely to be essential parts of an organisation's competitive success, because they play a key role in value creation and productivity.

7.6.1 Knowledge-based organisations

Many organisations focus primarily on the routine tangible and observable activities that they carry out on a daily basis.

However, knowledge-based organisations also focus on two related processes which underlie these primary processes: the effective **application of existing knowledge**, and the **creation of new knowledge** to produce economic benefits. The success of knowledge-based organisations relies on their intangible assets – such as research, design, development, creativity, learning and human capital.

Knowledge-based organisations have four goals in relation to knowledge management:

(a) to ensure that knowledge from one part of an organisation is applied to activities in other parts of the organisation

(b) to ensure that knowledge is shared over time so that the organisation benefits from past experience

(c) To facilitate people from different parts of the organisation collaborating to create new knowledge

(d) to provide opportunities and incentives for experimentation and learning

As we would expect, knowledge creation and knowledge sharing are key features of a knowledge- based organisation. And while it is important that knowledge is shared within an organisation – for example, between different departments – organisations are also realising that knowledge can often also be gathered and shared as a result of interactions with customers, suppliers, and possibly even competitors.

In this respect, initiatives such as **co-creativity** and **crowdsourcing** have been important for the creation and sharing of knowledge, as we discussed in the chapter Strategic marketing and brand management.

However, while technological developments such as data warehousing and data mining have already helped to promote knowledge management in organisations, and the advent of Web 2.0 technologies has helped organisations gather more information about customers, the volume and variety of 'big data' means it has the potential to act as a source of even more knowledge to be created and shared.

In this context, big data presents two key challenges to organisations:

(a) **Information strategy –** Organisations need to harness the power of information. Big data is providing new ways to leverage information, but organisations need to be able to take advantage of them if they are going to be able to use the information to generate growth. How will they harness big data to improve strategic decision making – for example in evaluating potential new investments?

Organisations also have to have the infrastructural capacity to manage the **volume**, **variety and velocity** of the data available – and to process it – in order to maximise its value as a source of information and knowledge.

(b) **Data analytics** – Organisations need to be able to draw insights from large and complex datasets, in order to understand and predict customer behaviours, and to improve customer satisfaction or to drive innovation.

7.6.2 Digitisation and business value

When considering the potential impact of digitisation on business, it is also important to acknowledge the point highlighted in McKinsey & Company's report: *Finding your digital sweet spot*. This report argues that 'while online sales, social networking and mobile applications have received most of the buzz when it comes to digital', the greatest bottom-line impact may come from cost savings and changes beyond the interface between company and customer. If organisations focus too narrowly on the impact of digitisation on distribution channels only and the end-user customer interface, they will only gain a small proportion of the value that digitisation could provide.

As such, McKinsey's report suggests that technology (and **digital transformation**, which we discussed earlier in this chapter) can drive business value in four different ways:

- Enhanced connectivity
- Automation of manual tasks
- Improved decision making
- Product or service innovation

Tools such as big data analytics, apps, workflow systems and cloud platforms – all of which can create business value – are too often applied selectively to certain parts of the organisation only, often around sales and marketing. However, they could also be used, equally beneficially, across a much wider range of activities:

(a) Connectivity with customers, colleagues and suppliers

- Customer experience – Seamless, multi-channel experience; 'Whenever, wherever' service proposition
- Product and service innovation – New digital products and services; co-creation of new products

(b) Decision making – based on big data and advanced analytics

- Enhanced corporate control – Improved real-time MISs, supporting improved decision making

(c) Innovation of product, business models and operating models

- Distribution – Digital augmentation of traditional distribution channels

- Marketing and sales – Digital marketing; improved targeting with greater customer insights
- Fulfilment – Digital fulfilment; virtual servicing and administration

7.6.3 Digitisation and the internet of things

More generally, there are a number of ways in which organisations can make use of digitisation and the information being created through the 'internet of things', as we have discussed in the chapter Strategic analysis.

Information and analytics

Monitoring behaviour – When products are embedded with sensors, companies can track the movements of the products and business models can be fine-tuned to take advantage of this detail. For example, insurance companies offer to install location sensors in customers' cars. These capture data on how well a car is driven, as well as where it travels. As a result, prices can be customised to the actual risk associated with operating the car, rather than being based on proxies such as a driver's age, gender and place of residence.

Similarly, **RFID tags** on products moving through the supply chain can be used to improve inventory management and to reduce working capital and logistics costs.

Situational awareness – Logistics managers for freight companies can use information about traffic patterns, weather conditions and vehicle locations to make adjustments to their vehicles' routes in order to reduce the risk of delays.

Decision analytics – Retail companies can monitor data from thousands of shoppers as they move through stores. Sensor readings and videos note how long shoppers spend at individual displays, while the stores also have information about what customers ultimately buy. Simulations based on the sensor readings, coupled with purchase records, can be used to increase revenues by optimising store layouts.

In relation to healthcare, sensors can be fitted to patients with heart problems and can monitor key indicators such as heart rate, rhythm and blood pressure which could give medics early warning of conditions which could otherwise lead to unplanned hospitalisation and emergency treatment costs.

Automation and control

Process optimisation – Sensors on production lines provide data about temperature, pressure or ingredient mixtures (for example) to computers which analyse the data and then send signals back to the production line to adjust the process – for example, to reduce the temperature if it has become too high. Similarly, sensors can be used to adjust the position of an object as it moves down an assembly line to prevent damage to the object, or the process jamming.

Resource consumption – 'Smart meters' can provide energy customers with visual displays showing their energy usage and the real-time costs of providing it. Equally, however, 'smart meters' provide suppliers with real-time information about the amount of energy being used across a network, so that they can manage the network more effectively in response to levels of demand.

8. *Using digital skills to develop competitive advantage*

Section Overview

In the digital era, organizations increasingly depend on data and information systems to support decision-making, innovation, and competitive advantage. An effective information strategy ensures that information resources are properly managed, accessible, and aligned with

organizational goals. However, the success of such strategies largely depends on the availability of appropriate digital skills among individuals and teams.

Digital skills are no longer limited to basic technical abilities

- *They encompass a broad range of competencies, including information literacy, data analysis, communication, and critical thinking.*
- *These skills enable organizations to transform raw data into meaningful insights and strategic actions.*

8.1 Digital Skills

Digital skills refer to the abilities required to use information and communication technologies (ICTs) effectively, safely, and critically. They involve not only technical proficiency but also cognitive and social competencies.

According to research, digital skills are a combination of:

- *Technical skills (use of tools and software)*
- *Cognitive skills (critical thinking and information evaluation)*
- *Social skills (communication and collaboration*

The European DigComp framework identifies five major areas of digital competence:

- *Information and data literacy*
- *Communication and collaboration*
- *Digital content creation*
- *Security*
- *Problem-solving*

8.2 Key Digital Skills Required for Competitive Advantage

(a) ***Data Literacy and Analytics Skills:*** *This is one of the most critical skills in modern organizations. This includes the ability to collect, interpret, and analyze data, the use of tools like dashboards and analytics software, and turning raw data into actionable insights.*

For example, companies like Netflix use data analytics to recommend content, improving user experience and retention.

(b) ***Digital Communication and Collaboration Skills:*** *Effective communication using digital tools helps organizations operate efficiently, such as the use of platforms like email, cloud collaboration tools, and project management software, the real-time information sharing across departments, and supporting remote and global teamwork. This leads to faster decision-making and improved productivity.*

(c) ***Technical and ICT Skills:*** *These include the ability to use and manage digital systems and technologies, an understanding of software, databases, and cloud computing, the ability to work with enterprise systems (ERP, CRM), and basic knowledge of automation and AI tools.*

Companies such as Microsoft leverage strong technical expertise to maintain innovation and market leadership.

(d) ***Cybersecurity Awareness and Skills:*** *Protecting information is essential for maintaining trust and avoiding losses. These include understanding data protection principles, identifying and preventing cyber threats, and ensuring secure use of digital systems. Strong cybersecurity reduces risks and strengthens the company's reputation.*

(e) Digital Content Creation Skills: *Organizations need to present information effectively, such as creating reports, dashboards, and presentations, producing digital marketing content (social media, videos, websites) and designing user-friendly digital interfaces. This skill improves customer engagement and brand value.*

(f) . Critical Thinking and Problem-Solving Skills: *Employees must be able to evaluate information and make informed decisions, i.e., analyze complex situations, identifying reliable vs. unreliable information, and solve business problems using digital tools. This skill is essential for strategic planning and innovation.*

(g) Adaptability and Continuous Learning: *Technology changes rapidly, so employees must learn new tools and systems quickly, adapt to digital transformation, and stay updated with industry trends. Companies that learn faster gain a long-term competitive edge.*

(h) AI and Automation Skills: *Modern competitive advantage increasingly depends on advanced technologies, such as understanding artificial intelligence and machine learning basics, using automation tools to improve efficiency, and enhancing decision-making with predictive analytics. For instance, Tesla uses AI-driven systems to innovate in autonomous driving and manufacturing.*

8.3 Digital Skills for Achieving Competitive Advantage

Digital skills have become a critical factor in helping organizations gain and sustain a competitive advantage in today's fast-changing business environment. These skills enable companies to effectively use information, adopt new technologies, and respond quickly to market demands.

(a) Enhances Data-Driven Decision Making:

- *Digital skills allow employees to collect, analyze, and interpret data accurately.*
- *Organizations can make informed strategic decisions*
- *Reduces uncertainty and business risks*
- *Improves forecasting and planning*

For example, companies like Amazon use data analytics to understand customer behavior and optimize their services.

(b) Increases Operational Efficiency: *With strong digital skills, employees can:*

- *Automate routine tasks*
- *Use advanced software and systems*
- *Reduce errors and save time*

This leads to higher productivity and lower operational costs, giving companies a competitive edge.

(c) Supports Innovation and Digital Transformation: *Digital skills encourage organizations to adopt new technologies such as AI, cloud computing, and automation.*

- *Enables the development of new products and services*
- *Improves business models and processes*
- *Helps organizations stay relevant in competitive markets*

For instance, Tesla continuously innovates using digital technologies.

(d) Improves Customer Experience: *Digital skills help organizations better understand and engage with customers.*

- *Personalized services and recommendations*
- *Faster communication through digital platforms*
- *Improved service quality*

Companies like Netflix use digital skills to enhance user experience through personalized content.

(e) ***Strengthens Communication and Collaboration:*** *Digital tools enable seamless communication within and outside the organization.*
- *Enhances teamwork and knowledge sharing*
- *Supports remote work and global operations*
- *Speeds up decision-making processes*

This improves overall organizational performance.

(f) ***Ensures Cybersecurity and Risk Management:*** *Digital skills help protect sensitive data and systems.*
- *Reduces the risk of cyberattacks*
- *Ensures data privacy and compliance*
- *Builds trust with customers and stakeholders*

A secure organization is more reliable and competitive.

(g) ***Enables Faster Adaptation to Change:*** *In a rapidly evolving digital environment, adaptability is key.*
- *Employees can quickly learn new technologies*
- *Organizations respond faster to market changes*
- *Maintains long-term competitiveness*

Context Example: Amazon

Amazon, one of the world's largest e-commerce companies, relies heavily on digital skills to implement its information strategy effectively.

The company collects vast amounts of data from customer purchases, browsing history, product reviews, and logistics operations. To manage and utilize this data, Amazon depends on employees with strong digital competencies.

First, Amazon's data scientists and analysts use data literacy and analytical skills to process large datasets. They apply advanced analytics and machine learning techniques to understand customer behavior and predict future purchasing patterns.

Second, employees use technical and ICT skills to operate cloud-based systems such as AWS (Amazon Web Services), which store and manage huge volumes of information securely and efficiently.

Third, Amazon teams apply communication and collaboration skills through digital platforms to share insights across departments such as marketing, supply chain, and customer service. This ensures coordinated decision-making.

Fourth, digital content creation skills are used to design personalized product recommendations, advertisements, and user interfaces that enhance customer experience.

Fifth, cybersecurity skills play a critical role in protecting sensitive customer data, ensuring privacy, and maintaining trust.

Finally, managers use critical thinking and decision-making skills to turn data insights into strategic actions, such as dynamic pricing, inventory optimization, and targeted marketing campaigns.

As a result, Amazon's strong digital skill base enables it to:

Deliver personalized customer experiences

Optimize operations and logistics

Maintain a competitive advantage in the global market

8.4 Role of Digital Skills in Implementing Information Strategy

Digital skills support information strategy in several ways:

- *Improved decision-making: Skilled employees can analyze data and generate insights*
- *Efficiency: Automation and digital tools streamline information processes*
- *Innovation: Digital competencies enable adoption of emerging technologies such as AI*
- *Competitive advantage: Organizations with strong digital capabilities perform better in dynamic markets*

Recent studies highlight a growing digital skills gap, where organizations struggle to find employees with the necessary competencies, affecting productivity and competitiveness

8.5 Challenges in Developing Digital Skills

Despite their importance, several challenges exist:

- *Digital divide (unequal access to technology and skills)*
- *Rapid technological change*
- *Lack of training and education programs*
- *Organizational resistance to change*

Study shows that differences in digital skills can significantly impact performance and access to opportunities

Digital skills are a key driver of competitive advantage in today's business environment. Organizations that invest in data literacy, technical expertise, cybersecurity, and innovation-oriented skills can make better decisions, improve efficiency, enhance customer experience, and stay ahead of competitors. Without these skills, even the best strategies may fail to deliver results

Summary

	Tick off
Information systems are necessary for an organisation to have the information it needs for strategic planning, and then to assess whether its objectives and targets are actually being met.	☐
Although IS/IT are often used to support business strategy, they may help determine business strategy (eg, e-commerce) and provide a source of competitive advantage in their own right (eg, data mining and CRM; analytics and inventory management; knowledge management and organisational learning).	☐
The amount of data and information organisations are processing means they need increased storage capacity. This has been one of the main factors behind the growth of cloud computing. Cloud computing provides organisations with the flexibility to increase their data capacity as required. It also improves accessibility (to data) for employees around the world.	☐
Managers need information for three main reasons: to make effective decisions, to control the activities of their organisation, and to coordinate the activities of the organisation.	☐
Management accounts are an important source of information for managers. For example, management accounts can provide information on costs and costing, prices and pricing, budgets, and variances.	☐
Management accounting tools can also be valuable in evaluating business strategies or elements of business strategies. Useful tools include: breakeven analysis, limiting factors, expected values, activity-based costing, transfer pricing, and the balanced scorecard.	☐
A key feature of an organisation's management information systems is that they enable the organisation to measure its performance in relation to its critical success factors (CSFs). An organisation should set its KPIs to measure performance in relation to its CSFs.	☐
Different types of information system must meet the information requirements of different levels of management within an organisation. Transaction processing systems (TPS) provide operational level data. Management information systems (MIS) summarise TPS data at a tactical level.	☐
Executive information systems (EIS), management information systems (MIS) and decision support systems (DSS) could be used to provide summarised information at strategic level.	☐

	Tick off
An important consideration for managers is how to integrate information from different systems. Enterprise applications (eg, enterprise resource planning systems, supply chain management systems, or customer relationship management systems) span different functional areas and focus on executing business processes across the organisation.	☐
Although information in general is seen as a valuable resource for organisations, when evaluating a new information system an organisation should consider the value of the information compared to the costs associated with using it (ie, cost-benefit analysis).	☐
Information is ultimately only useful to managers or staff if it adds to their understanding of a situation. The qualities of good information can be summarised by the mnemonic 'ACCURATE': accurate, complete, cost-beneficial, user-targeted, relevant, authoritative, timely and easy to use.	☐
The increased reliance on information systems in organisations increases the potential impact of cyber risks (eg, viruses; hacking). In turn, this increases the strategic importance of cyber security within organisations, and the need for either internal or external assurance over the security of their systems and the integrity of the information produced by them.	☐

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, complete the following knowledge diagnostic and check you are able to confirm you possess the following essential learning from this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
1.	What are the benefits and problems of digital transformation? (Topic 1)
2.	What do managers use management accounting information for? (Topic 2)
3.	What are the strategic benefits of management information systems? (Topic 3)
4.	What assurance might an organisation seek over its information and information systems? (Topic 5)
5.	What practical steps can an organisation put in place to better manage cybersecurity risk? (Topic 6)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
1 MMM	In this question you are asked to read through the scenario to understand the information requirements of the company in light of recent over or under-charging of clients. Although the company currently provides management accounting information to the board, you are asked to identify additional information to help plan and control the business. Review the characteristics of board level information in this chapter and then apply these to the scenario to identify specific information requirements.
2 FDS Co	This is a multi-part question looking at different aspects of information. The scenario focuses on discussions around the introduction of a customer profitability reporting system. The CEO wants to know where the information will come from and how it will be collected, as well as what the costs and benefits of the reporting system would be. As is often the case with SBM&L questions, the emphasis is on demonstrating application of knowledge to the scenario and commercial awareness.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your ueries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Self-test questions

Answer the following questions.

1 MMM

MMM is an advertising agency specialising in work for the hotel industry. MMM has no formal mission statement or strategy. However, the management board agrees that MMM should grow, while remaining profitable as it always has been.

MMM's market niche is small, and competition with the sector is intense. MMM is unaware of the total size of its market niche, but MMM believes that the market is growing. Within its market, BBB estimates it is the second or third largest company. MMM thinks it wins most of its work because of the high quality of its output, but thinks sometimes price is the determinant for securing a new client.

MMM employs 15 staff, but it has always found it difficult to attract sufficient staff. MMM sometimes has to turn down work due to a lack of staff. It passes such work on to other advertising agencies. When this happens, MMM earns commission on the work it has referred. However, due to the seasonal nature of the hotel industry, there are times when MMM has surplus capacity. The management board believes MMM could increase its profit if it increased the number of its staff in order to accept some of the work that it currently turns down.

MMM's accountant provides management accounting information to the management board to support planning and decision making. This consists of budgetary control and standard costing information. The accountant produces budgetary control monthly reports which are very detailed and show every expenditure over £25. The annual budget is flexed each month to reflect that month's level of activity. The accountant produces very detailed monthly variance reports relating to labour, variable overheads and fixed overheads. The accountant produces a monthly profit figure.

Work undertaken for clients is priced by adding a standard uplift to total cost. A blanket overhead recovery rate is used in arriving at total cost. On occasion, some of MMM's clients have complained that they have been charged too much. However, on other occasions MMM believes that it may have undercharged its client.

The management board has stated that it 'urgently needs additional information to support its planning and decision making'. A member of the management board attended a recent seminar which discussed benchmarking and is investigating whether this technique could assist MMM.

Requirement

Advise MMM's management board what additional information it needs to support its planning and decision making.

2 FDS Co

FDS Co installs irrigation systems. Its customers include farmers, local government bodies, sports centres and building contractors. Its annual sales turnover is currently £25 million, and annual pre-tax profit is £1.2 million. The company is currently working at close to capacity, and its activities are restricted by a shortage of skilled engineers to install and maintain the pumping equipment for the irrigation systems.

Prices for the installation of irrigation systems are negotiated between customers and sales representatives of FDS. The sales representatives have authority to offer discounts on price in order to win large contracts or in return for more favourable payment arrangements.

The installation of irrigation systems typically takes several months for large contracts, and FDS usually sets up a site office on the customer's premises with a compound for holding system parts and other inventory. Delivering inventory to a site can be difficult, especially when the customer is a farmer in a remote location.

Sports centres often insist on minimum disruption to sporting activities during the installation of a new irrigation system. This can limit the size of the site office and inventory compound, which sometimes delays progress on a contract when the installation team has to wait for more inventory to be delivered. The installation teams fill in timesheets on a daily basis.

The management accountant of FDS is not satisfied with current reporting arrangements and thinks that some types of contract are much more profitable than others. It seems likely, for example, that farmers negotiate much better prices than local government bodies (such as local councils). Some contracts are more complex and difficult to negotiate than others, and winning a contract from a local government body can take much longer than contract negotiations with other customers.

The management accountant thinks that the company would benefit from the introduction of a customer profitability reporting system, where the profitability of each type of customer is measured and assessed separately. A benefit of this type of reporting system is that FDS should be able to put more resources into selling to more profitable types of customers, thereby helping to increase the company's profitability.

The management accountant is particularly concerned that as FDS is working close to its capacity there is a danger it could turn down contracts (due to lack of capacity) which would have been more profitable than contracts it has accepted.

This issue has been highlighted recently as FDS has been asked to undertake three new installations, but only has sufficient resources to carry out two of them.

	Sports centre	Farm	Building contractor
Basic price (£)	200,000	200,000	200,000
Discount negotiated	4%	5%	3%
Installation team time budgeted (@ £1,500 per day)	56 days	50 days	45 days

The inventory delivery costs for the sports centre and the building contractor's installations were expected to be the same, but the costs for the farm installation were expected to be £25,000 higher. All other costs were expected to be the same for the three contracts. FDS has not yet decided which two contracts to accept.

Despite the management accountant's concerns about the current reporting arrangements, the CEO is not convinced that a customer profitability reporting system would be useful for FDS. The CEO wants to know where the information will come from and how it will be collected, as well as what the costs and benefits of the reporting system would be.

Requirements

2.1 Discuss the information requirements for a customer profitability reporting system within FDS and where the data for this system might be obtained.

2.2 Discuss how the information in a customer profitability reporting system might improve management control and decision making within FDS.

2.3 Explain what the direct and indirect costs of obtaining data and producing information for this system might be, and how the value of such a system should be assessed.

3 CMA Supermarkets

The board of CMA Supermarkets is considering an upgrade of its company-wide IT system. The company has been opening new supermarket stores at a rapid rate in recent years, and has ambitions to rival the established supermarket chains in its country. The board believes that CMA could gain a significant competitive advantage from having a unified corporate database and from replacing its bar code technology with RFID, the radio frequency technology for labelling and identifying inventory.

The management accountant at CMA has been asked to explain to the board how a new network system and RFID technology may help to improve the management accounting system within the company, and also the company's performance.

At the moment, management accounts are prepared for each individual store, and monthly sales and profitability reports are presented to the store manager. Regional reports and a national report on company performance are also prepared each month, and presented to senior management and the board.

Because the company is trying to increase market share rapidly, it keeps its prices as low as possible and, whereas rival companies achieve a gross profit margin of about 55% on its sales, CMA's average gross margin is slightly below 50%.

In addition, a number of CMA's stores have reported an increase in out of stock products in recent weeks. The store managers are concerned that if customers keep finding items out of stock they will stop shopping at CMA and will revert to one of the established supermarket chains.

The management accountant thinks that new technology will help to improve profitability and will influence the nature of performance reporting systems. However, the benefits of improved IT will only be obtained if there is a radical rethinking of how information is used within CMA.

Requirements

3.1 Explain how RFID technology for tracking inventory and inventory movements may help to improve the quality of management information in CMA Supermarkets, and may also help to improve CMA's operational performance.

3.2 Assess the changes which may be required to performance reporting in CMA Supermarkets in order to obtain the greatest value from a new IT system and RFID technology.

3.3 Suggest how the IT system for a supermarket might exploit information about individual customers to improve sales and profits.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 **Strengths**

Operations – The owners have developed a considerable income stream by charging for their respected consultancy advice about antiques.

Service – The owners have developed a national reputation in the antiques trade and their experience has allowed them to build up a loyal following of repeat customers.

Firm infrastructure – The business operates from a large modern shop in a fashionable area, which makes its premises very valuable. If the owners sold the shop they could realise a substantial capital gain.

Human resource management – The business is run by the owners, and their experience in the antiques trade, and their reputation as experts, is a valuable asset for GLS.

Weaknesses

Inbound logistics – The inconvenience of the shop's location may mean that people with antiques to sell may take them to another antiques dealer rather than bringing them to GLS.

Operations – The location of the shop is not convenient for potential customers (antique traders and collectors) to visit, and the people who live nearby are not interested in buying antiques.

The owners are unable to attend antiques fairs because they need to be physically at the shop to keep it open.

Outbound logistics – The inconvenience of the location may also be a weakness in relation to customers buying antiques; for example, a collector may choose not to buy an item due to the difficulty of transporting home, or GLS may have to bear the cost of transporting customers' purchases to their homes.

Security costs have also increased significantly in recent years, and these costs are presumably related to the cost of storing antiques at the shop. If these security costs continue to rise, storing antiques in the shop may become a weakness.

Marketing and sales – The owners themselves believe that the location of the shop restricts the success of the business.

However, at the moment GLS's website is very basic, and although a number of people have visited it there is no scope for them to buy anything through the website.

Firm infrastructure – The fixed costs relating to the shop premises have increased sharply over the last five years. This is likely to be a contributing factor to the fact that the business made a loss for the first time in the last year.

Human resource management – There appear to be no succession plans in place for when the owners retire. Given that they have been running the business for over 30 years it is likely they will be approaching retirement age soon.

Technology development – The business has only recently set up a very basic website. As we have already noted, it does not have any e-commerce capabilities.

1.2 **Note:** A useful way of approaching this question would be to consider how the introduction of e-commerce will help address some of the weaknesses of GLS's operations which were identified in part (a), or to build on some of its existing strengths. However, note that, as in part

(a), the question requires you to link your answer to the value chain, so it would be sensible to use value chain functions as headings for your answer.

Operations – The main impact of introducing e-commerce is that GLS is no longer reliant on its current shop as a physical site for their business. GLS will still need a site where it can store antiques, and it is likely to retain some kind of shop or showroom where the owners can make face to face sales. However, this can be moved to new, cheaper premises, for example an out of town location which should also be easier for antique traders to get to.

Outbound logistics – The new location should provide a more convenient base from which to distribute the antiques, and it should also help to reduce the insurance and security costs relating to storing the antiques.

Marketing and sales – E-commerce should increase the geographical reach of the business, and so should increase sales. Potential customers who previously couldn't visit GLS's shop (perhaps even international customers) can now search the website to look for items they may want to buy (and then also buy them online if they want to).

Firm infrastructure – The reduction in overheads and potential for increased sales should allow GLS to return to profit. The owners will also receive an injection of cash from the proceeds of the sale of the shop, which should cover the costs of the website upgrade.

Human resource management – Once the website is running, the owners will be able to reduce the opening hours of their 'bricks and mortar' shop, which will give them more opportunity to attend the antiques fairs which they have previously been unable to attend.

Technology – Once the website starts becoming a source of income for the business, then technology becomes a much more important aspect of its value chain. In effect, the website could become GLS's main 'shop'.

Procurement – As the website increases the geographical reach of GLS's customers, it may mean that the supply of antiques into the business also increases as more and more people become aware of it.

Answer to Interactive question 2

2.1 WORKINGS

(1) **Fixed production costs absorbed per unit**

$$= \frac{\text{Budgeted fixed production costs}}{\text{Budgeted production}}$$

$$= \frac{110{,}000}{2{,}200}$$

= £50 per unit

Therefore full cost (absorption cost) per unit = £50 + 70 = £120

(2) Under-/over-absorption of overheads

	July	August
Actual production	2,000	2,500
Expected production	2,200	2,200
(Under-)/over-production	(200)	300
Fixed costs per unit		£50 £50
(Under-)/over-absorption	(£10,000)	£15,000

(a) **Income statements – absorption costing**

		July		August
	£	£	£	£
Sales (£150 per unit)		225,000		450,000
Cost of sales:				
Opening inventory	Nil		60,000	
Production	240,000		300,000	
Less closing inventory	(60,000)		Nil	
		(180,000)		(360,000)
Gross profit		45,000		90,000
(Under-)/over-absorption		(10,000)		15,000
		35,000		105,000

(b) **Income statements – marginal costing**

		July		August
	£	£	£	£
Sales		225,000		450,000
Cost of sales				
Opening inventory	Nil		35,000	
Production	140,000		175,000	
Less closing inventory	(35,000)		Nil	
		(105,000)		(210,000)
Contribution		120,000		240,000
Less fixed costs		(110,000)		(110,000)
Net profit		10,000		130,000

2.2 **Difference in profits**

	July	August	
	£	£	
Absorption cost profit	35,000	105,000	
(Increase)/decrease in inventory			
(Increase)/decrease in inventory and fixed costs charged against sales	(25,000)	25,000	(500 × £50)
Marginal cost profit	10,000	130,000	

Profits are different under each method due to the fixed costs that are included in closing inventory with absorption costing.

Answer to Interactive question 3

3.1

Overhead cost	Cost driver
Set-up costs	Number of set ups
Materials handling costs	Number of times materials handled
Ordering costs	Number of orders
Engineering costs	Number of spare parts required

3.2

$$\frac{\text{[Number of set - ups per product/Total number of set - ups} \times \text{Total set - up costs]}}{\text{Number of units produced}}$$

$$\text{Alpha} = \frac{[(20/100) \times 25{,}000]}{500} = £10.00/\text{unit}$$

$$\text{Brave} = \frac{[(15/100) \times 25{,}000]}{300} = £12.50/\text{unit}$$

$$\text{Echo} = \frac{[(30/100) \times 25{,}000]}{400} = £18.75/\text{unit}$$

$$\text{Oscar} = \frac{[(35/100) \times 25{,}000]}{200} = £43.75/\text{unit}$$

Material handling costs per unit =

$$\text{Alpha} = \frac{[(3/15) \times 69{,}000]}{500} = £27.60/\text{unit}$$

Bravo = £61.33/unit

Echo = £23.00/unit

Oscar = £138.00/unit

$$\text{Ordering costs per unit} = \frac{[\text{No of orders per product/Total no of orders}) \times \text{Total ordering costs}]}{\text{Number of units produced}}$$

$$\text{Alpha} = \frac{[(11/64) \times 32{,}000]}{500} = £11.00/\text{unit}$$

Bravo = £20.00/unit

Echo = £20.00/unit

Oscar = £62.50/unit

Engineering costs/unit = [(No of space parts per product/Total spare parts) × Total engineering costs / Number of units produced.

$$\text{Alpha} = \frac{[(15/60) \times 45{,}000]}{500} = £22.50/\text{unit}$$

Bravo = £50.00/unit

Echo = £18.75/unit

Oscar = £56.25/unit

Total cost per unit

	Alpha	Bravo	Echo	Oscar
	£	£	£	£
Direct costs:				
Material	60.00	42.00	80.00	100.00
Labour	32.00	20.00	35.00	50.00
Other costs:				
Set-up costs	10.00	12.50	18.75	43.75
Material handling costs	27.60	61.33	23.00	138.00
Ordering costs	11.00	20.00	20.00	62.50
Engineering costs	22.50	50.00	18.75	56.25
	163.10	205.83	195.50	450.50

Note: Remember to include the direct costs that were given in the question when calculating the total cost per unit. This is a common mistake in exam situations.

Answer to Interactive question 4

4.1

	A	B	C
1		Current	After marketing campaign
2	Sales price per pair	55	65
3	Variable cost per pair	25	25
4	Contribution per pair	30 [1]	40
5	Fixed costs:		
6	Marketing	75,000	115,000
7	Salaries	200,000	200,000
8	Other	150,000	150,000
9	Total fixed costs	425,000	465,000
10	Annual sales volume (pairs)	20,000	20,000
11	Breakeven point (pairs)	14,167 [2]	11,625

[1] =B2-B3. The formula can be copied in C4.

[2] =B9/B4. The formula can be copied in C11.

4.2

	A	B	C
1		Current	After marketing campaign
2	Sales price per pair	55	65
3	Variable cost per pair	25	25

4	Contribution per pair	30 [1]	40
5	Fixed costs:		
6	Marketing	75,000	115,000
7	Salaries	200,000	200,000
8	Other	150,000	150,000
9	Total fixed costs	425,000	465,000
10	Annual sales volume (pairs)	20,000	20,000
11	Breakeven point (pairs)	14,167 [2]	11,625
12	Margin of safety (pairs)	5,833 [3]	8,375
13	Margin of safety %	29% [4]	42%

[1] =B2-B3. The formula can be copied in C4.

[2] =B9/B4. The formula can be copied in C11.

[3] =B10-B11. The formula can be copied in C12.

[4] =B12/B10. The formula can be copied in C13.

4.3

	A	B	C
1		Current	After marketing campaign
2	Sales price per pair	55	65
3	Variable cost per pair	25	25
4	Contribution per pair	30 [1]	40
5	Fixed costs:		
6	Marketing	75,000	115,000
7	Salaries	200,000	200,000
8	Other	150,000	150,000
9	Total fixed costs	425,000	465,000
10	Annual sales volume (pairs)	20,000	20,000
11	Breakeven point (pairs)	14,167 [2]	11,625
12	Margin of safety (pairs)	5,833 [3]	8,375
13	Margin of safety %	29% [4]	42%
14	Target profit		25,000
15	Required sales revenue		796,250 [5]

[1] =B2-B3. The formula can be copied in C4

[2] =B9/B4. The formula can be copied in C11

[3] =B10-B11. The formula can be copied in C12

[4] =B12/B10. The formula can be copied in C13

[5] =(C9+C14)/(C4/C2)

Answer to Interactive question 5

The first step is to calculate contribution per unit:

	McEnrova	Grafassi	Federdal
Total sales units	8,000	6,000	4,000
Total contribution (£)	144,000	132,000	120,000
Contribution per unit	£18	£22	£30

The lowest common denominator for the sales mix is 4:3:2.

Total contribution for the standard batch is £198 (4 × £18 + 3 × £22 + 2 x £30).

$$\text{Breakeven point in batches} = \frac{306{,}900}{198} = 1{,}550 \text{ batches}$$

The number of units of each model that must be sold in order to break even is:

McEnrova:	1,550 × 4	= 6,200
Grafassi:	1,550 × 3	= 4,650
Federdal:	1,550 × 2	= 3,100

The breakeven point in terms of sales revenue is:

	£	
McEnrova	248,000	(£40 × 6,200)
Grafassi	279,000	(£60 × 4,650)
Federdal	310,000	(£100 × 3,100)
	837,000	

Check:

	McEnrova	Grafassi	Federdal	Total
Sales units	6,200	4,650	3,100	
Contribution per unit	£18	£22	£30	
Total contribution (£)	111,600	102,300	93,000	306,900
Total fixed costs (£)				306,900
Profit/(loss)				Nil

Answer to Interactive question 6

MEMORANDUM

To:	Board members
From:	Accountant
Date:	[today's date]
Subject:	Purposes and benefits of a Management Information System

KLL requires a new Management Information System to provide more detailed information on the various activities of the company. The existing MIS is limited in the information that it can provide, and the directors have identified additional information that would help them control and develop the business. This memo summarises the purposes and benefits that can be obtained from a modern Management Information System.

(1) Purposes of a Management Information System

An MIS is a system to convert data from internal and external sources into information and to communicate that information, in an appropriate form, to managers at all levels in all functions to enable them to make timely and effective decisions for planning, directing and controlling the activities for which they are responsible. The MIS is therefore established in a company to satisfy the information needs of management.

Within KLL, the directors will already be aware of this objective of an MIS because the company already has an MIS. The limitations of that MIS are now apparent, however, because activities cannot be split between those that are profit making and those that are loss making.

(2) Benefits of an MIS

The benefits of an MIS are summarised below, focusing particularly on the requirements of KLL.

(a) Provision of financial information

The existing MIS can provide some financial information, although the limitations of this information have been recognised by the directors. This limitation may well be a function of an older MIS being designed to produce specific reports rather than holding the data in some form of database and then different reports being generated from that data as required.

A new MIS should store data in a less rigorous format, enabling different reports to be produced as required. Details of profit- and loss-making sports can therefore be obtained.

(b) Provision of more timely information

The current MIS produces reports on a monthly basis. It is not clear whether this is a system limitation or whether reports have not been requested on a more frequent basis. However, monitoring the profitability of individual sports activities may benefit from more frequent provision of information. For example, if a competitor starts pricing activities below the price charged by KLL, then an immediate response will be required rather than waiting up to a month to amend prices.

A modern MIS should be able to provide information on a daily, if not real-time, basis to enable the directors to make quicker and more effective decisions.

(3) Provision of summary information

The Managing Director is concerned about the inappropriate level of detail being provided by the MIS. If the detail cannot be interpreted (per the question), then it is likely that the MIS is producing information at an operational level rather than a strategic or tactical level. The detail is available, but this has not been summarised appropriately. It is possible, for example, that income from individual games of squash can be seen, but not the total income for each court or for the sport squash itself for each week or month.

The new MIS will provide a summary of income initially, with the ability to provide more operational information as necessary using the 'drill down' ability of many information systems. Focusing the information at the strategic level first, rather than the operational, should provide the Managing Director with the appropriate level of detail.

(4) 'Better' information

The Managing Director is also concerned about the lack of 'good' information. This appears to be linked to the comment concerning the limited technical knowledge of staff and poor support from the software company. It is therefore possible that staff either have a lack of training in the use of the MIS or they are producing bespoke reports, and are not receiving the support from the supplier to help them do this. The board is not receiving good information because reports are not sufficiently focused on the activities of KLL.

Whether the situation actually needs a new MIS to resolve it remains unclear. It is possible that appropriate training or support would enable staff to provide the appropriate reports for the board. Alternatively, more recent MIS programs normally provide an easy to use report generator so staff should find it easier to produce the necessary reports.

Alternatively, data can be exported into a spreadsheet package for additional analysis and production of visual aids such as charts and graphs as necessary.

(5) Staff morale

Providing a new MIS will have other benefits for the company, such as increased staff morale and a better working environment. Staff are likely to be more motivated because the company is providing the software that is needed to carry out their job.

Answer to Interactive question 7

Issues for the board to consider

There is an allegation that Hiway's infotainment systems are vulnerable to cyberattack. Although Hiway has strenuously denied the claim, it does not appear to have any processes in place for monitoring cybersecurity – hence its inability to demonstrate these to Fova.

If Hiway does not have a cybersecurity policy in place, this represents a significant weakness in company's risk management regime, in general terms. However, there are potentially also two more specific implications in this case:

- Just because Hiway's infotainment systems have not yet suffered a cyberattack does not provide any guarantee they will not be attacked in the future. Hiway needs some more robust evidence to support their defence against the claims made in the report.
- Moreover, Fova's response indicates the seriousness which Hiway's customers place on system security, and the importance of ensuring cybersecurity throughout their supply chains. If Hiway is not able to demonstrate that its products are safe – and potentially, the information systems throughout the company as well – this is likely to cause significant damage to its reputation and could result in it losing customers.

Actions to take

As an urgent response to the report, Hiway should hire an independent expert to test the security of the infotainment system in question – so that, either, Hiway can substantiate their defence against the claims in the report or, if there are weaknesses in the infotainment system, these can be addressed as soon as possible.

More generally, however, the board needs to consider Hiway's cybersecurity risk management strategy – beginning with the fact that the board itself needs to take responsibility for cybersecurity within the company, and to agree how principal risks need to be mitigated or managed.

As well as ensuring the security of its products, Hiway also needs to protect the company's information systems and information assets against cyberthreats. In this respect, the board needs to establish a strategy for monitoring all Hiway's ICT systems and networks, and then implement that strategy. An important, on-going element of the strategy should be monitoring

and testing security controls within the company, and addressing any weaknesses in them.

Although the testing of the infotainment system will need to be a separate, specific exercise, Hiway should consider obtaining an ISO 27001 certification to demonstrate that its overall information security management system is functioning properly.

Answer to Interactive question 8

8.1 **CSFs and performance management** – PBB's CSFs should identify the areas which are central to its future success and therefore where it needs to perform well if it is to be successful overall.

CSFs and KPIs – Once PBB has identified which aspects of performance are crucial to future success, it needs to be able to measure how well it is performing in relation to them. It can use KPIs to measure whether or not its CSFs are being achieved.

Examples – So, for example, measuring the number of students who do not complete each year of a course, or measuring the percentage of students who drop out each year, will allow PBB to monitor student performance.

8.2 **Improved student performance** – One of the aims of PBB's knowledge management strategy should be to gather, organise and share knowledge and experience about areas of its business which will contribute to its future success.

In this respect, if the staff in the departments that currently have high student drop-out rates can learn more about how to improve student performance, this should enable them to reduce the drop-out rates. In turn, this should allow the university to improve its performance in this area.

Reduce staff turnover – Currently PBB has an abnormally high level of staff turnover. Its staff turnover levels are double those of KLN, which has a culture of knowledge sharing.

Although there is no guarantee that introducing a knowledge management system will reduce staff turnover, if sharing knowledge helps improve performance this, in turn, should help improve the motivation and job satisfaction of PBB's staff. If staff value working for PBB because it is a successful organisation, this should help reduce staff turnover levels.

Increased collaboration between colleagues and departments – Currently very few of PBB's staff share knowledge with any of their colleagues. However, this lack of collaboration is likely to be stifling **innovation and creativity** within the university.

By contrast, if PBB implemented a knowledge management system, this should encourage staff to share ideas and experience with their colleagues, which in turn could lead to new opportunities for collaboration and innovation across the university. These should not only be focused on new opportunities for how to improve student performance, but could also extend into areas of academic research as well.

8.3 **Data mining** – Data mining is concerned with analysing large pools of data to highlight previously unknown patterns and relationships in that data.

Predicting future behaviour – One of the key benefits which accrue from data mining is the ability to use the identified patterns and relationships to predict future behaviour. In PBB's case, the aspect of behaviour it wants to be able to predict is students dropping out of their courses.

Identify factors affecting drop-out rates – PBB could use data mining software to analyse data to try to identify which factors appear to be influencing student drop-out rates.

Because data mining software is designed to analyse large pools of data, PBB shouldn't be restricted to analysing only the performance of its own departments. It could also look at data for students in other UK universities to see what factors appear to be affecting their drop-out rates.

Importantly, if PBB can get a better understanding of the issues which are causing students to drop out of their courses, it can then develop plans to address those issues and thereby hopefully improve student retention rates.

Preventing drop outs – If PBB can identify, in advance, the groups of students which are most at risk of dropping out, it can then offer them additional support or guidance to try to prevent them dropping out of their courses.

For example, data mining might identify that foreign students who perform poorly in mock exams also fail their end of year exams and therefore cannot continue with their courses. In this case, action could be taken to offer extra tuition to students who fail their mock exams to help them improve their chances of passing their end of year exams.

Answers to Self-test questions

1 MMM

The current management accounting information provides a very detailed analysis of MMM's internal financial performance, but MMM does not appear to have any information about non-financial aspects of its performance, or any external information.

External information – It would be useful if the board had additional information about MMM's market niche and its competitors, for example, about the size of the market, and MMM's share of that market.

Market size – MMM is currently unaware of the size of its market niche and whether or not that niche is growing. Information about the size of the market and the rate at which it is growing (or declining) will be very useful for MMM because it can help identify how much scope there is for it to grow as well.

Market share – The fact that MMM has to 'estimate' that it is the second or third largest company in its niche indicates that it doesn't accurately know the size of its competitors or its market share.

However, this information would be useful not only for monitoring MMM's current performance, but also for evaluating possible future growth plans or strategies.

Basis of competition – MMM is not sure whether it is the high **quality** of its output or its **prices** which are most important in securing new clients. This lack of information about what its customers value makes it harder for MMM to determine what its most effective competitive strategy should be. If customers value the high quality of its output, a differentiation strategy would be appropriate; if customers are attracted by low prices, this suggests a cost leadership strategy could be more appropriate.

Competitor pricing – It seems that MMM does not compare its prices against the prices which its competitors are charging for similar jobs. Consequently, MMM's clients sometimes complain they have been charged too much, while on other occasions MMM feels it has charged too little. However, both these examples suggest that MMM would benefit from being aware of its competitors' prices so that it can ensure that it sets its own prices at a competitive level.

Industry forecasts – MMM could also look at industry forecasts for the travel and tourism sectors as a whole. The levels of demand which the sector is expecting, and therefore how well hotels might expect to be performing, could affect how much they are prepared to pay for advertising, or are able to pay.

Costing methods – MMM's approach to pricing work for clients accentuates the internal focus of MMM's planning and decision making, but it also helps to explain why there are concerns that MMM is charging too high or too low a price for items.

By adopting an inflexible approach to costing and pricing, MMM may be setting uncompetitive prices in relation to competitors' prices, or in relation to what its customers want. This again highlights the importance of MMM obtaining additional information about customers and competitors. For example, before setting a price for a client, MMM should try to find out what price the customer has paid for similar pieces of work previously, or what price competitors have charged for similar pieces of work.

Historical focus – An additional problem the board faces is that the accountant's monthly figures appear to only show actual performance against budget.

Forecast – However, it would be useful if they also showed a **forecast**. This could allow the board to highlight any shortfalls between the forecast position and their desired future position. In this way, if the forecast highlights any expected profit gaps, it could alert the board that they need to plan for ways to reduce the gap.

Forecasting could also help with **human resources planning**. For example, forecasting could alert BBB to potentially busy periods when it would be beneficial for BBB to hire additional, freelance designers to satisfy the levels of demand.

2 FDS Co

2.1 A customer profitability reporting system for FDS will presumably categorise customers into four types: farmers, local government bodies, sports centres and building contractors. It would be possible to have a project costing system whereby costs and revenues are attributed to individual projects, and the profitability of each category of customers would then be calculated as the total profits of all the projects for that customer type.

Sources of information

The revenue for each project can be found from the contract agreed with each the customer or from the invoices raised for the work done.

Direct costs

The information for the costs directly associated with each project should also be obtainable from internal sources.

(1) The cost of **materials** for each project would be recorded from the documents for requisitioning materials to the project, and materials would be priced either from inventory records or from purchase invoices from suppliers.

(2) The cost of labour working on site for each project should be obtained from **timesheets** and payroll records.

(3) The cost of plant hire from each project should be obtained from **invoices from plant hire companies**. If FDS owns its own plant, records should be kept of the use of the plant on each project and a charge for depreciation can be made to a project on the basis of the time that the plant is on the project site.

Overhead costs

To establish a customer profitability reporting system, it will probably be necessary for FDS to record some costs that in the past may have been accounted for as general overheads.

(1) The cost of the sales effort to win contracts should be recorded and charged to contracts. For customer profitability reporting, the cost of salespeople's time and expenses should include the cost of unsuccessful sales effort as well as successfully negotiated contracts. A system of recording time spent selling, and related expenses, will be required. Timesheets can be used to record time, and expense claims should indicate which expenses were incurred on particular projects or types of customer.

(2) Irrigation systems presumably have to be planned, and some contracts are more complex than others. The time spent by planners on the design of the system for each contract should be recorded on timesheets.

(3) Costs of delivering inventory to the customer's site should be recorded. The costs will include the time of drivers and their assistants, together with fuel costs. It is possible that an average cost per tonne-kilometre carried may be used as a standard delivery cost. This could be estimated from delivery records in the transport department. If external delivery firms are used to deliver inventory, invoices can be attributed directly to individual projects.

(4) The cost of holding inventory may be considered a significant cost. If so, a cost of holding

inventory can be calculated for each project from the cost of the inventory on site, the time from purchase of the inventory to payment by the customer, and an appropriate cost of interest or capital for FDS.

(5) If some maintenance costs are included within the price of a contract (for example if maintenance is provided free for a time after installation), maintenance times for engineers and any direct expenses should be recorded and charged to each contract.

(6) If there are miscellaneous directly attributable costs to projects, a system may be required to capture this cost data and attribute it to individual projects or customer types.

2.2 **Highlight differences in profitability** – The purpose of a customer profitability reporting system should be to provide information that will help management to monitor and control the profitability of contracts for each type of customer. However, this information will only have value if costs differ significantly between different types of customers.

It appears that farmers often negotiate lower prices than some other groups, which might initially suggest that the profitability of farming contracts will be lower than for other contracts.

However, negotiations with local government bodies may be lengthy and the contracts may be complex, suggesting that selling and planning times and costs may be high for this type of customer.

Equally for sports centres, there is often difficulty delivering materials to the customer's site which means that delivery costs of materials to site and costs of completing projects may be higher than for other types of customers.

Therefore, there is a degree of uncertainty about how profitable each type of contract actually is, if all the relevant costs are included. However, as the management accountant has suggested, it is important the FDS understands the profitability of different types of contracts; either so that it can focus its attention on the most profitable contracts, or so that it can increase the profitability of other types of contracts (for example, by reducing the level of discounts it is prepared to give to farmers).

Decision making – The potential differences in the profitability of different types of contracts become even more important as FDS gets closer to full capacity. FDS needs to ensure that it accepts the contracts which will generate the highest contributions to profit.

Of the three installations FDS has recently been asked to undertake, the 'Farm' installation generates a significantly lower contribution to profit than the 'Sports Centre' or the one for the 'Building Contractor'. This would suggest that FDS should not accept the 'Farm' contract.

	Sports centre	Farm	Building contractor
Discounted price (£)	192,000	190,000	194,000
Installation team costs	(84,000)	(75,000)	(67,500)
Additional delivery costs		(25,000)	
Contribution to profit (£)	108,000	90,000	126,500

Improved pricing – Better information about each type of customer may help FDS to price its contracts differently, or to resist demands for lower prices from farmers if profit margins on their contracts seem too low.

There is a shortage of skilled engineers for installation and maintenance; therefore, another possible use of customer profitability analysis is to make decisions about which type of contract should be preferred.

FDS might even introduce some kind of limiting factor analysis which looks at contract revenues and profitability in relation to engineer hours, so that preference (and more selling effort) can be given to the type of contract or customer which generates the highest profit per engineer hour.

2.3 Establishing costs

The costs of establishing a system for measuring and reporting customer profitability are difficult to estimate. The costs of establishing the system can be estimated as the costs of the time of managers (including the management accountant) in designing and testing the system. There may be external software development costs that would be directly attributable to the system design.

The costs of operating the system would also be very difficult to measure since the data records would be originated by different individuals. If a cost or management accountant is employed to collect and input data and produce profitability reports, the cost of his or her time would be directly attributable.

Most of the costs of the information system would probably be 'lost' in general overheads, however, and the benefits of monitoring the costs are doubtful. The only significant decision affecting the cost of the system is whether the cost of developing and introducing the system is justified by the expected benefits.

Expected benefits

The benefits of the information system will depend on whether the profitability reports would be likely to affect decision making by management. Specifically, would it affect decision making about the pricing of contracts, or would it affect decisions about which projects to undertake when there is insufficient skilled engineers' time to meet all current demand?

If the potential benefits are considered significant, the performance measurement system should be introduced. Unfortunately, it will not be possible to estimate the benefits with certainty until after the system has been introduced.

3 CMA Supermarkets

3.1 IT technology in supermarkets commonly uses bar codes to identify the products that are sold, and bar code readers enable a supermarket to monitor the quantities of items that it sells as well as to price them for customers. Bar code reader systems are therefore quite sophisticated.

Inventory tracking – Radio frequency ID systems replace bar codes with a chip, and the chip on each item of inventory can hold information in addition to product identification data. This means that RFID readers are able to detect where an item of inventory is at any time and can track inventory movements.

In the case of a supermarket such as CMA, RFID readers could track the movement of inventory from a central stores depot to a supermarket store room, from the store room to the shelves in the supermarket and from the shelves to the customer checkout.

Inventory management – In an industry where fast throughput of items is a critical aspect of success, the ability to monitor the movement of items in such detail, and the time between receiving stores items and selling them, may be of operational value by helping management to adjust purchasing and deliveries in order to speed up sales.

From an accounting perspective, RFID may also be used for inventory counts. An RFID reader can gather information about all the product items held in store at any time, without the need for detailed manual counting.

The additional benefit of having up to date 'real time' data about inventory will depend on the company's management information systems. In principle, a company-wide IT system should be capable of comparing throughput times for different types of product, and comparing the operational performance of different supermarket outlets. The ability to locate stores items may help management to transfer items from where they are turning over slowly to where customer demand is stronger.

Operational performance – CMA's store managers are rightly concerned that the number of out of stock products appears to be rising.

This is a problem for two reasons:

- If an item is out of stock customers cannot buy it, and so this will reduce CMA's revenue (unless the customers find a direct replacement they are prepared to buy instead).
- If customers keep finding that the items they want to buy are out of stock they will stop shopping at CMA. This is potentially a bigger problem for CMA because not only will its revenue and market share fall as a direct result of the lost customers, but it could find it harder to recruit new customers if it develops a reputation for not having items in stock.

Reducing out of stock products – RFID tags should lead to fewer out of stock products. In turn, keeping CMA's shelves fully stocked should lead to increased sales and profits, and a more positive shopping experience for customers (leading to higher customer retention).

Inventory ordering – Because RFID technology provides real-time information it will enable CMA to manage its supply chain more efficiently, both between its stores and its warehouses, and from its stores back to suppliers.

Using RFID tagging, whenever a product is scanned through a till, inventory levels for that product are updated. However, perhaps more importantly, CMA could also use RFID tagging to inform product suppliers of sales and inventory levels. For example, CMA's RFID system could send inventory messages to suppliers whenever their products are scanned through its tills. In this way, the suppliers are aware of the up to date inventories at the stores and can ship additional products as necessary. (If CMA pursued this approach, it might be able to switch to a vendor management inventory relationship with its suppliers, which could be used to help improve product availability.)

However, for CMA to maximise the benefit it can get from using RFID in the short term, it will need its supplier to implement RFID technology on all the products they supply to it. It is not clear how many of CMA's suppliers can currently do this, or even whether it will be possible to tag all the products CMA sells (for example, fresh fruit and vegetables).

3.2 **Integrated system** – The greatest value may be obtained from a new IT system by integrating it across the company. The same system should record data for and report on the performance of central inventory and distribution depots as well as individual supermarket outlets. The system should also record information about costs (which could be held within the data on RFID chips) and selling prices, so that information can be reported about gross profits of stores and product groups within each store. Unexplained losses (due to theft by customers) could also be monitored as a cost item.

Real-time information – The system should also operate in real time, so that users of the system are able to access information they want at any time. Stores managers, for example, should be able to obtain information about sales and gross profits for the store, and then if required drill down for further information about the profitability of product ranges, or manufacturers' brands or even individual product items.

Similarly, rather than having to wait until the end of the month for summary performance reports, senior management could receive summary trading updates at the end of each day or week.

Dashboards and drill downs – At the moment, individual store accounts are prepared for each store and then presented to the store manager, alongside summary reports for the senior management team. It would be more useful for CMA if this information was available electronically, in a way that allowed managers to drill down from summary information to more detailed information for regions and then individual stores.

Having an integrated system in this way should help senior management make comparisons between different central inventory and distribution depots or between different supermarket

outlets. Non-financial performance information (for example speed of product throughput) as well as financial information should be provided all within the same system.

The system should also include external data, although much of this may have to be input by the company's own staff. For example, employees who visit rival supermarkets to check competitors' prices should be able to insert their current prices into the system, so that supermarket managers and senior management can monitor competitive pricing and respond to rivals' price changes.

Data mining – The information that is gathered about product sales should also be used to extract sales data and analyse it to produce information that might help the company to improve sales further – such as information about what products sell well in different areas and at different times of the day, week or year. Data analysis can also be used to 'mine' for data about individual customers.

3.3 **Identify buying habits** – In order to exploit data about individual customers, CMA needs to obtain data about a customer's buying habits. For online sales (for home delivery), the system identifies individual customers and it can prompt them to buy items (by presenting a list of the items they regularly buy as a proforma shopping list) or it can try to encourage them to buy more with discount vouchers.

Loyalty card schemes – For other customers, some supermarket groups use a loyalty card scheme. Customers who present their loyalty card at a store checkout may be awarded bonus points which eventually build up into money-off vouchers. The customer is identified through the loyalty card, and sales details are recorded by the checkout system. Through loyalty cards, supermarkets are able to gather data about what individual customers buy, how much they spend and when they do their buying. Offers to encourage customers to buy more can be related to the known buying preferences of the individual.

If required, a supermarket should be able to calculate the gross profit contributed each period by each 'loyal' customer. This may help the company to target particular types of customers who are more profitable than others.

Chapter 10
Human resource management

Introduction

Learning outcomes

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Chapter study guidance

Models of HRM

Learning topics

1 Strategic human resource management (HRM)

2 The impact of HRM on business strategy

3 Appraisal and performance management

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Answers to Self-test questions



Introduction

Learning outcomes

- Assess, explain and advise on the role of human resource management in implementing strategy
- Demonstrate and explain how human resource management can contribute to business strategy, including flexible workforce management
- Identify the impact of remuneration structures on organisational behaviour and other aspects of human resource management, and show the corporate reporting consequences
- Demonstrate and explain the role and impact of human resource management in change management

Examination context and syllabus links

Human resource management (HRM) plays a vital role in underpinning strategy. Successful strategic implementation requires the effective recruitment, training and organisation of people, coupled with effective leadership and performance management.

As with any other resources, it is crucial that an organisation's human resources are appropriate for the strategy it is pursuing.

There are a number of links between elements of this chapter and topics we have covered earlier in this Workbook. In the chapter "Strategic implementation", we looked at organisational structure and noted that the leadership and management styles which are appropriate for an organisation will depend on the context of the organisation. This idea of contingency (obtaining a fit between an organisation's human resources (HR) strategy and its business strategy) is an important strand of HRM.

Also, it is important that an organisation's remuneration and reward systems are aligned to the organisation's objectives and its critical success factors.

For example, if highly skilled employees are critical to an organisation's competitive success, then its reward system should be designed to try to retain these staff, rather than trying to keep staff costs as low as possible.

In the chapter "Strategic performance management" we looked at the relationship between employees' remuneration/reward and performance. Remuneration structures – and the impact they have on employee behaviour – are key issues within HRM, so there are close links between the performance measurement/management issues we discussed in the chapter "Strategic performance management" and those we will discuss in this chapter.

Aspects of HRM (such as setting performance objectives and measuring performance against objectives through methods like staff appraisal) also play an important role in the performance management and control of the organisation as a whole. In this respect, HRM follows a similar control model to that used for the overall strategic and operational control of an organisation.

More generally, employee performance is vital to the overall performance of most organisations (especially service organisations). Therefore HRM's role in leveraging people's capabilities is critical to achieving sustainable competitive advantage.

Change management, which we also discussed in the chapter "Strategic implementation", is another aspect of strategy implementation which has an important HRM element.

For example, in order to manage change successfully an organisation will need to handle staff concerns about such change, and to ensure it has the appropriate number, and quality, of staff to operate effectively once the change has been introduced.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Strategic human resource management** Successful strategy implementation requires the effective recruitment, training and organisation of people, coupled with effective leadership and performance management. Equally, if an entity is looking to reduce costs, it might look to outsource certain functions, or relocate them (possibly internationally). Such changes could have significant HRM implications.	**Approach** Sections 1 and 2 of the chapter are closely linked and so should be read together if possible. Section 1 looks at what HRM is and what its aims are. **Stop and think** To what extent does HRM contribute to an entity's strategic competitive advantage?	In the exam you may be required to assess the role which human resource management could play in enabling an organisation to implement a strategy successfully, and advise the organisation about this.	**IQ1: ScanTech** This question relates to a human resource plan and asks how it would assist in the growth of the company discussed in the scenario.
2	**The impact of HRM on business strategy** HRM strategy needs to be relevant to, and supportive of, the business strategy. Different HRM strategies are appropriate at	**Approach** Section 2 looks at the impact HRM has on business strategy. Consider the strategy of the business and also how the structure of the workforce could influence its performance – this is	Exam questions may ask you to demonstrate how human resource management can contribute to an organisation's corporate strategy, including in relation to flexible workforce	**IQ2: Financial service companies** This question develops your commercial awareness by asking you to explain the impact of changes in financial services on various

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	different stages of an organisation's life cycle. The HRM strategy will also vary depending on whether an entity is a cost leader or a differentiator.	one of the key considerations in relation to workforce flexibility. **Stop and think** What impact has the Covid-19 pandemic had on workforce flexibility?	management.	elements of human resource management.
3	**Appraisal and performance management** Appraisal is fundamental to performance management, connecting the achievement of strategy with individual performance. However, if not carefully designed, appraisal systems and performance management targets can also have an adverse effect on performance.	**Approach** Work through the material about appraisals relatively quickly, the issues around performance targets and behaviour are more important. **Stop and think** What are the potential impacts on employees and on the business as a whole of setting challenging targets for employees?	In the exam you may need to assess the effectiveness of the appraisal process and performance measurement systems, in motivating employees. You may also need to recommend appropriate performance measures to support strategic objectives.	**IQ3: Appraisals** In this short question you are asked to use the scenario to discuss reasons why the appraisal system objectives are not being met. This is a good knowledge building example and you can also make use of your own business experience to generate ideas.
4	**The impact of remuneration and reward packages** The strategic importance of staff should be clear from the number of entities who refer to their staff as their key assets. As with any other assets or resources, an entity needs to ensure the quality and quantity of its staff are appropriate to the strategies it is	**Approach** One of the key themes in this section is the impact that remuneration structures have on organisational behaviour. We looked at the relationship between employees' remuneration/reward and performance in the chapter 'Strategic performance management', but	The content within this topic could be examined in several ways. You may be asked to assess the impact that remuneration structures have on organisational behaviour. You may also need to identify the potential corporate reporting consequences of the remuneration structures in an	**IQ4: Reward packages** In this question you are provided with information about a consultancy business hoping to change its remuneration policy. You are asked to evaluate the directors' proposal to reduce consultants' salaries but to increase the commission paid. Remember that an

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	pursuing. For example, if an entity wants to differentiate itself from its competitors on the basis of quality, then it is likely to need highly skilled and highly trained staff, and its reward and remuneration packages need to be attractive to those staff.	since remuneration and reward are key issues in HRM they are revisited here. Decisions about remuneration structures (in particular, pensions policies, and share-based payments) can also have a direct impact on an entity's financial statements, as highlighted in this section. **Stop and think** What is the impact of share based payments on the financial statements?	organisation. As part of your exam preparation ensure that you prepare your open book materials so that you are able to easily find the relevant content in the exam.	evaluation is asking you to look at arguments in favour of *and* against the proposal.
5	**HRM and change management** In order for strategic implementation to be successful the way change is communicated to employees, and the way those people subsequently respond to change, must be carefully managed.	**Approach** You have studied models of change in Business Strategy and Technology so quickly recap this topic. Focus more on HRM and due diligence since assurance will appear in some form in every exam. **Stop and think** Consider a change that has occurred in your life. Did you resist it and why? Did you learn to accept the change and what helped to you to achieve this?	In the exam you may be required to explain the importance of human resource management in the context of change management. Change management models can be used to provide structure to your answer although will not be awarded marks in their own right, as is the case with all theoretical models in SBM&L.	

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

Models of HRM

In the course of this chapter, we refer to a number of models of HRM. You will not be expected to discuss the theory of these models by name in your exam, but rather to apply the ideas in a practical context.

For example, Fombrun, Tichy and Devanna's model illustrates how HRM activities link together. One of the linkages it highlights is that between staff reward and performance. Therefore, if a question scenario indicates there are concerns around performance levels and the scenario also gives details of the reward system (eg, performance targets and bonuses), you should consider whether the reward system is contributing to the problems in performance and, therefore, whether the reward system could be revised.

Similarly, Guest's model highlights the link between HRM and an organisation's performance (including its financial performance). So, again, if a question scenario highlights that an organisation's financial performance is deteriorating, you could use the ideas of the model as a general framework to assess how far staff issues might be contributing to the performance issues, and whether any changes to HRM practices might help improve overall performance.

In a similar vein, consider how the references to Theory X and Theory Y, and hard and soft approaches to HRM, could be relevant to a scenario. For example, if you are told a manager is trying to manage highly skilled professional staff in an autocratic way without any scope for consultation, this should raise concerns. In effect, the manager is employing a 'hard' (Theory X) approach to deal with staff who are likely to respond much better to a 'soft' (Theory Y) approach.

1 Strategic human resource management (HRM)

Section overview

HRM emphasises that employees are crucial to achieving sustainable competitive advantage, and that HR practices need to be integrated with the corporate strategy.

1.1 HRM

The concept of HRM and its goals were covered in the Business Strategy and Technology syllabus. HR strategy involves two interrelated activities:

- identifying the number and type of people needed by an organisation to enable it to meet its strategic business objectives
- putting in place the programmes and initiatives to attract, develop and retain appropriate staff

HRM includes all the activities management engage in to attract and retain employees, and to ensure that they perform at a high level and contribute to achieving organisational goals.

For some companies, particularly service companies, human resources may be a source of strategic advantage in their own right. After all, how many times are we told that a company's people are key assets?

Components of HRM

Within the overall aims of attracting and retaining employees, and ensuring they perform at a high level, we can identify five major components for an organisation's HRM systems:

(a) **Recruitment and selection** – Attracting and hiring new employees who have the ability, skills and experience to help an organisation achieve its goals.

(b) **Training and development** – To ensure that all staff develop the skills and abilities which will enable them to perform their jobs as effectively as possible in the present and in the future. In the context of knowledge management and learning organisations, the idea of learning and development should become increasingly important.

(c) **Performance appraisal and feedback** – Appraisals serve two different purposes in HRM: judgement (in order to make decisions about pay, promotion and work responsibilities) and development (assessing employees' training and development needs, and supporting their performance).

(d) **Pay and benefits** – The level of pay and benefits offered to staff has to be appropriate to retain staff. By rewarding high-performing staff with pay rises, bonuses etc managers can increase the likelihood that an organisation's most valued human resources are motivated to continue their high levels of performance, and are more likely to stay with the organisation. Equally, offering attractive pay and benefits should help an organisation fill vacant positions with talented people.

(e) **Labour relations** – Labour relations encompass the steps that managers take to develop and maintain good working relations with unions that may represent their employees' interests.

The following short example highlights the importance of HRM and employee engagement, in terms of staff motivation and performance.

Context example: John Lewis

The retailer John Lewis is often named among the best organisations to work for in the UK, and this reflects its strong focus on people.

One of the key ways John Lewis highlights the value it places on its employees is by calling them 'partners'. Employees are not only given a share of the company's profits, they're also given a say in how the company is run.

Equally, however, by treating its staff as 'partners' the company is recognising their role in delivering excellent service to customers, and empowers them to create the best experience they can for customers.

In turn, employees feel motivated to drive the company's performance. They believe they can make a difference, and this engages them to want to perform and to help the company succeed.

Definitions

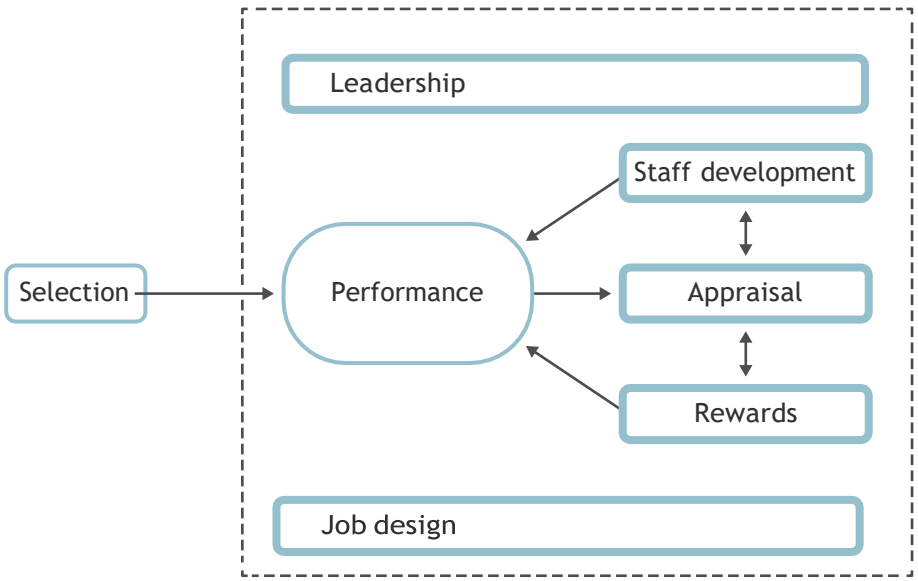
Human resource management (HRM): 'A strategic and coherent approach to the management of an organisation's most valued assets: the people working there who individually and collectively contribute to the achievement of its objectives for sustainable competitive advantage.' (Armstrong)

Human resource management (HRM): 'A strategic approach to managing employment relations which emphasises that leveraging people's capabilities is critical to achieving sustainable competitive advantage, this being achieved through a distinctive set of integrated employment policies, programmes and practices.' (Bratton and Gold)

Figure 10.1 below, adapted from Fombrun, Tichy and Devanna's model of HRM, is a useful way of illustrating how HRM activities link together. Try to keep this diagram (and the linkages between the activities) in your mind as you read through this chapter and when answering a question about HRM in your exam.

Remember also HRM's role in business strategy overall as illustrated by the definitions above.

Figure 10.1: Elements of Human Resource Management (after Fombrun, Tichy & Devanna)



1.1.1 Goals of strategic HRM

- Serve the **interests of management**, as opposed to employees
- Suggest a **strategic approach to personnel issues**
- Link **business mission to HR strategies**
- Enable **HR development to add value** to products and services
- gain **employees' commitment** to the organisation's values and goals

It is important to recognise that the HR strategy has to be related to the business strategy.

For many businesses, staff are key assets, and this reiterates the importance of HRM. HRM emphasises that employees are crucial to achieving sustainable competitive advantage, but also that HRM practices need to be integrated with the corporate strategy.

In this respect, it is important to understand the links between HRM and an organisation's ability to achieve its objectives and its critical success factors (CSFs). For example, if an organisation identifies excellent customer service as a CSF, then its recruitment process, training, appraisal and reward systems should all be geared towards promoting customer service skills in its staff.



Interactive question 1: ScanTech

ScanTech is a rapidly growing high-technology company which specialises in producing electronic scanners. It currently has 110 employees, but aims to double in size over the next three years. The company was set up by two researchers from a major university who now act as joint Managing Directors. However, they intend to leave ScanTech once the growth objectives are achieved and the company is large enough to be sold.

The sophisticated imaging devices which ScanTech makes are used by the airline security and health industries. These two markets are very different in terms of customer requirements, although they use the same basic technology.

In recent years, ScanTech has seen a significant increase in sales from exports, and as a result its strategic plan anticipates a foreign manufacturing plant being set up within the next three years.

ScanTech's current managers are all staff who joined in the early years of the company, and their primary expertise is in research and development. The future growth of the company will require additional staff in all parts of the business, particularly in manufacturing and sales and marketing.

Sue Franklin is HR Manager at ScanTech. She is annoyed that HR is the one management function not involved in the strategic planning process shaping the future growth and direction of the company. She feels trapped in a role traditionally given to HR specialists, that of simply reacting to the staffing needs brought about by strategic decisions taken by other parts of the business. However, she feels that HR also has a strategic role to play in helping ScanTech deal with the challenges over the next three years.

Requirement

Discuss how a human resource plan could help support ScanTech's growth strategy.

See **Answer** at the end of this chapter.

1.2 Roles of HRM

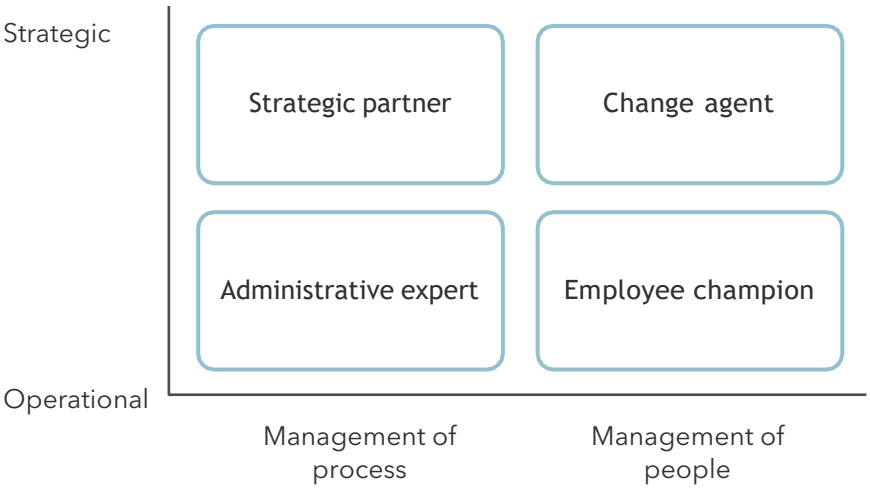
Dave Ulrich has identified four elements of human resources (HR) activity within an organisation:

HR role	Elements of activity
Strategic partner	Aligning HR with business strategy and business requirements Manpower planning Environmental monitoring
Administrative expert	Running the organisation's HR processes and 'shared services': • Payroll • Appraisal • Recruitment and selection • internal communications
Employee champion	Listening and responding to employees Providing resources and training to employees Conciliation Grievance procedures
Change agent	Managing transformation and change in the organisation Ensuring capacity for change and development in the organisation – for example, through management development and performance appraisal

However, it is important to recognise that these different roles are inter-reliant. For example, there is little point in trying to change an organisation's culture and structure from an 'individualist' to a 'team-based' approach without also providing training and changing reward procedures. For example, if performance appraisals still focus on individual results rather than team performance there will be little incentive to move towards a team-based approach.

The different elements highlighted in Ulrich's model identify that HRM is important at both a **strategic level** and an **operational level** within organisations, and also that it involves processes as well as people:

Figure 10.2: HR activities within organisations (based on Ulrich's model)



HRM and personnel management

It is important to distinguish between HRM (as a strategic activity) and personnel management.

Personnel management deals with day to day issues such as hiring and firing, and industrial relations. Unlike HRM, it does not play a strategic role in an organisation.

1.3 Becoming an employer of choice

One area in which HRM can play a particularly important role in an organisation is helping it become an employer of choice.

Many organisations strive to become employers of choice. The status of being an 'employer of choice' implies that people will want to seek employment with the organisation and, once there, will contribute sustained high performance by remaining motivated and committed to the organisation.

For the employer, being an employer of choice should help to attract a high number of well-qualified, suitable and able candidates for any vacancies. Equally, if employees are committed to the organisation and its objectives, this should improve corporate performance and make it a good place to work.

2 The impact of HRM on business strategy



Section overview

- HRM is based on the assumption that the management and deployment of staff is a key strategic factor in an organisation's competitive performance.
- However, to be successful, the HR strategies an organisation pursues need to be aligned to the organisation's overall business strategy.

2.1 Approaches to HRM

Marchington and Wilkinson suggest that there are two main approaches to the relationship between business strategy and HRM strategy: the best fit (contingency) approach and the resource-based approach.

2.1.1 Best fit (contingency) approach to HRM

This approach suggests that HRM strategy needs to be relevant to, and supportive of, the business strategy. This means the strategies need to 'fit' with the internal and external contexts of the organisation.

So, for example, as an organisation moves through its life cycle, different HRM strategies become necessary to support the business strategies at each stage. Thus, HRM strategies are contingent on the life cycle stage.

Similarly, an organisation's HRM strategy will be dependent on its competitive strategy – whether it is pursuing a cost leadership or differentiation approach.

2.1.2 Resource-based approach to HRM

The resource-based approach takes the opposite perspective.

The best-fit approach argues that HRM must be flexed to align an organisation with outside factors in order to deliver effective performance. But the resource-based approach argues that HR activity itself can be strategic, and activities such as training and development can directly influence organisational performance. Therefore, HRM can be used strategically in its own right as one of the resources available to an organisation.

In this way, the resource-based approach to HRM encourages organisations to identify those parts of the workforce which have the greatest impact on performance, and then focus attention on how those staff should be used within the organisation (for example, if there is any need to change processes or practices to increase the value they can add to the organisation).

The resource-based approach also encourages organisations to look at the ways interpersonal and team relationships develop within the organisation, and how this can affect performance. In effect, this also highlights the importance of 'culture' in an organisation, helping staff to work productively and efficiently.

Importantly, however, the resource-based approach does not contend that HRM strategy should only consider factors internal to an organisation. There is still a need to consider influences outside an organisation, and how they can affect HRM strategy: for example, education levels, and economic conditions in a country.

2.1.3 Guest's model of HRM

David Guest has also developed a model which aims to show the link between HRM and organisational performance, and how developing an integrated set of HRM practices can improve the performance of individual staff and, in turn, the organisation as a whole.

Guest's model acknowledges the link between HR strategy and general business strategies. However, the central hypothesis of the model is that HR practices should be designed to lead to HRM outcomes of **high employee commitment, high quality** and **flexible employees**.

High employee commitment is seen as a critical HR outcome because it is concerned with the goals of binding employees to the organisation and obtaining behavioural outcomes of increased effort, cooperation and organisational citizenship.

Quality highlights that employee behaviour has a direct impact on the quality of goods and services. Flexibility is concerned with employees' receptiveness to innovation and change.

The right-hand side of the model focuses on the link between HR practices and performance. A critical assumption in Guest's model is that only when all three HR outcomes – commitment, quality and flexibility – are achieved can an organisation expect to generate superior performance outcomes (both financially and non-financially).

Guest's six components:

(1) HRM strategy	(2) HRM practices	(3) HRM outcomes	(4) Behavioural outcomes	(5) Performance outcomes	(6) Financial outcomes
Differentiation (innovation) Focus (quality)	Selection Training Appraisal	Commitment Quality Flexibility	Effort Motivation Cooperation	**High:** Productivity Quality	Profits Return on investment
Cost (cost reduction)	Rewards Job redesign Involvement Status and security		Involvement Organisational citizenship	Innovation **Low:** Absenteeism Employee turnover Conflict Customer complaints	

2.2 HR and generic strategies

As Guest's model acknowledges, there is a close link between HR strategy and overall business strategy.

The generic business strategy which an organisation pursues is likely to have a significant impact on HRM.

For example:

(a) Cost leadership is often to be associated with terms such as: Theory X (autocratic) management, role culture, tall narrow organisational structures, task specialisation, close direction and control, repetitive tasks, and top-down information flows.

(b) Differentiation is often to be associated with terms such as: Theory Y (participative) management, task culture, wide flat organisational structure, multi-skilled employees, autonomy and self- direction, unique and creative tasks, and multi-directional information flows.

The contrast between Theory X and Theory Y can also be illustrated by the contrast between 'hard' and 'soft' approaches to HRM:

Issue	Hard	Soft
Goals	Meet organisational objectives Workers are a resource to use to achieve those objectives	Develop HR as asset Workers are valuable to a company as assets
Behavioural assumption	Theory X Financial focus on wages (emphasis on gaining work efficiencies)	Theory Y Develop employees
Management style	Imposed, top-down Tell workers what to do	Consultative, participative Listen to employees' views, and encourage involvement and commitment
Training and development	Training only given to meet needs of current position, in order to fully utilise labour resources	HRM is about developing resourceful people; personal development as well as career development are encouraged
Organisational structure	Centralised	Devolved, delegation, autonomy

As we will see later in this chapter, the system of rewards management and remuneration used by an organisation will also vary according to the nature of its business and the generic strategy it is employs.

2.3 HR and Porter's five forces analysis

HR can play an important role in reducing the adverse effect of the five forces for an existing member of the industry. Some forces are more susceptible to HRM than others.

Barriers to entry

To enter an industry, an organisation needs staff of the right quality, ability and cost. Assuming that entry is not achieved by takeover, then almost certainly recruitment will be required, and this can be difficult if there is a shortage of suitable candidates either already trained or wanting to train.

Increasingly, many industries have a higher technical content in their jobs today and more qualified employees are therefore needed. Additionally, many economies have moved from manufacturing to service industries and, because more employees have dealings with customers, employees with better interpersonal skills might be needed. Remember, if a poor product is manufactured it can be identified through the quality assurance process. Poor service is often delivered instantly and can cause immediate damage to customer relations.

Suppliers

Supplier power can derive from various factors such as the number of suppliers in the market, how specialised their goods are, geographical proximity and the fact that the organisation requires goods of a certain standard in a certain time.

HRM can help to erode supplier power by ensuring that buyers have comprehensive knowledge about suppliers and their products and have good negotiating skills. Additionally, partnership sourcing is becoming more common. This is where a purchaser and supplier build a long-term relationship involving mutual trust and recognise that they need each other if they are to be successful. (This builds on the ideas of supply chain management and networks we have considered earlier in this Workbook.)

Bargaining power of customers

The bargaining power of customers is high if there are only a few of them and each buys a large quantity, and also if customers buy mainly on price. The bargaining power of customers can be reduced by finding more customers and by trying to lock in all customers. HR, for example through suitable training, can do this by:

(a) Raising switching costs in both cash terms, and in terms of operational inconvenience. An example is where staff provide exceptionally good service, or design, manufacturing and dispatch are excellent. If a customer is looked after well, there will be a reluctance to switch to a new supplier. Even if the new supplier is not worse, there will be a period of disruption until the new supplier provides exactly what the customer wants. Therefore the recruitment, training and retention of good staff will help to retain customers.

(b) Finding more customers. Recruitment of and training a good sales team and, increasingly, the establishment of a good internet site (again often dependent on recruiting skilled people) are essential for this.

Rivalry/competition

Winning over competitors depends on the nature of the competition. For example, winning a cost leadership battle depends on keeping costs down, and winning a differentiation battle depends on producing goods and services that are really appreciated by customers. Appropriate HRM techniques will help with these objectives. Cost levels will be affected by wage levels, recruitment and training. Differentiation will be affected by recruiting talented people and managing them in an appropriate way.

Substitutes

The emergence of substitute products and services is difficult to predict. Substitutes are often caused by technological breakthroughs. For example, mobile phones are a substitute for fixed line phones. HRM can, however, play important roles as follows:

(a) **The adoption of substitutes**. For example, it can be argued that cheap airlines have become a substitute for some car and train journeys. If an airline wants to win part of that type of business it will have to adopt HR policies that are consistent with the cheap airline model with regards to wages, job flexibility and working hours.

(b) **The invention of substitutes**. The marketing, research and development and production departments can all contribute here. Recruiting and developing creative people in an environment conducive to conceiving new products and services is vital.

2.4 HRM implications of business strategy

When looking at strategic decisions an organisation is considering, it may initially seem as if they do not have much to do with HRM. However, this is not the case. Almost all objectives, and almost every issue facing an organisation, have HRM implications.

The following tables illustrate this in two simple examples:

(a) Business strategy – to grow market share in a new country

Area of HRM	Application to scenario
Organisation and culture	New sales team needed, based in the country
Recruitment and selection	Recruit local sales people Appoint experienced international manager
Training and development	Sales training Cultural awareness of the new country
Pay and benefits	Local packages, salary and benefits

(b) Strategic issue – reducing the number of customer complaints in a retail store

Area of HRM	Application to scenario
Organisation and culture	Assess general culture towards customers Analyse complaints to see if caused by organisational process inefficiencies (such as stock-outs) or by staff issues Are there conflicts between the needs and objectives of individual stores vs head office targets? Are jobs suitably designed?
Recruitment and selection	Are there sufficient staff in the store? Are shift patterns appropriate?
Learning and development	Is any customer-focused training required? Is any other personal skills training needed for in-store staff?
Communications and employee relations	Ensure all staff are aware of the volume and reasons for customer complaints Institute 'service quality' groups
Pay and benefits	Are rewards a source of employee dissatisfaction? Could there be an element of performance- related pay, based on customer feedback?
HR policies	Review why staff leave Are any policies (such as overtime requirements, shift patterns) a source of discontent?

All strategic decisions have implications for human resource management, just as changes to

Professional skills focus: Assimilating and using information

human resource policies can impact upon successful strategic implementation. Read the scenario exhibits carefully, making sure you gather all the relevant information, and be aware of the potential HRM impacts of any recommendations you make.

Interactive question 2: Financial service companies

Since the beginning of the 1990s financial services institutions (eg, banks and insurance companies) have pursued the following business strategies:

- Shift to telephone and internet-based servicing of customer accounts, leading to reductions in the total number of high street branches
- Expansion in range of financial products offered at branches
- Introduction of 'customer service ethos' with emphasis on providing advice and selling products while increasing reliance on electronic technologies to handle routine transactions
- increasing use of 'offshore' call centres and transactions processing centres

Requirement

What impact would these changes have had for the following factors?

- Forecast human resource demand
- Forecast human resource supply
- Training and development
- external recruitment

See **Answer** at the end of this chapter.

2.5 HR and the knowledge economy

A knowledge economy is one in which knowledge is the prime source of competitive advantage. Remember, the source of competitive advantage is either cost leadership or differentiation (each with or without focus). Examples of businesses in the knowledge economy include:

(a) Engineering, such as Rolls-Royce jet engines. The engines are the result of very advanced research, design, development, testing and incremental modification.

(b) Software, such as HP Autonomy. This company specialises in pattern recognition in unstructured data: speech, faces and car licence plates.

(c) Hardware, such as Apple Inc. The invention and design of new concepts and knowing how to produce them (mainly by subcontracting to suitable companies) reliably and efficiently.

(d) Biotechnology, such as GlaxoSmithKline plc or Convergence Pharmaceuticals and the development of new drugs and vaccines.

The knowledge economy poses the following challenges to HR managers:

(a) Finding and recruiting enough people with the right skills. The skills and abilities required can be very high and very scarce.

(b) Providing an environment in which employees' skills and abilities are used to their maximum potential.

(c) Motivating and developing employees.

(d) Retaining employees. When an employee leaves, valuable knowledge can be lost and can be transferred to a competitor business.

2.5.1 HR and big data

In the chapters "Data analysis" and "Information Strategy" of this Manual, we have highlighted how big data and big data analytics could provide opportunities for organisations to improve their business insight and their decision-making. However, one of the key challenges which big data presents is getting from the underlying data to interesting and useful information.

The head of ICAEW's IT Faculty, Richard Anning (quoted in the February 2016 edition of economia) has suggested that exploiting big data requires:

- skills in computing and the manipulation of data;
- statistical skills to build models; and
- knowledge of the business areas to ask the right questions and enable interpretation of the results.

Anning continues by suggesting that this usually requires teamwork between specialists in IT, data science and different business functions. However, if organisations do not currently have sufficient staff with the necessary skills, they will either need to recruit additional staff, or train existing staff, before they are able to take advantage of big data. This could also apply to machine learning and AI, which – in effect – could be seen as applications of big data.

2.5.2 Automation

Organisations could also consider the potential for automation/robotic process automation (RPA), and the impact this could have on its HR model. As we highlighted in the chapter "Strategic analysis", RPA could reduce the amount of staff required to deal with rules-based processes, leaving staff to focus on more complex processes instead.

More generally, technological advances (automation, and algorithms used in AI and machine learning) are likely to reduce the number of workers required for certain tasks.

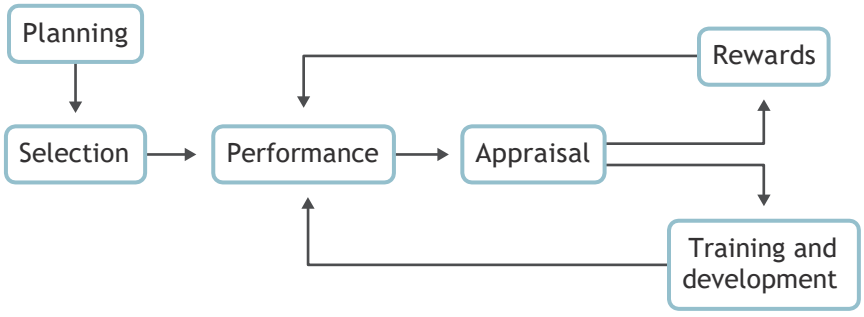
On the other hand, there is increasing demand for new, specialist roles related to emerging technologies: AI specialists; big data specialists; process automation experts; information security analysts; and human-machine interaction designers.

An important issue for HRM in organisations will be identifying the skills that it needs to be successful and take advantage of new opportunities, and ensuring that its workforce is equipped with these skills, either by retraining and upskilling existing staff, or by recruiting new workers.

2.6 The human resource cycle

In section 1 of this chapter, we looked at Fombrun et al's model of the HR cycle (Figure 10.1) but that model can be adapted to include HR planning, which is a key element of the HR cycle:

Figure 10.3: HRM and HR planning



HR planning

HR planning must link back to the organisation's strategic plan. The number of staff needed with given skills will depend on business plans (for example, any plans to develop new products, or expand into new markets); the skills needed might depend on whether strategic advantage relies on cost leadership or differentiation.

Personnel information systems can be of great help at this stage, if not essential, because these systems should hold information about current employees and their skills and be able to predict how many employees might be needed in the future.

For example, assume that every branch of an organisation requires, ideally, one manager, two sales people, three part-qualified engineers and two qualified engineers. If the strategic plan says that 20 new branches are to be opened in the next three years, then the personnel information system can compare current numbers of personnel who have appropriate skills with the number that will be needed after three years. The personnel information system will also be able to identify any current employees who are due to retire, estimate the number who might leave, forecast how many part- qualified engineers should attain full qualification and also how many employees might achieve management status. From these calculations a recruitment and development budget can be identified.

	Managers	Sales personnel	Part-qualified engineers	Qualified engineers
Current employees (100 branches)	100	190	310	195
Due to retire	(5)	(1)	0	0
Estimated leavers	(10)	(30)	(25)	(70)
Promotion			(100)	100
	85	159	185	225
Employees needed (120 branches)	120	240	360	240
To be recruited	**35**	**81**	**175**	**15**

HRM and gap analysis

In the context of business strategy, we have seen that managers can use gap analysis to identify gaps between forecast performance and target performance, with a view to finding new markets, launching new products or finding other ways to close the gap.

However, the idea of gap analysis can also be applied to HR, as in the example above. The example shows that there is a shortfall between the future forecast number of engineers (if no additional recruitment takes place) and the number of engineers who will be needed to support the organisation's planned branch openings. In turn, this identifies the number of additional engineers who need to be recruited as a result of the planned openings. An alternative approach to filling a resource gap would be to consider whether some jobs which are currently done manually could be automated. In this way, an organisation might be able to use IT as a substitute for labour to overcome a staffing resource constraint.

Skills gap

Instead of looking purely at the numbers of people employed, organisations also need to consider whether their staff members have the skill sets necessary to deliver a strategy. If there is a 'gap' between the current skill set of a person or group compared to the required skill set, this represents a skill gap. A strategy then needs to be devised to mitigate that gap; for example, by training staff to use a new software program.

2.6.1 The impact of increased job mobility on HR planning

Historically, many employees looked for 'a job for life' and low employee turnover was expected. That pattern still exists in some economies, such as Japan, but in many countries moving from job to job, gaining skills as you go, is the norm. If more people leave, then there is a greater burden on recruitment to replace them. Furthermore, unless the organisation's knowledge management has been successful, people leaving take with them valuable knowledge and this impoverishes the organisation.

2.7 HR planning – overview

HR planning should be based on the organisation's strategic planning processes, with relation to analysis of the labour market (internal and external), forecasting of the external supply and internal demand for labour, job analysis and plan implementation. HR planning concerns the acquisition, utilisation, development and return of an enterprise's human resources. HR planning may sometimes be referred to as 'workforce planning' or 'workforce strategy'. HR planning involves:

- budgeting and cost control
- recruitment
- retention (company loyalty, to retain skills and reduce staff turnover)
- downsizing (reducing staff numbers)
- training and retraining to enhance the skills base
- dealing with changing circumstances

Human resources are hard to predict and control:

(a) **Demand**. Environmental factors (eg, the economy) create uncertainties in the demand for labour.

Estimating demand: Planning future HR needs requires accurate forecasts of turnover and productivity (eg, if fewer staff are required for the same output). The demand can be estimated from the markets a company plans to operate in, the products and services it plans to offer, and its strategy for offering them (for example, pursuing a cost leadership model or differentiation; and/or how many functions to perform in-house as opposed to being outsourced).

(b) **Supply**. Factors such as education and the demands of competitors for labour create uncertainties in the supply of labour.

The available **supply** of labour, competences and productivity levels may be forecast by considering internal and external factors.

Internal factors

The competences, skills, trainability, flexibility and current productivity level of the existing workforce

The structure of the existing workforce in terms of age distribution, skills, hours of work, rates of pay and so on – The likelihood of changes to the productivity, size and structure of the workforce

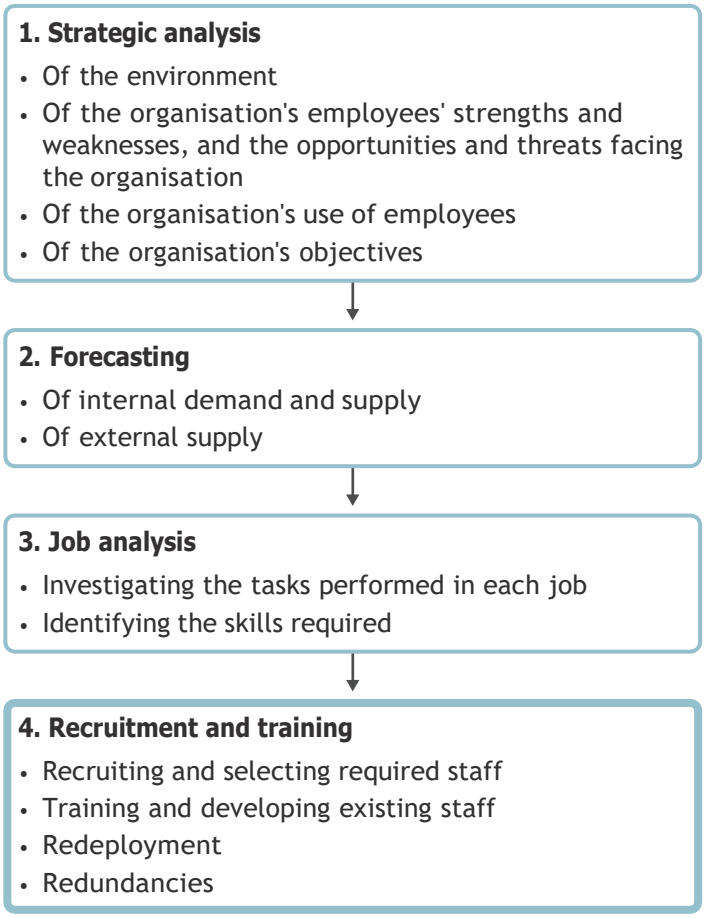
External factors

The present and potential future supply of relevant skilled labour in the **external labour market** will be influenced by a range of factors. These could include general economic conditions, competitor activity, demographic changes, government policy (eg, relating to labour mobility) and the changing nature of work (leading to changes in the skills required).

(c) **Goals**. Employees have their own personal goals, and make their own decisions about whether to undertake further training. When large numbers of individuals are involved, the pattern of behaviour which emerges in response to any change in strategy may be hard to predict. There can sometimes be powerful resistance to change.

(d) **Constraints**. Legislation as well as social and ethical values constrain the ways in which human resources are used, controlled, replaced and paid.

Figure 10.4: Stages of HR planning



Points to note based on the process of HR planning

HR strengths and weaknesses – An organisation's HR strengths and weaknesses need to be analysed so as to identify skills and competence gaps. HR planning is important not simply for analysing overall numbers of staff, but also for looking at the mix of skills within the workforce. While headcount numbers are always likely to be a concern, it is equally important to consider whether the current workforce has the skills and attitudes required to sustain organisational success in the future.

Efficiency – An organisation needs to know how effectively it is using its staff (for example, utilisation statistics, idle time). If staff are currently not fully utilised, how far can future growth be staffed by the existing staff, rather than having to recruit additional staff?

Timescale – If an organisation can identify a 'gap' in advance, this will allow recruitment and training to be planned in advance. However, an unplanned 'gap' which needs filling immediately will require instant recruitment.

The importance of HR planning

It is arguable that forecasting staff and skill requirements has become more difficult in recent times because of the increasing uncertainty and rate of change in the business environment. However, it has also arguably become more necessary, because the risks of 'getting it wrong' (particularly in an era of global economic recession) are correspondingly greater.

In this respect, HR planning can be seen as a form of **risk management**. It involves realistically appraising the present and anticipating the future (as far as possible) in order to get the **right people** into the **right jobs** at the **right time** and managing employee behaviour, organisational culture and systems in order to **maximise the human resource capability** in response to anticipated opportunities and threats.

An attempt to look beyond the present and short-term future, and to prepare for contingencies, is increasingly important. Some manifestations of this are outlined below.

(a) Jobs in innovative and fast-changing contexts may require experience and skills which cannot easily be bought in the marketplace, and the more complex the organisation, the more difficult it will be to supply or replace highly specialised staff. The need will have to be anticipated in time to initiate the required development programmes. The decline of the 'job for life' and the common desire to gain wide and rounded experience have contributed to higher rates of employee attrition. Leavers must be replaced with suitable staff. At senior levels, succession planning should identify potential replacements, internal or external, for those expected to retire or simply move on.

(b) Employment protection legislation and increasing public demand for corporate social responsibility make downsizing, redeploying and relocating staff (eg, in response to economic recession) a slow and costly process.

(c) Rapid technological change is leading to a requirement for human resources that are both more highly skilled and more adaptable. Labour flexibility is a major issue, and means that the career development and retraining **potential** of staff is at least as important as their actual qualifications and skills. Thus, 'trainability' is now a major criterion for selection.

(d) Organisations that differentiate themselves in the market through superior customer service or other people-related activities need to place people at the centre of their corporate strategy. If their people are to be the difference they need to invest time and effort in finding and developing the right ones.

(e) The scope and variety of markets, competition and labour resources are continually increased by environmental factors such as the expansion of the European Union, the globalisation of business and the explosive growth of e-commerce.

(f) Information and Communication Technology has made available techniques which facilitate the monitoring and planning of HR over fairly long time spans: accessing of demographic and employment statistics, trend analysis, 'modelling' of different scenarios and variables, and so on.

(g) Labour costs are a major proportion of total costs in many industries and must be carefully controlled. Cost control action will involve carefully planned remuneration schemes, strict control of headcount and avoidance of waste in such forms as overstaffing and unnecessary activity. Business process re-engineering and the de-skilling of jobs may lead to redundancies, especially among overqualified staff.

Armstrong sums up the aims of HR planning as follows:

(a) To attract and retain the number of people required with the necessary skills, expertise and competences

(b) To anticipate potential surpluses or shortfalls which will need to be adjusted

(c) To develop a well-trained and flexible workforce which will support organisational adaptation to external changes and demands

(d) To reduce dependence on external recruitment to meet key skill shortages (by formulating retention and development strategies)

(e) to improve the utilisation of people (most notably by developing flexible working systems)

2.7.1 HR analytics

In the chapter "Data analysis" we looked at how organisations can use the insights gained from data analytics to manage their performance more effectively.

Equally, though, organisations could also use HR analytics to help them recruit, retain and reward people more effectively. For example, data and analytics could help organisations assess whether their workforce is matched with future demand (in terms of both employee numbers and skills). Similarly, analytics could help organisations identify their key staff (in terms of the value they create) and how likely it is that they will leave the company in the near future.

However, in the same way that we highlighted the need for managers to be able to trust operational data as the basis for data analytics, so organisations need to collect and maintain accurate HR data in order for them to apply HR analytics.

2.8 Flexible workforce management

It has been suggested that **long-range, detailed people planning** is a necessary form of risk management, preparing businesses for foreseeable contingencies.

However, organisations are also regularly searching for opportunities to reduce costs and/or to increase the productivity of their workforces.

Increasingly, traditional staffing models – based around full-time employees, working a standard number of hours each week – are proving inefficient in terms of matching demand and supply. In peak times, there is too much work for the employees available (or large amounts of overtime have to be paid) while employees are under-utilised in less busy periods.

As a result, organisations are looking to implement workforce models which match employees to work more efficiently, and they are doing this through flexible workforce management.

Equally, flexible working can be attractive to employees, for example by helping them to manage their work/life balance, or combining responsibilities as a parent of young children with their careers. As such, offering flexible working arrangements can help organisations to attract and retain staff. In its 2017 report, *Flexible working: a talent imperative*, Timewise identified that 63% of full-time employees in the UK already worked flexibly in some way, while 87% of full-time employees either already worked flexibly in some way, or said they wanted to.

2.8.1 Elements of workforce flexibility

There are three main elements of workforce flexibility: **where** someone works; **when** they work; and
how much they work:

Flexibility in where someone works

This could involve:

- Working from home (remote working) for some or all of the time
- Working in several different offices or on several different sites within the same organisation
- working remotely from a number of different sites

When we discussed cloud computing in the previous chapter, we noted that 'accessibility' was one of the key benefits of a cloud-based system. If data is stored in the cloud, employees can access it from anywhere there is internet connectivity. As such, cloud-based systems help to support the flexibility around where people can work.

Flexibility in when someone works

This could involve:

- Flexible start or finish times (eg, working 8-4 or 10-6, rather than having to work 9-5)
- Compressed work (eg, working 4 days of 9 hours each, and then having 3 days off, instead of working 5 days of 7 hours each, and then having 2 days off)
- shift work (particularly important for organisations which operate 24 hours a day)

An important consideration when flexing when someone works will be trying to match employees' availability with workload peaks and troughs. Doing this could be a more cost-effective way of managing resources than having to hire additional temporary staff, or having to pay large amounts of overtime, to meet workload demands in busy periods.

Note: We saw an example of this in the chapter "Strategic analysis", Section 4.6, when we discussed how the gas supply network Wales & West Utilities had changed its engineers' shift patterns to fit more accurately with demand for their services.

Flexibility in how much someone works

This could involve:

- Working part time
- job-sharing

2.8.2 Multi-skilling

An additional way in which organisations can increase flexibility in their workforce is through multi- skilling. If employees are multi-skilled, the organisation has the flexibility to arrange them in the way which best fits its needs at any given time – for example, by asking them to work in an area of the business that requires additional resource at a particular time.

In this respect multi-skilling can also help reduce an organisation's employee costs, because a business with a multi-skilled labour force may be able to operate with fewer employees than one whose employees are not multi-skilled. For example, workers who are only skilled in one area may have periods of idle time, waiting for work to become available. However, multi-skilled workers may be able to move to different areas, reducing the amount of idle time, and meaning the overall level of work can be carried out by fewer people.

Multi-skilling could also increase job satisfaction among employees – by increasing the variety in their work, and giving them the opportunity to learn and apply new skills.

2.9 Remote working (Home working)

One of the major changes in the way workforces are structured has been the increasing number of remote workers or home workers.

An important factor in the development of remote working is the beneficial impact it can potentially have on employees' work–life balance. Employees value the time and money savings which remote working offers them, compared to having to commute to work. Some employees may also value the autonomy and independence which working at home affords them. Since companies have increasingly recognised the importance of attracting and retaining talented staff, they have also realised the need to offer staff flexibility in their working arrangements.

Remote working could also improve productivity. On the one hand, if remote working leads to higher job satisfaction, higher retention and lower absenteeism, these factors could contribute to increased productivity. On the other hand, remote workers could also be more productive than 'office-based' colleagues because they are able to work without interruption, for example from office meetings.

Nevertheless, while some employees value remote working, others may not want to work at home; for example, because they feel it isolates them from colleagues and shared knowledge; or because they would prefer to keep their 'private' lives separate from their 'work' lives.

IT and remote working

Developments in technology have been crucial in facilitating the growth of home working. For example, remote workers need a laptop or personal computer, with reliable internet access, and secure remote access to a company's internal networks and internal messaging systems (eg, sharepoints) in order to work from home effectively.

Video-conferencing (for example, through Skype or Google Video Chat) can also be valuable for contacting remote employees, particularly in relation to important matters which it may not be appropriate to discuss by email.

In general, technology has been essential for the development of remote working, because it provides opportunities for exchanges of information between employees working remotely and their colleagues or managers in a different location.

Communication and remote working

Nonetheless, poor workplace communication is often seen as the biggest disadvantage of remote working. In this respect, the growth of remote working presents new issues for managers in relation to managing staff.

Socialisation and relationship building are important aspects for helping remote employees to feel included within an organisation. As such, managers' communication skills and relationship building skills will be important in ensuring that remote workers do not feel isolated. While many remote workers value the greater independence which they gain from working at home, it remains important for managers to communicate regularly with these workers in order to build trust and to maintain a relationship with them.

Equally, managers will need to provide the employees with clear goals and expectations for their work, as for any other employee in the company. In order to manage remote employees effectively, managers should adopt a 'management by objectives' approach, as opposed to managing by observation. This will involve setting goals and action plans, and then evaluating employees' performance based on the outputs or results.



Context example: The perils of home working

An article in *The Guardian* newspaper highlighted the findings from research which indicated that home working can take a 'heavy psychological toll' where workers are encouraged to adopt an 'always on' mentality, where the lines between work and home life are increasingly blurred. The article (Jowit, 2016) highlighted that "working away from the office can isolate employees from social networks and career opportunities while fostering a 'grazing' instinct that keeps dangerous stress hormones at persistently high levels".

'Grazing' is the term used to describe the compulsion to work outside of normal office hours. The article highlighted that working ever longer days increased stress and depression, and led to poor diet and sleep among home workers. Research carried out by the University of Bedfordshire found that "work has become more intense as new technology enables, and even forces, people to work faster, do more, and multi-task". This in turn has led to 'presenteeism' where people continue to work even when they are off ill. According to an Ofcom report, on

average adults in the UK now [2016] 'spend more time using technology than sleeping each day'. (Jowit, 2016)

Source: Jowit, J. (2 January 2016) 'Work-life balance: flexible working can make you ill, experts say'.
The Guardian. [Online]. Available from: www.theguardian.com [Accessed 28 April 2017]

2.10 Contract workers and freelancers

Perhaps the most significant change in workforce management in recent years has been the increased use of freelance or contract workers in place of permanently employed staff.

In this way, the role of HRM is to ensure that the appropriate people are brought together to complete a specific project or task. However, such a model fundamentally changes the nature of job vacancies, compared to a model in which organisations employ staff on a full-time basis.

The freelancing model can offer benefits for organisations and workers alike. For organisations, particularly ones involved in project work, the use of short-term contracts enables particular skills to be brought in as and when required. This means the organisations only incur employee costs when work is required, rather than having to pay regular salaries (and associated costs) for full-time members of staff.

Equally, however, the freelance model can have benefits for workers, because they can be selective about the work they undertake, based on the opportunities they are most interested in (subject to there being sufficient work available to provide them with an income to live on).

Gig economy

The increased use of freelance workers also gives rise to the concept of a gig economy.

A gig economy is a labour market characterised by the prevalence of short-term contracts or freelance work instead of permanent jobs. Workers get paid for the 'gigs' they do, instead of receiving a regular wage. The gig economy is prevalent among couriers (eg, Deliveroo) and ride- hailing drivers (eg, Uber).

Proponents of the gig economy claim it offers flexibility for companies and workers. Companies only pay when work is available, and do not incur staff costs at times when there is no demand. Similarly, workers can benefit from flexible hours, and can control how much time they work, and how they juggle work around other priorities in their lives.

However, the gig economy has also proved contentious because workers in the gig economy are classed as independent contractors, rather than employees. This means they have no rights to receive the minimum wage, paid holiday, or sickness pay, and they have no protection against unfair dismissal.

A number of labour disputes have arisen in this respect, with people who work for companies such as Uber and Deliveroo fighting to have their status upgraded from that of an independent contractor, to that of a worker for the company (employee). In a high profile case which could have significant implications for the gig economy, a 'contractor' who had worked exclusively for the plumbing company Pimlico Plumbers for six years won a ruling that he should be classed as a worker, rather than being classified as self-employed.

2.11 The people plan

Once the analysis of HR requirements has been carried out, and the various options for fulfilling them considered, the **people plan** will be drawn up. This may be done at a strategic level. It will also involve tactical plans and action plans for various measures, according to the strategy that has been chosen. Typical elements might include the following:

a) **The resourcing plan**: approaches to obtaining skills/people within the organisation, and by external recruitment

b) **The internal resource plan**: availability of skills within the organisation; plans to promote/redeploy/develop

c) **The recruitment plan**: numbers and types of people, and when required; sources of candidates; the recruitment programme; desired 'employer brand' and/or recruitment incentives

d) **The training plan**: numbers of trainees required and/or existing staff who need training; training programme

e) **The redevelopment plan**: programmes for transferring or retraining employees

f) **The flexibility plan**: plans to use part-time workers, job sharing, home working, outsourcing, flexible hours arrangements and so on

g) **The productivity plan**: programmes for improving productivity, or reducing manpower costs; setting productivity targets

h) **The downsizing plan**: natural wastage forecasts; where and when redundancies are to occur; policies for selection and declaration of redundancies; redevelopment, retraining or relocation of employees; policy on redundancy payments, union consultation and so on

i) **The retention plan**: actions to reduce avoidable labour wastage

The plan should include budgets, targets and standards. It should allocate responsibilities for implementation and control (reporting, monitoring achievement against plan).

2.12 People and strategic success

Bratton and Gold's definition of HRM (see section 1 above) highlights that human knowledge and skills are a strategic resource for an organisation, and that they can play a vital role in achieving sustainable competitive advantage.

The **strategic significance** of having the right people working effectively increases as technology becomes more complex, the importance of knowledge work increases and strategy relies more and more on the talents and creativity of human beings.

An important aspect of HRM, therefore, consists of the various activities that attempt to ensure the organisation has the people it needs when it needs them. These activities include **recruitment**, **retention** and, when necessary, **reduction** of headcount.

However, aspects of HRM (such as setting **performance objectives** and **reward management**) also play an important role in the performance management and control of the organisation. In this respect, HRM follows a similar control model as is used for the overall strategic and operational control of an organisation:

Step 1 Goals are set

Step 2 Performance is measured and compared with target

Step 3 Control measures are undertaken in order to correct any shortfall

Step 4 Goals are adjusted in the light of experience

However, it is crucial to recognise that these goals link to both strategic and operational success. Effective performance management requires that the strategic objectives of the organisation are broken down into layers of more and more detailed sub-objectives, so that **individual performance** can be judged against personal goals that support and link directly back to corporate strategy.

2.13 People and operational success

Recruitment and selection

Operational success relies on people's ability to do their jobs properly. This could include their ability to perform a range of activities, such as being able to operate machinery correctly, use computers, manage others, or perform specific technical routines. In this respect, operational success requires the proper **recruitment** and **selection** of people with the right skills for the particular job, and the provision of further training as the requirements may dictate.

An organisation's staff are a very important resource, and they are likely to play a crucial role in an organisation achieving its strategic objectives. Therefore, it is vital that an organisation has the right number (quantity) and the right quality of staff to achieve its objectives.

In this respect, HR planning is very important – not only in forecasting the numbers and levels of staff an organisation is likely to need, but also in deciding whether, for example, the staff should all work 'in house' or whether it might be more appropriate to outsource some functions, or to move to a more 'network' based organisation rather than using a more formally structured one.

In this way, recruitment and HR planning play a vital role in ensuring that organisations have the necessary quantity and quality of staff to facilitate their success.

Objectives and performance targets

Staff should also have individual work **objectives** and **performance targets** (for example the number of sales calls made) and their performance should be measured against these objectives. These individual objectives and targets should be derived from department and organisation objectives. This should mean that, in theory, if every individual achieves their objectives then their department will achieve its objectives, and if every department achieves its objectives then the organisation as a whole will achieve its objectives.

Two factors which play an important role in determining whether employees achieve their objectives are **management** and **motivation**. We will look at a number of aspects of employee performance management later in this chapter, but in general terms we can highlight the link between performance and motivation by reference to the following equation (after Vroom):

Performance = Ability × Motivation

(where Motivation = Desire × Commitment)

In this equation, desire is seen as enthusiasm for a task, and commitment is about putting in effort. Therefore, as well as ensuring that employees have the necessary abilities to carry out their jobs, managers need to make sure that their staff have the desire and commitment to do so efficiently and successfully.

Staff retention

Keeping staff motivated can also help an organisation retain staff more effectively, and in doing so can reduce the costs associated with **staff turnover**. These include the time and costs spent in advertising for and recruiting new staff; time and costs spent training new staff, and the 'learning curve' associated with new staff getting up to speed with their jobs; and the loss of organisational knowledge which occurs when individuals (particularly key employees) leave an organisation.

3 Appraisal and performance management



Section overview

- Appraisal is fundamental to performance management, forming a link between individual members of staff and an organisation's overall strategy.
- However, within this overall setting, appraisal has two different purposes – judgement and development – and there is an inherent conflict between the two which has never satisfactorily been resolved.
- The choice of targets selected for performance measurement systems can also have a significant impact on the effectiveness of those systems. It is possible that unintended consequences of performance targets could end up having an adverse effect on performance.

While the need for some kind of performance assessment is widely accepted, appraisal systems are frequently criticised as bureaucratic, ineffective and largely irrelevant to the work of the organisation. Partly as a response to this view, modern approaches attempt to enhance the relevance of appraisal by linking it to organisational strategy and objectives. This emphasises the use of appraisal as an **instrument of control over the workforce**. However, running in parallel with this trend is an awareness, among HR professionals at least, that appraisal systems are fundamental to the aspirational model of HRM outlined above and to the cooperative psychological contract.

3.1 The purpose of appraisal

Appraisal is a process that provides an analysis of a person's overall capabilities and potential. An important part of the appraisal process is assessment – collecting and reviewing data on an individual's work.

The purpose of appraisal is usually seen as the **improvement of individual performance**, but it may also be regarded as having close links to a wide range of other HR issues, including discipline, career management, identifying training and development opportunities, motivation, communication, selection for promotion and determining rewards. It is also fundamental to the notion of **performance management**, which may be regarded as trying to direct and support individual employees to work as effectively and efficiently as possible so that the individual's goals are aligned with the organisation's goals and business strategy.

Within this wider view, regular appraisal interviews can be seen as serving two distinct purposes:

(a) **Judgement**: Judgemental appraisals are undertaken in order for decisions to be made about employees' pay, promotion and work responsibilities.

These decisions have to be made on the basis of judgements about the appraisee's behaviour, talent, industry and value to the organisation. Such judgements can be uncomfortable for both appraiser and appraisee and lead to hostility and aggression.

(b) **Development**: The focus of developmental appraisals is to assess employees' training and development needs.

Development appraisal can contribute to **performance improvement** by establishing individuals' development needs, progress and opportunities. This is the more supportive aspect of appraisal, but still requires the appraiser to make decisions about the appraisee.

> "The tension between appraisal as a judgemental process and as a supportive development process has never been resolved and lies at the heart of most debates about the effectiveness of appraisal at work." (Bratton and Gold)

Feedback on performance has been widely regarded as an important aspect of the participative style of management which, in turn, has been promoted as having potential to motivate higher performance. However, the link between feedback and motivation is not simple and an important aspect of the judgemental part of appraisal is its potential to **demotivate**.

The classic study which highlighted this was carried out by Meyer et al at the General Electric Company in 1965. Gold suggests that their findings are still relevant and provides a summary:

(a) Criticism often has a negative effect on motivation and performance.

(b) Praise has little effect, one way or the other.

(c) Performance improves with specific goals.

(d) Participation by the employee in goal-setting helps to produce favourable results. (Don't forget the whole point of performance management is to improve performance!)

(e) Interviews designed primarily to improve performance should not at the same time weigh salary or promotion in the balance.

(f) Coaching by managers should be day to day rather than just once a year.

From their research in the 1980s into the effectiveness of self-appraisal as a means of evaluating performance, Campbell and Lee pointed out the ways in which discrepancies may arise between people's own opinions of their performance and their supervisors' opinions of their performance.

(a) **Information**. There may be disagreement over what work roles involve, standards of performance and methods to be used.

(b) **Cognition**. The complexity of behaviour and performance leads to different perceptions.

(c) **Effect**. The judgemental nature of appraisal is threatening to the appraisee and, possibly, to the appraiser.

Since Meyer *et al*'s study there has been a long search to find a way of appraising employees which reduces the feeling that feedback is about criticism.

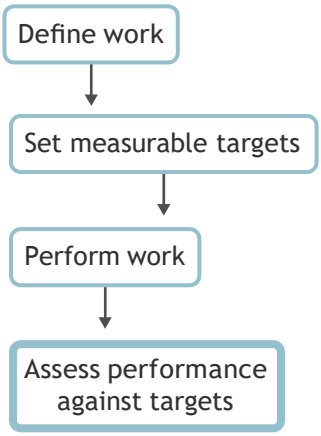
One approach to mitigating the undesirable effects of judgemental appraisal has been the use of **multisource feedback**, including 360 degree appraisal, in order to provide a demonstrably more **objective** review. Such approaches have tended to be used principally for appraisal of managers. Multisource feedback can be seen as empowering for staff. It may also be seen as reinforcing for good management behaviour (since it shows managers how they are seen by others) and likely to improve the overall reliability of appraisal. However, research has shown that the effects can vary significantly.

3.1.1 Appraisal as control or development?

The last of Meyer *et al*'s findings, "coaching by managers should be day to day rather than just once a year", highlights the role of managers in the **development** of their staff on a continual basis.

However, any shift towards a more developmental view of appraisal sits uncomfortably with the traditional management objectives of having a means of measuring, monitoring and controlling performance. Most appraisal schemes are still ultimately **performance control schemes**, as illustrated in Figure 10.5 below.

Figure 10.5: Appraisal as a performance control system



This somewhat rigid approach, based on the drive for rationality and efficiency in organisations, highlights what Mintzberg has called '**machine bureaucracy**'. According to this approach, getting organised, being rational and achieving efficiency are the best bases on which to structure an organisation.

This mechanistic view of organisations will, almost inevitably, mean that employees will view appraisals as control systems, and employees will feel they are being controlled by appraisal systems. Such a situation is unlikely to motivate employees or to generate trust, commitment and high productivity, though.

Employees' trust and commitment to an organisation will come about through management creating a culture that supports people's long-term development. Assessment and appraisal could play a key part in this shift, but only if HR managers can convince organisations that, while control remains important, development needs to play a much greater role in the appraisal process.

Interactive question 3: Appraisals

The Jackson Business Centre (JBC) provides professional courses for students of accounting, law and marketing.

JBC has operated a formal performance appraisal system, supported by standardised procedures and paperwork, for a number of years. The system has clear organisational objectives, which are based on staff development and improved performance, rather than being a basis for pay reviews or paying individual annual bonuses. However, the scheme is not well regarded by either managers or staff and its objectives are not being met.

Senior managers complain about the amount of time that is taken up holding appraisal interviews and then completing the necessary paperwork.

Exit interviews are conducted whenever someone leaves JBC, and a review of a sample of recorded comments indicates staff feelings on the scheme very clearly: "appraisal is just a paper exercise", "a joke", "a waste of time and effort".

Requirement

Discuss the possible reasons why the objectives of the formal appraisal system are not being met.

See **Answer** at the end of this chapter.

3.2 HRM and performance management

Performance management systems attempt to integrate HRM processes with the strategic direction and control of the organisation. Remember the cycle of control we mentioned earlier:

Step 1 Goals are set

Step 2 Performance is measured and compared with target

Step 3 Control measures are undertaken in order to correct any shortfall

Step 4 Goals are adjusted in the light of experience

You should be familiar with this kind of management control in business organisations, where the balanced scorecard, for example, is often used as the basis for such an approach.

Performance management requires that the strategic objectives of the organisation are broken down into layers of more and more detailed sub-objectives, so that individual performance can be judged against personal goals that support and link directly back to corporate strategy.

The performance management system, although it emphasises the control aspects of appraisal, must also allow for the **development** aspect of appraisal, providing for coaching and training where needed.

3.3 Target selection

We have noted how performance management acts as a control system in measuring people's achievement against targets. However, in order for performance management to be beneficial, it is important to select the right measures or targets at the outset when setting performance goals.

The adage "what gets measured, gets done" is often used in relation to corporate performance management, but it is equally relevant here. If the 'wrong' performance measures or targets are set, this could lead to staff behaviour being different to that originally intended, and ultimately adversely affecting performance.

We looked at the **behavioural implications of performance targets** in section 4.13 of the chapter "Strategic performance management", but they are also important here, in highlighting the impact that remuneration and reward structures can have on organisational behaviour.



Context example: Bankers' bonuses

In the aftermath of the global financial crisis of 2008–09, a lot of media attention focused on bankers' bonuses. A number of investment banks link employees' annual bonuses to the amount of money they earn in that year, a short-term approach which can influence employees' decision-making.

Critics argue that the bonuses encourage risky behaviour that maximises profits in the short term but could potentially be loss-making in the longer term.

The sub-prime mortgage crisis in the US in 2007 was a good example of this. The mortgage bond market proved extremely profitable for the banks in the short term, but once mortgage-holders started defaulting on their loans the banks had to foreclose them, causing the loans to be written off.

During the bull market (before 2007) certain financial packages made a great deal of money for the banks in the short term, resulting in their staff receiving large bonuses. However, those same financial packages failed shortly afterwards, triggering the financial crisis.

The individual performance measures selected should be relevant to the overall objectives of the organisation. Individuals' objectives must reflect the overall strategic initiatives management are taking. For example, if management is focusing on quality, performance measures must reflect this by measuring employees on their contribution to achieving quality targets.

This sort of situation represents a **contingency approach** to reward: that the organisation's strategy is a fundamental influence on its reward system, and in turn the reward system should support the organisation's chosen strategy.

Targets and motivation

Some employees respond well to difficult targets and are motivated to attain them. Others may find the targets daunting and feel they are unachievable, and indeed there may be valid reasons why they believe this. For example, in an economic downturn, a number of businesses reduce the amount they spend on their IT budgets. Therefore if a salesperson in an IT company was given a target of increasing sales 25% on the prior year they would appear to be justified in thinking this target is unachievable.

Equally, care must be taken when using certain measures, for instance numbers of sales, as the basis for rewarding employees. As an example, here are some possible negative consequences of using sales numbers as a primary performance measure:

(a) The salesperson might offer potential customers large discounts in order to make the sale (but with the effect that the company makes a loss on the sale).

(b) The salesperson is concerned solely with the immediate sale, which may lead to poor after-sales service, low customer satisfaction levels and poor customer retention.

(c) The salesperson might use expensive promotions that actually generate less in sales value than they cost, but which allow the salesperson to register a number of sales.

(d) Once a salesperson has reached their target figure for a period they might look to defer future sales into the next period.

It may be better to use a balanced mix of targets – for example, setting customer care and customer profitability targets as well as the number of sales made.

It is also important to make sure whatever goals are set that they are capable of being controlled by the individual, otherwise the individual is likely to become demotivated.

In addition, if processes are being redesigned and job roles are changing, performance measures must be adapted to reflect the new jobs and responsibilities.

However, it is important that people are not given too many objectives and targets. There is a danger that people could become overwhelmed by the sheer number of goals they are expected to meet, but with the result that they do not know what their priorities are or what aspects of their work they should give most attention to.

Finally, it is useful to remember the acronym SMART when setting performance targets: are the targets specific, measurable, achievable, relevant and time-bound?



Professional skills focus: Structuring problems and solutions

Performance targets should be challenging for employees but if they are unachievable the individual is likely to be demotivated. Targets should be relevant to the overall objectives of the organisation but too many measures can be overwhelming for employees. Demonstrate your professional skills by anticipating such problems when developing solutions and making recommendations about a company's performance targets.

4 The impact of remuneration and reward packages

Section overview

- A reward system should fulfil three key behavioural objectives: supporting staff recruitment and retention; motivating employees to high levels of performance; and promoting compliance with workplace rules and expectations.
- A contingency approach to reward accepts that an organisation's strategy is a fundamental influence on its reward system, and that its reward system should support its chosen strategy.
- Equally, however, an effective reward system should align individuals' goals with an organisation's strategic goals.

4.1 Reward

Employment is fundamentally an economic relationship; the employee works as directed by the employer and, in exchange, the employer provides reward. The relationship inevitably generates a degree of tension between the parties, since it requires **cooperation** if it is to function, but it is also likely to give rise to **conflict** since the employee's reward equates exactly to a cost for the employer.

Definition

Reward: All the monetary, non-monetary and psychological payments that an organisation provides for its employees in exchange for the work they perform.

Rewards may be seen as **extrinsic** or **intrinsic**.

(a) **Extrinsic rewards** derive from the **job context**; such extrinsic rewards include pay and other material benefits as well as matters such as working conditions and management style.

(b) **Intrinsic rewards** derive from **job content** and satisfy higher-level needs such as those for self- esteem and personal development.

The organisation's reward system is based on these two types of reward and also includes the policies and processes involved in providing them.

Reward is a fundamental aspect of HRM and the way an organisation functions. It interacts with many other systems, objectives and activities.

- It should support the overall strategy.
- It is a vital part of the psychological contract.
- It influences the success of recruitment and retention policies.
- It must conform to relevant laws and regulations.
- It consumes resources and must be affordable.
- It affects motivation and performance management.
- It must be administered efficiently and correctly.

The dual nature of reward mentioned earlier – a benefit for the employee, a cost for the employer – means that the parties in the relationship have divergent views of its purposes and

extent. Employees see reward as fundamental to their standard of living: inflation, comparisons with others and rising expectations put upward pressure on their notion of what its proper level should be. Employers, on the other hand, seek both to control their employment costs and to use the reward system to influence such matters as productivity, recruitment, retention and change.

Context example: Tesco – Remuneration strategy

As we saw in an example in the chapter "Strategic analysis", Tesco's Annual Report highlights the principal risks the company thinks it is facing. One of the risks highlighted in the 2018 Annual Report related to people: failing to attract and retain staff with the capabilities required to make the business a success, and failing to develop the company's culture, will impact on its ability to achieve its strategic goals.

The Governance section of the report then outlines the company's reward principles for its staff.

In order to build an inclusive culture, where everyone feels welcome, it is important that staff feel fairly rewarded. Tesco's reward principles identify that staff can expect:

- A total reward package (salary, bonus, share scheme, pension, benefits) that provides flexibility and choice, and is competitive in the markets in which Tesco operates and from which it recruits for talent
- To share in Tesco's success through variable pay (eg, bonuses) that is transparent, and rewards performance
- The opportunity to plan and save for the future (eg, through the pension scheme)
- To make wellbeing and lifestyle choices, having access to a range of benefits and flexibility over working hours and place of work
- to have access to career opportunities and accredited training to develop their potential

4.2 A reward management model

An effective reward system should facilitate both the **organisation's strategic goals** and also the goals of **individual employees**.

Within this, an organisation has to make three basic decisions about monetary reward:
(a) How much to pay
(b) Whether monetary rewards should be paid on an individual, group or collective basis
(c) how much emphasis to place on monetary reward as part of the total employment relationship

However, there is no single reward system that fits all organisations. Bratton proposes a model of reward management based on five elements.

(a) The **strategic perspective**
(b) Reward **objectives**
(c) Reward **options**
(d) Reward **techniques**
(e) Reward **competitiveness**

Professional skills focus: Concluding, recommending and communicating

One of the professional skills tested in the CA exams is the ability to apply professional experience and evidence to support reasoning. When advising on the appropriateness of

reward schemes use your personal experience of organisational reward schemes as well as technical and commercial knowledge to support your recommendations.

4.2.1 The strategic perspective

A **contingency approach to reward** accepts that the organisation's strategy is a fundamental influence on its reward system and that the reward system should support the chosen strategy.

Thus, for example, cost leadership and differentiation based on service will have very different implications for reward strategy (and, indeed, for other aspects of HRM). This is because each strategy needs a reward which is appropriate for it. The closer the alignment between the reward system and the strategic context, the more effectively the organisation can implement its strategy. The following example illustrates this.

Example of strategic perspective

Bratton and Gold in their text *Human Resource Management* provide an illustration of how two different businesses with different generic strategies have completely different rewards systems.

The first business produces high-quality, custom-made machine tools for a high-tech industry. The production process is complex and workers are highly skilled and capable of performing various jobs. They all work in self-managed teams.

In contrast to the industry norm, these skilled machine operators are not paid an hourly wage, but instead receive a base salary which is increased as they learn new skills. The employees receive an excellent benefits package and profit-sharing bonuses. Not surprisingly, staff turnover is very low.

Labour costs at this company are above the industry average, but the company is successful nonetheless because its reward system is aligned to its strategy. It is following a differentiation strategy, and its reward system encourages commitment from its staff. The system also encourages higher productivity than its competitors because of the increased functional flexibility of having multi-skilled staff. The incentive of their salary increasing as they learn new skills encourages the staff to become multi-skilled. In turn, having a multi-skilled workforce reduces machine downtime and

scrap rates. Because the teams are self-managed, the company does not need to employ supervisors or quality inspectors (the teams self-regulate their own quality). Because staff turnover is low, recruitment and training costs are similarly low.

Therefore, although the company's labour costs are above the industry average, these additional costs deliver benefits elsewhere and support its differentiation strategy.

Against this, Bratton and Gold contrast a production process producing frozen food. The work is low- skilled and monotonous, and requires little employee commitment. The production line is automated and managers – not workers – control the speed of the line.

The workers are paid an hourly wage marginally above the minimum wage, and there are no additional payments or benefits. Not surprisingly, labour turnover is very high.

However, again this company is successful, because its reward system is aligned to its strategy. It is following a cost leadership strategy and so low-cost production is essential. The high labour turnover is not a problem because unskilled workers are easy to recruit and training costs are low. Therefore, the company's policy of paying near-minimum wage only is appropriate to a strategy in which little commitment or loyalty is required from the employees.

It is vital that reward systems are aligned to an organisation's objectives and its critical success factors, as well as to the job in question. As the scenarios above illustrate, if the organisation has highly skilled employees who are crucial to its competitive success, then the reward system should be designed to try to retain such staff.

However, it is also important to recognise the impact that implementing a reward system can have on employees' day to day performance. Once again, the idea that "what gets measured, gets done" is relevant here. For example, if a reward system is based primarily around individual performance, then staff will focus on their own individual results and teamwork could suffer as a result.

More generally, if a reward system is not appropriate for the context in which it is used, there is a danger it could have a negative impact on an organisation's performance.

4.2.2 Reward objectives

The reward system should pursue three behavioural objectives:

(a) It should support **recruitment and retention**

(b) It should **motivate** employees to high levels of **performance**. This motivation may, in turn, develop into commitment and a sense of belonging, but these do not result directly from the reward system.

(c) It should promote **compliance** with workplace rules and expectations

Recruitment and retention

The reward system should support **recruitment and retention**. Several influences are important here. Employees will certainly assess their pay and material benefits against what they believe to be the prevailing market rate. They will also take account of disadvantageous factors, such as unpleasant working conditions, in their assessment of the degree of equity their reward achieves for them.
Finally, they will be very sensitive to comparisons with the rewards achieved by other employees of the same organisation. Failure to provide a significant degree of satisfaction of these concerns will lead to enhanced recruitment costs.

Motivation

The reward system should **motivate employees** to high levels of performance.

Despite the apparently tenuous link between performance and level of pay (for example, with Herzberg arguing that pay is a hygiene factor rather than a motivating factor), traditional pay systems have featured incentives intended to improve performance; there has also been a tendency for UK and North American companies to adopt systems of individual performance-related pay intended to support overall organisational objectives rather than simply to incentivise individual productivity.

Compliance

The reward system should **promote compliance** with workplace rules and expectations. The psychological contract is complex and has many features, including material rewards. The incentives included in the reward system play an important role in **signalling to employees the behaviour that the organisation values**. It is also an important contributor to the way employees perceive the organisation and their relationship with it.



Context example: Barclays Bank – bonuses for reward or retention?

Consider the following case, and whether you think the bonus payments are justified.

In 2013, despite a fall in annual pre-tax profits of 37% in Barclays' investment bank, and return on equity falling from 12.7% to 8.2%, bonuses increased by 13%. Barclays' investment bankers enjoyed bonuses of £1.6 billion compared to £1.4 billion in 2012. Total bonus pay across the Barclays group as a whole increased from £2.2 billion in 2012 to £2.4 billion in 2013.

Announcing these figures, Barclays' chief executive said: "At Barclays, we believe in paying for performance and paying competitively".

Barclays is trying to compete directly, on a worldwide basis, with JPMorgan and Goldman Sachs to recruit and retain the investment bankers it values most highly. As the company's chief executive pointed out: "We compete in global markets for talent. If we are to act in the best interests of our shareholders, we have to make sure we have the best people in our firm".

The chief executive insisted that the bonuses needed to be paid, because Barclays has no control over market-led pay. In other words, if Barclays pays less than rival banks, it will not be able to retain its staff.

As such, Barclays' chief executive argued that the increase in the incentive pool was required in order to build the business in the long-term interests of shareholders. This suggests that it is more important to retain talented employees to support the business's future growth than to tie annual rewards to annual performance.

But is this really how performance-related pay should work? Critics argued that, instead of rewarding past performance, Barclays bonuses are more akin to a reward for performance that the bank hopes will occur in the future – provided that JPMorgan (and others) do not increase their bonuses even higher, and in the hope that trading bonds becomes interesting again.

Similarly, critics have argued that these bonuses are not consistent with the general principle that bonuses are meant to reflect financial performance – that is, bonus awards increase as financial performance increases, and go down if financial performance worsens.

Moreover, after the results were announced, Barclays faced a revolt among its institutional investors, and 34% of its shareholders voted against the remuneration report at the Group's annual meeting in June 2014.

In its 2012 annual report, Barclays had pledged to take 'a different approach to the balance between directors' and employees' remuneration, and returns for shareholders'.

However, the corporate governance director at the Institute of Directors was very critical of the impact of this new balance. In the light of the bank proposing to pay £2.4 billion in bonuses, compared with £860 million in dividends to shareholders, he questioned who Barclays was being run for – its staff or its owners?

Based on: Pratley, N. (2014), 'Barclays' bonuses: back to the bad old days', *The Guardian*, 11 February 2014

Treanor, J. (2014), 'Barclays condemned over £2.4bn bonuses', *The Guardian*, 11 February 2014)

4.2.3 Methods of reward

Material reward may be divided into three categories:

(a) **Base pay** is a simply established reward for the time spent working.

(b) **Performance-related pay** is normally added to base pay and is intended to reward performance, learning or experience.

(c) **Indirect pay** is made up of benefits such as health insurance and child care, and is provided in addition to base pay or performance pay.

Note: Reward schemes and their implications were discussed in Section 4 of the chapter "Strategic performance management".

Share options

One further type of reward option we should consider is share options (or employee share option plans).

Share options give directors – and possibly other managers and staff – the right to purchase shares at a specified exercise price after a specified time period in the future.

The options will normally have an exercise price that is equal to, or slightly higher than, the market price on the date that the options are granted. The time period (vesting period) that must pass before the options can be exercised is generally a few years. If the director or employee leaves during that period, the options will lapse.

In this respect, share options can be seen as a way of rewarding directors and employees for remaining with a company. In turn, this could mean that they are concerned with the longer-term success of the company, rather than simply focusing on short-term performance.

Share options will generally be exercisable on a specific date at the end of the vesting period. In the UK, the Corporate Governance Code states that shares granted, or other forms of remuneration, should not vest or be exercisable in less than three years. Directors should be encouraged to hold their shares for a further period after vesting or exercise. If directors or employees are granted a number of options in one package, these options should not all be able to be first exercised at the same date.

If the price of the shares rises so that it exceeds the exercise price by the time the options can be exercised, the directors will be able to purchase shares at lower than their market value, which is clearly advantageous for the directors exercising the options. Share options can therefore be used to **align management and shareholder interests**, because the directors have an interest in ensuring that the share price increases over time such that it is higher than the exercise price when the options come to be exercised. This is particularly relevant for options held for a long time when value is dependent on long-term performance.

However, the main danger with share options is that they could give directors an incentive to manipulate the share price if a large number of options are due to be exercised.

Alternatively, granting options could be used as a way of encouraging cautious (or risk-averse) directors to take positive action to increase the value of the company.

Again, this could help align the interests of directors and shareholders, if the directors would not otherwise be prepared to accept the same risks which the shareholders would tolerate by themselves.

The upside risk of share options is unlimited because there is no restriction on how much the share price can exceed the exercise price. However, there is no corresponding downside risk for the directors. If the share price is less than the exercise price, the intrinsic value of options will be zero and the options will lapse. In these circumstances it will make no difference how far the share price is below the exercise price.

If directors hold options, the value of their options will rise if a strategic investment succeeds and they will not suffer any loss on their options if the investment fails. Therefore, granting the options might encourage the directors to take actions they would not otherwise be prepared to take.

Risk, reward and performance

Although we have noted that share options could encourage cautious directors to be less cautious, it is equally important that reward structures do not encourage directors and managers to take excessive risks.

Since the collapse of Northern Rock bank in 2007, and throughout the ensuing financial crisis, there has been much political and media interest in the issue of reward management. This has focused on the role which reward structures were perceived to have played in encouraging excessive risk taking in the financial services sector and, in turn, what role this risk taking played in the problems which have affected the sector.

Additionally, there has been increasing concern about the extent to which the level of remuneration given to senior executives reflects (or does not reflect) the value their companies generate for their shareholders.

In the UK, in a speech to the High Pay Commission and the Institute for Public Policy Research (January 2012), the MP Chuka Umunna highlighted the extent to which the value of incentive packages for executives has risen disproportionately to improvements in company performance. In the first decade of the 21st century, FTSE 350 firms increased their pre-tax profits by 50% and their earnings per share by 73%, while year-end share prices fell by 5%. Over the same period, bonuses for executives in these companies rose by 187% and long-term incentive plans by 254%.

And, as Mr Umunna pointed out, in the worst cases "you end up with perverse incentive structures which encourage the wrong kind of decision-making, as the failures in many financial institutions in the wake of the 2008/9 financial crises so clearly illustrated".

Another issue which causes increasing anger and frustration among shareholders is the level of bonuses being awarded by companies that were rescued by taxpayer funds.

This is perhaps symptomatic of a potentially wider issue: the extent to which companies are perceived to be **rewarding failure**. The senior executives of failed companies often walk away with significant payouts, while large numbers of other managers and staff lose their jobs and their incomes**.**

Critics have argued that if companies are serious about improving performance, then they need to stop rewarding failure.



Context example: Rewards for failure – Tesco

Following pay outs of more than £2 million to two former directors of Tesco in February 2015, City investors are looking at ways to reduce contractual payments to departing company bosses.

Philip Clarke and Laurie McIlwee, formerly chief executive and finance director of Tesco respectively, received the payments despite the £263 million accounting scandal which was uncovered by Mr Clarke's successor in September 2014.

Tesco had previously sought to withhold the payments – £1.2 million to Mr Clarke and £970,800 to Mr McIlwee – but in February 2015 it announced it would have to pay them, after being told it had no legal grounds to continue withholding them. Tesco said it was contractually obliged to make the payments to the former directors unless it could "legally establish a case of gross misconduct" against them.

Following Tesco's announcement, investors have argued it is time to review the length of directors' contracts. Old Mutual – a leading investment group – has called for directors' contracts to be 'substantially shorter' than the current norm of 12 months. The head of UK stewardship and governance at Old Mutual Global Investors said that 12-month contracts, and the requirement to give departing directors a year's pay, were 'an anachronism' in today's business environment.

Source: Treanor, J. (2015) 'City bosses' 12-month contract rule should be reviewed, say investors', *The Guardian*, 17 February 2015.

4.2.4 Reward techniques

Reward systems must attempt to achieve **internal equity**. This means when employees make comparisons between their own rewards and those of others, they see the overall structure as fair. If internal equity is not achieved, employees will conclude that the psychological contract has been breached and their behaviour will be affected. They may become less cooperative or they may leave.

Three techniques contribute to the establishment of internal equity.

Job analysis

Job analysis is the "systematic process of collecting and evaluating information about the tasks, responsibilities and the context of a specific job" (Bratton). The data collected during job analysis identifies the major tasks performed by the job holder, the outcomes that are expected, and how the job links to other jobs in the organisation. This data is used to prepare job descriptions, job specifications and job performance standards. (Note that in practice the terms job description and job specification may be used loosely and a **job specification** is often referred to as a **person specification**.)

This information is useful in itself for a range of HRM purposes, including recruitment and training needs analysis, and it also forms the basis for **job evaluation**.

Note also that job analysis is an important aspect of quality and process redesign initiatives and is almost certainly required when e-business methods are adopted.

Job evaluation

Job evaluation is a systematic process designed to determine the relative worth of jobs within a single work organisation. The process depends on a series of subjective judgements and may be influenced by organisational politics and personal preconceptions. In particular, it can be difficult to separate the nature of the job from the qualities of the current incumbent.

Evaluation may be carried out in four ways.

(a) **Ranking** simply requires the arrangement of existing jobs into a hierarchy of relative value to the organisation.

(b) **Job grading** starts with the definition of a suitable structure of grades in a hierarchy. Definitions are based on requirements for skill, knowledge and experience. Each job in the organisation is then allocated to an appropriate grade.

(c) **Factor comparison** requires the allocation of monetary value to the various factors making up the content of a suitable range of benchmark jobs. This method is complex and cumbersome.

(d) **Points rating** is similar to factor comparison, but uses points rather than monetary units to assess the elements of job content.

Whichever method is used, the end point of a job evaluation exercise is the production of a **hierarchy of jobs** in terms of their relative value to the organisation. The **pay structure** is then set by reference to this hierarchy of jobs.

Performance appraisal

Performance appraisal has already been discussed in section 3 earlier in this chapter.

4.2.5 Reward competitiveness

The level of rewards an organisation offers will inevitably be subject to factors external to the organisation:

(a) The **labour market** as it exists locally, nationally and perhaps globally, as relevant to the organisation's circumstances

(b) The pressure for **cost efficiency** in the relevant industry or sector

(c) **legislation** such as the level of any applicable minimum wage

4.3 Setting reward levels in practice

Many companies use commercially available **survey data** to guide the overall level of the rewards they offer. This approach can be combined with the reward techniques outlined above.

An element of **flexibility** must be incorporated to reflect both the different levels of skill, knowledge and experience deployed by people doing the same work and their effectiveness in doing it.

Governments influence pay levels by means other than outright legislative prescription:

(a) They affect the demand for labour by being major employers in their own right.

(b) They can affect the supply of labour by, for example, setting down minimum age or qualification requirements for certain jobs.

(c) Their fiscal and monetary policies can lead them to exert downward pressure on public sector wage rates.

4.3.1 Problems with reward systems

Reward systems are subject to a range of pressures that influence their working and affect the psychological contract.

(a) Where **trade unions** are relatively weak, as in the UK, employers have more freedom to introduce performance-related pay.

(b) **Economic conditions** may prevent employers from funding the rewards they might wish to provide in order to improve commitment. The result would be disappointment and dissatisfaction.

(c) Performance pay systems are prone to **subjective and inconsistent** judgement about merit; this will discredit them in the eyes of the employees.



Context example: Failure of reward systems

Why reward systems fail to deliver IT transformation

(This example is based on a short article on the technology website www.zdnet.com which illustrates some failings in reward systems in relation to IT projects.)

An organisation had a plan for an enterprise-wide service-orientated approach which was well thought through and should have worked well. But when the project was implemented it turned out to be a failure. One of the reasons for the failure was the way IT professionals and managers were rewarded, highlighting the importance of rewarding the right behaviour in any IT-driven transformational project.

The article highlights four common misconceptions in reward systems:

- **Rewarding programmers for lines of code produced, or based on programme complexity**. This type of reward system will encourage programmers to develop more complex or difficult programmes without considering what the organisation needs. It may not need – or want – complex or difficult programmes.
- **Rewarding developers based on long hours worked**. There is a danger with this kind of measure that **quantity** gets rewarded rather than **quality**. A programmer may end up working very long days simply because they did a poor job of estimation and planning up front, or the long hours could be an indication that there is a lot of code rewriting going on, to correct mistakes which the programmer had made initially.
- **Rewards based on salary surveys**. Basing IT salaries on industry averages means that some of the competitor companies in the market are paying more (although some are also paying less). However, if you simply pay an average rate, as soon as the economy becomes more buoyant and demand for workers heats up, programmers will defect and move to higher-paying rival companies.

- **Rewarding people based on the number of problem statements they close**. This is problematic because some people will solve multiple problems with one problem statement, while others will open and solve as many problem statements as they can to inflate the number of problems solved.

(Based on: McKendrick, J. (2010) 'Why reward systems fail to deliver IT transformation', www.zdnet.com)

4.4 Benefits and adverse consequences of linking reward schemes to performance measurement

4.4.1 Benefits for the organisation

It is clear how objectives set at higher levels can be translated into individual goals, thereby linking strategy to outcomes for the individual. This is illustrated in Bratton's model where the strategic perspective explains that the reward system should support strategy, and the two should be closely aligned.

A reward scheme should also provide an incentive to achieve a good level of performance, and the existence of a reward scheme can help to attract and retain employees who make favourable contributions to the running of the organisation.

A reward scheme can also help emphasise the key performance indicators of the business, if these are incorporated into the performance measures which underpin the scheme. This will help reinforce to employees the key aspects of their performance which contribute most to the organisation's success.

4.4.2 Drawbacks for the organisation

However, the financial crisis of 2007–08 showed the dangers of linking reward schemes to performance measures if those **performance measures are poorly designed**. We highlighted this in the case study about bankers' bonuses earlier in the chapter, suggesting that the bonus culture encouraged a focus on short-term decision-making and risk taking.

A European Commission report into the financial crisis suggested that

"Excessive risk taking in the financial services industry… has contributed to the failure of financial undertakings… Whilst not the main cause of the financial crises that unfolded… there is widespread consensus that inappropriate remuneration practices… also induced excessive risk taking". (European Commission, Commission recommendation on remuneration policies in the financial sector)

In this case, there appears to be a direct link between the profit measures (short-term profitability) and the **risk appetite of employees**. Employees were prepared to take greater risks in the hope of making higher profits and therefore getting larger bonuses.

However, a second potential drawback for an organisation arises if it is unable to reward individuals for good performance (for instance, due to a shortage of funds) because then the link between reward and motivation may break down.

4.4.3 Benefits and drawbacks for the individual

If an individual's goals are linked to the objectives of the organisation, then it is clear to the individual how their performance is measured and why their goals are set as they are. However, on occasion there may be a problem in linking individual rewards directly to organisational outcomes, especially if the outcomes are uncertain.

Another drawback is that in striving to meet targets some individuals may become cautious and reluctant to take risks given they have a stake in the outcome. Conversely, other individuals may choose riskier behaviour especially if reward is linked to, say, revenue generation or levels of output.

4.4.4 Risk and reward

Overall, a reward system needs to achieve a balance between risk and reward:

Recruitment and retention: Rewards need to be structured in such a way that they attract and retain key talent. If an organisation's reward system is not deemed to be attractive, then there is a risk it will not be able to attract or retain the staff it needs to be successful.

Alignment with business strategy and culture: If reward strategy is not aligned to organisational goals then there is a risk the organisation will not achieve those goals. Equally, the reward system needs to encourage styles of behaviour that fit with the organisation's culture.

Reputation/brand: If the organisation's reward systems generate negative press coverage (as has been the case with some banks in the recent financial crisis) there is a risk this will adversely affect the organisation's reputation or brand.

Interactive question 4: Reward packages

The Superior Business Consultancy (SBC), based in Jayland, provides clients with a range of business consultancy services as well as IT services and support.

Currently, SBC pays all its consultants a fixed salary. However, some of the IT consultants are unhappy that their salaries are lower than those earned by the other types of consultant. Recently, four of SBC's longest-serving IT consultants resigned to go and work for rival consultancies. All of them said that the reward packages available had played a significant part in their decisions.

The directors are worried about the prospect of more consultants leaving SBC and joining rival consultancies. As a result, the directors are reviewing SBC's reward packages. The directors are aware that all the major software providers in Jayland pay a commission to consultancy firms if the firm recommends their software to a client. Currently, this commission is payable to SBC as a whole, but the directors are considering whether it should be paid to individual consultants. They are also considering a proposal under which the IT consultants would receive a lower basic salary, but would then be entitled to receive any commissions earned from the software providers.

Requirement

Evaluate the directors' proposal to revise the way SBC's IT consultants are paid.

See **Answer** at the end of this chapter.

4.5 Remuneration and corporate reporting

Professional skills focus: Applying judgement

When recommending remuneration policies, you should consider their impact on the financial statements of the business as well as on employees and the overall business strategy. Use your judgement to identify and recommend the most appropriate remuneration policy in light of the specific business objectives and challenges presented in the question scenario.

In addition to considering the impact of proposed remuneration policies on employees and their organisations, and how shareholders and other stakeholders might react to any proposed

benefit packages, we need to consider how employee benefits will be accounted for in an organisation's financial statements.

Two accounting standards are relevant here: IAS 19, *Employee Benefits*, and IFRS 2, *Share-based Payment*.

We have already looked at these two standards in the chapter "Strategic performance management", where we noted the concern which British Airways' large pension deficit caused in relation to the merger between British Airways and Iberia.

Crucially, British Airways' main pension plan was a defined benefit plan. As section 4.5.1 below explains, the corporate reporting consequences of operating a defined benefit plan are significantly different from operating a defined contribution plan.

Therefore, when deciding what type of pension plan to offer employees (as part of their reward package) it will also be important for an organisation to consider the corporate reporting implications of that decision.

Many UK defined-benefit schemes are heavily in deficit, with these deficits being driven by a fall in equities and bond yields (which have reduced the value of plan assets). At the same time, falling mortality rates and increasing life expectancy are also a problem – if people are living longer, the defined benefit obligations (and hence the deficit) also increase.

More generally, a pension fund deficit can also be an important issue in corporate valuation – particularly in relation to a potential acquisition. The need to address the funding gap in any such schemes will be likely to reduce the amount a potential purchaser is prepared to pay to acquire a company.

Alternatively, as we saw in the chapter "Strategic performance management" in the example of Pearson's sale of the FT to Nikkei, a potential purchaser (Nikkei) who is prepared to contribute to the funding gap may persuade a company (Pearson) to sell an asset (FT) they had previously been reluctant to sell.

4.5.1 IAS 19, *Employee Benefits*

The objective of this standard is to prescribe the accounting and disclosure for employee benefits, where employee benefits are all forms of consideration, for example cash bonuses, retirement benefits and private healthcare, given to an employee by an entity in exchange for the employee's services.

The standard requires an entity to recognise:

(a) A liability when an employee has provided service in exchange for employee benefits to be paid in the future

(b) An expense when the entity consumes the economic benefit arising from service provided by an employee in exchange for employee benefits

However, accounting issues could arise due to:

(a) The valuation problems linked to some forms of employee benefits

(b) The timing of benefits, which may not always be provided in the same period as the one in which the employee's services are provided

Two elements of IAS 19 are particularly relevant to remuneration structures:

(a) **Short-term employee benefits** (falling due within 12 months from the end of the period in which the employees provide their services) such as wages, salaries, bonuses and paid holidays, non- monetary benefits such as private medical care and company cars.

These benefits should normally be treated as an expense, with a liability being recognised for any unpaid balance at the year end.

(b) **Post-employment benefits** such as pensions and post-retirement health cover.

Pension plans can either be **defined contribution** or **defined benefit** plans. The accounting for defined benefit plans is much more complex than for defined contribution plans.

Defined contribution plan

Contributions by an employer into a defined contribution plan are made in return for services provided by an employee during the period. The employer has no further obligation for the value of the assets of the plan or the benefits payable.

(a) The entity should recognise contributions payable as an expense in the period in which the employee provides services (except to the extent that labour costs may be included within the cost of assets).

(b) A liability should be recognised where contributions arise in relation to an employee's service, but remain unpaid at the period end.

4.5.2 Defined benefit plans

These are defined by IAS 19 as all plans other than defined contribution plans. Characteristics of a defined benefit plan are as follows:

- The amount of pension paid to retirees is defined by reference to factors such as length of service and salary levels (ie, it is guaranteed).
- Contributions into the plan are therefore variable depending on how the plan is performing in relation to the expected future obligation (ie, if there is a shortfall, contributions will increase and vice versa).

Contribution levels

The actuary advises the company on contributions necessary to produce the defined benefits ('the funding plan'). It cannot be certain in advance that contributions plus returns on investments will equal benefits to be paid.

Formal actuarial valuations will be performed periodically (eg, every three years) to reveal any surplus or deficit on the scheme at a given date. Contributions may be varied as a result; for example, the actuary may recommend a contribution holiday (a period during which no contributions are made) to eliminate a surplus.

Risk associated with defined benefit plans

As the employer is obliged to make up any shortfall in the plan, it is effectively underwriting the **investment** and **actuarial risk** associated with the plan. Thus in a defined benefit plan, the employer carries both the investment and the actuarial risk.



Definitions

Investment risk: This is the risk that, due to poor investment performance, there will be insufficient funds in the plan to meet the expected benefits.

Actuarial risk: This is the risk that the actuarial assumptions such as those on employee turnover, life expectancy or future salaries vary significantly from what actually happens.

4.6 Accounting for defined benefit plans

As we noted in the previous section, contributions to defined benefit plans will vary depending on whether the actuary assesses the value of the plan to be adequate to meet future obligations.

In some instances there will be a shortfall, in which case the actuary will advise increased contributions. In other instances there may be a surplus, in which case the actuary may recommend a contributions holiday. Contributions will therefore vary substantially from year to year.

For this reason, it is inappropriate to apply the accounting treatment for defined contribution plans and expense contributions through profit or loss.

IAS 19 instead requires that the defined benefit plan is recognised in the sponsoring entity's statement of financial position as either a liability or asset depending on whether the plan is in deficit or surplus.

The value of the pension plan is calculated in its simplest form as:

	£
Present value of the defined benefit obligation at the reporting date	X
Fair value of plan assets at the reporting date	(X)
Plan deficit/surplus	X/(X)

4.6.1 Present value of the defined benefit obligation



Definition

Defined benefit obligation: The defined benefit obligation is the present value of all expected future payments required to settle the obligation resulting from employee service in the current and prior periods.

Expected future payments

Expected future payments are based on a number of assumptions and estimates, such as:

- the final benefits payable under the plan (often dependent on future salaries, as benefits are often quoted as a percentage of the employee's final salary); and
- the number of members who will draw benefits (this will in turn depend on employee turnover and mortality rates).

Discounting to present value

Once determined, the expected future benefits should be discounted to present value (including those which may become payable within 12 months) using a discount rate determined by reference to:

- market yields on high-quality fixed-rate corporate bonds at the reporting date, or where there is no market in such bonds:
- market yields on government bonds

The corporate or government bonds should be denominated in the same currency as the defined benefit obligation, and be for a similar term.

Note: The examples of discount rates used later in this section are merely to illustrate relevant calculations and may therefore be rather higher than would currently be found in practice.

4.6.2 Fair value of plan assets



Definition

Plan assets: Plan assets are defined as those assets held by a long-term benefit fund and those insurance policies which are held by an entity, where the fund/entity is legally separate from the employer and assets/policies can only be used to fund employee benefits.

Investments owned by the employer which have been earmarked for employee benefits but which the employer could use for different purposes are not plan assets.

Definition

Fair value: Fair value is the price that would be received to sell an asset in an orderly transaction between market participants at the measurement date. (IFRS 13)

Guidance on fair value is given in IFRS 13, *Fair Value Measurement*. Under IFRS 13, fair value is a market-based measurement, not an entity-specific measurement. It focuses on assets and liabilities and on exit (selling) prices. It also takes into account market conditions at the measurement date.

4.6.3 Actuarial assumptions

Actuarial assumptions are needed **to estimate the size of the future (post-employment) benefits** that will be payable under a defined benefit plan. The main categories of actuarial assumptions are as follows.

(a) **Demographic assumptions** are about mortality rates before and after retirement, the rate of employee turnover, early retirement, claim rates under medical plans for former employees, and so on.

(b) **Financial assumptions** include future salary levels (allowing for seniority and promotion as well as inflation) and the future rate of increase in medical costs (not just inflationary cost rises, but also cost rises specific to medical treatments and to medical treatments required given the expectations of longer average life expectancy).

The standard requires actuarial assumptions to be neither too cautious nor too imprudent: they should be **'unbiased'**. They should also be based on **'market expectations'** at the year end, over the period during which the obligations will be settled.

4.6.4 Accounting for the movement in defined benefit plans

Both the present value of the defined benefit obligation and the fair value of plan assets, and therefore the overall plan surplus or deficit, will change from year to year. This movement is broken down into its constituent parts and each is accounted for separately.

The opening and closing obligation and plan assets can be reconciled as follows:

	PV of defined benefit obligation	FV of plan assets
	£	£
B/f at start of year (advised by actuary)	(X)	X
Retirement benefits paid out	X	(X)
Contributions paid into plan		X
Interest on plan assets		X
Interest cost on obligation	(X)	
Current service cost	(X)	
	(X)	X
Gains/losses on remeasurement (balancing figure)	X/(X)	X/(X)
C/f at end of year (advised by actuary)	(X)	X

Note that while the interest on plan assets and interest on obligation are calculated separately, they are presented net and the same rate is used for both.

4.6.5 Outline of the method

There is a **four-step method** for recognising and measuring the expenses and liability of a defined benefit pension plan.

An outline of the method used by an employer to account for the expenses and obligation of a defined benefit plan is given below.

Step 1 Measure the deficit or surplus:

(a) An **actuarial technique** (the **projected unit credit method**) should be used to make a reliable estimate of the amount of future benefits employees have earned from service in relation to the current and prior years. The entity must determine how much benefit should be attributed to service performed by employees in the current period, and in prior periods. Assumptions include, for example, levels of employee turnover, mortality rates and future increases in salaries (if these will affect the eventual size of future benefits such as pension payments).

(b) The benefit should be **discounted** to arrive at the present value of the defined benefit obligation and the current service cost.

(c) The **fair value** of any **plan assets** should be deducted from the present value of the defined benefit obligation.

Step 2 The surplus or deficit measured in Step 1 may have to be adjusted if a net benefit asset has to be restricted by the asset ceiling. (The asset ceiling is a threshold established by IAS 19 to ensure that any defined benefit asset (ie, a pension surplus) carried at no more than its recoverable amount. In simple terms, this means that any net asset is restricted to the amount of cash savings that will be available to the entity in future.)

Step 3 Determine the amounts to be recognised in profit or loss:

(a) Current service cost

(b) Any past service cost and gain or loss on settlement

(c) Net interest on the net defined benefit liability (asset)

Step 4 Determine the remeasurements of the net defined benefit liability (asset), to be recognised in other comprehensive income (items that will not be reclassified to profit or loss).

(a) Actuarial gains and losses

(b) Return on plan assets (excluding amounts included in net interest on the net defined benefit liability (asset))

(c) Any change in the effect of the asset ceiling (excluding amounts included in net interest on the net defined benefit liability (asset))

4.6.6 Retirement benefits paid out

During an accounting year, some of the plan assets will be paid out to retirees, thus discharging part of the benefit obligation. This is accounted for by:

DEBIT	PV of defined benefit obligation	X	
CREDIT	FV of plan assets		X

Note that there is no cash entry, as the pension plan itself rather than the sponsoring employer pays the money out.

4.6.7 Contributions paid into plan

Contributions will be made into the plan as advised by the actuary. This is accounted for by:

DEBIT	FV of plan assets	X	
CREDIT	Cash		X

4.6.8 Return on plan assets



Definition

Return on plan assets: Is defined as interest, dividends and other revenue derived from plan assets together with realised and unrealised gains or losses on the plan assets, less any costs of administering the plan and less any tax payable by the plan itself.

Accounting for the return on plan assets is explained in more detail below.

4.6.9 The statement of financial position

In the statement of financial position, the amount recognised as a **defined benefit liability** (which may be a negative amount, ie, an asset) should be the following.

(a) the **present value of the defined obligation** at the year end; **minus**

(b) the **fair value of the assets of the plan** as at the year end (if there are any) out of which the future obligations to current and past employees will be directly settled.

The earlier parts of this section have looked at the recognition and measurement of the defined benefit obligation. Now we will look at issues relating to the assets held in the plan.

4.6.10 Plan assets

Plan assets are:

(a) Assets such as stocks and shares, held by a fund that is legally separate from the reporting entity, which exists solely to pay employee benefits

(b) Insurance policies, issued by an insurer that is not a related party, the proceeds of which can only be used to pay employee benefits

Investments which may be used for purposes other than to pay employee benefits are not plan assets.

The standard requires that the plan assets are measured at fair value, as "the price that would be received to sell an asset in an orderly transaction between market participants at the measurement date". This is consistent with IFRS 13, *Fair Value Measurement*.

IAS 19 includes the following **specific requirements**:

(a) The plan assets should exclude any contributions due from the employer but not yet paid.

(b) Plan assets are reduced by any liabilities of the fund that do not relate to employee benefits, such as trade and other payables.

4.6.11 The statement of profit or loss and other comprehensive income

All the gains and losses that affect the plan obligation and plan assets must be recognised. The **components of defined benefit cost must be recognised as follows** in the statement of profit or loss and other comprehensive income:

Component	Recognised in
(a) **Service cost**	Profit or loss

Component	Recognised in
(b) **Net interest on the net defined benefit liability**	Profit or loss
(c) **Remeasurements of the net defined benefit liability**	Other comprehensive income (not reclassified to profit or loss)

4.6.12 Service costs

These comprise:

(a) **Current service cost**; this is the increase in the present value of the defined benefit obligation resulting from employee services during the period.

(b) **Past service cost**; this is the change in the obligation relating to service in **prior periods**. This results from amendments or curtailments to the pension plan.

(c) Any **gain or loss on settlement**.

Changes in the terms of membership of a defined benefit plan may give rise to a plan amendment, curtailment or settlement. If this occurs, the current service cost and the net interest for the period after the remeasurement must be determined using the assumptions used for the remeasurement, rather than the assumptions determined at the beginning of the period.

4.6.13 Net interest on the net defined benefit liability (asset)

In section 4.6.5 we looked at the recognition and measurement of the defined benefit obligation. This figure is the discounted **present value** of the future benefits payable. Every year the discount must be 'unwound', increasing the present value of the obligation as time passes through an interest charge.

4.6.14 Interest calculation

IAS 19 requires that the interest should be calculated on the **net defined benefit liability (asset)**. This means that the amount recognised in profit or loss is the net of the interest charge on the obligation and the interest income recognised on the assets.

The calculation is as follows:

Net defined benefit balance × Discount rate

The **net defined benefit liability/(asset)** should be measured as at the **start** of the accounting period, taking account of changes during the period as a result of contributions paid into the scheme and benefits paid out.

4.6.15 Discount rate

The **discount rate** adopted should be determined by reference to **market yields on high-quality fixed-rate corporate bonds**. In the absence of a 'deep' market in such bonds, the yields on comparable government bonds should be used as reference instead. The maturity of the corporate bonds that are used to determine a discount rate should have a term to maturity that is consistent with the expected maturity of the post-employment benefit obligations, although a single weighted average discount rate is sufficient.



Worked example: Interest cost

In 20X8, an employee leaves a company after working there for 24 years. The employee chooses to leave his accrued benefits in the pension plan until he retires in seven years' time (he now works for another company).

At the time of his departure, the actuary calculates that it is necessary at that date to have a fund of £296,000 to pay the expected pensions to the ex-employee when he retires.

At the start of the year, the yield on high quality corporate debt was 8%, and remained the same throughout the year and the following year.

Requirement

Calculate the interest cost to be debited to profit or loss in Years 1 and 2.

Solution

		£
Year 1:	Discounted cost b/f	296,000
	Interest cost (profit or loss) (8% × £296,000)	23,680
	Obligation c/f (statement of financial position)	319,680
Year 2:	Interest cost (profit or loss) (8% × £319,680)	25,574
	Obligation c/f (statement of financial position)	345,254

4.6.16 Remeasurements of the net defined benefit liability

Remeasurements of the net defined benefit liability/(asset) comprise:

(a) actuarial gains and losses;

(b) the return on plan assets (excluding amounts included in net interest on the net defined benefit liability/(asset)); and

(c) any change in the effect of the asset ceiling (excluding amounts included in net interest on the net defined benefit liability/(asset)).

The gains and losses relating to points (a) and (b) above will arise in every defined benefit plan. The asset ceiling is a complication that is not relevant in every case.

4.6.17 Actuarial gains and losses

Actuarial gains and losses arise for several reasons, and IAS 19 requires these to be recognised in full in other comprehensive income.

At the end of each accounting period, a new valuation, using updated assumptions, should be carried out on the obligation. Actuarial gains or losses arise because of the following.

- **actual events** (eg, employee turnover, salary increases) differ from the actuarial assumptions that were made to estimate the defined benefit obligations
- The effect of **changes to assumptions** concerning benefit payment options
- **estimates are revised** (eg, different assumptions are made about future employee turnover, salary rises, mortality rates, and so on)
- The effect of changes to the **discount rate**

Actuarial gains and losses are recognised in **other comprehensive income**. They are **not reclassified to profit or loss**.

4.6.18 Return on plan assets

The return on plan assets must be calculated.

A new valuation of the plan assets is carried out at each period end, using current fair values. Any difference between the new value, and what has been recognised up to that date (normally the opening balance, interest, and any cash payments into or out of the plan) is treated as a 'remeasurement' and recognised in other comprehensive income.

Note: In the examples in this chapter, it is assumed that cash from contributions is received and pensions paid out at the end of the year, as no interest arises on it. In practice, it is more likely that contributions would be paid in part way through the year, and pensions paid out part way through the year or evenly over the year.



Worked example: Remeasurement of the net defined benefit liability

At 1 January 20X2 the fair value of the assets of a defined benefit plan was £1,100,000 and the present value of the defined benefit obligation was £1,250,000. On 31 December 20X2, the plan received contributions from the employer of £490,000 and paid out benefits of £190,000.

The current service cost for the year was £360,000 and a discount rate of 6% is to be applied to the net liability/(asset). After these transactions, the fair value of the plan's assets at 31 December 20X2 was £1.5 million. The present value of the defined benefit obligation was £1,553,600.

Requirement

Calculate the gains or losses on remeasurement through other comprehensive income (OCI) and the return on plan assets and illustrate how this pension plan will be treated in the statement of profit or loss and other comprehensive income and statement of financial position for the year ended 31 December 20X2.

Solution

	Assets	Obligation
	£	£
Fair value/present value at 1.1.X2	1,100,000	1,250,000
Interest (1,100,000 × 6%)/(1,250,000 × 6%)	66,000	75,000
Current service cost		360,000
Contributions received	490,000	
Benefits paid	(190,000)	(190,000)
Return on plan assets excluding amounts in net interest (balancing figure) (OCI)	34,000	–
Loss on remeasurement (balancing figure) (OCI)	–	58,600
	1,500,000	1,553,600

The following accounting treatment is required.

In the **statement of profit or loss and other comprehensive income**, the following amounts will be recognised.

	£
In **profit or loss**:	
Current service cost	360,000
Net interest on net defined benefit liability (75,000 – 66,000)	9,000
In **other comprehensive income** (34,000 – 58,600)	24,600

In the **statement of financial position**, the net defined benefit liability of £53,600 (1,553,600 – 1,500,000) will be recognised.

Interactive question 5: Defined benefit pension plan

Penn Co operates a defined benefit pension plan for its employees conditional on a minimum employment period of six years. The present value of the future benefit obligations and the fair value of its plan assets on 1 January 20X1 were £110 million and £150 million respectively.

The pension plan received contributions of £7 million and paid pensions to former employees of £10 million during the year.

Extracts from the most recent actuary's report show the following:

Present value of pension plan obligation at 31 December 20X1	£116m
Fair value of plan assets at 31 December 20X1	£140m
Present cost of pensions earned in the period	£11m
Yield on high quality corporate bonds at 1 January 20X1	10%

On 1 January 20X1, the rules of the pension plan were changed to improve benefits for plan members. The actuary has advised that this will cost £10 million.

Requirement

Produce the extracts for the financial statements for the year ended 31 December 20X1. Assume contributions and benefits were paid on 31 December.

See **Answer** at the end of this chapter.

4.7 IFRS 2, *Share-based Payment*

We have already mentioned IFRS 2, *Share-based Payment* briefly in the chapter "Strategic performance management" in the context of performance management, recognising that share options could be used to encourage directors to focus on the longer-term performance of their companies and not just on short-term results.

However, if a company considers offering share options to its directors, it is important to weigh the potential impact these could have on its financial position.

IFRS 2 requirements

Prior to the publication of IFRS 2 there appeared to be an anomaly to the extent that if a company paid its employees in cash, an expense was recognised in profit or loss, but if the payment was in share options no expense was recognised.

IFRS 2 resolved this anomaly by requiring an expense to be recognised in profit or loss in relation to share-based payments.

However, the introduction of the IFRS and the requirement to recognise share-based payments as an expense caused huge controversy, with opposition especially strong among hi-tech companies. The arguments over expensing share-based payments polarised opinion, especially in the US.

(a) The main argument **against** recording an expense was that no cash changes hands as part of such transactions, and therefore there is no true expense.

(b) The main argument **for** recording an expense was that share-based payments are simply another form of compensation that should go into the calculation of earnings for the sake of transparency for investors and the business community.

Practical application of IFRS 2

In practice, the implementation of IFRS 2 has resulted in earnings being reduced, sometimes significantly. It is generally agreed that as a result of the IFRS companies now focus more on the earnings effect of different rewards policies.

Following the adoption of IFRS 2, some companies have admitted that they are re-evaluating the use of share options as part of employee remuneration.

Impact on earnings and financial position

In the financial statements, entities should disclose information that enables users of the financial statements to understand the effect of share-based payment transactions on the entity's **profit or loss for the period** and on its **financial position**.

(a) The **total expense recognised for the period** arising from share-based payment transactions, including separate disclosure of that portion of the total expense that arises from transactions accounted for as equity-settled share-based payment transactions

(b) For **liabilities** arising from share-based payment transactions:

  (1) The **total carrying amount** at the end of the period

  (2) the **total intrinsic value** at the end of the period of liabilities for which the counterparty's right to cash or other assets had vested by the end of the period

(c) Although not mentioned specifically by IFRS 2 in the context of disclosures, share-based payment transactions will have an impact on equity, being the other half of the double entry:

DEBIT	Expense
CREDIT	Equity (if equity settled)

Impact of share-based payments on earnings per share (EPS)

IAS 33, *Earnings per Share* requires that for calculating diluted earnings per share (EPS) all dilutive options need to be taken into account. Employee share options with fixed terms and non-vested ordinary shares are treated as options outstanding on grant date even though they may not have vested on the date the diluted EPS is calculated. All awards which do not specify performance criteria are treated as options.

4.7.1 Share-based transactions with employees (share options)

Transactions with employees are normally:

- Measured at the **fair value** of equity instruments granted **at grant date**
- Spread over the vesting period (often a specified period of employment)

In accordance with IFRS 2, Share-based Payment (paragraphs 16 and 17), where a transaction is measured by reference to the fair value of the equity instruments granted, fair value is based on market prices where available. If market prices are not available, the entity should estimate the fair value of the equity instruments granted using a suitable valuation technique (such as the Black- Scholes model, the Binomial model or Monte Carlo simulation).



Worked example: Employee transactions – equity-settled

A company provides each of 10 key employees with 1,000 share options on 1 January 20X7. Each option has a fair value of £9 at the grant date, £11 on 1 January 20X8, £14 on 1 January 20X9 and £12 on 31 December 20X9.

The options do not vest until 31 December 20X9 and are dependent on continued employment. All 10 employees are expected to remain with the company.

Requirement

What are the accounting entries to be recorded in each of the years 20X7, 20X8 and 20X9?

Solution

The changes in the value of equity instruments after grant date do not affect the charge to profit or loss for equity-settled transactions.

Based on the fair value at grant date, the remuneration expense is calculated as follows.

Number of employees × number of equity instruments × fair value of equity instruments at grant date = 10 × 1,000 × £9 = £90,000

The remuneration expense should be recognised over the vesting period of three years. An amount of £30,000 should be recognised for each of the three years 20X7, 20X8 and 20X9 in profit or loss with a corresponding credit to equity.

It is important to remember that the expense recognised in each year of the vesting period should be based on the best available estimate of the number of equity instruments expected to vest.

Therefore if any employees are expected to leave within the vesting period, this should be taken into account when calculating the expense.



Worked example: Employees leave

Dexter Co issues 10,000 options to each of the 50 directors and senior managers on 1 July 20X1.

The exercise price of the options is £4.50 per share. The scheme participants have to stay with the company for four more years before being able to exercise their options.

At 31 December 20X1, it is estimated that 75% of the current directors and senior managers will remain with the company for four years or more. The estimated figure is 70% at 31 December 20X2.

The fair value of an option is £3 at the grant date.

Requirement

Identify the journal entries required to record the share-based payment transaction in each of the years ended 31 December 20X1 and 20X2.

Solution

The remuneration expense in respect of the options for the year ended 31 December 20X1 is calculated as follows:

Fair value of options expected to vest at grant date:

(75% × 50 employees) × 10,000 options × £3 = £1,125,000

Annual charge to profit or loss therefore £1,125,000/4 years = £281,250

Charge to profit or loss for y/e 31 December 20X1 = £281,250 × 6/12 months = £140,625 The accounting entry for the year ending 31 December 20X1 is:

		£	£
DEBIT	Remuneration expense	140,625	
CREDIT	Equity		140,625

In 20X2 the remuneration charge is for the whole year, and is calculated as: (70% × 50 employees) ×10,000 options × £3 = £1,050,000
Charge to date is £1,050,000 × 1.5/4 years = £393,750 Therefore charge for the year is £393,750 – £140,625 = £253,125 The accounting entry is:

DEBIT	Remuneration expense	253,125	
CREDIT	Equity		253,125

4.7.2 Modifications and repricing

Equity instruments may be modified before they vest. For example, a downturn in the equity market may mean that the original option exercise price set is no longer attractive. Therefore the exercise price is reduced (the option is 'repriced') to make it valuable again.

Such modifications will often affect the fair value of the instrument and therefore the amount recognised in profit or loss.

The accounting treatment of modifications and repricing is:

(a) Continue to recognise the **original** fair value of the instrument in the normal way (even where the modification has reduced the fair value).

(b) Recognise any **increase** in fair value at the modification date (or any increase in the number of instruments granted as a result of modification) spread over the period between the modification date and vesting date.

(c) If modification occurs after the vesting date, then the additional fair value must be recognised immediately, unless there is, for example, an additional service period, in which case the difference is spread over the additional period.

4.7.3 Cancellations and settlements

An entity may settle or cancel an equity instrument during the vesting period. Where this is the case, the correct accounting treatment is:

(a) To immediately charge any remaining fair value of the instrument which has not been recognised to profit or loss (the cancellation or settlement accelerates the charge and does not avoid it).

(b) Any amount paid to the employees by the entity on settlement should be treated as a buy-back of shares and should be recognised as a deduction from equity. If the amount of any such payment is in excess of the fair value of the equity instrument granted, the excess should be recognised immediately in profit or loss.

4.7.4 Market-based and non market based vesting conditions

When share-based payments are offered to employees, there are likely to be some conditions (vesting conditions) which have to be satisfied before the employee is entitled to receive the share- based payment.

IFRS 2 distinguishes between two different types of vesting conditions:

Market-based vesting conditions

Market-based performance or vesting conditions are conditions linked to the market price of the shares in some way. Examples include vesting dependent on achieving:

- a minimum increase in the share price of the entity
- a minimum increase in shareholder return
- a specified target share price relative to an index of market prices

Non market based vesting conditions

These are conditions other than those relating to the market value of the entity's shares. Examples include vesting dependent on:

- the employee completing a minimum period of service (eg, remaining with the company for a further three years) – also referred to as a service condition
- achievement of minimum sales or earnings target
- achievement of a specific increase in profit or earnings per share
- successful completion of a flotation
- completion of a particular project

The distinction between the two different types of vesting conditions has important implications for the accounting treatment of the shares:

Market-based vesting conditions

- These conditions are taken into account when calculating the fair value of the equity instruments at the grant date.
- They are not taken into account when estimating the number of shares or share options likely to vest at each period end.
- If the shares or share options do not vest, any amount recognised in the financial statements will remain.

Non market based vesting conditions

- These conditions are taken into account when determining the expense which must be taken to profit or loss in each year of the vesting period. (They are not taken into account when calculating the fair value of the equity instruments at the grant date.)
- Only the number of shares or share options expected to vest will be accounted for.
- At each period end (including interim periods), the number expected to vest should be revised as necessary. The movement in cumulative expense is charged to profit or loss.
- On the vesting date, the entity should revise the estimate to equal the number of shares or share options that do actually vest.

Vested options not exercised

If after the vesting date options are not exercised or the equity instrument is forfeited, there will be no impact on the financial statements. This is because the holder of the equity instrument has effectively made that decision as an investor.

The services for which the equity instrument remunerated were received by the entity and the financial statements reflect the substance of this transaction. IFRS 2 does, however, permit a transfer to be made between reserves in such circumstances to avoid an amount remaining in a separate equity reserve where no equity instrument will be issued.

4.8 Executive remuneration and remuneration reports

In chapters "Strategic implementation" and "Strategic performance management" of this Workbook we mentioned the principal-agent problem, and the importance of aligning directors' (agents') interests with those of their company's shareholders (principals).

Granting share options to directors is one way of aligning directors' interests with those of the company's shareholders. More generally, however, there has been increasing pressure on companies to be more transparent about the way directors' pay is set, and to improve accountability to shareholders

In the context of Strategic Business Management, the linkage between remuneration and performance is also particularly important.

As the Foreword to the Department of Business, Innovation & Skills (BIS) discussion paper *Executive Remuneration* (2011) noted:

> "Executive remuneration that is well-structured, clearly linked to the strategic objectives of a company, and which rewards executive directors who contribute to the long-term success of that company, is important in promoting business stability and growth. Shareholders want to see remuneration being used effectively to attract, incentivise and appropriately reward executives, so that the value of the companies they invest in increases over time."

Generous rewards can be justified when a company has delivered strong long-term performance and growth. However, as the BIS discussion paper highlights, the financial crisis in 2008–09 made shareholders, wider stakeholders (such as employees and customers) and the public at large more aware of the apparent disconnect between pay and performance. For example, one area of concern is that the remuneration of the highest paid executives in large companies seems to rise virtually every year, regardless of the performance of the company. The link between strategy, pay and performance should therefore be an important factor for shareholders to consider when assessing remuneration proposals.

The BIS discussion paper proposed that companies should provide a clearer statement on how executive remuneration relates to a company's achievement of its strategic objectives over the previous year.

However, as well as justifying its current payments, it is equally important for a company to describe its pay policy for the year ahead, so that shareholders can gauge how effectively future pay is linked to company strategy and performance.

In the UK, the Directors' Remuneration Report Regulations (2002) already required quoted companies to produce a detailed annual directors' remuneration report, and to hold a shareholder vote on that report. However, these Regulations were substantially amended by the **Directors' Remuneration Report (DRR) Regulations (2013)** which had far-reaching implications for the information which companies have to disclose in directors' remuneration reports.

The aim of the 2013 DRR Regulations was to enable investors to be better informed about how much directors have been – and will be – paid, and how this relates to company performance.

The DRR Regulations also sought to increase transparency over directors' remuneration by requiring a company to separate its reports about Directors' Remuneration into two parts:

(a) **Policy Report** – A forward-looking report setting out the company's future policy on directors' remuneration. The policy report should cover the company's approach to setting salary, pensions, bonus and incentive awards, as well as payments for loss of office and recruitment packages for new directors.

    The policy report must disclose a description of each element of pay, and how it links to the company's strategy. The report must also provide performance scenarios showing fixed pay, total remuneration when performing in line with expectations, and maximum total remuneration.

(b) **Implementation Report** – A retrospective report, setting out the actual payments made to directors in the financial year being reported on. The DRR Regulations also require a company to include, within the implementation report, a **single total figure** of remuneration for each director, along with tables showing that total figure broken down into its component parts (base salary; taxable benefits; bonuses; the value of share and share options awards; and any pension-related benefits).

    The Regulations also increase shareholder power by allowing shareholders a **binding vote** on the Policy Report at least once every three years.

    A company is then prohibited from making any remuneration payment which is not covered by an approved remuneration policy, or which is inconsistent with that policy.

The implementation report is subject to an advisory vote each year. Although this vote isn't binding, it provides an opportunity for shareholders to indicate whether they are satisfied with how the policy has been implemented.

4.8.1 Pay gaps

The increased focus on the relationship between executive pay and company performance also reflects increasing concerns about the disparity ('pay gap') between the rewards paid to senior executives and the remuneration received by the workforce at large (which we discussed in chapter "Strategic performance management").

Shareholders are becoming increasingly sensitive to the idea that it is unfair for executives to get significantly higher pay increases than everyone else. Similarly, shareholders are also starting to ask whether the package of rewards received by each director is fair and justified in the context of their contribution to the business. Both these issues should also serve as a warning that shareholders could vote against executive pay deals if they believe them to be unfair or unjustified.

However, the increased focus on performance and rewards could also pose some questions for the performance measures which are used to determine rewards, and how weightings are assigned to different measures. For example, should directors be rewarded for meeting environmental targets in a year when profits have decreased?

Another area where pay gaps have been a source of concern is in pay gaps between male and female employees. Following publication of the salaries of its top earners in the summer of 2017 the BBC was criticised for the substantial pay gap between male and female staff.

In the light of this, the BBC engaged PwC, in conjunction with the law firm, Eversheds, to conduct an equal pay audit of graded staff levels within the Corporation. This review found there was 'no evidence' of gender bias in pay decisions at the BBC, although this conclusion has been disputed by some employees within the Corporation.

5 HRM and change management



Section overview

- We discussed the importance of change management in strategy implementation in the chapter "Strategic implementation" of this Workbook. However, the link between change management and HRM is important because 'change' will also inevitably affect the people in an organisation.
- Therefore, in order for the change to be implemented successfully, the way the change is communicated to the people in an organisation and the way those people respond and adapt to the change will also have to be managed effectively.

5.1 HRM and change agents

One of the four roles for HRM highlighted in Dave Ulrich's model (see Figure 10.2) is that of the change agent.

HR departments in nearly every business have a key role in managing change effectively. Although some changes occur as clearly defined episodes in response to external environmental factors, change can also be a continuous process within organisations (as

implied by the notion of the learning organisation). Learning new knowledge could itself be a catalyst for change.

Moreover, change comes in different forms and can occur at different levels within an organisation:

- Individuals
- Structures and systems
- Organisational climate

Changing individuals involves changing their skills, values, attitudes and behaviours. Any such individual changes have to support the overall organisational changes required. However, ultimately organisational changes can only be achieved if the individual people working for an organisation change as necessary.

Changing structures and systems involves changing the formal and informal organisational structures in place: for example, changing business processes, or changing roles, responsibilities and relationships.

Changing the organisational climate involves changing the way people relate to each other in an organisation; the management style; and the overall culture of the organisation. For example, this might involve creating a culture of high interpersonal trust and openness between staff.

The presence of these three levels means a change manager needs to ensure that appropriate methods exist in order that the desired change is achieved at each level.

5.1.1 HRM and organisational change

Despite the range of possible change scenarios which might arise in an organisation, HR functions can still play a central role in the change process. Key activities might include:

- the recruitment and/or development of people with the necessary leadership skills to drive change, and staff with the necessary technical and operational skills to deliver the change
- Advising project leaders about reward and job design
- Communicating the benefits and effects of change to staff, and encouraging staff involvement in the change process
- Identifying the appropriate medium of communication to reach different stakeholder groups
- Understanding staff (or other stakeholders') concerns about changes, and helping to deal with them
- Negotiating and dealing with conflict; engaging with various stakeholders; understanding stakeholder concerns in order to anticipate problems with change programmes
- Assessing the impact which changes in one business area/department/location could have on other parts of an organisation
- providing a structured framework for change, and helping people cope with change
- Constructing reward systems which underpin the change process and motivate staff to support the changes

The Chartered Institute of Personnel and Development has identified seven areas of activity ("the seven Cs of change") which increase the probability of change programmes being successful. HR practitioners have a key role to play in each of the seven:

- choosing a team
- crafting the vision and path
- connecting organisation-wide change
- consulting stakeholders
- communicating

- coping with change
- capturing learning

Alongside these specific areas of activity we could also suggest, more generally, that HR managers can play a key role as **change agents** in leading change.

The Workbook for Business Strategy and Technology identifies that the role of the change agent could include:

- defining the problem
- examining what causes the problem and considering how this can be overcome
- suggesting possible solutions
- selecting an appropriate solution
- implementing the change
- communicating information about the change throughout the organisation
- gaining support from all involved to deliver the solution

In order to be effective, a change agent should have the following skills and attributes:

a) Communication skills – and the ability to communicate effectively with people at all levels within an organisation

b) Networking skills to establish and maintain contacts, both within and outside an organisation

c) Negotiation and 'selling' skills – negotiating with stakeholders in the business to obtain resources for a project, or to resolve conflict; selling the vision of change to key stakeholders to increase support for a change programme. A change agent also needs to have influencing skills, to be able to convince potential sceptics about the benefits of a change programme, and thereby to overcome their resistance to it.

d) An awareness of organisational 'politics'

e) Sensitivity to the impact changes will have on different stakeholders, and sensitivity in dealing with different stakeholders

f) An understanding of the relevant processes

g) Financial analysis skills: to assess the financial impacts of proposed changes, or to be able to look at how changes to operations and systems can deliver a desired financial goal

h) Flexibility to be able to respond to shifts in project goals or objectives, or to adapt in response to internal or external factors which affect the change process

i) An important point that Kanter highlights is that a change agent needs to be able to adapt to cope with the complexities of modern organisations

In particular, a change agent needs to:

(a) Be able to work across a range of business units and functions, and across a network of different stakeholders

(b) Be an effective collaborator, able to work in ways that enhance collaboration across different functions and divisions

These skills should resonate with the HRM competences of an organisation.

Importantly, though, a change agent should not be selected just because they have good general project management skills. The change agent must be **directly involved in the change process** and must see clear linkages between their future success in an organisation and the effective implementation of the change.

5.2 Models of change

Lewin's stage model of change

We have already looked at change management in the chapter "Strategic implementation" of this Workbook, but it is worth noting how the elements of 'unfreeze', 'change' and 'refreeze' contain elements of HRM. In particular, a key element of the 'refreeze' stage is the use of positive reinforcement to reward and validate successful change, which could be linked to bonus and reward schemes where the bonuses are dependent on staff members adopting new approaches or methodologies.

Kotter's eight step model of change

Another widely cited model of change is Kotter's eight 'lessons' which organisations need to address when implementing change. (We have already looked at this model in the chapter "Strategic implementation" in relation to implementing digital strategy.) Note, however, that the 'lessons' in Kotter's model highlight the importance of 'people' in successful change management, as well as highlighting the importance of having a 'felt need' for change in an organisation, and the importance of communication throughout the change process.

(a) **Establish a sense of urgency** – Discuss the current competitive position and look at potential future scenarios. Increase the 'felt need for change' (in other words, promote the driving forces for change).

(b) **Form a powerful guiding coalition** – Assemble a powerful group of people who can work well together to promote the change.

(c) **Create a vision** – Build a vision to guide the change effort, together with strategies for achieving it.

(d) **Communicate the vision** – The vision, and accompanying strategies and new behaviours, need to be communicated. Kotter stresses that effective communication is crucial in change management.

(e) **Empower others to act on the vision** – This includes getting rid of obstacles to change such as unhelpful structures and systems. People need to be allowed to experiment.

(f) **Plan for and create short-term wins** – Look for and advertise short-term visible improvements because these will help sustain the driving forces for change. Kotter suggests that these short- term wins should be planned into the change programme, and people should be publicly rewarded for making improvements.

(g) **Consolidate improvements and produce still more change** – Promote and reward those who are able to promote and work towards the vision. Maintain the energy behind the change process by introducing new projects, resources and change agents.

(h) **Institutionalise new approaches** – Ensure that everyone understands that the new behaviours and systems will lead to corporate success.

Kotter's model can also be used to analyse the reasons why change initiatives have been unsuccessful – in effect, where the eight 'lessons' have not been followed successfully:

Reason for failure	Possible antidotes
Not enough sense of urgency	**Establish a sense of urgency by:** • examining market or competitive pressures • identifying potential crises or major opportunities • ensuring levels of dissatisfaction with current position or perception of future threat are sufficient to kick-start the change and maintain momentum

Reason for failure	Possible antidotes
Failure to create a powerful support base	**Form a powerful guiding coalition by:** • assembling a group with enough power to **lead the change effort** (without an effective change management team, any change management project is likely to fail)
	• encouraging the group to work together as a team • ensuring that key stakeholders are engaged
Vision not clearly developed	**Create a vision by:** • having a clear understanding of what the change needs to achieve • developing clear strategies for achieving the vision
Vision poorly communicated	**Communicate the vision by:** • using a variety of media to communicate the new vision and strategies. Within this, it will also be important to **highlight the benefits** of the changes • teaching new behaviours by the example of the guiding coalition of senior management • ensuring people have a shared understanding and commitment to the direction of the change
Obstacles block the vision	**Empower people to act on the vision by:** • senior management demonstrably tackling obstacles to change • ensuring that all the people who are needed to make the change happen have the necessary resources and authority to achieve their goals
Failing to create short- term wins	**Plan for and create short-term wins by:** • planning for visible performance improvements • identifying smaller goals along the way to the ultimate target so that success can be demonstrated; being able to demonstrate success will maintain momentum • recognising and rewarding employees involved in improvements
Systems, policies and skills not aligned	**Consolidate improvements and produce more change by:** • changing systems, structures and policies that don't fit the vision • hiring, promoting and developing employees who can implement the vision • building on improvements in the organisation as and when they occur to continue to move the change forward
Failing to anchor changes in the corporate culture (not refreezing)	**Institutionalise new behaviours by:** • explaining how the new behaviours will deliver corporate success • developing the means to ensure leadership development and succession • ensuring knowledge about the new approaches is captured and shared

(Adapted from: Cameron & Green, 'Making sense of change management' and based originally on an article by Kotter in the Harvard Business Review)

One additional factor which could jeopardise the success of a change management project is a **lack of change management/implementation expertise** and skills within an organisation's senior management team. Change does not just happen on its own; management need to define the change programme, ensure the necessary resources are allocated to it, and drive it forward.

However, for example, if the senior management team do not have any previous experience of change programmes and do not allocate sufficient resources to a change programme, this could jeopardise its success.

5.3 Maintaining internal consistency

When planning or implementing change in an organisation, it is important that the proposed changes 'fit' with the existing context of the organisation.

Successful change management requires more than simply recognising a change trigger and acting on it. Instead, successful exploitation of a change situation requires:

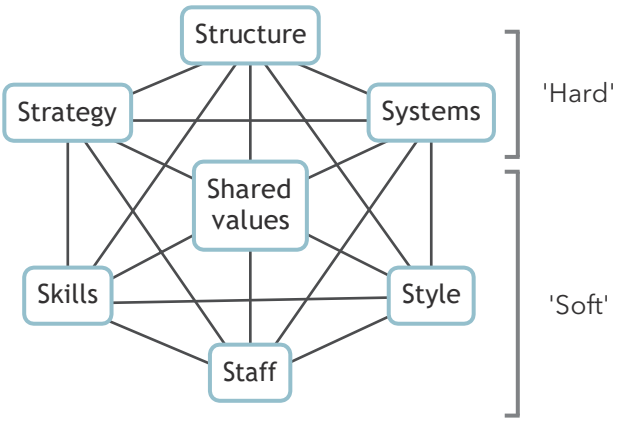
- knowledge of the circumstances surrounding a situation
- understanding of the interactions in that situation
- awareness of the potential impact of the variables associated with the situation

McKinsey 7-S model

The McKinsey 7-S model provides a framework for looking at an organisation as a set of interconnected and interdependent subsystems. This interdependence highlights that strategies adopted in any one area of an organisation (or changes to the strategies pursued in any area of the organisation) will have an impact on other parts of the organisation.

Therefore, when considering changes in an organisation, it could be useful to think about how the proposed changes fit with the 7 S's:

Figure 10.6: McKinsey 7-S model



There are three '**hard'** elements of business behaviour:

(a) **Structure**. The organisation structure refers to the formal division of tasks in the organisation and the hierarchy of authority from the most senior to junior.

(b) **Strategy**. How the organisation plans to outperform its competitors, or how it intends to achieve its objectives. This is linked to shared values.

(c) **Systems**. These include the technical systems of accounting, personnel, management information and so forth. These are linked to the skills of the staff.

These 'hard' elements are easily quantified and defined, and deal with facts and rules.

'Soft' elements are equally important.

(a) **Style** refers to the **corporateculture** that is the shared assumptions, ways of working, attitudes and beliefs. It is the way the organisation presents itself to the outside world.

(b) **Shared values** are the guiding beliefs of people in the organisation as to why it exists. (For example, people in a hospital seek to save lives.)

(c) **Staff** are the people in the organisation.

(d) **Skills** refer to those things that the organisation does well. For example, the UK telecom company BT is good at providing a telephone service but, even if the phone network is eventually used as a transmission medium for TV or films, BT is unlikely to make those programmes itself.

All elements, both hard and soft, **must pull in the same direction for the organisation to be effective**.

For example, an organisation will not benefit if it installs the most sophisticated, up to date management information systems, yet its managers continue to want to receive the same reports as they always have because they don't understand or trust the new technology. In this simple example, there is a **mismatch** between systems and staff/skills.

5.4 Leadership and change

Change management is a comprehensive effort to lead an organisation through transformation. To be successful, the transformation effort must be actively led and managed with a clear set of objectives and an agreed plan for achieving these objectives.

A crucial problem organisations have to address is how they can **manage change** in the fast-moving environment of contemporary business while also **maintaining control** and their **core competences**.

Designing, evaluating and implementing successful change strategies depends to a significant extent on the quality of the senior management team, and in particular that team's ability to design the organisation in a way to facilitate the change process.

5.4.1 Who leads change?

Although a CEO plays an important role in leading strategic change, leading change is not only the responsibility of the CEO.

Whetten and Cameron have pointed out:

> "… the most important leadership demonstrated in organisations usually occurs in departments, divisions, and with teams and with individuals who take it upon themselves to enter a temporary state of leadership …"

Inevitably, though, leading change will also require competences in influencing and conflict handling, because people may need to be persuaded of the value and benefits of change.

Change may create conflict between individuals and their environment – and, often, within individuals themselves. Change can often make people uncomfortable, which is why many people resist it. Organisational change also creates potential conflict between management (who may be identified as the causes or agents of change) and employees (who often feel like the 'victims' of it).

Managing change is in essence a process of facilitating internal and external conflict resolution. So change leaders have to play a dual role of not only leading a business forward, but also resolving any conflicts which are created during the course of the change process.

5.5 Strategy, change management and HRM

Ultimately, change is inevitable in any progressive organisation. Any business that wants to thrive in an ever-changing world needs to adapt to its environment. One of the key responsibilities for an organisation's management is to detect trends inside and outside the organisation to identify changes that are needed and then to initiate a change management process to introduce those changes.

The change management process can be summarised in three steps:

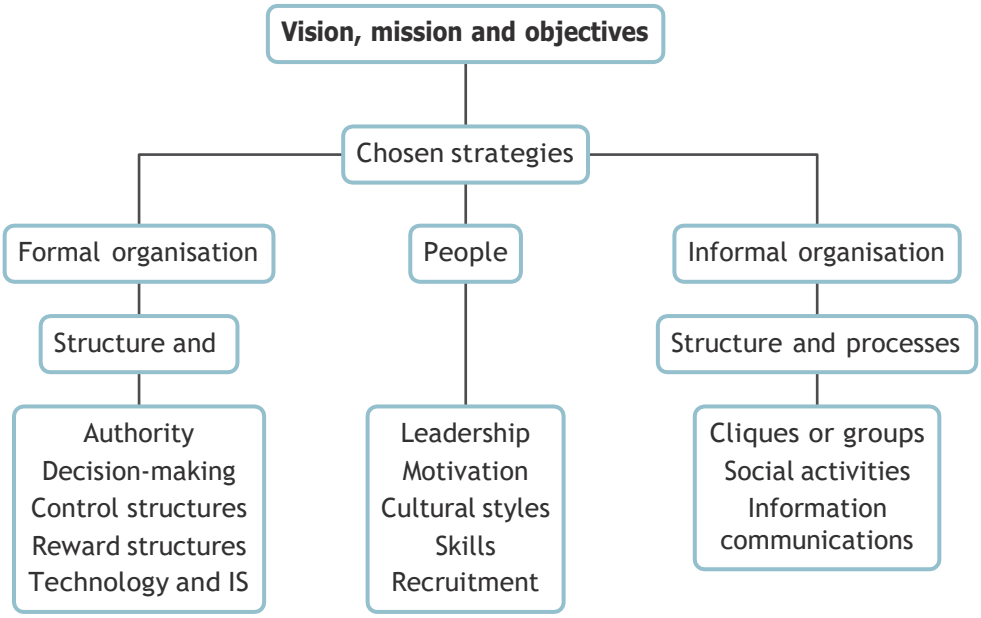
(a) **Strategic planning and design**: Form a change management team, define the vision and strategy, design a programme from which to manage the change and determine the tools needed for implementation

(b) **Strategy implementation**: Communicate the vision and implementation to staff, manage staff responses and lead them through the change; maintain momentum

(c) **Evaluation and readjustment**: Look at the results, track performance against targets, modify structure if necessary, plan for the future but continue to monitor performance

However, it is important to remember that change can affect all the aspects of an organisation and, in turn, how implementing a business strategy could require changes in all the aspects of an organisation. As the diagram below illustrates, people are likely to be central to strategy implementation and change management within organisations. Consequently, HRM also needs to be considered as an important element of change management.

Figure 10.7: HRM and strategy implementation



5.6 HRM and acquisitions

Although we have so far looked at the role of HRM in managing internal changes within an organisation, HRM is also a crucial part of an acquisition or merger.

Managing the 'human' side of an acquisition or merger is critical for maximising the value of the deal, and a number of the key issues involved in any such deal relate directly to HR issues:

- Determining the organisational structure of the new company
- Integrating the organisational cultures of the different companies
- Retaining key talent and key managers

- Communicating to staff in both companies, and addressing any concerns they may have
- Dealing with any redundancies which may be necessary
- Aligning the remuneration and reward systems of both companies
- deciding on HR policies and practices for the new company

5.6.1 HRM and due diligence

Not only could protecting and developing the rights and interests of human resources be crucial to a successful acquisition, but there may also be legal obligations associated to it (for example, obligations relating to a pension plan, or obligations arising if employees' terms and conditions of employment are protected when a business is transferred from one owner to another).

In this respect, HR due diligence will be an important element of a takeover. The functions which are relevant to HR due diligence are likely to include the following:

- HR audit:
  - Benefits and compensation programmes
  - Recruitment process
  - Practices for recruitment and dismissal
  - Exit procedures
  - Employee contracts and employee handbooks
  - Personnel files
  - Organisation charts
  - Performance reviews, and guidelines for how employee performances are evaluated
  - Training and education programmes
  - HR strategy
- obligations under the pension plan
- union contracts/union memberships
- Evaluation of synergies, gaps and duplications in numbers and skills
- review of potential redundancies (and redundancy costs, including senior management compensation plans) and cost savings post-acquisition
- Talent retention
- Legal compliance (eg, group insurance, employment legislation, health and safety)

As we have already noted, the pre-acquisition process also needs to review organisational structure (eg, number of management layers, centralisation vs decentralisation) and the organisational 'fit' (culture and values) between the two companies.

5.7 HR and ESG

We have highlighted the increasing importance of ESG considerations for business a number of times already. However, it is also necessary here to consider how ESG is important in relation to human resource management.

The 'S' in ESG focuses on an organisation's social impacts and dependencies, including those relating to its workforce. As such, the way an organisation's managing issues relating to its workforce could be important in helping an organisation boost its ESG credentials. Areas where HR issues contribute to ESG include:

Executive remuneration *– Executive remuneration is addressing ESG concerns, with increasing numbers of executive bonus schemes now having an ESG component (as we noted in the chapter, Strategic performance management).*

Pay *– In addition to executive remuneration, the remuneration of employees across the workforce can be a key factor in relation to the 'social' aspect of ESG. Increasing attention is being focused on not only ensuring that organisations pay their staff above the legal minimum wages, but also that they are committed to paying a living wage to their workforce (to ensure that employees are able to meet their everyday needs).*

Diversity and inclusion *– The importance of ensuring organisations promote a diverse and inclusive working environment is widely acknowledged; for example, in relation to employee engagement and retention. Therefore, it will be important that organisations measure and monitor key metrics in relation to diversity and inclusion (eg, percentages of employees from monitored groups). Issues around diversity and inclusion can also overlap with pay: for example, the extent to which there is pay inequality in an organisation based on gender or race.*

Pay ratio *– Pay is also an area which could be considered by investors, and potential employees, as an indicator of the value placed on the workforce. In this context, the ratio between CEO's pay and that of the average (median) worker could be seen as an indicator of the fairness of an organisation's pay structures – for example, by comparing the pay ratio in that organisation against industry benchmarks or competitors. In particular, organisations will be scrutinised to ensure that executives are not receiving excessive salaries and/or bonuses compared to the rest of the workforce and relative to the organisation's performance.*

Modern slavery *– Businesses need to ensure that they do not engage in practices which amount to modern slavery or human trafficking and have measures in place to prevent, identify and remedy any issues that may arise. Investors and customers will want to ensure that businesses have sound practice in place to safeguard against modern slavery, not only in their own business, but also in the wider supply chain. The reference to the wider supply chain is significant here because the complex supply chains that some businesses have will make it harder to spot exploitation and abuses in their supply chain. Nonetheless, organisations need to be accountable for the treatment of workers across their supply chains.*

Health and safety *– Health and safety in the workplace are vital for the physical and mental wellbeing of staff. Therefore, organisations need to identify, and mitigate, health and safety risks in their operations and their supply chains. Failure to uphold health and safety standards, leading to accidents in the workplace could have various financial implications; compensation claims and legal liabilities; disruption and loss of production; and – more widely – reputational damage and loss of trust among employees, customers, investors and other stakeholders.*

Training *– It should go without saying that employees who receive relevant training are likely to be able to perform better in their jobs than employees who do not receive training. However, as well as the immediate benefits of training in terms of improving an employee's skills and knowledge, it can have wider benefits in terms of improving employee engagement and motivation, and in improving employee loyalty and retention (by showing to employees that the company is investing in their development). Therefore, metrics relating to the amount of training and development an organisation provides its staff (for example, expenditure on training per employee per year) could be a valuable metric in relation to the 'social' element of ESG.*

Summary

	Tick off
Successful strategic implementation requires the effective recruitment, training and organisation of staff, coupled with effective leadership and performance management.	☐
Human resource management is vital in ensuring that an organisation attracts, and retains, the number and quality of staff it needs in order to achieve its organisational goals.	☐
To be successful, an organisation's HRM strategies need to be aligned to its business strategy. The systems of reward management and remuneration used by an organisation will vary according to its business and the generic strategy it is pursuing.	☐
Performance measurement and management are equally important in the context of HRM as they are in relation to the overall strategic and operational control of an organisation. Appraisals are a key part of performance management, and provide a link between an individual's performance and an organisation's overall strategy.	☐
It is important to recognise the potential behavioural implications of performance targets. Poorly designed performance targets may lead to an adverse impact on employees' performance and, consequently, on an organisation's ability to implement its strategy successfully.	☐
Reward systems should have three overall objectives: to support staff recruitment and retention; to motivate employees to high levels of performance; and to promote compliance with workplace rules and expectations.	☐
When formulating employee benefit packages, an organisation also needs to consider the corporate reporting implication of the decision. This has been a particular issue for defined benefit plans (in relation to IAS 19, *Employee Benefits*) in which pension scheme deficits have created large liabilities in companies statements of financial position.	☐
Human resources departments are likely to have a key role in managing change effectively. The majority of changes will be internal within an organisation, but HRM will also be crucial in relation to an acquisition or merger.	☐

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, complete the following knowledge diagnostic and check you are able to confirm you possess the following essential learning from this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
1.	What are the main components of an organisation's Human Resource Management systems? (Topic 1)
2.	What does HR planning involve? (Topic 2)
3.	How should an organisation set performance targets to best motivate employees? (Topic 3)
4.	What factors should an organisation take into consideration when setting its remuneration and reward policy? (Topic 4)
5.	How should an organisation account for employee benefits in the annual financial statements? (Topic 4)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
2 Elegard	In this scenario question you are required to assess the deficiencies of the current reward management scheme and to analyse the limitations of the performance measures proposed by the HR director. Note the wording of the question - you are specifically requested to look at deficiencies and limitations so do not waste time looking at the positive aspects of either the current scheme or the proposed measures. Ensure your answer applies to the scenario company, since application of knowledge to the scenario is always required in SBM&L exams.
4 Della Co	This is a short question to test your corporate reporting knowledge, which represents approximately 15-20% of the SBM&L syllabus. You are asked to show the accounting entries for three years in three different circumstances. CR style questions in the SBM&L exam are often overlooked by students so it is important to include such practice in your exam preparation. Note that SBM&L is not a corporate reporting exam and the emphasis is on the business implications arising from how items are treated in the financial statements.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your ueries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Technical reference

1 IFRS 2, *Share-based Payment*

- The objective of IFRS 2 is to specify the financial reporting by an entity when it undertakes a share- based payment transaction. IFRS 2 requires an entity to recognise share-based payment transactions (such as shares granted, or share options) in its financial statements. This includes transactions with employees or other parties to be settled in cash, other assets, or equity instruments of the entity.

2 IAS 19, *Employee Benefits*

- Outlines the accounting requirements for employee benefits, including short-term benefits (eg, wages and salary, annual leave); post-employment benefits (eg, retirement benefits); and termination benefits. The standard requires that the cost of providing employee benefits should be recognised in the period in which the benefit is earned by the employee, rather than when it is paid or payable. The standard also outlines how each category of employee benefits are measured, and it provides detailed guidance about post-employment benefits.

3 IAS 33, *Earnings Per Share*

- The objective of IAS 33 is to prescribe principles for determining and presenting earnings per share (EPS) to improve performance comparisons between different entities in the same reporting period, and between different reporting periods for the same entity. IAS 33 sets out how to calculate both basic EPS and diluted EPS. The calculation of basic EPS is based on the weighted average number of ordinary shares outstanding during the period, whereas diluted EPS also includes dilutive potential ordinary shares (such as options and convertible instruments) if they meet certain criteria.

Self-test questions

Answer the following questions.

1 Connie Head

Connie Head is the recently appointed HR manager in a medium-sized accounting firm. Her appointment was a belated recognition by the senior partners of the firm that their ambitious corporate growth goals were linked to the performance of the individual business units and the accountants working in those units. Connie is convinced that performance management and an appraisal system are integral elements in helping the firm achieve its strategic objectives. This reflects her experience of introducing an appraisal system into the corporate finance unit for which she was responsible. The unit had consistently outperformed its growth targets and individual members of the unit were well motivated and appreciative of the appraisal process.

However, the senior partner of the firm remains unconvinced about the benefits of appraisal systems. He argues that accountants, through their training, are self-motivated and should have the maximum freedom to carry out their work. His experience of appraisal systems to date has shown them to lack clarity of purpose, be extremely time consuming, involve masses of bureaucratic form filling and create little benefit for the supervisors or their subordinates. Certainly, he is resistant to having his own performance reviewed through an appraisal system. Connie, however, is convinced that a firm- wide appraisal system would be of major benefit in helping the achievement of growth goals.

Requirements

1.1 Evaluate the extent to which an effective appraisal system could help the accounting firm achieve its goals.

1.2 Using models where appropriate, assess the contribution, if any, of performance management to the strategic management process.

2 Elegard

Elegard offers warranties for electrical and electronic equipment to both business and household customers. For a fixed annual fee, the company will provide a free fault diagnosis and repair service for equipment covered by the warranty. A warranty lasts for one year and customers are invited to renew their warranty one month before it expires. Elegard employs 350 full-time engineers around the country to undertake these repairs. It costs about £7,000 to train a newly recruited engineer.

When equipment breaks down the customer telephones a support help line where their problem is dealt with by a customer support clerk. This clerk has access to the work schedules of the engineers and an appointment is made for a visit from an engineer at the earliest possible time convenient to the customer. When the engineer makes the visit, faults with equipment are diagnosed and are fixed free of charge under the terms of the warranty.

Elegard is extremely concerned about the relatively high labour turnover of its engineers and has commissioned a report to investigate the situation. Some of the findings of the report are summarised in the following table (Table 1). It compares Elegard with two of its main competitors.

Table 1

Company	Labour turnover*			Average days holiday/year	Performance related pay	Average training spent per year per engineer
		£				£
Elegard	12%	32,000	No	21	No	1,000
Safequip	8%	30,000	Yes	24	Yes	1,500
Guarantor	7%	29,500	Yes	26	Yes	1,250

*Labour turnover is the number of engineers leaving in the last year as a percentage of the number of engineers employed at the beginning of the year.

An exit survey of engineers leaving the company recorded the following comments:

(1) "This is the first place I have worked where learning new skills is not encouraged. There is no incentive to improve yourself. The company seems to believe that employees who gain new skills will inevitably leave, so they discourage learning."

(2) "There is no point in doing a good job, because you get paid no more than doing an ordinary one. Average work is tolerated here."

(3) "The real problem is that the pay structure does not differentiate between good, average and poor performers. This is really demotivating."

The HR director of Elegard is anxious to address the high turnover issue and believes that quantitative measurement of employee performance is essential in a restructured reward management scheme. He has suggested that the company should introduce two new performance- related pay measures. The first is a team-based bonus based on the average time it takes for the company to respond to a repair request. He proposes that this should be based on the time taken between the customer request for a repair being logged and the date of the engineer attending to fix the problem. He argues that customers value quick response times and so the shorter this time the greater the bonus should be for the whole team.

In addition, he proposes an individual bonus. This will be based on the average time taken for an engineer to fix a reported fault and complete the job once they have arrived at the customer's address and started work on it. He argues that the company values quick repair time as this increases business efficiency and so the quicker the fix the greater the bonus should be for the individual.

Requirements

2.1 Assess the deficiencies of Elegard's current reward management scheme.

2.2 Analyse the limitations of the proposed performance measures suggested by the HR director.

3 Grateley plc

Grateley plc (Grateley) is a listed company which manufactures clothing. About 60% of its output is sold to Bloomsdale plc (Bloomsdale), a major UK-based chain of clothes stores. Clothes are sold under Bloomsdale's own label and are regarded as being in the mid- to upmarket range. The clothes are manufactured at Grateley's three factories, all of which are in the UK and are of approximately equal size.

The workforce at Grateley is largely unskilled or semi-skilled. There is poor morale, low motivation and a high staff turnover. There is little opportunity for career progression as manual employees are all at the same level, reporting directly to section managers. Trade unions frequently complain about both the repetitive nature of the production line work and the low pay. There have been three strikes at Grateley's factories in the last five years.

The management philosophy of Grateley is prescriptive and top-down, with the imposition of budgets and quotas. Little training or staff development is given, with the major focus on the achievement of output and quality targets. Employees are, however, given bonuses which are based on two different targets. First, when monthly factory output achieves predetermined levels; and second, if quality thresholds are satisfied based on the monthly number of items returned by customers as defective. On average, these targets are achieved only one month in every three.

Bloomsdale has been a major customer of Grateley for about 30 years, but a new management team has now taken over at Bloomsdale. It informed the board of Grateley that a new annual contract is to be arranged which would involve a major reduction in prices offered, and that the volumes purchased by Bloomsdale from Grateley next year would be only half that of previous years. It was also made clear that further price reductions would need to take place in future years if the contract is to be maintained at the new lower volumes.

As employees became aware of the increasingly competitive conditions, the possibility of factory closure emerged and there was increasing unrest at all three factories.

At the crisis meeting the board of Grateley identified two options:

Option 1	Close one factory and attempt to cut costs at the other two factories through a policy of efficiency improvements and redundancies.
Option 2	Close two (UK) factories and open a new factory in North Africa where labour costs are significantly lower than in the UK. Efficiency improvements and redundancies would also take place at the sole remaining UK factory.

Requirement

As the human resources director of Grateley, write a memorandum to the board evaluating the human resource management issues which may arise under each of the two strategic options identified.

Refer to relevant studies or evidence where appropriate.

4 Della Co

Della Co grants 100 share options on its £1 shares to each of its 500 employees on 1 January 20X5. Each grant is conditional upon the employee working for the entity over the next three years. The fair value of each share option as at 1 January 20X5 is £15.

On the basis of a weighted average probability, the entity estimates on 1 January 20X5 that 20% of employees will leave during the three-year period and therefore forfeit their rights to share options.

Requirement

Show the accounting entries which will be required over the three-year period in the event of the following:

- 20 employees leave during 20X5 and the estimate of total employee departures over the three- year period is revised to 15% (75 employees).

- 22 employees leave during 20X6 and the estimate of total employee departures over the three- year period is revised to 12% (60 employees).
- 15 employees leave during 20X7, so a total of 57 employees left and forfeited their rights to share options. A total of 44,300 share options (443 employees × 100 options) are vested at the end of 20X7.

You may assume that the entity maintains a share-based payment reserve in equity.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

Human resource planning and strategy

ScanTech's growth plans envisage the company doubling in size over the next three years. This will require the employment of extra staff, particularly in marketing, sales and manufacturing. The ambitious planned rate of growth and the high technology base of ScanTech's business mean that these extra staff must be of very high quality. Human resource (HR) management is thus an essential component of the company's business strategy and so should be integrated with its development. The alternative is increased potential for serious shortages of staff and mismatches between job requirements and staff availability.

The proposed opening of a foreign manufacturing plant will complicate all HR issues significantly and will demand very careful consideration.

The following elements of HR planning could be useful at ScanTech:

An **audit of existing staff** should reveal those with potential for promotion or employability in new specialisations. It would also indicate where shortages already exist.

Concurrently, an analysis of **likely future staff requirements** could be carried out. It seems inevitable that ScanTech will need to employ more staff in the areas already mentioned, but it does not seem to know how many will be required, whether other functions will need to be increased in size, or if more support and administrative staff will be needed. There are also the related and sensitive issues of **management succession** and **internal promotion** to consider. In particular, ScanTech needs to consider the eventual replacement of the existing joint Managing Directors, who are expected to leave once the current growth objective has been achieved.

These two elements of analysis should help ScanTech to identify the gaps that it will need to fill if it is to have the staff required for its overall growth strategy.

Recruitment will be the logical next step once the analysis of resource requirements have been completed – in the sense of attracting applicants, and **selection** from within the pool of applicants. Recruitment work is often **outsourced** (to a recruitment agency) and it will be necessary for ScanTech to decide whether the **expertise** and **economies of scale** offered by outsourcing outweigh the need for deep familiarity with our operations on the part of the recruiters.

Reward policy must be considered. At the moment, ScanTech's staff profile is heavily biased towards people with a background in research and development. Different types of people will be required in the future and their expectations for their rewards and remuneration are likely to vary from those of the existing staff.

A doubling in size to, say, 220 employees is likely to take the company into an area of HR complexity in which a formal reward policy and structure is required. Informal decisions about pay and benefits will not be satisfactory.

Increasing size is also likely to require the establishment of a policy on **appraisal and performance management**. This should be linked to a programme of **training and development**. It is likely that ScanTech will continue to hire well-qualified technical staff, but there will be a need for development of staff in other functions and for management development in particular.

Answer to Interactive question 2

Forecast human resource demand

- Reduction in numbers of staff required in the future
- Roles – amalgamation of management roles due to reduction in number of branches
- Skills – need for staff with customer skills rather than bureaucratic and professional banking skills
- Availability – requirement for more flexible working practices, eg, late/weekend opening, 24/7 cover of call centres
- Location shift – staff will be required in off-shore centres with reductions in UK centres

Forecast human resource supply

- Potential excess supply of staff internally as internal jobs contract and external opportunities diminish
- Increased availability of staff on external labour market due to downsizing by other banks may make this a cheap source of staff – eg, on short-term contracts
- Need to consider the forecast supply in off-shore locations

Training and development

- Change in the skills and competences away from professional qualification of Chartered Institute of Bankers or Chartered Insurance Institute
- Requirement for some staff to pass compulsory regulatory exams to sell new financial products
- Regular updating of staff in new products and legal/regulatory issues

External recruitment

- Potential for cheaper sources of staff externally than internally
- Reduction in intake of school-leavers and graduates due to surplus of staff internally
- Need to access new sources of staff to cover new technologies (eg, IT recruitment)
- Recruitment will also take place off-shore

Answer to Interactive question 3

There is only limited information available about JBC's appraisal system. However, it is clear that the organisation has taken a formal approach using standardised forms with clear objectives for staff development and performance improvements.

Problems with the system can be considered under two headings – firstly inherent problems with the design and implementation of the system, and secondly problems concerning its operation.

Design and implementation problems

The system may have been poorly designed in the first place. For example, it may be based on systems used by other organisations and no thought given to whether it is suitable for JBC.

The design of the system may have reflected the needs of the organisation at that time but is no longer relevant because the company has 'moved on'.

There may have been a lack of consultation and communication with senior managers when the system was being developed. They may view it as being imposed on them and therefore are not interested in making it work.

Appraisal schemes should provide **benefits** which justify the cost and effort put into them. Senior management comments such as 'a waste of time and effort' indicate that there is an imbalance between what is put into the scheme and what comes out – for example, whether or not any staff development needs which are identified during an appraisal are actually subsequently addressed.

This imbalance may have been caused by the system being put into place because senior management thought they should be seen to have an appraisal system, rather than it being a genuine method of improving staff development and performance.

Operational problems

Senior managers may have **insufficient time** to conduct the appraisal process properly. This may reduce the scheme to a form-filling exercise just to meet HR requirements, missing the point of the scheme and its objectives.

The scheme focuses on staff development needs. This is likely to involve some **additional training costs**, and may also reduce the amount of time that academic staff are available for teaching (if they are attending training courses of their own). Therefore, managers may not see it as being in their interest to have staff undergo training. This, of course, is a short-sighted view, as properly structured training should improve JBC's performance in the long run. However, managers may not wish to wait for such benefits to materialise, preferring to focus on short-term issues and performance instead.

The scheme is **not linked to annual bonuses**. Employees are likely to act in a manner that maximises their bonus, which may be at odds with the objectives of the appraisal system.

Standard procedures indicate a **bureaucratic** or mechanical approach to appraisals. Senior managers will be faced with a large volume of identical paperwork that needs to be processed in addition to

their existing workload. There is likely to be a temptation to rush through the process with not much thought to the objectives.

Appraisal schemes often involve **subjective judgements** and **opinions** by senior managers over their staff. There is a risk that employees are not assessed correctly or consistently, meaning that some staff who do not require training are offered it while others who need help to improve their performance are not.

Answer to Interactive question 4

Staff retention – It appears that SBC's current reward package for its IT consultants is not as competitive as that offered by some of its rivals. If this continues, then SBC's staff turnover could increase further, which is likely to be costly for SBC both in terms of having to recruit and train new staff and also in terms of the loss of knowledge which occurs when consultants leave.

If the new proposal means that the overall value of the consultants' salary increases, then this could help to reduce staff turnover which should be beneficial to SBC.

Value of commissions – However, it is not clear what impact the proposed changes will have on the consultants' salaries. The scenario does not indicate how much lower the new basic salary will be than the consultants' current salaries, nor does it indicate the size of the commissions received from the software companies.

It is possible that the proposal could actually end up reducing the consultants' salaries, which will have the opposite effect to what the directors are trying to achieve.

Impact on SBC's profits – Equally, however, the directors will need to ensure that the changes are not too generous in favour of the consultants because they are likely to reduce SBC's profit margins, for example because the commissions will no longer be income for the company.

Moreover, if the commission system doesn't stimulate higher sales revenue, the effect of the commissions will be to reduce profits overall. Therefore, a key issue surrounding the acceptability of the proposal is whether it will result in higher revenues being generated.

Other consultants – The directors also need to consider how the other types of consultant will respond. Again, it is not clear how much communication there is between the three types of

consultant but, if advertising and recruitment consultants find out the IT consultants have had their rewards schemes revised, they may want something similar themselves.

Risk to customers – When SBC's clients are looking to select a new software system, a key factor in their choice should be how well the system fits their requirements.

Advice about the suitability of different systems is likely to be one of the key pieces of advice they want from the consultants. However, the new system could compromise the consultants' ability to give this advice impartially.

Under the current system, it appears that the consultants have no incentive to recommend one software supplier over another. However, under the proposed new system, consultants may be tempted to advise clients to buy the system that will earn them the highest amount of commission rather than the system which is best for the client.

Such practices could be damaging to SBC's reputation and future revenues.

If clients install software systems on SBC's advice which do not meet their requirements effectively, then they are unlikely to use SBC in future.

Alternative bonus/reward scheme – It appears that the commission scheme is the only option which the directors have looked at so far. However, rather than only looking at one scheme, they should also consider whether there are any alternative schemes which may be more appropriate.

For example, it is not clear whether SBC currently has any kind of performance-related pay scheme, or bonus scheme; a scheme which rewards consultants for their performance in relation to a range of targets, linked to SBC's overall objectives, may be more appropriate than the current proposal.

Answer to Interactive question 5

Statement of profit or loss and other comprehensive income notes

Defined benefit expense recognised in profit or loss	**£m**
Current service cost	11
Past service cost	10
Net interest on the net defined benefit asset (10% × (110 + 10)) – (10% × 150)	(3)
	18

Other comprehensive income (items that will not be reclassified to profit or loss) **Remeasurement of defined benefit plans**	**£m**
Actuarial gain on defined benefit obligation	17
Return on plan assets (excluding amounts in net interest)	(22)
	(5)

Statement of financial position notes

Net defined benefit asset recognised in the statement of financial position

	31 December 20X1	31 December 20X0
	£m	£m
Present value of pension obligation	116	110
Fair value of plan assets	(140)	(150)
Net asset	(24)	(40)

Changes in the present value of the defined benefit obligation	£m
Opening defined benefit obligation	110
Interest on obligation (10% × (110 + 10))	12
Current service cost	11
Past service cost	10
Benefits paid	(10)
Gain on remeasurement through OCI (balancing figure)	(17)
Closing defined benefit obligation	116

Changes in the fair value of plan assets	£m
Opening fair value of plan assets	150
Interest on plan assets (10% × 150)	15
Contributions	7
Benefits paid	(10)
Loss on remeasurement through OCI (balancing figure)	(22)
Closing fair value of plan assets	140

Answers to Self-test questions

1 Connie Head

1.1 The Senior Partner and Connie emphasise the aspects of appraisal schemes that **support their own favoured policies**. Such schemes should support the organisation's overall objectives without incurring excessive administrative and management costs.

In an organisation such as an accounting practice, the professional staff should indeed be highly **self-motivated**, able to judge the effectiveness of their own performance and bring to their work a commitment to high professional standards. On the other hand, it is inevitable that their **talents and performance will vary** and they will need **guidance and help with their future development**. Dealing with these issues would be the role of an appraisal scheme.

The overall aim of such a scheme would be to **support progress towards the achievement of corporate objectives** and it would do this in three ways: performance review, potential review and training needs review.

Performance review. Performance review should provide employees with an **impartial and authoritative assessment of the quality and effect of their work**. Individuals should have personal objectives that support corporate goals via intermediate objectives relevant to the roles of their work groups. A reasoned assessment of performance can have a **positive motivating effect** simply as a kind of positive, reinforcing feedback. It can also provide an opportunity for analysing and addressing the **reasons for sub-optimal performance**.

Potential review. Any organisation needs to make the best use it can of its people. An accountancy practice is typical of many modern organisations in that its people are its greatest asset and future success depends on managing them in a way that makes the best use of their skills and aptitudes. An important aspect of this is **assessing potential for promotion and movement into positions of greater challenge and responsibility**.

Training needs review. A further aspect of the desirable practice of enabling staff to achieve their potential is the provision of training and development activities. The appraisal system is one means by which **training needs can be assessed** and training provision initiated.

In this context, the reviews within the appraisal system would seem to be a supportive developmental process.

However, there is a tension at the heart of an appraisal system between appraisal as a **judgement process** and a **developmental process**. Whereas development will help motivate, the judgemental aspect of appraisal may **demotivate** and this will hinder the firm in trying to achieve its goals.

The appraisal system

An appraisal system must be properly administered and operated if it is to make a proper contribution to the organisation's progress.

The appraisal cycle. Formal appraisal, with interviews and written assessments, is typically undertaken on an **annual cycle**. This interval is commonly regarded as too long to be effective because of the speed with which individual roles can evolve and their holders can develop, so the annual appraisal is often supplemented with a less detailed review after six months. Sometimes the procedure is sufficiently simplified that the whole cycle can be done at intervals of six months. Much modern thinking on this topic is now suggesting that any frequency of periodic appraisal is unsatisfactory and that it should be replaced by a **continuous process of coaching and assessment**.

This aspect of 'continuous improvement' will be more likely to help the firm achieve its goals.

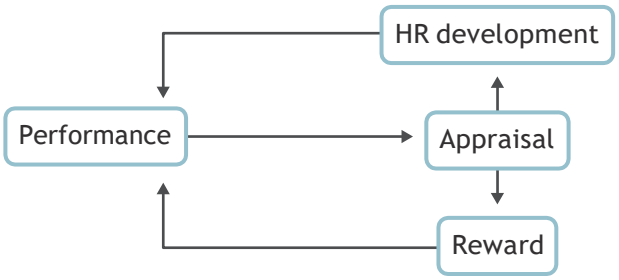
Objectivity and reliability. Appraisal involves an element of direct personal criticism that can be stressful for all parties involved. If the system is to be credible, its outputs must be seen as objective and reliable. These outputs should be used to motivate individuals to achieve their goals.

Setting targets. Past performance should be reviewed against **objective standards** and the performance against these targets should form the basis of the employee's reward.

If the approval system acts to motivate the staff by rewarding them for achieving their **individual goals** and these individual goals are properly **aligned to the firm's goals**, then the appraisal can be effective in helping the firm achieve its goals.

However, the goals in themselves must be **realistic and achievable**. If they are not, then staff will become demotivated and the appraisals will be counter-productive.

However, overall an effective appraisal system can help manage the workforce in a rational way, as through the feedback look illustrated below, thereby helping the firm to achieve its goals.



1.2 **Performance management** involves the establishment of clear, agreed individual **goals and performance standards**; continuous leadership action to both **motivate and appraise subordinates**; and a **periodic review** of performance at which the goals and performance standards for the next cycle are set.

Performance management is an application of the **rational model** of strategic management, in that individual goals are intended to form the lowest echelon of a **hierarchy of objectives** that builds up to support the **overall mission** of the organisation. It is an essential aspect of the system that individual goals should be **agreed and internalised** so that true **goal congruence** is achieved.

This overall approach was first described by Peter Drucker, and is seen most clearly in the system of **management by objectives** (MbO). MbO as a management system has fallen somewhat out of favour with the rise of quality management methods that emphasise processual and procedural conformance rather than the attainment of overall performance goals. Nevertheless, it has much to offer.

MbO and strategic analysis

Under a formal MbO system, the process of setting goals is part of the **implementation phase** of strategic management and follows consideration of resources, overall objectives and SWOT analysis. In this way, MbO resembles the strategic analysis stage of the rational planning model.

Strategic choice

Top level subordinate goals are agreed for heads of departments, divisions or functions: these goals should be specific, measurable, attainable, relevant and time-bound (SMART). It is particularly important that the achievement of a goal can be established by objective **measurement**. There may be different timescales for different objectives, with short-term goals

supporting longer-term ones. Again there is a parallel here to the notions of suitability, acceptability and feasibility of the rational planning model.

Strategic implementation

Departmental heads then agree SMART goals for their subordinates in discussion with them, that support their own personal goals, and so on down the hierarchy to the level of the individual employee. All members of the organisation thus know what they are expected to achieve and how it fits into the wider fabric of the organisation's mission.

Periodic **performance review** is based on the objective appraisal of success against agreed goals, the agreement of goals for the next period and an assessment of the resources, including training, that the reviewee may require to reach those goals. The MbO system thus closes the **feedback loop** in the corporate control system.

2 Elegard

2.1 Impact of the reward management scheme on staff turnovers

The engineers at Elegard are very important workers to the company. Their motivation, their commitment and their proficiency in undertaking repair work are all critical success factors for Elegard because they will influence how customers perceive the company and, therefore, whether customers will renew their warranties rather than moving to a competitor.

Elegard's high labour turnover suggests its engineers are not as motivated and committed to the company as they could be, and this is a significant problem.

High labour turnover is also a problem because of the costs incurred in training newly recruited engineers – about £7,000 each – in addition to the costs of advertising jobs and arranging interviews.

Consequently, the extent to which the current reward management scheme contributes to this high labour turnover among Elegard's engineers suggests there are a number of problems with the scheme.

Too much focus on base pay – Elegard's rewards scheme focuses on base pay with little attention given to performance pay or indirect pay. The focus on basic pay is unlikely to encourage motivation among skilled staff, like the engineers. Although Elegard's basic pay is higher than its competitors, it has a higher staff turnover rate than its competitors. This suggests that base pay alone is not an effective reward.

Lack of performance-related pay – Safequip and Guarantor both offer their engineers performance-related pay. This is likely to act as a motivating factor for their engineers, knowing they can gain extra pay by virtue of doing their jobs well. By contrast, Elegard's engineers have no such incentive. An exit interview with one of the engineers reinforces this point: 'The real problem is that the pay structure does not differentiate between good, average and poor performers. This is really demotivating'.

The HR director has recognised this weakness in the current reward management scheme, which is why he has suggested two new performance-related pay measures.

Current scheme does not promote organisational goals – The lack of performance related pay means there is little incentive for the engineers to do a good job. Given the key role the engineers play in the success of the company, this is a major business risk. If customers do not feel they are getting a good service from Elegard, they are unlikely to renew their warranties. Again, one of the exit interviews stresses the problem here: 'There is no point in doing a good job, because you get paid no more than [for] doing an ordinary one. Average work is tolerated here'.

Absence of profit share scheme – Overall organisational performance can be supported through profit sharing schemes, provided individuals' goals are properly aligned to corporate objectives.

If employees benefit from the profitability of their company, then they have an incentive to try to maximise that profitability. Both Safequip and Guarantor offer a profit sharing scheme for their engineers, but Elegard does not. This is likely to reinforce the attitude among Elegard staff that there is no point trying to do a good job, because they will get no benefit from doing so.

Levels of indirect pay – Indirect pay (or benefits) such as pension plans and private healthcare can form a valuable part of an organisation's total rewards package.

Two measures which could indicate Elegard's approach to indirect pay are the number of days of holiday staff are offered per year, and the average amount of training they are given. In both these measures, Elegard performs worse than its competitors.

Low average training spend – The relatively low amount which Elegard spends on training is a particular concern. It suggests that Elegard views training as a cost rather than as an investment in human capital.

One of the exit interviews highlights the impression this view is giving to the staff: 'This is the first place I have worked where learning new skills is not encouraged'. Elegard seems to view training as a risk, thinking that once staff gain new skills they will inevitably leave. Ironically,

however, the lack of opportunities for training and development seems to be one of the reasons prompting staff to leave.

2.2 General problems

Ability of employees to influence performance measures – The key logic behind performance-related pay is that the incentive of an increased income will motivate employees to improve their performance. However, if the employees cannot influence the performance targets they are being measured against then performance-related pay will not be a motivating factor for them. Unfortunately, it seems that the performance targets the HR director is proposing are largely outside the scope of the employees' influence.

Goal congruence – Performance measures should be designed so that individuals' goals are aligned with organisational goals. If a scheme encourages employees to work in a way that maximises their individual income, but in doing so reduces the performance or profitability of their organisation as a whole, this will be a problem for the organisation. The HR director's focus on speed may create problems in this respect.

Limitations of proposed team-based bonus scheme

Response time measures outside employees' control – The HR director has proposed that the bonus should be based on the **time between a customer logging a repair request and the date the engineer arrives** to fix the problem. This correctly reflects that customers value quick response times, but it overlooks that the measure is influenced by factors outside the team's control.

The date an engineer can attend to fix the problem **depends on the availability of an engineer**. This could be influenced by the **number of engineers Elegard chooses to employ** rather than necessarily the efficiency of the engineers.

Customers can dictate visit dates – Also, Elegard's policy is to schedule visits "at the earliest possible time **convenient to the customer**". However, if domestic customers are out at work and cannot immediately take time off to be at home for a service visit, this 'convenient time' may be quite a long way in the future. The team cannot control this timescale, making it an unsuitable basis for a performance measure.

Limitation of proposed individual bonus scheme

The individual bonus will be based on the average time taken for an engineer to fix a fault once they have arrived at the customer's premises. The HR director's logic for this is that quick response time increases business efficiency.

To an extent, the engineer can control the time taken to fix a fault, but there are still some significant problems with this measure.

Repair time depends on the complexity of the problem – An engineer called out to fix a complex problem will inevitably take longer than an engineer who has to fix a simple problem. This measure would therefore penalise engineers working on complicated problems, which ironically could be the most important jobs for Elegard to do well.

Trade-off between speed and quality – The performance measure might encourage engineers to perform a **quick fix** (to get the job signed off) rather than to sort the underlying problem properly.

Consequently, the **measure could actually increase the volume of repairs** Elegard has to undertake, whereas the business model is based on the need to minimise calls and repairs.

In this respect, the HR director's proposal would create a **problem with goal congruence**. By performing low-quality quick fixes individual engineers can boost their own incomes, but their doing so will damage the profitability of the company as a whole.

Inaccurate job reporting – The measure could also encourage engineers to misrepresent the time they actually spent on a job. The bonuses are based on the time taken to **fix a fault once the engineer has arrived at the customer's premises**, so if an engineer claims it took longer to get to a client than it did the engineer can **artificially reduce the time reported against the job**. Again, the measure is promoting behaviour which is unhelpful for the company as a whole. For example, if customers' warranty fees are calculated according to the time taken to fix faults, an understated time could result in Elegard charging a warranty that is too low, which in turn could cause restriction of Elegard's profits.

Focus only on time – As well as these specific issues around goal congruence, the HR director's proposals suffer through focusing exclusively on time. While speed is important to the customer (and so is an important performance measure), the proposals would benefit by including other measures which address quality, skills or training. The lack of focus on quality and training are key issues behind the current high staff turnover, yet these proposals do nothing to address this.

3 Grateley plc

MEMORANDUM

To:	The Board of Grateley plc
From:	HR Director
Date:	[today's date]
Subject:	HRM implications of strategic options

In broad terms, approaches to human resource management can be classified into 'hard' and 'soft' approaches, and these represent opposite ends of the spectrum.

Hard approach. Emphasises **resources** element of HRM. Human resources are planned and developed to meet the wider strategic objectives of the organisation, as with any other resource. This involves managing the functions set out below to maximise employee effectiveness and control staff costs.

Soft approach. Emphasises **human** element of HRM. This is concerned with employee relations, the development of individual skills and the welfare of staff.

Grateley currently adopts the hard view of HRM but this needs consideration given the magnitude of change elsewhere in the organisation. The implications of this approach and of the proposed changes themselves can be seen in all the HRM functions within the company.

Personnel planning and control. This is the analysis of the organisation's future need for employee resources, with respect to quantity and skills, given the nature of the labour market.

The most obvious feature in the current circumstances is to allow for the loss of business from Bloomsdale. The reduction in future employee levels will need to correspond with future demand for Grateley's output. If Bloomsdale is halving the volume of its purchases from Grateley and these currently account for 60% of revenue, then total revenue in future will be around 70% of its current level (ie, 60% × ½ from Bloomsdale, and 40% from other customers). This would indicate that at least one factory will need to be closed unless Grateley is able to attract any significant new customers, and even this is dependent on Bloomsdale not cutting its purchase further in future.

If there is to be a new factory in North Africa then appropriate planning is necessary to determine the optimal quantity of new labour. This may not be the same as for a UK factory because of differences in labour productivity, different levels of capital investment, different procedures and employee agreements, different motivation and incentives, and different labour costs.

Recruitment and selection. Choosing the right person for the posts specified in the job design.

There is unlikely to be recruitment in the UK, but under strategic option 2 there will be significant recruitment in North Africa and this would be a major HRM exercise in an unfamiliar labour market.

Remuneration. Preparing remuneration packages for employees to provide appropriate incentives while controlling costs in the circumstances of the organisation and the labour market. This may involve participation in collective bargaining with trade unions.

It would appear that labour rates are relatively low for the UK but are likely to be much lower in North Africa.

The two issues here are the amount of remuneration and the form it takes in terms of incentives or conditions.

In terms of total remuneration per employee, the company is restricted by the need to control costs and to meet Bloomsdale's demands on price. This in turn is determined by competitors' prices in the clothing market. The other constraint, however, is the UK labour market whereby it might not be possible to attract the appropriate quality and quantity of labour in the long term if pay is too low.
Other conditions, such as the minimum wage and union agreements, also constrain the ability to reduce the remuneration per employee.

However, while there are a number of constraints on remuneration per employee, labour cost savings can be made by labour efficiency gains. There has been strong union resistance to this in the past but, given the external threat from Bloomsdale, there is an improved prospect of managing redundancies at those UK plants remaining open as part of a reorganisation package.

In North Africa, labour costs would be much lower but the workforce is likely to be, initially at least, unskilled and there would be a need to develop the right corporate culture in order to maintain quality of output. Transportation costs (inward and outward), set-up costs and inventory holding costs would partially offset any labour cost advantage.

Incentives. The incentives schemes presently in operation in the UK do not appear to be working, given the failure to meet targets in two months out of three. Low motivation and morale also seem to be apparent. Studies argue, however, that financial reward can only have a limited effect on motivation (eg, Maslow, McGregor Theory X and Herzberg). Thus, while the form of remuneration should be reconsidered, other aspects of HRM should also be examined to try to improve morale and motivation at existing UK factories and to develop them at the North African factory if option 2 is selected.

Employee communication and counselling. Developing communication channels to and from individuals, groups and all employees collectively, but also participation in operations, such as communication procedures.

The top-down management approach may be partly responsible for the lack of motivation. The nature of the work, culture of instruction/imposition and the lack of opportunity to advance would all appear to reduce motivation for employees. There might therefore be a key role for HRM both in changing management style in existing plants and also in developing a new management style for the potential new plant. This could lead to increased motivation and improved output and efficiency.

A study by Blake and Mouton analysed management style within a two-dimensional matrix (or managerial grid). The two factors identified were: (1) concern for people; (2) concern for production or task.

In the context of this matrix, the management style at Grateley could be regarded as high in terms of (2) and low in terms of (1). A new style of participation might be regarded as more balanced between the two factors. Indeed, to the extent that participation may lead to greater motivation and increased production, it could be viewed as high in respect of both of Blake and Mouton's factors.

Studies (eg, the Hawthorne Experiments) have also indicated the potential for motivation through consultation. In the Hawthorne Experiments, the work carried out by staff was repetitive and boring. A series of changes were introduced after consultation with employees. After the changes, productivity rose in almost every case. The rise in productivity was as much due to the fact that employees had been consulted (and so felt appreciated) as due to the nature of the changes themselves.

A particular issue in Grateley's case will be the likely cultural differences between the UK and most North African countries. This may result in different types of motivation, different work methods and different managerial styles being appropriate.

Job design. Producing a job description and job specification in terms of experience, skills and education. In setting up a new workforce in North Africa, a major exercise in job design and specification will be necessary.

Training and development. Involves the analysis of training needs and the organisation of the provision of training and staff development to meet those needs. Includes new and existing staff, and all levels of management.

In terms of the new jobs in North Africa, training all new employees simultaneously is a major undertaking.

Even in the UK, however, the scale of the labour reductions under either option is likely to involve reorganisation of the workforce, which could have implications for training needs.

Compliance with legal and other standards. Involves informing and advising managers of employment, contract and other relevant law with respect to employees, and setting up procedures to comply with such legislation and other codes of conduct, agreements and ethical standards.

In North Africa, the law and codes of conduct are likely to be different to the UK and a learning process will be needed. Local expertise will be necessary in employment law.

Moreover, even if employment laws in North Africa are less strict than in the UK, in terms of being seen as a socially responsible company, Grateley will need to consider whether its own policies are more stringent than required by local law.

Other HRM issues are likely to include performance appraisal, disciplining employees, grievances and disputes, and workforce diversity.

The closure of one or more of the UK factories will also lead to specific HRM issues around redundancies. Grateley will need to be seen to act fairly and ethically in terms of any redundancies in order to protect the company's reputation and to avoid any legal claims being brought against it, particularly given the unionisation of the workforce.

4 Della Co

20X5	**£**
Share-based payment reserve c/f and P/L expense ((500 – 75) × 100 × £15 × 1/3)	212,500

	£	**£**
DEBIT Staff costs	212,500	
CREDIT Equity – share-based payment reserve		212,500

20X6	**£**
Share-based payment reserve b/f	212,500
∴ Profit or loss expense	227,500
Share-based payment reserve c/f ((500 – 60) × 100 × £15 × 2/3) =	440,000

	£	**£**
DEBIT Staff costs	227,500	
CREDIT Equity – share-based payment reserve		227,500

20X7	**£**
Share-based payment reserve b/f	440,000
∴ Profit or loss expense	224,500
Share-based payment reserve c/f (443 × 100 × £15) =	664,500

	£	**£**
DEBIT Staff costs	224,500	
CREDIT Equity – share-based payment reserve		224,500

Chapter 11
Finance awareness

Introduction

Learning outcomes

Knowledge brought forward

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Examination context

Chapter study guidance

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2 Impact of financial crises

3 Developments in the Eurozone

4 Other current issues in finance

5 Social responsibility and environmental matters

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Demonstrate and explain the financing alternatives for projects and assets, and make informed choices as to which alternative is the most compatible with the overall financial strategy of the entity, showing the corporate reporting consequences relating to presentation, disclosure, recognition and measurement of projects and their financing
- Assess and explain current and emerging issues in finance
- Identify social responsibility, sustainability and environmental factors for a range of financial stakeholders, including UN sustainable development goals, natural capital and green finance, and assess assurance and corporate reporting issues relating to such factors
- Explain how financial crises, which have occurred in the past over a long time period, may impact on approaches and attitudes to financial risk and may inform corporate reporting practice
- Explain the role of boards in determining and evaluating an entity's policy for social responsibility, sustainability and environmental matters and advise on corporate reporting disclosures relating to these policies
- Explain the role and impact of the finance function as a business partner

Knowledge brought forward

We start with a reminder of the key finance decisions that a business needs to make.

Syllabus links

The last part of section 1 is a very important part of the financial strategy section of this Workbook, since it relates the main requirements of financial reporting standards to their business context. This reflects one of the main themes of the syllabus, which is to cover the implications for corporate reporting of business decisions.

Examination context

The quick list of factors affecting the choice of financial strategy at the start of section 1 may be a useful checklist for questions.

Scenarios will include real-life issues and concerns related to finance, and also new developments in finance; the middle sections in this chapter set these scenarios in context.

The final section brings together corporate social responsibility issues that range across all aspects of the syllabus. You were introduced to these issues in Business Strategy & Technology, and some of them are mentioned briefly earlier in this Workbook, while certain topics receive more coverage in the chapters Investment appraisal and Ethics. Section 5 of this chapter addresses the possibility that a question could focus on social responsibility, including business strategy, costs, corporate reporting and assurance issues.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Financial Strategy** This chapter covers many of the major influences on financial strategy. They include recent developments, particularly the economic problems encountered in recent years. As an accountant, you will be expected to synthesise the different areas of knowledge you have, linking business and financial strategy decisions with their impact on corporate reporting.	**Approach** Go quite slowly through the first few paragraphs of section 1 as a lot of the themes about financial strategy and financial objectives summarised here will be developed later in the manual. Also remember that the SBM&L syllabus specification indicates that around 15–20% of the marks in the exam will relate to the corporate reporting implications of an entity's strategic decisions. This makes the sections on corporate reporting in the later part of section 1 particularly important, as they highlight the links between corporate reporting and strategic decision making. Being up to date with current developments in reporting standards is important. **Stop and think** What are the three key financial management decisions?	In the examination, you may be required to advise organisations on the financial strategy decisions they have to make, taking into account the most significant influences on those decisions. You may also be required to demonstrate the consequences for corporate reporting of the financial strategy decisions that organisations take. Finally, from this section you may be asked to explain the role and impact of the finance function as a business partner	**IQ1: Financial Strategy** This question considers the impact of business expansion on a company's financial strategy. **IQ2: Developments in financial reporting** This brief question asks you to consider the general impact that developments in financial reporting have on financial strategy.
2	**Impact of financial crises** Financial crises impact on all businesses as financial institutions	**Approach** You can go through this section more quickly, focusing on the impacts that the economic	You may be asked to discuss how organisations can respond to the current economic environment and	

CHAPTER 11

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	reduce lending to limit their own exposures and comply with stricter regulations. Businesses may be able to learn lessons from history to limit the damage to their own position.	environment has on business. **Stop and think** How much can businesses do to mitigate the consequences of a long-term economic slowdown?	the difficulties it places upon them.	
3	**Developments in the Eurozone** The financial position of many firms has been affected by the credit crunch and problems in the Eurozone, particularly the debt crisis. You will need to be aware of the business environment and how developments in the Eurozone are affecting (or will affect) the businesses you deal with.	**Approach** You can cover this brief section quickly. Further developments in the euro and the Eurozone is inevitable. You should monitor the financial news for developments. **Stop and think** Are you up to date with recent events? Make sure that you review the business press and other suitable sources to ensure that you are aware of topical developments in the global economic environment.	Scenarios in the exam will include real-life issues and concerns related to finance.	
4	**Other current issues in finance** This section covers some of the current issues in finance such as the credit crunch, dark pool trading and Islamic finance.	**Approach** Cover this brief section quickly, focussing mainly on understanding any new terminology. **Stop and think** What are the main advantages of Islamic finance?	Scenarios in the exam may include some new developments in finance.	
5	**Social responsibility and environmental matters**	**Approach** Corporate social responsibility is	In the exam you may be asked to discuss the impacts on	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	It is now very difficult to avoid businesses facing up to their responsibilities as corporate citizens, particularly in regard to carrying on business in a sustainable way. The UN Sustainable Development Goals are a notable example of how business activities will be closely scrutinised. Issues around accounting for 'natural capital' are also a highly topical area. The financial decisions businesses make will have often very significant impacts on the content of their corporate reports. This includes effects on the information disclosed in the accounts, as well as the additional reports that businesses publish. Today's accountants need to be fully versed in green finance, they need to be able to consider and report on the impact that strategic decisions may have on the environment.	significant to this syllabus. You need a good awareness of how stakeholder pressures may influence business decisions and the impact of sustainability on management control systems and corporate reporting. This was also examined in the chapter Strategic performance management. The examples of how companies are reporting CSR issues should aid your understanding in this area. Make sure that you note the developments in relation to natural capital. Note the links between this and Chapter 4: the attention being given to environmental and social responsibility means that entities are placing emphasis on their performance. **Stop and think** To what extent are the decisions of finance providers influenced by a business's corporate social responsibility stance?	financial stakeholders of the corporate social responsibility stances that organisations take and evaluate an organisation's corporate social responsibility policies.	

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Financial strategy

Section overview

- We begin coverage of the financial strategy element of the syllabus by briefly revising the key financial decisions a business must make and the interaction between them. All these decisions will be covered in more detail in later chapters, but you should keep them in mind throughout this section of the Manual.
- We also begin looking at the financial standards that cover areas affected by financial strategy decisions. Coverage of the standards is selective – we only include relevant financial reporting issues. Comprehensive coverage of all standards is contained in the Corporate Reporting Manual.

1.1 Financial strategy decisions

Definition

Strategic financial management: The identification of the possible strategies capable of maximising an organisation's net present value, the allocation of scarce capital resources among the competing opportunities and the implementation and monitoring of the chosen strategy so as to achieve stated objectives.

We discussed strategic objectives generally in the chapter Strategic analysis, but here we focus on financial objectives and the strategies for achieving them.

1.1.1 Financial objectives

It is often assumed that the overall financial objective of a company should be to increase, or even maximise, the wealth of its shareholders.

The financial strategies for achieving this objective involve strategies for investing and returns, and for dividends. If a company's shares are **traded on a stock market**, the wealth of shareholders is increased when the **share price goes up**. The price of a company's shares should go up when the company is expected to make **additional profits**, which it will pay out as dividends or reinvest in the business to achieve future profit growth and dividend growth.

Maximising the wealth of shareholders generally implies maximising profits consistent with long- term stability. Sometimes short-term gains must be sacrificed in the interests of the company's long-term prospects. In addition, to increase the share price the company should achieve its profits without taking excessive **business risks** and **financial risks** that worry shareholders.

1.1.2 Non-financial objectives

Non-financial objectives that may significantly impact on financial strategy include the following:

Non-financial objectives	
Customer satisfaction	A key target, because of the adverse financial consequences if businesses switch suppliers

Non-financial objectives	
Welfare of employees	Competitive wages and salaries, comfortable and safe working conditions, good training and career development
Welfare of management	Competitive salaries, company cars and other benefits
Welfare of society	Concern for environment
Provision of service to minimum standard	For example, regulations affecting utility companies (water, electricity providers)
Responsibilities to suppliers	Not unscrupulously exploiting power as buyer
Leadership in research and development	Failure to innovate may have adverse long-term financial consequences
Maintaining competitive position and market share	Preventing rivals from becoming too large and enjoying benefits of size such as economies of scale

Professional skills focus: Concluding, recommending and communicating

You are expected to be able to make evidence-based recommendations which can be justified by reference to supporting data and other information. When recommending an appropriate financial strategy, you must consider both the company's financial and non-financial objectives.

1.2 Factors affecting choice of financial strategy

The following general considerations will have significant impacts on the strategies chosen.

1.2.1 Profits

The effect of **strategies chosen on profits** will be a critical influence on how strategies are perceived, as a business's accounts present the public face of its financial strategies.

1.2.2 Cash

Though financial strategies should hopefully enhance long-term profitability, **cash flow considerations** will often be shorter term but possibly more pressing. It is easy to just concentrate on operating and investing cash flows and ignore financing cash flows (interest and dividends).
However, a critical, and often finely balanced, question will be whether operating cash flows will be sufficient to pay the servicing costs of finance.

1.2.3 Shareholders

Enhancing shareholder value is of course a key objective. Although this is connected with dividend policy, you also need to assess the effect on **value of shares** (which we shall see later is not just a function of current and future dividends).

1.2.4 Other stakeholders

Many financial strategies will have major impacts on stakeholders other than shareholders with whom the organisation wishes to maintain good relations. These include:

(a) **Managers and employees**, who will be affected by any fundamental decision about a business's direction such as a merger or acquisition.

(b) **Other suppliers of finance**, who will be affected by decisions on capital structure. Suppliers of finance such as banks and bondholders, together with the company's shareholders, are the main **financial stakeholders** in a company.

(c) The **Government**, who might be concerned that major investment decisions have an adverse impact on the outside environment (not just the natural environment but also the competitive environment).

(d) **Customers.** Businesses need to be sensitive not only to actions which directly affect customers (product pricing) but how other actions may indirectly impact (for example redundancies meaning that fewer staff are available to deal with customer demands).

1.2.5 Economic and market considerations

Business decisions are never taken in isolation in a perfect world. Therefore you have to evaluate the impact of factors such as **key economic indicators**. For this paper you also need to consider the **international implications**, including exchange rates and conditions for investment abroad.

In addition, the organisation's achievements may be assessed from distorted viewpoints that place excessive emphasis on certain factors and do not have all the necessary information. Thus an assessment of **market efficiency** will often be helpful. Assessing market efficiency involves considering the information to which the market is responding.

1.2.6 Restrictions

Bear in mind that a business may not always be able to do what it wants. **Legal** restrictions may be a powerful restraining force, as may restrictions imposed by **suppliers of finance** (restrictions imposed by loan arrangements, or the need to maintain dividend levels to keep shareholders happy).
Particularly for small companies, there may also be **restrictions** on the **funds available** to the business.

1.2.7 Risks

Remember that it is not necessarily just financial risk you are considering. Another area where risks should be considered is investment in information technology.

Firms also need to be aware of **risk issues** such as **limited data** and how the **changing environment** can be **measured**, and also the impact of **uncertainty** (how difficult it is to measure the likelihood of outcomes). **Various measures** of risk may need to be considered (possible outcomes, likelihood of an unsatisfactory outcome, worst possible outcome).

The organisation's (and its directors') **attitude to risk** will also be important. It may be a good textbook answer to say that the business should obtain debt finance, but the current directors may be unwilling to put the business at risk by taking on extra debt.

1.2.8 Timescale

We've already mentioned timescale, but it's worth considering separately:

(a) Strategy is primarily concerned with long-term direction. **Businesses** have to maximise shareholder value in the **long term**. However, you also should consider what objectives or conditions have to be fulfilled in the **short term** (such as making enough money to carry on trading!).

(b) Decisions relating to a particular strategy are not always made at a single time. In the chapter Investment appraisal, we shall look at **real option theory**, which takes into account that businesses may choose one course, and then later decide on a different course of action depending on how the first course of action has gone.

Professional skills focus: Assimilating and using information

When suggesting an appropriate source of finance, you will need to make an informed choice as to which alternative is most compatible with the overall financial strategy of the entity. You will also need to consider the corporate reporting consequences of the various financing options. It is important that you can filter the information provided in the scenario and select the most appropriate source of finance while giving consideration to the impact on the entity's various stakeholders.

1.3 Financial management decisions

We discussed the main decisions (**investment financing** and **dividend** decisions) in the chapter Strategic analysis. We shall look at investment decisions in greater detail in Chapters 16 and 17, sources of finance and financing decisions in chapters Financial instruments and financial markets and Financial structure and financial reconstruction and dividend strategy in the chapter Financial structure and financial reconstruction.

1.4 Financial strategy decision making

In the chapter Strategic choice, we revised the checklist for assessing strategic options, provided by Johnson, Scholes and Whittington:

- **Suitability**: Is this option appropriate considering the strategic position and outlook of the business?
- **Acceptability**: Will the option gain stakeholder support?
- **Feasibility**: Does the firm have the resources and competences required to carry out the strategy? Are the assumptions of the strategy realistic?

Here we shall look briefly at how these elements apply to financial strategy decisions. In the chapter Financial structure and financial reconstruction, we discuss in detail their application to the capital structure decision.

1.4.1 Suitability

Suitability relates to the **strategic logic** of decisions. Financial strategy should fit the situation of the business. The financial decisions taken should help businesses **generate and maintain competitive advantages**, **seize opportunities** and **exploit company strengths and distinctive competences**.

When considering the suitability of an option, you might also think about whether there are any better **alternatives** not specified in the question. For example, if a business is looking to acquire another, would it be better off trying to grow internally instead? It may be necessary to create a **framework** for ranking alternatives.

1.4.2 Acceptability

Shareholders will obviously be concerned with how far the chosen strategy contributes to meeting the dominant objective of increasing shareholder wealth. Other suppliers of finance will be concerned with whether the organisation is able to meet its commitments to them.

All stakeholders may be concerned with the potential **risks** of the strategies.

Financial acceptability will need to be **demonstrated** by **investment appraisal** and by **analysis** of the effect on **profitability**, **liquidity** and **gearing** indicators. These may need to be supplemented by risk analysis.

Do not forget the other conditions (legal, loan covenant etc) imposed by other stakeholders. Acceptability may not be just a matter of fulfilling a set of rules, but also being a good citizen (acting sensitively towards the natural environment, paying suppliers on time).

1.4.3 Feasibility

Specification of required **resources** is a key aspect of clear strategy exposition. These resources have to be available in order for strategies to be **feasible**. Obviously for financial strategy decisions, financial resources will be critical, but businesses will also need to have access to **technology**, **materials** and **other natural resources**.

Businesses must also be able to **generate sufficient returns** to compensate resource providers for the resources consumed. They will need to be able to **supply sufficient goods and services** and be able to withstand competitor threats.

Again it will be important to demonstrate **financial feasibility** with supporting financial data. **Cash and funds flow forecasts** will be central to this.

Feasibility also relates to the **conditions** that must be fulfilled if a strategic decision is to be successful. If, for example, the success of an acquisition depends on the key staff of the acquired company staying, how feasible is it that they will?

1.4.4 Clarity

Financial strategies must also be clear, which means they need to specify certain things.

Benefits

The benefits to the **organisation** and **principal stakeholders** of strategies need to be spelt out – for example, higher revenues or falling interest commitments meaning more funds available for distribution to shareholders.

Direction

The activities required to **implement the strategy** must be **directed** towards providing the benefits. Direction also means setting **financial targets**, enabling better control when the strategy is implemented.

Resource specification and management

The **financial resources needed** to implement investment strategies should be specified in detail. Remember also the need to ensure that **resources** are **managed carefully**. In your previous studies you encountered the concept of **overtrading**, where a business expands too quickly and fails to manage its working capital and cash flows properly.

Changes in the environment

The strategy chosen should enable the business to cope with **environmental changes** and **complexity**. If environmental changes are likely to be a significant factor, it may be important to specify the responses that may be required, for example obtaining contingency funding.

Timescale

The strategy chosen should be **relevant for the long term** that is going beyond short-term profit targets. What constitutes long term will vary, but for an investing strategy it is likely to mean the **lifetime of the investment**, and for a financing strategy perhaps the **duration of long-term debt instruments**.



Professional skills focus: Applying judgement

There is unlikely to be a financial strategy that is acceptable to all stakeholders. Use your judgement to identify the key stakeholders and recommend the most appropriate strategy to

satisfy their needs, whilst recognising any stakeholder conflict that may arise as a consequence.



Interactive question 1: Financial strategy

As a general illustration of the principles we've just discussed, consider the following outline scenario.

Getfitquick operates leisure clubs in a number of countries in Europe. The company is a major presence in the leisure club market, specialising in high care, good facility operations. However, this market is highly competitive and over the last few years a number of chains that have not been growing have been taken over. Getfitquick's performance over the last couple of years has been stagnant, with falling membership in existing clubs and newly opened clubs taking time to become profitable.

The company is listed on a major European stock exchange, with a significant percentage of its shares being held both by its founders and by institutional investors. It also has significant loan finance, with loan creditors holding title deeds to some of the clubs, and in the past some of its loan commitments have had to be rescheduled.

Requirement

Getfitquick is considering whether to open a number of new clubs over the next few years. Draft notes suggesting how an expansion strategy would be judged against the criteria outlined above.

See **Answer** at the end of this chapter.

1.5 Strategic cash flow planning

Strategic cash flow planning ensures that **sufficient funds** are available for investment and that **surplus funds** are used to best advantage.

In order to survive, any business must have an adequate inflow of cash. Cash flow planning at a strategic level is similar to normal cash budgeting, with the following exceptions:

(a) The **planning horizon** (the furthest time ahead plans can be quantified) is longer.

(b) The **uncertainties** about future cash inflows and cash outflows are much greater.

(c) The business should be able to respond, if necessary, to an **unexpected need** for cash. Where could extra cash be raised, and in what amounts?

(d) A company should have planned cash flows that are consistent with:

- its dividend payment policy
- its policy for financial structuring, debt and gearing

1.5.1 New investments and product developments

Investments in new projects, such as new product developments, use up cash in the short term. It will not be for some years perhaps that good profits and cash inflows are earned from them.

One aspect of strategic cash flow planning is to try to achieve a balance between the following:

(a) **Making and selling products** that are still in their **early stages of development**, and are still 'soaking up' cash

(b) **Making and selling products** that are **cash cows** – ie, established products that are earning good profits and good cash inflows

1.5.2 Cash surpluses

A company should try to plan for adequate cash inflows and be able to call on 'emergency' sources of cash in the event of an unforeseen need, but it might be unwise to hold too much cash.

When a company is **cash rich** it can invest the money, usually in short-term investments or deposits such as the money market, to earn interest. However, for companies which are not in financial services or banking, the main function of money is to be spent. A cash-rich company could do one of the following:

(a) **plan to use the cash**, for example for a project investment or a takeover bid for another company

(b) **pay out the cash** to shareholders as **dividends** and let the shareholders decide how best to use the cash for themselves

(c) repurchase its own shares

1.6 Strategic fund management

Strategic fund management involves asset management to make assets available for sale if cash deficiencies arise.

Strategic fund management is an extension of cash flow planning that takes into consideration the ability of a business to overcome unforeseen problems with cash flows, recognising that the assets of a business can be divided into three categories:

(a) Assets that are needed to carry out the **'core' activities** of the business. A group of companies will often have **one or several main activities**, and in addition will carry on several peripheral activities. The group's strategy should be primarily to develop its main activities. There has to be enough cash to maintain those activities and to finance their growth.

(b) Assets that are **not essential for carrying out** the main activities of the business and could be **sold off** at **fairly short notice**. These assets will mainly consist of short-term marketable investments.

(c) Assets that are not essential for carrying out the main activities of the business and could be **sold off to raise cash**, although it would probably take time to arrange the sale and the amount of cash obtainable from the sale might be uncertain. These assets would include: long-term investments (for example, substantial shareholdings in other companies); subsidiary companies engaged in 'peripheral' activities, which might be sold off to another company or in a management buyout; and land and buildings.

If an unexpected event takes place which threatens a company's cash position, the company could also meet the threat by:

(a) **working capital management** to **improve cash flows** by reducing stocks and debtors, taking more credit, or negotiating a higher bank overdraft facility

(b) changes to dividend policy

1.7 The finance function as a business partner

Business partnering is about contributing to organisational performance. The involvement of finance professionals in business decision making, strategy development and driving performance is not something that is new. However, involving the finance function as a business partner looks at whether finance professionals can make a valuable contribution to

decision making and successfully contribute to organisational performance outside of core work such as financial reporting. In smaller organisations it is almost impossible for the most senior finance professional not to be a business partner. They will be directly involved in making sound finance, investment and dividend decisions with the focus of increasing shareholder wealth. In larger organisations, finance professionals are more likely to be centralised at headquarters and are often regarded as collators of reports and restrictors of innovation by imposing controls on capital spending and setting budgets.

Decentralising the finance function and locating accountants close to operational managers can increase the acceptance of the finance function as a business partner. Embedding finance in all functions will lead to accountants taking a more active role and becoming involved in decision making and strategy development. Business partners are better positioned to support businesses when they report to local managers and are located alongside them; they will bring an expert financial perspective which may challenge other perspectives.

Finance business partners will need to understand the organisation's strategy, competitive environment, commercial drivers, systems and culture. Labelling the finance function as a business partner can help to break down the stereotypes of what accountants and the finance department do. For example, rather than simply identifying that a proposed investment project does not reach the required hurdle rate to be accepted, finance business partners will work with the business to find ways of increasing the return and keeping an organisation focussed on its strategy. The finance function as a business partner will consider the key problems facing an organisation and use financial knowledge and techniques to suggest solutions. For example, activity-based costing and customer profitability analysis to look for cost efficiencies, target costing to make products more competitive and consideration of all available sources of finance to overcome financial constraints. To be successful, finance business partners will need to constantly adapt to the needs of the organisation and consistently demonstrate a positive impact on organisational performance.

Source: https://www.icaew.com/-/media/corporate/files/technical/business-and-financial-management/finance-direction/business-partnering.ashx?la=en [Accessed 10 July 2019]

1.8 Corporate reporting implications of investment and financing decisions

The investment and financing decisions will have a number of consequences for the contents of its accounts. In this chapter we provide an overview of how investment and financing decisions relate to the requirements of financial reporting standards. In later chapters, where relevant, we will cover some of the corporate reporting consequences in more detail. The consequences of many decisions will be significant for the financial statements, equity and profits of a company, affecting in turn analysts' measures of return and gearing that investors will use to make financial decisions themselves.

As mentioned in the Section overview at the start of this section, the Corporate Reporting Workbook discusses fully the accounting issues that this Strategic Business Management Workbook treats selectively. This section also revises topics that were covered in the Financial Accounting and Financial Reporting *Workbook*s.

For the purposes of this chapter, we shall group the accounting requirements under the following headings:

- Investment in entities
- Other investment issues
- Financing
- Other issues

Professional skills focus: Structuring problems and solutions

One of the professional skills assessed in the CA exams considers your ability to identify and apply relevant technical knowledge and skills to analyse a specific problem. The SBM&L syllabus specification indicates that around 15–20% of the marks in the exam will relate to the corporate reporting implications of an entity's strategic decisions. It is therefore important that you can identify relevant corporate reporting issues from a given scenario and apply your technical knowledge to resolve any financial reporting issues you are presented with.

1.9 Investment in entities

The economic reality is that many businesses take on complex forms, often establishing themselves as a number of companies that act coherently, or are centrally directed and managed. Large groups of companies emerge. Investors need to know, if their investments are at group level, exactly how the group as a whole is performing. Financial statements normally set out the financial position and performance of a single entity. If that entity is controlled by another entity (its parent) and there are intragroup transactions, then its financial statements may not reveal a true picture of its activities. As a consequence, consolidated financial statements are prepared by aggregating the transactions, assets and liabilities of the parent and all its subsidiaries on the basis that the group is a single economic entity.

Without consolidated financial statements, the shareholders in the parent entity would receive its single entity financial statements, which would recognise the income of subsidiaries only to the extent of the dividends receivable from them. The information provided to such shareholders would not reflect the economic realities. Consolidated financial statements provide useful information about all the activities carried out by the management of the parent entity. Their preparation is an application of the IASB's *Conceptual Framework* requirement that transactions are accounted for in accordance with their substance, not just their legal form.

The introduction to IFRS 12, *Disclosure of Interests in Other Entities* highlights the importance that financial statement users place on disclosure of interests in other entities to help identify the profit or loss and cash flows available to the investor company.

As you will remember from earlier studies, the extent of **control** that an investor has over its investment will determine how it is treated in the investor's accounts.

1.9.1 Subsidiaries

If an investment has subsidiaries, it is subject to the requirement of IFRS 10, *Consolidated Financial Statements* to prepare consolidated financial statements. A parent-subsidiary relationship is based on control, which an investor has if it has:

(a) **power over the investment –** existing rights that give the investor the ability to direct the activities of the investment that significantly affect its returns; these may come through voting rights, the power to govern financial and operating policies or the power to appoint and remove a majority of the board of directors

(b) **exposure or rights to variable returns** from the investment

(c) ability to **use the power over the investment to affect the returns it obtains**

1.9.2 Joint arrangements

Investments can take a number of different forms. It is not always appropriate to acquire the majority of the voting rights of another entity in order to gain control. In some industries and in particular circumstances, it is more beneficial to share such investment and control with other parties. By sharing the investment each investor brings with them different skills. Consequently

the arrangement may benefit all parties through reduced costs. Such arrangements are commonly known as joint ventures.

If a firm makes an investment jointly with another party, it may be subject to the requirements of IFRS 11, *Joint Arrangements* to recognise the rights and obligations arising from the investment.

The standard only applies to **joint arrangements** where the parties have **joint control**. Joint control is the contractually agreed sharing of control where strategic decisions require the unanimous consent of the parties sharing control (majority control, where decisions require the consent of a majority of owners, is not joint control).

Joint arrangements are of two types, depending on the rights each party has:

(a) **Joint operations** include, but are not confined to, arrangements that are not structured through a separate entity (a separately identifiable financial structure, for example a limited liability company). The parties share their activities and pool their resources. In joint operations the parties provide and have rights to their own assets, and obligations for the liabilities relating to the arrangement. The accounts include the reporting entity's own assets and liabilities, its share of the revenues and the expenses it incurs relating to its interest in the joint operation (known as gross accounting).

(b) **Joint ventures** are arrangements where the parties that have joint control have rights to the net assets of the arrangement. The arrangement holds assets and incurs liabilities on its own account and the parties to the arrangement are only liable to the extent of their investment. Each party treats its interest in the joint venture as an investment in the arrangement using the equity method of accounting, in accordance with the requirements of IAS 28, *Investments in Associates and Joint Ventures*.

1.9.3 Associates

Outright control is not always the most appropriate form of investment for an entity to have. There may be circumstances where although an entity does not control another entity, the business operated by that other entity is still of significant importance to it. In such circumstances an investor may obtain sufficient ownership of the entity to have the power to **influence** decisions of its governing body (eg, its board of directors) but not to have control over it.

Investments that meet these criteria will generally be classed as associates. As the investor has significant influence over such an investee, it is appropriate to report its share of the investee's results rather than just the dividends receivable. After all, it is partly answerable for the investee's performance. Such accounting requirements provide more useful information about the financial performance and position of the investor.

Interests in associates are common in real life, particularly where a controlling interest is built up over time rather than being purchased in one step. As you will remember, the accounting treatment of associates differs from subsidiaries, even if the investor eventually wishes to gain control.

IAS 28 also applies to investments in associates. Remember that holding 20% of voting power creates a presumption of significant influence.

The equity method means that the investment will be recognised initially at cost and then adjusted for the **post-acquisition change** in the share of the investment's net assets. Distributions will reduce the carrying value of the investment.

A firm may make an associate investment in a business whose viability and success will play an integral part in an entity's own success. For example, where an entity uses distributors or agents in foreign countries, it may wish to take an equity stake in them. This strengthens the relationship between the parties and allows a level of influence and communication greater than could be obtained through a contractual agreement alone. Alternatively, a non-

controlling stake may be the initial stage of a larger plan to obtain control of the target entity. The investor is able to gain board representation and strengthen its business relationships with the target. If the relationship is successful, further investment can be facilitated.

1.9.4 Business combinations

Entities may increase their market share, diversify their business or improve vertical integration of their activities in a number of ways, including by organic growth and through acquisitions. The acquisition of a competitor may offer rapid expansion or access to new markets and is often seen as more attractive than expansion through organic growth.

To ensure that users of financial statements are able to distinguish between organic growth and growth through acquisition, detailed financial reporting requirements should be applied and comprehensive disclosures should be presented.

The process of acquisition will mean that the investor is subject to IFRS 3, *Business Combinations*. The accounting consequences will be that the investor recognises the assets acquired and liabilities assumed, including those that may not have been recognised by the entity being acquired, such as brand names. Since the assets and liabilities being acquired are recognised at **fair value**, the investment is included at an amount that reflects the market's expectation of the value of its future cash flows.

We shall look at the requirements of this standard in more detail in the chapter Business and security valuation.

1.9.5 Discontinued operations

A key business event arises when an entity closes or discontinues a part of its overall business activity. Management will have assessed the impact of the closure on its future profitability, but users of the financial statements, be they investors or other stakeholders, will want to make their own assessment. IFRS 5, *Non-current Assets Held for Sale and Discontinued Operations* provides an analysis not of future profit but of the contribution of the discontinued element to the current year's profit (or loss), ie, the part that will not be included in future years' profits.

Showing separate information about discontinued operations allows users of financial statements to make relevant future projections of cash flows, financial position and earnings-generating capacity.

In addition, an asset is held for sale when the entity does not intend to use it as part of its ongoing business, but instead intends to sell it. The separate identification of assets that are held for sale, rather than to generate continuing economic benefits for the entity on an ongoing basis, substantially improves the information made available to users as it provides information on the entity's plans and likely future performance.

If at the end of the accounting period some business assets are no longer needed, they will be classified as held for sale in accordance with IFRS 5 if their carrying amount will be recovered principally through a sale transaction, they are available for immediate sale and the sale is highly probable. Abandoned assets cannot be classified as held for sale. If the assets fulfil the definition of held for sale, they will be subject to the requirements of IFRS 5 and not depreciated. They will be held at the lower of fair value less costs to sell and carrying amount.

1.9.6 Disclosure of interests

Investments in any subsidiaries, joint arrangements, associates or unconsolidated structured entities will mean that the investor has to apply the requirements of IFRS 12. As well as interests (including non-controlling interests in subsidiaries) IFRS 12 has other disclosure requirements depending on the nature of the investments. These include disclosure of restrictions on assets, liabilities and the movement of funds and details about the nature of, and changes in, risks associated with the investments.

1.9.7 Separate financial statements

In jurisdictions where an entity must present separate financial statements in addition to the consolidated financial statements, or if an entity chooses to do so voluntarily, the requirements of IAS 27, *Separate Financial Statements* apply. The entity making the investment in subsidiaries, joint ventures or associates must account for these:

- at cost;
- in accordance with IFRS 9, *Financial Instruments;* or
- using the equity method specified in IAS 28

1.9.8 Segments

Most large entities sell different products and services in a number of different markets or geographical locations, which may be regarded as different business segments. The total profitability of the entity will depend on the performance of each of these segments. For some entities the key segments will be based on products and services; for others it will be by geographical area. In each case, the separate management and performance measurement of individual segments is essential, as while one product or geographical area may be performing well, another may be failing.

While management will have access to management accounting performance data on each separate part of the business, the published accounts, in the absence of segmental reporting, would only report the aggregate performance of the entity. Overall profitability is important, but additional information is needed by external users to fully understand the separate businesses hidden below this top-level reporting.

IFRS 8, *Operating Segments* provides a link between the business operations and the main components of the financial statements by requiring information to be disaggregated. Investors can therefore make better assessments of the performance of each part of the business, leading to a better understanding of the business as a whole.



Definition

Operating segment: This is a component of an entity:

(a) That engages in business activities from which it may earn revenues and incur expenses (including revenues and expenses relating to transactions with other components of the same entity)

(b) Whose operating results are regularly reviewed by the entity's chief operating decision maker to make decisions about resources to be allocated to the segment and assess its performance

(c) for which discrete financial information is available

Significant new investments, for example a large diversification into a new geographical area or product, may mean that a listed company becomes subject to the requirements of IFRS 8. The standard will apply if there is separate financial information available about the investment and senior management regularly evaluates it as part of the overall process of deciding how to allocate resources and also when assessing performance.

Under IFRS 8 information must be given about determination of the operating segments, the products and services they provide, profit or loss, significant income and expense items and segment assets and liabilities.

1.10 Other investment issues

Even if a company does not invest in another entity, a major investment may include substantial investment in non-current assets.

1.10.1 Tangible assets

Businesses operating in certain industries, for example manufacturing, typically have ownership of substantial property, plant and equipment (PPE). PPE is tangible assets such as freehold and leasehold land and buildings, plant and machinery, fixtures and fittings, which are held for use in the production or supply of goods or services or for administrative purposes. The management of these resources underpins the continued viability of businesses and therefore represents a key feature of business prosperity. It is important that users of financial statements understand how businesses use their PPE and how such assets are accounted for.

Any investment in tangible assets will be subject to the requirements of IAS 16, *Property, Plant and Equipment* relating to capitalisation, depreciation, revaluation and measurement of fair value and impairment. In order to be capitalised, it has to be probable that future economic benefits associated with the item will flow to the entity. Therefore if the asset purchase is part of a project, the entity has to expect that the project will be successful.

Under IAS 16 costs include:

- **purchase price**
- **costs directly attributable to bringing the asset to the location and condition required for operation** including employee costs in construction or acquisition, site preparation costs, initial delivery and handling costs, installation and assembly costs, testing costs and professional fees
- **costs of dismantling and removing the item and site restoration costs**

Depreciation policy will depend on the use to which the asset is put, based on the expectations of how economic benefits will be derived from it. Factors that will determine the useful life that is used in calculating depreciation include:

- expected usage of the asset
- expected physical wear and tear
- possible technical or commercial obsolescence
- limits on the use of the asset

1.10.2 Intangible assets

Brand names, such as Coca-Cola and Microsoft, are in many cases an entity's most valuable asset but they are extremely difficult to value when they have been generated internally and over a long period of time. Brands are one example of an intangible asset. If they cannot be reliably measured, they cannot be recognised in the financial statements. Such a problem calls into question the quality of information financial statements contain and, consequently, potentially diminishes the power of financial statements in providing information for a business on which economic and investment decisions are made.

However, an entity that has acquired, as opposed to internally generated, an equally valuable brand will recognise it, since a fair value can be attributed to it. An intangible asset may be recognised by the acquirer **that was not recognised by the acquiree,** for example an internally generated brand may be recognised by the acquirer. This is because the acquisition provides sufficient evidence that:

- There will be future economic benefits attributable to it – otherwise, why would the acquirer buy it?

- The cost can be measured reliably; the acquirer will have built up the total purchase consideration by estimating values (= costs) for each asset.

This inconsistent treatment has led to difficulties in valuing entities and assessing their performance.

An investment in a major IT project, for example, may involve investment in intangibles such as computer software and licences. These will be subject to the requirements of IAS 38, *Intangible Assets*. They can only be carried on the statement of financial position if it is probable there will be future economic benefits from the assets.

Their useful life may be determined by the life of the investment. It may be difficult to decide whether the firm has control of some intangible assets and it may also be problematic to assess their fair value or impairment, particularly if there is no active market for them and they are part of a specific investment.

1.10.3 Impairment of assets

Asset availability and usage in a business is one of the key drivers in business success. Any assessment of assets, therefore, should consistently reflect their worth to the business and financial reporting of their value should be accurate, particularly when there exists the possibility that an asset may have diminished in value. At the very least, an asset recognised in the statement of financial position should not be reported at a value above the amount that could be recovered from it. During periods where general prices increase there is an assumption that recoverability of the reported value of such assets will not be an issue. However, it is important that all the assets are considered in relation to business decisions that are made by the entity. For example, some intangible assets which are highly technical in nature may attract premium valuations which may not be recoverable as technology continues to be developed or competitors enter the market.

Assets will be subject to the requirements of IAS 36 *Impairment of Assets* if their carrying amounts exceed the amount expected to be recovered from their use or sale. The firm must reduce the carrying amount of the asset to its recoverable amount and recognise an impairment loss.

The recoverable amount is the higher of fair value less costs of disposal and value in use.



Definition

Value in use: The present value of the future cash flows expected to be derived from an asset or cash-generating unit.

The following elements should be reflected in the calculation of an asset's value in use:

- An **estimate of the future cash flows** the entity expects to derive from the asset
- Expectations about **possible variations in the amount or timing** of those future cash flows
- The **time value of money** (represented by the current market risk-free rate of interest)
- The price for bearing the **uncertainty inherent in the asset**
- **other factors** that market participants would reflect in pricing the future cash flows the entity expects to derive from the asset

The standard could therefore apply if the expected future cash flows associated with the asset are worse than was forecast when the initial investment was made. The cash flow projections used to assess whether impairment has occurred should be the most recent management-approved budgets or forecasts, generally covering a maximum period of five years. The net cash flows expected to arise on the asset's ultimate disposal should be taken into account.

Indicators of impairment include the following:

External

- Significant decline in market value of the asset below that expected due to normal passage of time or normal use
- Significant changes with an adverse effect on the entity in:
  - technological or market environment
  - economic or legal environment
- Increased market interest rates or other market rates of return affecting discount rates and thus reducing value in use
- Carrying amount of net assets of the entity exceeds market capitalisation

Internal

- Evidence of obsolescence or physical damage
- Significant changes with an adverse effect on the entity include:
  - the asset becoming idle
  - plans to discontinue/restructure the operation to which the asset belongs
  - plans to dispose of an asset before the previously expected date
  - reassessing an asset's useful life as finite rather than indefinite
- Internal evidence available that asset performance will be worse than expected

1.11 Financing

A number of standards are linked to the decisions managers make about how operations and investments should be financed.

1.11.1 Financial instruments

All companies have financial instruments in their statements of financial position. Receivables and cash are, after all, financial instruments. In particular, there are trillions of dollars of financial instruments outstanding in the financial markets. Prior to the standards on financial instruments, many were off balance sheet until the gain or loss became realised. Users can now better judge the impacts on future profits and liquidity and the risks businesses face as a result of the following standards being implemented:

- IAS 32, *Financial Instruments: Presentation*
- IAS 39, *Financial Instruments: Recognition and Measurement**
- IFRS 7, *Financial Instruments: Disclosures*
- IFRS 9, *Financial Instruments*

Note: IFRS 9 is now the examinable standard, but an awareness is needed of the hedging requirements of its predecessor, IAS 39, as these hedging rules can be applied for a portfolio of financial assets and liabilities and as an accounting policy choice.

Relevant requirements of these standards are covered in more detail in chapters Financial structure and financial reconstruction to International financial management, but for now you should keep the following principal points in mind.

(a) The **requirement to split financial instruments into financial assets, financial liabilities and equity instruments**, with the distinction between liabilities and equity being that liabilities are instruments that embody an obligation to deliver cash whereas equity instruments contain residual interests in net assets.

(b) **Classification as a liability or equity** determines whether interest, dividends, losses or gains are treated as income and expense (liability) or as changes in equity.

(c) **Special hedge accounting requirements**, recognising the offsetting effects of changes in value of a hedged item and an instrument that hedges it may apply to hedges of asset or liability fair value, hedges of variable cash flows and hedges of net investments.

(d) Under IFRS 7, the firm needs to explain the **significance of financial instruments** for its financial position and performance, and provide qualitative and quantitative disclosures about exposures to risk and risk management.

1.11.2 Foreign investment and finance

Business is becoming increasingly international in terms of trading goods and services and in the operation of capital markets. One measure of the significance of globalisation is that most developed countries have external trade in the range 15% to 30% of their gross domestic product.

International activity can vary enormously from relatively straightforward import and export transactions through to financing arrangements in multiple currencies or maintaining operations overseas, for example, in the form of a subsidiary or branch.

Operating in multi-currency locations presents a number of accounting challenges, including:

- **conversion** – accounting for transactions where one currency has been physically changed into another currency
- **translation** – restating assets/liabilities initially recognised in more than one currency into a common currency
- **exchange gains and losses** – where relative currency values change, gains and losses arise which need to be appropriately measured and accounted for

Carrying on any trade abroad, making investments or seeking finance in other currencies will mean that the firm is subject to IAS 21, *The Effects of Changes in Foreign Exchange Rates*. We looked at some requirements of this standard in the chapter Strategic choice and will return to it again in the chapter International financial management. For now, the standard allows exchange differences on monetary items (loans) relating to a net investment in a foreign operation to be treated as other comprehensive income in the statement of comprehensive income. They are treated as a component of equity and only taken to profit or loss on disposal of the investment.

1.11.3 Borrowing costs

Inventories that require a number of manufacturing processes and the construction of non-current assets such as manufacturing plant and investment properties can take a significant time to complete. These activities may be financed by borrowings on which the entity incurs finance costs during the manufacturing/construction periods. These finance costs can be considered as part of the cost of the asset.

The requirements of IAS 23, *Borrowing Costs* may apply to this type of investment (qualifying asset). If borrowing costs can be directly attributed to the acquisition, construction or production of a qualifying asset, then the borrowing costs should be **capitalised**. Borrowing costs eligible for capitalisation are those that would have been **avoided** otherwise. Judgement will be required when the firm uses a range of debt instruments for general finance. In these circumstances the amount of borrowing costs to be capitalised should be calculated by reference to the weighted average cost of the general borrowings. When the qualifying asset is ready for its intended use or sale, any remaining borrowing costs intended to finance it are treated as part of the general borrowings pool.

1.11.4 Leasing

Businesses may obtain financing from a number of different sources. Such financing arrangements may vary significantly in nature, from a simple bank overdraft to a complex sale and leaseback transaction.

Leases can be a major source of finance to a business. The accounting treatment adopted must provide sufficient information for users of the financial statements to be able to understand the substance of such transactions. The accounting treatment for leases previously caused much debate among national standard setters, with important issues such as gearing and off balance sheet (ie, unrecognised) financing at the centre of the debate.

IFRS 16 abolished the distinction between finance leases (where substantially all the risks and rewards of ownership of an asset are transferred to the lessee) and operating leases (where those risks and rewards remain with the lessor) for lessees, although not for lessors. Previously, when a lease was deemed to be a finance lease, the asset and liability were recognised in the financial statements of the lessee. When a lease was deemed to be an operating lease, lease payments were recognised in profit or loss, but no lease obligation was recognised – making the transaction 'off- balance sheet'. IFRS 16 introduced a single lessee accounting model and requires a lessee to recognise assets and liabilities for all leases with a term of more than 12 months, unless the underlying asset is of low value.

Leasing is covered in more detail in the chapter Financial instruments and financial markets.

1.12 Other issues

A number of other standards may be of general relevance, whatever strategic decisions the firm makes.

1.12.1 Fair values

Investments and financial liabilities are now subject to the requirements of IFRS 13, *Fair Value Measurement*.

Generally fair value should be determined on a market basis, making maximum use of observable market prices. It may alternatively be determined using a cost approach or income approach. The cost approach reflects the amount that would currently be required to replace the service capacity of an asset. A number of techniques could be employed if the income approach is used, for example present value techniques, option pricing models and the multi-period excess earnings model.

1.12.2 Events after the reporting period

In assessing business performance, pertinent information sometimes arises following the cut-off date for which financial statements are prepared and may have important implications for judging financial position and performance in the year just gone. The end of the reporting period is a cut-off date and events that happen after this point in time should not generally be recognised in the financial statements of the period just ended. However, information that comes to light after the reporting date sometimes provides additional information about events that actually occurred before the end of the reporting period, in which case it is appropriate to recognise it.

The objective is to prepare a set of financial statements that reflect the most up to date information about events that existed at the end of the reporting period. However, it is sometimes difficult to establish whether the event happening after the end of the reporting period is new information about an existing event or a new event. Guidance is therefore provided to ensure there is consistency of treatment of such events across all financial statements.

IAS 10, *Events After the Reporting Period* includes a number of examples of investment and financing issues for which accounts would need to be adjusted or for which disclosure would need to be made. These disclosures provide investors with additional important information that may significantly impact on investment decisions.

Examples of issues which would require accounts to be adjusted include:

- subsequent evidence of impairment of assets
- subsequent determination of the costs of assets

Examples of events requiring disclosure include:

- a major business combination
- announcing a plan to discontinue an operation
- major purchases of assets
- destruction of assets
- abnormally large changes in asset prices or foreign exchange rates

1.12.3 Provisions and contingencies

One of the key creative accounting devices used in the past by businesses that wish to manipulate their financial results has been to 'smooth' earnings and thus provide a false indication that the business was more stable than was the case. One way this has been undertaken in the past has been through the creation of a provision. For example, in periods where performance exceeded expectations an entity might be tempted to make what has been commonly referred to as a 'rainy day' provision. The provision set up during prosperous times would be released in periods where results were not quite up to expectations. This provided management with some flexibility over the smoothing of results. Also a number of unrelated provisions could be grouped together to form a 'big bath' provision which provided some flexibility over its release or reversal.

As a consequence IAS 37, *Provisions, Contingent Liabilities and Contingent Assets* restricted entities from making large 'general' provisions which can have a significant impact on the results of an entity. Guidance is provided on the type of provisions that can be made and on the general principles surrounding recognition. In so doing, the requirements of IAS 37 limit the provisions that can be made.

No provision can be made for future losses of an investment, as these do not represent obligations of the firm at the period end. However, expectations of losses may be an indication of an impairment of value of assets used in an investment.

If the firm is to undertake a major restructuring of an investment, it can only make a provision for that restructuring if it has a detailed formal plan and has either started to implement the plan or announced its main features to those affected.

Contingent liabilities or assets that relate to investments should be disclosed in the accounts.



Interactive question 2: Developments in financial reporting

What general impacts might developments in financial reporting have on financial strategy?

See **Answer** at the end of this chapter.

2 Impact of financial crises



Section overview

Inevitably, financial crises impact on other businesses as financial institutions reduce lending to limit their own exposures and comply with stricter regulations. However, businesses may be able to learn lessons from history to limit the damage to their own position.

2.1 Impact of financial crises

Many financial crises are not only followed by a period of retrenchment but a period of increased regulation, reining back what has become regarded as excesses, particularly in the financial sector. The Wall Street crash resulted in the Glass-Steagall Act of 1933, limiting commercial bank activities in securities markets and affiliations between commercial banks and securities firms. The Enron crisis prompted the Sarbanes-Oxley Act of 2002 in the US, increasing compliance costs for internal controls and, some argue, reducing flexibility in corporate decision making. In the UK, the financial crisis of 2007–2008 prompted the Vickers report (2011) and subsequent legislation, the Banking Reform Act 2013.

The Banking Reform Act:

(a) introduces a 'ring-fence' around the deposits of individuals and small businesses, to separate retail banking from their wholesale and investment operations, including the trading floor (and in doing so, protecting taxpayers when things go wrong)

(b) imposes higher standards of conduct on banks by means of a criminal sanction for reckless misconduct that leads to bank failure

2.2 Impact on business of the global financial crisis

As banks became uncertain about the credit status of other banks during the financial crisis of 2007– 2008, they became reluctant to lend to each other without security, even short term, and the interbank market – which had been very liquid – dried up for a time.

Banks also came under pressure from financial regulators to strengthen their balance sheets and improve their capital base. A consequence was that banks became more reluctant to lend to business, contributing to economic recession.



Context example: The impact of low interest rates

Even though the Bank of England raised interest rates in early August 2018 by 0.25% to 0.75%, a report from the Institute for Public Policy Research (IPPR) had already suggested that monetary policy would be unable to fulfil its normal function should the economy falter. In all three of the last recessions dating back to the early 1980s, interest rates were cut by 4.5%–5% in order to sustain economic demand. In the downturn following the most recent financial crisis, the Bank had to go a step further with additional stimulus from quantitative easing, pumping £445 billion into the economy by buying government bonds from the financial industry to help consumers and companies keep on spending. The effect of QE is to create new money and increase the money supply in the UK, in the hope of stimulating spending and encouraging banks to lend money.

The IPPR said an interest rate cut of that size would not be available now, given the current low level of rates. Meanwhile, quantitative easing would be unreliable because it boosted the wealth of homeowners and shareholders at the expense of pensioners and young people renting homes.

Source: https://www.theguardian.com/business/2018/apr/22/bank-of-england-dangerously-ill- equipped-for-next-recession-says-ippr [Accessed 10 August 2018]

2.3 Inflation and interest rate rises 2021–2023

Monetary policy and rising interest rates became a topical issue again at the beginning of the 2020s.

Increases in demand triggered by fiscal stimulus measures to help economies recover from Covid- 19, coupled with supply chain problems in the aftermath of the pandemic, led to increasing inflation from the middle of 2021. This was then exacerbated by Russia's invasion of Ukraine at the beginning of 2022. Russia is a major supplier of oil, gas and metals, whilst Ukraine is of wheat and corn. The reduced supply of these commodities has driven their prices up, resulting in a knock-on effect across the supply chain.

To try to control inflation, central banks have tightened monetary policy and raised interest rates. The rationale for raising interest rates is to reduce demand in the economy (and to encourage consumers to save rather than to spend). So, rising interest rates can hurt businesses by reducing demand for goods and services. For example, for households, the interest rate rises have made it much more expensive to borrow for major purchases (such as cars, furniture or electrical appliances) so demand for these would be expected to fall.

Rising interest rates can affect business in other ways:

Reduced ability to service debt – Rising interest costs with no corresponding increase in revenues to offset reduce a company's ability to service debt. Increasing the amount of capital they have to use for paying off debt instead of investing in new products or services is likely to reduce their potential for growth. And, ultimately, if companies have to use too much of their capital for paying off highinterest debt they could face a higher risk of collapsing.

Context example: Real life example

In June 2023, Thames Water, the UK's largest water company – which serves a quarter of the UK population – announced it was in urgent talks to secure additional funding amid fears that it could collapse.

Central to the company's problems is the amount of debt it has, which is estimated to be around £14 billion. Interest payments on more than half of this debt are linked to the rate of inflation, so as inflation has soared in 2022–2023 the company has struggled to generate the cash it needs to cover its interest payments.

Following claims of underinvestment in the company, in 2022, Thames Water's owners – a consortium of institutional investors – put £500m into the business and pledged a further £1bn to help turn things around. But investors are reluctant to take on the risk of further investment due to fears it will not be repaid.

Since 2016, the company's profits have not covered the cost of paying interest on its debt, investment costs and dividend payments.

In addition to the concerns about Thames Water, Ofwat (the water industry regulator in the UK) has also said it is concern about other water firms, which are facing similar pressure due to higher interest payments on their debts, as well as rising costs including higher energy and chemical prices.

Source: BBC, (2023) Thames Water in urgent funding talks amid fears of collapse, 29 June, Available online at: https://www.bbc.co.uk/news/business-66039170 [Accessed 10 July 2023]

***Slower growth** – As we have already mentioned, higher interest rates are expected to reduce consumer demand for goods and services, thereby slowing demand-led growth. However, higher interest could also reduce companies' own opportunities to expand. Accessing capital is often vital for companies when they are looking to expand and grow, but when interest rates rise, the cost of that capital could become prohibitive. (For example, higher interest rates will mean higher discount factors are used in investment appraisal, which could lead to the net present value of potential projects becoming negative.)*

Lower valuations *– Higher interest rates also lower companies' net worth by cutting equity valuations, as future cashflows are discounted with higher discount factors. In turn, the reduction in net worth reduces the amount of collateral companies can offer up for their loans. Consequently, lenders could perceive there to be greater risk in lending to the companies, so lenders will charge a premium to compensate for the extra risk, raising borrowing costs. This will translate into fewer loans being approved or taken out. Tighter lending standards could have a particularly significant impact on businesses with little collateral to offer up as insurance against default. This tends to mean, smaller, younger firms.*

Asset values *– Rising interest rates will also increase a company's weighted average cost of capital which could pose a concern to companies in their year-end reporting. For example, the carrying amount of an asset or cash generating unit may not be recoverable, and so may need to be impaired to lower value in use or fair value less costs of disposal. In turn, the impairment loss will also lower the company's net profit.*

Bond prices *– Rising interest rates also lead to falling bond prices (because of demand for bonds with lower rates falls). When demand for bonds declines, issuers of new bonds need to offer higher yields to attract buyers. However, the existence of these new bonds with higher yields further reduced.*

Context example: Silicon Valley Bank

In March 2023, Silicon Valley Bank (SVB) in the US collapsed, with the increase in interest rates proving a key trigger in its collapse.

SVB had invested heavily in long-term US treasury bonds, but – in doing so – the bank had taken an enormous, unhedged gamble on interest rates staying low. When interest rates started rising in 2022, and the bank's depositors demanded higher returns, SVB was forced to sell some of its bonds at a loss.

Most of SVB's clients came from the tech sector and held deposits greater than the $250,000 amount guaranteed by the Federal Deposit Insurance scheme. When news of its bond sale was publicised on social media, investors panicked, fearing that they would lose their deposits, and so withdrew their funds. This triggered a run on the bank, which led to it collapsing very rapidly.

3 Developments in the Eurozone

Section overview

The problems faced by various countries in the Eurozone have led to speculation about whether it can survive in its current form.

3.1 Introduction of the euro

3.1.1 Maastricht treaty 1991

The main points of the Maastricht Treaty on Economic and Monetary Union (EMU) were:

- Agreement to EMU by 1999 for countries which fulfilled the economic criteria
- The establishment of the European Central Bank (ECB)

The Maastricht Treaty established three basic principles regarding fiscal policy:

(a) No excessive budget deficits

(b) No monetary financing of budget deficits, ie, unlimited credit from the central bank

(c) No bailouts of bankrupt governments

These principles were intended to ensure that fiscal mismanagement by one or more member states did not happen in the future. It was accepted that fiscal profligacy could undermine the monetary union.

3.2 Brexit

On 23 June 2016, the British electorate voted to leave the European Union. The so-called 'Leave' campaign highlighted concerns about levels of immigration and what it saw as excessive bureaucracy as key reasons for Britain to leave the EU and negotiate its own way in the world. Those who supported the 'Remain' campaign pointed out the benefits of EU membership, such as the easier selling of goods and services to other EU countries, and the economic benefits provided by the migrant workforce. Staying in the EU made it easier for UK businesses to send money, people, products and services around the world.

'Remainers' also claimed that Britain's status would be damaged by leaving, and that being forced to 'go it alone' would only cause economic harm.

Immediately following the Brexit vote, there was a fall in the value of the pound against the dollar, and in share prices. Britain lost its last AAA credit rating. However, since then share prices in the UK have recovered, with the FTSE 100 trading higher than before the referendum. The biggest effect of Brexit has probably been a strong sense of uncertainty about the economic future of Britain, and the future of the European Union itself, if more member states seek to leave.

The effect on sterling was more obvious, with it having fallen in value against both the euro and the dollar. Firms have pointed out that some goods become more expensive. On the other hand, UK exporters may see increased orders because of the fall in the value of the pound.

The British government invoked Article 50 of the Treaty on European Union on 29 March 2017, giving formal notification of the UK's intention to quit the EU. The UK left the EU on 31 January 2020. The transition period that was in place – during which nothing changed – ended on 31 December 2020. The rules governing the new relationship between the EU and UK took effect on 1 January 2021.

3.3 The future of the euro

3.3.1 Possible developments

Perhaps the central flaw in the euro and Eurozone is the exchange rate and interest rate philosophy/policy of 'one size fits all', which was always going to pose a problem among diverse member states. The reality has been that unit labour costs have risen much less in most countries of northern Europe (most notably Germany) than they have in southern countries such as Italy, Spain, Portugal and Greece. Without the possibility of exchange rate depreciation, Germany's wealth

increased whereas Greece and other southern European countries became poorer through their deteriorating balance of payments current account positions.

At the moment, Eurozone countries share the same monetary policy, controlled by the ECB. The only ways in which governments can control their economy are through fiscal policy (which differs between countries) and through austerity measures.

3.3.2 Exit from the Eurozone

With large external debts, a lack of competitiveness and a decline in economic activity, some peripheral Eurozone countries have two options: stay in, which means austerity and high unemployment for the foreseeable future with all important economic decisions affecting them taken in Brussels or Berlin, or default on their debt, exit the euro and go back to their own national currency.

At the moment it seems more likely that countries will remain within the euro, in spite of the short- term problems.

However, the situation is by no means settled, and further developments in the euro and the Eurozone are inevitable. You should monitor the financial news for developments.

4 Other current issues in finance



Section overview

- Most companies have faced the ramifications of the credit crunch over recent years, with banks being too nervous to lend money and charging a higher rate of interest if they do.
- Other recent developments include dark pool trading (trading without displaying quotes publicly) and the increased availability of Islamic finance.

4.1 Access to credit

The problems in the banking sector that began in 2007 resulted in banks themselves finding it more difficult to borrow money, with the result that they have had less money to lend. Regulators require UK banks to hold more than twice the capital they were holding before the credit crunch.

Small businesses have faced particular problems in accessing bank finance in recent years, and these are discussed further in the chapter Financial structure and financial reconstruction.

4.2 Dark pool trading

Dark pools are off-exchange facilities, operated by banks, that allow secondary market trading of large blocks of shares. They allow brokers and fund managers to place and match large orders anonymously to avoid influencing the share price. The transactions are only made public after the trades have been completed. Their popularity has increased as electronic trading has resulted in the reduction of the average size of trades. Traders placing large orders on the transparent exchanges risk signalling that they are large buyers or sellers. Such signals could cause the markets to move against them and put the order at risk.

The main problem with dark pool trading is that the regulated exchanges do not know about the transactions taking place until the trades have been completed. As a result, the prices at which these trades are executed remain unknown until after the event.

Such a lack of information on significant trades makes the regulated exchanges less efficient. Although the prices in dark pools are based on those in the regulated exchanges, the dark pool trades do not contribute to the changes in prices as their liquidity is not displayed.

Transactions in dark pools can almost be viewed as 'over the counter' as prices are not reported and financial risks are not effectively managed. There is the danger that such risks

spread in a manner similar to those attached to credit default swaps and collateralised debt obligations, which triggered the global financial crisis.

Dark pools and their lack of transparency arguably defeat the purpose of fair and regulated markets with large numbers of participants and threaten the healthy and transparent development of these markets. They could lead to a two-tier system whereby the public would not have fair access to information regarding prices and volumes of shares that is available to dark pool participants.

4.3 Islamic finance

Islamic finance has undergone rapid growth over recent years up to the point today where it is an industry worth more than $1 trillion. Islamic financing is not only the preserve of Islamic banks, but is becoming an important revenue stream for some of the world's biggest lenders. Many of the major conventional banks, including HSBC and Standard Chartered, have Islamic banking arms also known as 'Islamic Windows'.

Islamic finance may be used for either cultural/religious or commercial reasons. Commercial reasons include the fact that Islamic finance may be available when other sources of finance are not. Islamic finance may also appeal to companies due to its more prudent investment and risk philosophy.

Conventional banks aim to profit by taking in money deposits in return for the payment of interest (**Riba**) and then lending money out in return for the payment of a higher level of interest. Islamic finance does not permit the charging of interest and instead invests under arrangements which share the profits and losses of the enterprises.

Taken from the perspective of Sharia'a (Islamic religious law and moral code), the taking of deposits which are subsequently lent out for interest which is paid whether or not the project is profitable is not justifiable. The Islamic bank arranges its business in such a way that the bank's profitability is closely tied to that of the client. The bank stands to take profit or make loss in line with the projects it is financing and as such must be more involved in the investment decision making. The bank acts in some ways more like a fund manager than a conventional lending institution. Speculation is not allowed and conventional derivative products are deemed to be un-Islamic.

The main advantages of Islamic finance are as follows:

(a) Gharar (uncertainty, risk or speculation) is not allowed, reducing the risk of losses.

(b) Excessive profiteering is also not allowed; only reasonable mark-ups are allowed.

(c) Banks cannot use excessive leverage and are therefore less likely to collapse.

The use of Islamic finance does not remove all commercial risk. Indeed there may even be additional risk from the use of Islamic finance. There are the following drawbacks from the use of Islamic finance:

(a) There is no international consensus on Sharia'a interpretations, particularly with innovative financial products.

(b) There is no standard Sharia'a model for the Islamic finance market, meaning that documentation is often tailor-made for the transaction, leading to higher transaction costs than for the conventional finance alternative.

(c) Due to governmental and Sharia'a restrictions, Islamic finance institutions are subject to additional compliance work which can also increase transaction costs.

(d) Islamic banks cannot minimise their risks in the same way as conventional banks as hedging is prohibited.

5 Social responsibility and environmental matters



Section overview

- *The interaction of financing decisions and corporate responsibility is stressed in the Strategic Business Management syllabus. Directors must be aware of their legal responsibilities and the interests of stakeholders.*
- *ESG issues may have a number of impacts on management and financial accounts. We have already discussed (in the chapters Strategic analysis and Strategic performance management) the increasing importance of ESG and sustainability in relation to corporate performance and performance reporting, including the significance of the United Nations (UN) 2030 Agenda for Sustainable Development, and its global sustainable development goals. We will also consider (in the chapter Investment appraisal) how social and environment costs may affect investment appraisal decisions.*
- *Sustainability is relevant to a wide range of corporate stakeholders such as investors and providers of finance who have a key interest in business risks and opportunities and how they affect enterprise value.*
- *Sustainability is also relevant to an organisation due to its responsibility to understand and manage its impact and dependencies on the environment and society.*

5.1 Corporate responsibility

Businesses face a number of pressures to widen the scope of their accountability. A key challenge is encouraging senior managers to evaluate the extent to which their business objectives create societal value.

(a) Stakeholder pressures

Businesses face pressures from different stakeholder groups to consider their wider responsibilities. Stakeholders include communities (particularly where operations are based), customers (product safety issues), suppliers and supply chain participants and competitors. Issues such as plant closures, pollution, job creation and sourcing can have powerful **social effects** for good or ill on these stakeholders. Without the support of stakeholders a business will find that its ability to operate is impaired and this will damage performance. Stakeholders also include governments facing political pressures, and international organisations (such as the UN) which emphasise the importance of organisations considering the environmental and social context within which they operate. Adopting corporate responsibility voluntarily may be more flexible, and in the end less costly, than having it imposed by statute.

(b) Corporate reputation

Increasingly a business must have the reputation of being a **responsible business** that enhances long-term shareholder value by addressing the needs of its **stakeholders** – employees, customers, suppliers, the community and the environment. Sponsorship and community involvement can reflect well on the business and attract ethical customers.

(c) Staff motivation

A commitment to corporate responsibility helps establish values and mission within the organisation, which may help attract and retain staff.

(d) Business issues

Corporate responsibility initiatives may provide opportunities to enter new markets or build new core competences. Businesses can achieve lower costs through using resources more efficiently, and not having to incur costs of remediation if they have negatively affected the environment.

5.1.1 Scope of corporate responsibility

The scope of corporate responsibility varies from business to business. Factors frequently included are:

(a) **Health and safety** – this includes workplace injury, customer and supplier injury and harm to third parties.

(b) **Environmental protection** – energy use, emissions (notably carbon dioxide), water use and pollution, impact of product on environment, recycling of materials and heat.

(c) **Staff welfare** – issues such as stress at work, personal development, achieving work/life balances through flexibility, equal opportunities for disadvantaged or minority groups.

(d) **Customer welfare** – through content and description of products, non-exclusion of customer groups, fair dealing and treatment.

(e) **Supply-chain management** – insisting that providers of bought-in supplies also have appropriate corporate responsibility policies, ethical trading, elimination of pollution and unrecycled packaging, eliminating exploitative labour practices among contractors.

(f) **Ethical conduct** – staff codes for interpersonal behaviour, prohibitions on uses of data and IT, management forbidden from offering bribes to win contracts, ensuring non-exploitation of staff.

(g) **Engagement with social causes** – this includes secondment of management and staff, charitable donations, provision of free products to the needy, involvement in the local community, support for outreach projects such as cultural improvement and education.

5.1.2 Strategic approaches to corporate responsibility

An organisation can adopt different strategic approaches to corporate responsibility:

Proactive strategy	A business is prepared to take full responsibility for its actions. For example, a company which discovers a fault in a product and recalls the product without being forced to, before any injury or damage is caused, acts in a proactive way.
Reactive strategy	This involves allowing a situation to continue unresolved until the public, Government or consumer groups find out about it.
Defensive strategy	This involves minimising or attempting to avoid additional obligations arising from a particular problem.
Accommodation strategy	This approach involves taking responsibility for actions, perhaps when one of the following occurs: Encouragement from special interest groups Perception that a failure to act will result in government intervention

5.1.3 Corporate responsibility and stakeholders

Pressures on organisations to widen the scope of their corporate public accountability come from **increasing expectations of stakeholders** and **knowledge** about the **consequences of ignoring such pressures**. The South African King report stresses the importance of

engagement with external stakeholders, and individual workers and stakeholders being able to communicate openly.

Whatever the organisation's view of its stakeholders, certain problems in dealing with them on corporate social responsibility may have to be addressed.

(a) Collaborating with stakeholders may be **time consuming** and **expensive**.
(b) There may be **culture clashes** between the company and certain groups of stakeholders, or between the values of different groups of stakeholders with companies caught in the middle.
(c) There may be **conflict between company and stakeholders** on certain issues when they are trying to collaborate on other issues.
(d) **Consensus** between different groups of stakeholders may be difficult or impossible to achieve, and the solution may not be economically or strategically desirable.
(e) Influential stakeholders' **independence** (and hence ability to provide necessary criticism) may be compromised if they become too closely involved with companies.
(f) Dealing with certain stakeholders (eg, public sector organisations) may be complicated by their being **accountable to the wider public**.



Context example: Solar power

For many companies, adoption of renewable energy sources plays a large part in sustainability goals. Offsetting energy costs through increased use of renewables is not only cost effective, but promotes companies' CSR credentials.

According to a 2015 study conducted by Cone Communications/Ebiquity, '91 percent of global consumers expect companies to do more than make a profit', such as 'operate responsibly to address social and environmental issues'. The study notes that companies that support social and environmental issues maintain large numbers of consumers that perceive said companies in a positive light (93%), self-identify as loyal to the company (88%), and generally trust the company and its purposes (90%).

As an example, the cloud-based software company Salesforce made headlines for achieving its net- zero carbon emissions' goal decades before its goal date of 2050. Salesforce worked on increasing the energy efficiency of its infrastructure, servers, and buildings and finally reached its goal after purchasing power through wind farms in Texas and West Virginia.

Solar power is one of the cheapest and easiest forms of renewable energy available. A 200-kW solar array can offset carbon emissions by roughly 250 tons per year: the equivalent of taking 47 passenger vehicles off the road for one year, or not burning 120 tons of coal.

Asics, the popular footwear company, recently announced the installation of a solar array at one of its distribution centres, which will power 20% of the energy usage at the location.

MGM Resorts International has the largest rooftop solar array in the US, with 8.3 MW spanning 23 acres on the roof of the Mandalay Bay resort. The system will be able to account for 25% of the hotel's total energy usage. As well as installing the system, it paid US$87 million to stop purchasing electricity from its previous supplier, a cost which is expected to be recouped within seven years.

Source: https://www.renewableenergyworld.com/ugc/articles/2017/05/23/corporate-social-responsibility-how-renewables-have-expanded-the-field.html [Accessed 13 August 2018]

5.2 Corporate citizenship

Definition

Corporate citizenship: is: 'Business strategy that shapes the values underpinning a company's mission and the choices made each day by its executives, managers and employees as they engage with society. Three core principles define the essence of corporate citizenship, and every company should apply them in a manner appropriate to its distinct needs: minimising harm, maximising benefit, and being accountable and responsive to stakeholders.' (Boston College Carroll School of Management Center for Corporate Citizenship)

However, some argue that corporate citizenship should neither be seen as a business strategy nor a way of developing the business. Instead, it should be seen as a matter of fact. A corporation has a legal personality and operates within a society which places rights and obligations on its members (recognising them as citizens). Consequently, these rights and obligations also extend to corporate members.

Much of the debate in recent years about corporate responsibility has been framed in terms of corporate citizenship, partly because of unease about using words like ethics and responsibility in the context of business decisions. Discussion of corporate citizenship also often has political undertones, with corporations acting instead of governments that cannot – or will not – act to deal effectively with problems.

Context example: Corporate citizenship at The Carlyle Group

From The Carlyle Group's 2018 Corporate Citizenship Report:
At Carlyle, we possess strong internal capabilities, including a Chief Sustainability Officer and external specialists in specific Environmental, Social and Governance (ESG) issues who help us identify the material types of ESG issues and provide insight for our investment teams during the due diligence process. In the last year, we augmented our existing investment process by adding the Sustainability Accounting Standards Board (SASB) sector guidelines as a resource. These guidelines identify material factors likely to create or destroy value in 11 sectors and 79 industries. For example, if we explore an investment in an agricultural products company, SASB points to several material sustainability issues to consider including water use, fertilizer consumption and food safety. These sector guidelines, combined with our in-house expertise, allow us to approach ESG strategically and focus on those issues that matter.
The second pillar of our ESG strategy – governance and transparency – ensures alignment between Carlyle's interests and those of our investors, and places a premium on effective communication with our investors and other stakeholders. We follow a rigorous process around ESG issues, with mandatory inclusion of these issues in Investment Committee materials in most asset classes, and annual board-level review of our Guidelines for Responsible Investment at all of our controlled Corporate Private Equity companies. We also publish this annual citizenship report and regularly brief and consult with investors on ESG issues. We are excited to announce that in 2017, one of our largest fund families, Carlyle Europe Partners, further advanced Carlyle's ESG policies and practices by moving from fund-level reporting to reporting, on a company-by- company basis, ESG Key Performance Indicators (KPIs). This enhanced framework gives our investors greater insight into the true ESG performance of our portfolio companies.
Carlyle's third ESG pillar is to continue providing leadership on ESG issues, as we have for ten years. In 2008, we created our Guidelines for Responsible Investment, which also became the basis of the American Investment Council's guidelines in 2009. We published the first sustainability report in the sector in 2010 and hired our first Chief Sustainability Officer in 2014.

We are pleased to announce that Carlyle, across our 31 global offices and the activities of our roughly 1,600 employees, is now carbon neutral. Over the past year, we conducted our first-ever carbon footprint analysis with the assistance of environmental consultants with expertise in carbon measurement and estimation, and continued efforts to reduce our carbon emissions through efficiency projects. We then used our carbon data to invest in an offset project with The Carbon Fund focused on reducing emissions from long-haul trucks, a service used by many of our portfolio companies. We are not aware of any other global private equity firm that has taken this step.

Another area in which we aim to provide leadership is diversity and inclusion. Pursuing a diverse workplace not only reflects our values as a firm, but also enables us to better serve our investors because teams with diverse perspectives and experiences make better decisions.

Source: https://www.carlyle.com/sites/default/files/reports/carlyleccr2018.pdf [Accessed 13 August 2018]

5.3 Sustainability

Much of the discussion about corporate responsibility has focused on businesses' commitment to sustainability, ensuring that development meets the needs of the present without compromising the ability of future generations to meet their own needs.

The influential Brundtland report of 1987 emphasised that sustainability should involve developing strategies so that the organisation only uses resources at a rate that allows them to be replenished (in order to ensure that they will continue to be available). At the same time, emissions of waste should be confined to levels that do not exceed the capacity of the environment to absorb them.

The Brundtland report defined sustainable development as 'not a fixed state of harmony, but rather a process of change in which the exploitation of resources, the direction of investments, the orientation of technological development and institutional change are made consistent with future as well as present needs'.

A key challenge facing businesses is being able to demonstrate the links between 'sustainability' and business success.

5.3.1 The triple bottom line

One approach to sustainability is known as the **triple bottom line** (or TBL, 3BL, or People, Planet, Profit) approach.

- **People** means balancing up the interests of different stakeholders and not automatically prioritising shareholder needs.
- **Planet** means ensuring that the business's activities are environmentally sustainable.
- **Profit** is the accounting measure of the returns of the business.

A similar approach to thinking about sustainability issues is to differentiate three different types of sustainability:

Issues	Examples
Social	Health and safety, workers' rights (in the business itself and its supply chain), pay and benefits, diversity and equal opportunities, impacts of product use, responsible marketing, data protection and privacy, community investment and bribery/corruption

Environmental	Climate change, pollution, emissions levels, waste, use of natural resources, impacts of product use, compliance with environmental legislation, air quality
Economic	Economic stability and growth, job provision, local economic development, healthy competition, compliance with governance structures, transparency, long-term viability of businesses, investment in innovation/new product development

5.3.2 The United Nations Sustainable Development Goals

The UN Sustainable Development Goals (SDGs), described in the chapter Strategic performance management, constitute 17 global goals for sustainable development, with the aims of ending poverty, combatting climate change, and fighting injustice and inequality. Overall goals are supported by a range of targets and indicators, which provide a quantifiable framework for assessing whether or not the goals are being achieved.

5.3.3 Natural capital

As we first saw in in the chapter Strategic performance management, natural capital will become a prominent business concern and in order for organisations to incorporate 'natural capital' into their business cases and decision making they need to be able to measure and value it. PwC highlight three key steps in the valuation process:

- Quantify resource use (dependency) or environmental emissions (impacts) in biophysical units (kilograms, litres, etc.)
- understand how the resource use or environmental emissions cause changes in the natural environment (for example, through water pollution or changes in air quality)
- value the impacts on people associated with these changes in the natural environment – also called outcomes (eg, impacts on health, or on agriculture or fisheries)

5.3.4 Managing natural capital

Various methodologies have been proposed for managing the value of natural capital elements. When valuing natural capital, it is important to consider the **value perspective**:

Value perspective	Typically used to
Business value	Assess how natural capital impacts positively or negatively, the financial performance of the company.
Societal value	Understand the significance of the entity's natural capital impacts on external stakeholders
Both value perspectives	A comprehensive natural capital assessment. Assessing societal vales, in particular the entity's future impacts on society. This will enable all business values to be considered as well.

The valuation can be done with a specific **business application** or a combination of them:

- assess risks and opportunities
- compare option
- assess impacts on stakeholders
- estimate total value and/or net impact
- communicate internally or externally

An appropriate valuation technique will be selected based on whether the intention is to assess values in qualitative, quantitative or monetary terms.

- **Qualitative valuation techniques** are used to inform the potential scale of costs and/or benefits expressed through qualitative, non-numerical terms (eg, increase in air emissions, decrease in social benefits of recreation)
- **Quantitative valuation techniques** focus on numerical data which are used as indicators for the costs and/or benefits (eg, changes in tons of pollutants, decrease in number of people benefitting from recreation).
- **Monetary valuation techniques** translate quantitative estimates of costs and/or benefits into a single common currency.

The choice of valuation technique depends on the chosen value perspective (business, societal, or both), the ultimate objective of the assessment, and the time and resources available. There are various valuation techniques that can be used:

Technique	Description
Replacement costs	The cost of replacing natural capital with an artificial substitute
Damage costs avoided	The potential cost of property infrastructure, and production losses due to natural capital degradation, treated as a saving or a benefit from observing natural capital.
Hedonic pricing	Based on the observation that environmental factors are one of the determinants of the market price of certain goods (eg the environmental quality of a neighbourhood affects the prices of properties locate there).
Contingent valuation	Infers ecosystem values by asking individuals their maximum willingness to pay (or willingness to accept compensation) for a specific change in the relevant non-market good or service from natural capital.

Shadow pricing

Shadow pricing is one way to account for natural capital costs. Shadow pricing adds a hypothetical surcharge to market prices for goods or services that involve significant carbon emissions in their supply chain. This can be seen as the cost on society or the social cost of carbon valuation technique. A growing number of companies are putting a shadow price on carbon to reduce their carbon footprint and inform business decisions.

Procurement and supply-chain management

For many companies, most environmental risks lie in their supply chain rather than in the company's own operations. Practical ways to apply natural capital accounting could be through the use of supply-chain risk assessments, supplier relationship management, and the implementation of sustainable procurement strategies.

This could involve requesting data on suppliers' carbon emissions or water use, and rejecting those suppliers who do not comply with standards. Companies could switch to certified sustainable suppliers for commodities such as timber or cotton.

Product design

Another way to use natural capital valuation is in the product development and design process. Lifecycle analysis (LCA) can quantify and reduce impacts associated with sourcing, manufacturing, using and disposing of products.

Scenario planning

Companies can use natural capital valuations to help with decisions on where to grow and invest capital, or withdraw, or for example when considering online retailing versus investing in bricks and mortar stores.

5.3.5 Role of assurance procedures for sustainability goals

There is a significant role for the accountancy profession in providing high quality reporting, audit and assurance of organisations' performance in relation to sustainability goals, alongside 'traditional' business decisions. According to ICAEW, sustainability needs to be 'measured, reported and assured'. Organisations need to be proactive in this area, pre-empting sustainability issues and finding new business opportunities. Chartered Accountants are well-placed to assist with the availability of accurate and reliable information, and with assurance of non-financial data. They could expect to be involved in a range of activities:

- Implementing energy efficiency (and cost-saving) measures
- Formulating and implementing corporate responsibility policies
- Designing management information systems
- assurance of supply chain processes and procedures

5.4 Green Finance

5.4.1 Need for green finance

Many businesses that are at the forefront of innovation to develop solutions to climate change are financially constrained, and this potentially threatens the ability of society to achieve goals such as the Paris Agreements of 2015. As a result of this, a market for green finance has grown in the last decade, in some cases supported by governments. In particular, the market for green bonds, which is discussed in in the chapter Financial instruments and financial markets. Financial constraints refers to a situation where firms' access to finance is limited and they cannot therefore obtain sufficient finance to achieve their objectives.

Green finance has many different definitions but can be thought of as the financing of investments that provide environmental benefits, as part of a broader context of encouraging environmentally sustainable development. This could include green crowdfunding for small-scale, community schemes or green bond issuance for major infrastructure projects. Some areas are usually accepted as 'green' with little argument. These include renewable energy, energy efficiency in construction, green transport, recycling, pollution prevention and water conservation. Other areas are more controversial such as nuclear energy and fossil fuel efficiency.

When the UK government launched its Green Finance Strategy in July 2019, it described its green finance strategy as aiming 'to align investment and cashflows generated by the private sector with sustainable growth'. The Green Finance Strategy has two main objectives. Firstly, the strategy aims to align investment and cashflows generated by the private sector with sustainable growth. The second objective is to ensure that any green initiatives lead to a strengthening of UK financial sector competitiveness. There are three pillars to the government strategy;

(a) Greening finance - ensuring that financial decision making takes account of the risks and opportunities generated by climate and environmental issues. For example, decisions need to take account of potential risk of asset damage, arising from an increase in extreme weather events.

(b) Financing green - encouraging public and private investment in sustainable energy and other projects to help achieve carbon targets. For example, between 2015 and 2021 the UK government will have invested nearly £1.5 billion in schemes to provide charge point infrastructure for electric vehicles.

(c) Capturing the opportunity - ensuring UK financial services capture the commercial opportunities generated by the increase in climate data and analytics as well as from new green financial products and services.

(Great Britain. Department for Business, Energy and Industrial strategy., (2019). Green Finance Strategy. Transforming Finance for a Greener Future. London: HMSO.)

In response to the government's strategy ICAEW has signed up to the Green Finance Education Charter to ensure that today's accountants are fully versed in green finance and are able to consider and report on the impact that strategic decisions may have on the environment.

5.4.2 Types of green finance

The following table summarises the main types of green finance:

Methods of 'financing green'	
Green loans	Some banks, often with government backing, specialise in financing green projects. For example, in the UK the Green Investment Bank (GIB) was launched in 2012 to finance investment in projects that have a specific green purpose such as reducing greenhouse gas emissions, promoting efficiency in the use of natural resources, protection of the environment, and promoting environmental sustainability. The GIB was privatised in 2017, but retains the same goals. The market for green loans is growing rapidly, and often lenders offer better loan terms to borrowers that can show they are reducing their environmental impact. The Green Loan Principles (GLP) set out a clear framework, enabling all market participants to clearly understand the characteristics of a green loan.
Sustainability linked loans	Sustainability Linked Loans (SLLs) are not the same thing as green loans. They are loans for any purpose (whether 'green' or not), but with an in-built pricing mechanism meaning that the loan is cheaper if the borrower achieves certain sustainable or ESG (environmental, social and governance) related targets. The use of SLLs is growing rapidly.
Green bonds	A green bond is a type of fixed-interest bond used to raise money for climate and environmental projects. These bonds are typically secured and have the same credit rating as a company's other debt obligations. Green bonds may come with tax incentives to enhance their attractiveness to investors. Over the past decade green bonds have increased in popularity dramatically. To qualify for green bond status, they are often verified by a third party such as the Climate Bonds Standard Board, which certifies that the bond will channel finance to environmentally beneficial projects.
Green funds	Many stock markets produce an index of firms that satisfy social and environmental criteria. For example in the UK the London Stock Market has the FTSE4Good index. Companies are excluded if, for example, they are involved in the tobacco, weapons or coal power industry. To be included in the FTSE4Good Index, companies must, for example, support human rights, make progress to become environmentally sustainable, and ensure good labour standards for their own company and for companies that supply them as well.

Methods of 'financing green'	
	This helps investors to target investments in companies with higher standards of behaviour in terms of social responsibility.

5.5 Control systems

Many companies have been forced to act on environmental issues because of shocks such as environmental disasters and attention from pressure groups. To reduce the chances of these happening, organisations must not only monitor their internal performance, but also include within their monitoring of the external situation assessment of the impact of environmental issues. It will be particularly important to monitor:

- emerging environmental issues
- likely changes in legislation
- changes in industry best practice
- attitudes of suppliers, customers, media and the general public
- activities of environmental enforcement agencies
- activities of environmental pressure groups

In *Accounting for the Environment*, Gray and Bebbington listed the functions that environmental management systems should cover. These tie in with the aims of natural capital valuation described above.

Environmental review and policy development	A first review of environmental impacts of materials, issues and products and of business issues arising, also the development of a tailored in-house policy or measures to ensure adherence to external standards
Objectives and target development	As with all business objectives and targets, it is preferable that those set be unambiguous and achievable. Initiatives such as the World Wildlife Fund initiative encourage quantified targets within a specified time period eg, reducing carbon dioxide emissions by X% within a specified time period
Life cycle assessment	This aims to identify all interactions between a product and its environment during its lifetime, including energy and material usage and environmental releases. • Raw materials used have to be traced back to the biosphere and the company recognise impact on habitat, gas balance, the energy used in the extraction and transportation and the energy used to produce the means of extraction • For intermediate stages, emissions, discharges and co-products • At the consumer purchase stage, the impact of manufacture and disposal of packaging, transport to shops and ultimately impacts of consumers using and disposing of the product
Establishment and maintenance of environmental management systems	Key features of environmental management systems (as with other management systems) including information systems, budgeting, forecasting and management accounting systems, structure of responsibilities, establishment of an environmentally friendly culture, considering impact on human resource issues such as education and performance appraisal

Regulatory compliance	Making sure that current legal requirements are being fulfilled and keeping up to date with practical implications of likely changes in legislation
Environmental impact assessment	A regular review of interactions with the environment, the degree of impact and an environmental SWOT analysis, also the impact of forthcoming major investments
Eco-label applications	Eco-labelling allows organisations to publicly identify products and services that meet the highest environmental standards. To be awarded an eco-label requires the product to be the result of a reliable quality management system
Waste minimisation	Whether waste can be minimised (or better still eliminated), possibility of recycling or selling waste
Pollution prevention programmes	Deciding what to target
Research, development and investment in cleaner technologies	How to bring desirable features into product development, bearing in mind product development may take several years, and opinion and legal requirements may change during that period. Desirable features may include minimum resource usage, waste, emissions, packaging and transport, recycling, disassembly and longer product life
Environmental performance and issues reporting	Consideration of the benefits and costs of reporting, how to report and what to include (policies, plans, financial data, activities undertaken, sustainability)

5.6 Management accounting and performance issues

Most **conventional accounting systems** are unable to apportion **environmental costs** to products, processes and services and so they are simply **classed as general overheads**. Environmental management accounting (EMA), on the other hand, attempts to make all relevant significant costs visible so they can be considered when making business decisions.

The major areas for the application of EMA are 'in the assessment of annual environmental costs/expenditures, product pricing, budgeting, investment appraisal, calculating costs and savings of environmental projects, or setting quantified performance targets'.

5.6.1 Input/output analysis

Input/output analysis records material flows and balances them with outflows on the basis that what comes in must go out, or be stored. This approach is similar to process costing where all materials in a process are accounted for either as good output or as scrap/waste. This forces the business to look at how it uses its resources and focuses it on environmental cost.

The difficulty with adopting this technique is putting monetary values on waste, non-accounted materials and scrap if these previously haven't been accounted for. It also requires additional reporting of factors included, such as water use and energy, which may be difficult to attribute to individual units.

5.6.2 Flow cost accounting

Flow cost accounting takes material flows and combines them with the organisational structure. It evaluates material flows in terms of physical quantities, cost and value. Material flows are classified into material, system and delivery and disposal. The values and costs of each of these are then calculated. This system requires additional reporting which may be unavailable on existing systems and time consuming to accomplish.

5.6.3 Environmental activity-based costing

Traditional activity-based costing allocates all the internal costs of a business to cost centres and cost drivers on the basis of the activities that caused the costs. Environmental activity-based costing distinguishes between environment-related costs and environment-driven costs.

Environment-related costs are costs specifically attributed to joint environmental cost centres, such as a sewage plant, or a waste filtration plant.

By contrast, **environment-driven costs** are hidden in general overhead costs and do not relate specifically to a joint environmental cost centre, although they do relate to environmental drivers. For example, a company may shorten the working life of a piece of equipment in order to avoid excess pollution in the later years of its working life. As a result, the company's annual depreciation charge will increase. This is an environment-driven cost.

In order for environmental activity-based costing to provide 'correct' information, the choice of allocation basis is crucial. The difficulty in allocating costs correctly could be a major complication in using this method.

Four main bases of allocation are:

- volume of emissions or waste
- toxicity of emissions or waste
- environmental impact added volume of the emissions treated
- the relative costs of treating different kinds of emissions

5.6.4 Life cycle costing

Life cycle costing records the complete costs of a product 'from cradle to grave' taking into account the environmental consequences across the whole life of the product. Organisations need to have the recording systems to capture all costs, especially those incurred **prior to production** and **after production ceases** (for example, the costs of cleaning and decontaminating industrial sites when they are decommissioned). It is important that potential decommissioning costs and other post- production costs are identified at the start of a project, so that they can be included in the investment appraisal (or similar cost-benefit analysis) to determine whether or not to undertake the project.

5.6.5 Environmental concern and performance

Martin Bennett and Peter James (authors of *The Green Bottom Line: Management Accounting for Environmental Improvement and Business Benefit*) looked at the **ways in which a company's concern for the environment can impact on its performance**.

(a) **Short-term savings** through waste minimisation and energy efficiency schemes can be substantial.

(b) Companies with poor environmental performance may face **increased cost of capital** because investors and lenders demand a higher risk premium.

(c) There are a number of **energy and environmental taxes**, such as the UK's landfill tax.

(d) **Pressure group campaigns** can cause damage to reputation and/or additional costs.

(e) Environmental legislation may cause the **'sunsetting'** of products and opportunities for **'sunrise' replacements**.

(f) The cost of processing input which becomes **waste** is equivalent to 5–10% of some organisations' revenue.

(g) The phasing out of CFCs has led to markets for alternative products.

5.6.6 Achieving business and environmental benefits

Bennett and James went on to suggest six main **ways in which business and environmental benefits can be achieved**.

(a) **Integrating the environment into capital expenditure decisions** (by considering environmental opposition to projects which could affect cash flows, for example). There is a feeling that most companies do not know about the extent of their environmental costs, and so tend to underestimate them. This can lead to distorted calculations in investment decisions.

(b) **Understanding and managing environmental costs**. Environmental costs are often 'hidden' in overheads and environmental and energy costs are often not allocated to the relevant budgets.

(c) **Introducing waste minimisation schemes**.

(d) **Understanding and managing life cycle costs**. For many products, the greatest environmental impact occurs upstream (such as mining raw materials) or downstream from production (such as energy to operate equipment). This has led to producers being made responsible for dealing with the disposal of products such as cars, and government and third-party measures to influence raw material choices. Organisations therefore need to identify, control and make provision for environmental life cycle costs and work with suppliers and customers to identify environmental cost reduction opportunities.

(e) **Measuring environmental performance**. Business is under increasing pressure to measure all aspects of environmental performance, both for statutory disclosure reasons and due to demands for more environmental data from customers.

(f) **Involving management accountants in a strategic approach to environment-related management accounting and performance evaluation**. A 'green accounting team' incorporating the key functions should analyse the strategic picture and identify opportunities for practical initiatives. It should analyse the short-, medium- and long-term impact of possible changes.

5.7 Corporate reporting implications

Businesses face legal requirements in many jurisdictions to report on environmental matters, and environmental issues may impact on a number of areas in their financial reports.

5.7.1 Business review in UK

The UK Companies Act 2006 requires directors to report on environmental issues in the business review within the directors' report. This should include reporting the impact of the company's business on the environment as well as information about the company's employees and social and community issues. The main aspects of these disclosures are risks and uncertainties, policies and effectiveness and key performance indicators, including non-financial indicators. Previous guidance has suggested disclosures on spillage, emissions and waste.

5.7.2 Tangible assets

The valuation of tangible assets may be reduced by contamination, physical damage or non-compliance with environmental regulations. Their carrying amounts should be reduced to value in use or net realisable value.

Measurement of an environmentally impaired asset can be affected by:

- delayed disposal of the asset, due to the need to decontaminate it, resulting in clean-up costs and interest charges
- uncertainties surrounding changes in technology or legislation

- reputation risks including the risks of deterring potential purchasers resulting in a restricted market

5.7.3 Intangible assets

Intangible assets are subject to an impairment test on their carrying value if they exceed the recoverable amount from use or realisation. Goodwill, for example, may be impaired by environmental issues.

IFRIC 3, *Emission Rights* required businesses to treat emission allowances as intangible assets, recorded at fair value. Actual emissions then would give rise to a liability. When allowances were given by the government for less than fair value, the difference would be treated as a government grant. IFRIC 3 was withdrawn by the IASB, but the accounting treatment it recommends remains acceptable. Other possible methods for accounting for emission rights include cost of settlement approach based on initial market value and cost of settlement approach where provision is only made for the costs of buying emission rights not covered by allowances. Whatever method is used, the market value of purchased emissions may become lower than their cost if there is a glut of allowances in the market.

5.7.4 Inventories

Inventories may be affected by environmental issues such as physical leakage and deterioration. Their valuation needs to be written down to net realisable value.

5.7.5 Provisions

Provisions may be required under IAS 37, *Provisions, Contingent Liabilities and Contingent Assets* as a result of activities connected with the environment, including waste disposal, pollution, decommissioning and restoration expenses. Companies in certain industries such as vehicle manufacture may be involved in activities relating to that industry that give rise to the need for provisions.

5.8 Environmental and social reporting

As well as legal requirements, businesses face peer pressure to provide increasing amounts of information about environmental and social performance. Organisations such as Trucost benchmark companies' environmental disclosures.

Environmental and social reports generally include narrative and numerical information about impact. Narrative information includes objectives, explanations and reasons why targets have or have not been achieved. Reports can also address concerns of specific internal or external stakeholders.
Useful numerical measures can include pollution amounts, resources consumed and land use.



Context example: Environmental and social reporting

BT has established a number of key performance indicators (KPIs) for its sustainability policies. It reports on these each year, comparing targets for the KPI with actual performance, and whether the target has been reached. The KPIs include targets for:

- increasing accessibility to fibre-based products and services to over 9 in 10 people by 2020
- improvements in an employee engagement benchmark
- reducing the sickness absentee rate among employees
- improving supplier relationships
- reviewing ethical trading in the supply chain, with a target of 100% follow-up within three months of all suppliers considered medium to high risk

- reducing CO_2 emissions
- improving the quality of customer service

For each KPI, a measurable target is established and actual performance in each year is compared with the target, and reported.

5.8.1 Global Reporting Initiative

The Sustainability Reporting Standards issued by the Global Reporting Initiative (GRI) are the best- known example of a global, voluntary code for corporate responsibility and sustainability reporting. A GRI-based report typically includes economic, social and environmental performance information and sets out the organisation's direct and indirect impacts. An updated version of the standards was issued during the year 2021 and applies to reports published on or after 1 January 2023.

There are three 'universal' standards (GRI 1, GRI 2 and GRI 3) that apply to all organisations that choose to apply GRI standards. These are supported by various sector standards that help organisations to identify which topics are likely to be material to organisations operating in those industries (eg, GRI 11 relates to the oil and gas sector). Topic standards provide guidance on what disclosures should be made for topics that are material to the organisation (eg, GRI 204 deals with disclosures relating to the organisation's impact on biodiversity).

GRI 1 Foundation 2021 sets out the overall structure of GRI guidance and provides key concepts for sustainability reporting, such as the concept of a stakeholder.

GRI 2: General disclosures requires disclosure about the organisation, including information on the following topics:

- The organisation's legal name and form. Information should also be provided about all entities that are included in the sustainability report (eg, subsidiaries)
- The organisation's activities, value chain and other business relationships
- Information about the organisation's employees, giving details such as analysis by gender, the proportion of employees who are not given guaranteed hours of work and the proportion of permanent and temporary employees
- Information about the governance arrangements of the organisation, such as structure of the board of directors, details of board committees, the role of the board with respect to sustainability and details of any conflicts of interest. It includes guidance about board remuneration
- Strategies of the organisation, including sustainable development strategies and processes to remediate the negative impacts that the organisation has

GRI 3: Material Topics 2021 provides guidance to organisations on identifying material topics that should be included in their sustainability report. A topic is material if the organisation's activities have a severe impact on it. Examples of topics include occupational health and safety or water and effluents. If a topic is material to an organisation, then it should apply the topic standard for that topic.

GRI 3 also provides its own disclosure requirements relating to material topics:

- The process used in determining its material topics
- A list of its material topics
- For each material topic:
  - the actual and potential negative impacts on the economy, environment and people (and whether the impact is through its own activities or as a result of business relationships)
  - its policies regarding the material topic

- the actions taken to manage the topic (eg, actions to mitigate the potential impact and actions to manage positive impacts)
- how the effectiveness of the actions is tracked and measured
- how engagement with stakeholders has informed the actions taken

The following brief outlines give a flavour of two of the topic standards:

- GRI 302: Energy requires disclosure of the organisation's energy management (as per GRI 3). In addition, the organisation is required to disclose total fuel consumption from non-renewable and renewable sources, analysed by type of energy; energy consumption outside of the organisation, by upstream and downstream organisations, related to the main activity of the organisation (eg, energy consumption by businesses transporting the organisation's goods); and any reduction in energy consumption over time as a result of efficiency or conservation.
- GRI 304 Biodiversity requires organisations to provide information about how the organisation manages its biodiversity-related impacts. This includes details of sites located next to protected areas and significant impacts of the organisation's activities, products or services on the biodiversity. Organisations are also required to disclose any activities taken to protect or restore habitats. It should also disclose the impact that the organisation has on protected species.

5.8.2 Integrated reporting

An integrated report should explain, using both quantitative and qualitative information, how an organisation creates value.

The *Integrated Reporting Framework* looks at the way value is created, maintained and increased through the development of four 'capitals': economic; human; social and natural capital.

Integrated reporting was explained in more detail in the chapter Strategic performance management.

5.8.3 International Sustainability Standards Board (ISSB)

The International Sustainability Standards Board (ISSB) was set up in November 2021 by the IFRS foundation. Its objective is 'to deliver a comprehensive global baseline of sustainability-related disclosure standards that provide investors and other capital market participants with information about companies' sustainability-related risks and opportunities to help them make informed decisions'. (*IFRS org)*

It took over the work of two other bodies, the Climate Disclosure Standards Board (CDSB) and the Value Reporting Foundation, thereby taking over responsibility for the Integrated Reporting Framework. It has also signed a joint cooperation agreement with GRI in which the ISSB and GRI will seek to coordinate their work programmes and standard-setting objectives.

The ISSB issued two exposure drafts for comment in 2022: IFRS S1 – General disclosure requirements, and IFRS S2 – Climate related disclosures. These exposure drafts are expected to be updated for comments and issued at the end of 2022. Both standards build on the structure used in the TCFD standards (see the chapter Strategic performance management).

5.9 Auditing and assurance on environmental issues in UK

As well as audits and assurance reviews that focus specifically on environmental matters, auditors are likely to consider environmental matters in several areas of the audit of the financial statements.
Guidance on the audit work required has been provided by ICAEW and the Environment Agency in *Environmental issues and annual financial reporting*.

Environmental issues should be considered as part of audit planning if they are likely to have a significant impact on the financial statements. Environmental indicators may be used within initial analytical review processes. Potential impacts on the financial statements may arise from:

- the application of laws and regulations
- participation in a carbon trading scheme
- activities and processes of the business, particularly processes involving pollution, the use of hazardous substances or the avoidance of hazardous waste
- holding an interest in land or buildings that could have been contaminated by a previous user
- climate change and potential flood risks
- fines or penalties
- dependence on a major customer or segment whose business is threatened by environmental pressures

To obtain an understanding of the business, auditors may need to consider:

- extent to which business is based on the use of environmentally sensitive materials
- environmental obligations arising from laws and regulations affecting specific businesses
- risks of carrying out construction work in flood plains
- effects of market issues such as compliance with environmental legislation, customer perceptions of environmental performance and the impact of environmental pressures on suppliers

As part of the auditors' review of controls and systems, auditors should consider whether management is **competent** to deal with environmental issues and systems and processes designed to identify and monitor environmental risks. What constitutes adequate control systems will vary by company. Where exposure to environmental risk is high, companies may operate a specific environmental management system or deal with environmental issues within an integrated control system.

Although some environmental issues have clear financial outcomes, others will be dependent on the judgement of the directors. Work on **estimates** relating to environmental matters could therefore be a significant part of the audit.

Auditors will be particularly concerned with **environmental laws and regulations** if non-compliance would be central to the core operation of the business and irregularities could result in closure, either due to the loss of an operating licence or a huge fine.

If a company participates in an emissions trading scheme, the auditors will need to see that the company complies with the requirements of the scheme, systems are operating to measure emissions and process data and measurement and disclosures comply with international financial reporting standards.

Auditors may face a number of difficulties in assessing the effects of environmental matters in the financial statements:

- Delays between activities causing environmental issues such as contamination of a site due to industrial activity and its identification by the business or regulator.
- Accounting estimates not having an established pattern and exhibiting a wide range of reasonableness because of the number and nature of assumptions underlying them.
- Interpretation of evolving environmental laws may be difficult or ambiguous.
- Liabilities can arise out of voluntary agreements as well as legal or contractual obligations.

5.9.1 Environmental auditing

An environmental audit is an evaluation of how well an entity, its management and equipment are performing, with the aim of helping to safeguard the environment by facilitating management control of environmental practices and assessing compliance with entity policies and external regulations.

Environmental auditing is also used for auditing the truth and fairness of an environmental report rather than the organisation itself.

An environmental audit may be undertaken as part of obtaining or maintaining external accreditation, such as the BSI's ISO 14001 standard.

In practice, environmental audits may cover a number of different areas. The scope of the audit will depend on each individual organisation. Often the audit will be a general review of the organisation's environmental policy. On other occasions the audit will focus on specific aspects of environmental performance (waste disposal, emissions, water management, energy consumption) or particular locations, activities or processes.

There are other specific aspects of the approach to environmental auditing which are worth mentioning.

(a) **Environmental Impact Assessments (EIAs)**

These are required, under an EU directive, for all major projects which require planning permission and have a material effect on the environment. The EIA process can be incorporated into any environmental auditing strategy.

(b) **Environmental surveys**

These are a good way of starting the audit process, by looking at the organisation as a whole in environmental terms. This helps to identify areas for further development, problems, potential hazards and so forth.

(c) **Environmental SWOT analysis**

A 'strengths, weaknesses, opportunities, threats' analysis is useful as the environmental audit strategy is being developed. This can only be done later in the process, when the organisation has been examined in more detail.

(d) **Environmental Quality Management (EQM)**

This is seen as part of Total Quality Management and it should be built into an environmental management system. Such a strategy has been adopted by companies such as IBM, Dow Chemicals and by the Rhone-Poulenc Environmental Index, which has indices for levels of water, air and other waste products.

(e) **Eco-audit**

The European Commission has adopted a proposal for regulation of a voluntary community environmental auditing scheme, known as the eco-audit scheme. The scheme aims to promote improvements in company environmental performance and to provide the public with information about these improvements. Once registered, a company will have to comply with certain ongoing obligations involving disclosure and audit.

(f) **Eco-labelling**

Developed in Germany, this voluntary scheme will indicate those EU products which meet the highest environmental standards, probably as the result of an EQM system. It is suggested that eco-audit must come before an eco-label can be given.

(g) **BS 7750 Environmental Management Systems**

BS 7750 also ties in with eco-audits and eco-labelling and with the quality BSI standard BS 5750. Achieving BS 7750 is likely to be a first step in the eco-audit process.

(h) **Supplier audits**

These ensure that goods and services bought in by an organisation meet the standards applied by that organisation.

5.9.2 Environmental audit stages

There are three main stages in most environmental audits.

Establishing the metrics

The greater the variety of metrics, the more information provided. However, measuring against a number of metrics could result in a costly audit.

Measuring planned or desirable performance against actual performance

This is an important aspect of a system. Some metrics will be objective, for example the level of carbon emissions or plastic bag issues can be measured. However, other aspects, for example public perceptions, cannot be measured objectively and may therefore be difficult to measure precisely.

Reporting the results of the audit

Important decisions will include the form of the report and how widely it should be distributed, in particular whether the organisation's annual report should include a report by the auditors.

5.9.3 Auditor concerns

(a) Board and management having good understanding of the environmental impact and related legislation of the organisation's activities in areas such as buildings, transport, products, packaging and waste

(b) Adoption and communication of adequate policies and procedures to ensure compliance with relevant standards and laws

(c) Adoption of appropriate environmental information systems

(d) adoption and review of progress against quantifiable targets

(e) Assessment of whether progress is being made economically and efficiently

(f) Implementation of previous recommendations of improvements to processes or systems

(g) True, fair and complete reporting of environmental activities

Summary

	Tick off
Financial strategy is based around investment, financing and dividend decisions, with strategic cash flow and fund management ensuring sufficient funds are available.	☐
Financial strategy decisions, particularly investments in other entities, can have numerous financial reporting consequences.	☐
The financial position of many firms has been affected by the credit crunch and problems in the Eurozone, particularly the debt crisis.	☐
Increasingly business strategy is determined by corporate responsibility and sustainability issues, with firms wishing to demonstrate in their corporate reporting that they are good corporate citizens.	☐

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, complete the following knowledge diagnostic and check you are able to confirm you possess the following essential learning from this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
1.	Do you know what factors affect the choice of a company's financial strategy? (Topic 1)
2.	Do you know how to correctly account for investments in other entities? (Topic 1)
3.	Is your knowledge of the financial reporting implications of financial instruments up to date? (Topic 1)
4.	What are the main issues that should be covered in an environmental and social report? (Topic 5)
5.	Do you know the different strategic approaches to corporate responsibility? (Topic 5)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
1 Zelo	This is a good introductory question covering environmental and social reporting. Work through this question carefully before attempting exam standard questions.
2 Burtchester	This question covers corporate social responsibility and the corporate reporting implications of a possible legal action. Before attempting this question make sure you are aware of the main features to be included in a CSR report and the requirements of IAS 37.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your ueries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Technical reference

1 IFRS 12, *Disclosure of Interests in Other Entities*

- IFRS 12 is a disclosure standard requiring a wide range of disclosures about an entity's interests in subsidiaries, joint arrangements (joint operations or joint ventures), associates and other unconsolidated 'structured entities'. The objective of IFRS 12 to require an entity to disclose information which enables users of financial statements to evaluate the nature of, and risks associated with, its interests in other entities; and the effect of those interests on its financial position, financial performance, and cash flows.

2 IFRS 10, *Consolidated Financial Statements*

- Outlines the requirements for the preparation and presentation of consolidated financial statements, with entities being required to consolidate other entities they control.

3 IFRS 11, *Joint Arrangements*

- Details the issues which should be considered when determining the appropriate financial reporting treatment for joint ventures and joint operations and specifies the accounting treatment for both kinds of joint arrangements.

4 IAS 28, *Investments in Associates and Joint Ventures*

- Outlines the accounting for investments in associates, where an associate is an entity over which an investor has significant influence but not control or joint control or a joint venture as determined in IFRS 11.

5 IFRS 3, *Business Combinations*

- Outlines the accounting when an acquirer obtains control of a business through an acquisition. These business combinations are accounted for using the 'acquisition method' which generally requires the assets acquired, and the liabilities assumed, to be measured at their fair values at the acquisition date.

6 IFRS 5, *Non-current Assets Held for Sale and Discontinued Operations*

- Outlines the accounting required for non-current assets held for sale (or for distribution to owners). In general terms, assets (or disposal groups) held for sale are not depreciated, are measured at the lower of carrying amount and fair value less costs to sell, and are presently separately in the statement of financial position.

7 IAS 27, *Separate Financial Statements*

- Outlines the accounting and disclosure requirements for separate financial statements, which are financial statements prepared by a parent, or an investor in a joint venture or associate, where those investments are accounted for either at cost, in accordance with IFRS 9, or using the equity method in IAS 28. The standard also outlines the accounting requirements for dividends.

8 IAS 32, *Financial Instruments: Presentation*

- IAS 32 outlines the requirements for the presentation of financial instruments, particularly the classification of such instruments into financial assets, financial liabilities and equity instruments. The standard also provides guidance on the classification of interest, dividends and gains or losses relating to the financial instruments.

9 IFRS 9, *Financial Instruments*

- IFRS 9 sets out the recognition and measurement requirements for financial instruments and some contracts to buy or sell non-financial items. It replaced IAS 39 with effect for annual periods beginning on or after 1 January 2018.

10 IFRS 7, *Financial Instruments: Disclosures*

- IFRS 7 prescribes disclosures about the significance of financial instruments to an entity's position and performance, and about the nature and extent of the risks arising from those financial instruments, in both quantitative and qualitative terms.

11 IFRS 8, *Operating Segments*

- IFRS 8 requires entities with publicly traded securities to disclose information about their operating segments, products and services; the geographical areas in which they operate, and their major customers. Segment information is based on internal management reports, both in the identification of operating segments and the measurement of disclosed segment information.

12 IAS 16, *Property, Plant and Equipment*

- Outlines the accounting treatment for most types of property, plant and equipment. The principal issues are the recognition of assets, the determination of their carrying amounts, and the depreciation charges and impairment losses to be recognised in relation to them.

13 IAS 38, *Intangible Assets*

- Outlines the accounting requirements for intangible assets, which are non-monetary assets which are without physical substance but which are identifiable (either being separable or arising from contractual or other legal rights). Intangible assets are capitalised and amortised on a systematic basis over their useful lives, unless an asset has an indefinite useful life, in which case it is not amortised.

14 IAS 36, *Impairment of Assets*

- Seeks to ensure that an entity's assets are not carried at more than their recoverable amount (being the higher of fair value less costs of disposal and value in use). An annual impairment test is required for goodwill and certain intangible assets, but for the majority of assets an impairment test is only required where there is an indication of impairment of an asset.

15 IAS 21, *The Effects of Changes in Foreign Exchange Rates*

- IAS 21 prescribes how to include foreign currency transactions and foreign operations in the financial statements of an entity, and how to translate financial statements into a presentation currency. The principal issues are which exchange rate(s) to use, and how to report the effects of changes in exchange rates.

16 IAS 23, *Borrowing Costs*

- The objective of IAS 23 is to ensure that borrowing costs directly attributable to the acquisition or production of a qualifying asset are included in the cost of the asset. Other borrowing costs are to be recognised as an expense. Borrowing costs include interest on bank overdrafts and loans, finance charges on leases and exchange differences on foreign currency borrowing where they are regarded as an adjustment to interest costs.

17 IFRS 16, *Leases*

- Effective from 1 January 2019 IFRS 16 effectively abolishes the distinction between operating and finance leases for lessees, although not for lessors. It introduces a single lessee accounting model and requires a lessee to recognise assets and liabilities for all leases with a term of more than 12 months, unless the underlying asset is of low value.

18 IFRS 13, *Fair Value Measurement*

- Defines fair value as the price that would be received to sell an asset or would be paid to transfer a liability in an orderly transaction between market participants at the measurement date. IFRS 13 defines fair value on the basis of an 'exit price' notion and uses a 'fair value hierarchy' which results in a market-based measurement rather than an entity-specific one.

19 IAS 10, *Events After the Reporting Period*

- IAS 10 prescribes when events after the end of the reporting period should be adjusted in the financial statements for that period. Adjusting events are ones which provide evidence of conditions existing at the end of the reporting period, whereas non-adjusting events are ones which are indicative of conditions arising after the reporting period.

20 IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*

- Outlines the accounting for provisions (liabilities of uncertain timing or amount), contingent assets (possible assets) and contingent liabilities (possible obligations or present obligations which are not probable or which cannot be measured reliably). Provisions are measured at the best estimate of the expenditure required to settle the present obligation. The objective of IAS 37 is to ensure that financial statements provide sufficient information to enable users to understand the nature, timing and amount of any provisions, contingent liabilities and contingent assets, but it is also underpinned by the key principle that a provision should only be recognised when a genuine liability exists.

Self-test questions

Answer the following questions.

1 Zelo plc

Zelo plc is a publicly quoted company. Its products are based on raw materials grown in tropical countries and processed either in these countries or in the eventual sales markets. Processing is undertaken partly by Zelo plc and partly by sub-contractors. The products are branded and sold worldwide, but mainly in the UK and North America. They are sold to consumers through a very large number of outlets.

The non-executive directors have for some time expressed concern that the company has not developed any systems of environmental or social reporting to shareholders, although many comparable companies already publish such information as part of their Annual Report. A government minister has now stated that legislation will be considered if all companies do not make progress on reporting on social and environmental policies.

The chief executive has always regarded reporting as ideally never exceeding legal requirements.

Requirements

1.1 Construct a case for including a report on environmental and social issues in the annual report.

1.2 Identify the main issues that could be covered in the environmental and social report.

1.3 Analyse the impact of business partners and other stakeholders on the content of the environmental and social report.

2 Burtchester

Burtchester is a privately owned company that manufactures chemicals. It is likely to seek a stock market listing in the next two years, and because of the higher profile that it will have once it is listed, its directors feel that the company must have a corporate social responsibility (CSR) programme. The chairman feels that the programme needs to concentrate in particular on manufacturing processes (which are noisy and smelly and produce a lot of waste), Burtchester's sourcing of raw materials from economically underdeveloped countries, the ethnic diversity of its workforce and compliance with the law.

Two years ago Burtchester acquired a manufacturing site, which now houses one of its largest plants. However, unlike its other factories which are situated on industrial estates, this factory is located in the middle of a residential area. Since it opened, there have been complaints from a group of local residents about the noise made by the factory's processes and by lorries loading and unloading at the start and end of the day. Two of the residents wrote to the Chief Executive three months ago to say that they intend to take legal action to stop the noise and claim compensation for distress previously caused. Nothing further has been heard from them since the original letter.

Requirements

2.1 Advise Burtchester on what the main features of its CSR policy should be, dealing with its manufacturing process, procurement policy, labour force and compliance with the law.

2.2 Briefly describe the contents of an environmental report that would be appropriate for Burtchester and briefly explain the main elements of the assurance work that could be carried out to verify the contents of this report.

2.3 Discuss the corporate reporting implications of the claim by the residents.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

Factors to consider

(1) Investment, finance and dividends

A major investment is planned. There may be financing problems, as equity holders may not subscribe if there is a prospect of low dividends.

(2) Profit effects

Growth in profit has to be sufficient to meet investor and market expectations. There may be problems initially with new clubs.

(3) Cash effects

There will be major expenditure on new clubs, paying finance providers, and possibly refurbishing old clubs. Again, new clubs may not generate very much cash flow just after opening.

(4) Shareholders

There may be conflicts between initial investors who will accept a long-term return and want to retain active involvement in the business, and institutional shareholders with shorter-term interests who may be willing to accept a reasonably priced takeover bid.

(5) Other stakeholders

Debt holders will be concerned whether investment will generate sufficient returns to guarantee sum lent and income. Managers will be worried about expansion/curtailment of their position. Employees will want to know how operational changes will affect their employment. Customers will be concerned with the fees charged and the services offered.

(6) Economic and market factors

The financial markets are not viewing the sector very positively. Investment in clubs that make initial losses may have an adverse effect on share price, if markets do not take potential long- term revenues into account.

(7) Restrictions

Takeover of another big sports chain may run into difficulties from competition authorities. Existing loan agreements may contain possible restrictions on taking on more debt or opening more clubs.

(8) Risks

Downturns in the economy may hit membership of sports clubs if it is perceived as a luxury item. The company may also be vulnerable to changes in property prices if it's investing in new sites, and changes in demand as leisure clubs become less popular. There may be some exposure to foreign exchange risks.

(9) Time frame

There is a potential conflict between the need for good short-term profits and cash flows, and longer-term returns.

Generating strategy

The strategy can be summed up as increasing profits and cash flows by opening X new clubs over the next X years throughout Europe.

(1) Benefits

Benefits may accrue to shareholders in terms of dividend and market price appreciation.

(2) Directions

Monitoring of investments should be by financial performance indicators for new and existing clubs and overall targets for returns on capital employed.

(3) Resources

The company will need to specify sources of finance, amounts expected to be raised and commitments (return to suppliers and security). Also the directors should consider the generation of increased returns from existing operations by increasing membership charges and savings in operating costs.

(4) Environment changes

The scale of investment in new clubs should be reconsidered if property prices rise beyond a certain level, or if demand for existing clubs falls beyond a certain level.

(5) Timescale

Short-term investments may be limited by current funding problems. Finance may be available on better terms only when the existing network has generated higher returns.

Judging strategy

(1) Suitability

A strategy of expansion will maintain the position as a market leader, necessary in a highly competitive market. Merely maintaining current position without any investment may leave the company vulnerable to takeover. However, there may be better ways of achieving its goals, taking over existing clubs or diversifying into other less costly areas.

(2) Acceptability

Will founder stakeholders maintain sufficient control, or will they prefer any combination of investment and financing to being taken over? Will institutional shareholders be satisfied with resultant dividend levels and movements in market prices? How will creditors view possible threats to their position? Will savings in operating costs adversely affect the service given to customers?

(3) Feasibility

Will required finance be available from all sources to support investments, or are the uncertainties or likely restrictions unacceptable? Will existing and new clubs generate enough cash flows to pay providers of finance and make their own contribution to future investment? Does the company have the management, personnel and systems needed to run a much- expanded network of clubs?

Answer to Interactive question 2

General impacts might include:

Relations with stakeholders

Companies have to consider how changes in standards will affect the key indicators for various stakeholders. Changes that affect the **calculation of profit** will **influence shareholders** if profits

become significantly more volatile as a result. Shareholders and lenders who are concerned with financial stability will be influenced by changes in the **treatment** and **classification of capital instruments**, for example changes that force companies to include previously off balance sheet items.

Other stakeholders may be affected by **new forms of accounting** or **increased disclosures**. Employees will be interested in developments in human resource accounting. Pressure groups and other interested stakeholders ('ethical investors') will be interested in how the organisation is reporting its impact on the natural environment (considered further below).

Undertaking and form of transactions

Changes in standards may also mean companies do **not undertake transactions** or stop doing what they previously have been doing. The introduction of more **volatility into accounts** may make certain transactions less attractive, for example the granting of employee share options. Many companies in the UK have alleged that IAS 19, Employee Benefits which introduced more onerous rules for accounting for pension liabilities, has forced them to close their final salary pension schemes and offer less generous arrangements instead. However, other factors would have had a major influence on this decision as well.

Alterations to standards can also affect the **form transactions take**. For example, the difference in the accounting treatment of finance leases and operating leases under IAS 17 provided **opportunities to structure transactions so as to achieve a particular lease classification**. IFRS 16, the new standard on leasing, was developed to avoid this issue. All leases are now recognised in the lessee's statement of financial position, with a right-of-use asset and financial liability that recognise more expenses in profit or loss during the earlier life of a lease.

Investments

Having diverse requirements internationally arguably allows companies in some jurisdictions to shelter behind **doubtful arrangements**. Evidence from the US suggests that having one set of strict standards forces companies to be more efficient and more competitive. It also lowers the risks of investment, reducing the cost of capital and hence making previously unattractive overseas investments feasible.

Costs of compliance

In particular the costs of fundamental changes such as the introduction of international accounting standards may be significant. Not only will this impact on companies' **financial accounting function**, but also may affect the **management accounting function's work** if compatibility between financial and management accounting is to be maintained.

Answers to Self-test questions

1 Zelo plc

1.1 Reasons for the inclusion of environment and social issues in the Annual Report Competitor disclosure

As the non-executive directors have observed, **many comparable companies** already **do include such issues** in their Annual Report and Zelo may attract adverse publicity if it does not follow this trend.

Attracting investments

There may be **positive benefits** to reporting such issues too. For example, an increasing amount of attention is devoted to environmental and social issues. **'Green' and 'ethical' fund managers** have more funds available for investment and Zelo may be able to demonstrate that it is suitable for such investment, thereby increasing its share price. Disclosure may also demonstrate that Zelo is concerned about **maintaining good relations with shareholders**. By providing lots of information, Zelo demonstrates its commitment to **transparency**. Environmental disclosures can also give shareholders **greater assurance** by providing information about the **environmental risks** that Zelo faces and how it is managing them.

Strategic considerations

Moreover, the board should be constantly scanning the environment in which the organisation operates, including social and environmental areas, in order to assess the impact of any **changes on the future of the business**. Given that there is substantial interest among consumers and investment fund managers in doing business with companies which are environmentally and socially responsible, the board may intend to report on environmental and social issues as part of its competitive strategy.

Government pressure

There have also been recent comments by a government minister that 'if all companies do not make progress', legislation on environmental and social issues will be introduced. If Zelo was to include such issues in its Annual Report it would **avoid being forced into providing damaging disclosures required by a more stringent regime**.

1.2 The range of environmental and social issues to cover Consumption of raw materials

The greatest focus from an environmental point of view is likely to be on **consumption of raw materials from tropical areas**. The directors may wish to consider the concept of **'sustainability'**– is Zelo replanting at a rate equal to or greater than that at which it is harvesting? If so, it is likely to be viewed favourably.

Costs of processing

The **costs of processing** should also be considered, in particular the percentage of **energy** coming from renewable and non-renewable sources and the steps taken to increase the efficiency with which Zelo uses energy.

Packaging

Packaging is of increasing concern to many consumers. The proportion of both Zelo's products and its packaging made from recycled material should be measured, as should the ease with which they can be **recycled after use**.

Social issues

Social issues to cover include minimum rates of pay, the minimum age of child labour, working conditions and living conditions, such as the availability of healthcare and education. The

public's interest in consumer markets tends to focus on the **discrepancies between 'living standards' in their affluent market compared to those in less developed countries**. In setting standards the board needs to gather data about these issues in the source countries. Rates of pay expressed in relation to UK earnings may seem derisory but when expressed in relation to the local average they may seem much more acceptable. Using children aged just 14 as part of the labour force may seem less offensive if local schooling is provided up to the age of 12 and Zelo provides additional education as part of its benefits package.

Nature and extent of reporting

How well actual performance compares with what the board considers to be acceptable standards will determine the nature and extent of any reporting on these issues. If the board believes Zelo's performance is above average it may well make extensive disclosures in order to gain maximum benefit. The poorer the performance, the less it may choose to disclose. If there are any single issues that would lead to adverse publicity were they disclosed, it may choose to make no disclosures until these issues are resolved to an acceptable level.

1.3 Business partners

Zelo also needs to consider **whether to report on the activities of Zelo alone or on those of all its business partners**, including those from whom Zelo sources its raw materials and the sub-contractors it employs during production. It could be argued that Zelo cannot control its sub-contractors, and therefore should not include their activities within its report. For example, it could be deemed unfair if Zelo was held responsible for contractors employing young children without the board's knowledge.

However, it is **unlikely that 'we didn't know' would be accepted as a defence were damaging information made public**. Once the range of performance benchmarks is established, Zelo should therefore provide it to all its sub-contractors and advise them that they are expected to conform to such standards. These could be included as a requirement in the **supplier tendering process**.

Impact of other stakeholders

Problems are likely to centre on **identifying those issues that will be of concern in the future to stakeholders**. Zelo will also need to balance the demands of shareholders for maintaining a profitable activity with the concerns of pressure groups over the activity in question.

Changing viewpoints

The **popularity** or otherwise of environmental and social issues moves constantly with **changes in public opinion and government policy**. The board should endeavour to **anticipate the demands of its stakeholders**, however, rather than appear to be simply reacting to the current 'popular' issues.

2 Burtchester

2.1 Burtchester's CSR policy needs to deal with the following matters:

Manufacturing process

Waste – The manufacturing process currently produces large quantities of waste, so Burtchester should aim to recycle as much of its waste as possible.

However, Burtchester should also try to reduce the levels of waste it produces, for example by making sure its manufacturing processes are as efficient as possible, and by reducing resource usage as far as possible.

Noise – The manufacturing process is currently very noisy, so the policy needs to identify how the noise levels may be reduced. In this respect, Burtchester needs to consider both the level of noise pollution for the local neighbourhoods around its manufacturing plants, and the noise levels within the plants where staff have to work.

Smell – The manufacturing process is smelly as well as noisy, so the policy needs to address smell in a similar way to noise.

Procurement policy

Raw material procurement – As Burtchester sources some of its materials from economically underdeveloped countries, its policy needs to define the basis on which it does business with companies in those countries. In particular, Burtchester needs to highlight the importance of ensuring that the price, terms and conditions it negotiates with suppliers are fair.

Labour force`

There are two issues in relation to the workforce which the CSR policy needs to address:
Equality – Ensuring that **all members** of the workforce are treated fairly and equally.

Working conditions – Improving the conditions the staff have to work under. As well as the manufacturing process being noisy and smelly, there will also be health and safety requirements. The CSR policy should therefore highlight Burtchester's commitment to workplace safety.

Compliance

Burtchester has always tried to comply with its country's laws, and it should continue to do so. However, CSR involves more than simply complying with basic legal requirements, so Burtchester's policy should establish its basic principle of complying with the law before going on to highlight the other principles that will enable it to be a good corporate citizen.

2.2 Reports

Reports are often referred to as 'triple bottom line reporting' on the following topics:

- **Economic** – Information provided should go beyond that required by law. It should demonstrate how Burtchester adds value in a wider sense, eg, by creating human capital.
- **Environmental matters** – The report should include information on how Burtchester's activities impact on the environment, eg, impact of sourcing chemicals, waste management, noise.
- **Social** – Information provided may be on a range of social issues – working hours, rates, policies on child employment, equal opportunities, disablement policies.

A number of companies are following the Global Reporting Initiative (GRI), which aims to deliver transparency, accountability, reporting and sustainable development.

Assurance

Suitable measures to monitor policies would include checking the appropriate documentation of policies and evidence that they are circulated to staff. The policies should, for example, form part of staff's induction training.

Burtchester should initiate internal audit procedures or employ external assurance providers to monitor implementation and check accuracy of reporting mechanisms and key performance indicators.

External evidence of, for example, the effectiveness of environmental policies should be obtained from expert third parties where appropriate.

2.3 Requirements of IAS 37

Although local residents have given notice that they would be taking legal action against Burtchester, no claim appears to have been made. As yet there is no present obligation on Burtchester to make reparations for the disturbance caused and so certainly no provision needs to be made. Whether disclosure is required is questionable. As no notice of a claim has yet been received, the probability of a claim can be regarded as remote and no disclosure required under IAS 37.

Chapter 12
Business and securities valuation

Introduction
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2 Acquisitions and mergers
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4 Valuation of debt
5 International Valuation Standards (IVS)
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Introduction

Learning outcomes

- Explain, advise on and demonstrate appropriate valuation methods for businesses and equity securities using: asset-based; adjusted-earnings-based; and cash-based methods (for example SVA, EVA®, VBM, MVA and other appropriate techniques)
- Critically appraise business and securities valuation methods in the context of specified complex scenarios, including the consideration of natural capital
- Explain and demonstrate appropriate valuation techniques in the context of acquisitions and mergers; assess the contribution of due diligence procedures; and show the impact on corporate reports arising from acquisitions for groups in consolidated financial statements
- Determine the value of debt and explain the techniques used
- Analyse, structure and assimilate historic and forecast data provided to evaluate the impact on valuations using data analysis, relevant statistical tools and spreadsheets, 1114ocusing1114ng the sensitivity of valuations to underlying assumptions and changes in estimates

Knowledge brought forward

This chapter contains more complex examples of valuation methods first covered in Financial Management.

Syllabus links

In the chapter Strategic choice, we covered why companies acquire or merge with other companies. We shall go on to look at debt as a means of finance in more detail in the next chapter.

Examination context

Particularly in the areas of acquisitions and mergers, a question on the material in this chapter could go right across the syllabus. You could be asked to value a target, but there may be various problems with the figures you are given, bringing in therefore the analysis skills covered in the chapter Data analysis or knowledge of accounting treatments. Due diligence work could well be required as part of the acquisition process. You may also need to discuss post-acquisition strategies and their associated risks, and also whether the governance arrangements for the combined entity appear to be workable.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Valuation methods** Valuations are subjective and affected by many factors. Understanding the various available techniques and how to apply them to different situations is an important skill.	**Approach** This chapter covers some quite complex techniques, although you should be familiar with the basic valuation methods from your Financial Management Studies. A key skill is to choose the most appropriate techniques to use in different circumstances. Read through all the sections carefully to make sure you understand the techniques and where their use may be appropriate. Make sure you understand which corporate reporting requirements are relevant to valuation and acquisition decisions. Fair value measurement, and the effect of acquisitions on group reporting, are important here. Note also the due diligence work that would be carried out, tying it in with what you've covered in the chapter Strategic choice	In the examination you may be required to advise directors on the possible value(s) of businesses.	**IQ1: Dividend valuation models** This question asks you to value a company using the DVM with changing growth. It is a good question to revise your knowledge from your Financial Management studies. **IQ2: Valuations** This is another good question to revise your knowledge of using the P/E ratio model to value a company. **IQ3: DCF basis of valuations** This is a simple question that will allow you to revise the basics of the DCF basis of valuation.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
		Stop and think Which method(s) of valuation are more appropriate for valuing a small holding of shares; and which method(s) are more appropriate for valuing the entire share capital of a company?		
2	**Acquisitions and mergers** There are many reasons for valuing companies, or components of companies. Acquisitions, mergers, probate and tax purposes are just some of the reasons for doing so.	**Approach** This section looks at the various ways in which the target company in an acquisition or merger situation can be valued, and how to value the combined company After the acquisition or merger takes place. The FCF model is important so make sure you work through this section carefully. Also pay particular attention to section 2.8 which covers the corporate reporting issues of business combinations. **Stop and think** Can you correctly identify when an entity should be consolidated using the principles of IFRS 10?	In the examination you may be required to recommend the due diligence work that should be carried out on an acquisition, recommend the most appropriate Method for a business to finance an acquisition and discuss the strengths and weaknesses of post- acquisition arrangements. You may also be asked to discuss the significance of differences between companies that are merging, for example their approaches to corporate social responsibility.	**IQ4: Valuing an acquisition using FCF model** This detailed question on FCF model is really good preparation before tackling exam standard questions. **IQ5: APV of acquisition** Make sure you go through the worked example on APV before tackling this question.
3	**Unquoted companies and**	**Approach** The same valuation	A scenario on the exam could include	**IQ6: Valuation methods**

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	start-ups While it is possible to value quoted companies given the availability of vast quantities of information, unquoted companies are more difficult to value. Start-ups can be valued using the models we have previously described, although there may be problems related to the uncertainties surrounding forecast figures used.	methods covered in section 1 will be used here but make sure you are aware of the additional complications with valuing unquoted companies and start-ups. **Stop and think** How do you value a company that has made losses recently? How do you value a company that has never made any profits?	an unquoted company or a start-up.	Go through the worked example on valuing a startup before attempting this question. The question includes a lot of information that you will need to filter through and is indicative of the type of question you could see in your exam.
4	**Valuation of debt** The basic idea behind debt valuation is that the value is the PV of all future cash flows to the investor, discounted at the debt investor's required rate of	**Approach** Cover this brief section quickly, 1117ocusing mainly on understanding any new terminology. **Stop and think** Do you know how to value convertible	In the examination you may be required to calculate the value of debt.	**IQ7: Valuation of redeemable debt** This brief question will test your understanding of the terminology covered in the final section of the chapter.
Topic	**Practical significance**	**Study approach**	**Exam approach**	**Interactive questions**
	return. Different types of debt (irredeemable, redeemable and convertible) are valued in different ways.	Debt?		
5	**International Valuation Standards (IVS)**	**Approach** Cover the Handbook on IVS	In the examination you may be required to apply IVS guidelines on	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	Chartered Accountants should be aware of the valuation standards applicable for valuation assignment.	published quickly, focusing mainly on broad level overview on valuation topics.	valuation.	

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Valuation methods

Section overview

- Accountants can be involved in the valuation of shares for numerous purposes.
- The purpose for which a share is valued will determine the value that is placed on the share

1.1 Why value a company and what is a company worth?

In the context of strategic business management, the need to value a company usually arises when:

(a) a would-be acquirer wants to take over another company: the would-be acquirer needs to decide on an offer price and the target company board and shareholders need to decide whether the offer price is worth accepting

(b) two companies are discussing a merger and so need to agree the relative value of each of the two companies

(c) a company is planning the sale of a part of its business, for example to a management buyout team

So what is a company worth? There is no precise answer to this question. The value of the company will often depend on the purpose for which it is valued.

Companies could be worth:

- the tangible assets
- the assets plus goodwill
- whatever someone is prepared to pay for them

Context example: BizStats.com

BizStats.com is a website that provides useful financial ratios and statistics for company valuation. Industry statistics can be used for benchmarking purposes when undertaking valuations. Some rules of thumb for valuing companies in different industries are given by BizStats.com as follows:

Type of business	'Rule of Thumb' valuation
Accounting firms	100–125% of annual revenues
Law practices	90–100% of annual revenues
Veterinary practices	70% of annual revenues + inventory
Coffee shops	40% of annual sales + inventory 15–20% of annual sales +
Petrol stations	equipment/inventory

See: http://www.bizstats.com/reports/valuation-rule-thumb.php

CHAPTER 12

1.1.1 What affects the value of a company?

Company valuations may be affected by asset valuations, or by estimates of future profits, dividends or cash flows.

Company value is not just affected by sales and profit forecasts. Value can be influenced by the type of industry the company operates in, the level of competition faced by the company and the customer base. If a company has a popular product or product range that the public will want to buy, its value will increase, provided of course that the people who want to buy it can actually afford to do so.

Remember also that valuations are affected by judgements and opinions. Views about what a company is worth can vary enormously due to differences of opinion or differences in expectations about future cash flows or earnings.

As we saw in the chapter Strategic performance management, the increasing focus on natural capital is starting to change the way in which organisations report on their activities, with many now attempting to place financial values on the natural assets that they use, to enable them to make more informed business decisions. This can be linked to materiality assessment. What is material to the business informs what should be measured, valued and taken into decisions.

The use of natural capital valuation (discussed further in the chapter Finance awareness) is likely to be increasingly significant in assessing a company's value, although at the moment the methodologies and approaches in this area are still evolving.

1.2 Distortions in accounting figures

Anyone undertaking a valuation using accounting figures will need to be aware of possible distortions in those figures. These may involve distortions in assets, liabilities, equity or earnings. Professional scepticism should be applied when reviewing data prepared for a valuation.

Scepticism does not mean that the users assume that the data or its conclusions must be wrong; rather it means being aware that data analysis is not always accurate for several reasons:

- There may be bias inherent in the data that is analysed. This may be intentional or unintentional.
- The data may have been intentionally manipulated during the analysis process.
- The data may have been analysed accurately, but the presentation of the data, or the conclusions drawn from it may be flawed or may have been designed to mislead the users.
- The data could be distorted due to omissions.

To combat problems with earnings, it may be necessary to produce a **normalised earnings** figure to reflect the sustainable earnings of a company. This figure excludes profits from discontinued operations, changes in estimates and fair values, profits/losses from the sale of non-current assets and start-ups, and restructuring and redundancy costs, among other adjustments.

1.3 Income and cash flow models

You should be familiar with the different methods of valuing the equity in a business and when they might be used. The rest of this section goes over some of the basic valuation methods briefly.

1.3.1 Dividend valuation model: constant annual dividends

This valuation model may be used to estimate a value of a small shareholding (minority shareholding) in a company where the shareholder has no influence over company profitability. This model assumes that to the shareholder, the share is worth the PV of all future dividends receivable in perpetuity.

There are two types of dividend valuation model – the constant annual dividends model and the constant growth model. The constant annual dividend model assumes that there will be no growth in dividends in the future, and annual dividend will be a constant amount in perpetuity

$$P_0 = \frac{D}{k_e}$$

Where: P_0 = Value of the share (ex dividend)

D = Annual dividend per share

k_e = Shareholders' required rate of return

1.3.2 Dividend valuation model: constant annual dividends

This valuation model may be used to estimate a value of a small shareholding (minority shareholding) in a company where the shareholder has no influence over company profitability. This model assumes that to the shareholder, the share is worth the PV of all future dividends receivable in perpetuity.

There are two types of dividend valuation model – the constant annual dividends model and the constant growth model. The constant annual dividend model assumes that there will be no growth in dividends in the future, and annual dividend will be a constant amount in perpetuity.

Dividend valuation model: dividend growth model

The dividend growth model assumes that annual dividends will grow by a constant percentage amount, g, every year for the foreseeable future (in perpetuity).

Constant growth model

$$P_0 = \frac{D_1}{k_e - g}$$

Where: P_0 = Share value (excluding any dividend currently payable)

D_1 = D0 (1 + g) = Expected dividend in one year's time

D_0 = Current year's dividend

g = Growth rate in earnings and dividends

k_e = Shareholders' required rate of return

1.3.3 Dividend valuation model: Gordon growth model

The Gordon growth model is a variation on the dividend growth model. It assumes that the value of g is the rate of return on capital employed applied to the proportion of earnings that is retained and reinvested in the business:

g = br

Where: b = the proportion of earnings (after-tax profits) that are retained in the business

r = the rate of return on capital employed by the company

1.3.4 Variations on the dividend valuation model

The dividend valuation model assumes that the value of a share is the discounted value of expected future dividends in perpetuity, discounting at the shareholders' required rate of return on investment. A valuation can therefore be made when future rates of dividend growth are not constant.

For example, suppose that annual dividends are expected to increase by 5% next year, 3% the year after that and then to remain constant at that annual level in perpetuity. The estimated share value, using the dividend valuation model, would be:

$$P_0 = \frac{D_0(1.05)}{(1+k_e)} + \frac{D_0(1.05)(1.03)}{(1+k_e)2} + \frac{D_0(1.05)(1.03) \times 1/k_e}{(1+k_e)2}$$

Interactive question 1: Dividend valuation models

Target has just paid a dividend of £250,000. The current return to shareholders of companies in the same industry as Target is 12%, although it is expected that an additional risk premium of 2% will be applicable to Target, being a smaller and unquoted company. Compute the expected valuation of Target, if:

Requirements

1.1 the current level of dividend is expected to continue into the foreseeable future

1.2 the dividend is expected to grow at a rate of 4% p.a. into the foreseeable future

1.3 the dividend is expected to grow by 4% per year for the next three years and then to remain constant thereafter

See **Answer** at the end of this chapter.

1.3.5 Income-based valuation: price/earnings (P/E) ratio model

With the price/earnings (P/E) ratio model, a share is valued at a multiple of its future expected annual earnings.

P_0 = Annual earnings per share × P/E ratio multiple

The P/E ratio is simply the ratio between a share price and the earnings per share.

$$\frac{\text{Market price of share}}{\text{EPS}} \text{ which is the same as } \frac{\text{Total market value of equity}}{\text{Total earnings}}$$

A P/E ratio can be calculated for every company with shares that are quoted on a stock market.

The value of the P/E ratio reflects the market's appraisal of the share's future prospects. It is an important ratio, as it relates two key considerations for investors – the market price of a share and its earnings capacity. An EPS ratio that changes substantially and frequently may indicate an earnings volatility issue that could increase the perceived risk of the company. This can also impact on periodic P/E ratios.

For the purpose of company share valuations, a P/E ratio valuation may be used to value:

(a) The shares of a quoted company, where it is recognised that the offer price for the shares in a takeover must be higher than the current market price, in order to gain acceptance of the offer from the target company's shareholders

(b) The shares of an unquoted company, where the P/E ratio multiple is derived from the current P/E ratios for similar quoted companies

IAS 33, *Earnings Per Share* prescribes principles for the determination and presentation of earnings per share (EPS). Before the EPS can be considered to be useful by analysts trying to value the company, the earnings figure must be adjusted to reflect the sustainable earnings of the company. The result of this process is a **normalised earnings** figure or **adjusted earnings** figure.

The P/E ratio valuation method for equity is conceptually very simple, but it has no 'scientific' rationale to justify its use. It is not a discounted cash flow (DCF) valuation, and there are many different assumptions that can be used to reach a valuation. As a result, many different valuations for a share can be estimated from the P/E ratio model.

- Earnings should be an estimate of future annual earnings, but how should future expected earnings be estimated?
- The valuation of shares is made by applying a P/E ratio to the earnings figure, but what is the appropriate P/E ratio figure?

Note: The P/E ratio is the reciprocal of the earnings yield. (The earnings yield = EPS/Share price, expressed as a percentage.) The earnings yield could possibly be used as a rough guide to the cost of equity when there is zero expected future dividends or earnings growth. In these circumstances, there would be a rational connection between a P/E ratio-based valuation and the dividend valuation model or a DCF-based valuation.



Interactive question 2: Valuations

Flycatcher wishes to make a takeover bid for the shares of an unquoted company, Mayfly. The earnings of Mayfly over the past five years have been as follows.

20X0	20X1	20X2	20X3	20X4
£50,000	£72,000	£68,000	£71,000	£75,000

The average P/E ratio of quoted companies in the industry in which Mayfly operates is 10. Quoted companies which are similar in many respects to Mayfly are:

(1) Bumblebee, which has a P/E ratio of 15, but is a company with very good growth prospects

(2) Wasp, which has had a poor profit record for several years, and has a P/E ratio of 7

Requirement

What would be a suitable range of valuations for the shares of Mayfly?

See **Answer** at the end of this chapter.

1.3.6 Income-based valuation – discounted cash flow (DCF) method

This method of valuation is based on the present value (PV) of future cash flows generated by the company.

(a) The value of a company is basically the PV of all its future expected cash flows, discounted at the company's weighted average cost of capital.

(b) The value of a company's equity is the PV of all future cash flows attributable to equity shareholders, discounted at the cost of equity.

This method of valuation may be appropriate when one company intends to buy the assets of another company and to make further investments in order to improve cash flows in the future.

Problems with this method of valuation are:

(a) Difficulties in estimating future cash flows, especially beyond the next few years: it may therefore be assumed that annual cash flows after a given future year will be constant.

(b) The time period over which a valuation should be made. If expected annual cash flows in perpetuity are used for the valuation, most of the company's valuation will come from the (uncertain estimates) about cash flows beyond the next few years.

For example, suppose that a company is considering the acquisition of an all-equity company and it is expected that the annual cash flows from the acquisition will be £100,000. The company's cost of capital is 10% and it usually expects to achieve payback on its investments within five years.

(a) The PV of an annuity at 10% for Years 1–5 is 3.791, so the PV of expected future cash flows in Years 1–5 after the acquisition will be £100,000 × 3.791 = £379,100.

(b) The PV of an annuity at 10% from Year 6 onwards in perpetuity is (1/0.10) − 3.791 = 6.209, so the PV of expected future cash flows from Year 6 onwards after the acquisition will be £100,000 × 6.209 = £620,900.

(c) The total valuation of the company will be £1,000,000, of which about 62% is value attributed to expected earnings from Year 6 onwards, and this estimate may be unreliable.

Another method of DCF valuation is to estimate a future disposal value of the company at the end of a given investment period, and discount this to a PV for inclusion in the equity valuation. In the previous example, it might be estimated that the disposal value of the company at the end of the fifth year will be £500,000: if so, this value should be discounted to a PV and included in the equity valuation.



Interactive question 3: DCF basis of valuation

Diversification plc wishes to make a bid for Tadpole Ltd. Tadpole makes after-tax profits of £40,000 a year. Diversification believes that if further money is spent on additional investments, the after-tax cash flows (ignoring the purchase consideration) could be as follows.

Year	Cash flow (net of tax)
	£
0	(100,000)
1	(80,000)
2	60,000
3	100,000
4	150,000
5	150,000

Requirement

The after-tax cost of capital of Diversification is 15% and the company expects all its investments to pay back, in discounted terms, within five years. What is the maximum price that the company should be willing to pay for the shares of Tadpole?

See **Answer** at the end of this chapter.

1.4 Asset-based valuation models

If the net asset method of valuation is used, the value of a share in a particular class is equal to the net tangible assets attributable to that class divided by the number of shares in the class.

The net assets basis of valuation might be used in the following circumstances:

(a) As a **'floor value'** for a business that is up for sale – shareholders will be reluctant to sell for less than the net asset value (NAV). However, if the sale is essential for cash flow purposes

or to realign with corporate strategy, even the asset value may not be realised. Similarly, a loss-making company that is close to insolvency, that cannot easily dispose of its assets separately, may struggle to sell the business for its NAV.

(b) **As a measure of the 'security' in a share value**. The **asset backing** for shares provides a measure of the **possible loss** if the company fails to make the expected earnings or dividend payments. Valuable tangible assets may be a good reason for acquiring a company, especially freehold property which might be expected to increase in value over time.

(c) **As a measure of comparison in a business combination**. For example, if company A, which has a low asset backing, is planning a merger with company B, which has a high asset backing, the shareholders of B might consider that their shares' value ought to reflect this. It might therefore be agreed that something should be added to the value of the company B shares to allow for this difference in asset backing.

There are a number of issues with the net asset basis of valuation that assurance work done on valuation will need to consider:

a) Do the assets need **professional valuation**? If so, how much will this cost?

b) Have the **liabilities** been accurately quantified, for example deferred taxation? Are there any contingent liabilities? Will any balancing tax charges arise on disposal?

c) How have the **current assets** been valued? Are all receivables collectable? Is all inventory realisable? Can all the assets be physically located and brought into a saleable condition? This may be difficult in certain circumstances where the assets are situated abroad.

d) Can any **hidden liabilities** be accurately assessed? Would there be redundancy payments and closure costs?

e) Is there an **available market** in which the assets can be realised (on a break-up basis)? If so, do the balance sheet values truly reflect these break-up values?

f) Are there any **prior charges** on the assets?

g) Does the business have a regular **revaluation and replacement** policy? What are the bases of the valuation? As a broad rule, valuations will be more useful the better they estimate the **future cash flows** that are derived from the asset.

h) Are there factors that might indicate that the **going concern valuation** of the business **as a whole** is **significantly higher** than the valuation of the individual assets?

i) **What shareholdings** are being sold? If a minority interest is being disposed of, realisable value is of limited relevance as the assets will not be sold.

1.4.1 Impact of IFRS 13, *Fair Value Measurement*

The figure attached to an individual asset may vary considerably depending on the valuation basis that is used. One criticism of asset-based measures has been their reliance on historical cost, not representing market value and incorporating future earnings potential. Under the fair value measurement approach, assets and liabilities are revalued periodically to reflect changes in their value, with the resulting change impacting either net income or other comprehensive income for the period. The result is a balance sheet that better reflects the current value of assets and liabilities.

IFRS 13, *Fair Value Measurement* provides guidelines that may be used in assessing the fair values of assets (and which should be used for the purpose of financial reporting). Proponents argue that fair value accounting reflects current market conditions and provides more timely, relevant and forward- looking information.

IFRS 13 defines fair value as '**the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date**'.

The standard requires that the following are considered in measuring fair value:

(a) The asset or liability being measured

(b) The principal market for the asset (ie, that where the most activity takes place) or, where there is no principal market, the most advantageous market (ie, that in which the best price could be achieved) in which an orderly transaction would take place for the asset or liability

(c) The highest and best use of the asset or liability and whether it is used on a standalone basis or in conjunction with other assets or liabilities

(d) Assumptions that market participants would use when pricing the asset or liability

IFRS 13 states that valuation techniques must be those which are appropriate and for which sufficient data are available. Entities should maximise the use of relevant **observable inputs** and minimise the use of **unobservable inputs**.

The standard establishes a three-level hierarchy for the inputs that valuation techniques use to measure fair value:

Level 1	Quoted prices (unadjusted) in active markets for identical assets or liabilities that the reporting entity can access at the measurement date. Investments for which there is an exchange listed price will often fall into Level 1. An example would be a stock which is traded on the London Stock Exchange.
Level 2	Inputs other than quoted prices included within Level 1 that are observable for the asset or liability, either directly or indirectly, eg, quoted prices for similar assets in active markets or for identical or similar assets in non-active markets. Other observable inputs that may be used for valuation include interest rates and yield curves, credit spreads and/or implied volatilities. Certain derivatives such as interest rate swaps or forward currency contracts, where inputs are mostly observable, might fall into Level 2.
Level 3	Unobservable inputs for the asset or liability, ie, using the best available information, which may include the entity's own data and assumptions about market exit value. Unquoted private equity or venture capital holdings might be expected to fall into Level 3.

The IFRS identifies three valuation approaches.

(a) **Income approach.** Valuation techniques that convert future amounts (eg, cash flows and income and expenses) to a single current (ie, discounted) amount. The fair value measurement is determined on the basis of the value indicated by current market expectations about those future amounts.

(b) **Market approach.** A valuation technique that uses prices and other relevant information generated by market transactions involving identical or comparable (ie, similar) assets, liabilities or a group of assets and liabilities, such as a business.

(c) **Cost approach.** A valuation technique that reflects the amount that would be required currently to replace the service capacity of an asset (often referred to as current replacement cost).

The requirements of IFRS 13 are particularly relevant for valuing a business in a business combination, as discussed below.

1.5 Value-based models: shareholder value analysis (SVA)

1.5.1 Value-based management (VBM)

Value-based management (VBM) starts with the philosophy that the value of a company is measured by its discounted future cash flows. Value is created only when companies invest capital at returns that exceed the cost of that capital. VBM extends this philosophy by focusing on how companies use the idea of value creation to make both major strategic and everyday operating decisions. So VBM is an approach to management that aligns the strategic, operational and management processes to focus management decision-making on what activities create value.

1.5.2 Value drivers

A value driver is any variable that affects the value of the company. They should be ranked in terms of their impact on value and responsibility assigned to individuals who can help the organisation meet its targets. Value drivers can be difficult to identify, as an organisation has to think about its processes in a different way. Existing reporting systems are often not equipped to supply the necessary information. A good way of relating a range of value drivers is to use scenario analysis. It is a way of assessing the impact of different sets of mutually consistent assumptions on the value of a company or its business units.

1.5.3 SVA

The SVA method of valuation of a business is based on the PV of estimated future free cash flows earned by the business. There are seven factors or 'drivers' that affect the valuation:

(a) **Sales growth**. Sales growth in the valuation model is usually expressed as a percentage annual revenue growth rate.

(b) **Operating profit margin**. Expected operating profit is expressed as a percentage of sales revenue.

(c) The **tax rate** on profits.

(d) **Changes in working capital**. With sales growth, there is likely to be an increase in working capital each year, and the incremental working capital investment (IWCI) reduces free cash flow.

(e) **ed capital investment**. This may be analysed into replacement fixed capital investment (RFCI) and incremental fixed capital investment (IFCI). RFCI is capital investment that is required to replace existing assets that have worn out and need replacement: it is often assumed that RFCI is equal to the annual depreciation charge. IFCI is incremental capital investment in addition to RFCI. Capital investment reduces free cash flow.

(f) The **cost of capital**. This is the discount rate for converting free cash flows to a PV.

(g) The **competitive advantage period**. This is the period of time in the future during which the business is expected to achieve growth in its current form or condition. After this period of time, which may be for three to five years or so, it is often assumed that annual free cash flows will be a constant amount.

Assumptions about all seven of these value drivers will affect the valuation of the business.

1.5.4 Steps in the SVA valuation method

There are three steps in the SVA valuation process.

Step 1 Calculate the PV of estimated future annual free cash flows

Step 2 Add the value of non-operational assets held by the business, such as short-term investments and cash

Step 3 Subtract the value of the company's debt to obtain a valuation for the company's equity

Annual free cash flow may be calculated as follows:

	£
Operating profit	X
Less tax on this profit (at the tax rate)	(X)
Assumed profit after tax	X
Add back annual depreciation charge	X
	X
Deduct:	
IWCI	(X)
RFCI	(X)
IFCI	(X)
Equals: Free cash flow	X

If it is assumed that replacement fixed capital investment equals the annual depreciation charge, these two items can be omitted from the free cash flow calculation.

Context example: SVA

The following information is available about a company.

Current year annual sales: £10m

Competitive advantage period: 3 years. Cost of capital: 12%.

Information about other value drivers:

	A	B	C	D	E
1		Year 1	Year 2	Year 3	Year 4 onwards
2	Sales growth (%)	10	10	10	0
3	Operating profit margin (% of sales)	20	20	20	15
4	Tax rate (%)	20	20	20	20
5	IFCI (as a % of sales revenue growth)	10	10	10	0
6	IWCI (as a % of sales revenue growth)	5	5	5	0

The annual depreciation charge in the current year is £5 million and it is assumed that replacement fixed capital investment will equal the annual depreciation charge in every year.

The company has cash and short-term investments together worth £6 million, and its debt liabilities are £8.5 million.

An SVA valuation can be made as follows.

Step 1: Calculate the PV of future free cash flows.

B12 =NPV(0.12,B11:D11)

	A	B	C	D	E
1		Year 1	Year 2	Year 3	Year 4 onwards
2		£m	£m	£m	£m
3	Sales revenue	11.000	12.100	13.310	13.310

4	Increase in sales revenue	1.000	1.100	1.21	
5					
6	Operating profit	2.200	2.420	2.662	1.997
7	Tax	0.440	(0.484)	(0.532)	(0.399)
8		1.760	1.936	2.130	1.597
9	IFCI	(0.100)	(0.110)	(0.121)	
10	IWCI	(0.050)	(0.055)	(0.061)	
11	Free cash flow	1.610	1.771	1.948	1.597
12	PV @12% T1-3	4.24 [1]			
13	Discount factor for perpetuity				0.712/0.12
14	PV year 4 onwards				£9.48m
15	Total PV of free cash flow (4.24 + 9.48)	13.72			

[1] =NPV(0.12,B11:D11)

Total valuation of free cash flows = £13.72m

Note: Depreciation charges and RFCI cancel each other out and so are ignored in the calculation.

Steps 2 and 3:

	£m
Value of future free cash flows	13.72
Value of non-operational assets: cash and investments	6.00
Value of debt	(8.50)
Value of equity	11.22

The value of the company's equity is £11.22 million.

Professional skills focus: Assimilating and using information

A question on business valuations will present you with a lot of information in the scenario. One of the professional skills assessed in the CA exams considers your ability to evaluate the relevance of information provided. When applying the different methods of valuations, it is important that you select the correct information from the scenario.

1.6 Value-based models: EVA® and MVA

1.6.1 Economic value added (EVA®)

Economic value added or EVA® is a trademark of the consultancy firm Stern Stewart & Co, now Stern Value Management. It is an alternative measure for calculating the value created in a given period of time (typically a year) by a business. It is based on the view that accounting

measures are not a reliable measure of value creation, and value creation should be based on economic principles rather than accounting methods.

In practice, annual EVA® is calculated by making adjustments to accounting costs and values to obtain an estimate of economic cost and value. It is also based on the idea that a business must cover both its operating costs **and** its capital costs.

EVA® = Net Operating Profit after Tax (NOPAT) – WACC × Capital Employed

OR

EVA® = (ROIC – WACC) × Capital Employed (where ROIC = Return on invested capital)

NOPAT should be a measure of economic profit, not accounting profit. The NOPAT figure is not therefore taken directly from the financial statements, but contains a number of adjustments to get to economic profit from accounting profit.

Capital employed should be the economic value of capital employed, not the accounting value of capital employed.

Examples of adjustments to accounting profit and asset values to derive economic profit and economic value of capital employed are as follows:

(a) **Spending on some intangible assets** is typically written off against accounting profit in the year that the expenditure is incurred. Examples are research expenditure, training expenditure and advertising expenditure which develops the value of a brand name. The approach is to capitalise all such expenditures on intangibles that create long-term value, and charge against profit only an amount for the consumption of the asset's economic value in the year. This adjustment affects both profit and capital employed.

(b) **Provisions or allowances** are not real costs and should be excluded from profit. For example, any increase in the allowance for irrecoverable receivables should be added back to both profit and capital employed.

(c) **Leases** should not be capitalised. The lease rental is a measure of the economic cost of the lease in the reporting period. The charge against profit should therefore be the lease rental charge, not depreciation on the leased asset plus a finance charge.

(d) The charge for **depreciation** should be an economic cost rather than an accounting charge. However, calculating economic depreciation from accounting figures can be a very complex process, and it is often assumed therefore that accounting depreciation is a reasonable approximation for economic depreciation; therefore no adjustment for depreciation charges is required.

It should be remembered that EVA® is not an estimate of cash flow. It includes non-cash items of cost in deriving NOPAT, and the cost of capital includes the cost of equity, which is not a cash flow item.



Context example: EVA®

The derivation of EVA® can be illustrated using an example.

The profits of a company in the year ended 31 December 20X5 were reported to management as follows:

	£m
Profit before interest and tax	20.0
Interest	(7.0)
	13.0
Tax at 20%	(2.6)
Profit after tax	10.4

The following information is also available:

(a) Dividends paid in the year were £6.6 million.

(b) During the year, the company invested in research and development (R&D) for the first time, spending £6 million. This cost was written off in full against profit. To measure EVA®, the spending should be capitalised and the amortisation charge for R&D in the year should be £2 million.

(c) The company increased its allowance for irrecoverable debts from £3 million to £3.8 million.

(d) The company's weighted average cost of capital is 9%.

(e) At the beginning of the year the company's economic capital employed was £160 million.

(f) Assume for the purpose of this calculation that the cost of capital is based on the value of capital employed at the beginning of the year.

NOPAT is calculated as follows:

	£m
Profit before interest and tax	20.0
Tax on profits	(2.6)
Tax relief on interest charge (20% × £7m)	(1.4)
	16.0
Adjustments to get to economic profit	
Increase in allowance for irrecoverable debts	0.8
R&D spending capitalised	6.0
R&D intangible asset amortised	(2.0)
NOPAT	20.8

EVA® is calculated as follows:

	£m
NOPAT	20.8
Capital employed × cost of capital (£160m × 9%)	(14.4)
EVA®	6.4

The economic value of capital employed at the beginning of the next financial year, 20X6, is as follows.

	£m
Economic value of capital employed at beginning of year	160.0
Adjustments for	
Increase in allowance for irrecoverable debts	0.8
R&D (£6m – £2m)	4.0
EVA®	6.4
Dividends	(6.6)
Economic value of capital employed at end of year	164.6

1.6.2 Market value added (MVA)

The market value added (MVA) of a company is a measure of how much the management of a company has added to the value of the capital contributed by the capital providers.

MVA may be measured for the company as a whole, or just from the perspective of the equity shareholders.

From the perspective of the company as a whole

MVA = Market value of the company (equity and debt) – Capital invested

Capital invested is measured as the book value of equity plus the book value of debt.

From the perspective of the company's equity shareholders

MVA = Market value of equity – Book value of equity

MVA is related to EVA because MVA is the PV of the future EVAs of the company.

If the market value and the book value of debt is the same, then the MVA simply measures the difference between the market value of common stock and the equity capital of the firm.

A firm's equity MVA is sometimes expressed as a market to book ratio:

$$\frac{\text{MVA}}{\text{book value of equity}}$$

The higher the MVA, the better for the company, as a high MVA indicates that the company has created wealth for its shareholders.

The main problem with MVA is that it does not take opportunity cost of capital into account, nor does it account for intermediate cash returns to shareholders.

Context example: MVA

A stock market company has issued 150 million shares which have a current market value of £4.50 per share as at the end of its financial year. The company's statement of financial position shows that the book value of the company's shares and reserves at this date was £420 million.

The company has debt capital of £200 million which has a par market value. MVA = (150 million × £4.50) – £420m = £255m.

2 Acquisitions and mergers

Section overview

This section looks at the various ways in which the target company in an acquisition or merger situation can be valued, and how to value the combined company after the acquisition or merger takes place.

2.1 Asset-based model

When a valuation of a company is made for the purpose of a merger or acquisition, it makes good sense to use a number of different valuation methods to obtain estimates of what a suitable valuation might be. The different estimated valuations can then be compared, to establish whether they are similar or whether they differ substantially. If they differ, the reasons for the differences may help to provide some insights into an appropriate valuation and give the buyer an idea of how high the offer price might reasonably be taken.

When making several valuations of a potential takeover target, a useful starting point is an assets- based valuation. Unless the target company is in serious financial difficulty, an assets-based valuation should indicate the absolute minimum price that shareholders in the target company are likely to accept.

The asset-based approach to valuation was reviewed above, but this section considers some issues in further detail.



Worked example: Asset-based model

The summary statement of financial position of Cactus is as follows.

Non-current assets		**£**	**£**
Land and buildings			160,000
Plant and machinery			80,000
Motor vehicles			20,000
			260,000
Goodwill			20,000
Current assets			
Non-current assets	**£**	**£**	**£**
Inventory		80,000	
Receivables		60,000	
Short-term investments		15,000	
Cash		5,000	
		160,000	
Current liabilities	**£**	**£**	
Payables	80,000		
Taxation	20,000		
		(100,000)	
			60,000
			340,000
12% loan notes			(60,000)
Deferred taxation			(10,000)
			270,000
Ordinary shares of £1			80,000
Reserves			140,000
			220,000
4.9% preference shares of £1			50,000
			270,000

Requirement

What is the value of an ordinary share using a net assets basis of valuation?

Solution

If the figures given for asset values are not questioned, the valuation would be as follows.

	£	£
Total value of assets less current liabilities		340,000
Less non-identifiable asset (goodwill)		20,000
Total value of assets less current liabilities		320,000
Less: preference shares	50,000	
loan notes	60,000	
deferred taxation	10,000	
		120,000
NAV of equity		200,000
Number of ordinary shares		80,000
Value per share		£2.50

However, it may be inappropriate to accept without question the figures in the statement of financial position. So an initial question should be to ask how reliable are the valuations of assets and liabilities in the target company's accounts.

2.1.1 Fair values

The accounting rules affecting acquisitions are particularly important in the context of asset valuations.

The presumption in IFRS 3, *Business Combinations* is that:

- An acquirer carefully reviews the value of the net assets to be acquired as part of the process of deciding how much to offer for the acquiree.
- The purchase consideration is the result of a transaction between a willing buyer and a willing seller in an arm's length transaction, so it is at fair value.
- This fair value can be reliably allocated across the identifiable assets and liabilities assumed.

The general rule under IFRS 3 is that the subsidiary's assets and liabilities **must be measured at fair value** except in **limited, stated cases**. The assets and liabilities must:

(a) meet the definitions of assets and liabilities in the *ConceptualFramework*

(b) be part of what the acquiree (or its former owners) exchanged in the business combination rather than the result of separate transactions

As discussed above, IFRS 13 provides extensive guidance on how the fair value of assets and liabilities should be established.

2.1.2 Examples of fair values and business combinations Non-current assets

For non-financial assets, fair value is decided based on the highest and best use of the asset as determined by a market participant.



Worked example: Land

Anscome Co has acquired land in a business combination. The land is currently developed for industrial use as a site for a factory. The current use of land is presumed to be its highest and best use unless market or other factors suggest a different use. Nearby sites have recently been developed for residential use as sites for high-rise apartment buildings. On the basis of that development and recent zoning and other changes to facilitate that development,

Anscome determines that the land currently used as a site for a factory could be developed as a site for residential use (ie, for high-rise apartment buildings) because market participants would take into account the potential to develop the site for residential use when pricing the land.

Requirement

How would the highest and best use of the land be determined?

Solution

The highest and best use of the land would be determined by comparing both of the following:

(1) The value of the land as currently developed for industrial use (ie, the land would be used in combination with other assets, such as the factory, or with other assets and liabilities)

(2) The value of the land as a vacant site for residential use, taking into account the costs of demolishing the factory and other costs (including the uncertainty about whether the entity would be able to convert the asset to the alternative use) necessary to convert the land to a vacant site (ie, the land is to be used by market participants on a standalone basis)

The highest and best use of the land would be determined on the basis of the higher of those values.

Restructuring and future losses

An acquirer **should not recognise liabilities for future losses** or other costs expected to be incurred as a result of the business combination.

IFRS 3 (revised) explains that a plan to restructure a subsidiary following an acquisition is not a present obligation of the acquiree at the acquisition date. Neither does it meet the definition of a contingent liability. Therefore, an acquirer **should not recognise a liability for a restructuring plan** as part of allocating the cost of the combination unless the subsidiary was already committed to the plan before the acquisition.

This **prevents creative accounting**. An acquirer cannot set up a provision for restructuring or future losses of a subsidiary and then release this to profit or loss in subsequent periods in order to reduce losses or smooth profits.

Contingent liabilities

Contingent liabilities of the acquiree are **recognised** in a business combination if their **fair value can be measured reliably**. A **contingent liability** must be recognised even if the outflow is not probable, provided there is a present obligation.

This is a departure from the normal rules in IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*, under which contingent liabilities are not normally recognised, but only disclosed.

After their initial recognition, the acquirer should measure contingent liabilities that are recognised separately at the higher of:

(a) the amount that would be recognised in accordance with IAS 37

(b) the amount initially recognised

2.1.3 Intangible assets

The acquiree may have **intangible assets**. IFRS 3 states that these can be recognised separately from goodwill only if they are **identifiable**. An intangible asset is identifiable only if it:

(a) is **separable**, ie, capable of being separated or divided from the entity and sold, transferred or exchanged, either individually or together with a related contract, asset or liability; or

(b) arises from **contractual or other legal rights**.

The problem with these principles is that they may exclude many intangible assets that are important to the acquirer. This renders this method unsuitable for the valuation of most established businesses, particularly those in the **service industry or industries** where intellectual capital may be a significant feature, as we discussed in the chapter Strategic marketing and brand management.

Remember the business's intellectual capital will include:

- patents, trademarks and copyrights
- franchises and licensing agreements
- R&D
- brands
- technology, management and consulting processes
- know-how, education, vocational qualification
- customer loyalty
- distribution channels
- management philosophy

Some of these will fulfil the IFRS 3 definition of intangible assets. Others will not, although they may be considered as separate valuable assets by the parties to the acquisition.

The valuation of brands, which may have a big influence on acquisition price, was also discussed in depth in the chapter Strategic marketing and brand management.

2.1.4 Exceptions to the IFRS 13 principles

(a) **Deferred tax**: Use IAS 12 values

(b) **Employee benefits**: Use IAS 19 values

(c) **Indemnification assets**: Measurement should be consistent with the measurement of the indemnified item, for example an employee benefit or a contingent liability

(d) **Reacquired rights**: Value on the basis of the remaining contractual term of the related contract regardless of whether market participants would consider potential contractual renewals in determining its fair value

(e) **Share-based payment**: Use IFRS 2 values

(f) **Assets held for sale**: Use IFRS 5 values

2.2 Market relative model (the P/E ratio)

The P/E ratio method produces an earnings-based valuation of shares. This is done by deciding a suitable P/E ratio and multiplying this by the EPS for the shares which are being valued. The EPS could be a historical EPS or a prospective future EPS. For a given EPS figure, a higher P/E ratio will result in a higher price. A high P/E ratio may indicate:

(a) Optimistic expectations

EPS is expected to grow rapidly in the years to come, such that a high price is being paid for future profit prospects. Many small but successful and fast-growing companies are valued on the stock market on a high P/E ratio. Some stocks (for example, those of some internet companies in the late 1990s) have reached high valuations before making any profits at all, on the strength of expected future earnings.

(b) Security of earnings

A well-established low-risk company would be valued on a higher P/E ratio than a similar company whose earnings are subject to greater uncertainty.

(c) Status

If a quoted company (the predator) made a share for share takeover bid for an unquoted company (the target), it would normally expect its own shares to be valued on a higher P/E ratio than the target company's shares. A quoted company ought to be a lower-risk company but, in addition, there is an advantage in having shares which are quoted on a stock market as the shares can be readily sold. The P/E ratio of an unquoted company's shares might be around 50% to 60% of the P/E ratio of a similar public company with a full stock exchange listing (and perhaps 70% of that of a company whose shares are traded on stock markets for smaller companies, such as the UK's Alternative Investment Market).



Worked example: Takeover offer price

Spider plc is considering the takeover of an unquoted company, Fly Ltd. Spider's shares are quoted on the stock exchange at a price of £3.20 and since the most recent published EPS of the company is 20p, the company's P/E ratio is 16. Fly Ltd is a company with 100,000 shares and current earnings of
£50,000, 50p per share.

Requirement

How might Spider plc decide on an offer price?

Solution

The decision about the offer price is likely to be preceded by the estimation of a 'reasonable' P/E ratio in the light of the particular circumstances.

(1) If Fly Ltd is in the **same industry** as Spider plc, its P/E ratio ought to be lower because of its lower status as an unquoted company.

(2) If Fly Ltd is thought to be **growing fast** so that its EPS will rise rapidly in the years to come, the P/E ratio that should be used for the share valuation will be higher.

(3) If the acquisition of Fly Ltd would **contribute substantially to Spider's own profitability and growth**, or to any other strategic objective that Spider has, then Spider should be willing to offer a higher P/E ratio valuation in order to secure acceptance of the offer by Fly's shareholders.

Of course, the P/E ratio on which Spider bases its offer will probably be lower than the P/E ratio that Fly's shareholders think their shares ought to be valued on. Some haggling over the price might be necessary.

Spider might decide that Fly's shares ought to be valued on a P/E ratio of 60% × 16 = 9.6, that is, at 9.6 × 50p = £4.80 each.

Fly's shareholders might reject this offer and suggest a valuation based on a P/E ratio of, say, 12.5, that is, 12.5 × 50p = £6.25.

Spider's management might then come back with a revised offer, say valuation on a P/E ratio of 10.5, that is, 10.5 × 50p = £5.25.

The haggling will go on until the negotiations either break down or succeed in arriving at an agreed price.

2.3 Free cash flow model

The procedure for valuing a target company on the basis of its predicted free cash flow is similar to that used for investment appraisal. The free cash flow model was illustrated earlier in the explanation of the SVA valuation method.

Step 1 **Calculate the free cash flow (FCF)**

The FCF to the firm is defined as:

FCF = Earnings before interest and taxes (EBIT)
Less tax on EBIT
Add back non-cash charges
Less capital expenditure
Less net working capital increases
Plus net working capital decreases
Plus salvage values received

Example

(Note: The tax rate is assumed to be 24% in this example.)

	£
Earnings before interest and taxes (EBIT)	80,000
Remove taxes (1 – Tax rate, T)	× 76%
Operating income after taxes	60,800
Depreciation (non-cash item)	14,000
Less capital expenditure	(9,000)
Less changes to working capital	(1,000)
FCF	64,800

Step 2 **Forecast FCF and terminal value**

For a time horizon of six years, the following values need to be estimated:



Step 3 **Calculate the weighted average cost of capital (WACC).**

The WACC is calculated using the cost of equity (k_e) and cost of debt (k_d).

$$WACC = k_e \times \frac{E}{E + D} + k_d \times \frac{D}{E + D}$$

Where: D = the value of debt E = the value of equity

(Note that k_d is the cost of debt **after** tax.)

Step 4 **Discount FCF at WACC to obtain value of the firm. Calculate equity value**

Step 5 Equity value = Value of the firm – Value of debe

Worked example: Calculating value per share

You have completed the following forecast of FCFs for an eight-year period, capturing the normal business cycle of Marathon plc.

Year	FCF
	£m
20X7	1,860
20X8	1,887.6
20X9	1,917.6
20Y0	1,951.2
20Y1	1,987.2
20Y2	2,016
20Y3	2,043.6
20Y4	2,070

Marathon has non-operating assets with an estimated PV of £400 million. Based on the PV of future interest payments, the PV of its outstanding debt is £3,000 million. FCFs are expected to grow at 3% beyond 20Y4. The WACC is assumed to be 12%. Marathon has 8,000 million shares in issue.

Requirement

What is Marathon's value per share?

Solution

B11 =NPV(0.12,B3:B10)

	A	B
1		FCF
2	Year	£m
3	20X7	1,860.0
4	20X8	1,887.6
5	20X9	1,917.6
6	20Y0	1,951.2
7	20Y1	1,987.2
8	20Y2	2,016.0
9	20Y3	2,043.6
10	20Y4	2,070.0
11	**NPV**	**9,680**

	£m
Total PV for forecast period (see above)	9,680
Terminal value = (£2,070 × 1.03)/(0.12 – 0.03) × 0.404	9,571
Value of non-operating assets	400
Total value of Marathon	19,651
Less value of debt	(3,000)
Value of equity	16,651
Shares outstanding	8,000
Value per share	£2.08



Interactive question 4: Valuing an acquisition using FCF model

The management of Atrium, a company in the DIY industry, is considering making a bid for Tetrion, a rival company. The current market price of Tetrion is $29 per share and a 20% premium will persuade Tetrion shareholders to sell. In order to convince its shareholders, a valuation of Tetrion is undertaken. The following information is used.

Statement of financial position

	20X4 $	20X3 $	Change $
Cash	300	100	200
Receivables	2,500	1,500	1,000
Inventory	2,600	1,400	1,200
Property, plant and equipment	5,800	4,000	1,800
Accum depreciation	750	500	250
Net property, plant and equipment	5,050	3,500	1,550
Total assets	10,450	6,500	3,950
Payables	3,600	2,560	1,040
Long-term debt	2,000	2,000	–
Equity	4,850	1,940	2,910
Total liabilities and owners' equity	10,450	6,500	3,950

Statement of profit or loss

	20X4 $
Sales	12,000
Cost of sales	3,500
Selling: General administrative	3,000
Depreciation	250

	20X4
	$
Total expense	6,750
Interest	100
Income before tax	5,150
Taxes (24%)	1,236
Profit for the year	3,914

FCFs

	20X4
Cash flows – Operations	$
Revenue	12,000
Cash expenses	(6,500)
Taxes	(1,236)
Total	4,264
Cash flows – Investments	
Working capital	1,360
Non-current assets	1,800
Total	3,160
FCF	1,104

The following predictions are made by the management of Atrium for the next five years:

	20X5	20X6	20X7	20X8	20X9
FCFs	$	$	$	$	$
Sales	12,000	15,000	18,750	23,438	29,297
Operating costs excluding depreciation	6,500	8,125	10,156	12,696	15,869
EBITD	5,500	6,875	8,594	10,742	13,428
Depreciation	250	300	360	432	519
EBIT	5,250	6,575	8,234	10,310	12,909
Less tax on EBIT @ 24%	1,260	1,578	1,976	2,474	3,098
Plus depreciation	250	300	360	432	518
Less capital expenditure	1,800	2,160	2,592	3,110	3,732
Less additions to working capital	1,360	1,700	2,125	2,656	3,320
FCF	1,080	1,437	1,901	2,502	3,277
Terminal value (3,277/0.12)					27,308
Total FCF	1,080	1,437	1,901	2,502	30,585

Requirement

If the WACC is 12% and there are 100 shares outstanding, what is the value of each share of Tetrion plc?

At the current market price of Tetrion of $29 per share, should the shareholders of Atrium agree to the acquisition of Tetrion?

See **Answer** at the end of this chapter.

2.4 EVA® approach

As we discussed above, EVA® is an estimate of the amount by which **earnings** exceed or fall short of the required minimum rate of return that shareholders and debt holders could get by investing in other securities of comparable risk. It can be relevant in valuing acquisitions, as an alternative to a P/E multiple valuation or an FCF valuation.

It has already been explained that estimates of future annual EVAs can be discounted to a PV to obtain a company valuation.

The PV of the company using EVA® is given by:

$$PV = C + \frac{EVA_1}{1 + WACC} + \frac{EVA_2}{(1 + WACC)_2} = \ldots$$

Where: C = the invested capital

The value of the company is equal to the value of the invested capital plus the PV of the economic profits from operations.

To arrive at the value of the equity, we need to subtract the value of debt from the value of the company.



Professional skills focus: Structuring problems and solutions

You are expected to be able to structure information from various sources into suitable formats for analysis. Questions on business valuations often ask you to apply various valuation methods. You will therefore need to identify the information that is relevant to each different valuation method to come up with a suitable range of values for an entity.

2.5 Adjusted present value approach

The adjusted present value (APV) approach to company valuation is another method based on discounting expected future cash flows to a PV.

It may be used to value large acquisitions, where the method of financing the acquisition means that the acquirer's WACC is inappropriate as the discount rate. For example, if an all-equity company proposes to make a large acquisition financed by one-third equity and two-thirds new debt, its existing cost of capital should not be used as a discount rate because it does not reflect the change in financing structure that the acquisition would entail.

An APV valuation may therefore be appropriate when an acquisition will require new financing, such as an increase in debt.

The APV approach is to divide the valuation into two stages.

(a) First, discount the expected future cash flows from the acquisition at an ungeared cost of equity. This is the cost of equity that would apply if the acquiring company were all-equity financed. Discounting the future cash flows at an ungeared cost of equity produces a 'base case NPV'.

(b) The base case NPV ignores the benefit of tax relief on any debt capital that will be used to finance the acquisition. It also ignores the costs of raising any new capital to finance the acquisition. These are the financing effects of the acquisition, and the PV of these financing effects should be calculated separately.

(c) The financing effects are likely to include the **benefits of tax relief on future interest payments**. These should be discounted to a PV. If in doubt, discount these financing benefits at the pre-tax cost of debt, because this is the cost that best reflects the systematic risk associated with these cash flows. You should not discount these future tax benefits at the ungeared cost of equity!

(d) If the company is planning to issue new capital to pay for the acquisition, a financing cost will be the issue costs for the new capital. You should generally assume that issue costs associated with new equity do not attract tax relief, whereas issue costs associated with the issue of new debt capital does attract tax relief. Since these costs will occur at the beginning of the investment they do not need to be discounted.

(e) The APV of the acquisition is the base case NPV plus (or minus, if negative) the PV of the financing effects of the acquisition. The APV is a valuation for the target company.

The APV approach is consistent with Modigliani and Miller's views of capital structure and the effect of gearing on the cost of capital.



Worked example: APV of acquisition

Suppose that the management of Xeron is considering the acquisition of Neron, an unquoted company as at the end of 20X7. The owners of Neron want £450 million for the business. The analysis of the prospects of Neron by Xeron is reflected in the following income statements and statements of financial position.

Proforma income statements and statements of financial position

	Current Years						
	20X7	20X8	20X9	20Y0	20Y1	20Y2	20Y3
	£m	£m	£m	£m	£m	£m	£m
Sales	620.00	682.00	750.20	825.22	907.74	998.52	998.52
Less cost of goods sold	(410.00)	(441.00)	(475.10)	(512.61)	(553.87)	(599.26)	(599.26)
Gross profit	210.00	241.00	275.10	312.61	353.87	399.26	399.26
Operating expenses	(133.00)	(144.30)	(156.53)	(169.78)	(184.16)	(199.78)	(199.78)
EBIT	77.00	96.70	118.57	142.83	169.71	199.48	199.48
Less interest expense	-	(32.00)	(26.88)	(20.19)	(11.73)	(1.27)	-
Earnings before tax	77.00	64.70	91.69	122.64	157.98	198.21	199.48
Less tax	(16.17)	(13.59)	(19.25)	(25.75)	(33.18)	(41.62)	(41.89)
Net income	60.83	51.11	72.44	96.89	124.80	156.59	157.59
Current assets	100.00	100.00	100.00	100.00	100.00	242.90	404.55
Non-current assets	400.00	378.00	354.00	328.00	300.00	270.00	238.00
Total assets	500.00	478.00	454.00	428.00	400.00	512.90	642.55
Debt	400.00	335.95	252.35	146.63	15.94	-	-
Equity	100.00	142.05	201.65	281.37	384.06	512.90	642.55
Total finance	500.00	478.00	454.00	428.00	400.00	512.90	642.55

Assumptions:

Growth rate for sales until 20Y3	10%
Assumed growth rate after 20Y3	Nil
Depreciation expense (20X7)	£40m
Depreciation charges will increase by £2 million per year between 20X7 and 20Y3	
Accounting depreciation equals tax depreciation	
Interest rate on debt	8%
Tax rate	21%
All debt is interest bearing	
Capital expenditures/year	£40m
All available cash flow is applied to repaying debt until repaid in full	
Ungeared beta	1.2
Terminal value in 20Y3 assumes that annual cash flows in perpetuity will be the same as in 20Y3	
Risk-free rate of return	6.0%
Market risk premium	7.5%

Requirement

Calculate the APV of Neron based on the expected future cash flows of the company, and determine on the basis of your answer whether Xeron should proceed with the acquisition.

Solution

In this example, we are required to calculate the APV of the company's cash flows. The financing effects are therefore simply the tax relief on the interest cost of the company's debt.

Step 1: Calculation of firm FCF

	A	B	C	D	E	F	G	H
1		20X7	20X8	20X9	20Y0	20Y1	20Y2	20Y3
2		£m	£m	£m	£m	£m	£m	£m
3	EBIT (1 – tax rate)	60.83	76.39	93.67	112.84	134.07	157.59	157.59
4	Depreci ation charge	40.00	42.00	44.00	46.00	48.00	50.00	52.00
5	Capital expendit ure	(40.00)	(40.00)	(40.00)	(40.00)	(40.00)	(40.00)	(40.00)
6	FCF	60.83	78.39	97.67	118.84	142.07	167.59	169.59

Step 2: Calculation of APV

The FCFs are discounted at the ungeared cost of equity capital which is:

$k_{eu} = r_f + ß_a (r_m - r_f) = 6 + 1.2 \times 7.5 = 15\%$

H3 =NPV(0.15,B3:G3)

	A	B	C	D	E	F	G	H
1		20X8	20X9	20Y0	20Y1	20Y2	20Y3	20Y4
2		£m	£m	£m	£m	£m	£m	£m
3	FCF	78.39	97.67	118.84	142.07	167.59	169.59	**458.03**

Step 3: Calculation of terminal value

The terminal value under the no growth assumption after 20Y3 is:

Value as at 20Y3: £169.59/0.15 = £1,130.60m

The PV of the terminal value is:

£1,130.60m/$(1.15)_6$ = £488.79m

Step 4: Calculation of PV of financing effects

Here the only financing cash flows are the tax relief on the company's tax payments.

The tax shield is discounted at the pre-tax cost of debt (8%).

	20X8	20X9	20Y0	20Y1	20Y2	20Y3	Total
	£m	£m	£m	£m	£m	£m	£m
Interest	32.00	26.88	20.19	11.73	1.27	0	
Tax savings at 21%	6.72	5.64	4.24	2.46	0.27	–	
PV tax saving @8%	0.926	0.857	0.794	0.735	0.681		**16.42**

Calculation of APV

	£m
PV of cash flows 20Y8 to 20Y3	458.03
PV of terminal value 20Y3	488.79
Base case NPV	946.82
PV of interest relief on debt	16.42
Total APV of company	963.24
Value of debt	(400.00)
Value of equity	563.24

The owners of the company are asking for £450 million for the company, therefore the value to Xeron of acquiring Neron is £563.24m – £450m = £113.24m.

The managers of Xeron should proceed with the acquisition.

Interactive question 5: APV of acquisition

Main plc is considering the acquisition of a profitable unquoted company Peek Limited. The directors of Peek have indicated that they would accept a cash offer of £54 million.

(1) If the acquisition goes ahead, it would be financed entirely by a 10-year loan at 6% interest. The costs of raising this money would be £1 million, so the total amount borrowed will be £55 million.

(2) It is estimated that the acquisition will increase the operating cash flows of Main plc, net of tax, by £4 million per year in perpetuity.

(3) The ungeared equity beta that would apply to this investment is 0.80. The risk-free rate of return is 5% and the market rate of return is 8%. Tax is at the rate of 21%.

Requirement

Using the APV method of valuation, suggest whether the purchase price demanded by the directors of Peek would be acceptable to Main plc.

See **Answer** at the end of this chapter.

2.6 Other aspects of valuation of a takeover

When a company is considering the acquisition of another company a number of factors may affect the valuation. These include:

(a) **Synergy**. This is considered below.

(b) **Risk**. An acquisition may affect the financial risk or business risk profile of the acquiring company. Changes in the risk profile should be considered and, where appropriate, the discount rate for a valuation should be adjusted.

(c) **Real options**. A real option is an option to take a new or different course of action in the future, that will arise if the current investment is undertaken. In a takeover, a real option may be the opportunity that the takeover creates for the company to develop its business in the future, for example by expanding into new geographical markets, if the opportunity would not exist if the takeover did not occur. Real options can have a positive value and the valuation of a target company may be affected by a valuation of any real options that may exist.

(d) **Financing**. It has already been explained that the method of financing an acquisition, and its cost, will affect the valuation of a takeover target.

2.7 Synergy

The existence of synergies may increase shareholder value in an acquisition. The identification, quantification and announcement of these synergies are essential as shareholders of the companies involved in the acquisition process may need persuasion to back the merger.

Unfortunately the actual value of synergy benefits from a takeover cannot be established until after the takeover has occurred, and even then results are often not as expected.

Context example: Bunnings and Homebase

Bunnings, the Australian DIY retailer that aimed to revamp Homebase in the UK, as part of its plans to expand its home improvement empire internationally, ended up writing off A$1 billion (£547 million) in February 2018 after some major mistakes.

Perth-based Wesfarmers, the parent company of Bunnings and one of Australia's biggest companies, bought Homebase for £340 million in 2016, but the heavy losses (nearly £100 million in the last six months of 2017) became untenable.

Wesfarmers admitted that the Bunnings management team had made mistakes, including the axing of the entire Homebase senior management team and about 160 middle managers as soon as the deal was completed. They also abandoned the home furnishings operation that had been a big draw for female shoppers, turning the stores into starkly industrial DIY warehouses selling power tools, large barbecues and other heavy equipment not necessarily suited to the British market. Added to this was the fact that the British home renovations sector has already been dampened by rising living costs and a weak housing market.

On 25 May 2018 Wesfarmers sold the UK and Ireland Bunnings/Homebase operation to the turnaround specialist Hilco for £1. The 24 stores that had been rebranded as Bunnings reverted to the Homebase name.

Claims by management that a takeover will provide synergy benefits may be unrealistic, and simply an excuse to justify a higher bid price for a takeover target, to increase the probability that the bid will be accepted.

Claims of synergy benefits resulting from a takeover should be scrutinised carefully, and with professional scepticism.

2.7.1 Revenue synergy

Revenue synergy exists when the acquisition of the target company will result in higher revenues for the acquiring company, higher return on equity or a longer period of growth. Revenue synergies arise from:

(a) increased market power

(b) marketing synergies

(c) strategic synergies

Revenue synergies are more difficult to quantify relative to financial and cost synergies. When companies merge, cost synergies are relatively easy to assess pre-deal and to implement post-deal. But revenue synergies are more difficult. It is hard to be sure how customers will react to the new situation, but revenue synergies must be identified and delivered. Customer Relationship Management and Product Technology Management are the two core business processes that will enable the delivery of revenue.

2.7.2 Cost synergy

A cost synergy results primarily from the existence of economies of scale. As the level of operation increases, the marginal cost falls and this will be manifested in greater operating margins for the combined entity. The resulting savings from economies of scale are normally estimated to be substantial.

2.7.3 Financial synergy Diversification

Acquiring another firm as a way of reducing risk cannot create wealth for two publicly traded firms, with diversified stockholders, but it could create wealth for private firms or closely held publicly traded firms. A takeover, motivated only by diversification considerations, has no effect on the combined value of the two firms involved in the takeover. The value of the combined firms will always be the sum of the values of the independent firms. In the case of private firms or closely held firms, where the owners may not be diversified personally, there might be a potential value gain from diversification.

Cash slack

When a firm with significant excess cash acquires a firm with great projects but insufficient capital, the combination can create value. Managers may reject profitable investment opportunities if they have to raise new capital to finance them. It may therefore make sense for a company with excess cash and no investment opportunities to take over a cash-poor firm with good investment opportunities, or vice versa. The additional value of combining these two firms lies in the PV of the projects that would not have been taken if they had stayed apart, but can now be taken because of the availability of cash.

Tax benefits

The tax paid by two firms combined together may be lower than the taxes paid by them as individual firms. If one of the firms has tax deductions that it cannot use because it is losing money, while the other firm has income on which it pays significant taxes, the combining of the two firms can lead to tax benefits that can be shared by the two firms. The value of this synergy is the PV of the tax savings that accrue because of this merger. The assets of the firm being taken over can be written up to reflect new market value, in some forms of mergers, leading to higher tax savings from depreciation in future years.

Debt capacity

The tax paid by two firms combined together may be lower than the taxes paid by them as individual firms. If one of the firms has tax deductions that it cannot use because it is losing money, while the other firm has income on which it pays significant taxes, the combining of the two firms can lead to tax benefits that can be shared by the two firms. The value of this synergy is the PV of the tax savings that accrue because of this merger. The assets of the firm being taken over can be written up to reflect new market value, in some forms of mergers, leading to higher tax savings from depreciation in future years.

Debt capacity

By combining two firms, each of which has little or no capacity to carry debt, it is possible to create a firm that may have the capacity to borrow money and create value. Diversification will lead to an increase in debt capacity and an increase in the value of the firm. It has to be weighed against the immediate transfer of wealth that occurs to existing bondholders in both firms from the shareholders. When two firms in different businesses merge, the combined firm will have less variable earnings and may be able to borrow more (have a higher debt ratio) than the individual firms.

2.8 Corporate reporting issues: business combinations

Acquisitions and mergers are subject to the requirements of IFRS 3, *Business Combinations* and IFRS 10, *Consolidated Financial Statements*. **These standards impose a number of significant requirements on the parties involved that may influence their thinking**.

2.8.1 Nature of acquisition

Remember that IFRS 3 only applies to the acquisition of businesses. All business combinations
should be accounted for by applying the **acquisition method**. This involves five key steps:

- **identifying the acquirer**
- Determining the **acquisition date**
- Measuring the consideration transferred
- Recognising and measuring the **identifiable assets acquired**, the **liabilities assumed** and any **non- controlling interest** in the acquiree
- Recognising and measuring **goodwill**, or a gain from a bargain purchase

Definition

Business: An integrated set of activities and assets capable of being conducted and managed for the purpose of providing:

(a) a return in the form of dividends; or

(b) lower costs or other economic benefits directly to investors or other owners.

A business generally consists of inputs, processes applied to those inputs, and resulting outputs that are, or will be, used to generate revenues. If goodwill is present in a transferred set of activities and assets, the transferred set is presumed to be a business.

If it is just some assets, and perhaps liabilities, which are acquired, then the consideration is allocated according to other IFRSs (eg, IAS 16 in the case of property, plant or equipment) across what has been acquired; no goodwill should be recognised.

IFRS 3 requires that one of the combining entities must be identified as the **acquirer**. Identifying the acquirer requires the assessment of all rights, powers, facts and circumstances. In a reverse acquisition, for example, the legal acquirer (the entity issuing the shares) may come under the control of the legal acquiree. Under IFRS 3 the legal acquiree will be treated as the acquirer.

2.8.2 Acquiring control

IFRS 10 states that an investor controls an investee if and only if it has all of the following:

(a) **power** over the investee (see below)

(b) Exposure, or **rights**, **to variable returns** from its involvement with the investee (see below)

(c) The **ability to use its power** over the investee to affect the amount of the investor's returns

(If there are changes to one or more of these three elements of control, then an investor should reassess whether it controls an investee.)

Power is defined as existing rights that give the current ability to direct the relevant activities of the investee.

Rights to variable returns may include rights to dividends, remuneration for servicing an investee's assets or liabilities or fees and exposure to loss from providing credit support.

Worked example: Control

1 Twist holds 40% of the voting rights of Oliver and 12 other investors each hold 5% of the voting rights of Oliver. A shareholder agreement grants Twist the right to appoint, remove and set the remuneration of management responsible for directing the relevant activities. To change the agreement, a two-thirds majority vote of the shareholders is required. To date, Twist has not exercised its rights with regard to the management or activities of Oliver.

Requirement

Explain whether Twist should consolidate Oliver in accordance with IFRS 10.

2 Copperfield holds 45% of the voting rights of Spenlow. Murdstone and Steerforth each hold 26% of the voting rights of Spenlow. The remaining voting rights are held by three other shareholders, each holding 1%. There are no other arrangements that affect decision-making.

Requirement

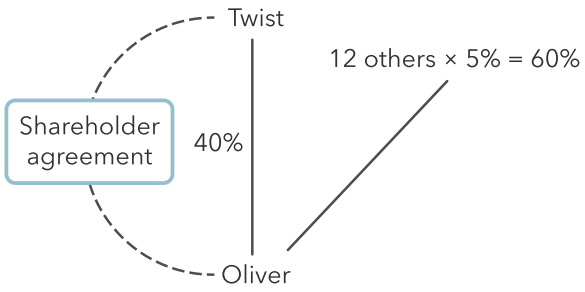
Explain whether Copperfield should consolidate Spenlow in accordance with IFRS 10.

3 Scrooge holds 70% of the voting rights of Cratchett. Marley has 30% of the voting rights of Cratchett. Marley also has an option to acquire half of Scrooge's voting rights, which is exercisable for the next two years, but at a fixed price that is deeply out of the money (and is expected to remain so for that two-year period).

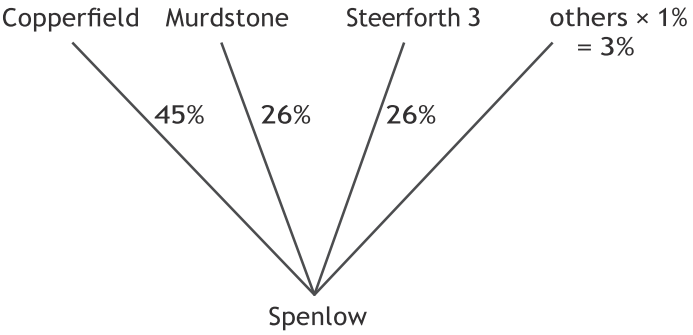
Requirement

Explain whether either of Scrooge or Marley should consolidate Cratchett in accordance with IFRS 10.

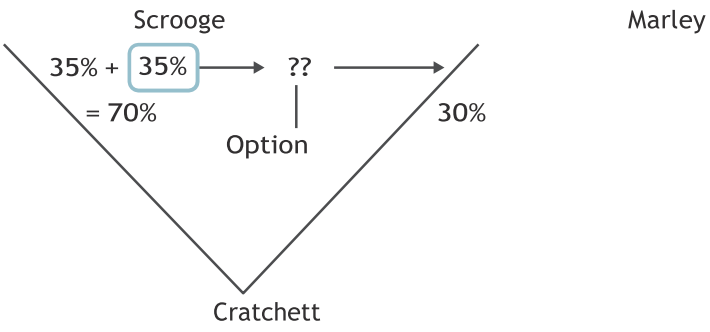
Solution



The absolute size of Twist's holding and the relative size of the other shareholdings alone are not conclusive in determining whether the investor has rights sufficient to give it power. However, the fact that Twist has **a contractual right to appoint**, **remove and set the remuneration of management** is sufficient to conclude that it **has power over Oliver**. The fact that Twist has not exercised this right is not a determining factor when assessing whether Twist has power. In conclusion, Twist does control Oliver, and should consolidate it.



In this case, the size of Copperfield's voting interest and its size relative to the other shareholdings are sufficient to conclude that Copperfield **does not have power**. Only two other investors, Murdstone and Steerforth, would need to cooperate to be able to prevent Copperfield from directing the relevant activities of Spenlow.



Scrooge holds a majority of the current voting rights of Cratchett, so is likely to meet the power criterion because it appears to have the current ability to direct the relevant activities. Although Marley has currently exercisable options to purchase additional voting rights (that, if exercised, would give it a majority of the voting rights in Cratchett), the terms and conditions associated with those options are such that the options are not considered substantive.

Thus voting rights, even combined with potential voting rights, may not be the deciding factor. Scrooge should consolidate Cratchett.

2.8.3 Fair value requirements

We discussed in the context of asset-based valuations above the basic requirement of IFRS 3 that the identifiable assets and liabilities acquired are measured at their acquisition-date fair value.

We have seen that the requirement to account for a business combination using fair value measurement will also mean that the acquisition is subject to the requirements of IFRS 13.

The consequences of the recognition of the acquiree's assets and liabilities at the acquisition date are that:

- The acquirer's consolidated statement of profit or loss should include the acquiree's profits and losses from the same date.
- The fair values of the acquiree's net assets form the basis of all the subsequent accounting in the consolidated financial statements. For example, depreciation should be based on the fair values of property, plant and equipment, which may not be the same as the carrying amount in the acquiree's statement of financial position.
- Any non-controlling interest in the acquiree is based on the non-controlling interest share of the net assets at their fair values.

2.8.4 Valuation of previously held stake

Sometimes an acquirer will make an acquisition, having previously held an equity interest in the company being acquired. If this happens, the previously held interest will be remeasured to fair value immediately before control is achieved and the gain or loss taken to profit or loss.

2.8.5 Valuation of non-controlling interest



Definition

Non-controlling interest: The equity in a subsidiary not attributable, directly or indirectly, to a parent.



Context example: NCI

X Ltd owns 75% of the ordinary shares of Z Ltd.

In this case, X Ltd **controls** 100% of Z Ltd as it owns more than 50% of the ordinary shares, but it only **owns** 75%. The **NCI owns the remaining 25%**.

In group accounts, the ownership interest of both X's shareholders and the NCI needs to be reflected, and the part of the group net assets in which X's shareholders do not have the ownership interest needs to be distinguished from that in which they do.

As both X's shareholders and the NCI own equity (in (X + Z) and Z, respectively), the sum of their respective ownership interests is described as **equity** in the consolidated statement of financial position.

The figure for **non-controlling interests (NCIs) in the statement of financial position** will be the appropriate proportion of the translated share capital and reserves of the subsidiary plus, where the NCI is valued at fair value, its share of goodwill translated at the closing rate. In addition, it is necessary to show any dividend declared but not yet paid to the NCI at the reporting date as a liability. The dividend payable should be translated at the closing rate for this purpose.

The **NCI in profit or loss** will be the appropriate proportion of profits available for distribution. If the functional currency of the subsidiary is the same as that of the parent, this profit will be arrived at after charging or crediting the exchange differences.

The **NCI in total comprehensive income** includes the NCI proportion of the exchange gain or loss on translation of the subsidiary financial statements. It does not, however, include any of the exchange gain or loss arising on the retranslation of goodwill.

2.8.6 Fair value of consideration

Part of the consideration for an acquisition may not pass to the acquiree's shareholders at the acquisition date but may be deferred until a later date.

Deferred consideration should be measured at its **fair value at the acquisition date**. The fair value depends on the form of the deferred consideration.

Where the deferred consideration is in the form of **equity shares**:

- its fair value should be measured at the acquisition date
- the deferred amount should be recognised as part of equity, under a separate heading such as 'shares to be issued'

Where the deferred consideration is payable in cash:

A **liability** should be recognised at the **PV** of the amount payable.



Worked example: Deferred consideration

The acquisition date for Winter plc's purchase of the whole of the share capital in Summer Ltd was 1 July 20X5. The consideration comprised 1,000,000 shares in Winter plc to be issued on 1 July 20X5, £10 million in cash payable on 1 July 20X5, 200,000 shares in Winter plc to be issued on 1 January 20X6 and £5 million in cash payable on 1 July 20X6.

The market price of one Winter plc share was 450p on 1 July 20X5, 475p on 1 January 20X6 and 480p on 1 July 20X6. A discount rate of 7% was appropriate.

Requirement

What is the total consideration for this acquisition?

Solution

	£m
1,000,000 shares issued on 1 July 20X5 at 450p	4.50
200,000 shares to be issued on 1 January 20X6 at 450p	0.90
Cash payable on 1 July 20X5	10.00
Cash payable on 1 July 20X6 – £5m/1.07	4.67
	20.07

Definition

Contingent consideration: is an obligation of the acquirer to transfer additional consideration to the former owners of the acquiree if specified future events occur or conditions are met.

The acquiree's shareholders may have a different view from the acquirer as to the value of the acquiree. They may have different views about the likely future profitability of the acquiree's business.

In such cases it is often agreed that additional consideration may become due, depending on how the future turns out. Such consideration is 'contingent' on those future events/conditions.

Contingent consideration agreements result in the acquirer being under a legal obligation at the acquisition date to transfer additional consideration, should the future turn out in specified ways. IFRS 3 requires contingent consideration to be recognised as part of the consideration transferred and measured at its fair value at the acquisition date.

Worked example: Contingent consideration

On 1 January 20X7 A acquired 100% of the shares of B when the fair value of B's net assets was £25 million. The consideration was 4 million shares in A issued at 1 January 20X7 when their market value was £6 per share and a cash payment of £6 million on 1 January 20X9 if the cumulative profits of B exceeded a certain amount by that date. At 1 January 20X7 the probability of B hitting that earnings target was such that the fair value of the possible cash payment was £2 million.

At 31 December 20X7 the probability that B would exceed the required profit level had risen and the fair value of the contingent consideration was judged to be £4 million.

At 31 December 20X8 it was clear that B had exceeded the profit target and the whole amount would be payable.

Requirement

Show calculations of the amounts to be recognised in the statements of financial position and in profit or loss for the two years ended 31 December 20X8.

Solution

The contingent consideration should be recognised at the acquisition date. It should then bere- measured at fair value each year end until ultimate settlement of the amount. Changes in fair value should be recognised in profit or loss.

Statement of financial position at 31 December 20X7

		£'000
Non-current assets – goodwill		
Consideration transferred	– 4 million shares × £6	24,000
– contingent at fair value		2,000
		26,000
Net assets acquired		(25,000)
Goodwill		1,000
Non-current liabilities – contingent consideration		4,000

	£'000
Profit or loss for year ended 31 December 20X7	
Additional contingent consideration	2,000
Statement of financial position at 31 December 20X8	
Non-current assets – goodwill (unchanged)	1,000
Current liabilities – contingent consideration	6,000
Profit or loss for year ended 31 December 20X8	
Additional contingent consideration	2,000

2.8.7 Goodwill on acquisition

IFRS 3 requires goodwill acquired in a business combination (or a gain on a bargain purchase) to be measured as:

	£
Consideration transferred: Fair value of assets given, liabilities assumed and equity instruments issued, including contingent amounts	X
NCI at the acquisition date	X
	X
Less total fair value of net assets of acquiree	(X)
Goodwill/(gain on a bargain purchase)	X/(X)

This calculation includes the NCI and is therefore calculated based on the whole net assets of the acquiree. Under IFRS 3 the acquirer is required to include a qualitative description of the factors that make up goodwill, such as expected synergies from combining operations and intangible assets that do not qualify for separate recognition.

Worked example: Worked example – goodwill

Abbot Ltd acquires 60% of Bishop Ltd on 31 December 20X5 for £80,000. The statement of financial position of Bishop Ltd at this date is as follows.

	£
Freehold land (fair value £30,000)	20,000
Goodwill arising on the acquisition of a sole trader	5,000
Sundry assets (carrying amount = fair value)	130,000
	155,000
Share capital	20,000
Retained earnings	85,000
Equity	105,000
Liabilities	50,000
	155,000

Requirement

Calculate the goodwill acquired in the business combination with Bishop Ltd.

Solution

The goodwill acquired:

Group structure

Abbot Ltd



Bishop Ltd

Net assets of Bishop Ltd

	Year end	=	Acquisition date
	£		£
Share capital			20,000
Retained earnings	85,000		
Add fair value uplift (30,000 – 20,000)	10,000		
Less goodwill	(5,000)		
			90,000
			110,000

Goodwill

	£
Consideration transferred	80,000
Non-controlling interest at acquisition (40% × 110,000 (W2))	44,000
Less FV of net assets at acquisition (W2)	(110,000)
Goodwill	14,000

2.9 Accounting policies and pensions

2.9.1 Accounting policies

Accounting policies can have a significant impact on the valuation of acquisitions. They can either work to inflate or depress the share price. **Optimistic accounting policies**, **valuing assets generously**, **bringing forward revenue recognition and delaying provisions** may inflate the company position and share price. On the other hand, **accelerating expenses or making very conservative estimates** of future earnings may depress share prices. There may be agency issues with both approaches. Directors who wish to retain their own jobs may attempt to boost earnings, and hence share prices to deter takeovers. Alternatively, directors who feel they can benefit if a takeover occurs may be tempted to depress a company's market valuation, even though shareholders may lose out as a result.

2.9.2 Pensions

In the chapter Strategic performance management, we raised the possibility that concerns over pension liabilities may impact on strategic decisions, and this includes acting as a deterrent to takeover. A pension parachute is a type of 'poison pill', preventing the acquiring firm in a hostile takeover from using surplus cash in the pension fund to finance the takeover. The arrangement ensures that the fund's assets remain the property of its participants.

A large deficit in a defined benefit pension fund can also be a deterrent in a merger or acquisition because of the risks involved. Under UK pension regulations, for example, employers operating schemes that have deficits must agree a plan with the scheme's trustees to pay off the deficit, generally by making extra payments. The requirement for three-yearly full valuations of pension schemes, taking into account new actuarial assumptions, can cause companies to reappraise their plans.

2.10 Due diligence

Due diligence should be carried out in two stages.

Preliminary due diligence should be conducted before an offer is made to acquire a target company. This will involve an investigation into the business on the basis of available information. An initial offer to acquire the target business will be made on the basis of this investigation.

If the board of the target company indicate a willingness to accept the offer, the two parties should sign heads of agreement, setting out the main terms of the acquisition, such as:

- what the purchaser has agreed to buy
- the purchase price and the form of the purchase consideration (for example cash and shares)
- preconditions that the buyer may insist on, such as a requirement that the target company's next half-yearly or annual accounts will show at least a stated amount of profit
- any warranties or indemnities that the sellers will provide

When heads of agreement have been signed, the purchaser should carry out a second stage of due diligence, which will be more detailed than the first. This may involve meeting with the target company's main customers or suppliers, studying the company's accounts for sales growth, profit margins and assets and liabilities. Due diligence may also include an employee audit, as well as legal due diligence (for example checking the legal ownership of certain assets, such as land and buildings and patents).

We covered due diligence in the context of acquisitions generally in the chapter Strategic choice. Here we focus on the financial and tax aspects.

Due diligence will attempt to achieve the following:

- Confirm the accuracy of the information and assumptions on which a bid is based
- provide the bidder with an independent assessment and review of the target business
- Identify and quantify areas of commercial and financial risk
- Give assurance to providers of finance
- place the bidder in a better position for determining the value of the target company

The precise aims will, however, depend on the types of due diligence being carried out.

2.10.1 Financial due diligence

Financial due diligence is a review of the target company's **financial position**, **financial risk and projections**. It differs from a statutory audit in a number of ways:

(a) Its nature, duties, powers and responsibilities are normally determined **bycontract or financial regulation** rather than by statute.

(b) Its purposes are **more specific** to an individual transaction and to particular user groups.

(c) There is normally a specific focus on **risk and valuation**.

(d) Its nature and scope is **more variable from transaction to transaction**, as circumstances dictate, than a statutory audit.

(e) The information being reviewed is likely to be different and **more future orientated**.

(f) The **timescale available is likely to be much tighter** than for most statutory audits.

The information which is subject to financial due diligence is likely to include the following:

- Financial statements
- Management accounts
- Projections
- Assumptions underlying projections
- Detailed operating data
- Working capital analysis
- Major contracts by product line
- Actual and potential liabilities
- Detailed asset registers with current sale value/replacement cost
- Debt/lease agreements
- Current/recent litigation
- property and other capital commitments

2.10.2 Tax due diligence

Information must be provided to allow the potential purchaser to form an assessment of the **tax risks and benefits** associated with the company to be acquired. Purchasers will wish to assess the robustness of tax assets, and gain comfort about the position in relation to potential liabilities (including a possible latent gain on disposal due to the low base cost).

2.10.3 Agreeing the final terms

Following detailed due diligence, some of the terms of the offer may be changed, and any such changes will have to be negotiated and agreed.

3 Unquoted companies and start-ups



Section overview

- While it is possible to value quoted companies given the availability of vast quantities of information, unquoted companies are more difficult to value.
- Shares are not quoted on the stock exchange, therefore the companies themselves have to be valued before the share price can be determined.
- Share price volatilities have to be determined before any share-based payments (such as share option schemes) can be valued.
- The size of the holding may affect the control of the company and this might have a significant effect on the valuation.
- Start-ups can be valued using the models we have previously described, although there may be problems related to the uncertainties surrounding forecast figures used.

3.1 Why value unquoted shares?

Unquoted shares are valued for all sorts of purposes, for example:

- o sell the company
- part of a divorce settlement
- or taxation purposes
- o raise capital from investors

The important thing to note is that as soon as you change the purpose of the valuation you will change the share price. Shares are valued on different bases for different purposes; there are many sources of information on which share values can be based.

3.2 How do we value unquoted shares?

The valuation of unquoted shares is not an exact science and there may be a range of possible values.

When trying to value an unquoted company's shares, one of the most frequent errors is to assume that this exercise can be conducted without taking into account the value of the entire enterprise. Sole reliance on such measures as the P/E ratio for comparable quoted companies can result in valuable information being omitted from the valuation process, leading to potential under- or overvaluation.

When unquoted shares are valued for tax purposes, the valuation method used tends to be – with a few exceptions – the estimated price that the shares would achieve if sold on the open market.

Various court decisions over the years have given guidance on how an imaginary market sale can be used to arrive at a value. For this purpose, the valuer should consider:

- the sustainable future reported earnings after tax
- the company's performance as shown in its financial statements, and any other information normally available to its shareholders
- the commercial and economic background at the valuation date
- the size of the shareholding and the shareholders' rights
- the company's dividend policy
- appropriate yields and P/E ratios
- the value of the company's assets
- gearing levels
- the existence of certain rights and obligations attached to preference shares, convertible bonds
- any other relevant factors

3.3 Valuing majority shareholdings

The valuation of a majority shareholding must reflect the extent to which a potential buyer of the shares can or cannot control – or influence – the company. There are many degrees of control which are usually determined by the voting power of a particular block of shares. These range from full control (including the power to liquidate the company) to a small or non-existent influence over the company's affairs.

Unless there are exceptional circumstances, if the degree of control is less than complete, the value of the shares will be less than a pro rata proportion of the overall value of the company.

In order to value a majority shareholding, consider the following:

(a) The rights and prospects attached to the shareholding – for example, the right to appoint directors.

(b) Whether restrictions apply to, for example, sale of shares or receipt of dividend.

(c) Whether control is in excess of 75%, meaning that the articles of association can be changed by the holder. A 75% holding also gives the holder the right to wind up the company.

(d) Whether minority shareholders have contractual rights that would be expensive for majority shareholders to buy out.

(e) Whether the minority shareholdings are concentrated on one individual or dispersed.

When a majority shareholding is valued, there is a control premium attached to it, meaning that the overall value will be in excess of the pro rata value to the value of the company as a whole. The extent of the premium depends on the answers to the above factors. As a crude guidance, the following table gives you a starting point for consideration:

Shareholding	Discount on 100% company value
75% +	Nil to 5%
> 50% but < 75%	10–15%
50%	20–30%
	(but a much greater discount if another party holds the other 50%)

3.4 Valuation of minority shareholdings of unquoted companies

While there are few differences between the value of a majority holding in a private company and that of a quoted company, there can be major differences between the values of minority holdings. This depends very much on the nature of the unquoted company.

Several factors should be considered when valuing minority shareholdings in an unquoted company:

- **Marketability** - This may be the only difference between the quoted company being used for comparison purposes and the unquoted company.
- **Size** - Private companies are often much smaller than quoted companies. Any size adjustment to valuations depends very much on specific circumstances. Although the growth prospects of an unquoted company may not be as attractive as those of a quoted company, smaller organisations operating in a niche market may be more dynamic and therefore the adjustment in valuation may be positive rather than negative.
- **Expectations of cash returns** – The marketability factor of unquoted companies is affected by the likelihood of minority shareholders receiving a future cash return, either in the form of dividends or due to the flotation of the company. Quoted companies' shareholders are more likely to receive cash returns by way of dividends or share buybacks. If there are little or no prospects of cash returns in the near future, there will be little or no value attached to minority shareholdings unless some significant influence can be exerted on the directors or other shareholders.

Rather than using DCF analysis and market comparability models for valuation purposes, it is more appropriate to value minority shareholdings of unquoted companies using the PV of anticipated dividend streams.

However, consideration should also be given to any contractual protection or rights pertaining to minority shareholdings (eg, through the Articles of Association or service contracts). Similarly, the question of which parties, and how many, hold the majority shares may be a key issue in valuing any tranche of minority shares. These circumstances would need to be considered on a case by case basis.

Professional skills focus: Applying judgement

Business valuations are not an exact science. You will therefore need to apply judgement when concluding on an appropriate value to use when acquiring a company.

3.5 Valuation of start-up businesses

Here we define a start-up business as an unquoted company that has not been in business for long. Start-ups may be attractive takeover propositions, not because of what they have achieved so far, but because of their future potential. The value of start-ups may lie in expectations of future revenue growth and profits, or in the value of some of the assets that it owns.

The valuation of **start-ups** presents a number of challenges for the methods we have considered so far, due to their unique characteristics which are summarised below:

(a) Most start-ups typically have little or no track record. Their revenues may be growing, but from a very small starting point.

(b) Because a start-up is earning relatively little revenue, it will probably be making losses and will also be cash flow negative.

(c) It will not yet have established a strong customer base.

(d) Its products may be insufficiently tested in the market. Its products may even still be under development, and not yet marketed at all. If so, its likely market acceptance and the volume of sales demand for the product will be unknown.

(e) Its management team may be inexperienced in business.

(f) Little may be known about the nature of competition, if the product is new.

3.5.1 Projecting economic performance

All valuation methods require reasonable projections to be made with regard to the key drivers of the business. The following steps should be undertaken with respect to the valuation of a high- growth start-up company.

Identifying the drivers

Any market-based approach or DCF analysis depends on the reasonableness of financial projections. Projections must be analysed in light of the market potential, resources of the business, the management team, financial characteristics of the guideline public companies, and other factors.

Period of projection

One characteristic of high-growth start-ups is that in order to survive they need to grow very quickly. Start-ups that do grow quickly usually have operating expenses and investment needs in excess of their revenues in the first years and experience losses until the growth starts to slow down (and the resource needs begin to stabilise). This means that long-term projections, all the way out to the time when the business has sustainable positive operating margins and cash flows, need to be prepared. These projections will depend on the assumptions made about growth. However, rarely is the forecast period less than seven years.

Forecasting growth

For most high growth start-ups the proportion of profits retained will be high, as the company needs to achieve a high growth rate through investing in R&D, expansion of distribution and manufacturing capacity, human resource development (to attract new talent), and development of new markets, products or techniques.

3.5.2 Valuation methods

Once growth rates have been estimated, the next step is to make a valuation. A number of different valuation methods may be used.

Asset-based method

The asset-based method may not be appropriate because the value of capital in terms of tangible assets may not be high. Most of the investment of a start-up is in people, marketing and/or intellectual rights that are treated as expenses rather than as capital.

Market-based methods

The market approach to valuation also presents special problems for start-ups. This valuation process involves finding other companies, usually sold through private transactions, that are at a similar stage of development and that focus on existing or proposed products similar to those of the company being valued. Complicating factors include comparability problems, differences in fair market value from value paid by strategic acquirers, lack of disclosed information, and the fact that there are usually no earnings with which to calculate price to earnings ratios.

Instead of making a valuation using a P/E ratio multiple, it may be appropriate to value the company on a multiple of its current or expected annual sales revenue (a price to revenue ratio multiple).

DCFs

A DCF method of valuation might be used, but only if the start-up company is expected to achieve positive cash flows in the near future, and if these cash flows can be estimated with reasonable confidence in their probable accuracy.

There is little point in making a DCF valuation when the cash flow estimates are uncertain and unreliable.

Using the DCF methodology, FCFs are first projected and then discounted to the present using a risk-adjusted cost of capital. For example, one could use the constant growth model which specifies that:

$$V = \frac{FCF}{r-g}$$

Where: r = required rate of return
g = growth rate of earnings

Our discussion of the high-growth start-up indicates that the growth rates of revenues and costs may vary. Since FCF = Revenue – Costs = R – C, the value of the company will be given by:

$$V = \frac{R-C}{r-g}$$

Now assuming that the growth rate of revenues g_R is different from the growth rate of costs g_C:

$$V = \frac{R}{r-g_r} - \frac{C}{r-g_c}$$

Entry cost valuation method

When an acquirer is seeking to enter a new sector of the market for the first time, it may consider that acquiring a start-up company will be a useful way of establishing its new business. Since a start-up has already occurred, it should be quicker to gain market entry by purchasing the start-up than by establishing a new business from scratch.

In these circumstances, a useful method of valuation might be to consider what it would cost to enter the market and establish a new business, and value the target start-up business on the basis of this entry cost. The entry costs for the target start-up might include:

- costs of raising the finance
- cost of assets acquired
- cost of product development
- costs of recruiting and training employees
- costs of building up the customer base

Having estimated this start-up cost, the acquirer should then consider any factors that would make it possible to establish a new business more cheaply, such as using better technology and locating operations in a less expensive area.

The entry cost valuation should then be based on the estimates of these cheaper alternatives, ie, the estimated entry costs incurred by the target start-up company should be reduced to allow for the cost-saving alternatives of starting the business from scratch.

First Chicago method

The First Chicago method is a variation of an approach to the valuation of high-risk high-growth start- ups known as the Venture Capital Valuation Method. This method can be summarised as follows:

(a) A would-be investor in a high-growth start-up business produces three sets of estimated future cash flows and returns from the investment. The three sets of returns are based on a best case scenario, a worst case scenario and a 'base case' scenario.

(b) For each scenario, estimates of investment returns are made for a given number of years, at the end of which the investor makes an assumption about the exit route for his investment, and the valuation of the investment at the exit date.

(c) The cash flows for each scenario are discounted to a PV, using a risk-adjusted discount rate.

(d) A probability estimate is made for the three scenarios, for example 40% worst case, 40% base case and 20% best case.

(e) A valuation is obtained by calculating the expected value of the PVs of the three scenarios.

(f) The resulting valuation is compared with other known information about business valuations, to assess the valuation for realism.

Standard industry methods of valuation

In some industries, it may be standard practice to value companies on a basis other than profits. For example, an estate agency (realtor) business may be valued on a multiple of outlets. A larger company of estate agents may offer to purchase a small company with 10 outlets, but making no profits, at a price of £20,000 per outlet. The buyer will be able to incorporate the acquired outlets into its operations and online services, and make profits through savings/synergy.

Similarly, a larger company may offer to buy a smaller competitor that is making no profits on the basis of an amount per regular customer.



Worked example: Valuing a start-up

QuickLeg is an internet legal services provider which next year expects revenue of £100 million and costs of £500 million. The revenues of the firm are expected to rise by 21% every year but costs will remain at the same level. The required rate of return is assumed to be 22%.

Requirement

What is the value of QuickLeg?

Solution

$$V = \frac{£100m}{0.22-0.21} - \frac{£500m}{0.22-0} = £7,727.27m$$

The above model seems to capture the phenomenon observed in many start-ups of high losses in the first year of operations with high values of the company.

A very important problem with the discounted flow approach is the sensitivity of the valuation model to the underlying assumptions. Changes in the growth rate induced by changes in demand, technology and management of other causes can have a dramatic effect on the value of the start-up. For example, suppose that the growth rate of revenues falls from 21% to 20%. The value of the company is now:

$$V = \frac{£100m}{0.22-0.20} - \frac{£500m}{0.22-0} = £7,727.27m$$

That is, the company has lost £5 billion.

Another problem with the DCF is that it cannot reflect managerial flexibilities and the strategic options to expand, delay, abandon or switch investments at various decision points. The best way of incorporating uncertainty into the DCF analysis for a new company, technology or product is to assign probabilities of success to each of the various possibilities.

3.5.3 Probabilistic valuation methods

In a probabilistic cash flow model, a number of scenarios are constructed for the drivers of value and derive a value under each scenario. The next step is to assign probabilities to each scenario and arrive at a weighted average value. The procedure to be followed is akin to the Monte Carlo methodology.

Interactive question 6: Valuation methods

Black Raven Ltd is a prosperous private company whose owners are also the directors. The directors have decided to sell their business and have begun a search for organisations interested in its purchase. They have asked for your assessment of the price per ordinary share a purchaser might be expected to offer. Relevant information is as follows.

Most recent statement of financial position

	£'000	£'000	£'000
Non-current assets (carrying amounts)			
Land and buildings			800
Plant and equipment			450
Motor vehicles			55
Patents			2
			1,307
Current assets			
Inventory		250	
Receivables		125	
Cash		8	
		383	

Non-current assets (carrying amounts) Current liabilities			
	£'000	**£'000**	**£'000**
Payables	180		
Taxation	50		
		230	
			153
			1,460
Long-term liability			
Loan secured on property			400
			1,060
Share capital (300,000 ordinary shares of £1)			300
Reserves			760
			1,060

The profits after tax and interest but before dividends over the last five years have been as follows:

Year	**£**
1	90,000
2	80,000
3	105,000
4	90,000
5 (most recent)	100,000

The company's 5-year plan forecasts an after-tax profit of £100,000 for the next 12 months, with an increase of 4% a year over each of the next 4 years. The annual dividend has been £45,000 for the last 6 years. As part of their preparations to sell the company, the directors of Black Raven Ltd have had the non-current assets revalued by an independent expert, with the following results:

	£
Land and buildings	1,075,000
Plant and equipment	480,000
Motor vehicles	45,000

The gross dividend yields and P/E ratios of three quoted companies in the same industry as Black Raven Ltd over the last three years have been as follows:

	Div. yield %	**Albatross plc P/E ratio**	**Div. yield %**	**Bullfinch plc P/E ratio**	**Div. yield %**	**Crow plc P/E ratio**
Recent year	12	8.5	11.0	9.0	13.0	10.0
Previous year	12	8.0	10.6	8.5	12.6	9.5
Three years ago	12	8.5	9.3	8.0	12.4	9.0
Average	12	8.33	10.3	8.5	12.7	9.5

Large companies in the industry apply an after-tax cost of equity of about 18% to acquisition proposals when the investment is not backed by tangible assets, as opposed to a rate of only 14% on the net tangible assets.

Your assessment of the net cash flows after interest and tax which would accrue to a purchasing company, allowing for the capital expenditure required after the acquisition to achieve the company's target five-year plan, is as follows.

	£
Year 1	120,000
Year 2	120,000
Year 3	140,000
Year 4	70,000
Year 5	120,000

Requirement

Use the information provided to suggest alternative valuations which prospective purchasers might make.

See **Answer** at the end of this chapter.

3.6 Valuation of loss-making businesses

Loss-making companies, whether start-ups or not, are also harder to value. Historic dividend and profit growth rates may be negative and the absence of a positive earnings figure to use in conjunction with a suitable P/E ratio makes it difficult to use the traditional valuation methods discusses earlier in this chapter. Losses can be caused by temporary factors or permanent (longterm) difficulties.

Temporary issues

can affect just one company, such as a disruption at the main production facility, or the entire market as was seen during the COVID pandemic in 2020. Longerterm problems may have to do with fundamental shifts in demand due to changing consumer preferences. This was the case with Blackberry's dramatic decline in 2013 due to the popularity of Apple and Samsung smartphones. Temporary issues If a company is making losses due to a one-off factor such as redundancy costs, then this factor could be stripped out of its earnings to create a normalised earnings figure which, if positive, would allow a P/E ratio valuation to be applied.

Alternatively, if a loss-making company still had a positive value for earnings before interest, tax, depreciation and amortisation (EBITDA), then the price to EBIDTA ratio of similar companies can help to place a value on the company.

In both cases it may be difficult to find a ratio for a company that is similar to the company being valued however industry specific ratios can be used.

Long-term difficulties

A company may have rapidly growing revenues and a rising number of customers, but it is loss making over a number of years, this is often seen with start-ups as discussed in the section above. P/E multiples are therefore not helpful in valuing this type of business. Many of today's high-growing, loss-making companies are new businesses and there are no established players to compare them to. High-growing, loss-making companies can still generate a cash flow even if they are not making profits and therefore the DCF technique can be used to estimate the value of the business. This issue with this technique is that it is difficult to estimate the growth rate due to uncertainties about the businesses future.

Context example: Real life example

Valuation of Uber

Uber Technologies Inc listed on the stock market in May 2019. At one point, Uber was valued at $120 billion by Wall Street analysts, which would have made it the largest company ever to debut on the stock market. After the IPO it was actually valued at about $69 billion, just over half of its high hopes IPO. This highlights the difficulties involved in valuing high-growth, loss-making companies. In its most recent annual earnings release, for 2022, Uber reported a total of 7.6 billion trips, $31.9 billion in revenue, and a net loss of $9.1 billion. Uber is a classic example of a high-growing, loss-making company. It's value is not based on historic losses but instead incorporates expectations of future cash flows that will be generated for investors due to the high demand for its services from customers.

3.7 Bargain purchases

Professional skills focus: Concluding, recommending and communicating

You are expected to be able to make evidence-based recommendations which can be justified by reference to supporting data and other information. A recommended bid value and method of payment must therefore be based on the information provided in the question.

Definition

Bargain purchase: The purchase of a company for an amount that is less than the fair value of the net assets acquired.

Occasionally a business may be acquired for a price that is less than the fair value of the net assets acquired. This could happen when the acquired company is in serious financial difficulties and is facing the risk of insolvency.

It might be supposed that in such circumstances, the owners should seek to wind up their business and realise the business assets. However:

- The realisable value of the assets may be less than the fair value at which the assets would be valued in an ongoing business.
- It may be difficult to find willing buyers for the assets, especially within a short time.

In these circumstances the business owners may agree to accept an offer price that is lower than the fair value of the net assets of the business.

IFRS 3 states that when a bargain purchase occurs, the acquirer should recognise the difference between the fair value of the acquisition and the actual purchase price as a **gain in profit or loss on the acquisition date**.

However, before recognising the gain the acquirer should reassess whether it has correctly identified:

- all the assets that have been acquired and the liabilities that have been assumed
- the NCI in the acquired business, if any
- the consideration transferred

If this reassessment indicates the need to adjust any of these amounts, the gain on the bargain purchase should be adjusted accordingly.

4 Valuation of debt

Section overview

- The basic idea behind debt valuation is that the value is the PV of all future cash flows to the investor (that is, interest plus capital redemption) discounted at the debt investor's required rate of return.
- Different types of debt (irredeemable, redeemable and convertible) are valued in different ways.

4.1 Notes on debt calculations

(a) Debt is usually quoted in £100 nominal units, or blocks (euro loans are usually quoted in €1,000 blocks); always use £100 nominal values as the basis to your calculations unless told otherwise.

(b) Debt can be quoted in % or as a value, eg, 97% or £97. Both mean that £100 nominal value of debt is worth £97 market value.

(c) Interest on debt is stated as a percentage of nominal value. This is known as the coupon rate. It is not the same as the redemption yield on debt or the cost of debt.

(d) Sometimes people quote an interest yield, defined as coupon/market price, but this is only a crude approximation unless the debt is irredeemable and there is no tax effect.

(e) Always use ex-interest prices in any calculations, unless the debt is specifically described as cum interest.

4.2 Irredeemable debt

For irredeemable debentures or loan stock, where the company will go on paying interest every year in perpetuity without ever having to redeem the loan, the valuation formula (ignoring taxation) is:

$$P_0 = \frac{i}{r_d}$$

Where: P_0 = the market price of the stock ex interest; that is, excluding any interest payment that might soon be due

I = the annual interest payment on the stock

r_d = the return required by the loan stock investors

Irredeemable (undated) debt, paying annual after-tax interest i (1 – T) in perpetuity, where P_0 is the ex-interest value:

$$P_0 = \frac{i\,(1\text{-}T)}{k_d}$$

Where: k_d = the after-tax cost to the company In practice, irredeemable debt is uncommon.

4.3 Redeemable debt

The valuation of redeemable debt owned by investors depends on future expected receipts and the yield required by the investors in the debt. The market value is the discounted PV of

future interest receivable, up to the year of redemption, **plus** the discounted PV of the redemption payment (ignoring tax).

Value of debt = (Interest earnings × Annuity factor) + (Redemption value × DCF factor)

Where the debt incurs interest at a variable rate, and if the variable rate payable is a market rate of interest, it can be assumed that the market value of the debt is its par value. For example, it is generally assumed that the market value of a variable rate bank loan is the face value/par value of the loan.

Discounting future income to obtain a market valuation is therefore required only for other types of debt, such as fixed interest debt and zero interest debt.

Worked example: Market value of debentures

Furry has in issue 12% debentures with par value £100,000 and redemption value £110,000 with interest payable quarterly. The redemption yield on the bonds is 8% annually and 2% quarterly. The debentures are redeemable on 30 June 20X4 and it is now 31 December 20X0.

Requirement

Calculate the market value of the debentures.

Solution

You need to use the redemption yield cost of debt as the discount rate, using the quarterly discount rate (8%/4).

To calculate the PV, the following variables need to be input to the PV function. The use of the PV function is covered in more detail in the chapter Spreadsheet formulae.

B5 =PV(B1,B2,B3,B4)

	A	B
1	Rate of return required over the period	0.02
2	Nper = the number of periods	14
3	Pmt = amount (interest) paid in single period	3,000
4	Fval = future value (amount paid at maturity)	110,000
5	Present value (issue price)	-119,685

Market value is £119,685.

Interactive question 7: Value of redeemable debt

A company has issued some 9% debentures, which are redeemable at par in 3 years' time. The most recent interest payment on the debentures has just been made. Investors now require an annual redemption yield of 10%.

Requirements

What will be the current market value of each £100 of debenture:
(a) if interest is paid annually?
(b) if interest is paid six-monthly?

See **Answer** at the end of this chapter.

4.4 Convertible debt

When convertible loan notes are traded on a stock market, the minimum market price will be the price of straight loan notes with the same coupon rate of interest. If the market value falls to this minimum, it follows that the market attaches no value to the conversion rights.

The actual market price of convertible loan notes will depend on:

- the price of straight debt
- the current conversion value
- the length of time before conversion may take place
- the market's expectation as to future equity returns and the associated risk

If the conversion value rises above the straight debt value then the price of convertible loan notes will normally reflect this increase.

Conversion value = $P_0 (1 + g)_n R$

Where: P_0 = the current ex-dividend ordinary share price
g = the expected annual growth of the ordinary share price
n = the number of years to conversion
R = the number of shares received on conversion

The current market value of a convertible bond where conversion is expected is the sum of the PVs of the future interest payments and the PV of the bond's conversion value.

Worked example: Convertible debt

What is the value of a 9% convertible bond if it can be converted in 5 years' time into 35 ordinary shares or redeemed at par on the same date? An investor's required return is 10% and the current market price of the underlying share is £2.50 which is expected to grow by 4% per annum.

Solution

Conversion value = $P_0 (1 + g)_n R = 2.50 \times 1.04^5 \times 35 = £106.46$

B5 =PV(B1,B2,B3,B4)

	A	B
1	Rate of return required over the period	0.10
2	Nper = the number of periods	5
3	Pmt= the amount (of interest) paid in any single period	9
4	Fval = the future value (the amount paid at maturity).	106.46
5	Present value (issue price)	-100.22

Current market value of convertible bond = £100.22

4.5 Floating rate debt

With floating rate debt, the coupon rate is reset at each payment date (or possibly even more regularly) to prevailing interest rates. The implication of this is that at a reset date the bond will be valued at par. In addition, the cash value of the next coupon has now been set. Though all subsequent coupons will still be reset at the next reset date if interest rates move, the next coupon will not be.

This resetting of the coupon at each payment date ensures that the price of floating rate debt is always close to par. Floating rate debt therefore safeguards the par value of the investment rather than showing great swings in value as interest rates fluctuate (as is demonstrated by normal bonds).

Coupon rates tend to be reset by reference to money market rates. As a result, pricing follows money market conventions, applying simple rates over periods less than one year.

To evaluate floating rate debt we have a choice of methods, specifically:

- first principles
- a more practical alternative

From **first principles**, floating rate debt can be evaluated using the relationship

$$P_0 = \frac{i_1}{(1+k_d)} + \frac{i_2}{(1+k_d)^2} + \frac{i_3}{(1+k_d)^3} + \ldots + \frac{i_n + R}{(1+k_d)^n}$$

The difficulty with this approach is that the future coupon rates are not known at present.

Worked example: Floating rate debt

The current interest rate is at 10% and is expected to remain at that level for the foreseeable future. A floating rate note (FRN) has just paid its most recent annual coupon, and future coupons have been reset to 10%.

Requirement

Calculate the current market value of the FRN.

Solution

$$P_0 = \frac{10}{1.10} + \frac{10}{1.10^2} + \frac{10}{1.10^3} + \ldots + \frac{110}{1.10^n} = £100.00$$

A more flexible alternative is to imagine that the debt may be sold at the next reset date at its par value. Though it is not necessary that the holder intends to sell at this date, it is always an option that is open and hence can validly be used to determine the current price. Based on this idea, the market value can be estimated as follows.

$$P_0 = \frac{\text{Next coupon} + \text{Par value}}{(1 + rc)}$$

Where: rc = rate for the period to the next coupon, ie, rc = r × Days/365 (assuming UK convention), so more generally:

$$P_0 = \frac{\text{Next coupon} + \text{Par value}}{1 + r \times \frac{\text{Days}}{365}}$$

Where: Days = time to the next coupon

Worked example: Floating rate debt 2

Calculate the current market value of the above floating rate debt if interest rates stay at 10%, and if interest rates increase to 11% and are expected to remain at that level.

Solution

Interest at 10% - Price = 10+100 1.10 =£100.00

Interest at 11% - In this situation, the next coupon in one year's time will remain at £10, its value determined at the last reset date. All further coupons will, however, be reset to £11 at the

next date (assuming no further interest rate changes), giving rise to a par value at that time. The current market value is therefore

$$\text{Price} = \frac{10 + 100}{1.11} = £99.10$$

5. International Valuation Standards (IVS)

ICAB Members and Students are expected to know that International Valuation Standards (IVS) are applicable for valuation assignments as it is adopted by Bangladesh Bank, the Bangladesh Securities and Exchange Commission (BSEC) and the Financial Reporting Council (FRC).

5.1 About IVS

IVS are developed with the aim of serving as a professional benchmark for valuers around the world with the "objective to increase the confidence and trust of users of valuation services by establishing transparent and consistent valuation practices".

IVS consist of mandatory requirements that must be followed in order to state that a valuation was performed in compliance with IVS. IVS are focused on the whole valuation engagement (i.e. process), not just the valuation report.

IVS are principles-based multi-disciplinary standards; hence, they include requirements for multiple asset classes - tangible assets (such as plant & equipment and real property interests), business valuation, intangible assets and financial instruments. IVS requirements apply to external as well as internal valuation engagements.

5.2 Structure of IVS

This version of International Valuation Standards is published on 31 January 2024, with an effective date of 31 January 2025. The IVSC permits early adoption from the date of publication. References hereafter to IVS will be to the IVS 2025 publication. IVS include a glossary, framework and seven general standards that apply to all valuation assignments, as well as eight asset standards that include requirements related to specific types of assets.

General Standards	Assets Standards
· IVS 100 – Valuation Framework · IVS 101 – Scope of Work · IVS 102 – Bases of Value · IVS 103 – Valuation Approaches · IVS 104 – Data and Inputs ·IVS 105 – Valuation Models ·IVS 106 – Documentation and Reporting	· IVS 200 – Businesses and Business Interests · IVS 210 – Intangible Assets · IVS 220 – Non-Financial Liabilities · IVS 230 – Inventory · IVS 300 – Plant, Equipment and Infrastructure · IVS 400 – Real Property Interests · IVS 410 – Development Property · IVS 500 – Financial Instruments

A separate Handbook is published by The Institute of Chartered Accountants of Bangladesh (ICAB) with respect to International Valuation Standards (IVS). The purpose of this publication is to raise awareness and understanding of IVS among ICAB Members, Registered Students and Corporates. This publication provides a high level factual overview of the structure of the IVS and the content therein that is relevant to valuation.

Summary

	Tick off
There are numerous ways in which companies (or components of companies) can be valued, including asset-based and cash-based measures.	☐
Different circumstances demand different valuation measures – there is no particular method that applies in all situations.	☐
Unquoted companies have the problem of limited information being available – a combination of DCF analysis and market comparability measures may have to be employed for majority shareholdings.	☐
Minority shareholdings in unquoted companies may have little value if there are no dividend payments forthcoming in the future.	☐
Different types of debt (irredeemable, redeemable and convertible) are valued in different ways.	☐
International Valuation Standards (IVS) are applicable for valuation assignments	☐

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, complete the following knowledge diagnostic and check you are able to confirm you possess the following essential learning from this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
1.	Can you value a business using discounted free cash flows? (Topic 1)
2.	Can you identify and explain the due diligence procedures that should be performed to support a business valuation? (Topic 2)
3.	Is your knowledge of the corporate reporting implications of business combinations up to date? (Topic 2)
4.	Can you apply each of the valuation models covered in this chapter to a start-up business? (Topic 3)
5.	Can you value redeemable, convertible and floating rate debt? (Topic 4)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed to the next chapter.

Question	Learning benefit from attempting this question
2 Wilkinson	This is a good introductory question covering business valuations, it is a revision of what you would have covered in your Financial Management studies. Work through this question carefully before attempting exam standard questions.
3 Cub	This question covers EVA. Before attempting this question make sure you have covered section 1.6 of the chapter in detail including the worked example on EVA. Having a good layout for your calculations will help you tackle this question.
4 Cromer	This is a good question to practice the corporate reporting treatment of a decommissioning liability under IFRS 13 and IAS 37. Fair value measurements are popular in SBM&L questions Refer back to your corporate reporting notes if you need to.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your ueries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Technical reference

1 IAS 33, *Earnings Per Share*

- The objective of IAS 33 is to prescribe the principles determining and presenting earnings per share (EPS) to facilitate performance comparisons between different entities in the same reporting period and between different reporting periods for the same entity. IAS 33 sets out how to calculate both basic EPS and diluted EPS. The calculation of basic EPS is based on the weighted average number of ordinary shares outstanding during the period, while diluted EPS also includes dilutive potential ordinary shares (such as options and convertible instruments) provided they meet certain criteria.

2 IFRS 13, *Fair Value Measurement*

- Defines fair value as the price that would be received to sell an asset or would be paid to transfer a liability in an orderly transaction between market participants at the measurement date. IFRS 13 defines fair value on the basis of an 'exit price' notion and uses a 'fair value hierarchy' which results in a market-based measurement rather than an entity-specific one.

3 IFRS 3, *Business Combinations*

- Outlines the accounting when an acquirer obtains control of a business through an acquisition. These business combinations are accounted for using the 'acquisition method' which generally requires the assets acquired, and the liabilities assumed, to be measured at their fair values at the acquisition date.

4 IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*

- Outlines the accounting for provisions (liabilities of uncertain timing or amount), contingent assets (possible assets) and contingent liabilities (possible obligations or present obligations which are not probable or which cannot be measured reliably). Provisions are measured at the best estimate of the expenditure required to settle the present obligation. The objective of IAS 37 is to ensure that financial statements provide sufficient information to enable users to understand the nature, timing and amount of any provisions, contingent liabilities and contingent assets, but it is also underpinned by the key principle that a provision should only be recognised when a genuine liability exists.

5 IAS 12, *Income Taxes*

- Outlines a method of accounting for income taxes which recognises both the current tax consequences of transactions and events, and the future tax consequences of the future recovery of the carrying amount of an entity's assets or the future settlement of its liabilities. Differences between the carrying amount and the tax base of assets and liabilities are recognised, with limited exceptions, as deferred tax liabilities or deferred tax assets.

6 IAS 19, *Employee Benefits*

- Outlines the accounting requirements for employee benefits, including short-term benefits (eg, wages and salaries; annual leave); post-employment benefits (eg, retirement benefits);

and termination benefits. The standard requires that the cost of providing employee benefits should be recognised in the period in which the benefit is earned by the employee, rather than when it is paid or payable. IAS 19 also outlines how each category of employee benefits is measured, and it provides detailed guidance about post-employment benefits.

7 IFRS 2, *Share-based Payment*

- The objective of IFRS 2 is to specify the financial reporting by an entity when it undertakes a share- based payment transaction. IFRS 2 requires an entity to recognise share-based payment transactions (such as shares granted and share options) in its financial statements. This includes transactions with employees or other parties, to be settled in cash, other assets, or equity instruments of the entity.

8 IFRS 5, *Non-current Assets Held for Sale and Discontinued Operations*

- Outlines the accounting required for non-current assets held for sale (or for distribution to owners). In general terms, assets (or disposal groups) held for sale are not depreciated, are measured at the lower of carrying amount and fair value less costs to sell, and are presently separately in the statement of financial position.

9 IFRS 10, *Consolidated Financial Statements*

- Outlines the requirements for the preparation and presentation of consolidated financial statements, with entities being required to consolidate other entities they control.

10 IAS 16, *Property, Plant and Equipment*

- Outlines the accounting treatment for most types of property, plant and equipment. The principal issues are the recognition of assets, the determination of their carrying amounts, and the depreciation charges and impairment losses to be recognised in relation to them.

Self-test questions

Answer the following questions.

1 Profed

Profed provides a tuition service to professional students. This includes courses of lectures provided on their own premises and provision of home study material. Most of the lecturers are qualified professionals with many years' experience in both their profession and tuition. Study materials are written and word processed in-house, but sent to an external printing company.

The business was started 15 years ago and now employs around 40 full-time lecturers, 10 authors and 20 support staff. Freelance lecturers and authors are employed in times of peak demand.

The shareholders of Profed mainly comprise the original founders of the business who would now like to realise their investment. In order to arrive at an estimate of what they believe the business is worth, they have identified a long-established quoted company, City Tutors, which has a similar business, although it also publishes texts for external sale to universities, colleges and so on.

Summary financial statistics for the two companies for the most recent financial year are as follows:

	Profed	City Tutors
Issued shares (million)	4	10
NAVs (£m)	7.2	15
EPS (pence)	35	20
Dividend per share (pence)	20	18
Debt: equity ratio	1:7	1:65
Share price (pence)		362
Expected rate of growth in earnings/dividends	9% p.a.	7.5% p.a.

Requirements

1.1 Compute a range of valuations for the business of Profed, using the information available and stating any assumptions made.

1.2 Comment on the strengths and weaknesses of the methods you used in (a) and their suitability for valuing Profed.

Notes

1 The net assets of Profed are the net book values of tangible non-current assets plus net working capital. However:

- A recent valuation of the buildings was £1.5 million above book value.

- Inventory includes past editions of textbooks which have a realisable value of £100,000 below their cost.
- Due to a dispute with one of their clients, an additional allowance for bad debts of £750,000 could prudently be made.

2 Growth rates should be assumed to be constant per annum; Profed's earnings growth rate estimate was provided by the marketing manager, based on expected growth in sales adjusted by normal profit margins. City Tutors' growth rates were gleaned from press reports.

3 Profed uses a discount rate of 15% to appraise its investments, and has done for many years.

2 Wilkinson plc

Wilkinson plc has identified Chris Limited as a potential acquisition target. It has approached you as its financial adviser to ask for assistance in valuing the company. You have obtained the following information about Chris Limited:

Statement of changes in equity for the years ended 31 December

	20X5	20X6	20X7
	£'000	£'000	£'000
Profit after tax	280	260	410
Dividends	(150)	(160)	(185)
Retained profit	130	100	225

Statement of financial position as at 31 December

	20X5	20X6	20X7
	£'000	£'000	£'000
Non-current assets	1,365	1,405	1,560
Working capital	810	870	940
	2,175	2,275	2,500
Share capital	100	100	100
Retained earnings	875	975	1,200
10% debentures 20Y2	1,200	1,200	1,200
	2,175	2,275	2,500

The non-current assets include an unused property which has a market value of £100,000. The debentures pay a semi-annual coupon and are redeemable at the end of 20Y2. The gross redemption yield on 20Y2 gilts paying a similar level of coupon is 11%.

The P/E ratio for the quoted company sector in which Chris Limited's activities fall is around 15 times and the sector's gross dividend yield is around 11%. The b of the sector is around 0.8 and the return on the market is around 21%.

Requirement

Estimate the value of Chris Limited, using four different methods of valuation. Explain the rationale behind each valuation, when it would be useful and why each method gives a different value. Discuss the limitations of your analysis and what further information you would require to gain greater assurance on the valuations.

3 Cub

The following data relates to Cub.

Profit or loss statement and dividend data

	20X1 $m	20X2 $m
Sales	995	1,180
Pre-tax accounting profit	204	258
Taxation	49	62
Profit after tax	155	196
Dividends	50	60
Retained earnings	105	136

Statement of financial position data

	20X1 $m	20X2 $m
Non-current assets	370	480
Net current assets	400	500
	770	980
Financed by:		
Shareholders' funds	595	720
Medium- and long-term bank loans	175	260
	770	980

This is taken after deducting the economic depreciation of the company's non-current assets.

Other relevant information

(1) Economic depreciation was $95 million in 20X1 and $105 million in 20X2.

(2) Interest expenses were $13 million in 20X1 and $18 million in 20X2.

(3) Other non-cash expenses were $32 million in 20X1 and $36 million in 20X2. (These should not be included in the EVA® calculation.)

(4) The tax rate in 20X1 and 20X2 was 24%.

(5) Cub had non-capitalised leases valued at $35 million in each year 20X0–20X2.

(6) The company's pre-tax cost of debt was estimated as 7% in 20X1 and 8% in 20X2.

(7) The company's cost of equity was estimated as 14% in 20X1 and 16% in 20X2.

(8) The target capital structure is 75% equity, 25% debt.

(9) Statement of financial position capital employed at the end of 20X0 was $695 million. There were no loans at that date.

Requirement

Estimate the EVA® for Cub for 20X1 and 20X2, stating any assumptions that you make and discussing what the calculations demonstrate about the performance of the company.

4 Cromer

Cromer assumes a decommissioning liability in a business combination. It is legally required to dismantle a power station at the end of its useful life, which is estimated to be 20 years.

Requirement

How would the decommissioning liability be measured under IFRS?

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 k_e = 12% + 2% = 14% (0.14) D_0 = £250,000

$P_0 = D_0 / K_e$ – £250,000 / 0.14 = £1,785,714

1.2 $P_0 = D_0(1+G) / k_e - g$ = £250,000(1.04) / 0.14 – 0.04 = £2,600,000

1.3 P_0 = £250,000(1.04) / (1.14) + £250,000$(1.04)^2$ / $(1.14)^2$ + £250,000$(1.04)^3$ x 1/0.14 / $(1.14)^2$

= £228,070 + £208,064 + £1,545,618 = £1,981,752

Answer to Interactive question 2

Suitable range of valuations for the shares of Mayfly:

(1) **Earnings**. Average earnings over the last 5 years have been £67,200, and over the last 4 years £71,500. There might appear to be some growth prospects but estimates of future earnings are uncertain.

A low estimate of earnings in 20X5 would be, perhaps, £71,500.

A high estimate of earnings might be £75,000 or more. This solution will use the most recent earnings figure of £75,000 as the high estimate.

(2) **P/E ratio**. A P/E ratio of 15 (Bumblebee's) would be much too high for Mayfly, because the growth of Mayfly's earnings is not as certain and Mayfly is an unquoted company.

On the other hand, Mayfly's expectations of earnings are probably better than those of Wasp. A suitable P/E ratio might be based on the industry's average of 10, but since Mayfly is an unquoted company and therefore riskier, a lower P/E ratio might be more appropriate; perhaps 60% to 70% of 10 = 6 or 7, or conceivably even as low as 50% of 10 = 5.

The valuation of Mayfly's shares might therefore range between:

high P/E ratio and high earnings: 7 × £75,000 = £525,000; and

low P/E ratio and low earnings: 5 × £71,500 = £357,500.

Answer to Interactive question 3

The maximum price is one which would make the return from the total investment exactly 15% over 5 years, so that the NPV at 15% would be 0.

B3 =NPV(0.15,C2:G2)

	A	B	C	D	E	F	G
1	Year	0	1	2	3	4	5
2	Cash flows, no purchase consideration £	(100,000)	(80,000)	60,000	100,000	150,000	150,000
3	PV@15% T1-5	201,895					
4	Less investment @T0	(100,000)					
5	NPV	101,895					

The maximum purchase price is £101,895

Answer to Interactive question 4

First we calculate the PV of the FCFs over the five-year planning horizon.

B4 =NPV(0.12,B3:F3)

	A	B	C	D	E	F
1		20X5	20X6	20X7	20X8	20X9
2		$	$	$	$	$
3	FCF	1,080	1,437	1,901	2,502	30,585
4	PV at 12%	22,408				

Total PV = $22,408

	$
PV @ 12% (see above)	22,408
Less debt	(2,000)
Equity value	20,408
Shares outstanding	100
Intrinsic value per share	$204.08
Market share price	$29.00
Price at which Tetrion shareholders will sell ($29.00 × 120%)	$34.80
Discount of selling price from value	83%

The shareholders of Atrium should buy Tetrion at the offered price.

Answer to Interactive question 5

Cost of ungeared equity = 5% + 0.80(8 – 5)% = 7.4%

Base case NPV (ignoring the purchase cost of Peek) = £4m/0.074 = £54.05m

Tax relief on interest Years 1–10 = £55m × 6% × 21% = £0.693m

PV of tax relief discounted at risk-free cost of debt 5% = £0.693m × 7.722 = £5.35m

Financing costs = £1m – (£1m × 21% × 0.952) = say £0.8m

Note: It is assumed that the costs of the debt issue are allowable for tax purposes.

Ignoring the purchase cost	**£m**
Base case NPV	54.05
PV of debt financing benefits	5.35
PV of financing costs	(0.80)
APV	58.60

This suggests that a purchase price of £54 million would be just acceptable financially, with an APV of +£4.6 million after deducting the purchase cost. However, the directors of Main should

try to negotiate a lower price if possible. They should also consider other factors, such as the risk and uncertainty in the estimated cash flows, before making a decision about whether or not to agree to the asking price.

Answer to Interactive question 6

Tutorial Note

This question provides comprehensive practice of valuation techniques. You need to make clear the basis of your calculations and the assumptions you are making (in (a) the assumption is that the purchaser will accept the valuation, in (b) that the last five years are an appropriate indicator and so on). Other important issues which this question raises include:

- valuation (if any) of intangible assets
- lack of likelihood that asset valuation basis would be used
- adjustment to P/E ratios used in calculations because company is unquoted

Do not take all the figures used in this answer as the only possibilities. You could for example have made adjustments to estimated earnings in (c) to allow for uncertainty, or used a different figure to 17% in (d).

Earnings-basis valuations

If the purchaser believes that earnings over the last five years are an appropriate measure for valuation, we could take average earnings in these years, which were:

$$\frac{£465{,}000}{5} = £93{,}000$$

An appropriate P/E ratio for an earnings basis valuation might be the average of the three publicly quoted companies for the recent year. (A trend towards an increase in the P/E ratio over three years is assumed and, even though average earnings have been taken, the most recent year's P/E ratios are considered to be the only appropriate figures.)

	P/E ratio	
Albatross plc	8.5	
Bullfinch plc	9.0	
Crow plc	10.0	
Average	9.167	(i)
Reduce by about 40% to allow for unquoted status	5.5	(ii)

Share valuations on a past earnings basis are as follows:

	P/E ratio	Earnings £'000	Valuation £'000	Number of shares	Value per share
(1)	9.167	93	852.5	300,000	£2.84
(2)	5.5	93	511.5	300,000	£1.71

Because of the unquoted status of Black Raven Ltd, purchasers would probably apply a lower P/E ratio, and an offer of about £1.71 per share would be more likely than one of £2.84.

Future earnings might be used. Forecast earnings based on the company's five-year plan will be used.

Expected earnings:	£
Year 1	100,000
Year 2	104,000
Year 3	108,160
Year 4	112,486
Year 5	116,986
Average	108,326.4 (say £108,000)

A share valuation on an expected earnings basis would be as follows:

P/E ratio	Average future earnings	Valuation	Value per share
5.5	£108,000	£594,000	£1.98

It is not clear whether the purchasing company would accept Black Raven's own estimates of earnings.

A dividend yield basis of valuation with no growth

There seems to have been a general pattern of increase in dividend yields to shareholders in quoted companies. It is reasonable to suppose that investors in Black Raven would require at least the same yield.

An average yield for the recent year for the three quoted companies will be used. This is 12%. The only reliable dividend figure for Black Raven Ltd is £45,000 a year gross, in spite of the expected increase in future earnings. A yield basis valuation would therefore be:

$$\frac{£45,000}{12\%} = £375,000 \text{ or } £1.25 \text{ per share}$$

A purchasing company would, however, be more concerned with earnings than with dividends if it intended to buy the entire company, and an offer price of £1.25 should be considered too low. On the other hand, since Black Raven Ltd is an unquoted company, a higher yield than 12% might be expected.

A dividend yield basis of valuation, with growth

Since earnings are expected to increase by 4% a year, it could be argued that a similar growth rate in dividends would be expected. We shall assume that the required yield is 17%, rather more than the 12% for quoted companies because Black Raven Ltd is unquoted. However, in the absence of information about the expected growth of dividends in the quoted companies, the choice of 12%, 17% or whatever is not much better than a guess.

$$P_0 = \frac{D0(1+g)}{(k-g)} = \frac{45,000(1.04)}{(0.17-0.04)} = £360,000 \text{ or } £1.20 \text{ per share}$$

The discounted value of future cash flows

The PV of cash inflows from an investment by a purchaser of Black Raven Ltd's shares would be discounted at either 18% or 14%, depending on the view taken of Black Raven Ltd's assets. Although the loan of £400,000 is secured on some of the company's property, there are enough assets against which there are no charges to assume that a purchaser would consider the investment to be backed by tangible assets.

The PV of the benefits from the investment would be as follows:

Year	Cash flow £'000
1	120
2	120
3	140
4	70
5	120
PV of cash flow @14%	395.87

A valuation per share of £1.32 might therefore be made. This basis of valuation is one which a purchasing company ought to consider. It might be argued that cash flows beyond Year 5 should be considered and a higher valuation could be appropriate, but a figure of less than £2 per share would be offered on a DCF valuation basis.

Summary

Any of the preceding valuations might be made but, since share valuation is largely a subjective matter, many other prices might be offered. In view of the high asset values of the company, an asset stripping purchaser might come forward.

Answer to Interactive question 7

(a) If interest is paid annually.

To calculate the market value, the following variables need to be input to the PV function:

B5 =PV(B1,B2,B3,B4)

	A	B
1	Rate of return required over the period	0.10
2	Nper = the number of periods	3
3	Pmt= amount (interest) paid in single period	9
4	Fval = future value (amount paid at maturity)	100
5	Present value (issue price)	-97.51

(b) Each £100 of debenture will have a market value of £97.51.

	A	B
1	Rate of return required over the period	0.05
2	Nper = the number of periods	6
3	Pmt= the amount (of interest) paid in any single period	4.50
4	Fval = the future value (the amount paid at maturity).	100
5	Present value (issue price)	-97.46

Each £100 of debenture will have a market value of £97.46.

If interest is paid six-monthly: Assumed six-monthly yield = 10%/2 = 5%

Answers to Self-test questions

1 Profed

1.1 The information provided allows us to value Profed on three bases: net assets, P/E ratio and dividend valuation.

All three will be computed, even though their validity may be questioned in part (b) of the answer.

Assets-based valuation

	£'000
Net assets at book value	7,200
Add increased valuation of buildings	1,500
Less decreased value of inventory and receivables	(850)
Net asset value of equity	7,850

Value per share = £1.96

P/E ratio

	Profed	City Tutors
Issued shares (million)	4	10
Share price (pence)		362
Market value (£m)		36.2
Earnings per share (pence)	35	20
P/E ratio (share price ¸ EPS)		18.1

The P/E for a similar quoted company is 18.1. This will take account of factors – such as marketability of shares, status of company, growth potential – that will differ from those for Profed. Profed's growth rate has been estimated as higher than that of City Tutors, possibly because it is a younger, developing company, although the basis for the estimate may be questionable.

All other things being equal, the P/E ratio for an unquoted company should be taken as between one-half to two-thirds of that of an equivalent quoted company. Being generous, in view of the possible higher growth prospects of Profed, we might estimate an appropriate P/E ratio of around 12, assuming Profed is to remain a private company.

This will value Profed at 12 × £0.35 = £4.20 per share, a total valuation of £16.8 million.

Dividend valuation model

The dividend valuation method gives the share price as:

$$\frac{\text{Next year's dividend}}{\text{cost of equity} - \text{growth rate}}$$

which assumes dividends being paid into perpetuity, and growth at a constant rate. For Profed, next year's dividend = £0.20 × 1.09 = £0.218 per share

While we are given a discount rate of 15% as being traditionally used by the directors of Profed for investment appraisal, there appears to be no rational basis for this. We can instead use the information for City Tutors to estimate a cost of equity for Profed. This is assuming the business risks are similar, and ignores the small difference in their gearing ratio.

Again, from the DVM, cost of equity = $\frac{\text{Next year's dividend}}{\text{Market price}}$ + Growth rate

$$\text{For City Tutors, cost of equity} = \frac{£0.18 \times 1.075}{£3.62} + 0.075 = 12.84\%$$

Using, say, 13% as a cost of equity for Profed (it could be argued that this should be higher since Profed is unquoted so riskier than the quoted City Tutors):

$$\text{Share price} = \frac{£0.218}{0.13 - 0.09} = £5.45$$

valuing the whole of the share capital at £21.8 million.

Range for valuation

The three methods used have thus come up with a range of values of Profed as follows:

	Value per share	Total valuation
	£	£m
Net assets	1.96	7.9
P/E ratio	4.20	16.8
Dividend valuation	5.45	21.8

1.2 Asset-based valuation

Valuing a company on the **basis of its asset values** alone is rarely appropriate if it is to be sold on a going concern basis. Exceptions would include property investment companies and investment trusts, the market values of the assets of which will bear a close relationship to their earning capacities.

Profed is typical of many service companies, a large part of whose value lies in the **skill, knowledge and reputation of its personnel**. This is not reflected in the net asset values, and renders this method quite inappropriate. A potential purchaser of Profed will generally value its intangible assets such as knowledge, expertise, customer/supplier relationships and brands more highly than those that can be measured in accounting terms.

Knowledge of the net asset value (NAV) of a company will, however, be important as a **floor value** for a company in financial difficulties or subject to a takeover bid. Shareholders will be reluctant to sell for less than the NAV even if future prospects are poor.

P/E ratio valuation

The P/E ratio measures the **multiple of the current year's earnings** that is reflected in the **market price** of a share. It is thus a method that reflects the earnings potential of a company from a market point of view. Provided the marketing is efficient, it is likely to give the most meaningful basis for valuation.

The market price of a share at any point in time is determined by supply and demand forces prevalent during small transactions, and will be dependent on a lot of factors in addition to a realistic appraisal of future prospects.

A downturn in the market, and economic and political changes can all affect the day to day price of a share, and thus its prevailing P/E ratio. It is not known whether the share price given for City Tutors was taken on one particular day, or was some sort of average over a period. The latter would perhaps give a sounder basis from which to compute an applicable P/E ratio.

Even if the P/E ratio of City Tutors can be taken to be **indicative of its true worth**, using it as a basis to value a smaller, unquoted company in the same industry can be problematic.

The status and marketability of shares in a quoted company have tangible effects on value but these are difficult to measure.

The P/E ratio will also be affected by **growth prospects** – the higher the growth expected, the higher the ratio. The growth rate incorporated by the shareholders of City Tutors is probably based on a more rational approach than that used by Profed.

If the growth prospects of Profed, as perceived by the market, did not coincide with those of **Profed's management** it is difficult to see how the P/E ratio should be adjusted for relative levels of growth. However, the earnings yield method of valuation could be useful here.

In the valuation in (a) a crude adjustment has been made to City Tutors's P/E ratio to arrive at a ratio to value Profed's earnings. This can lead to a very inaccurate result if account has not been taken of all the differences involved.

Dividend-based valuation

The dividend valuation model (DVM) is a **cash flow based approach**, which values the dividends that the shareholders expect to receive from the company by discounting them at their required rate of return. It is perhaps more appropriate for valuing a minority shareholding where the holder has no influence over the level of dividends to be paid than for valuing a whole company, where the total cash flows will be of greater relevance.

The practical problems with the DVM lie mainly in its **assumptions**. Even accepting that the required 'perfect capital market' assumptions may be satisfied to some extent, in reality, the formula used in (a) assumes constant growth rates and constant required rates of return in perpetuity.

Determination of an **appropriate cost of equity** is particularly difficult for an unquoted company, and the use of an 'equivalent' quoted company's data carries the same drawbacks as discussed above. Similar problems arise in estimating future growth rates, and the results from the model are highly sensitive to changes in both these inputs.

It is also highly dependent on the **current year's dividend** being a representative base from which to start.

The dividend valuation model valuation provided in (a) results in a higher valuation than that under the P/E ratio approach. Reasons for this may be as follows:

- The **share price** for City Tutors may be currently **depressed below its normal level**, resulting in an inappropriately low P/E ratio.
- The **adjustment** to get to an **appropriate P/E ratio** for Profed may have been too harsh, particularly in light of its apparently better growth prospects.
- The **cost of equity** used in the DVM was that of City Tutors. The validity of this will largely depend on the relative levels of risk of the two companies. Although they both operate the same type of business, the fact that City Tutors sells its material externally means it is perhaps less reliant on a fixed customer base.
- Even if business risks and gearing risk may be thought to be comparable, a prospective buyer of Profed may consider investment in a **younger, unquoted company** to carry **greater personal risk**. His required return may thus be higher than that envisaged in the DVM, reducing the valuation.

2 Wilkinson plc

Earnings-based valuation P/E ratio

The earnings-based valuation would be based on applying the P/E ratio for similar quoted companies to Chris Limited's earnings.

15 × £410,000 = £6,150,000

This value would be the approximate value of a quoted company in the sector. Research has indicated the acquisitions of private companies are typically priced at a discount of 30% to 40% to the quoted sector P/E. This discount probably reflects the lack of marketability of the private company shares compared to those of its quoted counterparts. Using a discount of 40%, the value of Chris Limited would be:

£6,150,000 × 60% = £3,690,000

This is the value of the earnings stream. On top of this, it should be possible to sell the unused property without affecting the earnings stream for around £100,000, giving a total valuation of £3,790,000.

Limitations of the calculation

The sector P/E is an average for the sector. It is not clear whether Chris Limited is an average company or whether it is in a more or less attractive part of the sector. It may be better to identify specific companies in exactly the same class of business as Chris Limited.

The sector P/E reflects the expected growth in earnings in the sector. It may be that Chris Limited's expected growth rate is better or worse than the sector average, meaning that a different P/E ratio would be more appropriate.

The sector P/E reflects the risk of the sector in general, including business and gearing risk. If Chris Limited has a different level of gearing or is in a part of the sector where the business risk is slightly different, then a different P/E ratio would be appropriate.

The calculation has assumed that Wilkinson plc acquires a controlling interest and forces the sale of the unused property.

When the calculation is useful

Use of earnings to value a company implies that the investor has control of the earnings and can dictate dividend policy, for example. It would therefore be suitable if Wilkinson plc is aiming to get control of the company.

Dividend-based valuation – Dividend valuation model

In order to use the dividend valuation model, the cost of equity of the company must first be estimated. This can be done with the capital asset pricing model: $k_e = r + \beta_e (r_m - r_f)$

Where: k_e = the company's cost of equity

r_f = the risk-free rate (estimated as the yield on gilts)

r_m = the return on the market

K_e = 11 + 0.8(21-11) = 19%

The market value (MV) of the company is given by the formula $MV = D_1 / (k_e - g)$ where D_1 is the prospective dividend (estimated as $D_0 \times \{1 + g\}$), k_e is the cost of equity and g is the expected growth rate in dividends. The value of g can be estimated from extrapolating the dividend growth of Chris Limited over the last few years and is 11% (√{185/150} – 1).

MV = 185 × 1.11/ (0.19 – 0.11) = £2,567,000

Since this valuation relies on the b of quoted companies and would give the value of a quoted company's dividend stream, this value should be discounted.

£2,567,000 × 60% = £1,540,000

Adjusting for the value of the unused property, the value of the business would be around **£1,640,000**.

Limitations of the calculation

As with the earnings estimate, use of the sector b implies that Chris Limited has comparable gearing and business risk to the sector average.

The assumption that past dividend growth of 11% will continue in the future may not be valid.

It has been assumed the company will dispose of the unused property even if Wilkinson plc only obtains a minority holding.

When the calculation would be useful

A minority investor who only receives dividends from the company will find this most useful and therefore it may be relevant if Wilkinson plc only intends to take a minority stake.

Dividend-based valuation – Dividend yield

$$185 \times \frac{1}{0.11} = 1{,}682$$

Discount for lack of marketability:

1,682 × 60% = 1,009,000

Including the proceeds from the assumed sale of the unused property, the valuation of Chris Limited would be **£1,109,000**.

Limitations of the calculation

As with the above methods, it assumes that Chris Limited is similar to the sector in terms of gearing and business risk.

It has been assumed that the company will dispose of the unused property, even if Wilkinson plc does not acquire a controlling shareholding.

When the calculation would be useful

As with the dividend valuation model, it is most useful for a minority investor who will only receive a dividend flow from the company.

Asset-based valuation

Net realisable value of assets

The net book value of the company's assets is £2,500,000. To establish the value available to equity investors, the market value of the loan stock must be deducted. This will be the PV of the loan stock's future cash flows, discounted at the investors' required rate of return. The best indication of required returns is given by the details on gilts for the same maturity with the same coupon in the question. A risk premium must be added on to the yield on the gilts to compensate for the additional risk of Chris Limited, say 3%. The resultant annual yield (11% + 3% = 14%) needs to be altered to a semi- annual rate of return of 6.8% (√1.14 – 1) since the coupon is semi-annual.

Note: The precise calculation here using the square root is probably over the top and gives a spurious level of accuracy. A six-monthly factor of 7% would be just as good.

Time	Cash flow	Discount factor	PV
	£	6.8%	£
1–10	60,000	7.09	425,400

10	1,200,000	$1/1.068^{10}$	621,539
			1,046,939

The total value of the company's equity is therefore £2,500,000 – £1,047,000 = **£1,453,000**.

Limitations of the calculation

The only data available is net book value, which may not represent realisable value due to potential revaluations, obsolescent inventories, costs of disposal, and so on.

The contents of each category of asset and liability are not known. For example, non-current assets may include intangibles which could not easily be sold at book value.

The value of intangibles such as brands and goodwill may not be included in the above valuation. The required yield on the loan stock has been estimated and the 14% may not be appropriate.

When the calculation would be useful

Net realisable value assumes the company's assets can be sold off.

This will only be appropriate if Wilkinson plc acquires a 75% interest and can force a compulsory liquidation. Alternatively, the company must obtain at the very least 50% to be able to force the disposal of any surplus assets.

Replacement cost of assets may be appropriate to use for a purchaser who is considering starting up an equivalent business from scratch. The problem will be in identifying the cost of replacing intangible assets such as goodwill and brands.

Reasons for differences between the valuations

The earnings-based valuation is based on an earnings figure that has grown dramatically in the final year. The very high earnings of £410,000 may not be representative and in subsequent years the earnings may fall back to a level similar to earlier years. Alternatively, if the earnings are going to grow from the current base, then the earnings-based valuation is likely to be the most appropriate of all the above.

The dividends of Chris Limited, by contrast, have grown at a reasonably constant rate over the three- year period regardless of earnings performance. Since the dividends have grown at a much slower rate than earnings, this explains the much lower valuations using dividend-based methods.

The dividend policy may give a clue as to the directors' expectations for future profitability and sustainable dividend growth. If this is the case, then the dividend-based methods are likely to give the most appropriate valuations.

The asset-based valuation is the lowest of the four figures. This is probably because the company derives its value not from its assets base but from its earnings stream and cash flows.

Additional information required

- Market value of assets
- Existence and value of intangibles
- Analysis of profits between ordinary recurring items and exceptional one-off items, which may have distorted the profits in individual years
- Details of costs or income that may be avoided or lost if the company is acquired, such as very high directors' remuneration
- Details on growth prospects

- More specific details on the company's business, its business risk and gearing risk and similar information for closely comparable quoted companies
- The shareholding Wilkinson plc intends to buy
- The possibilities for synergies
- Cash flow details, to obtain a more fundamental cash-based valuation
- Details of any comparable deals executed in the recent past

3 Cub

Economic value added®

Economic value added® = Net operating profit after tax (NOPAT) – (Capital employed × Cost of capital)

NOPAT

NOPAT is arrived at after making a number of adjustments.

	20X1	20X2
	$m	$m
Profit after tax	155	196
Add non-cash expenses	32	36
Add interest after tax charge (1 – 0.24)	9.9	13.7
NOPAT	196.9	245.7

Capital employed

Capital employed is on start of year figures

20X1 Capital employed	= Capital employed at end of 20X0 + Leases
	= 695 + 35
	= $730m
20X2 Capital employed	= Book value of shareholders' funds + Bank loans + Leases
	= 595 + 175 + 35
	= $805m

Weighted average cost of capital

20X1 cost of capital	= (0.75 × 14%) + (0.25 × (7% (1 – 0.24)))
	= 11.8%
20X2 cost of capital	= (0.75 × 16%) + (0.25 × (8% (1 – 0.24)))
	= 13.5%

Economic value added®

20X1 EVA®	= 196.9 – (0.118 × 730)
	= $110.8m
20X2 EVA®	= 245.7 – (0.135 × 805)
	= $137.0m

On this measure, the company has created significant value in both 20X1 and 20X2 and appears to be on a rising trend.

4 Cromer

Because this is a business combination, Cromer must measure the liability at fair value in accordance with IFRS 13, *Fair Value Measurement*, rather than using the best estimate measurement required by IAS 37, *Provisions, Contingent Liabilities and Contingent Assets*.

Cromer will use the expected present value technique to measure the fair value of the decommissioning liability. If Cromer were contractually committed to transfer its decommissioning liability to a market participant, it would conclude that a market participant would use all of the following inputs, probability weighted as appropriate, when estimating the price it would expect to receive.

(1) Labour costs

(2) Allocated overhead costs

(3) The compensation that a market participant would generally receive for undertaking the activity, including profit on labour and overhead costs and the risk that the actual cash outflows might differ from those expected

(4) The effect of inflation

(5) The time value of money (risk-free rate)

(6) Non-performance risk, including Cromer's own credit risk

As an example of how the probability adjustment might work, Cromer values labour costs on the basis of current marketplace wages adjusted for expected future wage increases. It determines that there is a 20% probability that the wage bill will be £15 million, a 30% probability that it will be £25 million and a 50% probability that it will be £20 million. Expected cash flows will then be (20% × £15) + (30% × £25m) + (50% × £20m) = £20.5 million. The probability assessments will be developed on the basis of Cromer's knowledge of the market and experience of fulfilling obligations of this type.

Chapter 13
Financial instruments and financial markets

Introduction

Learning outcomes

Knowledge brought forward

Syllabus links and examination context

Chapter study guidance

Learning topics

1 Financial instruments – introduction

2 Raising equity

3 Equity markets

4 Fixed interest securities and bonds

5 Bond markets

6 Bond valuation and yields

7 Credit risk

8 Leasing

9 Derivatives

10 Derivative markets

11 Financial reporting and financial instruments

Summary

Further question practice

Technical reference

Self-test questions

Answers to Interactive questions

Answers to Self-test questions



Introduction

Learning outcomes

- Assess and explain the types of equity securities, and evaluate the implications for disclosure, presentation, recognition and measurement in financial statements
- Appraise and explain the characteristics of equity markets and the financial institutions operating in these markets
- Analyse and evaluate the cost of equity, portfolio theory and the use of appropriate asset pricing models, applying principles of financial economics
- Explain the types of fixed interest securities and evaluate the implications for disclosure, presentation, recognition and measurement in financial statements
- Appraise and explain the characteristics of bond markets and the financial institutions operating in these markets
- Appraise and evaluate the use of bonds/loans as a method of finance, and explain the implications of terms included in loan agreements in a given scenario (for example, covenants and guarantees) and explain the procedures by which monitoring and assurance can be provided in respect of such agreements
- Explain and appraise bond valuation techniques and assess flat and gross redemption yields
- Explain and appraise yield curves, sensitivity to yield and components of the yield
- Appraise and evaluate credit risk and credit spread
- Explain the types of derivative securities and evaluate the implications for disclosure, presentation, recognition and measurement in financial statements
- Assess and explain the characteristics of derivative markets and the financial institutions operating in these markets
- Appraise and evaluate the characteristics of forwards, futures, options, swaps and credit derivatives

Knowledge brought forward

The chapter revises the main features of equity and debt finance, and then builds on your earlier knowledge of equity and debt finance and derivatives.

Syllabus links and examination context

We shall look at the choice between different sources of finance at the start of the next chapter. You should not necessarily expect to have to choose between debt and equity. It may be that all the choices available are forms of debt, and you may well have to consider corporate reporting implications. The choice may also be influenced by the financial markets that the business is able to use.

The use of derivatives to hedge interest rate and foreign exchange risks is covered in the chapter Financial risk management. The chapter revises the main features of equity and debt finance, and then builds on your earlier knowledge of equity and debt finance and derivatives.

Chapter study guidance

Use this schedule and your study timetable to plan the dates on which you will complete your study of this chapter.

Topic	Practical significance	Study approach	Exam approach	Interactive questions
1	**Financial instruments – introduction** Without finance, businesses would not exist. Therefore, a finance professional must have detailed knowledge of the various ways in which different methods of financing can help individual organisations. There has been rapid international expansion in the use of financial instruments. As we will see in this chapter, these vary from straightforward, traditional instruments such as bonds and equities, through to derivative instruments.	**Approach** In Chapter 1, we highlighted the linkage between business and financial strategy in relation to the questions of whether an entity has sufficient funds to support a proposed business strategy, or how can it raise the additional funds needed to support that strategy. In this chapter we now look at the different sources of finance available to entities. You should focus on the advantages and disadvantages of the different financing methods. It is worth taking time to make sure you are comfortable with the calculations in this chapter. **Stop and think** If a business has surplus cash, should this cash always be its first-choice source of finance for a new project?	In the examination you may be required to discuss the strengths and weaknesses of different sources of finance and the risks relating to them.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
2	**Raising equity** There are broadly three methods of raising equity: retentions, rights issues and new issues. Classification of financial instruments as debt or equity can have a significant effect on the financial statements. If a financial instrument is classified as equity, reported gearing will be lower than if it were classified as a financial liability.	**Approach** You can cover sections 2.1 and 2.3 quite quickly as this is revision from your Financial Management studies. Spend time on section 2.2 ensuring that you can distinguish between debt and equity in accordance with IAS 32, *Financial Instruments: Presentation*. **Stop and think** Why is it important to correctly classify a financial instrument as debt or equity?	In the exam you may be required to correctly classify a financial instrument as debt or equity.	
3	**Equity markets** The London Stock Exchange's Main Market is the world's most international market for the listing and trading of equity, debt and other securities. The Alternative Investment Market (AIM) and ICAP Securities and Derivatives Exchange (ISDX) are also available for smaller companies.	**Approach** Sections 3.6 and 3.7 on the factors affecting share price and the efficient market hypothesis are important. Make sure you work carefully through these two sections. **Stop and think** How has the increased use of automated trading platforms impacted on market efficiency?	In the examination you may be required to advise businesses of the features of different financial markets and which markets they should use.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
4	**Fixed interest securities and bonds** Bonds come in various forms, including redeemable, irredeemable, floating rate, zero coupon and convertible.	**Approach** The terminology in this section should be familiar to you from your earlier studies, but this section provides a worthwhile recap. You should focus on the advantages and disadvantages of the different types of bonds, aiming to gain awareness of the key features of each. **Stop and think** What are the main advantages to a borrower of using convertible bonds?	In the exam you may be required to suggest an appropriate type of bond to fund a new investment project.	**IQ1: Convertibles** This simple question will help with your understanding of the terms conversion value and conversion premium. **IQ2: Derecognition of financial assets** This question provides a good revision of your knowledge of IFRS 9.
5	**Bond markets** Large multinational firms with good credit ratings are able to access the international bond market. Smaller companies will have to use the less active domestic corporate bond market.	**Approach** This brief section covers the sterling domestic bond market and bond markets in other countries. **Stop and think** What are the advantages of issuing bonds on the international bond market?	In the exam you may be required to examine the impact on investment and financing decisions of developments in capital markets.	
6	**Bond valuation and yields** There is an inverse relationship between bond yields and bond prices. As interest rates rise, the price of bonds falls (and bond prices rise when yields fall). The yield that	**Approach** Go through each of the worked examples carefully to practice calculating the price of a bond and its yield. **Stop and think** What factors impact on the	Exam questions may ask you to calculate and compare the yields of different bonds.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
	investors require from a bond is made up of a number of elements, including a risk-free return plus return to compensate for credit and default risk, liquidity and marketability risk, issue-specific risk and fiscal risk.	required yield of a bond?		
7	**Credit risk** Credit risk is the risk for a lender that the borrower will default either on interest payments or on the repayment of principal on the due date, or on both.	**Approach** You can work through this brief section of the chapter quite quickly. Make sure you are familiar with the credit ratings noted in section 7.1.2 **Stop and think** What two factors determine the credit risk of a bond?	Exam questions may require you to interpret the credit rating attached to bonds.	
8	**Leasing** Leasing is a commonly used source of finance, especially for small or medium sized entities. It is a form of debt finance that can be very useful where capital is rationed. Rather than buying an asset outright, using either available cash resources or borrowed funds, a business may lease an asset.	**Approach** The financial reporting implications of leasing are covered in more detail in section 11. This section focuses on the lease or buy decision. Go through the two worked examples carefully. **Stop and think** What non-financial factors would a company need to consider when making the lease or buy decision?	Exam questions may ask you to assess the option of leasing an asset rather than buying it outright. You may need to consider the financial and strategic or operational implications of leasing rather than buying an asset.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
9	**Derivatives** Derivatives are instruments based on (or derived from) underlying assets such as bonds, shares, indices, commodities, currencies and property. They enable investors to reduce risk or enhance returns on these investments. Derivatives include forwards, futures, options and swaps.	**Approach** This section is key as an introduction to derivatives. There is lots of terminology and worked examples in this section so take your time working through each of them. **Stop and think** Why would a company choose not to use derivatives?	In the exam you may have to calculate the outcome from using derivatives to hedge risk. You may also need to discuss the advantages and disadvantages of using derivatives in the context of the scenario provided.	**IQ4: Black-Scholes formula** A simple question allowing you to practice the application of the Black-Scholes pricing model.
10	**Derivative markets** Derivatives can only exist when the market for the underlying asset or item allows buyers and sellers to freely influence the market price. Those underlying markets include the FX markets for exchange rates and money markets for short-term interest rates.	**Approach** Section 10 is relatively brief. The table at section 10.2 is a useful summary of the differences between OTC vs exchange traded markets. **Stop and think** How have the advances in technology changed how the derivates markets operate?	In the exam you may be required to consider the use of the derivatives market for a company's hedging strategy.	

Topic	Practical significance	Study approach	Exam approach	Interactive questions
11	**Financial reporting and financial instruments** The use of financial derivatives has implications for financial reporting and profitability. The three accounting standards that currently apply to financial instruments, including derivatives, are IAS 32, IFRS 9 and IFRS 7. IFRS 16 deals with accounting for leases.	**Approach** When going through this section remember that between 15–20% of the marks available in SBM&L could relate to financial reporting issues, so make sure that you are comfortable with the key elements of the financial reporting treatment of financial instruments. There are lots of numerical examples in this section so take time to work through each of them. **Stop and think** Is your corporate reporting knowledge of financial instruments up to date?	In the exam you may be required to discuss the financial reporting implications of proposed financing methods and describe the assurance work that may be required to support an application for new finance.	**IQ5: Derecognition of financial assets and liabilities** **IQ7: Lessee accounting** **IQ9: Sale and leaseback** **IQ10: Factoring with and without recourse** Each of these brief questions cover some of the key elements of the financial reporting treatment of financial instruments. Work through each of them carefully.

Once you have worked through this guidance you are ready to attempt the further question practice included at the end of this chapter.

1 Financial instruments – introduction

Section overview

- There has been rapid international expansion in the use of financial instruments. As we will see in this chapter, these vary from straightforward, traditional instruments such as bonds and equities, through to derivative instruments.
- Equity-based financial instruments represent ownership of an asset, while debt-based financial instruments represent a loan made by an investor to the owner of the asset

Financial instruments are discussed in the sections which follow and may be divided into two types:

Cash instruments

The value of such instruments is determined by the market, and includes equity shares (section 2), cash deposits, certificates of deposit and loans. They also include bonds in their various forms (section 4.1 – 4.10), warrants (section 4.11), commercial paper (section 4.12), medium term notes (section 4.13), repos (section 4.14) and Islamic bonds (section 4.15).

Derivative instruments

The value of a derivative instrument is based on its underlying components, such as assets or interest rates. Derivatives may be either **exchange traded** (where there are standardised contracts, an active secondary market accessible to participants and guarantees against default via a clearing house) or contracts entered directly between counterparties **over the counter** (on the OTC market).

Derivatives which are OTC traded include forwards (section 9.1), forward rate agreements (section 9.2), swaps (section 9.8), credit derivatives (section 9.9) and CDOs (section 9.10). Exchange traded derivatives include futures (section 9.3) and options (section 9.4 – 9.7).

Some important definitions are set out below.

Definitions

Financial instrument: Any contract that gives rise to both a financial asset of one entity and a financial liability or equity instrument of another entity.

Derivative: A financial instrument or other contract with all three of the following characteristics:

(a) its value changes in response to the change in an underlying variable such as a specified interest rate, financial instrument price, commodity price or foreign exchange rate

(b) it requires no initial net investment, or only a small initial net investment

(c) it is settled at a future date

2 Raising equity

Section overview

You will recall from earlier studies that there are various methods of issuing new equity, depending on whether the company is listed.

Definition

Equity: represents the **ordinary shares** in the business. Equity shareholders are the owners of the business and through their voting rights exercise ultimate control. Equity shares have the rights to participate in the distribution of residual assets after any fixed claims from loan holders or preference shareholders have been satisfied.

2.1 Raising new equity

There are broadly three methods of raising equity:

Method	Real world use
Retentions, ie, retaining profits flows rather than paying them out as dividends (sometimes mistakenly called financing through retained profits – but profits are not cash flows)	By far and away the most important source of equity
Rights issues, ie, an issue of new shares to existing shareholders	The next most important source
New issues to the public, ie, an issue of new shares to new shareholders	The least important source of new equity

In addition, new shares may be issued as all or part of a payment for an acquisition and some new shares are issued by companies that operate share option schemes for employees.

2.1.1 Retained earnings

The cash generated from profits earned by a business can either be paid out to shareholders in the form of dividends or reinvested in the business.

There is sometimes a misconception that because no new shares are being sold, using cash flows from retained earnings has no cost. There may be no issue costs but shareholders will still expect a return on the funds reinvested in the business.

Such retentions represent a very easy and important source of finance, particularly for young growing businesses where there may be a continual need for funds but where it is impractical to keep raising them using rights/new issues (and debt).

2.1.2 Rights issues

Definition

Rights issue: is an issue of new shares for cash to existing shareholders in proportion to their existing holdings.

Legally a rights issue must be offered to existing shareholders before a new issue to the public. Existing shareholders have rights of first refusal (pre-emption rights) on the new shares and can, by taking them up, maintain their existing percentage holding in the company. However, shareholders can, and often do, waive these rights by selling them to others. Shareholders can vote to rescind their pre-emption rights.

Companies need to consider the following factors when making rights issues:

(a) **Issue costs** – These have been estimated at around 4% on £2 million raised but, as many of the costs are fixed, the percentage falls as the sum raised increases.

(b) **Shareholder reactions** – Shareholders may react badly to firms continually making rights issues as they are forced either to take up their rights or sell them (doing nothing decreases their wealth). They may sell their shares in the company, driving down the market price.

(c) **Control** – Unless large numbers of existing shareholders sell their rights to new shareholders there should be little impact in terms of control of the business by existing shareholders.

(d) **Unlisted companies** – These often find rights issues difficult to use, because shareholders unable to raise sufficient funds to take up their rights may not have available the alternative of selling them if the firm's shares are not listed. This could mean that the firm is forced to use retentions or raise loans.

2.1.3 Issues in the market

These account for around 10% of new equity finance. When they occur they are often large in terms of the amount raised. They are often used at the time a firm obtains a listing and a quotation on the stock exchange, known as an initial public offering (IPO). An IPO is an offer to sell shares in a company to the public for the first time. The shares may be existing shares owned by the founders of the business who wish to realise profits from their investment, or new shares issued to raise additional capital to support growth, or a combination of both. The company is normally advised on its IPO by an investment bank.



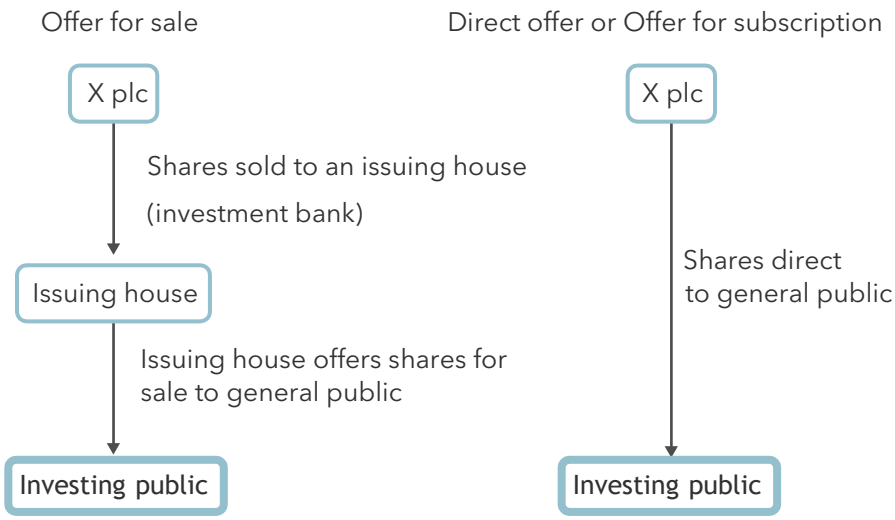
Definition

Initial public offering (IPO): An initial public offering (IPO) is an offer to sell shares in a company to the public for the first time.

Issues in the market are also popular when a firm wants to raise a very large amount and therefore needs a high-profile issue.

There are two methods of making a public offer, and they are best illustrated diagrammatically:

Figure 13.1: Methods of making a public offer



In practice the offer for sale is far more common; in either method the issue is likely to be underwritten. There is no restriction on the amount of capital raised by this method.

2.2 Accounting: recognition of equity

You will remember the definition of an equity instrument from your earlier financial reporting studies.

Definition

An equity instrument: is any contract that evidences a residual interest in the assets of an entity after deducting all its liabilities.

An instrument is an equity instrument if and only if:

(a) The instrument includes no contractual obligation to:
- deliver cash or another financial asset to an entity; or
- exchange financial assets or financial liabilities with another entity under conditions that are potentially unfavourable to the entity.

(b) If the instrument will or may be settled in the issuer's own equity instruments, it is:
- A non-derivative that includes no contractual obligation for the issuer to deliver a variable number of its own equity instruments
- A derivative that will be settled only by the issuer exchanging a fixed amount of cash or another financial asset for a fixed number of its own equity instruments

2.2.1 Classification as equity

Under IAS 32, *Financial Instruments: Presentation*, whether a financial instrument is classified as an equity instrument should be in accordance with the substance of the contractual terms and not with factors outside the terms. Terms may include whether the instruments are redeemable, whether the returns are mandatory or discretionary and whether they contain features such as put or call options that require the issuer to settle the instrument in cash or other financial assets.

The critical feature therefore in contractual terms is whether there is an obligation to deliver cash or another financial asset, or to exchange a financial asset or financial liability on

potentially unfavourable terms. If the issuer does not have an unconditional right to avoid delivery of cash or other financial assets, then the instrument is a liability.

The definition of residual interest is not confined to a proportionate interest ranking equally with all other interests; it may also be an interest in preference shares.

2.2.2 Debt/equity distinction in more detail

In practice, the debt/equity distinction may not be clear cut. Classification of financial instruments as debt or equity can have a significant effect on the financial statements. Guidance is provided in IAS 32, *Financial Instruments: Presentation* (and other standards for items outside IAS 32's scope), but there are sometimes areas where it is difficult to determine whether a transaction is debt or equity.

A contract is not an equity instrument solely because it may result in the receipt or delivery of the entity's own equity instruments. It is necessary to consider whether the settlement results in receipt or delivery of variable or fixed number of the entity's own equity instruments.

If the contract results in delivery of variable number of equity instruments, the fair value of the equity instruments equals to the amount of the fixed contractual right or obligation. Such a contract does not evidence a residual interest in the entity's assets after deducting all of its liabilities.

The following contracts require delivery of the entity's own equity instruments but are **classified as financial liability**:

- a contract to deliver as many of the entity's own equity instruments as are equal in value to £6,000
- a contract to deliver as many of the entity's own equity instruments as are equal in value to 200 ounces of gold

Contingent settlement provisions

A financial instrument may require the entity to deliver cash or another financial asset in the event of **occurrence or non-occurrence of uncertain future events** that are beyond the control of both the issuer and holder of the instrument. These are known **as contingent settlement provisions** and could relate to changes in stock market index, consumer price index, interest rate, issuer's future profits or revenues. Such instruments are classified as financial liability. This is because the issuer **does not have an unconditional right to avoid** delivering cash or another financial asset.

The financial instrument would be an equity instrument if it has equity-like features, for example, obligation arises only in the event of liquidation of the issuer.

2.2.3 Practical implications

The classification of financial instruments as debt versus equity is important as the presentation of the two items and associated financial effects are very different.

If a financial instrument is classified as a financial liability (debt) it will be reported within current or non-current liabilities. Non-current liabilities are relevant in determining an entity's leverage, or gearing, ie, the proportion of debt finance versus equity finance of a business, and therefore risk to ordinary equity holders.

Distributions relating to instruments classified as financial liabilities are classified as finance cost, having an impact on reported profitability.

If a financial instrument is classified as equity, reported gearing will be lower than if it were classified as a financial liability. However, classification as equity is sometimes viewed negatively as it can be viewed as a dilution of existing equity interests.

Distributions on instruments classified as equity are charged to equity and therefore do not affect reported profit.

It is important to note that it is not just the presentation of an arrangement that determines the underlying gearing of a business that an analyst will use to assess the business's risk, but the substance of the arrangement. Analysts often make their own adjustments to financial statements where they believe that the information reported under IFRSs does not show the underlying economic reality. IAS 32, *Financial Instruments: Presentation* strives to follow a substance-based approach to give the most realistic presentation of items that are in substance debt or equity, avoiding the need for analysts to make adjustments.

Classification of instruments can also have financial implications for businesses. For example, debt covenants on loans from financial institutions often contain clauses that reported gearing cannot exceed a stated figure, with penalties or call-in clauses if it does.

Companies with high gearing may also find it harder to get financing or financing may be at a higher interest rate.

High gearing is particularly unpopular in the current economic climate and there have been high profile cases of companies that have been pressed to sell off parts of their business to reduce their 'debt mountains', eg, Telefónica SA, the Spanish telecoms provider, selling O_2 Ireland to Hutchison Whampoa (the owner of the '3' telephone network).

Consider the following examples.



Worked example: Debt or equity?

1 Acquittie issued 40 million non-redeemable £1 preference shares at par value. Under the terms attaching to the preference shares, a dividend is payable on the preference shares only if Acquittie also pays a dividend on its ordinary shares relating to the same period.

2 Acquittie entered into a contract with a supplier to buy a significant item of equipment. Under the terms of the agreement the supplier will receive ordinary shares with an equivalent value of £5 million one year after the equipment is delivered.

3 The directors of Acquittie, on becoming directors, are required to invest a fixed agreed sum of money in a special class of £1 ordinary shares that only directors hold. Dividend payments on the shares are discretionary and are ratified at the Annual General Meeting of the company. When a director's service contract expires, Acquittie is required to repurchase the shares at their nominal value.

Solution

1 IAS 32 requires a financial instrument to be classified as a liability if there is **a contractual obligation** to deliver cash or another financial asset to another entity.

In the case of the preference shares, as they are **non-redeemable**, there is **no obligation to repay the principal**.

In the case of the **dividends**, because of the condition that preference dividends will only be paid if ordinary dividends are paid in relation to the same period, the **preference shareholder has no contractual right** to a dividend. Instead, the distributions to holders of the preference shares are at the discretion of the issuer as **Acquittie can choose** whether or not to pay an ordinary dividend and therefore a preference dividend. Therefore, there is **no contractual obligation** in relation to the dividend.

As there is no contractual obligation in relation to either the dividends or principal, the **definition of a financial liability has not been met** and the preference shares should be treated as **equity** and initially recorded at **fair value** ie, their par value of £40 million.

The treatment of **dividends** should be consistent with the classification of the shares and should therefore be charged directly to **retained earnings**.

2 The price of the equipment is fixed at £5 million one year after delivery. In terms of recognition and measurement of the equipment, the £5 million price would be discounted back one year to its present value.

The company is paying for the equipment by issuing shares. However, this is **outside the scope of IFRS 2**, *Share-based Payment* because the payment is not dependent on the value of the shares, it is fixed at £5 million.

This is an example of a contract that 'will or may be settled in an entity's own equity instruments and is a non-derivative for which the entity is or may be obliged to deliver a **variable** number of the entity's own equity instruments', ie, a financial liability.

It is the number of shares rather than the amount paid that will vary, depending on share price. Therefore it should be classed as a **financial liability** and initially measured at the **present value** of the £5 million.

Subsequently, as it is not measured at fair value through profit or loss (as it is not held for short- term profit-taking or a derivative), it should be measured at **amortised cost**.

As a result, **interest** will be applied to the discounted amount over the period until payment and recognised in **profit or loss** with a corresponding increase in the financial liability.

3 Most ordinary shares are treated as equity as they do not contain a contractual obligation to deliver cash.

However, in the case of the directors' shares, a **contractual obligation to deliver cash exists** on a specific date as the shares are **redeemable** at the end of the service contract.

The redemption is **not discretionary**, and Acquittie has no right to avoid it. The mandatory nature of the repayment makes this capital a financial liability. The financial liability will initially be recognised at its fair value, ie, the present value of the payment at the end of the service contract. It will be subsequently measured at amortised cost and effective interest will be applied over the period of the service contract.

Dividend payments on the shares are **discretionary** as they must be ratified at the Annual General Meeting. Therefore, no liability should be recognised for any dividend until it is ratified. When recognised, the classification of the **dividend** should be consistent with the classification of the shares and therefore any dividends are classified as a **finance cost** rather than as a deduction from retained earnings.

2.2.4 Interest, dividends, losses and gains

Again this is a summary of what you've studied before.

Interest, dividends, losses and gains arising in relation to a financial instrument that is classified as a financial liability should be recognised in profit or loss for the relevant period.

Distributions, such as dividends, paid to holders of a financial instrument classified as equity should be charged directly against equity (as part of the movement on retained earnings in the statement of changes in equity).

For compound instruments such as convertible bonds, the annual interest expense recognised in profit or loss should be calculated by reference to the interest rate used in the initial measurement of the liability component.

2.3 Cost of equity and portfolio theory

This section provides a brief revision of portfolio theory and the cost of equity. The cost of equity is the size of returns that investors expect to receive on their equity shares, measured as an annual percentage amount on the value of their investment.

Decisions by investors to buy or sell investments are based on an assessment of expected returns and risk. Risk is the possibility that actual returns could be higher or lower than expected, and risk can be measured statistically from historical data as the standard deviation of returns. A higher standard deviation around the mean expected return indicates higher volatility in returns and so greater investment risk.

Risk can be divided into two categories:

(a) **Systematic risk** is variability in returns that is caused by general market factors, such as changes in economic conditions. Systematic risk is a feature of the stock market as a whole; when the market as a whole suffers a fall in returns, the expected returns on all companies' shares in the market also fall. Similarly, expected returns for all companies tend to rise when market returns as a whole increase.

(b) **Unsystematic risk or diversifiable risk** is risk that is unique to individual companies. It represents variations in returns that are independent of market returns generally, for example due to changes in the perceived future profitability of a company.

Portfolio theory is based on the view that investors with well-diversified portfolios can ignore diversifiable risk because, by holding shares in a fairly large number of different companies, the diversifiable risk with different companies will tend to cancel each other out. Some companies will provide unexpected high returns and others will disappoint with unexpected low returns, due to factors that are unique to those companies.

Diversifiable risk can be 'diversified away' by having a sufficiently large portfolio of shares in different companies. If this is the case, systematic risk should be the only aspect of risk that concerns these investors. Investors should decide on the balance between risk and return that satisfies their investment preferences: do they want higher returns by accepting higher risks; or lower returns and lower risks?

2.3.1 Risks and returns in a portfolio

Investors in market securities (shares and corporate bonds) must accept that there will be some volatility in the returns they receive, due to systematic risk. If they invest in a portfolio of shares and other securities that represents the make-up of the stock market as a whole, they accept the average level of systematic risk in the market (the market risk).

They could choose to invest in companies with higher systematic risk, or in companies with lower systematic risk. Whatever they choose to do, their expected returns (based on systematic risk only and ignoring diversifiable risk) will vary up or down with market returns, by a larger or smaller amount than returns for the market as a whole.

Investors can also invest in risk-free securities. These are securities whose systematic risk is 0. The expected returns are predictable and do not change with changes in stock market returns. Risk-free returns can be defined as the return from government bonds (or other government securities) issued by a government with a high credit rating and in its domestic currency.

An investor can therefore seek to create a portfolio that meets their requirement for risk and return by:

(a) creating an investment portfolio that has a suitable mix of shares representing the stock market as a whole and risk-free securities; or

(b) creating a portfolio containing shares whose systematic risk is above or below the market average.

Investors will expect a return (ignoring diversifiable risk) that is the best obtainable for the amount of investment risk that they are prepared to accept.

2.3.2 Beta factors

Systematic risk can be measured by beta factors. When average stock market returns change, up or down, the beta factor of an individual security (including risk-free securities) is a measure of the expected amount by which the expected returns on that security will change.

(a) The beta factor for the market as a whole is 1.0, meaning for example that if expected market returns go up by 1% (100 basis points) due to improving economic conditions, the change in expected market returns will be + 1% × 1.0 = 1%.

(b) The beta factor for a risk-free security is 0, which means that if there is a change in the expected market returns, there will be no change in the expected returns from the risk-free security.

(c) The systematic risk for a company's shares may be higher than the systematic risk for the market as a whole, so that its beta factor is higher than 1.0. For example, if a company's shares have a beta factor of 1.5 and expected market returns rise by 1% (100 basis points), the expected returns from the company's shares will increase by + 1% × 1.5 = 1.5%.

(d) Similarly, the systematic risk for a company's shares may be lower than the systematic risk for the market as a whole, so that its beta factor is higher than 0 but lower than 1.0. For example, if a company's shares have a beta factor of 0.8 and expected market returns rise by 1% (100 basis points), the expected returns from the company's shares will increase by + 1% × 0.8 = 0.8%.

Beta factors can be used to calculate the expected returns on a company's equity shares (ignoring diversifiable risk) using the capital asset pricing model (CAPM).

$k_e = r_f + \beta_e (r_m - r_f)$

Where: k_e = cost of equity capital

r_f = risk free rate of return

r_m = the return from the market as a whole

β_e = the beta factor of the individual security

The CAPM is discussed in more detail in the chapter Financial structure and financial reconstruction.

3 Equity markets



Section overview

The London Stock Exchange's Main Market is the world's most international market for the listing and trading of equity, debt and other securities with AIM and ISDX also available for smaller companies.

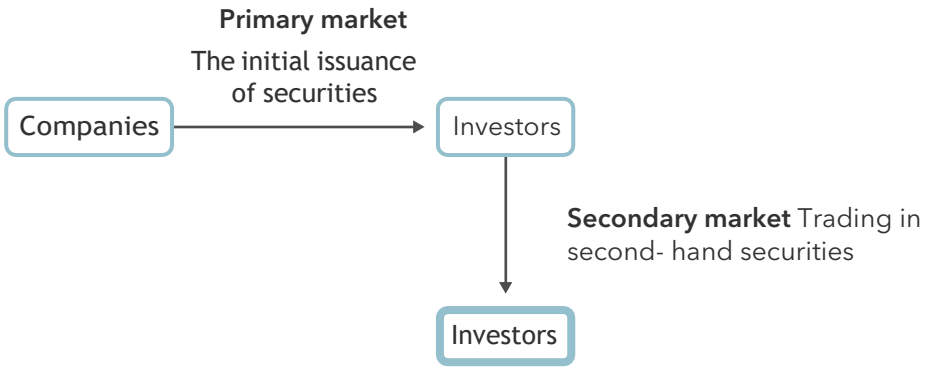
3.1 London Stock Exchange

The **London Stock Exchange** (LSE) is, above all else, a business. Its primary objective is to establish and run a marketplace in securities. In any economy there are savers and borrowers. The exchange acts as a place in which they can meet.

Initially, the companies (the borrowers) issue shares to the investing public (the savers). This is known as the **primary market**. The main role of the LSE's primary market is to enable a company to issue shares. Investors would not be willing to invest their money unless they could see some way of releasing it in the future.

Consequently, the exchange must also offer a **secondary market** trading in secondhand shares; this allows the investor to convert the shares into cash.

Overview of the Exchange's activities



Companies whose shares are traded on the LSE are generally known as **quoted companies**. Listed shares are those which have been admitted to the Official List by the **UK Listing Authority** (UKLA) (the title used by the Financial Conduct Authority in this area). Joining the Main Market of the LSE involves a two-stage process:

- The UKLA is responsible for the approval of prospectuses and the admission of companies to the Official List.
- The LSE is responsible for the admission to trading of companies to the Main Market.

3.1.1 Primary market

The primary market is the market for the issue of new securities. It is important to ensure that the primary market is selective. A poor-quality primary market will undermine the liquidity of the secondary market. Companies wishing to issue shares often find it useful to hire the services of an issuing house or sponsor (usually these are services available from the investment banks) which will guide them through the process of preparing for the new issue and advertising the fact to potential new investors.

3.1.2 Secondary market

The secondary market (markets in secondhand securities) exists to enable those investors who purchased investments to realise their investments into liquid cash.

In order to ensure the stability of the secondary market, the exchange must establish strong controls over the companies that are allowed onto the market. Equally, the exchange must establish rules concerning the activities of brokers in the secondary market in order to preserve the reputation of the market. However, the LSE has itself a number of rules concerning the restriction on the investments in which a member firm can deal. These rules are designed to complement the UKLA rules.

Most investors who buy or sell shares in a secondary market want a liquid market, in which there is always a number of other investors willing to buy or sell the shares. A liquid market should mean that a buyer or seller will always be able to get a fair price for the shares, and market prices will not be distorted by major share transactions (such as the sale of a large quantity of shares in a company by an investor) or by a temporary lack of willing buyers or sellers.

Although secondary market trading is carried out on the stock exchange because it can provide a liquid market, a significant amount of trading in shares occurs elsewhere. Some institutional investors want to enjoy confidentiality about their share dealing and to avoid reporting requirements for large transactions on the stock exchange. Some banks therefore operate their own private share dealing services for off-market transactions.

3.1.3 Role of stock exchange member firms

All member firms of an exchange are **broker-dealers**. Broker-dealers have **dual capacity**, giving them the choice to either act as **agents** on behalf of customers, or to deal for themselves as **principals**, dealing directly with customers.

Some broker-dealers can elect to take on board an additional level of responsibility through one of the following roles.

Market makers	Market makers are the focus of market activity and their obligation is to ensure that there is always a two-way price in the securities in which they are registered.
Inter-dealer brokers	Inter-dealer brokers (IDBs) provide market makers with an anonymous dealing service to unwind positions.
Stock Borrowing and Lending Intermediaries	Stock Borrowing and Lending Intermediaries provide a service allowing market makers to borrow securities in order to settle a 'short' transaction, ie, selling shares they do not own.

3.1.4 Trading methods of the London Stock Exchange (LSE)

Different types of shares, with varying levels of liquidity, trade via a variety of market systems. On the LSE, there are three trading systems, through which Main Market shares and Alternative Investment Market (AIM) shares are traded: the Stock Exchange Electronic Trading Service (SETS), the Stock Exchange Electronic Trading Service – quotes and crosses (SETSqx) and the Stock Exchange Automated Quotation System (SEAQ).

System	SETS	SETSqx	SEAQ
Nature	Order driven Electronic trading system	Combined order and quote driven Periodic electronic auction book combined with standalone non-electronic quote-driven market making	Quote driven Non-electronic trading system in which market makers quote prices that buyers/sellers must accept
Market makers?	**No** market makers	**At least one** market maker per stock	**At least two** market makers per stock
Full list stocks traded	All stocks classified as 'liquid' under the Markets in Financial Instruments Derivative (MiFID), ie, all FTSE All-Share Index stocks, Exchange Traded Funds, Exchange Traded Commodities, actively traded Irish stocks	All stocks classified as 'illiquid' under MiFID, ie, equities that are less liquid than shares traded on SETS	None

AIM stocks traded	Some traded AIM stocks	All AIM stocks in AIM EURO sectors not traded on SETS	All remaining AIM stocks

3.1.5 Order- and quote-driven systems

Where a market is **order driven**, the relevant electronic trading system will match buyers and sellers automatically, provided that they are willing to trade at prices compatible with each other. This is essentially the function which SETS performs. Prices of securities which trade on SETS are, therefore, purely driven by the buyers and sellers in the market themselves: for example, buyers and sellers can specify limits to the prices they are prepared to pay or accept.

A **quote-driven** market, such as SEAQ, requires certain market participants (market makers) to take responsibility for acting as buyers and sellers to the rest of the market so that there will always be a price at which a trade can be conducted. The market makers quote buying and selling prices for the securities in which they are prepared to trade. For such a market to operate efficiently, up to date prices at which market makers are willing to trade need to be made available to other market participants. This is the function performed by SEAQ. Unlike SETS, SEAQ does not provide a mechanism for trades to be executed automatically.

3.2 The criteria for listing

The UKLA's rules for admission to listing are contained in the Listing Rules. These detail the requirements that a company must meet prior to being admitted to the Full List. The basic conditions are as follows:

(a) The expected market value of shares to be listed by the company must be at least £700,000. If the company is to issue debt, the expected market value of any such debt is to be at least £200,000.

(b) All securities issued must be freely transferable.

(c) The company must have a trading record of at least three years. Its main business activity must have been continuous over the whole three-year period. In addition, there should be three years of audited accounts. This requirement is waived for innovative high-growth companies, investment companies and certain other situations.

(d) The shares must be sufficiently marketable. A minimum of 25% of the company's share capital being made available for public purchase (known as the free float) is normally seen to satisfy this requirement. The lower the free float, the fewer shares are available in the market, potentially leading to higher share price volatility.

Applicants for listing must appoint a UKLA-approved **sponsor** whose role is to:

- ensure the company and its directors are aware of their obligations
- ensure the company is suitable for listing and satisfy UKLA of this fact
- liaise with UKLA and submit documentation to it as required
- coordinate the listing process

Companies must generally produce a prospectus containing information on the company's past and present activities and performance, directors, capital structure and future prospects.

3.2.1 Premium and Standard listings

UK entities are permitted to apply for a 'Primary' listing.

(a) The **standard** listing allows UK companies to qualify through the less stringent requirements and standards that were previously only open to non-UK entities. This now provides a level playing field for UK and overseas companies.

(b) A **premium** listing is available for equity securities of UK and non-UK incorporated companies and investment entities. The issuer will have to meet the 'super-equivalent' standards.

Entities with a premium listing are subject to more extensive continuing obligations, such as the publication of an annual financial report and other information. They also need to comply with the Disclosure and Transparency Rules.

A premium listing is a prerequisite for admission to the FTSE UK index. Therefore, most larger companies are likely to continue to apply for premium listing despite the additional compliance costs and continuing obligations. Many institutional shareholders have indicated that they will expect 'premium' listings.

3.2.2 Continuing obligations

All applicants for membership agree to be bound by the continuing obligations of the Listing Rules which require the company to:

- notify the LSE of any price-sensitive information
- publish information about important transactions undertaken by the company under the Class Tests Rule
- inform the LSE of any changes in the important registers of ownership of the shares, such as notifiable interests and directors' shareholdings
- notify the LSE of dividends
- issue reports

3.3 Alternative Investment Market (AIM)

The LSE introduced the second-tier **AIM** in 1995. This forum for trading a company's shares enables companies to have their shares traded through the LSE in a lightly regulated regime. Thus smaller, fast-growing companies may obtain access to the market at a lower cost and with less regulatory burden.

An admission to trading on the AIM used to be regarded as a stepping stone towards obtaining a full listing. Companies now choose to remain on the AIM rather than progress to the Main Market. Recent years have seen an increase in traffic of companies transferring the other way – from the Main Market to the AIM – attracted by the lighter touch regulation and also the **tax advantages offered to investors**.

Context example: AIM

Since London AIM's launch in 1995, 3,700 companies have listed, raising an aggregate of £100 billion. The largest institutional investors in London's AIM include BlackRock, Invesco, J.P. Morgan, Schroders, Legal & General, Aviva, Standard Life, Barclays, M&G and AXA.

The 1,000 companies currently listed on London's AIM have an average market capitalisation of £81 million, with 61% falling between £10 million and £250 million. The most common reason an AIM listed company leaves the market is because of acquisition by a competitor, supplier or customer.

65% of the companies listed on London's AIM are British, with the balance coming from overseas jurisdictions such as the US, China, Australia, Ireland, India, South Africa, Israel, Canada and Russia.

Source: http://www.aimadvisers.com/ [Accessed 13 August 2018]

3.3.1 Conditions for admission to the AIM

A summary of the main conditions that companies must meet in order to secure admission to AIM is as follows:

(a) All securities must be freely transferable.

(b) AIM companies must have an LSE-approved Nominated Adviser (NOMAD) to advise the directors on their responsibilities and guide them through the AIM process.

(c) AIM companies must also have a broker to support trading of the company's shares.

(d) AIM companies must comply with ongoing obligations to publish price sensitive information immediately and to disclose details of significant transactions. The NOMAD has a duty to ensure the AIM company meets its ongoing obligations.

(e) Companies with a track record of less than two years must agree to a 'lock-in' whereby the directors, significant shareholders and employees with 0.5% or more of their capital agree not to sell their shares for a year following admission.

For AIM companies, there is no minimum level of free float (shares available for purchase by the public), no minimum market value for their securities and no minimum trading history.

However, they must produce a prospectus, which is considerably less detailed than for a full listing and is known as an Admission Document.

3.3.2 Comparison of UKLA and AIM rules

	Full List	AIM
Trading record	3 years	None
Percentage in public hands	25%	None
Minimum market value	£700,000 for equity £200,000 for debt	None
Free transferability	Yes	Yes
Requirement to produce a prospectus	Yes	Yes
Corporate governance requirements	Yes	Yes
Requirement to announce price-sensitive information without delay	Yes	Yes
Applicable rules	UKLA	LSE

As noted in the chapter Corporate governance, with effect from 28 September 2018 all AIM companies need to report on their application of a recognised corporate governance code. AIM companies will be required to include:

- details of a recognised corporate governance code that the board of directors of the company has decided to apply;
- how the company complies with that code; and
- where it departs from its chosen code, an explanation of the reasons for doing so ('comply or explain')

3.4 ISDX

ICAP Securities and Derivatives Exchange (ISDX) is a Recognised Investment Exchange and may be described as London's third stock market after the LSE Main Market and AIM.

The market is lightly regulated and was established to help smaller companies raise capital. However, it has struggled to attract listings, thanks in part to the success of AIM. The market had previously operated under the name PLUS, but due to financial difficulties it was acquired by ICAP in 2012. It targets smaller companies needing to raise between £2 million and £3 million. The average market value of companies on the ISDX is about £10 million.

3.5 International stock exchanges

3.5.1 US

The US is home to the world's largest stock market, with its constituent parts being the New York Stock Exchange, the American Stock Exchange, NASDAQ OMX, the Philadelphia Stock Exchange, the Boston Stock Exchange, the Chicago Stock Exchange, the Cincinnati Stock Exchange and the Pacific Exchange.

Trading on the NYSE revolves around **specialists** who receive and match orders via a limit order book. Specialists will also act as market makers where an order cannot be matched via the order book to ensure continuous liquidity in a stock. The primary regulator of the US equities market is the Securities and Exchange Commission (SEC).

NASDAQ OMX Group owns and operates the **NASDAQ** stock market, a screen-based, quote-driven market, and is monitored by the SEC. NASDAQ is the largest electronic screen-based securities trading market in the US, and is the second largest by capitalisation in the world. As the successor to over the counter trading of stocks and the world's first electronic stock market when it was set up in 1971, NASDAQ introduced new competition into the market and helped to lower spreads between bid and offer prices. It has specialised in providing a market for the more innovative companies and technology companies.

3.5.2 Japan

There are eight stock exchanges in Japan, the Tokyo Stock Exchange (TSE) being by far the largest. The TSE is order driven through the CORES dealing system. Settlement is on T+3 by book entry transfer through the Japanese Securities Clearing Corporation.

3.5.3 Hong Kong

The Stock Exchange of Hong Kong Limited (SEHK) is a subsidiary of the Hong Kong Exchanges and Clearing Limited. Trading is via AMS/3, an automated order matching system. The Central Clearing and Settlement System (CCASS) is the book-entry clearing and settlement system, operated by the depository, Hong Kong Securities Clearing Company Limited.

3.5.4 France

Trading on Euronext Paris is via the Nouveau Système de Cotation, the exchange's integrated electronic cash market trading platform. It is an order-driven market.

3.5.5 Germany

The Deutsche Börse is a fierce competitor with the LSE. Trading takes place through a computerised order book called Xetra. The Deutsche Börse's Xetra US stars trading scheme allows for the trading of US shares in Europe outside the normal market hours.

3.5.6 Emerging markets

There are various **bond and equity markets** in growing and **emerging markets**, each with its own unique characteristics. In many of these markets, bond and equity trading is OTC and is restricted to locally registered participants. Settlement systems in such markets tend to be run by central banks with no counterparty guarantees. In addition, non-electronic settlement systems and physical delivery are not uncommon.

Potential benefits from investing in emerging markets are as follows:

(a) Potential for high returns – in the long run, emerging markets may offer higher returns than developed markets.
   - There is a perception that foreign securities are mispriced, giving investment opportunities.
   - Faster expanding developing market economies give higher profits growth for companies.

(b) Potential for reduction in overall portfolio volatility – emerging markets have relatively low correlations with developed markets, giving diversification benefits.

Brazil, Russia, India and China are often referred to collectively as the **BRIC** countries. Brazil and Russia are rich in resources, particularly in oil. China is strong in manufacturing, while India has a well- developed information technology industry.

Additional factors motivating investment in emerging markets are as follows:

(a) Liberalisation of foreign markets encouraged foreign investment. The demand caused by foreign investors caused prices to rise, attracting more investors. Note that this has a risk of becoming a speculative bubble, ending with the market crashing after excessive investment by foreign and local investors.

(b) Deregulation (eg, abolishing price controls, foreign exchange controls) made entry into emerging markets easier.

(c) Privatisation of government industries gave a boost to emerging stock markets.

(d) Improved trade, communication and transport links make investing easier.

Investability describes whether it is possible to invest in an emerging market or not. There are various **restrictions on investability** which will make an emerging market less attractive:

(a) limitation on foreign holdings in local stocks, either to a maximum percentage or to a particular class of share

(b) small free float of shares eg, the Government is still a substantial shareholder in a privatised industry

(c) restrictions on remittance of funds to overseas

(d) taxes on foreign investors

(e) dual currency system and other foreign currency controls

(f) investment may only be permitted by authorised foreign investors

(g) lack of liquidity with holdings being too small to be meaningful, giving rise to excessive transaction costs (market impact)

(h) emerging market indices are often based only on investable shares

3.6 Factors affecting share prices

Share prices do not always react in the way that is expected of them. Investors buy ordinary shares mainly because they expect share values to increase. Over a long period, what causes share prices to rise is the increasing earning power of the companies and their ability to pay higher dividends out of these increasing earnings. However, in the short term, there are many other factors that can distort the picture.

3.6.1 Impact of information

Share prices are observed to move when new information is received, and will fairly reflect that new information. Since new information must, by definition, be unpredictable and random (otherwise it would not be new information), it follows that share price movements will also be random.

From these observations, the **efficient markets hypothesis (EMH)** (discussed below) was developed which hypothesised that, at any time, share prices are fairly valued on the basis of all existing known information.

3.6.2 Profit taking

Surprising though it may seem, a company's share price will often fall when it announces good profits. This may happen because speculators buy before the profit announcement is actually made, and then sell their shares in order to take the profit.

3.6.3 Press recommendations

If the press tips a particular share for whatever reason, the price of that share is likely to rise. It is worth noting, however, that this may be caused by buying or simply by the market makers pushing up the price in anticipation of likely buying.

3.6.4 Market makers' and speculators' manoeuvres

Market makers depend on high levels of activity in terms of buying and selling in order to make their profits. Thus, in periods of little activity, they may well try to stimulate the market artificially by moving their quoted prices, and possibly sell shares that they do not own.

Speculators may also employ this technique of selling shares they do not own if they anticipate that the share price will fall. However, if the market makers get wind of this activity, they may push up their quoted prices in order to force speculators to buy the shares, which they need to deliver, at a high price. These speculators who have traded in anticipation of a fall in the share price are known as bears.

3.6.5 Interest rate and currency movements

An increase in interest rates may cause share prices to fall for two separate reasons:

- The available alternative investments in, say, government stocks will become more attractive.
- The higher cost of borrowing is likely to damp down economic activity in general, thus having an adverse effect on company profits.

The effect of movements in sterling can also be fairly complicated. Although a fall in sterling may lead to a rise in interest rates, with its associated problems, it may also improve a country's competitive position relative to other countries and may thus give rise to greater profit potential.

3.6.6 New share issues

If a company raises funds by issuing a large number of new shares, the share price may well fall, as at that time there may not be enough willing buyers to absorb the number of shares in issue.

3.6.7 Political and other sentiment

In the run-up to an election, the anticipated result will often have an effect on share prices.

It has been known in the past for share prices to be affected by the result of a cricket match and also by more logical factors such as economic statistics and the behaviour of other markets in the world.

3.6.8 Takeovers

Rumours of a corporate action such as a major takeover tend to push up the share price of the company concerned and also quite possibly of the market as a whole.

3.6.9 Automated trading

Automated forms of trading have sometimes been blamed for major movements in stock prices.

(a) **Program trading** has been defined by the NYSE as an order to buy or sell 15 or more stocks valued at over US$1 million total. In practice, this means that computers are used to enter program trades. Such trades are often made to take advantage of arbitrage opportunities, which seek to exploit small price differences between related financial instruments, such as index futures contracts and the stocks underlying them.

(b) **Algorithmic trading** is a form of trading in which orders are entered to an electronic trading platform based on an algorithm which sets criteria for making the trade based on

aspects such as timing, price and quantity. Generally, orders will be executed without human intervention. A special type of algorithmic trading is **high frequency trading** or 'flash' trading, which seeks to respond to information received electronically before other human traders are able to process the information and place trades. Studies have found that over half the equity trading volume on major exchanges results from algorithmic trading.

3.7 Market efficiency

You covered the EMH in previous studies and we summarise it here.

The **random walk hypothesis suggests** that future share price movements cannot be predicted from details of historical movements. On the other hand, share prices are observed to move when **new information** is received, and will fairly reflect that new information. Since new information must, by definition, be unpredictable and random (otherwise it would not be new information), it follows that share price movements will also be random.

From these observations, the **EMH** was developed which broadly stated that, at any time, share prices are fairly valued on the basis of all existing information.

If securities are priced efficiently, their prices reflect forecasts of expected benefits from future cash flows capitalised at appropriate discount rates. Of course, individuals can disagree, and it is this disagreement which results in transactions. The aggregation and resolution of expectations in the transaction process produces an unbiased valuation in an efficient market.

Broadly speaking, the evidence suggests that efficient markets can be divided into three forms that are classified according to the degree to which information is reflected in share prices and the availability of that information.

3.7.1 Weak form efficiency

The **weak** form of the EMH states that all information that can be discovered from past price movements has already been incorporated into the current share price. This implies that it should not be possible to predict future price movements from past price movements or to produce superior returns on the basis of past information, since any such information available from past prices has already been taken account of in the current price.

3.7.2 Semi-strong form efficiency

The **semi-strong** form of the EMH states that current share prices not only reflect the information referred to in the weak form, but also incorporate any information that has been published about a company (all publicly available information). For example, release of preliminary figures by the company constitutes new information and the share price will move to reflect this.

This level of market efficiency has been investigated through a number of **events studies** investigating the reaction of the market to the release of new information. Observations in major markets have shown that prices do respond to the release of new information and, traditionally, many people have suggested that share markets tend to exhibit this level of efficiency.

3.7.3 Strong-form efficiency

The **strong** form of the EMH is based on the premise that share prices fully reflect all information, whether publicly available or private.

If markets are strong-form efficient, then no investor will ever be able to make better than average returns, except through luck. It would be impossible for even the most corrupt investor to find out anything which the share price did not already reflect.

The evidence on strong-form efficiency is not conclusive, but the seemingly widespread

incidence of insider traders (until they are caught) suggests that knowledge of secret information does give the investor a way of predicting share price movements before they have happened. The strong form of market efficiency cannot, therefore, hold.

3.7.4 Alternative theories

The global crisis of 2007–09 has brought renewed discussion of alternatives to the EMH, which include the following.

Behavioural finance. This area of study postulates that markets are driven by traders who are not rational in the classical sense. Instead, they display behavioural and cognitive shortcomings which psychologists have identified. For example, market participants show a tendency to avoid losses at all costs. They also tend to act like a herd, which will make occasional extreme market crashes more probable.

Adaptive evolution. Markets are complex 'ecosystems'. They may be in a stable equilibrium for extended periods of time. Occasionally, a disruptive event may upset the equilibrium, and new financial 'species' may emerge while others may become 'extinct'.

Chaos theory. It is hypothesised that markets do not behave in a random fashion. Market movements may be modelled in a similar way to the modelling of 'chaotic' systems such as wind turbulence and weather patterns.

4 Fixed interest securities and bonds



Section overview

Some of this section is revision material, but it also includes some types of bonds that you may not have come across before.

The term **bonds** describes various forms of long-term debt a company may issue. Bonds come in various forms, including redeemable, irredeemable, floating rate, zero coupon and convertible**.** The terminology should be familiar to you from your earlier studies but it is worthwhile to recap on it here.



Definitions

Fixed interest securities: are issued by companies or governments to borrow money from investors. Securities issued by the UK Government are known as 'gilts', while securities issued by companies are known as corporate bonds.

Face (or par or nominal) value: The amount of money the bondholder will receive when the bond matures (provided it is not redeemable at a premium or a discount). This is not the market price of the bond. If the market price is above par value, the bond is said to be trading at a premium; if price is below par value, the bond is trading at a discount.

Coupon rate: The amount the bondholder receives as interest payments based on the par value.

Maturity: The date at which the principal will be repaid. Maturities can range from one day to as long as 30 years (although it has been known for 100-year bonds to be issued).

Issuer: The issuer's stability is the bondholder's main assurance for getting repaid. For example, the UK Government is much more secure than any company. Hence government-issued bonds are known as risk-free assets and will have lower returns than company-issued bonds.

Fixed charge: The security given on the bond relates to a specific asset or group of assets, typically land and buildings. The company will be unable to dispose of the asset without providing a substitute asset or without the lender's consent.

Floating charge: The charge is on certain assets of the company and the lender's security in the event of a default of payment is whatever assets of the appropriate class the company then owns. The company would be able to dispose of the assets as it chose until a default took place.

4.1 Bonds

Unlike shares, debt is often issued at par, ie, with £100 payable per £100 nominal value. Where the coupon rate is fixed at the time of issue, it will be set according to prevailing market conditions and the credit rating of the company issuing the debt. Subsequent changes in market (and company) conditions that change the yield that investors require on the bond will cause the market value of the bond to fluctuate, although the coupon will stay at the fixed percentage of the nominal value (the fixed coupon rate).

4.2 Terms of a loan agreement

Covenants within the loan agreement are conditions attached to a bond issue. Bond covenants are legally enforceable rules that borrowers and lenders agree upon at the time of a new bond issue. Covenants enumerate what issuers are required to do (affirmative covenants) and what they are prohibited from doing (negative covenants).

For example, an affirmative covenant may require an issuer to maintain enough liquid assets to cover the principal of the bond. More commonly, a positive covenant requires the issuer to have a certain amount of insurance or submit to periodic audits.

A negative covenant may prevent an issuer from issuing more debt until the bond matures. More commonly, a restrictive covenant limits the dividends an issuer may pay to shareholders so as to reduce the risk to the bond.

A cross-default clause is included in most loan agreements and states that if the borrower defaults on any one of its loans, then this will constitute a default of this borrowing as well.

4.2.1 Typical loan covenants

Typical affirmative covenants:	Typical negative covenants:
The borrower agrees to maintain various kinds of insurance, such as property, liability and 'key man'	No pledge of any assets if doing so gives the lender less security. Prevents the borrower from issuing any debt in the future which would jeopardise a current claim
Quarterly or monthly financial statements must be submitted to the bank or lender's representatives. These may need to be audited if the loan is large.	No changes in management or the nature of business without the lender's permission
All shareholder loans to the business must be subordinate to the bank's loan	No distribution of profits without prior lender approval

The borrower agrees to maintain liquidity and performance ratios such as maximum gearing ratio, minimum interest cover and minimum business net worth. These financial covenants may state an express limit on the overall level of borrowing that can be undertaken.	No further loans from other sources to the company without lender approval or without meeting certain conditions. Such a condition is normally linked to key ratios such as interest cover
Any lender is granted the same level of security as given to any new lenders ('pari passu')	No sale of equipment without prior lender approval. This ensures that the company cannot sell assets unless it receives a fair market value and unless a specified percentage of the consideration is received in cash. A further extension to this clause is often that the monies received are used to: - repay senior debt - reinvest in the business within one year - repay the debt

4.2.2 Events of default

A breach of the covenants or the failure to pay the coupon or principal on the due dates are events of default. Within the agreement there may be allowance for 'grace periods', during which time the borrower is able to remedy the breach. If the borrower remains in default, the loan is immediately repayable.

In principle, an event of default gives the lender/bondholders the right to demand immediate repayment of the debt. In practice, a breach of covenant will lead to discussions between the borrower and the bondholders' representatives about appropriate measures to be taken.

Most bonds are constituted in such a way that there is a trustee who is responsible for ensuring the bondholder's interests are protected. Where there is no trustee, the bondholder is obliged to seek redress directly from the company.

4.3 Bond coupons

4.3.1 Predetermined coupons

The majority of bonds have a fixed coupon rate. On these bonds, the gross annual coupon (ie, the amount due to be paid in a one-year period, irrespective of the frequency of payment) is specified as a percentage of the nominal value of the bond. Sub-classes here include:

(a) **Straight/fixed coupon bonds** – where the coupon is at a set level for the entire life of the bond.

(b) **Stepped coupon bonds** – where the coupon increases in steps to pre-specified amounts as the bond moves through its life.

(c) **Zero-coupon bonds** – bonds that carry no coupon and simply redeem at face value at maturity. Investors obtain a yield on these bonds because the bonds are issued at a discount to face value, and continue to trade at less than face value in the entire period up to maturity and redemption.

4.3.2 Variable coupons

This category includes:

(a) **Floating rate bonds or notes** – where the coupon varies as the yield on a benchmark interest rate varies. For example, for sterling floating rate bonds that pay interest every six months, the benchmark rate is likely to be the six-month sterling SONIA (Sterling Overnight Index Average) or SOFR (Secured Overnight Financing Rate). Loan agreements

specify the interest rate (called reference or index rate to be used) and also the quoted margin, the rate above the reference rate that must be paid. The quoted margin will reflect the borrower's credit rating and size of issue, along with market conditions. Arrangements may include a floor (the rate below which the rate paid cannot fall) and a ceiling or cap (the rate above which the rate paid cannot rise).

(b) **Index-linked bonds** – where the coupon and redemption proceeds figures get scaled for the effects of inflation.

4.3.3 Coupon frequency

The frequency of the payment of the coupons is predetermined before issue, normally following the local market conventions. As a result, all investors will (or should) be aware of these dates.

Conventions regarding the frequency of payment differ between bond markets. Some markets have a convention of paying semi-annual coupons, as is the case in the UK and the US, whereas other markets, in particular the Eurobond market, France and Germany, pay coupons on an annual basis.

4.3.4 Zero-coupon bonds

Zero-coupon bonds are bonds that are issued at a discount to their redemption value, but no interest is paid on them. The investor gains from the difference between the issue price and the redemption value. There is an implied interest rate in the amount of discount at which the bonds are issued (or subsequently resold on the market).

(a) The advantage for borrowers is that zero-coupon bonds can be used to raise cash immediately. There is no cash repayment until redemption date. The cost of redemption is known at the time of issue, and so the borrower can plan to have funds available to redeem the bonds at maturity.

(b) The advantage for lenders is restricted, unless the rate of discount on the bonds offers a high yield. The only way of obtaining cash from the bonds before maturity is to sell them. Their market value will depend on the remaining term to maturity and current market interest rates.

4.4 Debentures



Definition

Debenture: A written acknowledgement of a debt by a company, usually given under its seal and normally containing provisions as to payment of interest and the terms of repayment of principal. A debenture is usually unsecured, which means that the investors rely on the creditworthiness of the borrower for repayment. In comparison, the term 'bond' is often used to mean a bond for which security has been given by the borrower.

4.5 Eurocurrency and Eurobonds



Definitions

Eurocurrency: Currency which is held by individuals and institutions outside the country of issue of that currency.

Eurodollars: US dollars deposited with, or borrowed from, a bank outside the US.

Eurobond: A bond sold outside the jurisdiction of the country in whose currency the bond is denominated.

The term 'euro-' is still occasionally used, but the terms 'currency deposits', 'currency loans' and 'international bonds' are now more common usage.

4.5.1 Eurocurrency

A UK company might borrow money from a bank or from the investing public in sterling. However, it might also borrow in a foreign currency, especially if it trades abroad, or if it already has assets or liabilities abroad denominated in a foreign currency. When a company borrows in a foreign currency, the loan is known as a eurocurrency loan or more simply as a currency loan. The eurocurrency markets are money markets. They involve the depositing of funds with a bank outside the country of the currency in which the funds are denominated and relending these funds for a fairly short term, typically three months, normally at a floating rate of interest.

4.5.2 Eurobonds

Eurobonds or international bonds are long-term loans raised by international companies or other institutions and sold to investors in several countries at the same time. Most Eurobonds are currently denominated in US dollars and, to a lesser extent, euros and yen.

In recent years, a strong market has built up which allows very large companies to borrow in this way, long term or short term. The market is not subject to national regulations but there is a self-regulatory association of bond dealers, the International Capital Market Association (ICMA).

Advantages of international bonds

(a) Most international bonds are 'bearer instruments', which means that there is no bond ownership register and the owner **does not have to declare his identity**.
(b) Interest is **paid gross** and this has meant that in the past Eurobonds have been used by investors to avoid payments of tax.
(c) International bonds can be used to create a liability in a foreign currency to **match** against a foreign currency asset.
(d) They may be **cheaper** than a foreign currency bank loan because they can be sold on by the investor, who will therefore accept a lower yield in return for this greater liquidity.
(e) They are typically issued by companies with **excellent credit ratings** and are normally unsecured, which makes it easier for companies to raise debt finance in the future. Companies with lower credit ratings that want to raise funds in the bond markets may do so through a national 'junk bond' market. There is a large junk bond market in the US.
(f) International bond issues are **not normally advertised** to the general investing public because they are placed with institutional investors and this reduces issue costs.

Disadvantages of international bonds

(a) Like any form of debt finance there will be issue costs to consider (perhaps about 2% of funds raised) and there may also be problems if gearing levels are too high.
(b) A borrower contemplating an international bond issue must consider the foreign exchange risk of a long-term foreign currency debt. If the money is to be used to purchase assets which will earn revenue in a currency different to that of the bond issue, the borrower will run the risk of exchange losses if the currency of the loan strengthens against the currency of the revenues out of which the bond (and interest) must be repaid.

4.6 Deep discount bonds



Definition

Deep discount bond: A bond offered at a large discount on the face value of the debt so that a significant proportion of the return to the investor comes by way of a capital gain on redemption, rather than through interest payment.

Deep discount bonds will be redeemable at par (or above par) when they eventually mature. For example, a company might issue £1,000,000 of bonds in 20X4 at a price of £50 per £100, and redeemable at par in the year 20Y9. For a company with specific cash flow requirements, the low servicing costs during the currency of the bond may be an attraction, coupled with a high cost of redemption at maturity.

Investors might be attracted by the large capital gain offered by the bonds, which is the difference between the issue price and the redemption value. However, deep discount bonds will carry a much lower rate of interest than other types of bonds. The only tax advantage is that the gain gets taxed in one lump on maturity or sale, not as amounts of interest each year.

4.7 Redeemable and irredeemable bonds

Bonds are usually redeemable. They are issued for a term of 10 years or more, and sometimes for up to 25 or 30 years. At the end of this period, they will 'mature' and become redeemable (at par or possibly at a value above par).



Definition

Redemption: Repayment of the principal amount (for example a bond) at the date of maturity.

Some redeemable bonds have an earliest and a latest redemption date. For example, 12% Loan Notes 20X9/Y1 are redeemable at any time between the earliest specified date (in 20X9) and the latest date (in 20Y1). The issuing company can choose the date. The decision by a company on when to redeem a debt will depend on how much cash is available to the company to repay the debt, and on the nominal rate of interest on the debt.

Some bonds do not have a redemption date, and are 'irredeemable' or 'undated'. Undated bonds might be redeemed by a company that wishes to pay off the debt, but there is no obligation on the company to do so.

Corporate bonds can have a variety of **redemption** terms.

4.7.1 Bullets

Many bonds issued are bullet issues, with a single redemption date when the full amount borrowed becomes repayable.

In practice, a company that has to redeem a bond in a bullet payment may seek refinancing, and may issue a new bond to raise money to redeem the maturing bond.

4.7.2 Sinking funds

The sinking fund or sinker is a process whereby a proportion of the bonds in issue are redeemed each year. The bonds to be redeemed in each year are selected by the process of 'drawing' the serial numbers. The serial numbers of the bonds drawn in this way are then published and the holders submit the bonds to the paying agent for redemption at par. The final repayment is normally larger than the others and is referred to as the balance or balloon repayment.

Sinking funds tend to come into operation towards the end of the bond's life and rarely start to redeem from the first coupon date.

4.7.3 Purchase fund

A purchase fund buys back the bonds in the secondary market and not at par. The obligation to repay is triggered by a condition specified in the offer document, normally the bond trading below par.

4.7.4 Serial notes

A serial note is one where a proportion of the capital is repaid each year together with the interest.

4.7.5 Optional redemption

The option to redeem a bond can be given to either side of the deal. A **call right** would give the issuer the right to seek an earlier redemption. An example would be the double-dated gilts in the UK market, where the Government has the right to redeem from the earlier of the two dates, but must redeem by the later date.

4.7.6 Callable bonds

Callable bonds are where the issuer has the right to redeem the bonds at an agreed price prior to its maturity. This price may be above the normal redemption price and the extra price paid is referred to as the call premium.

The call provision is valuable to the issuer, but is a disadvantage to the investor, since the issuer will only exercise the call if it suits the issuer to do so. As a result, the price at which a callable bond can be issued will be lower than for a comparable straight bond, and the interest rate it will need to pay will consequently be higher.

A call provision will reduce the expected time to maturity of the bond, since there is a possibility that the bond will be retired early as a result of the call provision being exercised.

Call provisions will be exercised when the issuer can refinance the issue at a cheaper cost due to interest rates having fallen. For example, if a bond were issued when interest rates were 15% and interest rates have now fallen to 5%, the issuer could issue a new bond at the currently low rate and use the proceeds to call back the higher coupon bond.

4.7.7 Puttable bonds

This is where the investor has a put option on the bond, giving him the right to sell the bond back to the company at a specified price (the put price). The put price is typically around par, given that the bond was issued at par.

The benefit to investors is that if interest rates rise after the bond is issued, they can sell the bond at a fixed price and reinvest the proceeds at a higher interest rate. As a result of the benefit to investors, puttable bonds are issued at higher prices or lower coupons than comparable non-puttable bonds.

4.8 Convertible bonds



Definition

Convertible debt: A liability that gives the holder the right to convert into another instrument, normally ordinary shares, at a predetermined price/rate and time.

Conversion terms often vary over time. For example, the conversion terms of convertible bonds might be that on 1 April 20X6, £2 of bonds can be converted into one ordinary share, whereas on 1 April 20X7 the conversion price will be £2.20 of bonds for one ordinary share. Once converted, convertible securities cannot be converted back into the original fixed return security.

The current market value of ordinary shares into which a unit of bonds may be converted is known as the conversion value. The conversion value will be below the value of the bonds at the date of issue, but will be expected to increase as the date for conversion approaches on the assumption that a company's shares ought to increase in market value over time.

Conversion value = Conversion ratio × Market price per ordinary share Conversion premium = Current market value – Current conversion value

Interactive question 1: Convertibles

The 10% convertible bonds of Starchwhite are quoted at £142 per £100 nominal. The earliest date for conversion is in four years' time, at the rate of 30 ordinary shares per £100 nominal. The share price is currently £4.15. Annual interest on the bonds has just been paid.

Requirements

1.1 Calculate the current conversion value.

1.2 Calculate the conversion premium and comment on its meaning.

See **Answer** at the end of this chapter.

Most companies issuing convertible bonds expect them to be converted. They view the notes as delayed equity. They are often used either because the company's ordinary share price is considered to be particularly depressed at the time of issue or because the issue of equity shares would result in an immediate and significant drop in earnings per share. There is no certainty, however, that the security holders will exercise their option to convert. Therefore the bonds may run their full term and need to be redeemed.

4.8.1 Price, coupon rate and premium

A company will aim to issue bonds with the greatest possible conversion premium so that, for the amount of capital raised, it will on conversion have to issue the lowest number of new ordinary shares. The premium that will be accepted by potential investors will depend on the company's growth potential and so on prospects for a sizeable increase in the share price.

Convertible bonds issued at par normally have a lower coupon rate of interest than straight debt. This lower yield is the price the investor has to pay for the conversion rights. It is, of course, also one of the reasons why the issue of convertible bonds is attractive to a company, particularly one with tight cash flows around the time of issue, but an easier situation when the notes are due to be converted.

When convertible bonds are traded on a stock market, their minimum market price or floor value will be the price of straight bonds with the same coupon rate of interest. If the market value falls to this minimum, it follows that the market attaches no value to the conversion rights.

The actual market price of convertible bonds will depend on:

- the price of straight debt
- the current conversion value
- the length of time before conversion may take place
- the market's expectation as to future equity returns and the risk associated with these returns

4.8.2 Mandatory convertibles

Sometimes the bondholder will be required to convert the nominal value into ordinary shares of the company at a set redemption date. These are known as mandatory convertibles.

The appeal for the investor of 'mandatories' is that the company takes the risk of the share price volatility in the period between issuance and conversion. Mandatories are often issued in conjunction with equity in an IPO to attract investors who would not otherwise be interested in the IPO.

4.8.3 Advantages and disadvantages of convertible bonds

Advantages

Compared to either a bond or share issue, convertibles offer the following advantages to the issuer:

- no immediate dilution of the current shareholders
- lower cost than normal bonds due to a lower coupon
- less dilution of earnings per share than either a normal share or bond issue as a consequence
- suitable for when assets are not available to secure straight finance
- suitable for finance projects with long payback periods

Disadvantages

The disadvantage to the issuer is that if the company fails to perform, it is obliged to make the coupon payments, and ultimately redeem the bond for cash if the holder chooses not to convert. Therefore, the firm cannot be sure that it is issuing deferred share capital when it issues a convertible.

4.9 Exchangeable bonds



Definition

Exchangeable bonds: Bonds that are convertible into the ordinary shares of a subsidiary or associate company of the issuer. This is therefore a different company to the issuer of the bond.

The use of exchangeable bonds can be to provide a way for a group to divest an unwanted investment. There is no dilution of control for the shareholders of the issuer.

4.10 Hybrid bonds



Definition

Hybrid bond: A security that combines characteristics of both debt and equity.

Fixed rate hybrids pay a predictable return or dividend until a certain date, at which point the holder has a number of options, possibly including converting the securities into the underlying share. The most common examples of hybrids are convertible bonds and convertible preference shares.

Though hybrids pay a fixed return like bonds, the payments are voluntary. A missed payment does not constitute a default and payments do not normally accumulate if deferred. Like equities, they tend to be irredeemable or very long dated and they rank behind other debt if the company becomes insolvent.

The motives behind corporate hybrid issues include the following:

(a) Consolidating credit ratings by buying back senior debt.

(b) Refinancing pension deficits by using the proceeds of the issue to increase the pension fund. This reduces the deficit and, if the hybrid is assessed as partly equity, replacing the pension liability debt will also have a positive impact on the company's credit rating.

(c) Financing corporate activity such as mergers and acquisitions, share buybacks and leverage buyouts, providing non-dilutive capital and flexibility.

For bond investors, hybrids offer higher yields than conventional debt but are not without risk. For equity investors, hybrids provide a tax-efficient, long-term source of finance to the company without either the diluting effect of an equity issue or the risk of extra gearing from a debt issue where the coupons cannot be deferred if circumstances dictate. Investors in the equity still enjoy the benefits associated with any improvement in company performance.

4.11 Warrants

Definition

Warrant: A right given by a company to an investor, allowing them to subscribe for new shares at a future date at a fixed, predetermined price (the exercise price).

Warrants are usually issued as part of a package with unsecured bonds. An investor who buys bonds will also acquire a certain number of warrants. The purpose of warrants is to make the bonds more attractive.

Once issued, warrants are detachable from the bonds and can be sold and bought separately before or during the 'exercise period' (the period during which the right to use the warrants to subscribe for shares is allowed). The market value of warrants will depend on expectations of actual share prices in the future.

4.11.1 Advantages of warrants

(a) Warrants themselves **do not involve the payment of any interest or dividends**. Furthermore, when they are initially attached to bonds, the interest rate on the bonds will be lower than for a comparable straight debt.

(b) Warrants **make a bond issue more attractive** and may make an issue of unsecured bonds possible where adequate security is lacking.

(c) Warrants provide a **means of generating additional equity funds** in the future without any immediate dilution in earnings per share. The cost will be the right that warrants holders have acquired to buy at the possibly reduced exercise price.

4.11.2 Disadvantages of warrants

(a) When exercised, they will result in the **dilution of share capital**.

(b) Warrants may be exercised when a business **does not need additional capital**.

(c) The company has **less control over the exercise of warrants** than it does over the exercise of share capital.

Professional skills focus: Assimilating and using information

An organisation in the exam scenario may have a variety of financial instruments available to it. Your task is to understand and review the business context in order to recommend the most appropriate financial instrument to use. You should not necessarily expect to have to choose between debt and equity. It may be that all the choices available are forms of debt, and you may have to consider corporate reporting implications. The choice may also be influenced by the financial markets that the business is able to use. It is therefore important that you can identify key business and financial issues from the scenario.

4.12 Commercial paper

Definition

Commercial paper (CP): Short-term unsecured corporate debt with maturity up to 270 days (US dollar CP) or 364 days (euro CP). The typical term of this debt is about 30 days.

As CP is unsecured debt, it can only be issued by large organisations with good credit ratings, normally to fund short-term expenditure on operating expenses or current assets. The debt is issued at a discount that reflects the prevailing interest rates, but the rates on CP are typically lower than bank rates.

CP is issued within a longer-term programme, which is managed on an issuer's behalf by a bank. For example, within a five-year CP programme, a company may be able to issue CP up to a maximum amount at any time, with each separate CP issue having a maturity of 30 days. A bank managing the programme arranges for each issue, including the sale of the paper to investors. Typically investors hold the paper until maturity and there is no secondary market.

CP is a cheaper alternative to bank credit, although the small difference in interest rates means that the saving on CP is only significant if large sums are being raised. However, many businesses still maintain bank lines of credit even if they use CP. The wide maturity gives greater flexibility, security does not have to be given and investors can trade their CP, although the market is less liquid than for bonds. However, issues are controlled, and banks that issue ordinary CP see a reduction in their credit limits.

A sub-sector of the CP market – the asset-backed CP market – is where the loans are backed by assets such as mortgages and credit card debt. The concern is that asset-backed CP could be held off balance sheet in special investment vehicles that use the funds raised to buy longer-term assets such as mortgage-based securities. As these vehicles are off balance sheet, their activities do not affect the banks' ability to lend money for other reasons.

4.13 Medium Term Notes

Medium Term Notes are not strictly a type of security. Instead, the term describes a facility enabling the issuer to issue a range of stock from one global facility. In the US, the facility can cover short-term CP out to 30-year debt.

The issuer sets up a shelf programme. When the programme is initially set up, relevant documentation is prepared to enable the issuer to issue a wide variety of different instruments in terms of:

- maturity (eg, from 1 week to 10 years)
- currency (sterling, US dollars, euro)
- coupon (coupon-paying or zero coupon)
- instrument (CP, long-term bonds)

The implication of this is that the costs of setting up the programme are relatively high compared to the cost of a large international bond issue. However, the benefit is that the programme offers greater flexibility and lower cost to the issuer in the long run. As a result, it is more suitable for issuers who need funding flexibility over a period of time. In contrast, a large issuer might find a oneoff large international bond issue more suitable.

4.14 Repos

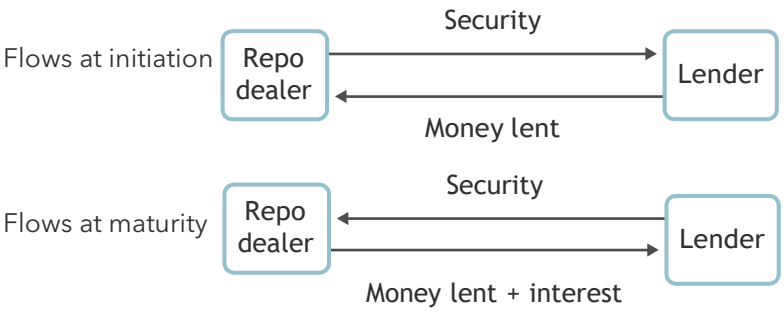


Definition

Repurchase agreement (repo): An agreement between two counterparties under which one counterparty agrees to sell a quantity of financial instruments to the other on an agreed date for an agreed price, and simultaneously agrees to buy back the instruments from the counterparty at a later date for an agreed higher price.

A repo is a loan secured by a marketable instrument, usually a treasury bill or a bond. The typical term is 1–180 days. A repo is an attractive instrument because it can accommodate a wide spectrum of maturities. The flows in a repo are shown in the following diagram.

Figure 13.2: Flows in a repo transaction



A repo is in effect a cash transaction combined with a forward contract. Repos can be:

- overnight (one-day maturity transactions)
- term (specified end date)
- open (no end date)

In effect, a repo is a short-term loan secured by a quantity of financial instruments, such as short- dated bonds and money market instruments. By selling securities and agreeing to buy them back at a higher price, an organisation is borrowing money and repaying it with interest at maturity.

A reverse repurchase agreement (reverse repo) is an agreement for the purchase of financial instruments with the simultaneous agreement to resell the instruments at an agreed future date and agreed price. In a reverse repo, the dealer purchases the securities initially and then sells them back at maturity. Because the two parties in a repo agreement act as a buyer and a seller of the security, a repo to one party is a reverse repo to the other.

4.14.1 Use of repos

Repos give buyers the chance to invest cash for a limited period of time, on a transaction where they receive collateral as security. Market liquidity and rates are generally good. Traders use repos to cover their positions and benefit from lower funding costs. The repo market is an important component of the overnight market.



Worked example: Repurchase agreements

Green Bank sells a bond carried on its statement of financial position to Red Bank for £1,000. Green Bank commits to buy back the bond in three months for £1,025.

Requirements

1 Has Green Bank transferred substantially all the risks and rewards of ownership?

2 Would your answer be different if Green Bank commits to buyback in three months at market price?

Solution

1 IFRS 9 rules on these situations:

This is a sale and repurchase transaction where the repurchase price is a fixed price or at the sale price plus a lender's return.
Green Bank has not transferred substantially all the risks and rewards of ownership and hence the bond is not derecognised.
Green Bank will recognise a loan liability of £1,000 and interest expense of £25 to reflect the collateralised borrowing.

2 This is a sale of a financial asset together with commitment to repurchase the financial asset at its fair value at the time of repurchase. Because any repurchase is at the then fair value, all risks and rewards of ownership are with the buying party and hence Green Bank will derecognise the bond and recognise at its fair value the liability implicit in its commitment to repurchase.



Interactive question 2: Derecognition of financial assets

2.1 Excel Bank sells an investment in shares, but retains a call option to repurchase those shares.

Requirement

Discuss if the investment in the shares should be derecognised in each of the following instances:

- The call option is deeply out of the money.
- The call option is deeply in the money.
- The call option is neither deeply in the money nor deeply out of the money.

2.2 Access Bank sells a bond and enters into a 'total return swap' with the buyer. Under the 'total return swap' arrangement, the buyer passes on interest payment cash flows along with increases in value of the bond to the seller. In return the seller compensates the buyer through a fixed payment and any decreases in value.

Requirement

Discuss whether the investment in the bond should be derecognised.

See **Answer** at the end of this chapter.

4.15 Islamic bonds

Islamic bonds or Sukuk are bond issues that satisfy Islamic principles and have gained Sharia'a approval. Western-style bonds do not satisfy certain Islamic principles, in particular the prohibition of speculation and the payment of interest or any charge simply related to time.

Though Islam frowns on speculation, it does approve of partnerships and risk sharing. Sukuk are based on partnership risk-sharing ideas. When investing in such bonds, the Sukuk holder is effectively becoming a part owner of the asset being financed, sharing in profits it generates or losses it incurs. The arrangement is achieved through a special purpose vehicle (SPV) which buys the asset to be financed with the funds raised and acts on behalf of all the Sukuk investors.

In any partnership there are at least two parties involved who may contribute capital, skill and effort, or both. With Sukuk, the parties are:

- Rab al Mal – provides capital
- Mudarib – provides skill and effort

The main types of issue are:

(a) **Mudaraba Sukuk** – This is the primary type of Sukuk contract where the SPV is the Rab al Mal and the company needing finance is the Mudarib. In such an arrangement, all profits are shared as defined in the Sukuk contract, and all losses are absorbed by the Rab al Mal. As part of the contract it is normal for the Mudarib to contract to buy the financed asset at maturity in order to redeem the Sukuk holders. The result of this arrangement can be very similar to that of Western asset-backed bonds depending on how the profit-sharing arrangements are structured.

(b) **Musharaka Sukuk** – Where all parties contribute both capital and effort. All profits are shared as defined in the Sukuk contract, and all losses are absorbed in proportion to capital introduced. This is very dissimilar to Western bonds, since Western bondholders would not expect to contribute anything other than capital.

(c) **Salam Sukuk** – Has a structure similar to a short-term zero-coupon bond which is achieved through a combination of spot and deferred payment sales.

(d) **Istisn'a Sukuk** – Used in construction contracts where the profits are generated from rents received and the ultimate sale of the property constructed.

(e) **Ijara Sukuk** – Has a structure similar to a leasing arrangement. This is the most common form of Sukuk structure.

Context example: Sukuk

In 2014, the UK became the first country outside the Islamic world to issue sovereign Sukuk (government bonds). The £200 million issue matures in 2019. The profit rate was 2.036%, similar to the yield available on UK government bonds (gilts) with the same maturity. The Sukuk has an ijara structure, and the profits for investors will be obtained from rentals on three government properties. The Sukuk issue was intended to demonstrate that the UK can be a successful base for Islamic finance.

Demand for the bond in 2014 was fierce from investors in the UK, Middle East and Asia. Later that year Luxembourg also issued a sovereign sukuk – the first AAA-rated government to issue a euro- based Islamic bond.

Hopes that the international sukuk market would enjoy exponential growth have not quite met with expectations. One reason is that the oil market has been depressed, and this has weighed on the wealth and exchange rates of many Muslim countries.

In a low interest rate environment, Western issuers are less interested by the diversification benefits of sukuk bonds and more worried about their lack of liquidity. The market is also being hampered by a lack of standardisation, with fundamental disagreement among Islamic scholars as to what constitutes an Islamic-compliant financial product.

Illustrating this problem is the ongoing Dana Gas case. The case rests on whether or not the United Arab Emirates-based energy company Dana Gas can, as it claims, restructure some $700 million of its debt based on the fact the bonds it originally sold are no longer, according to some scholars, Islamic- compliant. Its investors include Goldman Sachs and BlackRock.

The global sukuk market totals around $400 billion according to a range of estimates, and with 65 products worth a total of $48 billion listed on the London Stock Exchange, the UK is a major centre for Islamic bonds. Organisations such as the Islamic Financial Services Board should help the drive towards standardisation.

The Bank of England has recently announced plans for a fund-based, sharia-compliant deposit facility, and the UK Government plans to reissue its sukuk once the bond matures in 2019.

Source: https://www.telegraph.co.uk/business/2017/10/15/can-uk-make-sukuk-islam-compliant- bond/ [Accessed 13 August 2018]

4.16 Green Bonds

Definition

Green bonds: Green bonds are any type of bond instrument where the proceeds will be exclusively applied to finance or re-finance, in part or in full, new and/or existing eligible green

projects, and those which are aligned with the four core components of the Green Bond Principles (ICMA, 2018).

4.16.1 Green Bond Principles

The Green Bond Principles were issued by the International Capital Market Association (ICMA) in 2015 and were most recently updated in 2021. They are voluntary guidelines for issuers of green bonds. The four components of the Green Bond Principles are:

(a) **Use of Proceeds**: it should be stated in the legal documentation for bonds that the proceeds will be used for projects that should provide clear environmental benefits

(b) **Process for Project Evaluation and Selection**: there should be defined processes in place to determine which projects are eligible

(c) **Management of proceeds**: the proceeds of the issue of the bonds should be kept in a separate account or tracked by the issuer

(d) **Reporting**: the issuer should make and keep readily available information on the use of the proceeds

Environmental objectives that the GBP recognises as being eligible are **climate change mitigation**, **climate change adaptation**, **natural resource conservation**, **biodiversity conservation**, and **pollution prevention and control**.

The ICMA is also responsible for the Social Bond Principles, Sustainability Bond Guidelines and Sustainability-Linked Bond Principles.

4.16.2 Market for Green Bonds

The market for Green Bonds is growing exponentially according to research from Climate Bonds Market Intelligence. This organisation estimates that during the year 2021, more than $517 billion of green bonds were issued globally. This is up from around $4.2 billion in the year 2012.

In September 2021, the UK government issued its first Green Gilt raising £10 billion to finance green government projects, such as zero emission buses, offshore wind farms and schemes to decarbonise homes and buildings. The Gilts are issued in accordance with the UK government's Green Bond Framework, which is aligned with the four core components of the ICMA Green Bond Principles.

4.17 Sustainability linked bonds and loans

Investor appetite for ESG-linked instruments is growing. The terminology is vast, but this could include sustainability-linked bonds, 'green' bonds relating to climate impacts or 'social impact' bonds relating to investment in community projects. The principle is that the terms of the instrument are linked to meeting certain sustainability or ESG-led targets.

Sustainability Linked Bonds are not the same thing as Green Bonds. They are bonds for any purpose (whether 'green' or not), but with an in-built pricing mechanism, meaning that the bond is cheaper if the borrower achieves certain sustainable or ESG (environmental, social and governance) related targets. The use of Sustainability Linked Bonds and loans is growing rapidly.



Context example: Sustainability linked loans – Britvic

Britvic will benefit from a lower interest rate on its £400 million loan facility if it delivers on the following ESG targets by 2025:

- *50% of all plastic bottles to be recycled.*
- *Halve carbon emissions from 2017.*
- *Move 75% of drinks to low or no sugar.*

The agreement with their bank will drive real behavioural change amongst Britvic teams making commercial decisions.

4.18 Social Bonds

To qualify as a social bond the proceeds must finance or refinance social projects or activities that achieve positive social outcomes and/or address a social issue. In many cases social projects are aimed at target populations such as those living below the poverty line, marginalised communities, migrants, unemployed, sexual and gender minorities and people with disabilities.

Recently, a new type of social bond has emerged in the form of COVID-related bonds. The proceeds from these bonds are specifically used to help mitigate COVID-19-related social issues. Examples of project categories eligible for social bonds include:

- *Food security and sustainable food systems (eg, food banks)*
- *Affordable housing*
- *Access to essential services*

5 Bond markets



Section overview

Companies wishing to issue bonds have a choice. Large multinational firms with good credit ratings are able to access the international bond market. Smaller companies will have to use the less active domestic corporate bond market.

5.1 Sterling domestic market

The sterling domestic bond market is for UK companies and non-UK companies wishing to issue sterling denominated bonds in the UK market, for sale predominantly to UK investors. The UK domestic market progressively declined in importance since the 1970s.

However, as government bond markets have declined, corporate issuance has risen. In the UK, the size of the corporate bond market is now greater than the size of the government debt market. There are a number of reasons for this change:

(a) Banks have come under increasing regulation and consequently have been less able to lend from their own balance sheets. It therefore makes sense for them to arrange debt issuance.

(b) Companies have become increasingly concerned about the efficiency of their capital structure.

(c) Mergers and acquisition activity is further fuelling this need for new debt capital.

(d) As government bond markets contract, the investment opportunities shrink. Investors are therefore compelled to take an interest in corporate debt.

(e) Within the Euroland bloc, the advent of the single currency has removed an element of risk (forex risk arising from cross-border investment) and therefore reduced reward. Consequently, investors are seeking extra reward by moving down the credit curve.

(f) Individual Savings Accounts allow the creation of tax-free collective investment vehicles offering the enhanced returns that bonds offer over bank accounts without the risks of equity.

These factors have all come together in order to stimulate an increase in issuance.

The secondary market in corporate bonds involves the buying and selling of bonds after the initial offering. The corporate bond secondary market is almost entirely an OTC market.

Trading on unregulated OTC markets carries risks. Liquidity may be low. There may be a price disadvantage due to unfavourable price spreads. Brokers' commission must be taken into account. Consequently, the market is not particularly structured, with relatively little involvement of market makers.

In 2010, the Order Book for Retail Bonds (ORB) was launched. Rather than constituting a separate market, ORB is a trading platform for gilts and corporate bonds listed on the LSE.

5.1.1 Interbank rates

Key reference interest rates are produced from this primarily interbank market. The key UK reference rate is Sterling Overnight Index Average (SONIA) which is based on actual transactions greater than or equal to £25 million in value. SONIA is therefore set in arrears because it is based on actual transactions. SONIA reflects the average of the rates at which interest is paid on sterling short-term wholesale funds in circumstances where credit, liquidity and other risks are minimal. SONIA is used in a variety of ways eg, to calculate the interest paid on swap transactions and sterling FRNs. SONIA is used to value approximately £30 trillion of assets each year.

SONIA is a risk-free rate whereas LIBOR (phased out in the UK at the end of 2021) reflected a credit risk premium for interbank lending. SONIA is an overnight rate, based on actual market rates and reset on a daily basis in arrears. LIBOR included forward-looking term rates set at the start of a period. LIBOR also embedded a liquidity premium for longer dated rates.

There is industry discussion about creating a forward-looking 'term SONIA' rate. At the time of writing (May 2021), the Bank of England plans to publish overnight index swap (OIS) rate yield curves to longer maturities as soon as operationally possible.

LIBOR reform outside the UK

LIBOR in British pounds, euro, Swiss francs and Japanese yen plus 1-week and 2-month US dollar will cease in December 2021. Other major US dollar LIBOR tenors (overnight, 1-month, 3-month, 6- month and 12-month) will cease in mid-2023. US regulators are encouraging no new US dollar LIBOR contracts to be initiated after 31 December 2021, while allowing most legacy contracts to mature before US dollar LIBOR ceases.

In the US, the Secured Overnight Funding Rate (SOFR) is used from 2022. SOFR is administered by the Federal Reserve Bank of New York. SOFR is the average rate at which banks can borrow US dollars overnight while posting US Treasury bonds as collateral. SOFR is a secured borrowing rate, whereas LIBOR is a rate for unsecured borrowing.

5.2 International market

The international bond market is an international market in debt. Companies issuing debt in the market have their securities traded all around the world and are not limited to one domestic marketplace. The market only accepts highly rated companies, since international bonds are unsecured debt.

Primary market bond issuance is through the process of **placing**. A large borrower will have a relationship with a number of bond 'houses' (investment banks specialising in bonds). The decision to issue can come either from the borrower who perceives the need for finance or from the issuing house who perceives that there is an opportunity to issue a reasonably priced bond into the market.

Another key decision to be taken is whether or not to **list** the bond on one of the various exchanges. Almost inevitably a bond will be listed, normally on either the London or Luxembourg Exchange. The reason for this is that certain institutions and investors by their trust deeds are prohibited from investing in securities that are not listed on a formal exchange.

There is no formal marketplace for international bond trading. The market is telephone driven (or, increasingly, driven by electronic trading platform) and the main bond houses for international bonds are based in London.

Settlement is conducted for the market by two independent clearing houses, Euroclear and Clearstream. These clearing houses immobilise the stocks in their vaults and then operate electronic registers of ownership.

An effect of low government bond yields appears to have been to make corporate bond yields more attractive for both issuers and investors. Issuers can benefit from the low interest rates in the bond markets, and investors are able to obtain a higher yield on corporate bonds than on government bonds.

5.3 Bond markets in other countries

5.3.1 US

The US government bond market is the largest in the world, having expanded rapidly in the 1980s. This size is evident in both the quantity of issuance in the primary market and the volumes of activity in the secondary market. The US Treasury is responsible for the issuance of new securities. Issuance takes place on a regular calendar with weekly issues of three- or six-month Treasury bills, monthly issues of one-year bills and two- and five-year notes, and quarterly issues in set cycles of longer- dated stocks.

Central to the operation of the secondary market in government bonds are the **primary dealers**. These firms are the market makers. They are authorised to conduct trades by the Federal Reserve (the Fed) and are obliged to make markets in all issues.

Federal agencies are quasi-governmental institutions established to fill gaps in the US financial markets. They are backed by the guarantee of the US Government via the Fed and are therefore riskless.

Unlike the UK, the US corporate sector has been able to borrow substantial sums via the issue of debt securities. In part, this has been forced on them by the highly fragmented nature of the domestic banking market. Equally, however, investors are willing to hold corporate debt as part of their portfolios in a way UK investors are not.

A Yankee is a dollar-denominated bond issued in the US by an overseas borrower. Given that international dollar-denominated bonds may not be sold into the US markets until they have **seasoned** (a period of 40 days), there is still a divide between the international bond market and the domestic US bond market. Apart from the access to a different investor base, the market also allows issuers to raise longer-term finance, since the US domestic investors are prepared to accept longer maturities.

The US corporate bond market is very large and there is an active market in bonds with a low credit rating (even a non-investment grade credit rating – junk bonds). Investors are willing to accept the higher risk of default by borrowers in order to earn higher yields on their investment.

In the US, corporates are allowed to issue bonds into the private placement market without seeking full SEC registration. This access to the market is available to both domestic and foreign issuers, though the market is dominated by US domestic issuers who have chosen not to enter the public market. The buy side of the market is restricted to certain institutional investors (primarily insurance companies) who are predominantly 'buy and hold' investors.

Reflecting the nature of the market as in effect a one-to-one contract, new issue terms are negotiated between the issuer and the lender. This process can take up to eight weeks to complete. Equally, as a consequence, borrowers tend to have to pay a premium over the public market yields. However, they do not have the cost of seeking a credit rating, nor are they bound by the rules of the SEC.

5.3.2 Japan

The **bond market** in **Japan** is dominated by government bonds. This domination is not in terms of volume, where **Japanese government bonds (JGBs)** account for only half the market, but in the secondary market where they account for over 80% of the secondary market trading.

JGBs trade on both the TSE and the Broker-Broker (OTC) market. The market tends to focus on the 'benchmark' issues, with occasionally 90% of the volume taking place in that stock.

International borrowers are able to access the domestic pool of savings through the issue of Samurai (publicly issued yen bonds) and Shibosai (yen bonds issued via private placement).

5.3.3 France

The French bond market has developed into one of the key international bond markets, mainly due to the economic transformation that took place since the introduction of the 'Franc fort' policy in 1985. The development of MATIF (the French financial futures market, now part of Euronext) and the trading of the 'Notional' future into the French long bond were also vital components in the reform of the market.

The French Government issues three **types of bond**, each with a different maturity. All French government issues are now in book entry form with no physical delivery.

Over half the market is made up of debt issues from the **public corporations**. These stocks are not, for the most part, guaranteed by the Government, but the corporations concerned do possess strong credit ratings. They have established their own market structure in order to facilitate trading in their stocks. All issues are by way of a placing through a syndicate of mainly local banks.

5.3.4 Germany

The German bond market, including government bonds, domestic bonds and international bonds, is the third largest in the world and one of the largest in Europe (the second largest after Italy).

Unlike the UK, Germany has a strong **corporate debt market** and a relatively weak equity market. The bulk of finance for industry is still provided through the banks, either as lenders or shareholders, with debt securities (other than international bonds) being less significant. Within the context of debt issuance, banks issue bonds and then lend the money on to the corporate sector.

5.4 Islamic bond markets

Sukuk markets are relatively new and are not yet as developed and standardised as Western markets. They were originated by government entities, particularly that of Malaysia, and government entities still dominate the market, though corporate issues are gaining in importance.

Issuance is achieved in a similar manner to corporate bond issues in the UK, with investment banks underwriting and managing the issues.

Sukuk other than the short-term Salam Sukuk are traded by being listed on stock exchanges, though liquidity levels are currently quite low and many issues are held to maturity.

6 Bond valuation and yields

Section overview

- Bonds are an important source of finance for companies.
- Bonds can be priced at par, at a premium or at a discount when they are issued and throughout their lives.
- Returns on bonds can be measured using flat yield and/or gross redemption yield. Yield is a measure of the return on the bond as a percentage of the bond's market value.
- There is an inverse relationship between bond yields and bond prices. As interest rates rise, the price of bonds falls (and bond prices rise when yields fall).
- The yield curve is a graphical representation of the structure of interest rates, where the yield offered by bonds is plotted against maturity.
- The yield that investors require from a bond is made up of a number of elements, including a risk- free return plus return to compensate for credit and default risk, liquidity and marketability risk, issue-specific risk and fiscal risk.
- The volatility of a bond is the sensitivity of the bond to movements in the yield/interest rate.
- Duration is the weighted average length of time to the receipt of a bond's benefits (coupon and redemption value), the weights being the PV of the benefits involved.

6.1 Bond pricing

Bonds can be priced at par, at a premium or at a discount. Like shares, the price of a bond will depend on the market forces of supply and demand; the demand for bonds from investors depends on the yield that they require on the bond. The required yield will depend on factors such as the coupon rate on the bond, the credit rating of the bond issuer and the period to redemption (the maturity) of the bond.

6.1.1 General method

The price of a bond is the sum of the PVs of all expected coupon payments plus the PV of the redemption value at maturity – in other words, all you are doing is discounting the future cash flows of the bond at the rate of return (yield) required by the bond investors.

The PV spreadsheet function is used to calculate the market price (issue price) of a bond, this is covered in more detail in the chapter Spreadsheet formulae and illustrated in the example below.

Worked example: Bond pricing

A bond has a face value of £2,500 which is to be redeemed in 15 years' time at par. The coupon rate of the bond is 8% and there is a required yield of 10%. Coupon payments are made every 6 months and the next payment is due in six months' time.

Requirement

What is the price of the bond?

Solution

The first thing we have to do is calculate the number of coupon payments that will be made. As two payments are made every year for 15 years, there will be 30 payments in total.

What is the value of each coupon payment? As the payments are made twice a year, you should divide the coupon rate in half – which gives a rate of 4%. Each semi-annual coupon payment will be 4% of £2,500 = £100.

Likewise, with the required yield, divide by two, giving a yield of 5%. If we did not reduce the yield, the bond price would be too low.

We can now use the PV spreadsheet function to calculate the bond price:

To calculate the PV, the following variables need to be input to the PV function.

B5 =PV(B1,B2,B3,B4)

	A	B
1	Rate of return required over the period	0.05
2	Nper = the number of periods	30
3	Pmt= the amount (of interest) paid in any single period	100
4	Fval = the future value (the amount paid at maturity).	2,500
5	Present value (issue price)	-2,115.69

Bond price = £2,115.69

The bond is trading at a discount – the bond price of £2,115.69 is less than the par value of £2,500. This is because the required yield is greater than the coupon rate. Investors will have to be attracted by a discount in order to invest in the bonds, as they could earn greater interest elsewhere at the prevailing rate.

6.1.2 Zero-coupon bonds

The pricing of zero-coupon bonds is actually quite straightforward, all you have to do is calculate the PV of the redemption value at maturity.



Worked example: Zero-coupon bonds

What is the price of a zero-coupon bond that matures in 10 years' time, has a required yield of 7% and a redemption value of £3,200? Assume that coupon payments occur semi-annually.

Solution

As the coupon payments occur semi-annually, we have to adjust the yield to its equivalent semi- annual payment rate. This means that the number of periods for the zero-coupon bond will be doubled from 10 to 20.

The yield will also have to be adjusted, giving a yield of 3.5%.

B5 =PV(B1,B2,B3,B4)

	A	B
1	Rate of return required over the period	0.035
2	Nper = the number of periods	20
3	Pmt= the amount (of interest) paid in any single period	0
4	Fval = the future value (the amount paid at maturity).	3,200
5	Present value (issue price)	−1,608.21

= £1,608.21

Zero-coupon bonds always trade at a discount, otherwise there would be no chance for the investor to make any money and thus no incentive to invest in such bonds.

6.2 Flat yield

6.2.1 Calculation of the flat yield

The simplest measure of return in the bond market is the flat yield. This is also referred to as the running yield or the interest yield. It looks at the cash return generated by an investment over the cash price. In simple terms, what is the income that you generate on the money that you invest?

$$\text{Flat yield} = \frac{\text{Annual coupon payment}}{\text{Market price}} \times 100\%$$



Worked example: Flat yield

5% loan stock

Redemption date five years from now at par Current market price £97.25

Solution

The flat yield for the above would be: 5 / 97.25 x 100 = 5.14%

6.2.2 Limitations of the flat yield

This measure is of some use, particularly in the short term, but it has important drawbacks for the investment markets.

(a) With all equities and some bonds (eg, floating rate notes), the return in any one period will vary. If the coupon is not constant, then the measure is only of historical value unless the predicted return is used.

(b) In addition to the coupon flows, bonds will have return in the form of the payment at maturity (the 'redemption monies'). Where the bond has been purchased at a price away from par, this will give rise to potential gains or losses arising from the difference between the purchase price for the bond and its redemption value.

(c) The calculation completely ignores the time value of money. If an investor were to be offered the choice between the receipt of £10 now or in two years' time, the logical choice would be to take the £10 now, since the money could then be invested to generate interest.

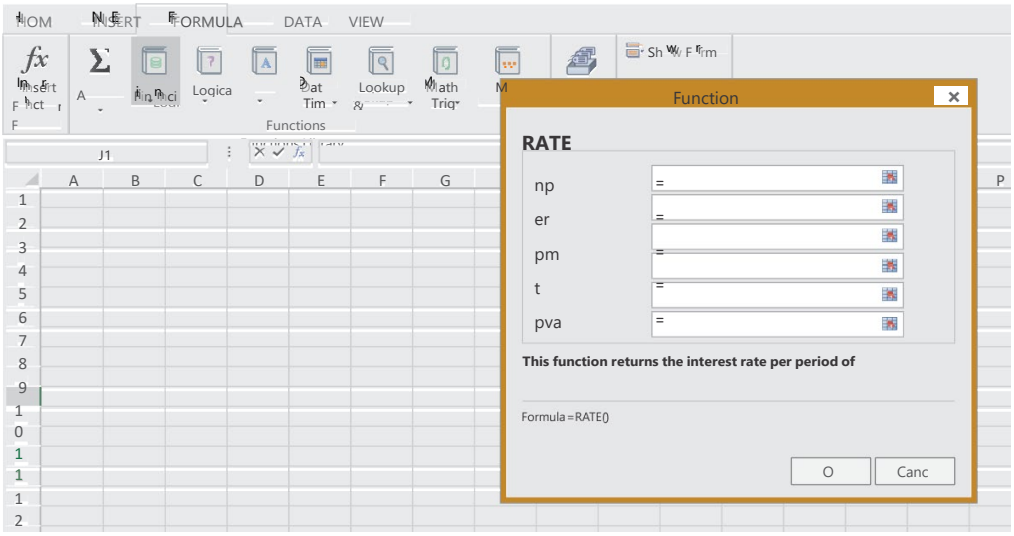
These limitations combine to make the flat yield of marginal use only.

6.3 Gross redemption yield (yield to maturity)

The gross redemption yield considers the flows of money arising from a bond.

Each flow is then converted into its PV by discounting the future flow. The RATE spreadsheet function is used to calculate the gross redemption yield. This is covered in more detail in the chapter Spreadsheet formulae.

The following information will need to be input into the RATE function:



Nper = the number of periods

Pmt= the amount (of interest) paid in any single period

Pval= the present value of the asset (its market price), inserted as a negative number Fval = the future value (the amount paid at maturity).
Type and guess can be left blank for the SBM&L exam.

This will give the gross redemption yield over a given period.

To calculate an annual yield the value will need to be multiplied by the number of periods in a year eg, if payments are made over 6 months, as there are two six-month periods in a year so multiply the RATE calculated by 2.

It can be seen that the GRY is simply the internal rate of return of the bond's flows, where the current price is treated as an outflow and the future interest and capital repayment flows are treated as inflows.

Worked example: GRY

5% loan stock (annual coupon)

Three years to redemption

Current market price £97.25

Solution

B5 =RATE(B1,B2,B3,B4)

	A	B
1	Nper = the number of periods	3
2	Pmt= the amount (of interest) paid in any single period	5
3	Pval= the present value of the asset (its market price)	−97.25
4	Fval = the future value (the amount paid at maturity).	100
5	Gross yield	0.063

The GRY of this bond is 6.03%.

6.3.1 Problems with the GRY

We have noted that the yield that has been calculated so far is in effect the internal rate of return of the flows generated by a bond.
As a measure of predicted return, the yield is limited, since it assumes that any coupon receipts are reinvested at the same rate as the current yield. If the investor is only able to reinvest the coupon at a lower rate, then the overall return generated by the bond will be lower than the yield.

6.4 The relationship between bond prices and interest rates

One of the fundamental correlations in bond markets is that as the interest rate rises and bond yields go up, the price of a bond will fall. When interest rates and bond yields fall, bond prices will rise.
There is an inverse relationship between yield and price.

This can be demonstrated either through the flat or the gross redemption yield. Suppose the flat yield was calculated as:

$$\frac{10}{97.25} \times 100 = 10.283\%$$

If the market interest rate were to rise to 12% then the holders of the bond would be encouraged to sell the bond and invest their money into assets yielding 12%. The result of this would be that the supply of the bond would rise and the demand fall, leading to a fall in the price. As the price falls to £83.33, the yield rises to 12% thereby removing the incentive to switch out of the bond:

$$\frac{10}{83.33} = 12\%$$

The GRY can prove the same result using the RATE function.

If the interest rate increases then the flows received in future periods will have a lower PV and consequently the value of the bond (its price) falls.

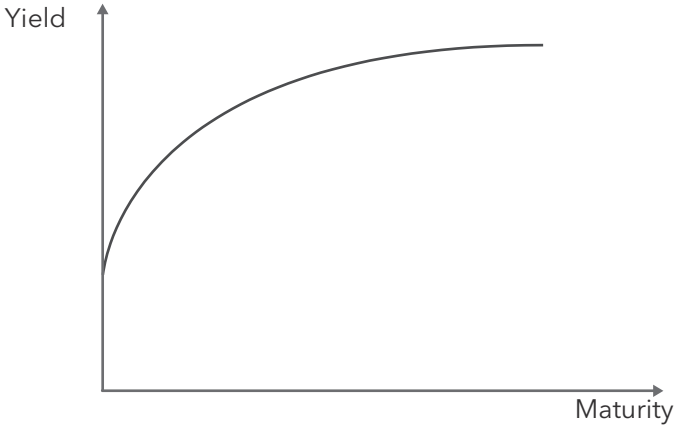
6.5 The yield curve

6.5.1 What is the yield curve?

The yield curve (Figure 13.3) is a graphical representation of the term structure of interest rates, where the yield offered by bonds is plotted against maturity. It is often calculated by reference to the GRY.

A 'normal' yield curve is upward sloping, which means that the yields required by investors are higher for longer-dated debt instruments.

Figure 13.3: The yield curve



6.5.2 The shape of the curve

Liquidity preference

If an investor's money is invested in longer-term (and therefore riskier) debt, then they will require a greater return – a risk premium. Short-term liquid debt carries a lower risk and therefore requires a lower return. This gives rise to the normal upward sloping yield curve.

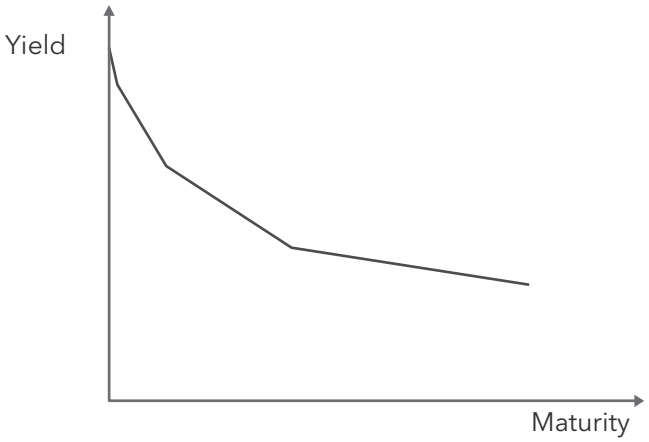
Expectations theory

Expectations theory states that the yield curve is a reflection of the market's expectation of future interest rates. If the market believes that the yield at the long end of the yield curve is high and is likely to fall then, in order to profit from the increase in prices that this will create, investors will buy long-dated stocks. As a result, the demand for these stocks will rise. This demand pressure will force the price to rise. As a consequence, the yield will fall, reflecting the expectation of a fall.

On the other hand, if the market believes that rates will have to rise, then the forces will work in the opposite direction and this will lead to a fall in the price and a rise in the yield.

The expectations of the market can clearly be seen in an inverted yield curve (Figure 13.4).

Figure 13.4: An inverted yield curve



Here, short-term rates are unusually high, but the market anticipates that this cannot last for long. The longer end of the market has anticipated this change by forcing yields down. This can lead to the anomalous situation where the long end of the market remains constant because it has anticipated change and the short end (which is technically the least volatile) exhibits all the movement.

Another key element of the market's expectations will be the expectation of inflation. If the market believes that inflation will rise in the future then the yields on the longer-dated stocks will have to rise in order to compensate investors for the fall in the real value of their money. The expectation of inflation is much more of a problem with the long – rather than the short – end.

Preferred habitat and market segmentation

Certain maturity ranges are appropriate to particular types of investors. In the UK, the short end of the market is dominated by the financial sector maintaining a proportion of their assets in liquid investments, whereas the long end is dominated by institutional investors such as pension funds. In effect, this gives rise to two markets and may be reflected in a discontinuity, or hump, in the yield curve.

Supply-side factors

The availability of debt in certain maturity ranges may lead to either an excess or a shortage of debt and consequently an anomalous yield on some debt.

6.5.3 Yield curve and interest rates

The yield curve on a range of long-dated bonds provides the best means of inferring market estimates of future interest rates. The steepness of the yield curve reflects market expectations about the change in forward rates.

6.6 Composition of the yield

The yield on any bond is made up of a number of elements.

(a) **The real return** – This is the real rate of return that the investment has to earn. Effectively, it represents the opportunity cost of saving over immediate consumption.

(b) **Inflation premium** – As inflation rises then the yield on a bond will have to increase in order to compensate the holder. Inflation is basically negative interest, eroding the value of savings, whereas interest adds to them.

Together, the real return and the inflation premium form the yield on government bonds and are therefore said to approximate to the interest rate. In the case of non-government bonds, the required return is a yield in excess of the risk-free rate (the rate obtainable on government bonds) to compensate investors for the additional investment risks:

(a) **Credit and default risk** – This is the risk of the issuer defaulting on its obligations to pay coupons and repay the principal. The ratings issued by commercial rating companies can be used to help assess this risk.

(b) **Liquidity and marketability risk** – This is the ease with which an issue can be sold in the market. Smaller issues are particularly subject to this risk. In certain markets the volume of trading tends to be concentrated in the 'benchmark' stocks, thereby rendering most other issues illiquid. Other bonds become subject to 'seasoning' as the initial liquidity dries up and the bonds are purchased by investors who wish to hold them to maturity.

(c) Even if an issuer has a triple A credit rating and is therefore perceived as being at least as secure as the Government, it will still have to offer a **yield above that offered by the Government** due to the smaller size (normally) and the thinner market in the stocks.

(d) **Issue specific risk** – For example, risk of call. If the company has the right to redeem the bond early, then it will only be logical for it to do this if it can refinance at a lower cost.

What is good for the issuer will be bad for the investor and, thus, the yield will have to be higher.

(e) **Fiscal risk** – The risk that withholding taxes will be increased. For foreign bonds, there would also be the risk of the imposition of capital controls locking your money into the market.

6.7 Sensitivity to yield

6.7.1 Introduction

All the risks of holding a bond come together in the yield. As we have already seen, there is an inverse relationship between the price of a bond and its yield. It is possible to define this relationship mathematically and predict the way in which a selection of bonds will perform. This is sometimes referred to as the volatility of the bond.

The sensitivity of any bond to movements in the yield/interest rate will be determined by a number of factors.

6.7.2 Sensitivity to maturity

Longer-dated bonds will be more sensitive to changes in the interest rate than shorter-dated stocks.

6.7.3 Sensitivity to coupon

Lower coupon stocks demonstrate the greatest level of sensitivity to the yield. It should be noted that the relationship between yield and price is not symmetrical. This is a relationship that is known as convexity.

6.7.4 The impact of the yield

If yields are particularly high, then the flows in the future are worth relatively little and the sensitivity is diminished. Conversely, if the yield is low then the value of flows in the future is enhanced and the bond is more sensitive to the changing GRY.

6.7.5 Summary of volatility

High volatility		Low volatility
Long dated	>	Short dated
Low coupon	>	High coupon
Low yields	>	High yields

While these simple maxims are good indicators of the likely sensitivity to fluctuations in the rate of interest, they do not allow for two bonds to be directly compared. For example, which of the following is likely to be the more sensitive to a rise in interest rates – a high-coupon, long-dated stock, or a low-coupon, short-dated stock? In order to enable two such bonds to be compared, a composite measure of risk known as the **duration** can be used.

6.8 Duration

6.8.1 What is duration?

A bond may be thought of as a combination of cash flows with different timings, each with its own sensitivity to interest rate movements. Hence, a measure of the sensitivity of a bond price to changes in interest rates must be a weighted average of the sensitivities of those individual cash flows. This measure is a bond's **duration**.

Definition

Duration: The weighted average length of time to the receipt of a bond's cash flows, the weights being the present value of each cash flow. The calculation gives each bond an overall risk weighting that allows two bonds to be compared. In simple terms, it is a composite measure of the risk expressed in years.

Mathematically, duration (also known as Macaulay duration) can be expressed using the following formula:

$$\text{Duration (D)} = \frac{\sum(t \times PT_t)}{\text{Price}}$$

Where: PV_t = present value of cash flow in period t (discounted using the redemption yield)
T = time period when cash flow occurs

Worked example: Duration

Calculate the duration of a three-year 9% annual coupon bond trading with a GRY of 8%.

Solution

Time	Cash Flow (£)	Discount Factor	Present Value (£)	Weighted (t × PVt)
1	9.00	1/1.08	8.33	8.33
2	9.00	$1/1.08^2$	7.72	15.44
3	109.00	$1/1.08^3$	86.53	259.59
			102.58	283.36

Using the above:

$$\text{Duration} = \frac{283.36}{102.58} = 2.7623 \text{ years}$$

Note that the duration on any bond that pays coupons will always be less than the maturity, because there is always some amount of the total payments that are going to arrive before the maturity date.

6.8.2 Modified duration

The measurement of duration can be converted into another measure, known as **modified duration**.

Definition

Modified duration: Used to determine the effect that a change in interest rates will have on the price of a bond. Modified duration = Duration/(1 + i), where i is the yield required by the bond investors.

To apply the modified duration of a bond to calculate the change in price that would occur with a change in interest rates, the following formula is used:

$-D_m \times P \times i$

D_m is the measure of modified duration and P is the current price of the bond before the price change. The longer the modified duration, the bigger will be the change in bond price given a change in the required yield.

Worked example: Modified duration

Continuing with our example above of a three year 9% bond paying annual coupons with a GRY of 8%, we can use modified duration to calculate the expected change in price if the yield were to rise to 9%.

Solution

Using the information from the above example:

$$\text{Modified duration} = \frac{2.7623}{1.08} = 2.5577$$

In this example, if the bond yield increases by 1% the change in the bond price will be:

–2.5577 × 102.58 × 0.01 = –2.62.

The bond price will fall from 102.58 to 99.96

It must be remembered that this is an inverse relationship. A rise in yields will lead to a fall in bond prices and vice versa.

6.8.3 Properties of duration

The basic features of sensitivity to interest rate risk will all be reflected by its measure of duration.

(a) **Longer-dated bonds** will have longer durations.

(b) **Lower-coupon bonds** will have longer durations. The ultimate low-coupon bond is a zero-coupon bond where the duration will be the maturity.

(c) **Lower yields** will give longer durations. In this case, the PV of flows in the future will rise if the yield falls, extending the point of balance, therefore lengthening the duration.

The duration of a bond will shorten as the lifespan of the bond decays. However, the rate of decay will not be at the same rate. For example, a five-year bond might have a duration of 4.157 years. In a year's time the bond might have a remaining life of four years and a duration based on the same GRY of 3.480 years. The lifespan has decayed by a full year, but the duration by only 0.677 of a year.

7 Credit risk

Section overview

- Credit risk is the risk for a lender that the borrower will default either on interest payments or on the repayment of principal on the due date, or on both.
- The credit spread is the premium required by an investor in a corporate bond to compensate for the credit risk of the bond.

7.1 Credit (default) risk

Definition

Credit risk: also referred to as **default risk**, is the risk for a lender that the borrower will default either on **interest payments** or on the **repayment of principal** on the due date, or on both.

7.1.1 Credit risk aspects

Credit risk arises from the inability of the borrower to fulfil its obligation under the terms of a contract. Creditors to companies such as corporate bondholders and banks are exposed to credit risk.

The credit risk of an individual loan or bond is determined by the following two factors:

(a) **The probability of default**
This is the probability that the borrower or counterparty will default on its contractual obligations to repay its debt.

(b) **The recovery rate**
This is the fraction of the face value of an obligation that can be recovered once the borrower has defaulted. When a company defaults, bondholders do not necessarily lose their entire investment. Part of the investment may be recovered depending on the **recovery rate**.

Definition

Loss given default (LGD): is the difference between the **amount of money owed** by the borrower and the **amount of money recovered**.

For example, a bond has a face value of £100 and the recovery rate is 80%. The **LGD** in this case is: LGD = £100 – £80 = £20

Definition

Expected loss (EL): from credit risk shows the **amount of money the lender should expect to lose** from the investment in a bond or loan with credit risk.

The **EL** is the product of the **LGD** and the **probability of default** (PD).
EL = PD × LGD
If the PD is, say, 10%, the EL from investing in the above bond is:
EL = 0.10 × 20 = £2 per £100 nominal value of the bond.
For all bonds, regardless of their credit rating, the PD increases with the future time period covered. For example, the PD for a bond may be a 0.05% PD within the next two months, a 1.2% PD within two years, a 2.0% PD within the next three years, and so on.

7.1.2 Credit risk measurement

The measurement of credit risk is quite complex. All the approaches concentrate on the estimation of the default probability and the recovery rate.
The oldest and most common approach is to assess the PD using financial and other information on the borrowers and assign a rating that reflects the EL from investing in the particular bond. This assignment of credit risk ratings is done by credit rating companies such as Standard & Poor's, Moody's Investor Services and Fitch. These ratings are widely accepted as indicators of the credit risk of a bond. The table below shows the credit rating used by the two largest credit rating agencies.

Standard & Poor's	Moody's	Description of category
AAA	Aaa	Highest quality, lowest default risk
AA	Aa	High quality
A	A	Upper medium grade quality
BBB	Baa	Medium grade quality
BB	Ba	Lower medium grade quality
B	B	Speculative
CCC	Caa	Poor quality (high default risk)
CC	Ca	Highly speculative
C	C	Lowest grade quality

S&P also use + and – symbols to refine its credit ratings. For example, the credit ratings below AAA are AAA– and then AA+, AA, AA–, A+ and so on. The lowest 'investment grade' credit rating is BBB–. Moody's uses the numbers 1, 2 and 3 to refine its credit ratings, for example Aa1, Aa2 and Aa3.

Both credit rating agencies estimate default probabilities from the empirical performance of issued corporate bonds of each category.

It would be very difficult for any company without a credit rating to raise money in bond markets, and the market itself is confined to dealing with high credit-worthy stocks exclusively in the investment grade.

A bond may be placed on credit watch as a prelude to a potential downgrade. If a bond is placed on credit watch then it is likely that its credit spread will widen with the consequence of a fall in price.

This impact is less pronounced for the higher rated (investment grade) bonds but becomes quite significant for non-investment grade bonds or those falling from investment grade to non-investment grade.

The more a bond gets downgraded, the fewer investors that will find it to be of interest. There are specialist high yield funds. However, as a general rule the major investors prefer the higher rated bonds.

One big step is when a bond falls from investment grade to non-investment grade, or high yield. At this stage there tends to be many forced sellers, and prices can be expected to decline dramatically.

7.1.3 Credit migration

There is another aspect of credit risk which should be taken into account when investors are investing in corporate bonds, beyond the PD.

A borrower may not default, but due to changing economic conditions or management actions the borrower may become more or less risky than at the time the bond was issued. As a result, the bond issuer will be assigned by the credit agency a different credit rating. This is called credit migration. The significance of credit migration lies in the fact that the assignment of lower credit rating will decrease the market value of the corporate bond.

7.2 Credit spreads and the cost of debt capital

Definition

Credit spread: is the premium required by an investor in a corporate bond to compensate for the **credit risk** of the bond.

The yield to a government bondholder is the compensation to the investor for forgoing consumption today and saving. However, corporate bondholders should require compensation not only for forgoing consumption, but also for the credit risk to which they are exposed. Assuming that a government bond such as the ones issued by the US and UK Governments is **free** of **credit risk**, the yield on a corporate bond will be:

Yield on corporate bond = Risk-free rate + Credit spread

Or in symbols $y = r + s$

Where: y = yield on the corporate bond
r = risk-free rate, ie, the rate on a government bond with no default risk
s = credit spread

Since **credit spreads** reflect the **credit risk** of a bond, they will be inversely related to the credit quality of the bond. Low credit quality bonds will be characterised by large spreads and high credit quality bonds will be characterised by low spreads.

7.2.1 The cost of debt capital

The cost of debt capital for a company will therefore be determined by the following:

- its credit rating
- the maturity of the debt
- the risk-free rate at the appropriate maturity
- the corporate tax rate

Cost of debt capital = (1 − Tax rate) (Risk-free rate + Credit spread)

Worked example: Cost of debt capital

Consider a corporate bond with a maturity of four years and a credit rating of BBB. The 4-year risk- free rate is 5% and the credit spread is 200 basis points. The tax rate is 24%. Find the cost of debt capital.

Solution

Cost of debt capital = (1 – 0.24) (5% + 2%) = 5.32%

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8 Leasing

Section overview

- Rather than buying an asset outright, using either available cash resources or borrowed funds, a business may lease an asset. This method of financing is extremely popular with airlines (which typically lease their aircraft) and companies with carpools; it has also been used extensively in the printing industry.
- The financial reporting implications of leasing are covered in more detail in section 11 of this chapter.

8.1 Leasing

Leasing is a commonly used source of finance, especially for small or medium sized entities. It is a form of debt finance that can be very useful where capital is rationed. Rather than buying an asset outright, using either available cash resources or borrowed funds, a business may lease an asset.

IFRS 16 abolished the distinction between operating and finance leases for lessees, although not for lessors. It introduces a single lessee accounting model and requires a lessee to recognise assets and liabilities for all leases with a term of more than 12 months, unless the underlying asset is of low value. The financial reporting implications of leasing are covered in more detail in section 11.

8.2 Sale and leaseback

Sale and leaseback occurs when a business that owns an asset agrees to sell the asset to a financial institution and then lease it back on terms specified in a sale and leaseback agreement. The business retains use of the asset and has the funds from the sale, but as a consequence has to pay rent in the future. Sale and leaseback arrangements are particularly appropriate for highly specialised assets that may have a low realisable value and therefore would be poor security for a more traditional loan.



Context example: Tesco

Tesco is one of the biggest corporate owners of property in the UK and the company has used leasing arrangements extensively for its properties. For example, in 2008 Tesco sold 113 of its stores and a depot to raise cash to fund its growth plans, and immediately leased them back under sale and leaseback agreements.

In 2014 Tesco obtained a new store in Sale, Manchester, under a 20-year leasing agreement. The agreement gives the tenant the right to break the agreement after 10 years. There are regular lease rental reviews with changes in payments linked to the Retail Prices Index, but with a collar of 1% and 4% limiting the scale of any annual increase.

8.3 Attractions of leasing

Attractions include the following.

(a) The supplier of the equipment is paid in full at the beginning. The equipment is sold to the lessor, and, apart from any guarantees, the supplier has no further financial concern about the asset.

(b) The lessor invests finance by purchasing assets from suppliers and makes a return out of the lease payments from the lessee. The lessor will also get capital allowances on his purchase of the equipment.

(c) Leasing may have advantages for the lessee:
   (1) The lessee may not have enough cash to pay for the asset, and would have difficulty obtaining a bank loan to buy it. If so, the lessee has to rent the asset to obtain use of it at all.
   (2) Leasing may be cheaper than a bank loan.
   (3) The lessee may find the tax relief available advantageous.

8.4 Lease or buy?

The decision whether to lease or buy an asset involves two steps.

(a) The **acquisition decision**: Is the asset worth having? Test by discounting project cash flows at a suitable cost of capital.

(b) The **financing decision**: If the asset should be acquired, compare the cash flows of purchasing against those of leasing or hire purchase arrangements. The cash flows can be discounted at an **after-tax cost of borrowing**.

The traditional method is complicated by the need to choose a discount rate for each stage of the decision. In the case of a **non-taxpaying organisation**, the method is applied as follows.

Step 1 The cost of capital that should be applied to the cash flows for the acquisition decision is the cost of capital that the organisation would normally apply to its project evaluations.

Step 2 The cost of capital that should be applied to the (differential) cash flows for the financing decision is the cost of borrowing.

- We assume that if the organisation decided to purchase the equipment, it would finance the purchase by borrowing funds.
- We therefore **compare** the **cost of borrowing** with the **cost of leasing** (or hire purchase) by applying this cost of borrowing to the financing cash flows.

In the case of a **taxpaying organisation**, taxation should be allowed for in the cash flows, so that the traditional method would recommend:

Step 1 **Discount** the cash flows of the acquisition decision at the firm's **after-tax cost of capital**.

Step 2 **Discount** the cash flows of the financing decision at the **after-tax cost of borrowing**.



Worked example: Lease or buy decisions 1

Mallen and Mullins Co has decided to install a new milling machine. The machine costs $20,000 and it would have a useful life of 5 years with a trade-in value of $4,000 at the end of the fifth year. Additional net cash generated from the machine would be $8,000 a year for 5 years. A decision still has to be taken about the method of financing the project. Two methods of finance are being considered:

- The company could purchase the machine for cash, using bank loan facilities on which the current rate of interest is 13% before tax.
- The company could lease the machine under an agreement which would entail payment of $4,800 at the end of each year for the next five years.

The company's weighted average cost of capital, normally used for evaluating projects, is 12% after tax. The rate of company tax is 21%. If the machine is purchased, the company will be able to claim an annual tax depreciation allowance of 20% of the reducing balance.

Requirement

Advise the management on whether to acquire the machine, on the most economical method of finance, and on any other matter which should be considered before finally deciding which method of finance should be adopted.

Solution

The traditional method begins with the **acquisition decision**. The cash flows of the project should be discounted at 12%. The first writing-down allowance is assumed to be claimed in the first year resulting in a saving of tax at Year 2.

Tax depreciation

Year		$
1	20% of $20,000	4,000
2	20% of $(20,000 – 4,000)	3,200
3	20% of $(16,000 – 3,200)	2,560
4	20% of $(12,800 – 2,560)	2,048
		11,808
5	$(20,000 – 11,808 – 4,000)	4,192
		16,000

Taxable profits and tax liability

Year	Cash profits	Tax-allowable depreciation	Taxable profits	Tax at 21%
	$	$	$	$
1	8,000	4,000	4,000	840
2	8,000	3,200	4,800	1,008
3	8,000	2,560	5,440	1,142
4	8,000	2,048	5,952	1,250
5	8,000	4,192	3,808	800

NPV calculation for the acquisition decision

	A	B	C	D	E
1	Year	Equipment	Cash profits	Tax	Net cash flow
2		$	$	$	$
3	0	(20,000)			(20,000)
4	1		8,000		8,000
5	2		8,000	(840)	7,160
6	3		8,000	(1,008)	6,992
7	4		8,000	(1,142)	6,858
8	5	4,000	8,000	(1,250)	10,750
9	6			(800)	(800)
10	PV @12% T1-6				27,880 [1]
11	Less outlay @T0				(20,000)
12	NPV				7,880

[1] The formula in cell E10 is =NPV(0.12,E4:E9)

Conclusion

The net present value (NPV) is positive, so the machine should be acquired, regardless of the method used to finance the acquisition.

The second stage is the **financing decision**, and cash flows are discounted at the after-tax cost of borrowing, which is at 13% × 79% = 10.27%, say 10%. The only cash flows that we need to consider are those which will be affected by the choice of the method of financing. The operating savings of $8,000 a year, and the tax on these savings, can be ignored.

- **The present value (PV) of purchase costs**

	A	B	C
1	Year	Item	Cash flow
2			$
3	0	Equipment cost	(20,000)
4		Tax savings from allowances	
5	1		0
6	2	21% × $4,000	840
7	3	21% × $3,200	672
8	4	21% × $2,560	538
9	5	21% × $2,048 +Trade-in value	4,430
10	6	21% × $4,192	880
11		PV@10% T1-6	4,814 [1]
12		Less outlay @T0	(20,000)
13		NPV	(15,186)

[1] The formula in cell C11 is =NPV(0.1,C5:C10)

- **The PV of leasing costs**

It is assumed that the lease payments are fully tax allowable.

Year	Lease payment	Savings in tax (21%)	Discount factor	PV
	$	$	10%	$
1–5	(4,800) p.a.		3.791	(18,197)
2–6		1,008 p.a.	3.446	3,474
			NPV of leasing	(14,723)

The cheaper option would be to lease the machine. However, other matters should be considered.

- **Running expenses**

  The calculations assume that the running costs are the same under each alternative. However, expenses like maintenance, repairs and insurance may differ between the alternatives.

- **The effect on cash flow**

  Purchasing requires an immediate outflow of $20,000 compared to nothing for leasing. This effect should be considered in relation to the company's liquidity position, which in turn will affect its ability to discharge its debts and pay dividends.

- **Alternative uses of funds**

  The proposed outlay of $20,000 for purchase should be considered in relation to alternative investments.

- **The trade-in value**

  The NPV of purchase is materially affected by the trade-in value of $4,000 in the fifth year. However, this figure may prove to be inaccurate.

8.5 Other issues

Once a company has made the decision to acquire an asset, the comparison of lease vs buy only needs to include the costs that differ; the costs that are common to each option need not be included.

A disadvantage of the traditional approach to making a lease or buy decision is that if there is a negative NPV when the operational cash flows of the project are discounted at the firm's cost of capital, the investment will be rejected out of hand. However, the costs of leasing might be so low that the project would be worthwhile provided the leasing option were selected. This suggests that an investment opportunity should not be rejected without first giving some thought to its financing costs.

Other methods of making lease or buy decisions are as follows.

(a) **Make the financing decision first**. Compare the cost of leasing with the cost of purchase, and select the cheaper method of financing; then calculate the NPV of the project on the assumption that the cheaper method of financing is used.

(b) **Combine the acquisition and financing decisions together into a single-stage decision**. Calculate an NPV for the project if the machine is **purchased**, and secondly if the machine is **leased**. Select the method of financing which gives the higher NPV, provided that the project has a positive NPV.

Context example: Lease or buy decisions 2

In the case of Mallen and Mullins Co, the NPV with purchase would be + $5,611. This was calculated above. The NPV with leasing would be as follows. A discount rate of 12% is used here.

Year	Profit less leasing cost $	Tax at 21% $	Net cash flow $	Discount factor 12%	PV $
1	3,200		3,200	0.893	2,858
2-5	3,200	(672)	2,528	2.528	6,856
6		(672)	(672)	0.507	(341)
					NPV = 9,373

Using this method, leasing is preferable, because the NPV is $3,762 higher.

Remember that the decisions made by companies are not solely made according to the results of calculations like these. Other factors (short-term cash flow advantages, flexibility, and use of different costs of capital) may be significant.

Professional skills focus: Applying judgement

When considering the lease or buy decision you will need to analyse not only the financial factors but also the strategic and operational factors impacted by the decision. You will therefore have to apply judgement when recommending whether a company should lease or buy an asset.

9 Derivatives

Section overview

- **Derivatives** are instruments based on (derived from) underlying assets such as bonds, shares, indices, commodities, currencies and property. They enable investors to reduce risk or enhance returns on these investments.
- Derivatives include **forwards,futures**, **options and swaps**.

Definition

Financial derivative: A financial derivative is a financial instrument or contract with the following characteristics:

(a) Its value is based on (derived from) the value of an underlying asset, such as a quantity of shares, bonds or foreign currency, or a bank deposit.

(b) Its value changes in response to the change in the price of the underlying item, such as a change in the share price, currency exchange rate or interest rate.

(c) It is a contractual agreement involving an agreement to buy or sell the underlying item, or to exchange payments based on the price of the underlying item.

(d) Acquiring a financial derivative requires no initial net investment, or only a small initial investment. Most of the eventual receipt or payment occurs at a future settlement date for the instrument.

(e) It is settled at a future date, specified in the agreement or contract creating the derivative.

Common **examples** of derivatives include:

(a) **Forward contracts**: agreements to buy or sell an asset at a fixed price at a fixed future date

(b) **Futures contracts**: similar to forward contracts except that contracts are standardised and traded on an exchange

(c) **Options**: rights (but not obligations) for the option holder to exercise at a pre-determined price; the option writer loses out if the option is exercised

(d) **Swaps**: agreements to swap one set of cash flows for another (normally interest rate or currency swaps)

In addition to financial derivatives, there are widely traded derivatives in commodities such as oil, wheat and precious metals.

You covered the basics of some derivatives in earlier studies. To recap:

At the heart of derivative products is the concept of deferred delivery. The instruments allow you, albeit in slightly different ways, to agree **today** the price at which you will buy or sell an asset at some time in the future. This is unlike normal everyday transactions. When we go to a supermarket, we pay our money and take immediate delivery of our goods. Why would someone wish to agree today a price for delivery at some time in the future? The answer is **certainty**.

Imagine a farmer growing a crop of wheat. To grow such a crop costs money for seed, labour, fertiliser and so on. All this expenditure takes place with no certainty that, when the crop is eventually harvested, the price at which the wheat is sold will cover these costs. This is obviously a risky thing to do and many farmers are unwilling to take on this burden. How can this uncertainty be avoided?

By using derivatives, the farmer is able to agree **today** a price at which the crop will ultimately be sold, in maybe four or six months' time. This enables the farmer to achieve a minimum sale price for his crop. He is no longer subject to fluctuations in wheat prices. He knows what price his wheat will bring and can thus plan his business accordingly.

Derivatives may either be:

(a) **Exchange-traded** – ie, where there is an active secondary market accessible to participants

(b) **Off-exchange** – ie, contracts entered directly between counterparties on the **OTC** market

The nature of derivatives often gives rise to **particular problems**. The **value** of a derivative (and the amount at which it is eventually settled) depends on **movements** in an underlying item (such as an exchange rate). This means that settlement of a derivative can lead to a very different result from the one originally envisaged. A company which has derivatives is exposed to **uncertainty and risk** (potential for gain or loss) and this can have a very material effect on its financial performance, financial position and cash flows.

Yet because a derivative contract normally has **little or no initial cost**, under traditional accounting it **may not be recognised** in the financial statements at all. Alternatively, it may be recognised at an amount which bears no relation to its current value. This is clearly **misleading** and leaves users of the financial statements unaware of the **level of risk** that the company faces. IASs 32 and 39 were developed in order to correct this situation.

Derivatives can be used for one of two purposes:

(a) **Speculation** – Speculation involves trading in derivatives for the purpose of attempting to make a gain from a favourable movement in the price of the underlying item. A speculator will seek to make a large gain from a relatively small initial investment in the derivative instruments.

(b) **Hedging** – Hedging involves using derivatives to reduce the market risk in an investment position (ie, the risk of an adverse movement in the price of the underlying item).

Interactive question 3: Loan agreement as derivatives

Two entities make loans to each other for the same amount and on the same terms except that one is based on a fixed rate of interest and the other on a variable rate of interest. There are no transfers of principal at inception of the transaction since the two entities have a netting agreement.

Requirement

Does the transaction fit the definition of a derivative?

See **Answer** at the end of this chapter.

9.1 Forward contracts

Definition

Forward contract: A forward contract is an off-exchange binding agreement to buy or sell an item for settlement at an agreed price on a future date.

Forward contracts allow businesses to set the price of an item in advance. They are particularly suitable in currency markets, and commodity markets such as gold, oil and agriculture where prices can be volatile.

Because a forward contract is an 'OTC' transaction between two parties, all terms of the contract can be tailored individually to meet the buyer's and seller's needs.

Context example: Commodity forward contract

A manufacturer needs to purchase a quantity of corn for future production in three months' time and wants to fix a price for it now.

The manufacturer can achieve this by agreeing with a grower to purchase a specified quantity for delivery at a specific date in the future at a price agreed now.

In July, when the price of a consignment of corn is (say) $50,000, the grower agrees a price with the manufacturer of $52,000 for delivery at the end of September. The price for both parties is now set – and thus whilst the market price in September may be higher or lower than the agreed price of $52,000, the benefit for both parties is that they have certainty, and so are better able to plan.

Forward contracts are tailor-made: the specific terms of a contract are agreed between the two parties and are not dealt on any exchange.

As a contract, they are binding on both parties. Whilst this is useful as a means of giving each party what they want, they are difficult for one party to cancel because agreement is needed from the counterparty. This is one of the reasons why a futures contract may be preferred to a forward contract (see below). Nevertheless, forward markets for some commodities flourish, and the forward market in foreign exchange is also large.

The key advantage of forwards is that they allow much greater flexibility to suit particular circumstances.

The main disadvantages are a **lack of liquidity** in comparison to exchange-traded futures and **increased credit risk for both buyer and seller**, as there is no clearing house to act as a central counterparty. This is known as counterparty risk.

The most common type of forward contract is the forward exchange contract.

9.2 Forward rate agreement (FRA)

A forward rate agreement or FRA is an OTC agreement between a company and its bank that fixes a future short-term interest rate. For example, a 3v9 FRA fixes an interest rate for a period beginning at the end of the third month from the date the FRA agreement is made and ending at the end of the ninth month. So it is an agreement that fixes a six-month interest rate for a period starting in three months' time. Similarly, a 2v5 FRA fixes a three-month interest rate for a period starting in two months' time.

Essentially, the two counterparties agree upfront to exchange a fixed rate of interest for a floating rate of interest on a predetermined notional deposit of cash, for a specific period of time in the future. One party to the FRA agrees to pay a fixed interest rate on the deposit and in return receive interest at a variable rate (a benchmark or reference rate that is specified in the FRA agreement). The other party agrees to receive the fixed interest rate on the deposit and in return pay interest at the variable rate. For example, in a 3v9 FRA, the benchmark variable rate may be the six-month SONIA rate. One party to the FRA is a bank, and the fixed rate in the FRA is determined by the bank.

An FRA is essentially a contract for difference. Rather than actually exchanging the interest payments at the variable and fixed rates, an FRA is settled by a payment of the net amount by one party to the other. For example, if the fixed rate in an FRA is 5.5% and the floating benchmark rate at settlement is 6.0%, the payer of the floating rate in the FRA will make a net payment to the other party for the difference between the fixed rate of 5.5% and the floating rate of 6%.

Because an FRA is settled immediately, at the beginning of the notional interest rate period, the amount payable/receivable is discounted from its end of interest period value to a PV as at the settlement date.

(a) **Buying an FRA**. The buyer of an FRA agrees to pay the fixed interest rate and receive the floating rate. Buying an FRA can fix the rate of interest on a future short-term loan. The FRA buyer will receive compensation when interest rates rise above the fixed FRA contract rate.

(b) **Selling an FRA**. This will provide compensation when interest rates fall below the FRA contract rate. It protects an implied lender.



Context example: FRA

A company has a variable rate loan of £10 million from its bank at SONIA + 1% and the fixing date for the next interest period is in six months' time. The company is concerned about the prospect of an increase in interest rates in the next six months and so buys a 6v12 FRA from its bank on notional principal of £10 million. The fixed rate in the FRA is 5%. This will fix the company's effective borrowing cost at 6% (5% + 1%).

At settlement date for the FRA, which coincides with the fixing date for the bank loan, the six-month SONIA rate might have risen to 7%. The company will have to pay interest at 8% on its bank loan. However, it will receive compensation under the terms of the FRA for the difference between the FRA fixed rate of 5% and the SONIA rate of 7% – ie, 2%. The net cost of borrowing will therefore be 8% – 2% = 6%.

Suppose instead that at settlement date for the FRA, the six-month SONIA rate has fallen to 3.5%. The company will pay interest at 4.5% on its bank loan. However, it will pay compensation under the terms of the FRA for the difference between the FRA fixed rate of 5% and the SONIA rate of 3.5% – ie, 1.5%. The net cost of borrowing will therefore be 4.5% + 1.5% = 6%.

9.3 Futures

Definition

Futures contract: A future is an exchange-traded **standardised contract** to **buy** or **sell** a **specific amount** of a commodity, currency or financial instrument at an **agreed price** on a **stipulated future date**. Futures are the oldest form of investing in commodities.

Futures are derivatives which have their origins in the markets for commodities such as wheat, coffee, sugar, meat, oil, and base and precious metals.

- The prices of all of these commodities **fluctuate seasonally** and are also subject to change because of unpredictable events such as storms, drought, wars and political unrest.
- Historically, in order to avoid the uncertainty arising from large swings in prices, buyers and sellers came together to agree quantities and prices in advance. This encouraged investment in production and helped both buyers and sellers by enabling them to plan.

Originally the buyer and seller would agree a forward price for settlement by actual delivery of an agreed amount of the commodity on an agreed date. As a protection against defaulting on the deal, both parties would put down a deposit. However, the commodity futures markets developed further when the contracts were **standardised** in terms of **delivery date** and **quantity**. This enabled futures contracts to be traded purely on the basis of price; this is now what separates the futures market from the physical commodity market.

There are two parties to a futures contract – a buyer and a seller – whose obligations are as follows.

- The buyer of a future enters into an **obligation** to buy the specified asset on a specified date.
- The seller of a future is under an **obligation** to sell the specified asset on a future date.

Futures contracts are traded on an exchange with the contract terms clearly specified by the rules of the exchange. The first futures market was established in the 19th century in Chicago. The largest group of futures exchanges in the world is still owned and operated by CME Group Inc (Chicago Mercantile Exchange). Futures based on the prices of agricultural commodities allow farmers to be sure of the profit they will receive from the commodities months in advance of the commodities being produced. The main derivatives exchange in the UK is NYSE Liffe. Although some exchanges trade the same or similar contracts, most futures contracts are unique to the exchange that trades them.

There is a large number of commodity futures, but financial futures are also traded in currencies, interest rates, bonds, shares and stock indices.

Futures markets are wholesale markets in risk – markets in which risks are transferred from the cautious to those with more adventurous (or reckless) spirits. The users fall into one of three categories – the **hedger**, the **speculator** and the **arbitrageur** – whose motivations are as follows:

- **Hedger** – someone seeking to reduce risk
- **Speculator** – a risk taker seeking large profits
- **Arbitrageur** – seeks riskless profits from exploiting market inefficiencies (mispricing)

9.3.1 The main elements of futures transactions

The contract size	The **contract size** is the fixed quantity which can be bought or sold with one futures contract. Dealing on futures markets must be in a whole number of contracts. The implication of this is that the amount of an exposure and the amount covered by a futures hedge may not match exactly.
The contract price	The **contract price** is the price at which the futures contract is bought or sold. Prices for futures change continually, which means that holders of positions in futures will make a gain or loss on their position when they eventually close it. The gain or loss on a position is the difference between the original buying or selling price to open the position and the price obtained to close it at a later date (or to settle the contract at the settlement date).
The settlement date	The **settlement date** (or delivery date, or expiry date) is the date when trading on a particular futures contract stops and all accounts are settled. For example, the settlement dates for all currency futures are at the end of March, June, September and December.
The initial margin	When futures contracts are bought or sold, a **deposit** known as the **initial margin** must be paid to the futures exchange. The size of this margin depends on the futures contract but may typically amount to about 5% of the value of contracts dealt in. This deposit is refunded when the contract is closed out. The objective of the initial margin is to cover any possible losses made from the first day's trading. Thereafter, any variations in the contract price are covered by a **variation** margin. Profits are advanced to the trader's account but losses must be covered by advancing further margin. This process is known as **marking to market**.
Basis	The price of a futures contract will normally be different from the spot price on any given date. This difference is called the **basis**. The exception is on the settlement date for the contract date, when the basis is zero (because the exchange settlement price and the spot market price are the same on this date). **Basis** = spot price – futures price The effect of basis is to **prevent** hedges from being 100% efficient, and the change in the spot **over a period** is not matched exactly by the change in the futures price. This means that there is basis risk for anyone hedging a position with futures. You will not be asked to calculate basis in the examination, but just to recognise that it causes hedges to be less than 100% efficient.
Hedge efficiency	The only risk to hedgers is that the futures market does not always provide a perfect hedge. (a) The first reason is that the value of the commodity being hedged (whether it is shares, currency or interest) must be **rounded to a whole number of contracts**, causing inaccuracies. (b) The second reason is **basis risk** as noted above. A measure of **hedge efficiency** compares the profit made on the futures market with the loss made on the cash or commodity market, or vice versa.

9.3.2 Long and short positions

(a) Someone who **buys futures opens a long position** in the contract, and will make a gain if the market value of the contract rises. This long position can be closed at any time before settlement date by selling an equal number of the same contracts.

(b) Someone who **sells futures opens a short position**, and will make a gain from any fall in the futures price. This short position can be closed at any time before settlement date by buying an equal number of the same contracts.

(c) Positions that are still open at the settlement date for the contract are settled by the futures exchange at a settlement price decided by the exchange. This will be the current spot price for the underlying item in the contract.

In practice, most positions in futures are closed before the settlement date for the contract.



Context example: Commodity futures – long position

1 July

(a) On 1 July the spot price of wheat is 410c per bushel.

(b) Miller plc knows that it wants to buy a consignment of 200,000 bushels of wheat on 30 September, which at the spot rate would cost $820,000. It decides to reduce its exposure to any price rise using the wheat futures market.

(c) It creates a long position on the futures market by buying three-month wheat futures contracts at 415c per bushel that expire on 30 September, to cost $830,000. (Wheat contracts are standardised at 5,000 bushels, so in this example Miller would buy 40 standard contracts in order to make up the total consignment of wheat that it requires.)

30 September

(a) Miller buys its physical consignment of wheat on 30 September from the wheat market. At this date the spot price is 420c per bushel, so the wheat will cost 420c × 200,000 bushels = $840,000.

(b) The futures contract in wheat is not settled by the physical exchange of wheat for cash. Instead, it is settled for the cash difference between the original price for buying the futures in July and the closing settlement price on the futures exchange at the end of September.

(c) The closing settlement price on the futures exchange will be the spot market price at 30 September. So Miller will sell its futures contracts for $840,000, having bought them for $830,000. It will make a gain of $10,000 (200,000 bushels × (420c – 415c)).

(d) This gain on the futures position can be set against the spot purchase it made in the wheat market.
Net cost = $840,000 – $10,000 = $830,000
This means that the effective net price that Miller has paid for the wheat is the $830,000 price that it contracted in July, when it opened its futures position.



Context example: Commodity futures – short position

1 July

(a) On 1 July the spot price of cotton is 70c per pound weight.

(b) Field plc knows that it wants to sell a consignment of 1,000,000 pounds weight of cotton on 30 September, which at the spot rate would give it $700,000. It decides to reduce its exposure to any price falls between July and September using the cotton futures market.

(c) It creates a short position on the futures market by selling 1,000,000 pounds weight of cotton in three-month cotton futures at 65c per pound weight that expire on 30 September, to receive $650,000. (Cotton contracts are standardised at 50,000 pounds weight, so in this example Field would sell 20 standard contracts in order to make up the total consignment.)

30 September

(a) Field sells its physical consignment of cotton on 30 September to the cotton market. At this date the spot price is 60c per pound weight, so the cotton will fetch 60c × 1,000,000 pounds = $600,000.

(b) The closing settlement price on the futures exchange will be the spot market price at 30 September. So Field will buy its futures contracts for $600,000, having sold them for $650,000. It will make a gain of $50,000 (1,000,000 pounds weight × (65c – 60c)).

(c) This gain on the futures position can be added to the spot sale it made in the cotton market. Net proceeds = $600,000 + $50,000 = $650,000

This means that the effective net price that Field has received for the cotton is the $650,000 price that it contracted in July, when it opened its futures position.

Context example: Currency futures

A company may wish to sell £740,000 in two months' time in exchange for US dollars. It could do this by arranging a forward contract with a bank. Alternatively it could deal in British £ sterling futures.

Each futures contract is for £62,500, so to sell £740,000 it would need to sell 11.84 contracts – ie, 12 contracts.

It might sell the contracts at a rate of, say, $1.5200 = £1, creating a short position of 12 futures contracts. After 2 months it will close the position by purchasing 12 contracts. Suppose the price to buy the contracts is $1.5050 = £1. It will make a gain on the futures dealing of $0.0150 per £1, or ($0.0150 × 62,500 × 12 contracts) = $11,250 in total.

It will sell the £740,000 in the spot FX market, at whatever the spot price happens to be, and will receive $11,250 in profit from the futures exchange (which takes on responsibility for the settlement of all futures contracts).

Worked example: Interest futures

Nicol Inc will shortly be making a short-term investment and wants to borrow $5 million for three months, starting in three months' time. However, interest rates are currently volatile and it is worried about adverse movements in these rates before it takes out the loan.

Nicol Inc is considering using interest rate futures to hedge the $5 million loan. It is now 1 March and the current futures prices are (for standard $1 million three-month contracts):

March delivery	97.00
June delivery	97.10
September delivery	97.20

These contracts will expire at the end of the quoted months. The spot 3-month SONIA rate is 2.5% and Nicol can borrow at a rate of approximately SONIA + 1%.

At 1 June, the price of the June futures contract is 96.07 and the price of the September futures contract is 96.10.

Requirement

Assuming that Nicol takes out a loan at a rate of SONIA + 1% on 1 June, illustrate the outcome of hedging with the futures contract if SONIA at that date is 3%.

Solution

Set up

Sell futures to hedge against the risk of an increase in the interest rate.

Choose the futures contract with the next delivery date/settlement date following the date when it is intended to close the position. In this case the company should sell June futures on 1 March.

$$\text{How many contracts? } \frac{\text{Exposure}}{\text{Contract size}} \text{ X } \frac{\text{Loan period}}{\text{Length of contract}} = (\$5\text{m}/\$1\text{m}) \times (3/3) = 5 \text{ contracts}$$

Closing the position on 1 June

Original sale price on 1 March	97.10
Buying price to close the position on 1 June	96.07
Gain per contract	1.03

Total gain on futures = 103 ticks × $25 per tick × 5 contracts = $12,875.

	%		$
Interest cost on loan (SONIA + 1%)	4.00	($5,000,000 × 4% × 3/12)	50,000
Gain on futures contract	(1.03)		(12,875)
Net cost	2.97		37,125

9.3.3 Index futures

Index futures are futures contracts on a portfolio of shares represented by a stock market index.

These can be used to protect investors against future falls in the value of a portfolio of shares. They can be of real importance to organisations with significant investments, such as pension funds.

There are futures on the FTSE100 share index. The **transaction costs** of investing in these futures are much less than the costs of actually selling and buying the underlying shares. Buying or selling futures is an alternative to adjusting a share portfolio, which may not be appealing because of the time and cost involved.

Each FTSE100 index futures **contract** is for a notional value of the futures price multiplied by £10. Thus, if the FTSE index futures price stands at 6,700, the notional value of a contract is £67,000.

Index futures can be useful to an investor in a number of ways, either for **speculative purposes** or as a **'hedge'** against risk of adverse movements in stock market prices generally.

Index futures are used by speculators because it is possible to take a very large position in share prices for a relatively small cost. With futures, there is a deposit to pay (the margin) for having an open or closed position, but the cost is much less than buying the underlying shares themselves. The speculator hopes to make a big profit from a favourable movement in the stock market index for a relatively small investment.

Worked example: Index futures I

The investment manager of Moonstar Pension Fund is concerned that share prices may fall over the next month and wishes to hedge against this using FTSE100 June stock index futures. The fund's pension portfolio comprises investments which have a market value of £7 million on 1 June 20X8.

On 1 June 20X8 the following prices are observed:

The prevailing value (ie, spot value) of the FTSE100 index is 7,000

The price quote for June FTSE100 index futures is 6,970

The face value of a FTSE100 index contract is £10 per index point.

Using the futures price of the FTSE100 index, this gives a contract value of 6,970 × £10 = £69,700

Requirement

Demonstrate what hedge should be undertaken to protect the portfolio against a fall in share prices.

Solution

Calculate number of contracts

Sell futures to protect the value of the portfolio.

If the stock market index falls, the investment manager will make a profit by closing the futures position and buying futures at a lower price than the original selling price. The gain on the futures position will offset the loss incurred from the fall in the stock market index (and so a fall in the value of shares in the pension fund's portfolio).

$$\text{Number of contracts} = \frac{\text{Market value of portfolio}}{\text{Value of one contract}}$$

= £7,000,000/£69,700

= 100.4 rounded to 100 contracts

Worked example: Index futures II

On 30 June 20X8, the settlement date for the June futures contract, the FTSE100 index and the futures settlement price had fallen to 6,700. The market value of the shares in the portfolio changed exactly in line with the change in the index ie, (6,700/7,000) × £7,000,000 = £6.7 million.

Requirement

Calculate the outcome of the hedge that Moonstar has undertaken.

Solution

Step 1

Position in spot market

Loss on portfolio = £6.7 million – £7 million

= £0.30 million

Step 2

Calculate gain or loss on futures

Buy futures at lower price than they were sold for (closing out). The price to close out the position is 6,700.

Gain on futures = (6,970 – 6,700) × £10 ×100 contracts

= £270,000

Step 3

Calculate net position

Net position = £270,000 gain on futures – £300,000 loss on share portfolio

= £(30,000) loss overall

Note: The hedge is less than 100% effective because of basis (ie, the 1 June FTSE index value and the futures price on 1 June are different) and the rounding of the number of contracts.



Worked example: Index futures III

Suppose in the previous example that on 30 June 20X8, the market value of the shares in the portfolio had risen to £7.2 million, and the FTSE100 index and the index futures settlement price were both 7,200 on that date.

Requirement

Explain what would happen as a consequence of the hedge.

Solution

Step 1
Position in spot market

Gain on portfolio = £7.2 million – £7 million = £0.2 million

Step 2

Calculate gain or loss on futures

Initially sold futures for 6,970

Now buy futures for 7,200
Loss on closing out futures (230) × £10 × 100 contracts = £230,000

Step 3

Calculate net position

Net position 200,000 gain on portfolio

(230,000) loss on futures

£(30,000) loss overall

9.3.4 The impact of a futures hedge

The objective of a futures hedge is to remove price risk by fixing the price in advance. The benefit is that downside risk is eliminated; the disadvantage is that any upside is also removed.

To summarise the above worked examples:

	If market rises	If market falls
Value of portfolio	7,200,000	6,700,000
Gain (loss) on futures market	(230,000)	270,000
Net effect	£6,970,000	£6,970,000

9.3.5 Forwards versus futures

The differences between forward contracts and futures contracts are summarised in the following table.

Attribute	Futures	Forward contracts
Traded	Exchange traded	OTC
Quality and quantity	Standardised by the exchange for all products	Specified in the contract
Delivery dates	Standard fixed settlement dates, but positions can be closed at any	Specified in the contract
	time before the settlement date	
Liquidity/ability to close out	Generally good liquidity/easy to close out	May be limited
Counterparty risk	None due to the workings of the clearing and settlement system	Default risk exists: risk of default by bank
Costs/margin	Relatively low initial costs (margin)	Arrangement costs specifically agreed; no payments for transaction until future settlement date
Regulation	Significant regulation and investor protection	Less regulated

Professional skills focus: Concluding, recommending and communicating

You are expected to be able to make evidence-based recommendations which can be justified by reference to supporting data and other information. A recommended hedging strategy must be based on your technical knowledge and the entity specific information provided in the question.

9.4 Options

An option is a contract that gives its holder/buyer the right, but not the obligation, to buy or sell an underlying asset at a given price (the exercise price or strike price) on or before a given date (the expiry date for the option). If the rights under the option agreement are not exercised by a stated time on the expiry date, the option lapses and cannot be exercised.

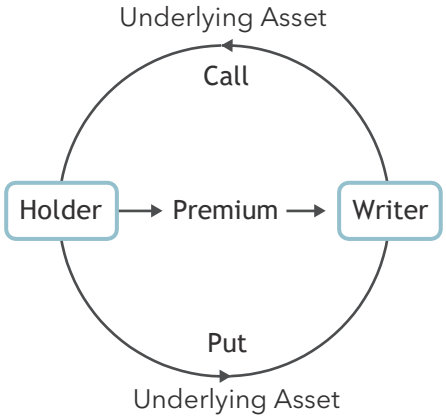
The right to buy the underlying item at the exercise price is known as a **call option**.

The right to sell the underlying item at the exercise price is known as a **put option**.

The rights to buy (call) or sell (put) are held by the person buying the option who is known as the **holder.**. The person selling an option is known as a **writer**. The following diagram shows the relationship between the holder and the writer.

Figure 13.5: The relationship between holder and writer in an option contract

Holder-Writer Relationship



The purchase cost of an option is known as its **premium**. It is paid by the holder to the writer: the amount of the premium will depend on the perceived value of the option rights that the option buyer is acquiring.

The holder is obliged to pay the premium to acquire the option, and the premium is payable even if the holder does not subsequently exercise the option. Depending on the option specification, the premium may be paid on the purchase of the option (**upfront**) or on exercise or expiry of the option (**onclose**).

In return for receiving the premium, the writer agrees to fulfil the terms of the contract, and sell the underlying item at the exercise price if a call option is exercised, and buy the underlying item at the exercise price when a put option is exercised. The holder has to pay a premium because the holder can take advantage of the upside risk without exposure to the downside risk.

Options are available in a range of different **exercise styles** which are specified when the options are traded. There are a number of potential styles:

(a) **American style options** – in which the option can be exercised by the holder at any time after the option has been purchased.

(b) **European style options** – when the option can only be exercised on its expiry date.

(c) **Asian style options** – when the option is exercised at the average underlying price over a set period of time. Similar to these are Traded Average Price Options. These are fixed strike price, but are exercised against an average underlying price.

A person describing an option will specify its:

- underlying asset
- expiry date
- exercise price
- call/put

For example, Cocoa (**Underlying asset**)

May (**Expiry date**)
600 (**Exercise price**)
Call (**Call/Put**)

9.4.1 Exercising options

The holder of a call option will exercise the option at (or before) expiry only if the exercise price for the option is more favourable than the current market price of the underlying asset.

When the exercise price is more favourable than the market price of the underlying item, the option is said to be 'in the money'.

When the exercise price is less favourable than the market price of the underlying item, the option is said to be 'out of the money'. An option will not be exercised if it is out of the money.

(a) A call option will be exercised if the exercise price is less than the current market price of the underlying asset, because the option holder can use the option to buy the asset for less than its current market value.

(b) A put option will be exercised if the exercise price is higher than the current market price of the underlying asset, because the option holder can use the option to sell the asset for more than its current market value.

9.5 Factors affecting option value

The option premium is the price paid to buy an option, but the value of an option changes over time between its creation and its expiry date. The value of a **call option** at any time up to expiry depends on:

(a) **Current asset price**. If the market price of the underlying asset rises, the value of a call option will increase. For currency options, the relevant 'price' is the spot exchange rate.

(b) **Exercise price of the option**. The higher the exercise price, the lower the value of a call option.

(c) **Asset price volatility or standard deviation of return on underlying share**. The higher the volatility in the market price of the underlying asset (ie, measured statistically as a standard deviation), the higher the value of a call option, because there is more likelihood that the asset price will rise above the option price.

(d) **Time to expiry of the option**. The longer the period to expiry, the higher the value of a call option because there is more time for the asset price to rise above the option's exercise price.

(e) **Risk-free rate of interest**. The higher the risk-free rate of interest, the higher the value of a call option. As the exercise price will be paid in the future, its PV diminishes as interest rates rise. This reduces the cost of exercise and thus adds value to the option.

9.5.1 Share options

The widespread use of derivatives involving options has resulted in much attention being paid to the valuation of options for financial reporting purposes and for measuring profits or gains. Exchange- traded share options (equity options) are a common form of option, giving the right but not the obligation to buy or sell a quantity of a company's shares at a specified price within a specified period.

The value of an equity option is made up of:

- 'intrinsic value'
- 'time value'

The intrinsic value of an option depends on:

- share price
- exercise price

If the option is in the money and the current market price of the underlying shares is higher than the exercise price of a **call option** on the shares, the intrinsic value of the option is the difference between the market price and the exercise price. If the option is out of the money and the exercise price is higher than the current market price of the underlying asset, the intrinsic value of a call option is zero.

The time value of an option is based on the probability that the option is affected by:

- time period to expiry
- volatility of the underlying security
- general level of interest rates

9.5.2 Summary of determinants of option prices

Summary of determinants of option prices		
↑ in	**Call price**	**Put price**
Share price	↑	↓
Exercise price	↓	↑
Volatility	↑	↑
Time to expiry	↑	↑
Risk-free rate of return	↑	↓

9.6 The Black-Scholes formula

The Black-Scholes formula for the valuation of European call options was developed in 1973 (European options are options that can only be exercised on the expiry date, as opposed to American options, which can be exercised on any date up until the expiry date). The formula is based on the principle that the equivalent of an investment in a call option can be set up by combining an investment in shares with borrowing the PV of the option exercise price.

The formula requires an estimate to be made of the volatility in return on the shares. One way of making such an estimate is to measure the volatility in the share price in the recent past and to make the assumption that this volatility will apply during the life of the option.

The formula is part of the wider **Black-Scholes-Merton model**, which models a financial market containing derivative investment instruments.

9.6.1 Assumptions of the Black-Scholes formula

In order to incorporate **volatility** and the **probabilities** of option prices into the formula, the following assumptions are needed:

- Returns are normally distributed.
- Share price changes are log-normally distributed.
- Potential price changes follow a random formula.
- Volatility is constant over the life of the option.

The Black-Scholes formula is also based on the following other important assumptions:

- Traders can trade continuously.
- Financial markets are perfectly liquid.
- Borrowing is possible at the risk-free rate.
- There are no transaction costs.
- Investors are risk neutral.

9.6.2 The Black-Scholes formula

The Black-Scholes formula for the value of a **European call option** is given by: $c = P_a N(d_1) - P_e N(d_2)e^{-rt}$

Where: P_a = current price of the underlying asset

P_e = exercise price

r = continuously compounded risk-free rate

t = time to expiration as a fraction of one year, (eg, t = 0.5 means 6 months)

e = base of the natural logarithms and is the constant value 2.7182818

N(x) = normal probability distribution up to a value of x

$$d_1 = \frac{In\left(\frac{P_a}{P_e}\right) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

Where: s = the volatility of share price movement (expressed as a standard deviation)

$In\left(\frac{P}{P_e}\right)$ is the natural logarithm of the spot price over the exercise price.

(A natural logarithm is the power to which 'e' would have to be raised, to equal the number in question. For example, the natural logarithm of 10 is 2.302585 because 10 = e2.302585.)

9.6.3 Value of European put options

The value of a **European put option** can be calculated by using the Put Call Parity.

$p = c - P_a + P_e e^{-rt}$

Where: p = value of the put option

c = value of the call option

To calculate the value of a put option, we can calculate the value of a corresponding call option, and then obtain the value of the put option using the put-call parity equation.

9.6.4 Value of American call options

Although American options can be exercised any time during their lifetime it is **never optimal** to exercise an option earlier. The value of an American option will therefore be the same as the value of an equivalent European option and the Black-Scholes formula can be used to calculate its price.

9.6.5 Value of American put options

Unfortunately, no exact analytic formula for the value of an American put option on a non dividend paying stock has been produced. Numerical procedures and analytic approximations for calculating American put values are used instead.



Worked example: Black-Scholes formula

In this example, the Black-Scholes formula is used to calculate the value of an option on shares.

Requirement

Consider the situation where the share price six months from the expiry of an option is $42, the exercise price of the option is $40, the risk-free interest rate is 10% p.a. and the volatility is 20% p.a.

This means $P_a = 42$, $P_e = 40$, $r = 0.10$, $s = 0.20$, and $t = 0.5$.

Solution

$$d_1 = \frac{\ln(42/40) + (0.1 + \frac{0.2^2}{2})0.5}{0.2\sqrt{0.5}} = 0.7693 = 0.77$$

$$d_2 = 0.7693 - 0.2 \times \sqrt{0.5} = 0.6279 = 0.63$$

and

$P_e e^{-rt} = 40e^{-0.1 \times 0.5} = 40e^{-0.05} = 38.049$

The values of the standard normal cumulative probability distribution: these can be found from normal distribution tables and are:

N(0.77) = 0.7794

N(0.63) = 0.7357

Hence if the option is a European call, its value is given by:

c = (42 × 0.7794) – (38.049 × 0.7357) = 4.76

If the option is a European put, its value is given by:

p = 4.76 – 42 + 38.049 = 0.81

The share price has to rise by $2.76 for the purchaser of the call to break even. Similarly, the share price has to fall by $2.81 for the purchaser of the put to break even.



Interactive question 4: Black-Scholes formula

The current share price of Cathlynn plc is £3.50. Using the Black-Scholes formula, estimate the value of a European call option on the shares of the company that has an exercise price of £3.30 and three months to run before it expires. The risk-free rate of interest is 8% and the variance of the rate of return on the share has been 12%.

See **Answer** at the end of this chapter.

9.6.6 Limitations of Black-Scholes formula

(a) The formula is **only designed** for the valuation of **European call options**.

(b) The basic formula is based on the assumption that **sharespay no dividends**.

(c) The formula assumes that there will be **no transaction costs**.

(d) The formula assumes knowledge of the **risk-free rate of interest,** and also assumes that the risk- free rate will be constant throughout the option's life.

(e) Likewise, the formula also assumes accurate knowledge of the **standard deviation of returns**, which is assumed to be constant throughout the option's life.

9.6.7 Accounting requirements

The pricing of options using Black-Scholes must be considered in the light of the requirements of IFRS 13, *Fair Value Measurement*.

The Black-Scholes formula contains assumptions that correspond to different levels of the IFRS 13 hierarchies. This is most likely to be an issue for volatility. Historical volatility, volatility derived from historical prices, would be classed as a Level 3 input. Historical volatility generally does not represent current market participants' expectations about future volatility. It may be possible to gain stronger,

Level 2, evidence, for example by observing prices for options with varying terms and extrapolating from those.

9.7 Monte Carlo simulation model

The Monte Carlo simulation model is widely used in situations involving uncertainty. It is particularly useful for non-traded derivatives, where the Black-Scholes formula is not appropriate. The method amounts to adopting a particular probability distribution for the uncertain (random) variables that affect the option price and then using simulations to generate values of the random variables.

To deal with uncertainty the Monte Carlo method assumes that the uncertain parameters or variables follow a specific probability distribution.

The basic idea is to generate through simulation thousands of values for the parameters or variables of interest and use these variables to derive the option price for each possible simulated outcome. From the resulting values we can derive the distribution of the option price.

The Monte Carlo simulation model is particularly relevant for IFRS 2, *Share-based Payment*. This standard requires that share-based transactions are recognised in the financial statements at their fair value at the grant date.

One of the major challenges of this statement is measuring the value of the equity instruments (such as share options). The statement requires such instruments to be included in the financial statements at fair value, which should be based on market prices if available. If market prices are not available, the Monte Carlo simulation model is often used to estimate the fair value of the instruments.

9.8 Swaps

A swap is an OTC contract that commits two counterparties to exchange, over an agreed period, two streams of cash or commodities. The contract defines the dates when the cash flows are to be paid and the way in which they are to be calculated. Usually the calculation of cash flows involves the future values of one or more market variables.

Examples of swaps include:

(a) **Interest rate** – where the two payments exchanged are calculated using different interest rates (discussed in the chapter Financial risk management).

(b) **Currency** – where two streams of interest payments in different currencies are exchanged (discussed in the chapter Financial risk management).

(c) **Equity** – where one cash flow is based on a reference interest rate such as SONIA, the other is based on the performance of an equity, a basket of equities or an equity index.

(d) **Commodity** – where one party offers at a fixed single price a series of cash-settled future contracts which the other party will buy or sell at an agreed date. Cash settlement is calculated on the difference between the fixed price and the price of an agreed index for a stated notional amount of the underlying asset.

(e) **Credit swaps** – discussed below.

9.9 Credit derivatives

Definition

Credit derivative: A derivative whose value is derived from the credit risk associated with an asset.

Participants in the credit derivatives market include commercial banks, investment banks, insurance companies, hedge funds, pension funds and other corporate entities.

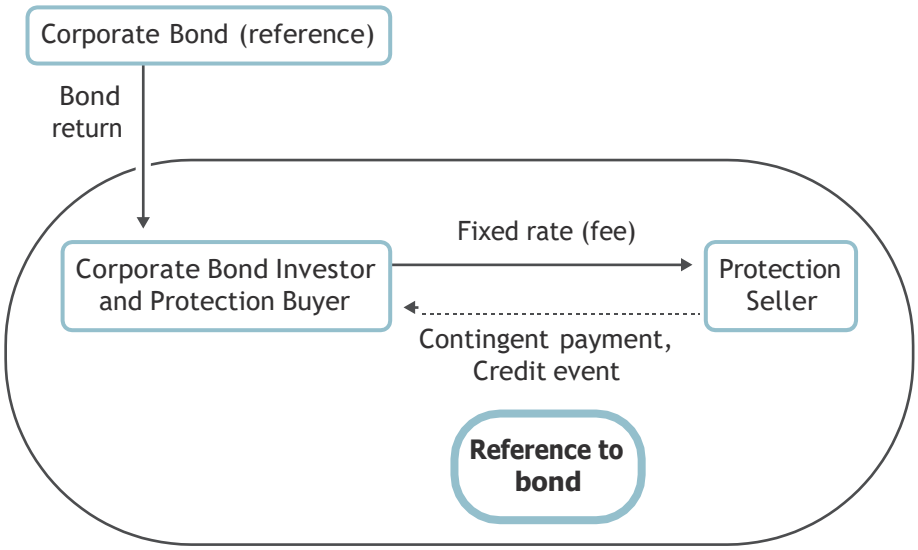
In its simplest form, a credit derivative is very similar to an insurance policy on an amount of debt. The buyer of the derivative pays a fee (in a lump sum or in periodic payments) to the seller. In return the seller undertakes to make a compensation payment to the buyer if there is a default on the underlying debt.

For example, an investor may buy a credit derivative on £5 million nominal value of a fixed rate bond issued by ABC plc. If ABC plc defaults on a payment of the bond before the expiry date of the credit derivative, the seller of the derivative must pay an amount in compensation to the buyer.

The credit derivative is therefore a way of buying protection against the risk of default on a debt. It is also a way of speculating that the borrower, in this case ABC plc, will default. An investor does not need to have lent any money to ABC plc in order to buy a credit derivative and speculate on default. This makes credit derivatives different from a traditional insurance policy.

The key concept underlying the credit derivative market is isolation. The buyer of a credit derivative wishes to 'isolate' the credit risk of an underlying security from its other key risks, interest rate and foreign exchange risk.

Figure 13.6: Credit derivatives



Credit derivatives are divided into two main categories: those which are funded and those which are unfunded. The funded instruments include credit linked notes and collateralised debt obligations. They are unique because the purchaser of the derivative is buying 'protection' and the cash flows from the underlying security together. In an unfunded credit derivative only the protection is purchased.

One of the most important aspects of credit derivatives is to define exactly what situations will result in the 'insurance' policy being triggered. These are known as 'default events'. The International Swaps and Derivatives Association (ISDA) has defined five 'default events' within its ISDA 2003 Credit Derivatives Definitions:

- **Bankruptcy** – Reference entity voluntarily or involuntarily files for bankruptcy or insolvency protection.
- **Failure to pay or default** – Failure of the reference entity to make aggregate payments due (at least US$1 million) following a grace period.

- **Restructuring** – Reference entity agrees to, or announces, a change in terms of an obligation (minimum US$10 million) as a result of deterioration in the financial condition of the reference entity.
- **Repudiation/Moratorium** – Any change in the owner status or independence of the asset or company is considered a credit event, as is any significant change in the asset ownership structure.
- **Obligation default** – A situation where one or more obligations have become capable of being declared due and payable before they would otherwise have been due and payable.

If a credit event does take place the seller of the 'protection' will be contractually obliged to settle the transaction by either taking '**physical delivery**' of the underlying asset from the buyer or via **cash settlement** which is paid to the buyer. An independent valuer will then assess the residual value of the reference asset with the difference between that value and nominal value being paid to the protection buyer.

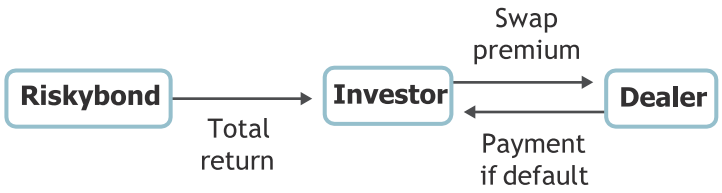
Another important aspect of credit derivatives is the pricing mechanism and how it changes as the credit quality of the underlying security changes. Market participants have developed models that track the credit rating of the underlying corporate and adjust the value of the derivative accordingly. As credit ratings rise, credit derivative values fall and vice versa. These **credit transitions** are pivotal to the effective valuing of the underlying derivative.

9.9.1 Credit default swaps (CDSs)

Credit default swaps (CDSs) are the most common form of credit derivative. They are mostly based on a single asset. Effectively a purchase of a credit derivative is buying an insurance policy against a credit event, such as bankruptcy, that will affect the value of the underlying security.

The spread of a CDS is the annual amount the protection buyer must pay the protection seller over the length of the contract (like an insurance premium), expressed as a percentage of the notional amount. The more likely the risk of default, the larger the spread. For example, if the CDS spread of the reference entity is 50 basis points (or 0.5%) then an investor buying $10 million worth of protection from a bank must pay the bank $50,000 per year. These payments continue until either the CDS contract expires or the reference entity defaults.

Figure 13.7: Credit default swaps



Unlike insurance, however, CDSs are unregulated. This means that contracts can be traded – or swapped – from investor to investor without anyone overseeing the trades to ensure the buyer has the resources to cover the losses if the security defaults.

CDSs are often used to manage the credit risk that arises from holding debt. For example, the holder of a corporate bond may hedge their exposure by entering into a CDS contract as the buyer of protection. If the bond goes into default, the proceeds from the CDS contract will cancel out the losses on the underlying bond.

CDS markets

An original CDS can go through as many as 15 to 20 trades. Therefore, when a default occurs, the so- called 'insured' party or hedged party does not know who is responsible for making up the default or indeed whether the end party has the funds to do so.

When the economy is booming, CDSs can be seen as a means of making 'easy' money for banks. Corporate defaults in a booming economy are few, thus swaps were a low-risk way of collecting premiums and earning extra cash.

The CDS market expanded into **structured finance** from its original confines of municipal bonds and corporate debt and then into the secondary market where **speculative investors** bought and sold the instruments without having any direct relationship with the underlying investment. Their behaviour was almost like **betting** on whether the investments would succeed or fail.

Context example: CDSs

At one time credit derivatives were a relatively obscure financial instrument for banks and bondholders. However, CDSs gained prominence during the global financial crisis due to the activities in the market of American International Group (AIG).

AIG – the world's largest insurer – could sell CDSs without putting up any real collateral as long as it maintained a triple-A credit rating. There was no real capital cost to selling these swaps; and there was no limit. Thanks to fair value accounting, AIG could book the profit from, say, a five-year CDS as soon as the contract was sold, based on the expected default rate. In many cases, the profits it booked never materialised.

When the financial crisis began in the US in 2007, the prospect of default by many banks and other corporate borrowers increased significantly. On 15 September 2007 the bubble in the CDS market burst when all the major credit rating agencies downgraded AIG, due to the soaring losses in its CDSs. The first big write-off came in the fourth quarter of 2007, when AIG reported an $11 billion charge. It was able to raise capital once, to repair the damage. But the losses kept growing. The moment the downgrade came, AIG was forced to come up with tens of billions of dollars in additional collateral, immediately. This was on top of the billions it owed to its trading partners. It didn't have the money. The world's largest insurance company was bankrupt.

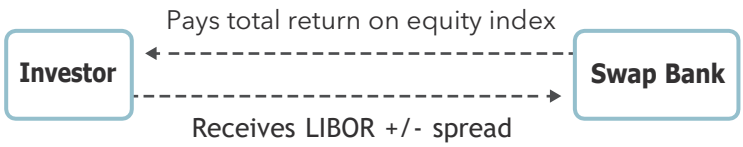
As soon as AIG went bankrupt, all the institutions that had hedged debt positions using AIG CDSs (or which had speculated on defaults by various borrowers) had to mark down the value of their assets. For lending banks holding AIG CDSs, this meant a reduction in their ability to lend. Investment banks had no ability to borrow, as the collapse in the CDS market meant that no one was willing to insure their debt.

9.9.2 Total return swaps (TRSs)

A total return swap (TRS) is a type of swap in which one counterparty agrees to pay a floating reference rate such as SONIA and in exchange will receive the total return, interest and capital gains, generated by an underlying asset. This transaction is specifically designed to transfer the credit risk and the market risk of an underlying security to the counterparty and effectively enables the party to derive an economic benefit from an underlying asset without actually buying it.

TRSs can be used with any underlying asset that generates a return but are most commonly used in the equity markets.

Figure 13.8: Total return swaps



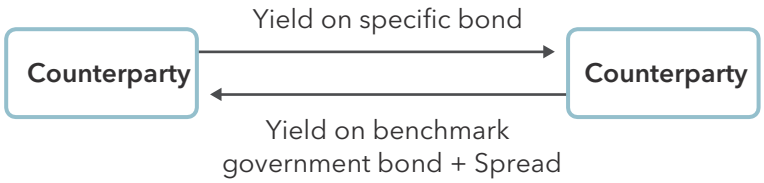
Note that if the equity markets actually fell in value, the receiver of the SONIA payments would be paying a negative amount of money to its counterparty. Hence the investor would actually be receiving money.

Bond markets also have two types of return associated with holding bonds: the price return and the coupon. TRSs are also increasingly popular in the credit derivatives market where the holder of a bond can not only swap the price exposure of a bond (based on coupon and price), but also the credit exposure can be swapped into the deal.

9.9.3 Credit spread swaps

A credit spread swap allows firms to trade smaller shifts in a borrower's credit rating short of outright default. One party makes payments based on the yield to maturity of a specific issuer's debt and the other makes payments based on the yield on a comparable sovereign bond plus a spread reflecting the difference in credit ratings between the bonds.

Figure 13.9: Credit spread swaps



If the above two flows are equal at inception of the swap, any change in credit rating of the specified bond relative to the benchmark will result in a payment being made from one party to the other.

This, like the total return swap, allows an investor to lock in a specific spread above a benchmark yield. One of the prime causes of the spread is the difference in credit rating between the corporate bond and the government bond.

Variations could see a swap between yields on a specific bond above a benchmark and the yield between two other bonds, or a swap on the spread on one security above a benchmark with a spread on another security above a benchmark.

9.10 Collateralised debt obligations (CDOs)

The overall market for asset-backed securities (ABSs) is known as the Collateralised Debt Obligation (CDO) market, where the ABS shifts credit risk on a pool of homogenous assets.



Definition

Collateralised debt obligation: is an investment-grade security that is backed or collateralised by a pool of actual bonds, loans or other assets.

CDOs are a common form of funded credit derivative. From the viewpoint of the issuer, a CDO is designed to remove debt assets (eg, mortgages, loans and credit card receivables), and their associated credit risk, from an originator's balance sheet in return for cash. The originator is typically a bank, seeking to remove assets from its balance sheet – either to reduce its regulatory capital requirements or to obtain more cash to make more loans.

To achieve this, the **assets (loans) are sold** by the originator to an **SPV** for cash, the cash having been raised by the SPV through the issue of bonds to investors. The interest on the SPV's bonds will be paid out of the cash flows from the debts that it has purchased from the originator.

Buyers of the SPV's bonds are therefore investing directly in a portfolio of banking assets (loans), but are not investing in the bank itself.

The value of the CDO should be directly linked to the value of the underlying securities that are used to collateralise it. Interest payments from the underlying securities will effectively be passed through to the CDO holders. Buyers of collateralised instruments can also participate in collateralised mortgage obligations and collateralised bond obligations, where the underlying assets held by the SPV are mortgage loans or bonds.

One of the most attractive aspects of investing in a CDO is that the issuer will normally issue a range of different classes of CDO for a single pool of assets. These classes are known as tranches and are split according to a credit rating. The higher the credit rating, the more senior the tranche. Losses will first affect the equity tranche, next the mezzanine tranches, and finally the senior tranche. Each tranche pays a periodic payment (the swap premium), with the junior tranches offering higher yields.

9.10.1 Synthetic CDOs



Definition

Synthetic collateralised debt obligation: An issued security like a CDO but instead of being backed by a pool of actual underlying assets it is backed by a pool of CDSs.

As discussed above, the value of a CDS is derived from the credit rating of the underlying asset and is therefore only a representation of that asset and not the asset itself. The CDO is therefore synthetic, as it is made up of a pool of these manufactured credit exposures.

From the issuer's viewpoint, the synthetic CDO acts to remove credit risk on assets that the issuer cannot or does not wish to sell. There may be adverse tax consequences from selling certain assets or legal restrictions on their sale, especially on certain foreign debt. A bank may not wish to sell a particular loan as it wishes to maintain a relationship with the client.

Like a cash CDO, the risk of loss on the CDO's portfolio is divided into tranches. A synthetic CDO issues various tranches of security to investors as with a normal CDO, but then invests any cash raised in a high-quality bond portfolio paying a safe, but low, return with minimal credit risk. They then boost this income by selling CDSs, thereby taking on the credit risk of the issuer's asset portfolio.

In a synthetic CDO, frequently some of the investors have an unfunded exposure to the risk of the credit default insurance provided, ie, they face exposure to the risk of the loss on an asset but have not contributed the associated capital which they may need to if the credit event arises. Arguably, the unfunded nature of many synthetic CDO structures has exacerbated the impact of debt defaults in a manner that would not arise with a funded CDO structure, thereby exaggerating the effects of the credit crisis.

9.11 Limitations of using derivatives

It might be assumed that derivatives should always be used to hedge interest rate and foreign exchange rate risk. However, in many instances organisations could have perfectly legitimate reasons for not using derivatives or forward contracts.

9.11.1 Cost

Many organisations will be deterred from using derivatives by the costs, including transaction costs and possibly brokerage fees. Companies may feel that the maximum losses if derivatives are not used are too small to justify incurring the costs of derivatives, either because the size of

transactions at risk is not particularly large, or because the risks of large movements in interest or exchange rates is felt to be very small. Using natural hedging methods, such as matching receipts and payments in the same currency as far as possible, avoids the costs of derivatives. Matching may be particularly important for countries trading in Europe with countries that use the euro.

9.11.2 Attitudes to risk

The desirability of hedging must also be seen in the wider context of the organisation's strategy and its appetite for taking risks. For example, directors may be reluctant to pass up the chance of making exchange gains through favourable exchange rate movements. They may therefore choose the riskier course of not hedging, rather than fixing the exchange rate through a forward contract and eliminating the possibility of profits through speculation.

There will also be occasions when a company's treasurer takes the view that exchange rates or interest rates will move favourably rather than adversely; or that they will remain stable for the foreseeable future. In these circumstances, a decision may be taken that hedging the risk is unnecessary, and the company should wait until it needs to make the transaction before dealing at the spot rate at that time.

9.11.3 Uncertainty over payments and receipts

There may not only be uncertainties over movements in rates. Uncertainties may also affect the amount and timing of settlement of the transactions being hedged. Uncertainties over timing may be particularly significant if the hedging transaction has to be settled on a specific date, for example an exporter having to fulfil a forward contract with a bank even if its customer has not paid.

9.11.4 Lack of expertise

If a business is making a lot of use of derivatives, or using complex derivatives, the potential losses from poor decision-making due to lack of expertise may be unacceptably high. These must be weighed against the costs of obtaining specialist external advice or maintaining an in-house treasury function, in order to reduce the risks of poor management of hedging.

For smaller companies with only limited knowledge of the derivatives markets, the lack of expertise will also mean that the time and effort required to get involved in derivatives is not worth the potential benefits, because the risks involved are not excessive.

10 Derivative markets



Section overview

Derivatives can only exist when the market for the underlying asset or item allows buyers and sellers to freely influence the market price. Those underlying markets include the FX markets for exchange rates and money markets for short-term interest rates.

10.1 Markets

Derivative products have developed out of a desire of users in the underlying markets to manage their continuing price exposure in those markets. The main derivatives markets are:

- **foreign exchange** (FX)
- **money markets** (short-term interest rate markets)
- **fixed income markets** (government and non-government)

- **equity markets**
- **commodity markets** (eg, crude oil, metals and soft commodities)
- **credit markets**

10.1.1 Floor versus voice versus electronic

For the Exchange Traded Derivatives Markets (the 'ETD' markets), trading takes place both electronically and via open outcry, on centralised physical trading floors. The trend is towards more electronic marketplaces with the evolution of technology and the desire for banks and brokerage to use the most cost-effective trading platforms. Some marketplaces adopt both.

The OTC derivatives market, by contrast, does not have centralised trading floors. It consists of a combination of telephone brokering and electronic trading. Just like the ETD markets, the shift is towards electronic broking and trading.

10.1.2 Quote driven versus order driven

For derivatives markets, price discovery is generated though two different types of market systems. A quote-driven market is one where prices are determined by designated market makers or dealers.
This is sometimes described as a 'price-driven' market. An order-driven market is one where the buy and sell orders placed in the trading system result in the best bid price and lowest offer price, resulting in a transaction taking place.

The difference between these two market systems lies in what is displayed in the market in terms of orders and bid and ask prices. The order-driven market displays all bids and asks, while the quote- driven market focuses only on the bids and asks of market makers and other designated parties. For example, NYSE Liffe, which is an organised ETD market, is a centralised, order-driven market on which continuous trading is supported by brokers.

OTC derivatives trade via a combination of quote-driven and order-driven systems.

10.1.3 Price taker versus price giver

All quote-driven transactions around the world involve one participant providing a two-way price on request (a price at which the market maker is willing to buy and a price at which they are willing to sell). Clearly their sell price is higher. A perfect day's trading for a market maker is one where he or she has simultaneously entered into buy and sell orders throughout the day with matching volumes. At the end of the day, they will have earned the difference between the buy price and the sell price on the volume that they have traded. The market maker is known as the price giver. Their counterparty to the deals is known as the 'price taker'. The price giver will always aim to make a profit that is equal to the difference between the buy price and the sell price that they are quoting. The price taker (known as the 'client') must accept that they will always pay the higher price quoted when they are buying and sell to the lower price quoted by the market maker when they are selling.

Central banks, investment banks, brokerage houses and insurance companies tend to be the typical price givers. Asset managers, corporations and private clients tend to be the typical price takers.

10.2 OTC vs exchange-traded markets

As discussed above, derivatives may be either **exchange traded** (where there is an active secondary market accessible to participants) or contracts entered directly between counterparties (on the **OTC** market). The table below summarises the differences between the OTC and exchange-traded market.

	Exchange-traded	OTC
Quantity and accuracy of hedging	The quantity covered by each contract is determined by the exchange. Only whole contracts may be traded	Can be tailored to meet the need of the investor exactly
Quality	Defined by the contract specification	Can be varied according to need
Delivery/ expiry dates	Only allowed on dates fixed by the exchange	Can be tailored to meet the customer's needs
Liquidity	Generally good, but dependent on the product	May be limited; market may be made by just one firm
Counterparty risk	No counterparty risk to the clearing member once contract registered	Risk exists of default, therefore credit rating of counterparty very important
Margin	Margin will normally be required	Normally no margin
Regulation and confidentiality	Subject to significant regulation and disclosure	Less actively regulated. May not be suitable for certain categories of customer

The differences between hedging using exchange-traded products and hedging on the open markets can be further summarised. (You may wish to refer back to this table when you have worked through the chapter Financial risk management, which deals with hedging.)

	Exchange-traded	OTC
Exposure characteristics	In the ETD market, it is much harder to match the exposure due to the standardisation of the contracts. Expiry dates are fixed, contracts have a minimum trading size and for commodities especially, the quality and location of the underlying delivery is fixed and may not match that of the exposure.	OTC hedges can be tailored to meet the exact exposure of the underlying position. This includes the size of the exposure, the timing of the exposure and also, if a commodity, the exact quality and location of delivery of the underlying commodity.
Cost of putting on the hedge/closing hedging	Fees are transparent with ETD contracts. A broker will outline exactly what they will charge a client for executing and clearing futures and options contracts.	With OTC transactions, the fees to transact them are disguised within the prices that are quoted. However, the more tailor-made the hedge, the higher the embedded costs will be to the hedger.
Ease of adjusting or closing the hedge	Successful futures contracts are extremely liquid and will have tight bid/ask spreads. Hence the cost of adjusting a hedge position or closing a hedge out is very small when executed on the ETD markets.	There are two options for a hedger to close down an OTC hedge position. Firstly and most obviously, they can go back to the counterparty that they initiated the hedge with. They are then at the mercy of the price offered by that counterparty.

	Exchange-traded	OTC
		In an illiquid marketplace, or if the hedge is very unique in its construction, then a hedger will have no option but to trade with this counterparty. Secondly, if the hedge is of straightforward construction, the trader will be able to ask for quotes from other banks and perhaps trade an equal and opposite OTC deal to their original hedge with the new counterparty. Having two counterparties will obviously increase the administration of the deals and use two credit lines which may not be attractive for the trader's company.
Counterparty exposure vs	Trading directly on the exchange as a clearing member effectively	All OTC transactions that are bilateral agreements involve counterparty
centralised clearing	eliminates any counterparty exposure as they have the clearing house as counterparty. Of course, if the trader is simply a client of a member of the exchange, they will have counterparty exposure to that clearing member.	exposure. The credit rating of the trader's company will thus have an impact on the prices that he is able to achieve with the OTC counterparties.
Price transparency	Exchanges sell real-time prices to quote vendors who then disseminate those prices into dealing rooms. Hence all dealers and traders know exactly what the current prices are at all times.	The OTC market has no formal price discovery process if a trader wishes to trade direct with a counterparty. A certain amount of price discovery exists when traders use OTC brokers/IDBs to ascertain the prices available for trading. However, those OTC brokers only collate and quote prices for 'standardised OTC' contracts. If a trader wants to trade a more complex hedge, they are at the mercy of the OTC counterparty for pricing.
Best execution	When a trader leaves an order with an ETD broker or dealer, they expect best execution.	Best execution can be achieved by OTC brokers for standardised OTC contracts. For anything more complex, a trader will simply have to compare quotations themselves from willing counterparties.

	Exchange-traded	OTC
Documentation	ETD documentation is more straightforward as the relationship with the exchange is standardised therefore no Master Agreements are required. Nevertheless, there still needs to be legal confirmation of each transaction and this is provided by contract notes, which are similar in content to the term sheets produced for OTC derivatives.	OTC transactions between counterparties are governed by the framework detailed in the Master Agreement. The individual transactions each have a trade confirmation or 'term sheet' which serves as a record.
Settlement mechanism	For exchange-traded derivatives all transactions are centrally cleared by the clearing house. In the UK LCH. Clearnet is the key clearer and facilitates the margin procedures and delivery/expiry processes. All transactions on exchange will be guaranteed by the clearing house.	Settlement of OTC transactions can be completed either bilaterally between the two counterparties or via a clearing house such as SwapClear. In the former, the counterparties deal directly with each other and are responsible for managing credit risk, overseeing cash flow transfers and facilitating settlement. In the latter, the entire process is managed and guaranteed by the clearing house.

10.2.1 Features of exchange trading

Futures and option contracts are traded on an organised exchange in a centralised location. These exchanges are also regulated by the local regulatory bodies in each country and are described as Recognised Investment Exchanges. They charge a membership fee and levies on contracts traded to cover the operational costs of the business.

10.2.2 Features of OTC markets

The process of agreeing an OTC transaction is normally conducted over the phone directly with the counterparty or via an IDB. Once verbally agreed the contract terms will be sent to each counterparty for confirmation. Historically, OTC derivatives have been time consuming and therefore expensive to process because each trade would be unique with different terms and conditions and therefore had to be confirmed individually, then settled and finally cash flows reconciled. However, the advent of technology and the semi-commoditisation of certain portions of the OTC derivatives markets has facilitated the introduction of several trade processing systems which are being used in the OTC market. These include MarkitWire, SwapClear and DTCC Deriv/Serv.

OTC products can present considerable **counterparty risk exposure**. To minimise this risk, participants normally undertake a lengthy due diligence investigation of their counterparty to avoid entering into contracts that may be subject to default. There is also a growing use of central counterparty clearing services provided by LCH. Clearnet, which accepts certain OTC contracts as intermediary, protected by a margining system. Credit risks are also minimised through the netting of cash flows in products such as interest rate swaps and, more broadly, by favouring products that are settled on a difference rather than by delivering the physical underlying.

Counterparties to an OTC product are at risk if either the paperwork does not operate as envisaged in the case of a default, or if one party is acting in a capacity for which it is not

authorised. The documentary risk is less intense than before, as the ISDA has produced a standard master agreement to cover most bilateral OTC derivatives transactions.

Valuing OTC positions can be difficult. While exchange-traded markets provide daily mark to markets against authoritative settlement prices, many OTC products are valued by the issuing bank. This may not provide the objectivity required by many trustees and custodians.

10.3 Participants in markets

For markets to operate in the most efficient and effective way possible (which is what market users are looking for), some marketplaces require additional market participants to help with access, execution and clearing services. These market participants include:

(a) **Intermediaries** – IDBs. These participants are the lifeblood of the OTC derivatives markets. They provide price discovery services and quotation for the buyers and sellers.

(b) **Primebrokers** – These are brokers who act as settlement agents, provide custody for assets, provide financing for leverage, and prepare daily account statements for its clients, who are money managers, hedge funds, market makers, arbitrageurs, specialists and other professional investors. They often also offer securities lending services for their hedge fund clients.

(c) **Futures commission merchants** – This is mostly a generic US term for an exchange traded clearing and/or execution broker. They must be registered by the Commodity Futures Trading Commission in the US.

(d) **Execution broker** – This is a brokerage corporation or individual who executes client orders on a derivatives exchange.
Note: The execution and clearing broker could be the same company or a separate company.

(e) **Clearing broker** – This is a brokerage/bank which is normally a member of an organised derivatives exchange. They offer clearing and settlement services internally and to clients who execute transactions on a derivatives exchange.

11 Financial reporting and financial instruments



Section overview

- The use of financial derivatives has implications for financial reporting and profitability. The three accounting standards that currently apply to financial instruments including derivatives are IAS 32, IFRS 9 and IFRS 7.
- IFRS 16 deals with accounting for leases.

There are three accounting standards on financial instruments:

(a) IAS 32, *Financial Instruments: Presentation*, which deals with:

(1) the classification of financial instruments between liabilities and equity

(2) presentation of certain compound instruments

(b) IFRS 7, *Financial Instruments: Disclosures*, which revised, simplified and incorporated disclosure requirements previously in IAS 32.

(c) IFRS 9, *Financial Instruments* replaced IAS 39, *Financial Instruments: Recognition and Measurement*. The standard covers:

(1) recognition and derecognition
(2) the measurement of financial instruments
(3) impairment
(4) general hedge accounting

IFRS 9 is mandatory for accounting periods beginning on or after 1 January 2018.

Professional skills focus: Structuring problems and solutions

You are expected to be able to identify and anticipate problems that may arise from a decision. When advising a company whether it should use hedging, you will need to consider not only the financial benefits that can be obtained from hedging but also the risks that a particular hedging strategy exposes the company to. The use of financial derivatives also has implications for financial reporting which must be considered when developing a hedging strategy.

11.1 IFRS 9 and the recognition of financial instruments

A financial asset or a financial liability should be recognised in the statement of financial position when the reporting entity becomes a party to the contractual provisions of the instrument.

For example, if a company takes out a forward contract to buy/sell an amount of currency at a future date, the contract is a financial derivative and should be recognised immediately, instead of waiting until the settlement date of the contract to recognise it in the accounts.

A financial instrument should initially be measured at cost. This is the fair value of the consideration given or received. Transaction costs under IFRS 9 are incremental costs associated with the acquisition, issue or disposal of a financial instrument. Transaction costs are added to the cost of financial assets classified as measured at amortised cost, or where an irrevocable election has been made to take all gains and losses through other comprehensive income, thereby increasing the cost of the asset. Transaction costs decrease the amount initially recognised as financial liabilities classified as measured at amortised cost. Transaction costs for other financial instruments are accounted for in profit or loss when they are incurred.

The value of financial instruments should be reassessed at each reporting date. The method of reassessment depends on the category of financial instrument.

When the carrying amount of a financial instrument is reassessed at fair value, there will be a difference between its previous carrying amount and its new carrying amount (new fair value). This difference is a gain or loss, which must be accounted for.

The method of accounting for the gain or loss on reassessment of the carrying amount depends on the class to which the financial instrument belongs. The following two tables summarise the accounting treatment of each category of financial asset and financial liability under IFRS 9, provided they have not been designated as hedged items. (The accounting treatment of hedged items follows different rules and these are discussed in the chapter Financial risk management.)

11.1.1 Summary of accounting treatments – financial assets and financial liabilities Financial assets

		Initial measurement (IFRS 9: para. 5.1.1)	Subsequent measurement (IFRS 9: paras. 4.1.2–4.1.5, 5.7.5)
(1)	**Investments in debt instruments** Business model approach: (a) Held to collect contractual cash flows; and cash flows are solely principal and interest	Fair value + transaction costs	Amortised cost
	(b) Held to collect contractual cash flows **and to** sell; and cash flows are solely principal and interest	Fair value + transaction costs	Fair value through other comprehensive income (with reclassification to profit or loss (P/L) on derecognition) NB: interest revenue calculated on amortised cost basis recognised in P/L
(2)	**Investments in equity instruments not 'held for trading'** (optional irrevocable election on initial recognition)	Fair value + transaction costs	Fair value through profit or loss by default If irrevocable election made on initial recognition, fair value through other comprehensive income (no reclassification to P/L on derecognition) NB: dividend income recognised in P/L
(3)	**All other financial assets** (and any financial asset if this would eliminate or significantly reduce an accounting mismatch)	Fair value (transaction costs expensed in P/L)	Fair value through profit or loss

Financial liabilities

		Initial measurement (IFRS 9: para. 5.1.1)	Subsequent measurement (IFRS 9: para. 4.2.1)
(1)	**Most financial liabilities** (eg, trade payables, loans, preference shares classified as a liability)	Fair value less transaction costs	Amortised cost
(2)	**Financial liabilities at fair value through profit or loss** • 'Held for trading' (short- term profit making) • Derivatives that are liabilities	Fair value (transaction costs expensed in P/L)	Fair value through profit or loss

		Initial measurement (IFRS 9: para. 5.1.1)	Subsequent measurement (IFRS 9: para. 4.2.1)
	• Designated on initial recognition at 'fair value through profit or loss' to eliminate/significantly reduce an 'accounting mismatch' • A group of financial liabilities (or financial assets and financial liabilities) managed and performance evaluated on a fair value basis in accordance with a documented risk management or investment strategy		

Example

A company purchases a financial investment for £25,000 on 1 May 20X5. At the end of its financial year, on 31 December 20X5, the fair value of the asset is remeasured as £31,000. The asset is subsequently sold on 16 March 20X6 for £35,500.

(a) If the asset is classified as an asset at fair value through profit or loss, a profit of £6,000 is recorded at 31 December 20X5 and the asset is revalued to £31,000. A further profit of £4,500 is recorded in the following year when the asset is sold.

(b) If the asset is classified as an asset at fair value through other comprehensive income, it is revalued to £31,000 on 31 December 20X5, and a gain of £6,000 is reported in other comprehensive income and taken to equity. When the asset is sold in March 20X6, a profit of £10,500 is recorded. The equity reserve is reduced by £6,000 and there is also a debit of £6,000 in other comprehensive income for 20X6, to avoid a double counting of the gain/profit of £6,000 that would otherwise occur.

Note that in the case of investments in equity instruments not held for trading where the irrevocable election has been made to report changes in fair value in other comprehensive income, all changes in fair value up to the point of derecognition are reported in other comprehensive income.

Therefore, a gain or loss in profit or loss will only arise if the investments in equity instruments are not sold at their fair value and for any transaction costs on derecognition.

11.1.2 Subsequent measurement: amortised cost

When financial assets or liabilities are subsequently measured at amortised cost, the asset or liability is revalued at the end of each accounting period as follows.

	Financial asset	Financial liability
	Opening balance	Opening balance
Add	Interest income recognised in profit or loss (using the effective interest rate)	Interest expense in profit or loss (using the effective interest rate)
Subtract	Interest actually received	Interest actually paid
Equals	Closing balance	Closing balance

Example

A company issues a two-year bond on the first day of its financial year. The bond has a coupon rate of interest of 4% and interest is payable annually. The bond is issued at a price of £96.00 and will be redeemed at par. The effective rate of interest on the bond is 6.1875%. The effective rate of interest is the rate that exactly discounts estimated future cash payments or receipts across the expected life of

the financial instrument to the net carrying amount of the financial asset or liability. You could be asked to calculate the effective rate of interest using the IRR technique.

Year	Opening balance £	Interest charge at 6.1875% (P/L) £	Interest paid £	Closing balance £
1	96.00	5.94	(4.00)	97.94
2	97.94	6.06	(4.00)	100.00
Redeemed				(100.00)
				0.00



Worked example: Classification and measurement of financial asset at fair value

DG acquired 500,000 shares in HJ, a listed entity, for £3.50 per share on 28 May 20X9. The costs associated with the purchase were £15,000 and were included in the cost of the investment. The directors plan to realise this investment before the end of 20X9. The investment is held within a business model whose objective is achieved by selling financial assets and was designated on acquisition as a financial asset at fair value through profit or loss. There has been no further adjustment made to the investment since the date of purchase. The shares were trading at £3.65 each on 30 June 20X9.

Requirement

Explain how the financial instrument should be classified, initially measured and subsequently measured. Prepare any journal entries required to correct the accounting treatment for the year to 30 June 20X9.

Solution

The investment is initially measured at fair value, in this case the cost of £1.75 million (500,000 shares × £3.50). Transaction costs are excluded from the initial fair value of a financial asset that is classified as held for trading. These costs are charged to profit or loss immediately, and not included in the cost of the investment. The investment is subsequently measured (at 30 June 20X9) at fair value of £1.825 million (500,000 shares × £3.65) with the gain of £75,000 (£1.825m – £1.75m) being recorded in profit or loss. The following adjustments are therefore required:

DEBIT	Administrative expenses	£15,000	
CREDIT	Financial asset		£15,000

Being the correction in respect of transaction costs

DEBIT	Financial asset	£75,000	
CREDIT	Gain on investment (P/L)		£75,000

Being the gain on the investment being credited to the statement of profit or loss

Worked example: Classification and measurement of financial asset at amortised cost

DG purchased a bond with a par value of £5 million on 1 July 20X8. The bond carries a 5% coupon, payable annually in arrears and is redeemable on 30 June 20Y3 at £5.8 million. The bond is held within a business model whose purpose is to hold financial assets in order to collect contractual cash flows and DG fully intends to hold the bond until the redemption date. The bond was purchased at a 10% discount. The effective interest rate on the bond is 10.26%. The interest due for the year was received and credited to investment income in the statement of profit or loss.

Requirement

Explain how the financial instrument should be classified, initially measured and subsequently measured. Prepare any journal entries required to correct the accounting treatment for the year to 30 June 20X9.

Solution

The bond purchased by DG should be classified as a financial asset at amortised cost, as DG intends to hold it to redemption, it is quoted (a bond) and it has fixed payments (5% interest, £5.8 million redemption value). It is initially recorded at the net cost of £4.5 million and then subsequently measured at amortised cost using the effective interest rate (10.26%). Only the interest received of £250,000 (5% × face value of £5m) has been recorded in the statement of profit or loss. The following adjustment is therefore required to bring the finance income up to the effective interest rate and to correct the carrying amount of the asset:

DEBIT	Financial asset	£211,700	
CREDIT	Finance income		£211,700

Being the additional finance income to be recognised in profit or loss

WORKING

	£'000
1 July 20X8 purchased (£5m × 90%)	4,500
Finance income (£4.5m × 10.26%)	461.7
Interest received (£5m × 5%)	(250)
30 June 20X9 balance c/d	4,711.7

The financial asset will be held at £4.7117 million and a further £211,700 (£461,700 – £250,000) will be credited to the statement of profit or loss.

Worked example: Investment in equity instrument

Atlantic Bank acquires an equity investment on 25 February 20X6 with the intention of holding it in the long term. The investment cost £100,000 and transaction costs were £6,000. Atlantic Bank has elected to present the equity investment at fair value through other comprehensive income.

At Atlantic Bank's year end of 31 December 20X6, the market price of the investment is £122,000.

Requirement

How is the asset initially and subsequently measured?

Solution

Asset measurement:

- The asset is initially recognised at the fair value of the consideration, being £100,000 plus transaction costs of £6,000. A total amount of £106,000 is initially recognised.
- At the period end it is re-measured to £122,000. This results in the recognition of £16,000 in other comprehensive income.

The gains or losses accumulated in other comprehensive income are never recycled to profit or loss. Only dividend income is recognised in profit or loss.



Worked example: Measurement of financial liability

On 1 January 20X3 Wallace issued £600,000 loan notes. Issue costs were £200. The loan notes do not carry interest, but are redeemable at a premium of £152,389 on 31 December 20X4. The effective finance cost of the loan notes is 12%.

Requirement

What is the finance cost in respect of the loan notes for the year ended 31 December 20X4?

Solution

The premium on redemption of the loan notes represents a finance cost. The effective rate of interest must be applied so that the debt is measured at amortised cost.

At the time of issue, the loan notes are recognised at their net proceeds of £599,800 (600,000 – 200). The finance cost for the year ended 31 December 20X4 is calculated as follows:

	B/f	Interest @ 12%	C/f
	£	£	£
20X3	599,800	71,976	671,776
20X4	671,776	80,613	752,389

The finance cost in respect of the loan notes for the year ended 31 December 20X4 is £80,613.

11.1.3 Financial reporting of derivatives: disclosure and measurement

As a reminder, a derivative is a financial instrument:

- whose value changes in response to the change in price of an underlying security, commodity, currency, index or other financial instrument(s)
- where the initial net investment is zero or is small in relation to the value of the underlying security or index
- that is settled at a future date

If a derivative is a financial asset, then it should be classified as a financial asset at fair value through profit or loss, being treated as a financial asset held for the purpose of selling in the short term.

If it is a financial liability, it will also be treated at fair value through profit or loss.

If there is no active market for a particular financial instrument, fair value should be determined using a valuation technique. Such techniques could include recent market transactions, transactions in other shares or securities that are substantially the same, discounted cash flow models and option pricing models. The inputs to such models should be market based.

Establishing a fair value for non-traded derivatives may be problematic. Their value will be very dependent on the assumptions used, particularly those relating to the timing and size of future cash flows.



Worked example: Derivatives – purchased option

On 31 December 20X0 Entity Theta purchases put options over 100,000 shares in Omega which expire on 31 December 20X2. The exercise price of the option is £2, the market price on 31 December 20X0, and the premium paid is £11,100.

Requirement

The intrinsic value of the option (ie, the exercise price less the price per share, times the number of shares specified in the option contract) is zero at acquisition. The cost of £11,100 reflects the time value of the option which depends on the time to expiration, the price of the stock and its volatility.

If the stock price falls below £2 the put becomes in the money by the amount below the £2 strike price times the number of option shares. For instance, if the price of Omega stock fell to £1.90, the intrinsic value gain on the put option is £0.10 per share. If the stock price rises and stays above £2 for the term of the contract, the put option expires worthless to the buyer because it is out of the money. The purchaser of the put option loses the premium which is kept by the seller (writer).

Solution

Economic assumptions

The value of the shares in Omega and the put options is shown in the table below. The value of the put option increases as the stock price decreases.

Omega shares	**31.12.20X0**	**30.6.20X1**	**31.12.20X1**
Price per share (£)	2.00	1.90	1.85
Value of put option (£)	11,100	13,500	15,000

On 31 December 20X1, Theta sells the option.

Accounting entries:

	Debit	**Credit**
	£	**£**
31 December 20X0		
Financial asset – put option	11,100	
Cash		11,100
(To record the purchase of the put option)		

30 June 20X1		
Financial asset – put option (13,500 – 11,100)	2,400	
Profit or loss – gain on put option		2,400
(To record the increase in the fair value of the put option)		
31 December 20X1		
Financial asset – put option (15,000 – 13,500)	1,500	
Profit or loss – gain on put option		1,500
(To record the increase in the fair value of the put option)		
Cash	15,000	
Financial asset – put option		15,000
(To record the sale of the put option on 31.12.20X1)		



Worked example: Derivatives – Forward contract

On 1 November 20X5, a company enters into a forward contract with its bank to sell US$750,000 on 1 February 20X6 in exchange for sterling, at a forward rate of $1.50 = £1. The company's year end is 31 December. At 31 December 20X5, the $/£ rate has changed to $1.55 and at settlement on 1 February 20X6 the spot exchange rate is $1.60.

Requirement

How should this forward contract be accounted for by the company?

Solution

The asset is classified as a financial asset or liability at fair value through profit or loss. On 1 November 20X5 it has no value as an asset or liability.

At 31 December 20X5, the financial instrument should be revalued to $750,000/1.55 = £483,871. The company will be receiving £500,000 on 1 February; therefore it has gained from the exchange rate movement. The gain of £16,129 (£500,000 – £483,871) should be recognised in profit for the year to 31 December 20X5.

On 1 February 20X6 when the forward contract is settled, the spot value of the dollars that the company must pay to receive £500,000 is now just £468,750 ($750,000/1.60). The total gain on the forward contract has been £31,250. As £16,129 was recognised in profit or loss for 20X5, a further £15,121 will be recognised in profit for the year to 31 December 20X6.

Accounting for derivatives where the derivatives are used for hedging is described in the chapter Financial risk management.

11.1.4 Embedded derivatives

Certain contracts that are not themselves derivatives (and may not be financial instruments) include derivative contracts that are 'embedded' within them. An embedded derivative is a derivative instrument that is included within a different type of contract, known as the 'host' contract.

An embedded derivative may be identified if contracts contain:

- rights or obligations to exchange at some time in the future
- rights or obligations to buy or sell

- provisions for adjusting the cash flows according to some interest rate, price index or specific time period
- options which permit either party to do something not closely related to the contract
- unusual pricing terms (eg, a bond which pays interest at rates linked to the FTSE 100 yield contains an embedded swap)

A key characteristic of embedded derivatives is that the embedded derivative cannot be transferred to a third party independently of the instrument. For example, a bond with a detachable warrant, which gives the right to the owner to exercise the warrant and buy shares while retaining the bond, is not a hybrid or combined instrument. The warrant is a separate financial instrument, not an embedded derivative.

Examples of host contracts

Possible examples include:

(a) An investment in a debt instrument that is convertible into ordinary shares of the issuer. The debt instrument is the host contract and conversion option is the embedded derivative.

(b) A debt instrument with an extension option with the interest rate in the extension period reset to 1.2 times the market rate. The debt instrument is the host contract with embedded extension option.

(c) A lease contract has a rent adjustment clause based on changes in the local inflation rate. The lease is the host contract with inflation-related rentals being the embedded derivative.

Examples of embedded derivatives

Possible examples include:

A bond which is redeemable in five years' time with part of the redemption price being based on the increase in the FTSE 100 Index:

'Host' contract	Bond	Accounted for as normal ie, amortised cost
Embedded derivative	Option on equities	Treat as derivative ie, remeasured to fair value with changes recognised in profit or loss

A construction contract priced in a foreign currency. The construction contract is a non-derivative contract, but the changes in foreign exchange rate is the embedded derivative.

Accounting treatment of embedded derivatives

IFRS 9 requires embedded derivatives that would meet the definition of a separate derivative instrument to be **separated** from the host contract **unless**:

(a) the economic characteristics and risks of the embedded derivative are **closely related** to those of the host contract;

(b) the hybrid (combined) instrument is measured at **fair value through profit or loss**;

(c) the host contract is a **financial asset** within the scope of IFRS 9; or

(d) the embedded derivative **significantly modifies the cash flows** of the contract.



Worked example: Convertible bond

QWE issued 10 million 5% five year convertible £1 bonds on 1 January 20X0. The proceeds of £10 million were credited to non-current liabilities and debited to bank. The 5% interest paid has been charged to finance costs in the year to 31 December 20X0.

The market rate of interest for a similar bond with a five-year term but no conversion terms is 7%.

Requirement

Demonstrate how this convertible instrument would be initially measured in accordance with IAS 32 **and** subsequently measured in accordance with IFRS 9 in the financial statements for the year ended 31 December 20X0.

Solution

A convertible instrument is considered part liability and part equity. IAS 32 requires that each part is measured separately on initial recognition. The liability element is measured by estimating the present value of the future cash flows from the instrument (interest and potential redemption) using a discount rate equivalent to the market rate of interest for a similar instrument with no conversion terms. The equity element is then the balance, calculated as follows:

	£
PV of the principal amount £10m at 7% redeemable in 5 yrs (£10m × 0.713)	7,130,000
PV of the interest annuity at 7% for 5 yrs (5% × £10m) × 4.100	2,050,000
Total value of financial liability element	9,180,000
Equity element (balancing figure)	820,000
Total proceeds raised	10,000,000

The equity element will not be remeasured; however, the liability element will be subsequently remeasured at amortised cost recognising the finance cost using the effective interest rate of 7% and deducting the coupon interest paid. The financial statements will include:

Statement of profit or loss and other comprehensive income for the year ended 31 December 20X0 (within profit or loss)

	£
Finance cost (7% × 9,180,000)	642,600

Statement of financial position as at 31 December 20X0	**£**
Equity	820,000
Financial liability (W)	9,322,600

WORKING

	£
Liability recognised 1.1.X0	9,180,000
Finance cost (7% × 9,180,000)	642,600
Interest paid (5% × 10,000,000)	(500,000)
At 31.12.X0	9,322,600

11.2 Assurance over fair values

ISA 540, *Auditing Accounting Estimates, Including Fair Value Accounting Estimates, and Related Disclosures* provides guidance on accounting estimates contained in financial statements and requires auditors to obtain sufficient, appropriate audit evidence for such accounting estimates. It treats fair values as a type of accounting estimate.

Definition

Accounting estimate: An accounting estimate is an approximation of a monetary amount in the absence of a precise means of measurement.

The ISA gives the following examples of fair value estimates:

- complex financial instruments, which are not traded in an active and open market
- share-based payments
- property or equipment held for disposal
- certain assets and liabilities acquired in a business combination, including goodwill and intangible assets

Directors and management are responsible for making accounting estimates included in the financial statements. The determination of fair value will generally be more difficult than determining historical cost, and it will certainly be more difficult to establish whether fair value is reasonable for complex assets and liabilities than for those more straightforward assets or liabilities which have an actively traded market. Such estimates are therefore often made in conditions of uncertainty regarding the outcome of events, and involve the use of judgement, or frequently the application of a formula based upon past experience. Others may have complex systems for determining fair value if they have a large number of assets and liabilities which they account for at fair value.

Not all financial statement items requiring measurement at fair value involve estimation uncertainty. In the chapter Business and securities valuation, we saw that in accordance with IFRS 13, entities should maximise the use of relevant observable inputs. For example, if using a Level 1 input (eg, the unadjusted quoted price in an active market of equity shares in a listed company), there is no estimation uncertainty.

For others, however, there may be moderate (eg, Level 2 inputs) or relatively high estimation uncertainty (eg, Level 3 inputs), particularly where they are based on significant assumptions, for example:

- fair value estimates for derivative financial instruments not publicly traded
- fair value estimates for which a highly specialised entity-developed model is used or for which there are assumptions or inputs that cannot be observed in the marketplace

ISA 540 requires auditors to assess the entity's process for determining fair value measurements and disclosures, and their related control activities. When a firm is engaged to undertake an assurance engagement over fair values of financial instruments (or in reviewing fair value measurements as part of an audit) the general principles and approach to assurance should apply. Assurance engagements have been described in some detail in the chapter Data analysis.

The general approach to an assurance exercise in relation to the valuation of financial instruments should be to:

- establish the method or model that has been used for the valuation of the financial instrument(s); and
- consider whether this selected model is suitable for the purpose of the valuation.

In practical terms, this would include considering the following:

- the valuation techniques adopted (ie, Level 1, 2 or 3)
- the market in which the transaction is assumed to have taken place
- the relevant control activities over the process (eg, controls over data and the segregation of duties between those committing the entity to the underlying transaction and those responsible for undertaking the valuations)
- the expertise and experience of those persons determining the fair value measurements

- the significant management assumptions used (particularly where Level 3 unobservable inputs are used)
- documentation supporting management's assumptions
- controls over the consistency, timeliness and reliability of the data used in valuation models

If the values that have been used for each of the variables in the model are considered suitable and appropriate, the financial instruments should be valued using the model to assess whether the model has been used correctly by the client. This valuation should be compared to the client's valuation.

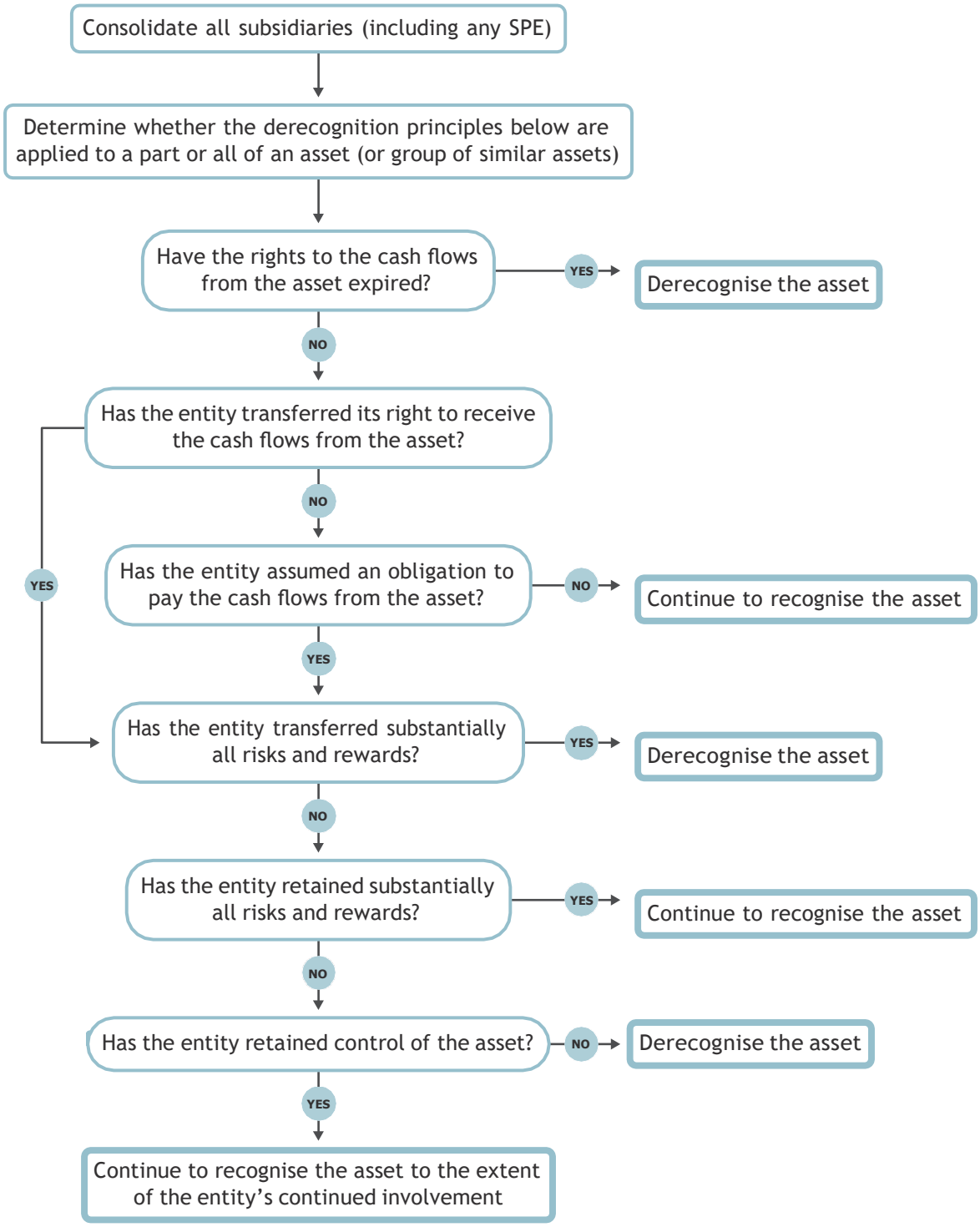
Reasonable assurance can be provided if the valuation obtained in the assurance exercise conforms in all material respects with the client's valuation.

11.3 Derecognition

The complexity of financial transactions and the difficulty of establishing whether the transfer of legal title leaves residual risk and reward exposure as well as control and involvement has prompted the IASB to produce a fairly prescriptive set of rules to aid companies in the derecognition of financial assets.

The following flowchart is included in the application guidance which accompanies IFRS 9 as an integral part of the standard. It summarises the evaluation of whether, and to what extent, a financial asset should be derecognised.

Figure 13.10: Flowchart



The following points relate to this flowchart:

(a) If the **contractualrights** to receive the cash flows from the asset have **expired** or have been wholly transferred, the whole of the asset should be derecognised. This is also the case if the contractual rights have been retained by the entity but it has assumed a contractual obligation to pay the cash flows to one or more recipients.

(b) If an entity has sold just a portion of the cash flows arising from an asset, only part of the asset should be derecognised.

(c) If substantially **all the risks and rewards** of ownership have been **transferred**, the financial asset should be derecognised; if they have not, it should not.

(d) If the entity has neither retained nor transferred all the risks and rewards of ownership, it should determine whether it has **retained control** of the financial asset. If it has, it continues to recognise the asset to the extent of its continuing involvement.

When a derivative is derecognised, the difference between the carrying amount and any consideration received should be recognised in profit or loss. Any accumulated gains or losses that have been recognised in other comprehensive income should also be reclassified to profit or loss on derecognition of the asset.

Where a financial asset such as a bond is sold with a simultaneous agreement to buy it back at some future date at a specified price, the substance of the transaction is that the risks and rewards of ownership have not been transferred. In effect, the proceeds of the sale are collateralised borrowing. The asset should continue to be recognised and the amount received recognised as a financial liability. Similarly, the asset should not be derecognised if it is sold but there is a total return swap transferring the market risk exposure back to the entity selling the asset.

Worked example: Derecognition

The Polyact Company purchased £60,000 of shares, which were classified as a financial asset at fair value through profit or loss. Later that year Polyact sold 50% of the shares for £40,000.

Requirement

What is the amount of the gain or loss on the disposal to be recognised in profit or loss?

Solution

On derecognition of the financial asset the difference between the carrying amount and any consideration received should be recognised in profit or loss. On the assumption that the asset was not remeasured during the year, the carrying amount of 50% of the asset is £30,000. The proceeds from the sale of 50% of the asset are £40,000, yielding a gain of £10,000 to be recognised in profit or loss.

For securitisations, whether the securitised assets will be derecognised depends on whether the SPV has assumed all the risks and rewards of the ownership of the assets and whether the originator has ceded control of the assets to the SPV. If the SPV is a mere extension of the originator and it continues to be controlled by the originator, then the SPV should be consolidated and any securitised assets should continue to be recognised in the group accounts. Securitisation will not in this case lead to derecognition.

Even when the SPV is not controlled by the originator, the risks and rewards of ownership will not have passed completely to the SPV if the lenders to the SPV require recourse to the originator to provide security for their debt, or if the originator retains the risk of bad debts.

For total return swaps, the originator is prohibited from derecognising the asset that it has transferred.

Interactive question 5: Derecognition of financial assets and liabilities Discuss whether the following financial instruments should be derecognised. **Requirements**

(a) AB Co sells an investment in shares, but retains a call option to repurchase those shares at any time at a price equal to the market value current at the date of repurchase.

(b) CD Co sells an investment in shares and enters into a 'total return swap' with the buyer. Under a 'total return swap' arrangement, the buyer returns any increases in value to the seller, and the seller compensates the buyer for any decrease in value plus interest.

(c) EF Co enters into a stock lending agreement where an investment is lent to a third party for a fixed period of time for a fee.

See **Answer** at the end of this chapter.

11.4 Accounting for equity

Accounting for equity (share capital and reserves) follows the requirements of company law and accounting standards, and IAS 32 and IFRS 9 do not contain many rules with regard to equity (with the exception of equity derivatives such as options).

It may be useful, however, to note the accounting treatment of treasury shares – buybacks of its own shares by a company, in the market or from shareholders. For example, a listed company may buy back 100,000 of its own shares in the market at a price of £4.80 per share. These treasury shares must be accounted for.

The basic rule is that no gain or loss must be recognised on the purchase or subsequent disposal or cancellation of treasury shares. The consideration paid for the shares should be recognised directly in equity. In the above, if a listed company buys back 100,000 of its own shares in the market at a price of £4.80 per share, the accounting entry should be to credit cash £480,000 and debit a Treasury shares account. In the statement of financial position, the balance on this account should be deducted from equity, and not included within assets.

11.5 Reclassification of financial assets

11.5.1 Reclassification of financial assets

IFRS 9 requires that when an entity changes its business model for managing financial assets, it should reclassify all affected financial assets. This is expected to be a rare event.

A change in an entity's business model will occur only when an entity begins or ceases to perform an activity that is significant to its operations – for example, when the entity has acquired, disposed of or terminated a business line.

The examples of **change in a business model** include the following:

- An entity has a portfolio of commercial loans that it holds to sell in the short term. The bank acquires an entity that manages commercial loans and has a business model that holds the loans in order to collect the contractual cash flows. The portfolio of commercial loans is no longer for sale, and the portfolio is now managed together with the acquired commercial loans and all are held to collect contractual cash flows.
- An entity decides to shut down its retail mortgage business. That business no longer accepts new business and the bank is actively marketing its mortgage loan portfolio for sale.

If an entity reclassifies financial assets, the **reclassification is applied prospectively** from the reclassification date. When an entity reclassifies a financial asset:

- Between amortised cost measurement category and the fair value through other comprehensive income measurement category, the recognition of interest revenue will not change and the entity continues to use the same effective interest rate
- Out of the fair value through profit or loss measurement category, the effective interest rate is determined on the basis of the fair value of the asset at the reclassification date

The following are not changes in business model:

- A change in intention related to particular financial assets (even in circumstances of significant changes in market conditions)
- The temporary disappearance of a particular market for financial assets
- A transfer of financial assets between parts of the entity with different business models

When financial instruments are reclassified, disclosures are required under IFRS 7.

11.6 Impairment

The impairment model in IFRS 9 is based on the premise of providing for expected losses. The financial statements should reflect the general pattern of deterioration or improvement in the credit quality of financial instruments within the scope of IFRS 9.

On initial recognition

The entity will **create a credit loss allowance/provision equal to 12 months' expected credit losses.** This is calculated by multiplying the probability of a default occurring in the next 12 months by the total lifetime expected credit losses that would result from that default. (This is not the same as the expected cash shortfalls over the next 12 months.)

Subsequent years

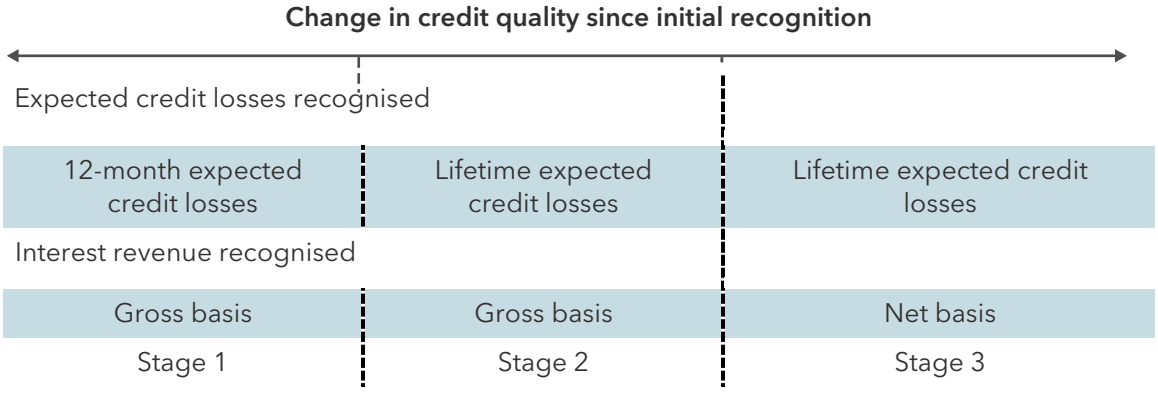
If the **credit risk increases** significantly since initial recognition this amount will be replaced **by lifetime expected credit losses**. If the credit quality subsequently improves and the lifetime expected credit losses criterion is no longer met, the 12-month expected credit loss basis is reinstated.

Certain financial instruments have a low credit risk and would not, therefore, meet the lifetime expected credit losses criterion. Entities do not recognise lifetime expected credit losses for financial instruments that are equivalent to 'investment grade', which means that the asset has a low risk of default. The model requires that there is a rebuttable presumption that lifetime expected losses should be provided for if contractual cash flows are 30 days overdue.

Amount of impairment

The amount of the impairment to be recognised on these financial instruments **depends on whether or not they have significantly deteriorated** since their initial recognition.

Figure 13.11: The impact of deterioration in credit quality



Stage 1: Financial instruments whose credit quality has not significantly deteriorated since their initial recognition

Stage 2: Financial instruments whose credit quality has significantly deteriorated since their initial recognition

Stage 3: Financial instruments for which there is objective evidence of an impairment as at the reporting date

For Stage 1 financial instruments, the impairment represents the present value of expected credit losses that will result if a default occurs in the 12 months after the reporting date **(12 months' expected credit losses)**.

For financial instruments classified as Stage 2 or 3, an impairment is recognised at the present value of expected credit shortfalls over their remaining life **(lifetime expected credit loss)**. Entities are required to reduce the gross carrying amount of a financial asset in the period in which they no longer have a reasonable expectation of recovery.

Interest

For Stage 1 and 2 instruments interest revenue will be calculated on their gross carrying amounts, whereas interest revenue for Stage 3 financial instruments would be recognised on a net basis (ie, after deducting expected credit losses from their carrying amount).

Simplified approach for trade and lease receivables

For trade receivables that do not have an IFRS 15 financing element, the loss allowance is measured at the lifetime expected credit losses, from initial recognition.

For other trade receivables and for lease receivables, the entity can choose (as a separate accounting policy for trade receivables and for lease receivables) to apply the three-Stage approach or to recognise an allowance for lifetime expected credit losses from initial recognition.



Worked example: Trade receivables provision matrix

On 1 June 20X4, Kredco sold goods on credit to Detco for £200,000. Detco has a credit period with Kredco of 60 days. Kredco applies IFRS 9, and uses a pre-determined matrix for the calculation of allowances for receivables as follows:

Days overdue	Expected loss provision
Nil	1%
311 to 1 to 30	5%
31 to 60	15%
61 to 90	20%
90 +	25%

Detco had not paid by 31 July 20X4, and so failed to comply with its credit term, and Kredco learned that Detco was having serious cash flow difficulties due to a loss of a key customer. The finance controller of Detco has informed Kredco that they will receive payment.

Ignore sales tax.

Requirement

Show the accounting entries on 1 June 20X4 and 31 July 20X4 to record the above, in accordance with the expected credit loss model in IFRS 9.

Solution

On 1 June 20X4

The entries in the books of Kredco will be:

DEBIT	Trade receivables	£200,000	
CREDIT	Revenue		£200,000

Being initial recognition of sales

An expected credit loss allowance, based on the matrix above, would be calculated as follows:

DEBIT	Expected credit losses	£2,000	
CREDIT	Allowance for receivables		£2,000

Being expected credit loss: £200,000 × 1%

On 31 July 20X4

Applying Kredco's matrix, Detco has moved into the 5% bracket, because it has exhausted its 60-day credit limit (note that this does not equate to being 60 days overdue!). Despite assurances that Kredco will receive payment, the company should still increase its credit loss allowance to reflect the increased credit risk. Kredco will therefore record the following entries on 31 July 20X4:

DEBIT	Expected credit losses	£8,000	
CREDIT	Allowance for receivables		£8,000

Being expected credit loss: £200,000 × 5% – £2,000



Worked example: Impairment of trade receivable

Included in Timpson's trade receivables at 31 October 20X8 is an amount due from its customer Thompson of £51,542,000. This relates to a sale which took place on 31 October 20X6, payable in three annual instalments of £20,000,000 commencing 31 October 20X7 discounted at a market rate of interest adjusted to reflect the risks of Thompson of 8%. Based on previous sales where consideration has been received in annual instalments, the directors of Timpson estimate a lifetime expected credit loss in relation to this receivable of £14.4 million. The probability of default over the next 12 months is estimated at 25%. For trade receivables containing a significant financing component, Timpson chooses to follow the three-stage approach for impairments (rather than always measuring the loss allowance at an amount equal to lifetime credit losses). No loss allowance has yet been recognised in relation to this receivable.

Requirement

How should the receivable be treated in the financial statements?

Solution

A loss allowance for the trade receivable should be recognised at an amount equal to 12-month expected credit losses. Although IFRS 9 offers an option for the loss allowance for trade receivables with a financing component to always be measured at the lifetime expected losses, Timpson has chosen instead to follow the three-Stage approach of IFRS 9.

The 12-month expected credit losses are calculated by multiplying the probability of default in the next 12 months by the **lifetime** expected credit losses that would result from the default. Here this amounts to £3.6 million (£14.4m × 25%).

Adjustment:

DEBIT	Expected credit loss	£3.6m	
CREDIT	Allowance for receivables (this is offset against trade receivables)		£3.6m



Worked example: Portfolio of mortgages and personal loans

Credito Bank operates in South Zone, a region in which clothing manufacture is a significant industry. The bank provides personal loans and mortgages in the region. The average loan to value ratio for all its mortgage loans is 75%.

All loan applicants are required to provide information regarding the industry in which they are employed. If the application is for a mortgage, the customer must provide the postcode of the property which is to serve as collateral for the mortgage loan.

Credito Bank applies the expected credit loss impairment model in IFRS 9, *Financial Instruments*. The bank tracks the probability of customer default by reference to overdue status records. In addition, it is required to consider forward-looking information as far as that information is available.

Credito Bank has become aware that a number of clothing manufacturers are losing revenue and profits as a result of competition from abroad, and that several are expected to close.

Requirement

How should Credito Bank apply IFRS 9 to its portfolio of mortgages in the light of the changing situation in the clothing industry?

Solution

Credito Bank should segment the mortgage portfolio to identify borrowers who are employed by clothing manufacturers and suppliers and service providers to the clothing manufacturers. This segment of the portfolio may be regarded as being 'in Stage 2', that is having a significant increase in credit risk. Lifetime credit losses must be recognised.

In estimating lifetime credit losses for the mortgage loans portfolio, Credito Bank will take into account amounts that will be recovered from the sale of the property used as collateral. This may mean that the lifetime credit losses on the mortgages are very small even though the loans are in Stage 2.



Worked example: Investment in loan notes

On 1 January 20X4 Pyllon acquired an investment in £600,000 8% loan notes. The investment is measured at amortised cost.

At 1 January 20X4 there is a 6% probability that the borrower will default on the loan during 20X4 resulting in a 100% loss.

At 31 December 20X4 there is 1% probability that the borrower will default on the loan before 31 December 20X5 resulting in a 100% loss.

At 31 December 20X5 the borrower is expected to breach its covenants as a result of cash flow problems. There is a 40% probability of the loan defaulting over the remainder of its term.

At 31 December 20X6 the borrower breached its covenants and there is a 70% probability of default over the remainder of the loan term.

Requirement

What impairment loss and interest revenue are recognised at initial recognition and in each of the years ended 31 December 20X4, 20X5 and 20X6?

Solution

	Impairment allowance	Interest revenue
1 January 20X4 (Initial recognition)	% × £600,000 = £36,000	-
31 December 20X4 (stage 1)	1% × £600,000 = £6,000	£600,000 × 8% = £48,000
31 December 20X5 (stage 2)	40% × £600,000 = £240,000	£600,000 × 8% = £48,000
31 December 20X6 (stage 3)	70% × £600,000 = £420,000	(£600,000 – £420,000) × 8% = £14,400

An impairment loss on a financial asset at amortised cost is recognised in profit or loss, with a corresponding entry to an allowance account, which is offset against the carrying amount of the financial asset in the statement of financial position.

1 January 20X4		**£**	**£**
DEBIT	Profit or loss	36,000	
CREDIT	Impairment allowance		36,000
31 December 20X4		**£**	**£**
DEBIT	Impairment allowance (36,000 – 6,000)	30,000	
CREDIT	Profit or loss		30,000
31 December 20X5		**£**	**£**
DEBIT	Profit or loss (240,000 – 6,000)	234,000	
CREDIT	Impairment allowance		234,000
31 December 20X6		**£**	**£**
DEBIT	Profit or loss (420,000 – 240,000)	180,000	
CREDIT	Impairment allowance		180,000



Interactive question 6: Particular defaults identified

Later in the year, more information emerged, and Credito Bank was able to identify the particular loans that defaulted or were about to default.

Requirement

How should Credito Bank treat these loans?

See **Answer** at the end of this chapter.

11.7 Reporting on leasing arrangements

As you learned in your Professional level studies, one way by which businesses can obtain the use of an asset is by a leasing agreement. A contract has to meet the definition of a lease contract to be in the scope of IFRS 16, *Leases*.

IFRS 16 was published in January 2016. The effective date of IFRS 16 is for annual reporting periods beginning on or after 1 January 2019. IFRS 16 takes the approach of a lessee measuring the 'right of use' of an asset, and recognising it on the statement of financial position.

Definitions

Lease: is a contract, or part of a contract, that conveys the right to use an asset, the underlying asset, for a period of time in exchange for consideration.

Underlying asset: is an asset that is the subject of a lease, for which the right to use that asset has been provided by a lessor to a lessee.

Right-of-use asset: An asset that represents a lessee's right to use an underlying asset for the lease term.

11.7.1 Control

The key to identifying a lease is confirming the right to control the use of the asset. The right to control the use of an identified asset depends on the lessee having:

(a) The right to obtain substantially all of the economic benefits from use of the identified asset; and

(b) the right to direct the use of the identified asset. This arises if either:

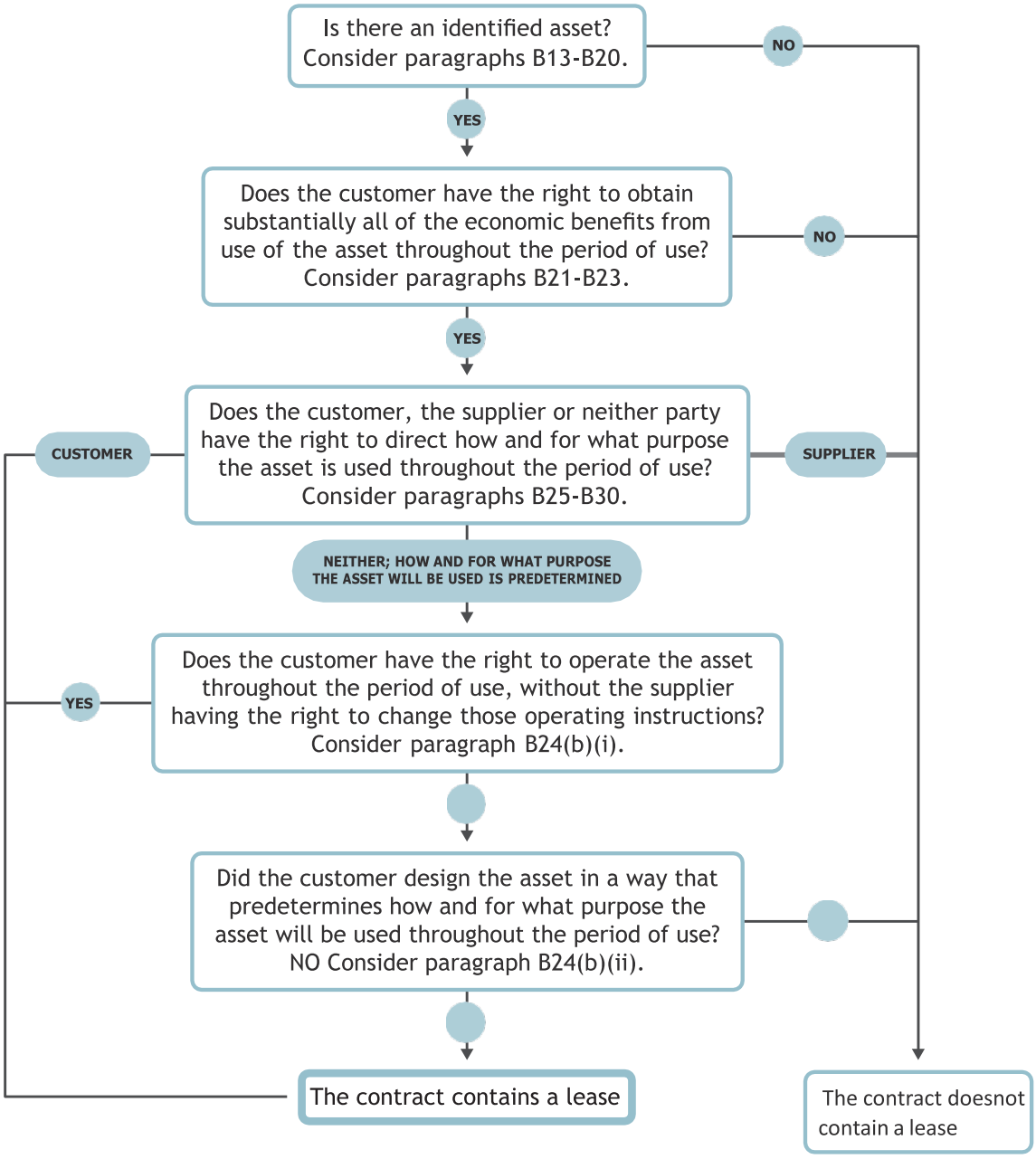
  (1) the customer has the right to direct how and for what purpose the asset is used during the whole of its period of use; or

  (2) the relevant decisions about use are predetermined and the customer can operate the asset without the supplier having the right to change those operating instructions, or the customer designed the asset in a way that predetermines how and for what purpose the asset will be used throughout the period of use.

A lessee does not control the use of an identified asset if the lessor can substitute the underlying asset for another asset during the lease term and would benefit economically from doing so.

The following flowchart, taken from IFRS 16, Appendix B, paragraph B31, may assist you in determining whether a lease should be identified:

Figure 13.12: IFRS 16, Appendix B flowchart



11.7.2 Non-lease components

Some contracts may contain elements that are not leases, such as service contracts. These must be separated out from the lease and accounted for separately.

Worked example: Is this a lease?

Outandabout Co provides tours round places of interest in the tourist city of Sightsee. While these tours are mainly within the city, it does the occasional day trip to visit tourist sites further away. Outandabout Co has entered into a three-year contract with Fastcoach Co for the use of one of its coaches for this purpose. The coach must seat 50 people, but Fastcoach Co can use any of its 50- seater coaches when required.

Requirement

Is this arrangement a lease within the scope of IFRS 16, *Leases*?

Solution

No. This is not a lease**.** There is no identifiable asset. Fastcoach can substitute one coach for another, and would derive economic benefits from doing so in terms of convenience. Therefore Outandabout should account for the rental payments as an expense in profit or loss.

11.7.3 Your exam

Recent students of the Financial Accounting and Reporting paper will be aware that lease accounting has changed significantly. Under IAS 17, *Leases*, the predecessor of IFRS 16, there was a distinction between an operating lease and a finance lease. The classification of the type of lease was essential to how the lessee and lessor would account for the transaction in their financial statements. Questions were asked on this distinction because accounting for an item as an operating lease removed it from the lessee's statement of financial position, so companies who wished to create a favourable gearing ratio, for example, were incentivised to account for leases as operating leases.

IFRS 16 has removed the differentiation between operating and finance leases from the lessee's point of view (although it remains for lessors - see below). However, the incentive to keep liabilities off the SOFP remains, and preparers of financial statements who wish to do this may argue that a transaction is not a lease when in fact it is, and therefore should appear on the SOFP. An exam question could present an ethical dilemma, where the auditor or junior accounting staff member is under pressure to accept this incorrect financial reporting treatment.

11.7.4 Lessee accounting

A single lease model is applied by all lessees, with no requirement to distinguish between operating and finance leases. At the commencement of the lease, the lessee recognises:

(a) a right-of-use asset, which represents the right to use the underlying asset; and

(b) a lease liability which represents the obligation to make lease payments.

11.7.5 Initial measurement of the right-of-use asset

The right-of-use asset is initially measured at cost, which is:

- the initial measurement of the lease liability
- **plus** any lease payments made before the commencement date
- **less** lease incentives received
- **plus** initial direct costs incurred by the lessee
- **plus** estimated dismantling and restoration costs that a lessee is obliged to pay at the end of the lease term



Worked example: Initial measurement of the right-of-use asset

Pyllon Ltd commenced a five-year lease to acquire plant on 1 January 20X8. Pyllon Ltd has paid an initial deposit of £1,000, and received lease incentives of £300. At the commencement date the present value of the future lease payments was £15,000. Pyllon Ltd has incurred legal costs of acquiring the lease of £1,500. The plant will produce a new product, the XP2, and marketing costs for the new product have been incurred of £250.

At the end of the lease, Pyllon Ltd will have to pay £800 (at present value) to remove the plant.

Requirement

What is the initial measurement of the right-of-use asset?

Solution

	£
PVFLP	15,000
Initial deposit	1,000
Direct costs	1,500
End of term costs to remove the plant	800
Less: lease incentives	(300)
Right-of-use asset	18,000

Note that the marketing costs for the new product to be made by the plant are not included as these costs are not directly attributable to the lease. This is consistent with IAS 16 (s.19(b)), where marketing costs cannot be capitalised as part of the non-current asset.

11.7.6 Subsequent measurement of the right-of-use asset

Subsequently, the right-of-use asset will be measured at cost less accumulated depreciation and impairment losses in line with IAS 16, *Property, Plant and Equipment*.

The right-of-use asset will be depreciated from the commencement date to the **earlier** of the end of its useful life or the end of the lease term **unless** the asset is expected to be transferred to the lessee at the end of the lease term. In that case, the asset will be depreciated over the useful life of the asset.



Worked example: subsequent measurement of the right-of-use asset

Taking the information from the previous example, Pyllon has estimated that the useful life of the plant will be six years. There is no option to purchase at the end of the lease term.

Requirement

What is the carrying amount of the right-of-use asset at 31 December 20X8?

Solution

Right-of-use asset is initially measured at £18,000

The depreciation period is the shorter of the useful life (six years) or the lease term (five years).

£18,000/5 years = £3,600 depreciation charge

Carrying amount of the right-of-use asset at 31 December 20X8 is £14,400 (£18,000 – £3,600)

11.7.7 Depreciating the asset

The asset should be depreciated over the shorter of the lease term or the asset's useful life

DR Depreciation expense

CR Accumulated depreciation

Points to note

(a) Depreciation policies adopted should be consistent with other non-current assets of the same class.

(b) If any impairment reviews are required, these must be conducted in accordance with IAS 36, *Impairment of Assets*.

(c) If there is reasonable certainty that the lessee will eventually own the asset, then it should be depreciated over its estimated useful life.

(d) The lease term comprises the period for which the lessee has contracted to lease the asset and any further terms for which there is reasonable certainty at the inception of the lease that the lessee will exercise the option.

11.7.8 Initial measurement of the lease liability

At the commencement of the lease, the lease liability is measured at the **present value** of **future lease payments,** including any payments expected at the end of the lease.

Lease liability will include:

- fixed payments less any lease incentives receivable
- variable lease payments that depend on an index (eg, the consumer price index) or rate (eg, market rental rates)
- amounts expected to be payable by the lessee under residual value guarantees
- purchase options (if reasonably certain to be exercised)
- penalties for early termination (if expected to be paid)

The discount rate used is the interest rate implicit in the lease, or if this is not available, the lessee's incremental borrowing rate.



Definition

Lease payments: Payments made by a lessee to a lessor in order to use an underlying asset during the lease term, less any lease incentives.

IFRS 16 requires lease payments to also include the exercise price of a purchase option if the lessee is reasonably certain to exercise that option, and any penalty payments for terminating the lease.



Context example: Present value of future lease payments

Alpha Ltd has commenced a four-year lease on 1 January 20X8. Payments are £10,000 pa in arrears and the interest rate implicit in the lease is 10%. The present value of the future lease payments can be calculated as follows:

B5 NPV(0.1,B1:B4)

	A	B
1	Lease payment £	10,000
2	Lease payment £	10,000
3	Lease payment £	10,000
4	Lease payment £	10,000
5	PV of future lease payments	31,699

Therefore, at 1 January 20X8, the lease liability is measured at £31,699.

11.7.9 Subsequent measurement of the lease liability

In future periods, the lease liability is amortised.

- Interest will accrue on the outstanding lease, at the rate stated in the lease contract.
- Payments made in respect of the lease by the lessee will reduce the outstanding liability.



Context example: Year 2 lease liability

Taking the example of Alpha Ltd in the previous example.

In the previous example, Alpha Ltd had calculated the initial lease liability to be £31,698 on 1 January 20X8.

Year	£
1.1.X8: Lease liability (PVFLP)	31,699
31.12.X8: Interest 10%	3,170
31.12.X8: Instalment in arrears	(10,000)
Liability at 31.12.X8	24,869

Therefore, at 1 January 20X9, the lease liability will be £24,869.

11.7.10 Setting up accounts in the statement of financial position

IFRS 16 requires that, when an asset is leased, the accounting treatment should reflect the **substance of the transaction**. In the lessee's books therefore:

The amount to be recorded in this way is the present value of the future lease payments

DR Right-of-use asset account

CR Lease liabilities

The **initial** deposit, if any, counts as one of the lease payments and hence is included in the cost of the asset.

Points to note

(a) The entries are made at the **commencement** of the lease term, with the **values** determined at the start of the lease.

(b) The present value of the future lease payments is derived by discounting them at the **interest rate implicit in the lease**. If it is not practicable to determine the interest rate implied in the lease, then the lessee's **incremental borrowing rate** can be used.

(c) Initial direct costs can be treated as part of the cost of the asset – provided they are directly attributable to activities performed by the lessee to obtain the lease.

(d) Although interest is payable under the lease, this is accrued over time. The justification is that the capital could, in theory, be paid off at any time, with cancellation charges. These charges could be avoided, so they are not a 'true' long term liability. **Interest is therefore recognised as it accrues**.

11.7.11 Making the payment

Every period, the payments are accounted for as follows.

Each lease payment is comprised partly of a repayment of capital and partly of an interest charge for the period.

DR Payables: Lease liabilities

CR Cash

11.7.12 Finance charge

The finance charge is dealt with as follows.

With the amount of the interest accrued over the period

DR Profit or loss: Finance cost

CR Lease liabilities

11.7.13 Allocating the interest charge to accounting periods

IFRS 16 requires the total finance charge to be allocated to each period during the lease term so as to produce a **constant periodic rate of interest** on the outstanding lease obligation.

As the lessee pays off the capital sum, the total capital owed falls from period to period. You would therefore also expect a reduction in the total interest payable, too, on the outstanding balance.

For example, if you owe £10,000 and pay 15%, the interest will be £1,500. After you have paid off, say, £8,000 of the capital, interest would be £300 (on £2,000). The monthly payments remain the same, but the mix of interest and capital changes over the life of the loan.

Therefore, in the earlier years, the finance charge (interest) is a higher proportion of the annual lease payments. Towards the end of the lease, the finance charge will be smaller as the outstanding lease liability is smaller.

11.7.14 Instalments in arrears



Worked example: Lease payments in arrears

A Ltd has a year-end of 31 December.

A lease commences on 1 January 20X1. Lease payments comprise three payments of £10,000 annually, commencing on 31 December 20X1.

There is no transfer of asset at the end of the lease and no purchase option.

The implicit interest rate is 10%.

Requirement

You are required to calculate the interest charge and the year-end liability for each year of the lease. Round your discounted amounts to the nearest £10.

Solution

The lease liability is measured at the present value of the three payments:

B4 NPV(0.1,B1:B3)

	A	B
1	Lease payment £	10,000
2	Lease payment £	10,000
3	Lease payment £	10,000
4	PV of future lease payments	24,869

	£
20X1 balance b/f	24,869
Interest 10%	2,487
Payment 31/12/X1	(10,000)
20X2 balance b/d	17,356
Interest 10%	1,735
Payment 31/12/X2	(10,000)
20X3 balance b/d	9,091
Interest 10%	909
Payment 31/12/X3	(10,000)
	–



Interactive question 7: Lessee accounting

Glenmore Ltd leases an asset on 1 January 20X1. The terms of the lease are to pay a non-refundable deposit of £500 followed by seven annual instalments of £2,000 payable in arrears. The asset is expected to have a useful life of seven years.

The present value of the future lease payments (excluding the deposit) is £9,736. There is no option to purchase the asset at the end of the lease term.
The interest rate implicit in the lease is 10%.

Requirement

Calculate the value of the lease liability and the right-of-use asset at 31 December 20X1 and 31 December 20X2.

(Round your answers to the nearest pound)

See **Answer** at the end of this chapter.

11.7.15 Instalments in advance

As we have seen in the examples above, interest accrues over time and is included in the payment at the end of each period of borrowing. However, where instalments are **paid in advance**:

- The first instalment **repays capital only** as no time has yet elapsed for interest to accrue.
- At the end of each accounting period the year-end liability will include **capital and interest** that has accrued to date but which has not been paid.
- The initial payment in advance (and/or any deposit) is **not included in the calculation of the lease liability. It is added to the right-of-use asset.**



Worked example: Lease payments in advance

[This is based on IFRS 16 Illustrative example 13.]

A lessee enters into a five-year lease of a building which has a remaining useful life of 10 years. Lease payments are £50,000 per annum, payable at the beginning of each year.

The lessee incurs initial direct costs of £20,000 and receives lease incentives of £5,000. There is no transfer of the asset at the end of the lease and no purchase option.

The interest rate implicit in the lease is not immediately determinable but the lessee's incremental borrowing rate is 5%.

Requirement

Calculate the initial lease liability.

Calculate the value of the right-of-use asset.

Solution

At the commencement date the lessee pays the initial £50,000, incurs the direct costs and receives the lease incentives.

The lease liability is measured at the present value of the **remaining four payments**, ie, the **future lease payments**:

B5 NPV(0.05,B1:B4)

	A	B
1	Lease payment £	50,000
2	Lease payment £	50,000
3	Lease payment £	50,000
4	Lease payment £	50,000
5	PV of future lease payments	177,297

Assets and liabilities will initially be recognised as follows:

		Debit	Credit
Right-of-use asset:		£	£
Initial payment	50,000		
Discounted liability	177,298		
Initial direct costs	20,000		
Incentives received	(5,000)		
		242,298	
Lease liability			177,297
Cash	(50,000 + 20,000 –5,000)		65,000
		242,298	242,298

The asset is subsequently measured at cost less accumulated depreciation and impairment losses unless the right-of-use asset is:

- a class of PPE that is measured using the revaluation model; or
- an investment property and the lessee applies the fair value model.

Where relevant, the asset is depreciated over the shorter of the useful life of the underlying asset and the lease term. Useful life is always the depreciation period when ownership of the underlying asset is transferred at the end of the lease term.

11.7.16 Simplified accounting

A lessee can elect to apply simplified accounting to a lease with a term of twelve months or less, or a lease for a low value asset. The election is made on a lease-by-lease basis for leases for low value assets, and by class of underlying asset for short-term leases.

In this case, the lessee recognises lease payments on a straight-line basis over the lease term.

Interactive question 8: Short lease

Oscar Co is preparing its financial statements for the year ended 30 June 20X6. On 1 May 20X6, Oscar made a payment of £32,000 for an eight-month lease of a milling machine.

Requirement

How should the transaction be recognised in the financial statements?

See **Answer** at the end of this chapter.

11.7.17 Lessor accounting

Lessor accounting is **largely unchanged** under IFRS 16. IFRS 16 retains the IAS 17 distinction between finance leases and operating leases.

Definitions

Finance lease: A lease that transfers substantially all the risks and rewards incidental to ownership of an underlying asset.

Operating lease: A lease that does not transfer substantially all the risks and rewards incidental to ownership of an underlying asset.

Lessor accounting	
Finance lease	**Operating lease**
Substance	
• Risks and rewards with the **lessee** (or other third parties)	• Risks and rewards with the **lessor**
Accounting treatment	
• Recognise a receivable equal to **'net investment in the lease'**. This is the gross investment (minimum lease payments plus any unguaranteed residual value accruing to the lessor) discounted at the interest rate implicit in the lease	• Asset retained in the books of the lessor and is depreciated over its useful life
• Initial direct costs incurred by the lessor are not added separately to the net investment, as they are already included in the discounted figures since they are included in the calculation of the interest rate implicit in the lease (reducing the return)	• Rentals are credited to profit or loss on a straight-line basis **over the lease term** unless another systematic basis is more representative
• Finance income is recognised reflecting constant periodic rate of return on the lessor's net investment outstanding	

11.7.18 Sale and leaseback transactions

Guidance in IFRS 15, *Revenue from Contracts with Customers* (see the chapter Strategic management and brand management) is applied in order to determine whether a sale has taken place.

Transfer is a sale

The seller (lessee) measures the lease asset at an amount equal to the right of use retained, ie, a proportion of the previous carrying amount.

A gain or loss on disposal is therefore calculated based only on the rights transferred to the buyer (lessor).

The buyer (lessor) accounts for the purchase of the asset by applying relevant IFRS and accounts for the lease by applying IFRS 16 guidance on accounting by lessors.

If transfer proceeds do not equal the fair value of the transferred asset or lease payments are not at market rate, the following accounting adjustments are made:

- below market terms are a prepayment of lease payments
- above market terms are additional financing provided by the buyer (lessor) to the seller (lessee)

Transfer is not a sale

The seller (lessee) should not derecognise the asset and instead recognises proceeds as a financial liability, accounting for it in line with IFRS 9.

The buyer (lessor) should not recognise the asset and instead recognises a financial asset, accounting for it in line with IFRS 9.

Technique for sale and leaseback

Measure the right-of-use asset arising from the leaseback at the proportion of the previous carrying amount of the asset that relates to the **right of use retained** by the seller/lessee. **This is calculated as:**

$$\frac{\text{Carrying amount} \times \text{discounted lease payments}}{\text{fair value}}$$

The discounted lease payments are calculated as for any other lease.

Recognise **only the amount of any gain or loss** on the sale that relates to the **rights transferred** to the buyer/lessor. Calculate in three stages:

Stage 1: Calculate gain = fair value (usually = proceeds) less carrying amount

Stage 2: Calculate gain that relates to rights retained:

$$\frac{\text{Gain} \times \text{discounted lease payments}}{\text{fair value}} = \text{Gain relating to rights retained}$$

Stage 3: Gain relating to rights transferred is the balancing figure:

Gain on rights transferred = total gain (Stage 1) less gain on rights retained (Stage 2)



Interactive question 9: Sale and leaseback

On 1 April 20X2, Wigton Co bought an injection moulding machine for £600,000. The carrying amount of the machine as at 31 March 20X3 was £500,000. On 1 April 20X3, Wigton Co sold it to Whitehaven Co for £740,000, its fair value. Wigton Co immediately leased the machine back for five years, the remainder of its useful life, at £160,000 per annum payable in arrears. The present value of the annual lease payments is £700,000 and the transaction satisfies the IFRS 15 criteria to be recognised as a sale.

Requirement

What gain should Wigton Co recognise for the year ended 31 March 20X4 as a result of the sale and leaseback?

See **Answer** at the end of this chapter.

11.7.19 Impact of IFRS 16

The new standard will have an impact on the statement of financial position, statement of profit or loss and statement of cash flows.

11.7.20 Impact on statement of financial position

Companies will be required to report larger amounts of assets and liabilities in their statements of financial position. This will particularly affect companies that currently have much of their leasing commitments off balance sheet in the form of operating leases. For example, many airlines lease their planes and show no assets or liabilities for their future commitments. Under IFRS 16, an airline entering into a lease for an aircraft would show an asset for the 'right to use' the aircraft and an equal liability based on the current value of the lease payments it has promised to make. Possibly, as a result, loan covenants may be breached and have to be renegotiated. This has resulted in some companies entering into shorter leases than previously. By entering into a shorter lease, the lease liability is lower as there are fewer future lease rental cash flows to discount to determine the amount of the lease liability. A lower lease liability therefore reduces the impact on the gearing ratio and loan covenants.

The diagram below, taken from the IASB's IFRS 16 *Project Summary and Feedback Statement*, illustrates the impact:

	IAS 17		IFRS 16
	Finance leases	Operating leases	All leases
Assets		–	
Liabilities	$$	–	$$$$$$$
Off balance sheet rights/ obligations	–	 $$$$$	–

11.7.21 Impact on statement of profit or loss

For companies that previously held material operating leases, the pattern of expenses will change. Specifically, the lease expense will be 'front-loaded'. This is because IFRS 16 replaced the usual straight-line operating lease expense for those applying IAS 17 with a depreciation charge for leased assets, included within operating costs and an interest expense on lease liabilities included within financing costs. While the depreciation charge is usually even, the interest expense is higher in the early years of the lease but reduces over the life of the asset. The total expense will generally be the same but there will be a **difference in expense profile** between IFRS 16 and IAS 17, with a consequent effect on EBITDA and operating profit, as seen in the diagram below, also taken from the IASB's IFRS 16 *Project Summary and Feedback Statement:*

	IAS 17		IFRS 16
	Finance leases	**Operating leases**	**All leases**
Revenue	x	x	x
Operating costs (*excluding depreciation and amortisation*)	–	*Single expense*	–
EBITDA			⇧⇧
Depreciation and amortisation	*Depreciation*	–	*Depreciation*
Operating profit			⇧
Finance costs	*Interest*	–	*Interest*
Profit before tax			⇔

11.7.22 Impact on statement of cash flows

While IFRS 16 does not change the amount of cash paid by the lessee to the lessor, or the total amount of cash flows reported, it will generally change the **presentation of cash flows** in respect of leases previously accounted for as operating leases.

Operating cash outflows, which is where operating lease payments were reported under IAS 17, will **reduce**. However, there will be a corresponding **increase in financing cash outflows**. This is because repayments of the capital component of a lease liability are included within financing activities.

11.8 Debt factoring

Debt factoring is widely used as a source of finance. In a factoring transaction, one party transfers the right to cash collected from its receivables to another party for an immediate cash payment. The main reason that companies choose to factor is that they want to receive cash quickly, rather than waiting the 30 to 60 days that it can take a customer to pay.

When a company factors an invoice, the factor advances a percentage of that invoice value, retaining the rest of the amount as a security for any bad debts. It will then pay the company the balance of the invoices, minus its fees, after it collects payment from the customer. The cash advance rate will vary depending on the industry, typically ranging from 80% of an invoice value to as much as 95%.

Such transactions need to be analysed to determine whether or not the receivables need to be derecognised in accordance with IFRS 9. When determining the recognition of these transferred receivables in the financial accounts of the transferor, there are three options:

- continuing recognition of the receivables
- partial derecognition of the receivables
- derecognition of the receivables (similar to recording a sale of the receivables)

Factoring arrangements are either '**with recourse'** or '**without recourse'**.

In **factoring with recourse**, the **transferor fully or partially guarantees** the performance of the receivables. The transferor has not therefore fully transferred the credit risk to another party. In most factoring with recourse transactions, the transferor does not allow the transferee to sell

the receivables, in which case the transferor still retains control over the asset. In this case the criteria for derecognition are not satisfied and the asset should not be derecognised; instead the amount advanced by the factoring company should be treated as a financial liability in the accounts (effectively a loan from the factor).

In **factoring without recourse**, the **transferor does not provide any such guarantees** about the performance of the receivables. The factoring company will bear the loss of any bad debts. In such a transaction, the entity has transferred the risks and rewards of the ownership of the receivables, and should derecognise them accordingly. Because the risk has been transferred from the transferor to the factor, 'without recourse' factoring is significantly more expensive than 'with recourse' factoring.

In practice, most factoring arrangements result in substantially all the risks and rewards being either transferred or retained, so the 'partial derecognition' noted above will not often apply.

When calculating the costs of factoring, there are two main costs involved: Interest – typical interest charges range from 1.5% to 3% over base rate.
Fees – covering administration and credit control, and typically 0.75% to 2.5% of turnover. In 'without recourse' agreements, there will also be a charge for the cost of bearing the risk of bad debts.



Context example: Factoring with recourse

Goods plc is a trading company. Faced with an urgent cash shortage, it decides to transfer its trade receivables of $300,000 to a factoring company for 80% of their value, for a fee of 8%. The factoring company has the full right to return the receivables back to Goods plc if they become uncollectible.

In this case, Goods plc has transferred the right to receive the receivables to the factoring company, but it still retains some of the risk resulting from the receivables to the factoring company. The credit risk associated with the clients was not transferred, because the factor has the right of return. As a result, Goods plc continues to recognise the receivables in the balance sheet, because the derecognition criteria in IFRS 9 have not been met. The receivables transferred to the factoring company are recognised as a financial liability (effectively, a loan from the factor).

Journal entries are:

- Debit Cash ($300,000 × 80%): $240,000
- Debit Profit or loss – fee: $24,000
- Debit Outstanding receivable from factor upon collection from customers: $36,000
- Credit Financial liability: $300,000

Assume that customers pay a total of $240,000 to the factor, representing a bad debt of $60,000. Journal entries under a with recourse agreement will be:

(a) Based on reports sent by the factoring company, Goods plc reduces its financial liability.
  - Debit Financial liability to the factor: $240,000
  - Credit Receivables: $240,000

(b) If the clients do not pay the remaining $60,000, the factoring company can apply its recourse right:
  - Debit Financial liability: $60,000 (the remaining amount of uncollected receivable)
  - Credit Outstanding receivable from factor: $36,000
  - Credit Cash: $24,000

(c) The uncollected $60,000 needs to be written off through the profit and loss account:
  - Debit Profit and loss – bad debts: $60,000 (the amount of uncollected receivable)
  - Credit Receivables: $60,000

Context example: Factoring without recourse

Assuming now a 'without recourse' agreement, the total transferred receivables again amount to $300,000, but Goods plc transfers them for 80% of their value and a fee of 12%.

Goods plc has transferred all the risks and rewards resulting from the receivables to the factoring company. As a result, it derecognises the receivables fully, because the derecognition criteria in IFRS 9 are met.

Journal entries are:

- Debit Bank account ($300 000 × 80%): $240,000
- Debit Profit or loss – fee: $36,000
- Debit Outstanding receivable from factor upon collection from customers: $24,000
- Credit Receivables: $300,000

If the clients do not pay the whole amount to the factoring company, the factoring company cannot apply to Goods plc for it to be reimbursed. On the other hand, the fees are higher than when factoring is 'with recourse'.

Interactive question 10: Factoring with and without recourse

On 1 January 20X7 Kitchen plc factored its accounts receivable of $200,000 at a fee of 8%. Under the terms of the agreement, the company received $165,000 in cash and the rest of the amount was retained by the factor as a security for any bad debts. Any excess of this security sum over the total bad debts was agreed to be returned by the factor on 31 December 20X7.

On 31 December 20X7 the full amount of the security sum was withheld because the actual bad debts totalled $20,000. Kitchen plc had already made a provision for doubtful debts of $25,000.

Requirement

Show the journal entries to record the transactions for Kitchen plc, both with and without recourse.

See **Answer** at the end of this chapter.

Summary

	Tick off
Retained funds are a more important source of equity funding than share (rights or new) issues.	☐
UK stock markets are the London Stock Exchange, AIM and the ISDX. Internationally the New York Stock Exchange, which has merged with Euronext, is most influential.	☐
Bonds are an important source of finance for companies. The UK domestic market in bonds has become more active in recent years, and larger firms can use Eurobonds, bonds available internationally, denominated in another currency.	☐
Bonds can be priced at par, at a premium or at a discount. As interest rates rise, the price of bonds falls and vice versa.	☐
Lenders are exposed to credit risks, which is the risk that borrowers will default on interest and/or principal repayments. Credit spread is the premium required by the lender to compensate for the credit risk.	☐
Rather than buying an asset outright, using either available cash resources or borrowed funds, a business may lease an asset.	☐
Derivatives include forward contracts, forward rate agreements, futures, options, swaps and credit derivatives. Derivative trading is carried out both over the counter and on exchanges.	☐
The use of financial derivatives has implications for financial reporting and profitability.	☐
Derivative markets include the FX markets for exchange rates and money markets for short-term interest rates.	☐

Further question practice

1 Knowledge diagnostic

Before you move on to question practice, complete the following knowledge diagnostic and check you are able to confirm you possess the following essential learning from this chapter. If not, you are advised to revisit the relevant learning from the topic indicated.

Confirm your learning	
1.	Can you correctly classify a financial instrument as debt or equity in accordance with IAS 32, *Financial Instruments: Presentation?* (Topic 2)
2.	Do you know the procedures for obtaining a listing on an international stock exchange such as the London or New York Stock Exchange? (Topic 3)
3.	Do you know the main elements of a futures transaction? (Topic 9)
4.	Do you know the main difference between hedging using exchange-traded products and OTC transactions? (Topic 10)
5.	Is your corporate reporting knowledge of financial instruments up to date? (Topic 11)

2 Question practice

Aim to complete all self-test questions at the end of this chapter. The following self-test questions are particularly helpful to further topic understanding and guide skills application before you proceed.

Question	Learning benefit from attempting this question
Question 2 - Bonds and covenants	This is a good introductory question covering bonds and covenants. It is a revision of what you would have covered in your Financial Management studies. Work through this question carefully before attempting exam standard questions.
Question 3 - IFRS 9	This question covers the revised approach to revenue recognition under IFRS 9, *Financial Instruments*. Before attempting this question make sure your knowledge of IFRS 9 is up to date.
Question 4 - Wallace plc	This is a good question to practice the treatment of leasing under IFRS 16, *Leasing*. Refer back to your corporate reporting notes if you need to.

Refer back to the learning in this chapter for any questions which you did not answer correctly or where the suggested solution has not provided sufficient explanation to answer all your ueries. Once you have attempted these questions, you can continue your studies by moving on to the next chapter.

Technical reference

1 IAS 32, *Financial Instruments: Presentation*

- IAS 32 outlines the requirements for the presentation of financial instruments, particularly the classification of these instruments into financial assets, financial liabilities and equity instruments. The standard also provides guidance on the classification of interest, dividends and gains or losses relating to the financial instruments.

2 IFRS 9, *Financial Instruments*

- IFRS 9 forms a complete replacement for IAS 39 from 1 January 2018. IFRS 9 sets out the recognition and measurement requirements for financial instruments and some contracts to buy or sell non-financial items.

3 IFRS 7, *Financial Instruments: Disclosures*

- IFRS 7 prescribes disclosure about the significance of financial instruments to an entity's position and performance, and about the nature and extent of the risks arising from those financial instruments, in both quantitative and qualitative terms.

4 IFRS 13, *Fair Value Measurement*

- Defines fair value as the price that would be received to sell an asset or would be paid to transfer a liability in an orderly transaction between market participants at the measurement date. IFRS 13 defines fair value on the basis of an 'exit price' notion and uses a 'fair value hierarchy' which results in a market-based measurement rather than an entity-specific one.

5 IFRS 2, *Share-based Payment*

- The objective of IFRS 2 is to specify the financial reporting by an entity when it undertakes a share- based payment transaction. IFRS 2 requires an entity to recognise share-based payment transactions (such as shares granted and share options) in its financial statements. This includes transactions with employees or other parties, to be settled in cash, other assets, or equity instruments of the entity.

6 IFRS 16, Leases

- Effective from 1 January 2019 IFRS 16 effectively abolishes the distinction between operating and finance leases for lessees, although not for lessors. It introduces a single lessee accounting model and requires a lessee to recognise assets and liabilities for all leases with a term of more than 12 months, unless the underlying asset is of low value.

Self-test questions

Answer the following questions.

1 Multimedia company

You have been appointed as the chief financial officer of a multimedia company which is financed by private equity. There is considerable public interest in the company and it continues a very rapid rate of growth under the leadership of its dynamic founder and chief executive officer Martin Pickle.

Martin Pickle owns over 30% of the company's equity and has also loaned the business substantial sums to sustain its overseas development. The balance of the other investors consist of some small shareholdings held by current and past employees and the remainder is in the hands of a private equity company which is represented by two directors on the board.

Recent financial information:

Year ending 31 December	Profit for the year after interest and tax	Investment in projects or capital expenditure	Dividend paid
	$m	$m	$m
20X5	18	–	9
20X6	21	30	6
20X7	30	–	15
20X8	33	45	6
20X9	48	–	24

After two months in the job you are called to a meeting with Martin Pickle and the company's non- executive chairman. In that time you have made significant progress in improving the financial controls of the business and the current year end, which is three weeks away, looks very promising. The company's underlying earnings growth promises to be in excess of 20% and its cash generation is strong. The CEO tells you that he would like you to put together a plan to take the company to full listing as a first step to him undertaking a substantial reduction in his financial stake in the business.

Requirements

1.1 Prepare a board paper, describing the procedure for obtaining a listing on an international stock exchange such as the London or New York Stock Exchange.

1.2 Prepare a briefing note, itemising the advantages and disadvantages of such a step for a medium-sized company.

2 Conflicts of interest

2.1 Discuss why conflicts of interest might exist between shareholders and bondholders.

2.2 Provide examples of covenants that might be attached to bonds, and briefly discuss the advantages and disadvantages to companies of covenants.

3 Mortgage

A bank makes mortgage loans to clients. Interest charged to these clients is SONIA (London Interbank Offered Rate) + 1%, reset monthly.

The bank recognises that in its portfolio of clients there will be some clients who will experience financial difficulties in the future and will not be able to keep up mortgage payments.

Under the mortgage agreement, the bank takes first legal charge over the mortgaged property and, in the event of a default where payments cannot be rescheduled, the property would be sold to cover unpaid debts.

Requirement

Discuss how revenue relating to the above would be accounted for under the revised approach to amortised cost and impairment in IFRS 9.

4 Wallace plc

Wallace plc obtains the use of an asset on a ten-year lease.

The terms of the lease require Wallace plc to make 10 annual lease payments of £25,000 in arrears. The discounted value of the lease payments at the implicit rate within the lease of 5% is £193,050.

There is an option to buy the machine on the final date of the lease for £20,000 (discounted value of £12,280), which Wallace plc is expected to exercise.

The useful life of the machine is 15 years.

Calculate the following values:

Requirements

4.1 The initial measurement of the right-of-use asset at the commencement of the lease on 1 January 20X1.

4.2 The carrying amount of the machine at 31 December 20X1.

4.3 The lease liability brought forward at 1 January 20X2.

Now go back to the Introduction and ensure that you have achieved the Learning outcomes listed for this chapter.

Answers to Interactive questions

Answer to Interactive question 1

1.1 Conversion ratio is £100 bond = 30 ordinary shares

Conversion value = 30 × £4.15 = £124.50

1.2 Conversion premium = £(142 – 124.50) = £17.50

$$\text{or } \frac{17.50}{124.50} \times 100\% = 14\%$$

The share price would have to rise by 14% before the conversion rights became attractive.

Answer to Interactive question 2

2.1 **Call option is deeply out of the money**

The investment in shares is derecognised, as the transferor has transferred substantially all the risks and rewards of ownership.

Call option is deeply in the money

Since the transferred asset can be called back by the transferor and the call option is deeply in the money, the transfer does not qualify for derecognition. The transferor has retained substantially all the risks and rewards of ownership.

Since the call option prevents transfer of financial assets from being accounted for as a sale, the call option is not separately recognised as a derivative asset as otherwise it would result in recognising the same rights twice.

Call option is neither deeply in the money nor deeply out of the money

If the asset is readily obtainable in the market, the asset is derecognised. This is because the bank has neither retained nor transferred substantially all the risks and rewards of ownership, and has not retained control.

If the asset is not readily obtainable in the market, derecognition is precluded to the extent of the amount of the asset that is subject to the call option because the entity has retained control of the asset.

2.2 Access Bank sells a bond and enters into a 'total return swap' with the buyer. As a result of the 'total return swap', the bank retains all the risks and rewards of ownership and hence the transfer does not qualify for derecognition. The 'total return swap' is not separately recognised, since it prevents transfer of the bond being accounted for as a sale.

Answer to Interactive question 3

In substance, the effect of the two transactions is an interest rate swap with no initial investment. This therefore meets the definition of a derivative since there is no initial net investment, an underlying variable is present and future settlement will take place. This would be the same even if no netting agreement existed because the definition of a derivative does not include a requirement for net settlement.

Answer to Interactive question 4

Find (d_1) and (d_2)

$$d_1 = \frac{\ln(3.50/3.30) + (0.25 \times 0.08)}{\sqrt{0.12}\sqrt{0.25}} + \frac{\sqrt{0.12}\sqrt{0.25}}{2}$$

= 0.4552 + 0.0866

= 0.5418

$$d_2 = 0.5418 - \sqrt{(0.12}\sqrt{0.25)}$$

= 0.5418 – 0.1732

= 0.3686

Find N (d_1) and N (d_2) using normal distribution tables

N (0.5418) = 0.5 + 0.2060 = 0.7060

N (0.3686) = 0.5 + 0.1438 = 0.6438

Using the Black-Scholes formula

$C_0 = (3.50 \times 0.7060) - ((3.30e^{-0.25 \times 0.08}) \times 0.6438)$

= 2.4710 – 2.0825

= 38.85p

Answer to Interactive question 5

(a) AB Co should derecognise the asset. Its option is to repurchase the shares at their then fair value, so it has transferred substantially all the risks and rewards of ownership.

(b) CD Co should not derecognise the asset, as it has retained substantially all the risks and rewards of ownership.

(c) EF Co should not derecognise the asset, as it has retained substantially all the risks and rewards of ownership. The investment should be retained in its books even though legal title is temporarily transferred.

Answer to Interactive question 6

The loans are now in Stage 3. Lifetime credit losses should continue to be recognised, and interest revenue should switch to a net interest basis, that is on the carrying amount net of allowance for credit losses.

Answer to Interactive question 7

Calculations

Calculation of the right-of-use asset

	£
PVFLP	9,736
Deposit	500
	10,236

Depreciate the asset over seven years

10,236/7 = £1,462 per annum

Carrying amount at 31 December 20X1: £10,236 – £1,462 = £8,774

Carrying amount at 31 December 20X2: £8,774 – £1,462 = £7,312

Calculation of the lease liability

Balance of the lease liability at 31 December 20X1: £9,260

Balance of the lease liability at 31 December 20X2: £8,186

	CR Bal b/f 1 January	CR Interest accrued at 10%	DR Payment 31 December	CR Bal c/f 31 December
	£	£		
20X1	10,236	1,024	(2,000)	9,260
20X2	9,260	926	(2,000)	8,186

Note: (You can do these calculations in vertical format (as in the chapter body) or horizontal, whichever is easiest. The principles are the same.)

Answer to Interactive question 8

The lease is for eight months, which counts as a short-term lease, and so it does not need to be recognised in the statement of financial position. The amount charged to profit or loss for the year ended 30 June 20X6 is therefore £32,000 × 2/8 = £8,000.

Answer to Interactive question 9

The answer is £12,973

Stage 1: Gain on sale: £740,000 – £500,000 = £240,000

Stage 2: Gain relating to rights retained = £(240,000 × 700,000/740,000) = £227,027

Stage 3: Gain relating to rights transferred = £240,000 – £227,027 = £12,973

Answer to Interactive question 10

1 January 20X7:

Without recourse, the accounts receivable from customers have been derecognised:

	Dr $	Cr $
Cash	165,000	
Expenses [8% × $200,000]	16,000	
Remaining due from factor	19,000	
Accounts receivable		200,000

With recourse, the liability to the factoring company is recognised:

	Dr	Cr
	$	$
Cash	165,000	
Expenses [8% × $200,000]	16,000	
Remaining due from factor	19,000	
Financial liability		200,000

31 December 20X7: The journal entries will differ under the two types of factoring.

Under factoring with recourse, Kitchen plc must pay the factor an additional $1,000, since the actual bad debts exceeded the amount initially retained by the factor.

Under factoring with recourse:

	Dr	Cr
	$	$
Provision for bad debts	20,000	
Remaining due from factor		19,000
Cash		1,000

There is no such remedy for the factor if the factoring is without recourse. Under factoring without recourse:

	Dr	Cr
	$	$
Provision for bad debts	19,000	
Remaining due from factor		19,000

Answers to Self-test questions

1 Multimedia company

1.1 Procedure for obtaining a listing on an International Stock Exchange – Paper prepared for the Board of Multimedia Company

This paper describes the necessary procedures for obtaining a listing on an international stock exchange.

Obtaining a listing on an international stock exchange such as the London or New York Stock Exchange consists of satisfying requirements in **three broad areas**: namely **registration, listing and admission to trading**.

In the UK, a firm seeking a listing must first **register** as a public limited company to ensure that it is entitled to issue shares to the public. This will require a change to the company's memorandum and articles of association agreed by the existing shareholders at a special meeting convened for the express purpose of agreeing this change.

The company must meet the regulatory requirements of the **Listing Authority** which is part of the Financial Conduct Authority in the UK. These regulatory requirements impose minimum size restrictions on a company and other conditions concerning length of time trading (normally audited accounts must exist for a three-year period). An issue of particular concern here is that the CEO has a controlling interest in the business – this will have to be addressed. Once these requirements are satisfied the company is placed on an official list and is allowed to make an initial public offering of its shares.

Once the company is on the official list it must then seek approval from the stock exchange to allow its shares to be traded. Exchanges such as the London Stock Exchange impose strict requirements which invariably mean that the applicant company will need the services of a sponsoring firm specialising in this area. The regulatory requirements of the London Stock Exchange include:

- compliance with corporate governance regulations (for example, 50% of the board should be independent non-executive directors)
- quality and experience of the executive directors and the business plan, all of which must be carefully laid out in a prospectus

This tends to be costly and therefore prohibitive for all but the larger companies. The restrictions (and costs) of obtaining a listing on a junior market (eg, AIM in the UK) may be lower.

1.2 Briefing note prepared for Martin Pickle, CEO

Subject – Advantages and disadvantages of a stock market listing for a medium-sized company

Following on from our discussion regarding a stock market listing for the company, I have detailed below the main advantages and disadvantages of taking such a step for a company of our size.

For a **medium-sized firm**, the principal advantage of obtaining a public listing is the **additional sources of finance** available. A listed company would have access to **equity capital** from both institutional and private investors, and the sums that can be raised are usually much greater than through private equity sources. In addition, the presence of a firm as a limited company

on a major exchange such as the London Stock Exchange enhances the **credibility** of the firm both to potential investors and to the general public as it has opened itself to a much greater degree of public scrutiny than a privately financed firm.

The **disadvantages** are significant; the distributed shareholding places the firm in the market for investors seeking corporate control and also increases the likelihood that the firm will be subject to a takeover bid. The higher degree of public scrutiny imposes a significant **regulatory burden** on the firm, as it must comply with a range of **disclosure requirements** and financial accounts must be prepared in accordance with relevant accounting standards. In the UK, this means in accordance with IFRS and the relevant GAAP as well as the Companies Acts. Under

the rules of the London Stock Exchange, companies must also comply with the governance requirements of the **UK Corporate Governance Code** and have an effective and ongoing business planning process in place. The requirement to comply or explain can impose a significant regulatory burden and can expose the company to critical comment.

2 Conflicts of interest

2.1 Conflicts of interest:

(1) Different attitudes to risk and return

Shareholders may want the company to undertake **risky projects** with correspondingly **high expected levels of returns**. Bondholders will want the company to undertake projects that guarantee sufficient returns to pay their interest each year, and ultimately to repay their loans.

(2) Dividends

Large (albeit) legal dividends may be preferred by shareholders, but may concern bondholders, because the payments leave low cash balances in the company and hence **put at risk** the **company's ability** to meet its commitments to the bondholders.

(3) Priority in insolvency

Bondholders may wish to take the company into **liquidation** if there are problems paying their interest, to guarantee their investment. Shareholders, however, may wish the company to **continue trading** if they expect to receive nothing if the company does go into liquidation.

(4) Attitudes to further finance

Shareholders may prefer the company to **raise additional finance** by means of loans, in order to **avoid having to contribute themselves** in a rights issue, or the risk of dilution of their shareholding and hence power if an open stock market issue is made. Bondholders may **not wish the company** to take on the burden of additional debt finance, because it may increase the risk that the **interest** that they are **due** will not be paid, or the company will have problems repaying its loans, particularly if the new loans rank above theirs.

(5) Restrictions imposed by bondholders

Restrictions imposed by bondholders to protect their loans, such as charges preventing the company from selling assets or covenants, may limit the company's ability to maximise returns for shareholders.

(6) Bankruptcy costs

If the **costs of bankruptcy**, such as receivers and lawyers' fees, are likely to be **significant**, bondholders may be much less willing than shareholders for the company to bear any risk. Significant bankruptcy costs may mean that there is insufficient money left to repay their loans.

2.2 A covenant in a loan agreement is an obligation placed on the borrower over and above repaying the loan according to terms.

The various types of covenant are:

Positive covenants

These require a borrower to do something, for example to **provide the bank** with its regular management accounts. This would allow the bank to **check** on the **financial performance** of the company, and to ensure that it is likely to be able to repay the loan as planned.

Loan covenants

The borrower may pledge not to take out further loans until the current loan has been repaid, or not to take further loans ranking above existing loans. The purpose of this is to protect the position of the lender, and to ensure that the risk of default is not increased, or the level of security diluted.

Asset covenants

There may be restrictions on the company's ability to acquire or **dispose of assets**.

Accounting covenants

These set **limitations on the borrower's financial position**. For example, the company might agree that its total borrowings should not exceed 100% of shareholders' funds.

The purpose of this is to keep the gearing, and hence the level of risk to the lender, within certain limits.

Dividend covenants

Covenants may **restrict the levels of dividends** borrowers can pay, or restrict the company's ability to purchase its own shares.

Investment covenants

Covenants may **limit the investments** the borrower can undertake, or prevent the borrower merging or making an acquisition.

Repayment covenants

The lender may be required to **build up a fund over time** to be drawn on to redeem bonds at the end of their life.

Advantages of covenants

(1) The main advantage of covenants is that lenders may be prepared to lend **more money** to the company if it provides the security of a covenant.

(2) Covenants may mean that the costs at which the company can **borrow money** are **lower**.

Disadvantages of covenants

(1) The main disadvantage of a covenant is that the company's actions may be **constrained**; it may not be able to raise further funds beyond the covenanted loans or undertake profitable investments.

(2) Covenants may require the borrower to bear **monitoring costs** such as provision of information, auditors' fees and trustee expenses.

3 Mortgage

The revenue associated with mortgage loan assets is the interest receivable. Under IFRS 9, the variable element of the interest is accrued on a time basis and the fixed element is reduced to reflect any initial transaction costs and to reflect a constant return on the balance outstanding.

4 Wallace plc

4.1 Right-of-use asset

	£
Discounted future lease payments	193,050
Option to purchase (discounted)	12,280
Lease liability and right-of-use asset	205,330

4.2 Carrying amount at 31 December 20X1

Right-of-use asset	205,330
Less Depreciation (W1)	(13,689)
Carrying amount	191,641

Working: 205,330/15 years = £13,689

The asset is depreciated over 15 years as there is an option to purchase the asset which Wallace plc is expected to exercise.

4.3 Lease liability brought forward at 1 January 20X2

Lease liability at 1.1.X1	205,330
Interest at 5%	10,267
Less payment	(25,000)
Liability	190,597

CA BANGLADESH

THE INSTITUTE OF CHARTERED ACCOUNTANTS OF BANGLADESH

CA Bhaban
100 Kazi Nazrul Islam Avenue
Dhaka 1215, Bangladesh

880 2 9115340, 9612612100
9117521, 9137847 (Off.)

880 2 9125266

ceo@icab.org.bd

facebook.com/ca.bangladesh

icab.org.bd